Real Analysis

— *EYNTKA* Series —

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August 5, 2026

Contents

The goal of this document is to encapsulate all the notes and intuition I will gain taking MAT457. The preliminary chapter will be a quick reminder on why we care to generization our notion of integrability.

Assumptions in reading this text

1. basic definition and notions in topology are assumed, covering at least:
   (a) open/closed sets, closure and interior, Hausdorff space, limit definition of closure (given the space if Hausdroff, or at least T_1), continuity
   (b) Compactness, connectedness, Heine-Borel Bolazno-Weiestass in $\mathbb{R}^n$, Path-connectedness
   (c) subspace topology, product topology, Quotient topology
   (d) Metric spaces, many equivalent metric on product spaces,
2. A good understanding of Riemann integration
3. A basic understanding of point-wise convergence and uniform convergence. For point-wise convergence, we'll use $f_n \to f$ notation. For uniform convergence, we'll use $f_n \rightrightarrows f$.
4. Defining and understanding basic properties of sup, inf, lim sup, and lim inf. In particular, remember that if S is a set and $M = \sup S$, for all $\epsilon > 0$, there exists an $s \in S$ such that

$$M < s + \epsilon$$

   And similarly for the infimum, with $s < m + \epsilon$.

Since we are doing analysis, the axiom of choice is assumed. There are some fascinating results we get by assuming the axiom of choice which we will point out.

What is Analysis?

Now that I've taken the course and have done some reserach in PDE's I feel like I've gained a perspective on how to think about analysis. Fundamentally, Analysis is about *estimates* and *approximation*, especially on functions; a lot of the fundamental results in anlysis is centered around these two concepts. Since we are working over $\mathbb{R}$, we have the privelge of having an *ordering*, allowing us make precise the notion of a number being bigger than another (something we can't do over $\mathbb{C}$ or many other fields). The following bullet points elaborates on this:

1. Many times, we will want to give a numerical value to a function or a space of functions. To accomplish this, we will often define a functional (or subfounctional) a space of functions. For example, the integral $\int : C^\infty(\mathbb{R}^n) \to \mathbb{R}_{>0}$ takes in functions and assigns them numerical value giving us some idea of how fast the function grows and the total "energy", "work", or "total value" the function has. Notie that the function $\int$ as we've defiend it above is not well-defined, there are function whose integral is infinite. Conversely, we are missing many integrable function (recall if a function is merely almost everywehre continuous, it is integrable). We will refine the domain in this book.

   The integral is not the only functional. We will soon find there are in fact many functions, letting us control many properties of a function such as:

(a) the total boundedness of the function (i.e. that it nowhere goes to infinity)

(b) the "total work" of the function (i.e. a finite integral)

(c) the speed decay of a function (i.e. that it goes to zero fast enough)

(d) the "frequency" of a function (i.e. that the function doesn't start oscillating too rapidly)

(e) the regularity of a function (i.e. that the derivative or derivatives dont' explode too fast or oscilate to fast)

2. interpolation: (I'm thinking of things like $L^p + L^q$ or $L^p \cap L^q$ and how useful this is and how it shows up)

3. "perfect approximatinos" via sequences: At some point, we all must have wondered how in the world can we find the area of a curved shape(?!). We found out when learning through Riemann integrals that the key idea is that we can approximate it with rectangles, and then make the approximation "perfect". This idea of being able to find some simpler objects and use them to approximate more complicated ones is at the heart of analysis. For example, those rectangles we used in Riemann integration can be thought of as characteristic functions. Thus, something is Riemann integrable if the "upper" and "lower" characteristic functions, when taking the limit of the partitions, are equal.

   More generally, we will often be working over many different types of functions (most of this book will in fact be about how to classify different function space). One key part of working with function spaces will often be to find a subset of those functions that we can use to approximate any other function: such a set is called a *dense set*. Dense sets are sort of like a basis in vector-spaces: just like any vector in a k-vector space is a k-linear combination of basis elements, elements in most function space will be the result of a *sequence of functions* where each function in the sequence comes from the dense set.

   Two major examples of this is in the theory of integration and Fourier analysis. In integration, we will

   approximate any integrable function via *characteristic functions*, and in the theory of Fourier transforms, we will approximate integrable functions via its *fourier series*, where each function in the fourier series will be part of a special dense set we will define later.

4. Estimates: having given numerical values to function and having interpreted them through interpolation or a good sequence, we can now often solve problems by bounding the "badness" of the thing we are measuring. A super-simple example is that if f is bounded, then there exists a constant C such that

   $$|f(x)| \leq C$$

   for all x in the codomain. Naturally, as we do more analysis, the estimates will be more sophisticated, but the idea remains the same: we will simplify our problems by bounding what we is bad and trying to "eliminate" it. One of the ways we will do this is by figuring out how common operations (such as $\frac{d}{dx}$, $\int$, $+, \cdot$, and so forth) which usually "complicate" our problem can be eliminated with some appropriate estiamtes. For example, let's say f is a bounded function, and we want to work with:

   $$\left| \int_D f \right|$$

   then this can be a hard value to find. Howver, we can eliminte the complexity by doing:

   $$\left| \int_D f \right| \leq \int_D |f| \leq \int_D C$$

which naturally now is much easier to work with. Of course, we might have over-simplified our problem with this estimate, which is why we will have a lot of different estimates playing different roles depending on our situation.

In terms of the theory of differentiation, real differentiable functions are very different from complex differentaible function. Real differentiable function are a strictly local property, while being complex-differentiable (holomorphic) entails many global properties (namely, they are *harmonic*, and give *closed forms*). Real smooth functions also out-number holomorphic functions (leading to their prominence in Distribution Theory), again due to the harmonic properties of holomorhpic functions being so restrictive (all holomorphic functions are locally a power series with coefficients given by contour integrals of any size, showing how much "global" information is captured in local representatives). This also means that holomorphic functions have some nicer properties (for example, they are all analytic functions), and lead to important results in PDE's with wave equations and algebraic geometry, and they can also be *factored* into an infinite polynomial up to an entire holomorphic function[1] (an impossible thoerem for real smooth functions due to their much greater flexibility). But these reaons also show why complex analysis is a separate field of study, moving more towards sheaf theory and analytic number theory rather than PDE's and harmonic analysis[2].

[1] see Weiestrass Factorization Theorem

[2] Though there will be some interesting motivation for harmonic analysis coming form elementary complex analysis

Preliminary Integration

In pre-university education, it is taught how to find the area of some simple shapes like rectangles, triangles, cubes, spheres, and so on. Later on, you learn how many of these shapes can be captured as the area under a particular graph (the graph usually being continuous). This means that the notion of area can be represented by functions (in particular the graph of a functions), and so it becomes a meaningful question to ask about how to find the area under the graph.

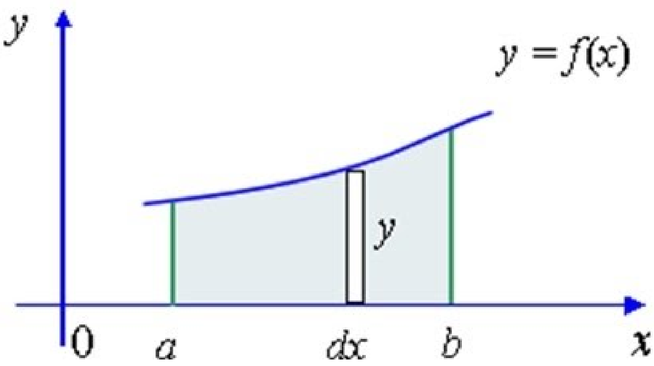


Figure 1: integrating under a graph

Let's say that f is the 1 (or more) dimensional continuous function over $\mathbb{R}$ (or $\mathbb{R}^n$), and we are trying to find the area under the graph. The first attempt at this might be the following (called Darboux sums): Given a partition the domain $P = \{x_1, x_2, ..., x_n\}$, we can make many rectangles which are all easy to measure. Then, as the partition gets finner (size$(P) \to 0$, $|P| \to \infty$), we will get a better an better approximation of the area.

If we take this approach, we will introduce some level of choice: there are many ways in which this partition can be formed, as well as a choice of the "height" of the rectangle. For example, you can choose the lowest or highest point:

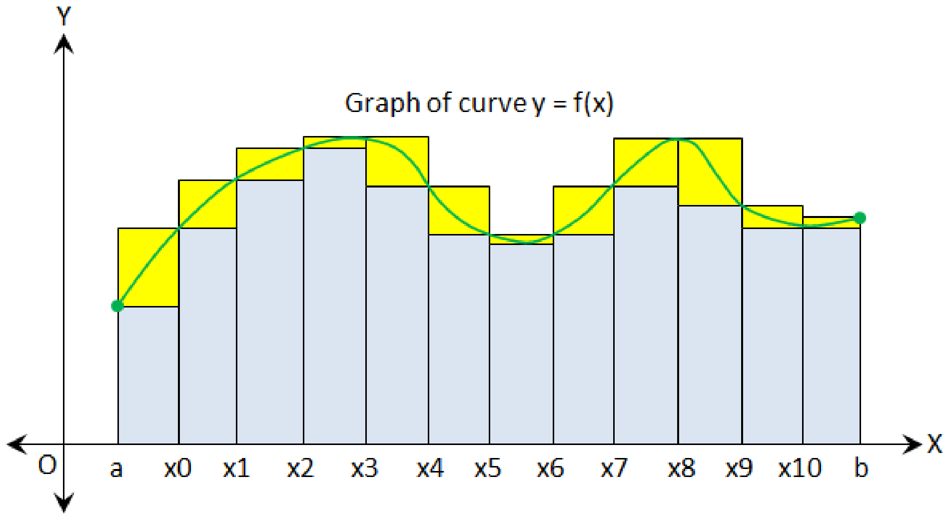


Figure 2: lower and upper sums

You can also choose any points in between, though what "in between" means starts to get hard to define in higher dimensional spaces where different orders can be put onto the "surfaces" (or hyper-surfaces) of the top of the rectangles (i.e. hyper-rectangles). However, maximal and minimal point is always well defined on these surfaces (or better, inf and sup always being well defined on these bounded surfaces), and even better, choosing the maximal point for each rectangle height upper-bounds all possible other choices of Darboux sums, and the minimal point for each rectangle's height lower-bounds all possible choices of Darboux sums. To be more precise, we'll replace the maximum and minimum condition with sup and inf, in which case the resulting sum is called the *Riemann sum.*

Intuitively, the upper and lower Riemann sums must match up and exist as size$(P) \to 0$ and $|P| \to \infty$. It would be strange if they don't match up, and so we can (as a start) define a function that is not "too weird" as one where the upper and lower Darboux sums match as size$(P) \to 0$, that is

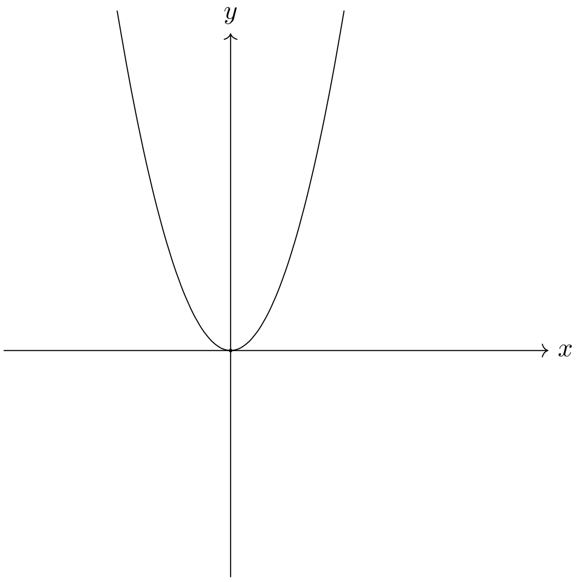
$$\lim_{n\to\infty} \sum_{i=1}^{n-1} \left(\inf_{[x_i, x_i+1]} f(x) \right) \cdot (x_{i+1} - x_i) = \lim_{n\to\infty} \sum_{i=1}^{n-1} \left(\sup_{[x_i, x_i+1]} f(x) \right) \cdot (x_{i+1} - x_i)$$

If this is the case, f is said to be *Riemann integrable.* Fortunately, it turns out that if f is Riemann integrable on one partition, it is Riemann integrable on all partitions, and so we don't need to "remember" our initial choice of partition when we say f is Riemann integrable (we don't say "f is Riemann integrable on partition p, while f is not Riemann integrable on partition q").

All continuous functions over a bounded domains will be integrable (so trigonometric functions, polynomials, log, $\sqrt{}$ over the appropriate domain, and so forth). Since any geometric shape learnt in pre-university education can be bounded by a continuous function, we have just generalized the idea of measuring those areas[3]. In fact, being continuous *almost* encapsulates all possible functions. To understand the generalization, consider

$$f_{\text{no }1} : [-1, 1] \to \mathbb{R}, \qquad f_{\text{no }1}(x) \begin{cases} x^3 & x \neq 1 \\ 0 & x = 1 \end{cases}$$

Notice that there is a gap in the line



however, this gap has "zero measure", we would think that the result should be the same. And in fact it is! The function $f_{\text{no }1}$ is also Riemann integrable (as can be checked) and will have the same value as x^3 on $[-1, 1]$. In general, if finitely (or even countably) many points are ommited, then f will still have the same measure. Since $f_{\text{no }1}$ is no longer continuous, but it as just off by some finite number of points, we say that f is continuous *almost everywhere* [4]. It should be reviewed that f is Riemann integrable if and only if f is continuous almost everywhere.

Stepping aside from measuring area under a graph for a moment, we turn to the study of another very common notion in analysis: limits. Limits are of fundamental importance in Analysis. They are the tool used to construct or define many different types of analytical objects (ex. $\mathbb{R}$, continuity, differentiability, Fourier transforms, sequences and series). More generally, the concept (and formalisation) of a limit is how mathematicians get to study infinites which are "well-defined". In my mind, limits are one of the distinguishing feature between algebra and analysis: many algebraic properties "break down" or are different when we try to bring in more than finitely many objects (ex. products of infinite groups cannot be direct products in **Grp** since $1 + 1 + \cdots$ is not defined (hence they are direct sums), the dimension of $\mathbb{Q}[x]$ and $\mathbb{Q}[[x]]$ as $\mathbb{Q}$-algebras differ, etc.). A good definition of limit is important to be able to define objects "at infinity" (or similarly, at the "infinitely small"), since if it is not well defined, then we can make the "result" non-unique.

Here is a silly example of that: what is $\infty - \infty$? Maybe we can say that the answer is 0? or ∞? If we are not careful with our definition, it can actually be any number. First, "notice" that

[3]though we won't go into the details of finding those areas

[4]Later, almost every continuous will mean continuous up to a zero measure set once we properly define a measure

$1+1/2+1/3+\cdots$ "$='' \infty$, or more precisely, the series $S_n = \sum 1/n$ is unbounded from above. Since these two are "equal", we can treat them as the same. Then assuming the usual algebraic properties of distributivity work over countably many components, we hvae

$$(1+1/2+1/3+\cdots)-(1+1/2+1/3+\cdots)=(1+1/2+1/3+\cdots)-1-1/2-1/3-\cdots$$

Now, pick any $x \in \mathbb{R}$. Then this sequence can be "rearranged" to converge to that number. Start by picking 1. If $x > 1$, pick $1/2$, if it's less pick -1. Keep doing this, and this will evidentially converge to x! Thus $\infty - \infty$ can be any real number we want!

This is why the idea of a limit is important: If the limit exists, we mean that there is a unique value for which the limit will converge to. This is why limits are used to define objects with some "countably infinite" properties [5].

One important consequence of limits being unique is when we use limits to construct new functions from a limit of functions! As review, you should check the construction of a *continuous but nowhere differentiable function*. Another example is with Fourier series and Fourier transformations which are used to encode and decode signals. Another example the definition of compact exhaustion for imperfect integrals. There are many more yet to come as well!

Now, returning to integrals for a moment, we can ask ourselves if integrals commute with limits, that is

$$\lim_{n\to\infty} \int f_n = \int \lim_{n\to\infty} f_n$$

where the limit here is *pointwise limit*. Note that uniform limits (if it exist) will sort of commute:

$$\text{unif}\lim_{n\to\infty} \int f_n = \int \lim_{n\to\infty} f_n$$

However, point-wise limit doesn't generally do

Example 0.0.1: Limit Doesn't Commute

Take $\chi_{\mathbb{Q}} : [0,1] \to [0,1]$ to be the indicator function on $\mathbb{Q}$, that is

$$\chi_{\mathbb{Q}}(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0,1] \\ 0 & \text{otherwise} \end{cases}$$

Notice that $\chi_{\mathbb{Q}}$ is nowhere continuous, and hence is not Riemann integrable. We can see this more clearly by noticing that the upper Riemann sum is 1 and lower Riemann sum is 0. However

$$g_n(x) = \begin{cases} 1 & \text{if } \frac{p}{q},\ p,q \in \mathbb{N}, p \leq n \\ 0 & \text{otherwise} \end{cases}$$

is integrable, and $g_n \to \chi_{\mathbb{Q}}$, that is $\lim_{n\to\infty} g_n = \chi_{\mathbb{Q}}$. Notice that each g_n are integrable, but $\chi_{\mathbb{Q}}$ is not, that is

$$\lim_{n\to\infty} \int g_n \neq \int \lim_{n\to\infty} g_n$$

Since the right hand side is not defined.

[5] Object with some "uncountably infinite" or larger properties use a generalisation of a limit called an *ultra-filter* which we don't need to study in this course

Thus, our notion of integrability does not work will with limits! Thus, since integrals gives us useful ways of analyzing our functions, and many functions are constructed as the limit of functions, we are going to spend lots of time upgrading our notion of integrability to make it commute (under appropriate circumstances) with limits.To upgrade the notion of integration, there must be some requirements we must keep in mind while finding a suitable replacement. In particular

1. The area under the graph should be the same as the Riemann Integral
2. Geometric intuition's should still be consistent, in particular
   (a) if $f \leq g$ then $\int f \leq \int g$
   (b) $\int af + bg = a \int f + b \int g$
3. A set A can be thought to be "integrable" if we define

$$\chi_A(x) \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

   and $\int \chi_A$ is defined.

This last criterion is actually very interesting as it allows us to generalize the question from finding the area's under graphs (or measuring the area under the graph) to figuring out how to measure sets in general! In fact, starting by defining how to measure sets and then defining how to integrate function turns out to be a fruitful order in which we can learn about this generalisation of integration. The reason this order is as actually an insight by Lebesgue: defining the properties of something being "measurable" will allow us to re-define the integral with more rigour. Even better, many different spaces can have different notions of a "measure" (for example, finite spaces where we measure points, or probability spaces where sets represent the probability of an event, or physics where we might want to measure the average density of a set), which the integral is the wrong concept (it's no longer strictly about "area"), but follows the same geometric intuitions as these other spaces (as we will show). Therefore, we will start by defining the spaces over which we will want to define a way to measure sets.

Part I

Measure Theory

(I feel like I should split the book in two, with all the measure theory/integration build up, followed by the more function-analysis stuff afterwards)

1

Measures

As a start, we might want to define a reasonable definition of what it means to measure any subset of $\mathbb{R}^n$. Such a function would be of the form $\mu : P(\mathbb{R}^n) \to [0, \infty]$. If μ were to exist, we would want our common geometric intuitions to hold, that is:

1. If $E_1, E_2, ..., E_n, ...$ is a countable collection of pair-wise disjoint subsets, then
$$\mu\left(\bigcup_i E_i\right) = \sum_i \mu(E_i)$$
2. If E and F are sets such that there is a function that only translates, rotates, and reflects E to get to F (i.e. a rigid motion), then $\mu(E) = \mu(F)$
3. If C is a unit cube in $\mathbb{R}^n$, then $\mu(C) = 1$

However, these three axioms are inconsistent with one another! We have actually shown this by showing if μ is defined on $P(\mathbb{R}^n)$ (even when $n = 1$), then we can construct *Vitali sets* that will have to be both 0 and ∞ in measure – a contradiction.

Theorem 1.0.1: Vitali Sets Are Unmeasruable

Vitali sets are un-measurable

***Proof*:**
Take $[0, 1]$. Partition this set as follows:
$$x \sim y \iff x - y \in \mathbb{Q}$$
In a sense, this is the set of cosets of $\mathbb{R}/\mathbb{Q}$ restricted to $[0, 1]$; we will rely on this intuition

to label the equivalence classes. Since $\mathbb{R}/\mathbb{Q}$ is the set of irrational numbers modulo rational numbers, let N_q be an equivalence class of contains x that is equivalent to the irrational number q. Notice that there are uncountably many such equivalence relations as there are uncountably many irrational that do not satisfy the equation $x - \frac{p}{q} = 0$.

Now, pick one $p \in N_q$ for each equivalence class, and let N be the collection of all these p's, so N contains one point from each equivalence class (notice that the axiom of choice must be used to create a non-empty N). We'll show that N is unmeasurable.

First, for each rational number $r \in \mathbb{Q} \cap [0, 1)$, define:

$$N + r = \{x + r \mid x \in N \cap [0, 1 - r)\} \cup \{x + r \mid x \in N \cap (1 - r, 1)\}$$

Then clearly, $N + r \subseteq [0, 1)$ for each $r \in \mathbb{Q} \cap [0, 1)$, and furthermore, for each $x \in [0, 1)$ there exists an r such that $x \in N + r$. To show this, let's say $y \in N$ is the element that was picked from the equivalence class of of $[x]$. Then by our choice of y, $x \in N + r$ since $r = x - y$ if $x \geq y$, or $r = x - y + 1$ if $x < y$. Next, notice that the $N + r$ also forms a partition: if $r \neq s$ and $(N + r) \cap (N + s) \neq \emptyset$ then $x - r$ (or $x - r + 1$) would equal $x - s$ (or $x - s + 1$), but they belong to distinct equivalence classes, for example:

$$0 = y - y = x - r - (x - s) = s - r \qquad \Rightarrow \qquad r = s \Rightarrow\Leftarrow$$

and so $(N + r) \cap (N + s) = \emptyset$.

We have now reached the point where we can declare our contradiction. If μ satisfies the three wanted properties, by (1) and (2):

$$\mu(N) = \mu(N \cap [0, 1 - r)) + \mu(N \cap (1 - r, 1)) = \mu(N + r)$$

for any $r \in \mathbb{Q} \cap [0, 1)$. Since $\mathbb{Q} \cap [0, 1)$ is countable and $[0, 1)$ is the disjoint union of the $N + r$'s:

$$1 = \mu([0, 1)) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(N + r)$$

where $1 = \mu([0, 1))$ comes from (3). Now, we have two possibilities:

1. Either $\mu(N) = 0$, in which case each $\mu(N + r) = 0$ but then $0 = 1$ – a contradiction
2. Either $\mu(N) \neq 0$ (say k, so $\mu(N + r) = k$, but then $0 = \infty$ – a contradiction

In other words, no mater what $\mu(N)$ is, we will run into a contradiction!

As you can verify, this construction required that we take advantage of the countability condition (the 1st property). You might ask if weakening it to only finitely many will fix the issue, but sort of crazily it does not! In 1924, Banach and Tarski proved the following paradox now known as the *Banach Tarski Paradox*:

> Let U and V be open subsets of $\mathbb{R}^n$, $n \geq 3$. Then there exists a $k \in \mathbb{N}$ and subsets $E_1, E_2, ..., E_k$, $F_1, F_2, ..., F_k$ such that
>
> 1. all E_j's are disjoint and their union is U
> 2. all F_j's are disjoint and their union is V

3. There exists a rigid motion from E_j to F_j for $j = 1, ..., k$

This is kind of insane: this means that we can take a pea-size ball, and make into the size of the earth! This means that the notion of "geometry" is not actually a consistent concept in general set theory!

Fortunately for us, The sets E_j and F_j are very strange sets; most sets that are in fact measurable. To be more precise, it is consistent to say that the axiom of choice fails and that all sets are Lebesgue measurable (that is, a measure on $\mathbb{R}^n$ that is translation and will be generated by the fact that $\mu([a, b]) = b - a$). For the purposes of this book, that means that any set we construct without the axiom of choice (and just doing normal ZF without any sneaky logical cheating) will be measurable. For more details, see

https://math.stackexchange.com/questions/1847052/most-functions-are-measurable

1.1 Sigma-Algebras and Properties

Since the axiom of choice is the source of the problem of unmeasurability in the previous problems, we start by defining a space on which we avoid unmeasurable sets, namely, we want to limit ourselves to *countable* choices and constructions. The definition should be pretty general, as it should accommodate measures many in more general spaces (finite, continuous, $\mathbb{R}^\infty$, probability spaces, function spaces, and so forth). It should also be "arithmetically expandable", as in we should be able to measure smaller parts of the space (perhaps in more convenient orientations) and add them all up. Breaking up a problem into smaller pieces that could be put together is a fundamental way the core principle of an algebra, hence we define the notion of (analytic) *algebra*[1]:

Definition 1.1.1: Algebra and σ-Algebra

Let X be an arbitrary set. Then let $\mathcal{A} \subseteq P(X)$ be a non-empty collection of subsets such that

1. for any finite set of sets $A_1, A_2, ..., A_n \in \mathcal{A}$, $\bigcup_i A_i \in \mathcal{A}$
2. For any $A \in \mathcal{A}$, $A^c \in \mathcal{A}$.

Then $\mathcal{A}$ is called an *algebra*. If it is closed under countable union, we call it a *σ-algebra* (or *σ-field*).

The σ-algebra will be our natural space in which we will show in the next section we can define a "measure function" without the previous contradiction.

I like to think of the finite closure as a more general way of saying that a space is closed under the operation $\cup$. Closure under compliment insures the "opposite" property of what A represents must always be within our space. Adding the fact that it's closed under countable union brings us further way from the realm of algebra, and hence it's a σ-algebra. The term sigma is very common to mean countable, for example, a space σ-compactness if it is the countable union of compact subspaces.

[1] Note to be confused with an algebra from [14]

Definition 1.1.2: Measurable Space

Let X be a set and $\mathcal{M} \subseteq P(X)$ be a σ-algebra. Then $(X, \mathcal{M})$ is called a *measurable space* and a set in $\mathcal{M}$ is called a *measurable set*

Proposition 1.1.3: Properties of σ-algebra

Some immediate properties of a σ-algebra:

1. It is closed under countable intersection, since $\bigcap_i X_i = (\bigcup_i X)^c$, and by difference, since $X \setminus Y = X \cap Y^c$.
2. $\emptyset$ and $X \in \mathcal{A}$, since for any $E \in \mathcal{A}$, $E \cap E^c = \emptyset \in \mathcal{A}$ and $E \cup E^c = X \in \mathcal{A}$
3. Any countable collection of subsets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ can be turned into a countable disjoint union, where
$$F_k = E_k \setminus \left(\bigcup_{i=1}^{k-1} E_i \right) = E_k \cap \left(\bigcup_{i=1}^{k-1} E_i \right)^c$$
and by what we've shown, every $F_k \in \mathcal{A}$, and by construction $\cup_i E_i = \cup_i F_i$. We will use this technique often very soon (note that not all sets are nonempty, that proof requires more set-theory manipulation).
4. Given any family of σ-algebras on X, their intersection is also a σ-algebra. Because of this, given $Y \subseteq X$, there is always a unique smallest σ-algebra $\mathcal{M}(Y)$ that contains Y (there is at least always one, $P(Y)$, and so this concept is well-defined). $\mathcal{M}(Y)$ is called the σ-algebra generated by Y.

A lemma we'll point out right now to simplify future proofs is the following:

Lemma 1.1.4

If $Y \subseteq \mathcal{M}(X)$, then $\mathcal{M}(Y) \subseteq \mathcal{M}(X)$

***Proof*:**
Since $\mathcal{M}(X)$ is a σ-algebra that contains Y, then it will contain all countable unions and compliment of Y, and hence it will contain $\mathcal{M}(Y)$ (i.e. proposition 1.1.3[4]).

Example 1.1.5: σ-Algebras

1. If X is any set, then $\{\emptyset, X\}$ and $P(X)$ are trivially σ-algebras. Once we define measurable function, these will be the initial and final object in measurable spaces.
2. If X is any set, then $P(X)$ is also a σ-algebra. Usually, we will use proposition 1.1.3[5] and other tools to construct a smaller σ-algebra to not work with $P(X)$ since it's too big for most cases.
3. If X is an uncountable set, define
$$\mathcal{A} = \{E \subseteq X \mid E \text{ is countable or } E^c \text{ is countable}\}$$

this set is called the σ-algebra of countable and co-countable sets. It is useful when we want to make sure that the countable union of sets is still countable (or it's compliment is) [a]

[a]Sneakily, we use the axiom of choice for countably many choices to prove this is true

Another common σ-algebra generated on a set X is related to a topology on X, that is, if $\mathcal{T}(X)$ is a topology, of X, we will define the following σ-algebra:

Definition 1.1.6: Borel σ-Algebra

Let X be a set and $\mathcal{T}(X)$ be a topology on X. Define the *Borel σ-algebra* to be

$$\mathcal{B}_X := \mathcal{M}(\{U \subseteq X \mid U \text{ is an open set contained in } X\}$$

The elements of $\mathcal{B}_X$ are called *Borel sets*

The set $\mathcal{B}_X$ will include the open sets, closed sets, countable union and intersection of closed sets, and the countable union of countable intersections of open/closed sets, et cetera.

Since the countable intersection of open sets and countable union of closed sets are not necessarily open or closed sets respectively[2], but are members in $\mathcal{B}_X$, we will give them names: A countable intersection of open sets is called a G_δ set (G for *Gebeit* in German), and a countable union of closed sets is called a F_σ set (F for *fermé* in french). The σ is for *Summe* (sum in German) since it's a union and the δ is for *Durchschnitt* (intersection in German) since it's an intersection. We can continue this pattern and define $G_{\delta\sigma}$ to be the countable union of countable intersection of open sets, $F_{\sigma\delta}$ to be the countable intersection of countable union, and so and so on and so forth for any length of interchanging σ's and δ's. In general, this "chain" of unions and intersections does not need to terminate, and a $F_{\sigma\delta}$ (as a simple example for a chain of length 2, $\mathbb{R} \setminus \mathbb{Q}$ is a $F_{\sigma\delta}$ set, but not a F_σ set; use Baire's Categories theorem). There will be cases in which it will terminate (in fact, the majority of measure's we'll work with will have this terminate after at just F_σ or G_δ "up to a zero set", see theorem 1.4.5, and example ref:HERE for a measure that does not terminate)

One of the most common Borel sets we will be working with is $\mathcal{B}_\mathbb{R}$ with the usual euclidean topology on $\mathbb{R}$. It is useful to have an of different generating sets for $\mathcal{B}_\mathbb{R}$:

Proposition 1.1.7: Generators For $\mathcal{B}_X$

Let $\mathbb{R}$ be the real numbers and $\mathcal{B}_\mathbb{R}$ be the Borel σ-aglebra on $\mathbb{R}$ given the standard euclidean topology $\mathcal{T}(\mathbb{R})$. Then the following generate $\mathcal{B}_\mathbb{R}$:

1. The set of open intervals
2. the set of closed intervals
3. the set of half-open intervals
4. the set of open rays
5. the set of closed rays

[2]it is possible that the countable intersection/union of open/closed sets are open/closed, and so the word "necessarily" is required

***Proof*:**
These proofs are quite easy to prove, and should be done as an exercise. A fact that you might need to recall is that every open set in $\mathbb{R}$ can be written as a countable union of open intervals (hint: $\mathbb{R}$ has a countable basis).

Products and σ-Algebras

Recall that the product between two sets is defined as $A \times B := \{(a, b) \mid a \in A, b \in B\}$, or more generally $\{X_\alpha\}_{\alpha \in A}$ for a collection of non-empty sets X_α[3], with $X = \prod_{\alpha \in A} X_\alpha$,. As we know from Category Theory, the product comes equipped with set functions π_α that satisfy the universal property of products (or more generally, this is a limit over the discrete category). Though we have yet to give a category where σ-algebras are the objects (we need to define what the morphisms are[4]), we can already construct our set $\prod_\alpha X_\alpha$ that will be the product of the X_α in the category of measurable spaces.

Definition 1.1.8: Product σ-Algebra

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of non-empty sets and $X = \prod_{\alpha \in A} X_\alpha$. Let

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in M_\alpha,\ \alpha \in A\})$$

to be the *product σ-Algebra* on $\{M_\alpha\}_{\alpha \in A}$, i.e., it is the σ-algebra generated by all of the pre-images of the sets in M_α.

It is not difficult to check that the product σ-algebra is indeed a σ-algebra. If $|A|$ is countable we usually write $\bigotimes_{i=1}^{\infty} M_i$, and if $|A|$ is finite, we can write $\bigotimes_{i=1}^{n} M_i$ or $M_1 \bigotimes M_2 \bigotimes \cdots \bigotimes M_n$. Notice that since we allow for countable unions, it is not so easy to represent $\bigotimes_{\alpha \in A} M_\alpha$ as the product of the elements of M_α, (just like for Groups in the category of **Grp**), and hence why we do not use the $\prod$ or $\bigoplus$ notation. On the other hand, the fact that only a countable union of elements is permitted will still limit the product in the number of non-trivial components, hence justifying the $\bigotimes$ notation instead of $\prod$. In fact, it is the same as $\bigoplus$, but instead of "all but finitely many", we have "all but countably many":

Proposition 1.1.9: Countable Product σ-algebra

Let $M_n = \mathcal{M}(\mathcal{E}_n)$ be a σ-algebra for $n \in \mathbb{N}$ where $\mathcal{E}_n \subseteq P(X_n)$. If $X_n \in \mathcal{E}_n$, then

$$\bigotimes_{n=1}^{\infty} M_n = \mathcal{M}\left(\left\{\prod_{i=1}^{n} E_i \;\middle|\; E_i \in M_i\right\}\right)$$

[3]If they were empty, then the product would be empty
[4]this is done in chapter 2

***Proof*:**
The $\supseteq$ direction is easy (Take $\bigcap_{i=1}^{n} \pi_i(E_i)$ for appropriate values). For $\subseteq$, Since $X_i \in M_i$, we have that every element is inside the right hand side. Make sure you see why $X_n \in \mathcal{E}_n$ makes a difference. By lemma 1.1.4, the result follows.

We can further simplify our representation of $\bigotimes_{\alpha \in S} M_\alpha$ given a generating set for each M_α (such a set always exists, namely M_α itself is always a generating set)

Proposition 1.1.10: Generating set for Product σ-Algebra

Let $\{M_\alpha\}_{\alpha\in A}$ be a non-empty collection of σ-algebras, and let $\mathcal{E}_\alpha \subseteq P(X_\alpha)$ be a generating set for M_α. If $\mathcal{F} = \cup_{\alpha\in A}\mathcal{E}_\alpha$, then:

$$\bigotimes_{\alpha\in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A\}) = \mathcal{M}(\mathcal{F})$$

Furthermore, if $|A|$ is countable and $X_\alpha \in \mathcal{E}_\alpha$ for each $\alpha \in A$, then

$$\bigotimes_{\alpha\in A} M_\alpha = \mathcal{M}\left(\left\{\prod_{\alpha\in A} E_\alpha \;\middle|\; E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A\right\}\right)$$

***Proof*:**
Dealing with the first case, the $\supseteq$ is clear since $E_\alpha \in \mathcal{E}_\alpha$ implies $E_\alpha \in M_\alpha$, so $E_\alpha \in \bigotimes_{\alpha\in A} M_\alpha$. For the $\subseteq$ direction, If $E_\alpha \notin \mathcal{E}_\alpha$, then since $\mathcal{M}(\mathcal{E}_\alpha) = M_\alpha$, some countable unions and intersections of elements of $\mathcal{E}_\alpha$ equal E_α, proving the $\subseteq$ direction.

The countable case follows proposition 1.1.9

Borel Spaces and Products

A topological space of much study in Analysis is metric spaces. As a reminder, if $\{X_\alpha\}_{i=1}^{n}$ is a collection of metric spaces, then $X = \prod_{i=1}^{n} X_i$ has an induced product metric structure. As was shown in [7], the following metrics on a product space all define the same product topology since they induce the same open sets (for notational simplicity, let's work with two metric spaces X and Y and consider $X \times Y$):

1. $d((x_1, x_2), (y_1, y_2)) = \max\{d_X(x_1, y_1), d_Y(x_2, y_2)\}$
2. $d((x_1, x_2), (y_1, y_2)) = \sqrt{d_X(x_1, y_1)^2 + d_Y(x_2, y_2)^2}$
3. $d((x_1, x_2), (y_1, y_2)) = d_X(x_1, y_1) + d_Y(x_2, y_2)$

We can ask about establishing Borel sets $\mathcal{B}_X$ on $X = \prod_{i=1}^{n} X_i$ with X having the product measure. It is tempting to say that this naturally splits into $\bigotimes \mathcal{B}_{X_i}$, however, this is not always the case! This is mainly due to the fact that not all topological spaces have a countable basis, and since we're allowing countable intersection of open sets, this will actually create more sets than the if we take the product of the Borel spaces:

Proposition 1.1.11: Borel sets on product Metric Space

Let $X_1, ..., X_n$ be metric spaces, and $X = \prod_{i=1}^n X_i$ be the product metric space. Then

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$$

If the spaces X_i are separable (i.e. has a countable dense subset, and since it's a metric space it means there is a countable basis), then

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} = \mathcal{B}_X$$

***Proof*:**
The proof of the $\subseteq$ direction rather simple. By proposition 1.1.10, $\bigotimes_{i=1}^n \mathcal{B}_{X_i}$ is generated by $\mathcal{E} = \left\{\pi_i^{-1}(U_j) \;\middle|\; U_j \text{ is open in } X_j,\ 1 \le j \le n\right\}$. Since the sets of $\mathcal{E}$ are open in X, by lemma 1.1.4, $\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$.
For the converse, since each X_j is separable, each have a countable dense subset. Since X_j is a metric space, we can form a countable basis with open balls with rational radii around each points in the countable dense subsets.

As a consequence, $\bigotimes_{i=1}^n \mathcal{B}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R}^n}$, since $\mathbb{R}$ has a countable dense subset.

Elementary Sets

This is a technical result needed for later (maybe introduce it as exercises?)

Definition 1.1.12: Elementary Family

Let X be a set and $\mathcal{E} \subseteq P(X)$. Then $\mathcal{E}$ is called an *elementary family* if

1. $\emptyset \in \mathcal{E}$
2. If $E, F \in \mathcal{E}$ then $E \cap F \in \mathcal{E}$
3. If $E \in \mathcal{E}$, then E^c is a finite disjoint union of elements of $\mathcal{E}$

Proposition 1.1.13: Elementary Families and Algebra

Let $\mathcal{E}$ be an elementary family. Then the collection of finite disjoint unions of element of $\mathcal{E}$ forms an algebra

***Proof*:**
(TBD) Let $U, V \in \mathcal{E}$, and consider $V^c = \coprod_{j=1}^{J} F_j$ (for disjoint $F_j \in \mathcal{E}$). Then

$$U \setminus V = U \cap V^c = \coprod_{j=1}^{J} (U \cap F_j)$$

notice that each $U \cap F_j \in \mathcal{E}$. So

$$U \cup V = (U \setminus V) \cup V = \coprod_{j=1}^{J} (U \cap F_j) \coprod V$$

where $\coprod_{j=1}^{J} (U \cap F_j) \in A$
Continuing inductively, can check Folland for proof, or just do it yourself.

(also, remember that separable means countable dense subset) Once the course starts

1. There is an interseting example ith logical statements!!

Remark. If A is a set. and $\varphi(x)$ is a logical statement about x. Then $\{x : \varphi(x) \text{ holds and } x \in A\}$ is a set.

Let $\mathcal{E} \subset \mathcal{P}(X)$, define:

$$\mathcal{E}_1 = \left\{ \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{\infty} E_{i,j} \ , \ \begin{array}{l} \text{either } E_{i,j} \in \mathcal{E} \\ \text{or } E_{i,j}^c \in \mathcal{E} \end{array} \right\}$$

$$\mathcal{E}_2 = \left\{ \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{\infty} E_{i,j} \ , \ \begin{array}{l} \text{either } E_{i,j} \in \mathcal{E}_1 \\ \text{or } E_{i,j}^c \in \mathcal{E}_1 \end{array} \right\}$$

etc.

$\bigcup \mathcal{E}_n$ is in general not a σ-alg. and there is no constructive way to understand $\mathcal{M}(\mathcal{E})$.

Figure 1.1: fascinating!

1.2 Measure

We can finally define the function which will give us an idea of the "size" of sets. The key idea behind the idea of notion is that the "bigger" a set is, the "larger" the the area. In other words, if μ is a measure on X, then (abusing notation a bit):

$$\mu(A \subseteq B) \;\Leftrightarrow\; \mu(A) \leq \mu(B) \tag{1.1}$$

In otherwords, a measure is an order-homomorphism from $(\mathcal{M}, \subseteq)$ to $([0, \infty], \leq)$. This also means that we want the codomain of a measure to be *totally ordered*, and since we want limits to work, it also have to be *complete*. Furthermore, if we have a countable number of disjoint areas, it should possible to measure them all separately and or all together at once and get the same measure (a sort of "invariance of space"). In fact, if we just assume $\mu(\emptyset) = 0$ (i.e. "nothing" should have no area of measure), then using this "invariance of space" notion, we can deduce equation (1.1), as we'll show in the next proposition, and hence we would want the codomain of a measure to be $[0, \infty]$ (more generally, a minimal complete totally ordered set embeds into $[0, \infty]$, see ref:HERE, showing this is a natural choice). We thus define a measure on these intuitions:

Definition 1.2.1: Measure

Let X be a set along with a σ-algebra $\mathcal{M}$. A function $\mu : \mathcal{M} \to [0, \infty]$ is called a *measure* on $\mathcal{M}$ (or $(X, \mathcal{M})$ or even X if the measure is clear from context) if

1. $\mu(\emptyset) = 0$
2. *(Countably Additivity)* If $\{E_i\}_{i=1}^{\infty}$ is a sequence of disjoint sets, then $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$

Given that $(X, \mathcal{M})$ is a measurable space, along with a μ we can say the space has a measure:

Definition 1.2.2: Measure Space

Let $(X, \mathcal{M})$ be a measurable space. Then if μ is a measure on X, then $(X, \mathcal{M}, \mu)$ is a *measure space*

If the 2nd condition of a measure was limited to a finite sequence (or more generally, all but countably many of the sets in the sequence or nonempty), then we would say that it respects finite additivity. If μ respects finite but not countable additivity, μ is called a *finite additive measure*.

Most measure's that we will work with have some finiteness condition associated with it:

Definition 1.2.3: Finiteness Condition on μ

Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. If $m(X) < \infty$, then μ is called a *finite measure*
2. if there exists sets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ such that $\bigcup_i^{\infty} E_i = X$ and $\mu(E_i) < \infty$ for all E_i, then μ is called a *σ-finite measure*. More generally, if $E = \bigcup_{i=1}^{\infty} E_i$ for some sequence of E_i, then E is called σ-finite on μ (some say that E is said to be σ-finite on μ)
3. if for for all $E \in \mathcal{M}$ such that $\mu(E) = \infty$, there exists a $D \subseteq E$, $D \in \mathcal{M}$ such that $\mu(D) < \infty$, then μ is called a *semi-finite measure*

If $\mu(X) < \infty$, then this implies for all $S \in \mathcal{M}$, $\mu(S) < \infty$ (as can quickly be checked). Naturally, all finite measure's are σ-finite, and all σ-finite measure are semi-finite, but the converses are not true (as we'll explore in the following examples). Note too that most measure's we will be working over will be σ-finite (think of $\mathbb{R}$ with the intuitive notion of length of an interval).

Example 1.2.4: Measures

1. Let X be nonempty, $\mathcal{M} = P(X)$, and $f : X \to [0, \infty]$. Then we can define

$$\mu_f(E) = \sum_{e \in E} f(e)$$

where the sum can be infinite (the domain of f is the positive part of the extended real numbers). Notice that f is semifinite if and only if $f(x) < \infty$ for all $x \in X$, and μ is σ-finite if and only if μ is semi-finite and $\{x \mid f(x) > 0\}$ if countable.

If $f(x) = 1$ for all $x \in X$, then μ_f is called the *counting measure*. On the other hand, if for some $x_0 \in X$, $f(x_0) = 1$ and $f(x) = 0$ if $x \neq x_0$, then μ_f is called a *point mass* or *Dirac measure* at x_0

2. Let X be an uncountable set an $\mathcal{M}$ be the σ-algebra of countable or co-countable sets (recall example 1.1.5). Define
$$\mu(E) = \begin{cases} 0 & E \text{ is countable} \\ 1 & E \text{ is co-countable} \end{cases}$$
Then μ is a measure!

3. An example of an finite additive measure that is not a measure is the following: let X be some infinite set and $\mathcal{M} = P(X)$. Define μ to be
$$\mu(E) = \begin{cases} 0 & E \text{ is finite} \\ \infty & E \text{ is infinite} \end{cases}$$
Then μ is an finite additive measure, but not a measure

Proposition 1.2.5: Properties of Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then

1. *(Monotonicity)* Let $E, F \in \mathcal{M}$ and $E \subseteq F$. Then $\mu(E) \leq \mu(F)$
2. *(Subadditivity)* if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$, then $\mu\left(\cup_i \mu(E_i)\right) \leq \sum_i \mu(E_i)$
3. *(Continuity from Below)* if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$, and $E_1 \subseteq E_2 \subseteq \cdots$, then
$$\mu\left(\cup_i E_i\right) = \lim_{i \to \infty} \mu(E_i)$$
4. *(Continuity from Above)*: if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ and $E_1 \supseteq E_2 \supseteq$ *and* $\mu(E_1) < \infty$. Then
$$\mu\left(\cap_i E_i\right) = \lim_{i \to \infty} \mu(E_i)$$

***Proof*:**
here. Do them as exercises! I also like this solution by Hytham Farah for continuity from above:

$$\begin{aligned}
\mu\Big(\bigcup_{n\geq 1} F_n\Big) &= \mu\Big(\bigcup_{n\geq 1} E_n\Big) \\
&= \mu\Big(\big(\bigcap_{n\geq 1} E_n^c\big)^c\Big) \\
&= \mu(X) - \mu\Big(\bigcap_{n\geq 1} E_n^c\Big) \\
&= \mu(X) - \lim_{n\to\infty} \mu(E_n^c) \\
&= \mu(X) - \lim_{n\to\infty} \big(\mu(X) - \mu(E_n)\big) \\
&= \lim_{n\to\infty} \mu(E_n) \\
&= \sum_{i=1}^{\infty} F_n.
\end{aligned}$$

Notice that for (d), we can instead limit it to $\mu(E_i) < \infty$ for some i, and the proof remains the same. We still require that some i exists such that $\mu(E_i) < \infty$, for it's possible that all $\mu(E_i)$ is infinite, but $\mu(\cap_i E_i) < \infty$, for example take the measurable space $(\mathbb{N}, P(\mathbb{N}))$ with the counting measure. Then the sets

$$E_i = \{n \in \mathbb{N} \mid n \geq i\}$$

then every E_i is infinite but their intersection is empty and thus has measure 0 in the counting measure.

zero measure sets

Frequently, there are sets in our measure that have a "trivial size" in the current measure. This is akin to when we integrate, the border has no effect on the value of the integral. Such sets are said to be zero sets or null set:

Definition 1.2.6: Null Set

Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$. Then if $\mu(E) = 0$, E is said to be a *null set* or *zero set*.

By countably subadditivity, the countable union of null sets is a null set, hence null sets are "invariant" under countable union. Due to this, we can work with measurable sets in $\mathcal{M}$ *up to null sets*. We shall often state in proofs that a statement is true *almost everywhere* (often abbreviated to a.e.) if it is true on the sets except for the null sets.

If we take $E \in \mathcal{M}$ such that $\mu(E) = 0$, then by monotonicity, if $F \subseteq E$ the $\mu(F) = 0$, *provided that* $F \in \mathcal{M}$. This is not always the case:

Example 1.2.7: Not Complete Space

Partition $[0, 1]$ into 10 (really any number) of subintervals, $\left[\frac{i}{n}, \frac{i+1}{n}\right]$ for $0 \leq i \leq 9$, and let each of their measures be 0. Take $\mathcal{M}$ to be the σ-algebra defined on this set. Then is clearly an uncountable number of sets that do not have a measure defined on them that are subset of zero-measure sets.

However, adding all of these F's that are subsets of null sets does not break our measure. A measure who's domain includes all subsets of null sets is a *complete measure* (i.e. the σ-algebra is closed under subsets of null sets). Proving that this is still a measure is useful to eliminate future "trivial obstructions" to our proofs:

Theorem 1.2.8: Completing a Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} \mid \mu(N) = 0\}$ and

$$\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subseteq N \text{ for some } N \in \mathcal{N}\}$$

Then $\overline{\mathcal{M}}$ is also a σ-algebra, and there is a *unique* extension $\overline{\mu}$ of μ that is a complete measure on $\overline{\mathcal{M}}$ that restricts to the measure of μ on $\mathcal{M}$

***Proof*:**

First of all, since $\mathcal{M}$ and $\mathcal{N}$ are both closed under countable union, so $\overline{\mathcal{M}}$ is closed under countable union. To show it's closed under compliments, consider some $E \cup F \in \overline{\mathcal{M}}$ ($E \in \mathcal{M}$, $F \subseteq N \in \mathcal{N}$). Without loss of generality, assume $E \cap N = \emptyset$ (if not, take $F \setminus E$ and $N \setminus E$ and continue the proof, then this will have he same value as our original $E \cap N$). Then

$$E \cup F = (E \cup N) \cap (N^c \cup F)$$

So we get

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$$

Since $E \cup N \in \mathcal{M}$, $(E \cup N)^c \in \mathcal{M}$. Since $N \setminus F \in \mathcal{N}$, Since it's closed under union, $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ-algebra.

Since $\overline{\mathcal{M}}$ is a σ-algebra, we can try to define a measure, and indeed

$$\overline{\mu}(E \cup F) = \mu(E)$$

is a well-defined measure. If

$$E_1 \cup F_1 = E_2 \cup F_2$$

Then $E_1 \subseteq E_2 \cup F_2$, and so by monotonicity:

$$\mu(E_1) \leq \mu(E_2 \cup F_2) \leq (E_2 \cup N_2) = \mu(E_2) + \mu(N_2) = \mu(E_2)$$

Similarly, $\mu(E_1) \geq \mu(E_2)$. Thus $\overline{\mu}(E_1 \cup F_1) = \overline{\mu}(E_2 \cup F_2)$, and so $\overline{\mu}$ is well-defined. Since $\overline{\mu}$ has all subsets of zero-sets in its domain, it is a complete measure.

Furthermore, if $\overline{\nu}$ was another complete measure on $\mathcal{M}$, then it is easy to see that $\overline{\nu} = \overline{\mu}$, (since if it were different, then some subset of a zero-set must have non-zero measure, a contradiction to monotonicity) showing that $\overline{\mu}$ is unique.

Definition 1.2.9: Completion of Measure

Let $(X, \mathcal{M}, \mu)$ be a measurable space. Then $\overline{\mathcal{M}}$ and $\overline{\mu}$ is called the *completion of* $\mathcal{M}$ *with respect to* μ and the *completion of* μ respectively

1.3 Outer Measure

We now move on to constructing a particular type of measure which is draws inspiration from how we measured area's when we calculated Riemann Integrals. In the abstract, when we tried defining a Jordan measure (or content[5]), we approximated the inner area of a shape E in $\mathbb{R}^n$ by the limit of sums of rectangles contained in E, and the outer-area of E in $\mathbb{R}^n$ by the limit of the sums of rectangles containing E. The two numbers are called the inner and outer area of E. If the two matched, we would call the result the area of E. This intuition can be generalized to the countable case, except we can replace rectangles with some other "basic sets" that we might want that form an algebra. Furthermore, the intuition of having the "inner area" and "outer-area' is actually a bit restricting to work with in the general measure case, and so that condition will be replaced with a different measureability condition we will soon introduce.

For the sake of generality, we will start by definition a more general notion of the measure called the outer-measure, and then build-up on how form the outer-measure, we can get a measure that satisfies this generalization of Riemann integration:

Definition 1.3.1: Outer Measure

A function $\mu^* : P(X) \to [0, \infty]$ is called an *outer measure* if

1. $\mu^*(0) = \emptyset$
2. if $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$
3. $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (even if the sets are disjoint!)[a]

[a] We will return to see when equality holds in definition 1.3.3

Note that we have not yet proven that an outer-measure is a measure; in particular notice that the domain is the powerset. The name "outer-measure" gets its name from the following general construction:

[5] since "Jordan measures" only allow finite additivity, some don't like to use the word measure and opt for the word content. In this book, we would say *finite additive measure* as has already been stated earlier

Proposition 1.3.2: Outer Measure Construction

Let $\mathcal{E} \subseteq P(X)$ where $\emptyset, X \in \mathcal{E}$, and let $\rho : \mathcal{E} \to [0,1]$ where $\rho(\emptyset) = 0$. For any $A \subseteq X$, define

$$\mu^*(A) = \inf\left\{\sum_1^\infty \rho(E_j) \;\middle|\; E_j \in \mathcal{E} \text{ and } A \subseteq \bigcup_1^\infty E_j\right\}$$

Then μ^* is an outer measure

***Proof*:**
First, this measure is well-defined since for any $A \subseteq X$, If we take $E_i = X$, then $A \subseteq \cup_i E_i$. Next, we verify the 3 properties

1. Clearly, $\mu^*(\emptyset) = 0$ by taking $E_i = \emptyset$ for all i
2. If $A \subseteq B$, then let $\mathcal{A}$ and $\mathcal{B}$ sets that correspond to $\mu^*(A)$ and $\mu^*(B)$. Then since all of the elements of $\mathcal{A} \supseteq \mathcal{B}$, by the property of the infimum $\inf \mathcal{A} \leq \inf \mathcal{B}$, which transferring notations gives us $\mu^*(A) \leq \mu^*(B)$
3. Let $\{E_i\}_{i=1}^\infty \subseteq P(X)$ be a collection of (not necessarily disjoint) subsets. Let $\epsilon > 0$. Consider $\mu^*(E_k)$ for some $E_k \in \{E_i\}_{i=1}^\infty$. By the property of the infimum, there must exist some $\{F_j^k\}_{j=1}^\infty$ such that $E_i \subseteq \cup_j F_j^k$ where $\sum_{j=1}^\infty \rho(F_j^k) \leq \mu^*(E_k) + \epsilon 2^{-k}$.

   Now, let $E = \cup_i E_i$. By construction, $E \subseteq \cup_{i,k=1}^\infty F_j^k$, and

   $$\sum_{j,k} \rho(F_j^k) \leq \mu^*(A) + \epsilon$$

   Since ϵ can be arbitrarily small, this completes the proof.

As mentioned before, an outer-measure is not necessarily a measure. What we will do to allow this to be a measure is single out sets that have the following property

Definition 1.3.3: μ^*-measureable

Let $E \subseteq X$ be a set and μ^* be an outer measure on X. Then E is called μ^**-measurable* if

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

In other word, if E can naturally "split" a set S into a part that's "inside" E or "outside" E. The set $S \subseteq X$ is called the *test set*.

By definition of the outer measure, since $S = (S \cap E) \cup (S \cap E^c)$:

$$\mu^*(S) \leq \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

Therefore, what is more important is to check the $\geq$ direction. Furthermore, if $\mu^*(A) = \infty$. Then $\geq$ holds trivially, so to show a set is μ^* measurable, it is equivalent to show that for all finite measure sets A, $\mu^*(S) \geq \mu(S \cap E) + \mu(S \cap E^c)$.

As mentioned in the definition, this gives us a sense of being able to measure the "inside" and "outside" of a set. If $E \subseteq A$, that is the μ^*-measurable set belong in a better set, then $\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \setminus E)$ which can be thought of as saying that we are measuring S from the "inside" of E and the "outisde" of E, and getting the same result. In fact, if $\mu^*(X) < \infty$, then we can define $\mu_*(E) = \mu^*(X) - \mu(E^c)$, and have that E is μ^*-measurable if and only if $\mu_*(E) = \mu^*(E)$ aligning with our intuition for inside equalling outside (see exercise ref:HERE). We do not do this method since it requires some finiteness condition on μ. We can conversely define μ_* in terms of supremums and need that E contains the collection of $\cup_i A_i$, and that the elements of $\{A_i\}_{i=1}^{\infty}$ must be disjoint (see exercise ref:HERE).

The big thing that is in the way of saying that an outer measure is a measure is that its domain is $P(X)$, which can cause problems when X is uncountable. However, all sets that satisfy μ^*-measureability turn out form a σ-algebra and make μ^* a complete measure!

Theorem 1.3.4: Carathéodory's Theorem

Let μ^* be an outer measure on X, and let $\mathcal{M}$ be the collection of μ^*-measurable sets. Then $\mathcal{M}$ forms a σ-algebra, and the restriction of μ^* to $\mathcal{M}$ is a complete measure

Notice that in the process of defining this measure, we started with the powerset, and then simply choose all sets that restrict (which we will show forms a σ-algebra). In this way proving a function is an outer-measure is a little less finicky since we don't need to be so careful in defining over which σ-algebra we want to define it over: the σ-algebra comes with μ^*

***Proof*:**
Let $\mathcal{M}$ be the collection of μ^*-measurable sets. Clearly, $\emptyset \in \mathcal{M}$ since for all $S \subseteq X$:

$$\mu^*(S \cap \emptyset) + \mu^*(S \cap X) = 0 + \mu^*(S)$$

implying μ^*-measureability. Next, $\mathcal{M}$ is closed under compliments, since if $E \in \mathcal{M}$, then

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \mu^*(S \cap E^c) + \mu^*(S \cap E)$$

which also shows that $X \in \mathcal{M}$. Finally, we need to show that $\mathcal{M}$ is closed under countable unions. We'll start by showing finite unions, which then means it suffices to show the union of two sets is closed (due to induction). Let $A, B \in \mathcal{M}$. As we remarked, it suffices to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Since A and B are μ^* measurable, we have

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) && E \text{ is the test set} \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) && E \cap A \text{ and } E \cap A^c \text{ are the test sets} \end{aligned}$$

Then notice we can write

$$A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$$

Thus, using subadditivity, we can re-write 3 of the terms in the previous expression as:

$$\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \geq \mu^*(E \cap A \cup B)$$

Since $E \cap A^c \cap B^c = E \cap (A \cup B)^c$, we get:

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

showing that $\mathcal{M}$ is closed under finite union. Since $\mathcal{M}$ is closed under compliment and contains $\emptyset$, $\mathcal{M}$ is an algebra. Even better, If we let A and B be disjoint, and let $A \cup B$ be the test set, then since A is μ^* measurable:

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B)$$

showing that $\mu^*|_{\mathcal{M}}$ is a finitely additive measure.

Next, we must show that $\mathcal{M}$ is a σ-algebra, which as a consequence will show that $\mu^*|_{\mathcal{M}}$ is a measure. Let $\{E_i\}_{i=1}^\infty \subseteq \mathcal{M}$. To simplify our task, re-construct $\{E_i\}_{i=1}^\infty$ to be the set of disjoint sets $\{D_i\}_{i=1}^\infty$. Let $B_n = \bigcup_{i=1}^n D_i$ and $B = \bigcup_{i=1}^\infty D_i$. Our goal is to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$. We will start by considering $\mu^*(E \cap B_n)$:

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap D_n) + \mu^*(E \cap B_n \cap D_n^c) && E \cap B_n \text{ is the test set} \\ &= \mu^*(E \cap D_n) + \mu^*(E \cap B_{n-1}) \end{aligned}$$

Notice that on the right hand side we have diminished to the case of $E \cap B_{n-1}$ from $E \cap B_n$. Using induction, we can see that the expression simplifies to

$$\mu^*(E \cap B_n) = \sum_{i=1}^n \mu^*(E \cap D_i)$$

Continuing, since B_n is measurable

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \overset{!}{\geq} \mu^*(E \cap B_n) + \mu^*(E \cap B^c) = \sum_{i=1}^n \mu^*(E \cap D_i) + \mu^*(E \cap B_n^c)$$

where $\overset{!}{\geq}$ comes from monotonicity after switching B_n^c to B^c. Since this equality holds true for all n, we get:

$$\begin{aligned} \mu^*(E) &\geq \sum_{i=1}^\infty \mu^*(E \cap D_i) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(\bigcup_i E \cap D_i\right) + \mu^*(E \cap B^c) && \text{subadditivity} \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c) && \text{by construction of } E \cap B \\ &\geq \mu^*(E) && \text{subadditivity} \end{aligned}$$

Since we got $\mu^*(E) \geq \mu^*(E)$, we see that in fact all inequalities are equalities, getting us that

$$\mu^*(E) = \mu^*(E \cap B) + \mu^*(E \cap B^c)$$

showing us that $B \in \mathcal{M}$. Furthermore, replacing $E = B$ in the previous proof shows us that

$$\mu^*(B) = \sum_{i=1}^\infty \mu^*(D_i)$$

giving us that μ^* is a measure on $\mathcal{M}$.

Finally, we must show that $\mathcal{M}$ is complete, that is, if $E \in \mathcal{M}$ such that $\mu^*(E) = 0$, then all subsets of E are in $\mathcal{M}$. Notice that since μ^* is defiend on all of $P(X)$, it is equivalent to show that if $\mu^*(A) = 0$, then $A \in \mathcal{M}$. Then for any test set E, by monotonicity:

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) = 0 + \mu^*(E \cap A^c) \leq \mu^*(E)$$

meaning equality holds, showing that $A \in \mathcal{M}$. Thus, $\mu^*|_{\mathcal{M}}$ is a complete measure, as we sought to show.

Carathéodory's Theorem is quite useful being able to state the existence of a measure given an outer-measure, but it's not good to construct a measure (for example, can you tell me a non-trivial element in $\mathcal{M}$ without directly appealing to the definition?). The following concept is used to more systematically construct the set $\mathcal{M}$ and the measure μ induced by μ^*:

Definition 1.3.5: Premeasure

Let $(X, \mathcal{A})$ be an algebra. Then μ_0 is called a *premeasure* if

1. $\mu_0(\emptyset) = 0$
2. **"Respects" countable union**: If $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ is a collection of disjoint such that $\cup_i A_i \in \mathcal{A}$, then $\mu_0(\cup_i A_i) = \sum_i \mu_0(A_i)$

Note that the premeasure is automatically finitely additive since we can let all but finitely many of the terms in (2) be $\emptyset$ (and hence are a stronger condition than finitely additive measure). However, it is not yet a measure since it's domain is not necessarily closed under countable union. However, if it is closed for countable union for a particular collection, that collection must satisfy the "countable" condition. Interestingly, by the definition of an algebra, this implies that the countable union can be represented as a finite union and intersection of elements from $\mathcal{A}$.

As before, we can define semifinite and σ-finite in a similar way. Using μ_0, we can define the following outer-measure:

$$\mu^* : P(X) \to [0, \infty] \quad \mu^*(E) = \inf \left\{ \sum_i^{\infty} \mu_0(A_i) \;\middle|\; A_i \in \mathcal{A},\; E \subseteq \cup_i A_i \right\}$$

Proposition 1.3.6: Premeasure To Measure

Let μ_0 be a premeasure on $\mathcal{A}$, and let μ^* be defined as above. Then

1. $\mu^*|_{\mathcal{A}} = \mu_0$
2. Every set in $\mathcal{A}$ is μ^*-measurable

Essentially, μ^* is an extension of μ_0 and the associated restriction of μ^* to a measure will contain $\mathcal{A}$

***Proof*:**

1. Let's show $\leq$ and $\geq$ for any $E \in \mathcal{A}$. The $\leq$ direction is obvious since $\mu^*(E) \leq \mu_0(E)$ since $E \subseteq \bigcup_{i=1}^{\infty} A_i$ where $A_1 = E$ and $A_i = \emptyset$ for $i > 1$, and so it is part of the set over which

we take the infimum, and so the result *has* to be smaller.

For the $\geq$ direction, let's say $E \subseteq \bigcup_{i=1}^{\infty} A_i$ for any $\{A_i\}_{i=1}^{\infty} \in \mathcal{A}$. Let $B_n = E \cap \left(A_n \setminus \cup_1^{n-1} A_i\right)$. Then the collection $\{B_i\}_{i=1}^{\infty}$ is a disjoint collection of members of $\mathcal{A}$ whose union is E. Thus, by the property of premeasure

$$\mu_0(E) = \sum_1^{\infty} \mu_0(B_i) \overset{!}{\leq} \sum_1^{\infty} \mu_0(A_i)$$

where the $\overset{!}{\leq}$ inequality comes from the fact that $\mu_0(B_i) \leq \mu_0(A_i)$. Since this is true for any arbitrary collection $\{A_i\}_{i=1}^{\infty}$, it follows that

$$\mu_0(E) \leq \mu^*(E)$$

Thus establishing equality

2. We need to show that for any $E \subseteq X$ and $A \in \mathcal{A}$, $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$. As mentioned earlier, it suffice to prove $\geq$ where $\mu^*(E) < \infty$.

   Let $\epsilon > 0$. By the property of the infimum, there exists a subset $\{B_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i B_i$ such that $\sum_i^{\infty} \mu_0(B_i) \leq \mu^*(E) + \epsilon$. By part (1), μ^* is [countably] additive on $\mathcal{A}$ (since it equals the pre-measure)

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{i=1}^{\infty} \mu_0(B_i \cap A) + \sum_{i=1}^{\infty} (B_i \cap A^c) && \text{additivity} \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) && \text{monotinicity} \end{aligned}$$

   and since ϵ was arbitrary, A must be μ^*-measurable, completing the proof

We can in fact be more precise with the measures that extend a premeasure:

Theorem 1.3.7: Premeasure Extensions

Let $\mathcal{A} \subseteq P(X)$ be an algebra, μ_0 a premeasure on $\mathcal{A}$, and $\mathcal{M} = \mathcal{M}(\mathcal{A})$ (i.e. $\mathcal{M}$ is the σ-algebra generated by $\mathcal{A}$). Let μ be the extension of μ_0 from proposition 1.3.6. Then if ν is another measure on $\mathcal{M}$ that extends μ_0, then

$$\nu(E) \leq \mu(E) \qquad \forall E \in \mathcal{M}$$

with equality when $\mu(E) < \infty$ (i.e., it possible that μ is infinite while ν is finite).
If μ_0 is σ-finite, then μ is the unique extension of μ_0 to a measure on $\mathcal{M}$

In other words, we can extend a pre-measure to a measure by knowing what it does on the algebra (which in most cases will be a collection of simpler sets, like intervals or rectangles), and then what happens on the rest of the measurable sets we use the infimum definition. If we define another extension of μ_0, then that measure will be *finer* than the μ given by μ^*, i.e, μ is the *coarsest* extension. If μ_0 is σ-finite, then this will in fact be the only extension of μ_0

***Proof*:**
Let's say ν is an extension of μ_0 so that $(X, \mathcal{M}(\mathcal{A}), \nu)$ is a measure space. To show that $\nu(E) \leq \mu(E)$, let's say $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i A_i$. Then since ν is a measure, it is sub-additive on non-disjoint sets, and so

$$\nu(E) \leq \sum_i \nu(A_i)$$

Since ν extends μ_0, by proposition 1.3.6 $\nu|_{\mathcal{A}} = \mu_0$, so

$$\nu(E) \leq \sum_i \nu(A_i) = \sum_i \mu_0(A_i)$$

Since $\{A_i\}_{i=1}^{\infty}$ was an arbitrary choice, this works for any collection, and so

$$\nu(E) \leq \mu(E) \quad [= \mu^*(E)]$$

We show that $\nu(E) = \mu(E)$ if $\mu(E) < \infty$. Let $A = \cup_i A_i$ for some collection $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$. Then

$$\nu(A) = \lim_{n\to\infty} \nu\left(\bigcup_{i=1}^{n}(A_i)\right) = \lim_{n\to\infty} \mu\left(\bigcup_{i=1}^{n}(A_i)\right) = \mu(A)$$

If μ is not σ-finite, then there can be more than one extension:

Example 1.3.8: 2 extensions of pre-measure

1. Let $\mathcal{A}$ be the collection of finite unions of the sets of the form $(a, b] \cap \mathbb{Q}$ with $-\infty \leq a \leq b \leq \infty$. Then $\mathcal{A}$ is an algebra on $\mathbb{Q}$

***Proof*:**
Recall proposition 1.1.13 where we can construct an algebra using an elementary family. Thus, we will show that the collection $\mathcal{A}$ satisfies the 3 properties of an elementary family:

(a) To get the empty-set in $\mathcal{A}$, we will understand the notion of "finite union" to also include "no union". Otherwise, since $\mathbb{Q}$ is dense in $\mathbb{R}$, all intervals $(a, b]$ intersected with $\mathbb{Q}$ is nonempty (by MAT157 knowledge). Since $a < b$, the singleton interval is not admissible, and so we will have to assume this "no union" property

(b) Let $E, F \in \mathcal{A}$. We want to show that $E \cap F \in \mathcal{A}$. By definition, $E = (a, b] \cap \mathbb{Q}$ and $F = (c, d] \cap \mathbb{Q}$. Then by some basic set-theory properties, we know that the intersection of two left-open/right-closed intervals is again a left-open/right-closed interval or $\emptyset$. If their intersection is $\emptyset$, then we're done. If not, let $(e, f] = (a, b] \cap (c, d]$. Then $E \cap F = (e, f] \cap \mathbb{Q}$. Thus, $E \cap F \in \mathcal{A}$

(c) Let $E \in \mathcal{A}$, so that $E = (a, b] \cap \mathbb{Q}$. Let $i \in \mathbb{N}$ ($0 \notin \mathbb{N}$) and:

$$E_i = (a - i, a - (i + 1)] \cap \mathbb{Q} \quad F_i = (b + (i - 1), b + i] \cap \mathbb{Q}$$

then each $E_i, F_i \in \mathcal{A}$, $E_i \cap E_j = F_i \cap F_j = \emptyset$ if $i \neq j$ and $E_i \cap F_j = \emptyset$ for all $i, j \in \mathbb{N}$. Let $E = \{E_i\}_{i=1}^{\infty}$ and $F = \{F_i\}_{i=1}^{\infty}$ and let $G = E \cup F$. Then

$$\bigcup_{g \in G} G = E^c$$

since G is a countable union of disjoint elements of $\mathcal{A}$, this completes the proof.

Thus, by proposition 1.7, $\mathcal{A}$ is an algebra.

2. The σ-algebra generated by $\mathcal{A}$ is $P(\mathbb{Q})$

***Proof*:**
Let $x \in \mathbb{Q}$ Notice that

$$\{x\} = \bigcap_{k=1}^{\infty} \left(x - \frac{1}{k}, x\right] \cap \mathbb{Q}$$

hence, every singleton is inside the σ-algebra of $\mathbb{Q}$. Since every subset of $\mathbb{Q}$ is countable, every subset is the countable union of singletons. Since every singleton is inside the σ-algebra, we have that the σ-algebra is $P(\mathbb{Q})$.

3. Define μ_0 on $\mathcal{A}$ by $\mu_0(\emptyset) = 0$ and $\mu_0(A) = \infty$ for $A \neq \emptyset$. Then μ_0 is a premeasure on $\mathcal{A}$, and there is more than one measure on $P(\mathbb{Q})$ whose restriction to $\mathcal{A}$ is μ_0

***Proof*:**
μ_0 is trivially a pre-measure. By definition, $\mu_0(\emptyset) = 0$. For any nonempty $A \in \mathcal{A}$, we have $\mu_0(A) = \infty$. When checking countable additivity, unless all the sets are empty, we get that both sides are infinite.

Thus, we can define μ to be the μ^* induced by μ_0 restricted to all measurable sets. By construction, all sets are measurable. If a sets is non-empty, then it will always be covered by a nonempty set, which has measure ∞. Thus, If $E \in P(X)$ for any nonempty test set $S \subseteq \mathbb{Q}$

$$\infty = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \infty$$

and if $S = \emptyset$

$$0 = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = 0$$

Thus, $\mathcal{M}_\mu = P(\mathbb{Q})$.

We can define another extension of μ_0 using the counting measure, in particular, let μ_C be the counting measure on $P(\mathbb{Q})$. Then since every $A \in \mathcal{A}$ has countable cardinality

$$\mu_C(A) = \infty$$

Thus, $\mu_C|_{\mathcal{A}} = \mu_0$. However

$$\mu_C(\{1\}) = 1 \neq \infty = \mu(\{1\})$$

Thus, we have defined two separate extensions of μ_0.

1.4 Borel Measure on the Real Line

The full title of this section should really be "Borel Measures on the Real line defined through Distributions", but that title seems to cumbersome. Nevertheless, this is exactly what we will be doing in this section. We will extensively concentrating on these types of measures, since they are, in a sense, the focus of "real analysis". From a high-level perspective, the fact that we are studying "real analysis" justifies why we will take so much time understanding this measure. More concretely, these measure are pervasive in analysis in general: they are central to defining integration, to the study of Fourier series, to studying L^p spaces, to linking differentiation to integration, to generalizing the notation of integration of groups (in particular, it will make sense to integrate locally compact topological groups), and so much more. Another way of looking at it is that the euclidean topology is central to the study of many fields of mathematics and physics, and is the basis for our notion of manifolds (which are locally euclidean). Therefore, getting a sense of how to define notions of "size" in $\mathbb{R}$ (and then generalize to $\mathbb{R}^n$, $\mathbb{R}^\infty$, or manifolds) gives us a solid first step in gaining an intuition on measures.

One way to define a measure in $\mathbb{R}$ is by using our intuition of the fact that the length of (a, b) is $b - a$. This is a good intuition and will be the basis to the generalization of this idea we'll do in order to be able to define a larger class of measures based on this idea which will also have more uses outside of $\mathbb{R}$. Here is how we will motivate the construction of these measures: let μ be a finite measure on Borel sets (a measure define on Borel sets is called the *Borel Measure*). Define:

$$F(x) = \mu((\infty, x])$$

F is sometimes called the *distribution function* of μ (this vocabulary will come back often in probability). Intuitively, it shows the "accumulated size" of the space. Clearly, F is an increasing function, and is right-continuous since

$$(\infty, x] = \bigcap_k (\infty, x + 1/k]$$

or any decreasing function, $x_n \searrow x$ (i.e. limits are preserved from the right). Furthermore, if we take $a < b$, then $(-\infty, b] = (-\infty, a] \cup (a, b]$, thus

$$\mu((a, b]) = F(b) - F(a)$$

Notice how similar this is to the usual result from Riemann integration! In the following, instead of defining F in terms of μ, we will define μ from an increasing right-continuous function F by showing how F define's a pre-measure and using Carathéodory. The special case of $F(x) = x$ will give us our usual notion of length in $\mathbb{R}$.

Throughout this section, we will use sets of the form $(a, b]$ where $a < b$, along with the edge cases of (a, ∞) and $\emptyset$. Let all intervals of this type be called h-intervals (h for half open). It is clear that the union or compliment of two h-intervals are h-intervals (or two disjoint h-intervals) and that h-intervals are closed under finite operations, and so h-intervals from a algebra $\mathcal{A}$. By proposition 1.1.7, the σ-algebra by the set $\mathcal{A}$ is $\mathcal{B}_{\mathbb{R}}$. Using this, we'll first show that we can define a premeasure μ based off of an F:

Proposition 1.4.1: Distribution and Premeasure

Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous. If $(a_i, b_i]$ is a set of disjoint h-intervals. If we define μ_0 to be:

$$\mu_0 \left(\bigcup_i^\infty (a_i, b_i] \right) = \sum_{i=1}^\infty F(b_i) - F(a_i)$$

and $\mu_0(\emptyset) = 0$, then μ_0 is a premeasure on the algebra $\mathcal{A}$

***Proof*:**
We will first show that μ_0 is well-defined, then we simply must show that $\mu\left(\bigcup_i (a_i, b_i]\right) = \sum_i \mu_0((a_i, b_i])$.

To show it's well-defined, we must show that for any finite representation of $(a, b]$, $\mu_0((a, b]) = F(b) - F(a)$. To that end, let $\{(a_i, b_i]\}_{i=1}^n$ be disjoint where $(a, b] = \coprod i = 1^n [a_i, b_i)$. Since the co-domain is commutative, we can re-arrange the order that we are taking the union (i.e., re-index the set) so that

$$a < a_1 < b_1 = a_2 < b_2 = a_3 < \cdots < b_n = b$$

Then the resulting μ_0 is simply an alternating series:

$$\sum_{j=1}^n F(b_j) - F(a_j) = F(b) - F(a)$$

showing that it is independent of representation of the intervals. Thus, if we have any disjoint collection $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^n$ such that $\cup_i I_i = \cup_j J_j$, since we can always form alternative series:

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j)$$

and so μ_0 is well-defined. Since it is well-defined, then we also get that μ_0 is finitely additive (essentially by definition), and since $\mu_0(\emptyset) = 0$, it is at least a finite measure.

Next, we need to show that it is a premeasure. Since $\mu_0(\emptyset) = 0$, it remains to check that if $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $\cup_i^{\infty} A_i \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu_0(A_i)$$

Since $\cup_i^{\infty} I_i = I \in \mathcal{A}$, there exists (at least one) subsets $\{A_i\}_{i=1}^{n} \subseteq \mathcal{A}$ such that $\cup_i^{\infty} I_i = \cup_i^{n} A_i$. Therefore the elements from I can be partitioned so that the union of each partition represents a single h-interval. By finite additivity of μ_0, it suffices to consider what happens at one of these partitions. So, let J be the union of one of the families so that $(a, b]J = \cup_{j=1}^{\infty} I_{i_j}$. We will show that $\mu_0(\cup_j^{\infty} J_j) = \sum_j^{\infty} \mu_0(J_j)$ by showing $\leq$ and $\geq$. For the $\geq$ direction, notice that

$$\mu_0(J) = \mu_0\left(\bigcup_j^{n} J\right) + \mu_0\left(J \setminus \bigcup_j^{n} J\right) \geq \mu_0\left(\bigcup_j^{n} J\right) = \sum_j^{n} \mu_0(J)$$

Thus, as $n \to \infty$ we have $\mu_0(J) \geq \sum_j^{\infty} \mu_0(J_j)$.

For the $\leq$ direction, we will take advantage of the right-continuity of F.

(Might not be the case TBD p.34 Folland) First, if either b is infinite, the result holds automatically. If a is infinite, then b would have to also be infinite, and so we have the zero-intervals. So, let a and b be finite.

Let $\epsilon > 0$. Then since F is right-continuous, there exists a δ such that $F(a + \delta) - F(a) < \epsilon$ or similarly $-F(a) < -F(a + \delta) + \epsilon$. Similarly, for the same epsilon, there exists a δ_j such that $F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^n}$. Notice that $(a_j, b_j] \subseteq (a_j, b_j + \delta_j)$, so the open intervals cover $[a, b]$, and since $[a, b]$ is compact, there exists a finite sub-cover! If $\{(a_j, b_j + \delta_j)\}_{i=1}^{N}$ is the finite subcover of $[a, b]$, if we discard all sets that are contained in a bigger set in this family, then we get that

$$b_j + \delta_j \in (a_{j+1}, \beta_{j+1} + \delta_{j+1}) \quad \text{for } j = 1, \ldots N - 1$$

With this, we get the following sequence of inequalities:

$$\begin{aligned}
\mu_0(J) &= F(b) - F(a) \\
&\leq F(b) - F(a+\delta) + \epsilon \\
&\leq F(b_N + \delta_N) - F(a_1) + \epsilon && \text{extrema of cover} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(a_{j+1}) - F(a_j))\epsilon && \text{fancy 0} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(b_{j+1} + \delta_j) - F(a_j))\epsilon && \text{expanding} \\
&< \sum_{j=1}^{N} [f(b_j - f(a_j) + \epsilon 2^{-n}] + \epsilon \\
&< \sum_{j=1}^{\infty} \mu_0(J_j) + 2\epsilon
\end{aligned}$$

and since ϵ was arbitrary, the result follows.

This result let's us prove that *any* increasing right-continuous function defines a measure!

Theorem 1.4.2: Distributions define Measure

If $F : \mathbb{R} \to \mathbb{R}$ is an increasing right-continuous function, there is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a, b]) = F(b) - F(a)$ for all $a, b \in \mathbb{R}$. This function is unique up to the following condition: if G is another such function, then $F - G$ is constant.
Conversely, if μ is a Borel measure that is bounded on all finite Borel sets, we can define

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((-x, 0]) & \text{if } x < 0 \end{cases}$$

Then F is an increasing right-continuous function

***Proof*:**
First, by the previous proposition, F induces a premeasure on $\mathcal{A}$. If $F - G$ is constant, then the sums will cancel out in $F(b) - F(a)$, and so $\mu_F = \mu_G$. Similarly, if the two are equal, F and G can only differ up to a constant and that these measures are σ-finite (recall that $\mathbb{R} = \bigcup_{-\infty}^{\infty} (-i, i+1])$.

For the converse, note that if μ is monotonic, then so is F. Next, above/bellow continuity of μ will imply right-continuity of F for $x \geq 0$ and $x < 0$. Therefore, by construction, $\mu = \mu_F$ on $\mathcal{A}$, and so by proposition 1.4.1, $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness in theorem 1.3.7.

Note that F need not be left-continuous. Something with the point-density measure that proves the counter-case.

We have been using intervals of the form $(a, b]$ and right-continuous function. The same theory can be developed with $[a, b)$ intervals and left-continuous functions. If we restrict μ to being a finite Borel measure, then we get $\mu = \mu_F$ where $F(x) = \mu((-\infty, x])$ (i.e. it differs up to a constant of the μ_F when μ is a Borel measure). Next, by the construction of the measure from the pre-measure, the resulting measure is in fact a complete measure. Even better, taking completion of μ_F and then applying Carathéodory is the same as just taking completion of μ after applying Carathéodory to μ_F, and so taking the completion on μ_0 or μ_F does not make a difference! The complete measure is also usually denoted by μ_F and called the *Lebesgue-Stieljes measure* of F.

As a consequence of the Lebesgue-Stieljes measure μ being complete, if we have a measure on a Borel space (which we usually call a Borel measure), then then then the set of measurable sets $\mathcal{M}^*$ might be larger than the number of Borel sets (in particular in that it may contain some zero sets that are not Borel sets).

The Lebesgue-Stieljes measure has a couple of nice properties worth exploring. Let μ be a complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F, and let $\mathcal{M}_\mu$ be the associated μ^*-measurable sets. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\left\{\sum_1^\infty F(b_i) - F(a_i) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i]\right\} = \inf\left\{\sum_1^\infty \mu((a_i, b_i]) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i]\right\}$$

Notice that the contribution of the right end points of every h intervals is clearly negligable and so we can replace it with open intervals

Lemma 1.4.3: h-Intervals to Open Intervals

Let μ be the complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\left\{\sum_1^\infty \mu((a_i, b_i)) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i)\right\}$$

***Proof*:**
This is the same epsilon trick you have seen before, right it down here as an exercise later

In fact, we can write down $\mu(E)$ in even simple terms

Theorem 1.4.4: Lebesgue-Stieljes in Terms of Open and Compact sets

Let μ be the complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to $\mathbb{F}$. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\{\mu(U) \mid U \supseteq E \text{ and } U \text{ is open}\}$$

$$\mu(E) = \sup\{\mu(K) \mid K \subseteq E \text{ and } U \text{ is compact}\}$$

***Proof*:**
By lemma 1.4.3, for all $\epsilon > 0$, there exists open intervals (a_i, b_i) such that $E \subseteq \cup_i^\infty (a_i, b_i)$ and $\mu(E) \leq \sum_i^\infty (a_i, b_i) + \epsilon$. If we let $U = \cup_i (a_i, b_i)$, we have that $\mu(U) \leq \mu(E) + \epsilon$. Conversely, since $E \subseteq U$, $\mu(U) \geq \mu(E)$, and so the result for the first statements follows.

For the 2nd equivalence, let's first suppose E is bounded. If E is closed, then it is compact and so E is contained in the set that we are taking the sup over, and so equality is immediate. If E was open, then we can choose an $\epsilon > 0$ such that $\overline{E} \setminus E \subseteq U$ and $\mu(U) \leq \mu(\overline{E} \setminus E) + \epsilon$. Set $K = \overline{E} \setminus U$. Then K is compact, and so

$$\begin{aligned}\mu(K) &= \mu(E) - \mu(E \cap U) \\ &= \mu(E) - [\mu(U) - \mu(U \setminus E)] \\ &\geq \mu(E) - [\mu(U) - \mu(\overline{E} \setminus E)] \\ &= \mu(E) - \mu(U) + \mu(\overline{E} \setminus E) \\ &\geq \mu(E) - \epsilon\end{aligned}$$

showing that any compact set arbitrarily approximates K.

If E is unbounded, let $E_i = E \cap (i, i+1]$ so that $E = \cup_i E_i >$ Then by what we've just shown, for all $\epsilon > 0$, there exists a $K_j \subseteq E_j$ such that $\mu(K_j) \leq \mu(E_j) - \frac{\epsilon}{2^n}$. Take $H_n = \cup_{-n}^n K_i$. Then H_n is compact, $H_n \subseteq E$, and

$$\mu(H_n) \geq \mu(\cup_{-n}^n E_i) - \epsilon$$

Since this equality holds when $n \to \infty$, the result follows

This gives us a strikingly easy representation of representing every measurable subset of $\mathbb{R}$!

Theorem 1.4.5: $E \subseteq \mathbb{R}$ Representation

Let $E \subseteq \mathbb{R}$. Then the following are equivalent

1. $E \in \mathcal{M}_\mu$
2. $E = V \setminus N_1$ where V is a G_δ set and $\mu(N_1) = 0$
3. $E = W \cup N_2$ where W is a H_σ set and $\mu(N_2) = 0$

***Proof*:**
Clearly, (2) and (3) imply (1). For (1) implies (2) or (3), I would highly recommend you try and solve this in the $\mu(E) < \infty$ case to jog your memory.

For the $\mu(E) = \infty$, TBD (exercise 25 p.37 Folland)

This means that all borel sets, or more generally sets in $\mathcal{M}_\mu$ (also contain all unions of borel sets with measure zero sets). Note that this theorem and proof would be much harrier if we did not assume the measure was complete. ‘

I will leave this section with another way of interpreting finite measurable sets that Folland left as an exercise (we'll use the notation of $A \triangle B = (A \setminus B) \cup (B \setminus A)$)

Proposition 1.4.6: Diminishing Difference

Let $E \in \mathcal{M}_\mu$ and $\mu(E) < \infty$. Then for every $\epsilon > 0$, there exists a set A which is the finite union of open intervals and

$$\mu(E \triangle A) < \epsilon$$

1.4.1 Lebesgue Measure

We will study the case where the measure μ is defined through the distribution $F(x) = x$, i.e., when F represents our usual intuition of length! This is arguably the most important measure on $\mathbb{R}$, and most certainly the most used. For that reason, it has some special symbology: we will denote the measure by m instead of μ and the set of measurable sets by $\mathcal{L}$. If we restrict m to $\mathcal{B}_{\mathbb{R}}$, we will also write it as m.

The first properties worth establishing is to see that measurable sets in $\mathbb{R}$ inherit some of our intuitions how a "shape" should act under translation and dilation, that it's invariant under translation (just like we assumed for the construction of Vitali sets!) and it scales linearly with dilation:

Proposition 1.4.7: Scaling and Dilation

Let $(X, \mathcal{L}, m)$ be a Lebesgue measure space. Define

$$E + s := \{e + s \mid e \in E\} \qquad rE := \{re \mid e \in E\}$$

Then $E + s, rE \in \mathcal{L}$ for all $r, s \in \mathbb{R}$, and:

$$m(E + s) = m(E) \qquad m(rE) = |r|m(E)$$

***Proof*:**
We essentially notice that open intervals are invariant under translation and operate in the form described in the proposition, then apply theorem 1.3.7 (which is a really cool application of this theorem!!)

For any $E \in \mathcal{B}_{\mathbb{R}}$ define $m_s(E) = m(E+s)$ and $m^r(E) = M(rE)$. Then m_s and m^r clearly agree with m and $|r|m$ on finite unions of intervals, and so by theorem 1.3.7 they agree on all of $\mathcal{B}_{\mathbb{R}}$. As a consequence, if $m(E) = 0$, then $m(E + s) = m(rE) = 0$. Thus, for all sets in $\mathcal{L}$ (i.e. all unions of borel sets and Lebesgue zero-sets) must be preserved and linearly scaled (respectively) under translation and dilation:

$$m(E + s) = m(E) \qquad m(rE) = |r|m(E)$$

as we sought to show

Since we are working over $\mathcal{M}(\mathcal{B}_{\mathbb{R}})$, the next interesting questions we may ask about m is how it interacts with the topological sets on $\mathbb{R}$. For starters, it is clear (or should be immediately proven) that any singleton has measure 0. Therefore, by countable additivity, every countable set has measure zero; in particular $m(\mathbb{Q}) = 0$. Notice that the rational are dense which is a "topologically large" concept, however it is measure-theoretically small. In fact, we can even have a collection of *open dense* sets, and still be measure-theoretically small. Take some enumeration $\{r_i\}_{i=1}^{\infty}$ of the rational numbers in $[0, 1]$ and let I be the indexing set for this enumeration $i \in I$. Let I_n be the interval of size $\frac{\epsilon}{2^n}$ which is centered at r_n. Then set $U = \bigcup_n I_n$. By construction, U must be dense and open in $(0, 1)$, however

$$\mu(U) \leq \sum_{i=1}^{\infty} \mu(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

and hence can be as measure-theoretically small as we want. From this we can also find the opposite, where we have something topologically small but measure-theoretically big, by taking $K = [0,1] - U$. By monotonicity

$$\mu(K) \geq 1 - \epsilon$$

and yet K is nowhere dense.

There is at least one intuition that can remain: a nonempty *open* set will have non-zero measure. For more on how open/closed sets interact with the measure, see the exercises.

Note that the converse that a non-zero measure set will have an open intervals, is not actually true! For that, we can explore the concept of *cantor sets*

Cantor Set

(was gonna look through: (commented so that this complies

Definition 1.4.8: Classical Cantor Set

Let $I = [0,1]$. From this, we'll construct the cantor set iteratively

1. Start by removing the middle third $I_1 = I - \left(\frac{1}{3}, \frac{2}{3}\right)$
2. Given I_i, remove the open middle third of each connected compoennt

Let $C = \cap_{n=0}^{\infty} I_n$ be the Cantor set. In particular, it is the *standard middle thirds Cantor set.* We'll usually refer to this set as "the" Cantor set.

Notice that C is measurable by continuity from above since $C_1 \subseteq C_2 \subseteq \cdots \subseteq C_n \subseteq$.

Proposition 1.4.9: Properties Of Cantor Sets

Give C the standard subspace metric of $[0,1]$. Then C is:

1. nonempty and uncountable
2. compact
3. nowhere dense ($\text{int}(\overline{A}) = \emptyset$ for all $A \subseteq C$
4. $m(C) = 0$
5. totally disconnected

***Proof*:**
Using some topological properties, since C is the intersection of compact spaces, it is compact.

Next, the cantor set is nonempty since $0 \in C_k$ for all k, and so $0 \in C$. To show C is uncountable, Notice that we can represent every element of the uncountable set $[0,1]$ in it's base 3 expansion, namely for appropriate $a_k \in \{0,1,2\}$:

$$x = \sum_{k=1}^{\infty} a_k 3^{-k}$$

Then take:

$$f : [0,1] \to C \qquad x = \sum_{k=1}^{\infty} a_k 3^{-k} \mapsto$$

To show total disconnectedness, let's take the same I as before. I is closed in $\mathbb{R}$, and thus in C^n. Note that I^c is the union of finite closed sets, and thus is also closed. Thus, I is clopen in C^n. Thus, $C \cap I$ is also a clopen neigborhood of x in C. Since x was arbitrary, C is totally disconnected.

Finally uncountablenss come from C being compact and complete, which makes C uncountable (exercise).

This construction does not rely on the size of the middle-third; in fact, we can make it converge to any value we want. To see this, first notice that To do this, we first prove a lemma:

Lemma 1.4.10

Suppose $\{\alpha_i\}_{i=1}^{\infty} \subseteq (0,1)$.

1. $\prod_1^{\infty}(1-\alpha_i) > 0$ if and only if $\sum_1^{\infty} \alpha_i < \infty$ (Compare $\sum_1^{\infty} \log(1-\alpha_i)$ to $\sum_1^{\infty} \alpha_i$)
2. Given $\beta \in (0,1)$, there always exists a sequence $\{\alpha_i\}_{i=1}^{\infty}$ such that $\prod_1^{\infty}(1-\alpha_i) = \beta$

***Proof*:**
First, we see that we can convert $\prod_i^{\infty}(1-\alpha_i)$ to $\sum_i^{\infty} \log(1-\alpha_j)$ by considering

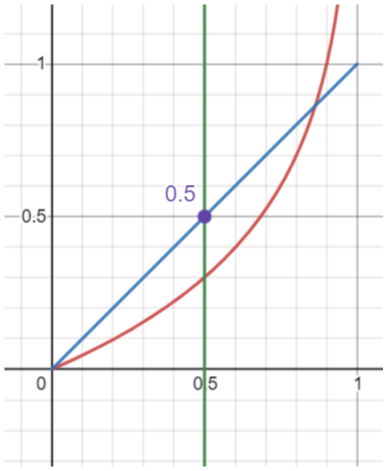
$$\begin{aligned}
\log\left(\prod_i^{\infty}(1-\alpha_i)\right) &= \log\left(\lim_{n\to\infty}\prod_i^{n}(1-\alpha_i)\right) \\
&\overset{!}{=} \lim_{n\to\infty} \log(\prod_i^{n}(1-\alpha_i)) \\
&= \lim_{n\to\infty} \sum_i^{n} \log(1-\alpha_i) \\
&= \sum_{i=1}^{\infty} \log(1-\alpha_i)
\end{aligned}$$

where $\overset{!}{=}$ comes from continuity of log for $1-\alpha_i > 0$. Notice that it's impossible that $1-\alpha_i \leq 0$ form some α_i terms since $\alpha_i \in (0,1)$, and so this proof works. Thus, we get the equivalent condition of

$$\sum_i^{\infty} \log(1-\alpha_i) > -\infty$$

With this setup, we proceed with the proof

($\Leftarrow$) Let $\sum_i^{\infty} \alpha_i < \infty$. Then we know that $\alpha_i \to 0$. Thus, pick an N such that for all $i > N$, $\alpha_i < 0.5$. Then notice that $\log(1-\alpha_i) < \alpha_i$. This is easier to see with a visual aid:



In particular, for all $0 < x < 1/2$, we have $|\log(1-x)| - x > 0$, showing that x is greater than $\log(1-x)$ for all $x \in (0, 1/2)$.

Then from here, this is a simple application of the Weiestrass M-test. Let $K = \sum_i^N \log(1-\alpha_i)$ to eliminate the irregular part. Then since $|\log(1-\alpha_i)| < \alpha_i$, for all $i > N$ and $\sum_i^\infty \alpha_i$ converges, by the Weiestrass M-test the $\sum_{i=N+1}^\infty \log(1-\alpha_i)$ converges. Let's say it converges to L. Then

$$\sum_{i=1}^{\infty} \log(1-\alpha_i) = K + L > -\infty$$

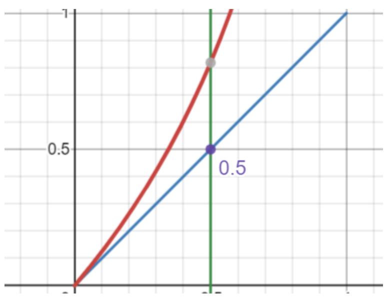
as we sought to show.

($\Rightarrow$) let $\prod(1-\alpha_i) > 0$ so equivalently $\sum_i^\infty \log(1-\alpha_i) > -\infty$ so $\sum_i^\infty \log(1-\alpha_i) = K$ for appropriate $K \in \mathbb{R}$. Then since the series converges, we have that

$$eK = e\sum_i^\infty \log(1-\alpha_i) = \sum_i^\infty e\log(1-\alpha_i)$$

i.e., we can scale the terms by e. We will show why this scaling is important in a moment; before we do we must setup a little more.

Since the series converges, it must be that $e\log(1-\alpha_i) \to 0$. So pick N such that for all $i > N$, $e\log(1-\alpha_i) < 1/2$. Then notice that $\alpha_i < e\log(1-\alpha_i)$. This is easier to see with a visual aid:



Or, to carry out some of the computations, if you raise $e\log(1-\alpha_i) = \log(1-\alpha_i)e$ to the power of e, you get

$$(e^{\log(1-\alpha_i)})^e = (1-\alpha_i)^e \geq \alpha_i^e \quad \forall \alpha_i \in (0, 1/2]$$

and so the inequality holds. Then the proof continues as for the $\Leftarrow$ direction by setting up an appropriate constant term to bound the irregularities, then using the Weiestrass M test bound every α_i by $\log(1-\alpha_i)$, and since $\sum_i^\infty \log(1-\alpha_i)$ converges, we get that $\sum_{i=N}^\infty \alpha_i$ also converge, and since it will be the sum of a constant plus the convergent value, $\sum_{i=1}^\infty \alpha_i$ converges, as we sought to show.

Now, let β be given. Let $x_i = \prod_{i=1}^n (1-\alpha_i)$. We'll construct a sequence where

$$x_1 = \beta + \frac{1-\beta}{2} \qquad x_2 = \beta + \frac{1-\beta}{4} \qquad \cdots \qquad x_n = \beta + \frac{1-\beta}{2^n} \cdots$$

to accomplish this, let

$$\alpha_i = 1 - \frac{\beta + \frac{1-\beta}{2^i}}{\beta + \frac{1-\beta}{2^{i-1}}} \tag{1.2}$$

This construction comes from solving the following equation:

$$\left(\beta + \frac{1-\beta}{2}\right) y_2 = \beta + \frac{1-\beta}{4} \;\Rightarrow\; y_2 = \frac{\beta + \frac{1-\beta}{4}}{\beta + \frac{1-\beta}{2}}$$

so $\alpha_2 = 1 - y_2$, and clearly $\alpha_2 \in (0,1)$ since the denominator is bigger than the numerator in the above calculations. Generalizing this pattern, we get the that α_i would be of the from presented in equation (1.2) and $\alpha_i \in (0,1)$. Thus, we get that

$$\lim_{n\to\infty} \prod_i^n (1-\alpha_i) = \lim_{n\to\infty} \beta + \frac{1-\beta}{2^n} = \beta$$

as we sought to show

Now, let's say we are starting with $[0,1]$ but on each iteration eliminates intervals that sum to α_i. Let K_i be the ith step in this itteration. Notice that:

$$m(K_{i+1}) = (1-\alpha_i)m(K_i)$$

Then by what we've just shown, we can make $K = \cap_i^\infty K_i$ be any value up to and including 0 and 1.

Proposition 1.4.11: All Cantor Set Homeomorphism

Let C_n be a cantor set where $m(C_n) = n$. Then $C_n \cong C_0$

***Proof*:**
section 9 of Pugh – will get back to it

You can also prove uncountability directly by making a string represent which side of the split interval you go to.

Next, you might've seen many uncountable sets which are dense in $\mathbb{R}$. Here we have an absolutely facsinating result: C is *nowhere dense in* $[0, 1]$.

Definition 1.4.12: Dense, Somewhere Dense, Nowhere Dense

If $S \subseteq M$ and $\overline{S} = M$, then S is dense in M. The set S is somewhere dense if there exists an open nonempty set $U \subseteq M$ such that $\overline{S \cap U} \supset U$. If S is not somewhere dense, then it is *nowhere dense*

Proposition 1.4.13: C Nowhere Dense

The Cantor set contains no interval and is nowhere dense in $\mathbb{R}$

***Proof*:**
To prove it doesn't contain an interval, assume it does, then chose an n large enough, that is $(\frac{1}{3})^n < b - a$, so that it will not be in the interval.

For nowhere dense, assume ti's somewhere then, then $\overline{C \cap U} \supset U \supset (a, b)$ – a contradiction.

We know present how Cantor sets are almost initial objects in compact metric space

Theorem 1.4.14: Cantor Surjection Theorem

Given a compact nonempty metric space M. there is a continuous surjection of C onto M

***Proof*:**
exercise

Finally, we present how all cantor sets are homeomorphic.

Theorem 1.4.15: Moore-Kline Theorem

We say M is a Cantor space if it is compact, nonempty, perfect, and totally disconnected (like the cantor set). Every cantor space M is homeomorphic to the standard middle-thirds Cantor set C.
In other words, those properties *define* Cantor sets.

***Proof*:**
Pugh. p.112

Corollary 1.4.16: Cantor And Fat Cantor Set

The fact Cantor set is homeomorphic to the standard Cantor set

***Proof*:**
Using 1.4.15, the result is immediate.

This is intersting, since it shows that meausre is *not* a topological poperty! This should makes sense with $(0, 1)$ and $\mathbb{R}$ being homeomorphic with the same measures, but different, but this is an even more extreme example.

Corollary 1.4.17: Product Of Cantor Sets

A cantor set is homeomorhic to its own cartesian product; $C \equiv C \times C$.

***Proof*:**

A fact that Folland added that I thought was cool is that since every subset of the cantor set is Lebesgue Measurable (since the cantor set has measure zero and the Lebesgue measure is complete), we have that

$$|\mathcal{L}| = |P(\mathbb{R})| > \aleph_1 \quad \text{but} \quad |\mathcal{B}_{\mathbb{R}}| = \aleph_1$$

so there are a *tone* of zero-measure sets, however, we can essentially ignore them because they are zero measure sets!!

For those who are Categorically inclined, Cantor sets are almost like the initial objects of the category of compact metric spaces. They are not initial because there does not exist a unique uniformally continuous function from the cantor set to any compact metric space, but there always will exist at least 1 such function.

Exercise 1.4.1

1. If $E \in \mathcal{L}$ and $m(E) > 0$, for any $\alpha < 1$, there is an open interval I such that $m(E \cap I) > \alpha m(I)$

2. If $E \in \mathcal{L}$, and $m(E) > 0$, the set $E - E = \{x - y \mid x, y \in E\}$ contains an interval centered at 0 (If I is as in the previous exercise with $\alpha > \frac{3}{4}$, then $E - E$ contains $(-\frac{1}{2}m(I), \frac{1}{2}m(I))$).

3. Let E be a Lebesgue measurable set (so $E \in \mathcal{L} = \overline{\mathcal{M}}(\mathcal{B}_{\mathbb{R}})$)

    1. If $E \subseteq N$ where N is a non-measurable set described in section 1.1, then $m(E) = 0$
    2. if $m(E) > 0$, then E contains a non-measurable set (it suffices to assume $E \subseteq [0, 1]$. In notation of section 1.1 $E = \cup_{r \in R} E \cap N_r$

2

Integration

In this chapter, we now work with function's between measure spaces that preserve measure structure! Measurable functions feel like the types of functions we "usually" work with, the same way measurable sets are the sets we "usually" work with. Being continuous can be a bit strong (ex. being a monotone function doesn't imply continuous), but measurability rectifies that by in a sense expanding what are the possible "pre-images" we allow to take (ex. Monotone functions *are* always measurable, as we will demonstrate).

After defining measurable functions, we will focus on real and complex functions (i.e. functions with codomain $\mathbb{R}$ or $\mathbb{C}$). In the real case, I like to think of real functions as given every point in the domain X a value or a weight, in contrast to a measure which is a real function which simply assigns a constant value at each point (this will become more clear as we introduce the definitions). For the complex case, each point is given a weight and a *phase* or *state*[1]. In terms of computation, we simply split the problem into the real and imaginary component. Then, we will define how to "measure" a real (resp. complex) measurable function. We then focus on measurable functions which have "finite measure", and will call them *(Lebesgue) integrable functions*; integrable functions will allow us to have some regularity in how we can combine measurable functions without running into some problems with infinities (as we'll soon show). One result of this is that we can think of the integral as being a "measure" for a function, and since integrable function will have "finite measure", it will make it more meaningful to see which functions are "close" (in the appropriate sense of the word close) to a certain function.

A word must be said about Lebesgue integrability vs. Riemann integrability. The key difference is the allowed measurable sets. For Riemann integrability, all our measurable sets are *rectangles*: every partition of the area of integration can be seen as a sum of indicator functions over pair-wise disjoint

[1]you can think of every input having a natural magnitude that periodically shifts. The phase $e^{i\theta}$ tracks this. So if $f(a) = 10$ and $f(b) = 10e^{-i\pi} = -10$, then a is at the beginning of its phase with magnitude 10, while b is half way through its phase which shifted it's magnitude to -10. Addition adds the interference (ex. $(2+i)+(-2-2i) = -i$) while

rectangles. On the other hand, Lebesgue integration is over *measurable* sets. The strategy shall shift so that instead of partitioning the domain based off of its geometry, we shall take partition it using $f^{-1}([-k,k])$ (or really any measurable covering of $\mathbb{R}$, and appropriately modified for $\mathbb{C}$) where we shall show the pre-image of a measurable set is measurable if f is measurable. This shall solve many headache the Riemann integral imposes (ex. any rectangle of $\chi_{\mathbb{Q}}$ on $[0,1]$ will have have rational and irrational points, making it unfeasible to measure).

After we have defined integrable functions, we will have to take a moment to compare different types of convergences, since at that stage we will have 4 different ways functions can converge. We will then move on to trying to bring the theory of the product of σ-algebras to measure and integrable functions more generally, allowing us to construct larger measures and integrable functions from smaller components. Importantly, we will show that we can find the a measure $\mu \times \nu$ by separating any problem to only measuring "slices" of the set with respect to μ, then "slices" of the set with respect to ν (this is Fubini's Theorem).

2.1 Measurable Functions

Recall that if $f : X \to Y$ is a function, then $f^{-1} : P(Y) \to P(X)$ is a function where $f^{-1}(E) = \{x \in X \mid f(x) \in E\}$ which preserves unions, intersections, and compliments. Thus, if $\mathcal{N}$ is a σ-algebra on Y, $f^{-1}(\mathcal{N})$ is a σ-algebra on X. If X and Y are measure space, if $f^{-1}(\mathcal{N})$ maps into $\mathcal{M}$, we will call f *measurable*:

Definition 2.1.1: Measurable Function

Let $f : X \to Y$ be a function and $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. Then f is called a $(\mathcal{M}, \mathcal{N})$-*measurable function* (or just measurable) if for all $E \in \mathcal{N}$

$$f^{-1}(E) \in \mathcal{M}$$

Just like in topology, this could be thought of as a surjection of the measurable space on X onto the measurable space on Y. It should be clear that the composition of two measurable functions is measurable (in particular, the composition of an $(\mathcal{M}, \mathcal{N})$-measurable function with $(\mathcal{N}, \mathcal{O})$-measurable function is $(\mathcal{M}, \mathcal{O})$-measurable), and that the identity function $\mathrm{id} : X \to X$ is clearly a measurable function, and so the collection of measurable functions and measurable spaces from a category, usually denoted **Meas**.

Just like lemma 1.1.4 was useful for simplifying proofs, we proof the following:

Lemma 2.1.2: Measurable Function and Generating Sets

Let $\mathcal{N}$ be generated by $\mathcal{E}$. Then $f : X \to Y$ is a $(\mathcal{M}, \mathcal{N})$-measurable function if and only if for all $E \in \mathcal{E}$, $f^{-1}(E) \in \mathcal{M}$

***Proof*:**
The $\Rightarrow$ direction is *a-fortiori* true by definition. The $\Leftarrow$ direction comes from f^{-1} preserving unions and compliments, so $\left\{E \subseteq Y \;\middle|\; f^{-1}(E) \in \mathcal{M}\right\}$ is a σ-algebra that contains $\mathcal{E}$, and so contains $\mathcal{N}$.

From this, there is a very nice corrollary when $\mathcal{M}$ is a collection of Borel sets:

Corollary 2.1.3: Continuous Function $\Rightarrow$ [Borel] Measurable Function

Let $f : X \to Y$ be a continuous function and Y have the borel σ-algebar. Then f is a measurable function.

***Proof*:**
Since f is continuous, the pre-image of an open set (in particular, open intervals) is open, and hence by proposition 1.1.7 and lemma 2.1.2 it is measurable

Naturally, in analysis, the most important measurable functions will be of the form

$$f : \mathbb{R} \to \mathbb{R} \qquad f : \mathbb{R} \to \mathbb{C}$$

This also means that the domain can either have the $\mathcal{B}_{\mathbb{R}}$ σ-algebra or $\mathcal{L}$ σ-algebra.

Example 2.1.4: Measurable Functions
Most functions you have worked with will in fact be measurable

1. As already shown, all continuous functions are measurable
2. All monotone functions are measurable: the pre-image of closed rays will either be open or closed rays, and so by proposition 1.1.7, they are measurable.
3. All indicator functions over measurable sets are measurable (hence, bump functions are essentially measurable)
4. All Riemann integrable functions are measurable (we will show this this in section 2.3.1)
5. We will soon show that the point-wise limit of measurable functions is also measurable, hence almost all examples of functions that shall be studied are measurable.

It is in practice rare to work with non-measurable functions unless a contrived example is used, like a measurable function purposefully built to map to non-measurable sets (like in the case of the composition of two Lebesgue measurable functions)

2.1.5 Convention e will *always* assume the codomain will have $\mathcal{B}_{\mathbb{R}}$ (resp. $\mathcal{B}_{\mathbb{C}}$) σ-algebara. In fact, if we have a function $f : X \to \mathbb{R}$ (resp. $f : X \to \mathbb{C}$), then we will sometimes say f is $\mathcal{M}$-measurable instead of $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable (resp. $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable). If the domain is also Borel, we will say that f is *Borel measurable* (and omit the Borel measure if it's $\mathbb{R}$ or $\mathbb{C}$). Similarly, if the domain is $\mathcal{L}$, we will call f Lebesgue measurable.

Because of the convention of the co-domain being Borel, it is possible that the composition of two Lebesgue measurable functions is *not* Lebesgue measurable: if $f : \mathbb{R} \to \mathbb{R}$, $g : \mathbb{R} \to \mathbb{R}$ are two Lebesgue measurable functions where $E \in \mathcal{B}_{\mathbb{R}}$, $g^{-1}(E) \in \mathcal{L}$ but $g^{-1}(E) \notin \mathcal{B}_{\mathbb{R}}$, it is possible that $f^{-1}(g^{-1}(E)) \notin \mathcal{L}$

Example 2.1.6: Composition Not Lebesgue Measurable

Let $f : [0,1] \to [0,1]$ be the cantor function form section 1.5, and let $g(x) = f(x) + x$.

1. g is a bijection from $[0,1]$ to $[0,2]$ and $h = g^{-1}$ is continuous from $[0,2]$ to $[0,1]$

   ***Proof*:**
   First, since f is increasing (though not injective) and x is strictly increasing, then $f + x$ is injective. Since f and x are both continuous, so is $g = f + x$. Hence the pre-image of a closed set must be closed. If we consider $g^{-1}([0,2]) = D$, then using the fact that g is a monotonic functions, since $g(0) = 0$ and $g(1) = 2$, it must be that $D = [0,1]$, showing that g is indeed surjective, and hence bijective.

   To show that $g^{-1} : [0,2] \to [0,1]$, is continuous, bijective, and the codomain is Hausdorff, then g is homeomorphic (recall that these imply that g is an open map: if $U \subseteq [0,1]$ then $S = [0,1] - U$ is closed, and since g is continuous $g([0,1] - U) = [0,2] - g(U)$ is closed, so $g(U)^c$ is closed, so $g(U)$ is open), and hence g^{-1} is continuous.

2. If C is the cantor set, then $m(g(C)) = 1$

   ***Proof*:**
   We will take advantage of translation invariance of the Lebesgue measure. Let C be the cantor set on $[0,1]$. Then C can be thought of as the interval $[0,1]$ minus all of the open sets that are eliminated during construction:

   $$C = [0,1] - I_1 - I_{21} - I_{22} - \cdots - I_{ik} - \cdots$$

   Let $I = \coprod_{i,k} I_{ik}$ collection of these sets so that $C = [0,1] - I$. Since $m(C) = 0$, $m(I) = 1$. Consider now $g(I)$. Since g is injective and I is the disjoint union of sets, then there exists sets $\{J_j\}_{j=1}^{\infty}$ such that $J_j = g(I_{ik})$. We now examine the measure of the sets J_j.

   By definition, I_{ik} does not contain any points of the cantor set, and so for appropriate $c_{ik} \in C$,

   $$g_{I_{ik}}(x) = f(c_{ik}) + x$$

   that is, g on the interval I_{ik} is of the form $x + f(c_{ik})$ for constant $f(c_{ik})$! Since the Lebesgue measure is translation invariant, we get:

   $$m(g(I_{ik})) = m(g(I_{ik}) - f(c))$$

   and so, since the I_{ik} intervals are all disjoint and g is injective:

   $$g(\coprod I_{ik}) = \coprod g(I_{ik})$$

   and so the measure of $g(I)$ is in fact exactly the measure of I, that is, $m(g(I)) = m(J) = 1$. Thus:

   $$2 = g(m([0,2])) = m(g(C) \coprod J) = m(C) + m(J) = m(g(C)) + 1$$

   and so $m(g(C)) = 1$, as we sought to show.

3. by exercise 29, $g(C)$ contains a Lebesgue non-measurable set A. Let $B = g^{-1}(A)$. Then B is Lebesgue measurable, but not Borel

***Proof*:**
Since $A \subseteq g(C)$, $B \subseteq g^{-1}(g(C)) = C$. Since $m(C)$ is zero, by completeness, $m(A) = 0$, and so A is Lebesgue measurable.

Next, to show that B is not Borel-measurable, For the sake of contradiction let's assume it was. Then since g^{-1} is continuous, it is Borel-measurable, meaning the pre-image (with respect to g^{-1}) of a measurable set is measurable. However, $g(B) = A$, which is not measurable, and hence B is not Borel measurable.

4. There exists a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable

***Proof*:**
Let F be the indicator function on B, I_B, and $G = g^{-1}$. Then F and G are clearly Lebesgue measurable, However, G^{-1} does not map all Lebesgue measurable sets to Lebesgue measurable sets (in particular, B maps to a Borel set), and so

$$(F \circ G)^{-1}(B) = (G^{-1} \circ F^{-1})(B) = (g(I_B^{-1}(B)) = g(B) = A$$

However, A is not Lebesgue measurable set, and so $F \circ G$ is not Lebesgue measurable.

Just like for Borel sets earlier, we have the following equivalence for measurable functions with codomain $\mathbb{R}$:

Proposition 2.1.7: Borel sets and Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $f : X \to \mathbb{R}$. Then the following are equivalent:

1. f is $\mathcal{M}$-measurable
2. $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
3. $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
4. $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
5. $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$

***Proof*:**
This follows from proposition 1.1.7 and lemma 2.1.2.

Similarly, we will soon be encountering co-domains which are product spaces (ex. $\mathbb{R}^n$ or $\mathbb{C}^n$). In particular, let's say we had some set X, and some family of measurable spaces $\{(Y_\alpha, \mathcal{M}_\alpha)\}_{\alpha \in A}$, and $f : X \to Y_\alpha$ is a map for each $\alpha \in A$. Then we can impose a unique smallest σ-algebra on X that makes each f_α a measurable function, in particular, $\mathcal{M}_X = \left\{ f_\alpha^{-1}(E_\alpha) \;\middle|\; E_\alpha \in \mathcal{M}_\alpha,\ \alpha \in A \right\}$ [2]. This σ-algebra on X is called the *σ-algebra generated by* $\{f_\alpha\}_{\alpha \in A}$.

[2] A similar strategy is used in the constrution of *weak* topologies. If you're more categorical, this constructions makes X satisfy the universal property of products in the category of **Meas**

Proposition 2.1.8: Product of Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $(Y_\alpha, \mathcal{N}_\alpha)$ be a collection of measurable spaces, so $Y = \prod_{\alpha \in A} Y_\alpha$, $\mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha$, and $\pi_\alpha : Y \to Y_\alpha$ is the coordinate map. Then $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable if and only if $f_\alpha = \pi_\alpha \circ f$ is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable.

***Proof*:**
Let's say f is $(\mathcal{M}, \mathcal{N})$-measurable. Then since the composition of two measurable functions is measurable, so are each f_α is measurable (since π_α is measurable by definition of $\bigotimes$).

Conversely, if every f_α is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable, then for each $E_\alpha \in \mathcal{N}_\alpha$, $f^{-1}(\pi_\alpha^{-1}(E_\alpha)) = f^{-1}(E_\alpha) \in \mathcal{M}$, and since these sets generate $\mathcal{M}$, we have that f is $(\mathcal{M}, \mathcal{N})$-measurable by proposition 2.1.7

This proposition also let's us define what it means to take the "product" of two functions. As a consequence, we can give an easy criterion for when complex functions are measurable:

Corollary 2.1.9: Complex Function Measureability

Let $f : X \to \mathbb{C}$ be a complex function. Then f is $\mathcal{M}$-measurable if and only if $\mathrm{Re} f$ and $\mathrm{Im} f$ are both measurable.

***Proof*:**
This is simply taking advantage of the topology of $\mathbb{C}$, namely

$$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} \stackrel{!}{=} \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$$

where the $\stackrel{!}{=}$ equality comes from the fact that $\mathbb{R}$ is separable (see proposition 1.1.11). And so proposition 2.1.8 with the fact that Re and Im can be considered projections completes the proof.

Sometimes, we want to work with the extended real line (or in general some compactification of a topological space). If $\overline{\mathbb{R}} = [-\infty, \infty]$, Then let

$$\mathcal{B}_{\overline{\mathbb{R}}} = \left\{ E \subseteq \overline{\mathbb{R}} \;\middle|\; E \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}} \right\}$$

Notice that is we put the metric ρ on $\overline{\mathbb{R}}$ to be $\rho(x, y) = |\tan^{-1}(x) - \tan^{-1}(y)|$, then $\mathcal{B}_{\overline{\mathbb{R}}}$ is indeed the usual definition of a Borel σ-algebra (exercise). It should be verified that $\mathcal{B}_{\overline{\mathbb{R}}}$ can be generated by open or closed rays, and so we will say $f : X \to \overline{\mathbb{R}}$ is $\mathcal{M}$-measurable if it is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}})$-measurable.

Next, if the codomain has some notion of $+$ and $\cdot$, we establish that the basic "$\mathbb{C}$-algebra operations" work on measurable functions:

Proposition 2.1.10: Arithmetics on Measurable Functions

Let $f : X \to \mathbb{C}$ and $g : X \to \mathbb{C}$ be $\mathcal{M}$-measurable functions. Then $f + g$, fg, and zf for $z \in \mathbb{C}$ are $\mathcal{M}$-measurable functions

Note that we have limited ourselves to $\mathbb{C}$. This still works with $\overline{\mathbb{R}}$ with the appropriate care of the $\infty - \infty$ and $0 \cdot \infty$ case (exercise 2 in book)

> ***Proof*:**
> The proof for zf being measurable is similar to that of $+$ and $\cdot$, so is left as an exercise.
>
> Since f and g, are measurable, then their "cartesian product" is measurable by proposition 2.1.8
>
> $$F : X \to \mathbb{C} \times \mathbb{C} \qquad F(x) = (f(x), g(x))$$
>
> In particular, F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}})$-measurable. Next, consider $p : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $p(x, y) = x + y$ and $t : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $t(x, y) = xy$ (p for plus, t for times). Then since p and t are continuous, by corollary 2.1.3 p and t are $(\mathcal{B}_{\mathbb{C}\times\mathbb{C}}, \mathcal{B}_{\mathbb{C}})$-measurable. Thus, $p \circ F$ and $t \circ F$ is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable, and by definition
>
> $$f + g = p \circ F \qquad fg = t \circ F$$
>
> completing the proof

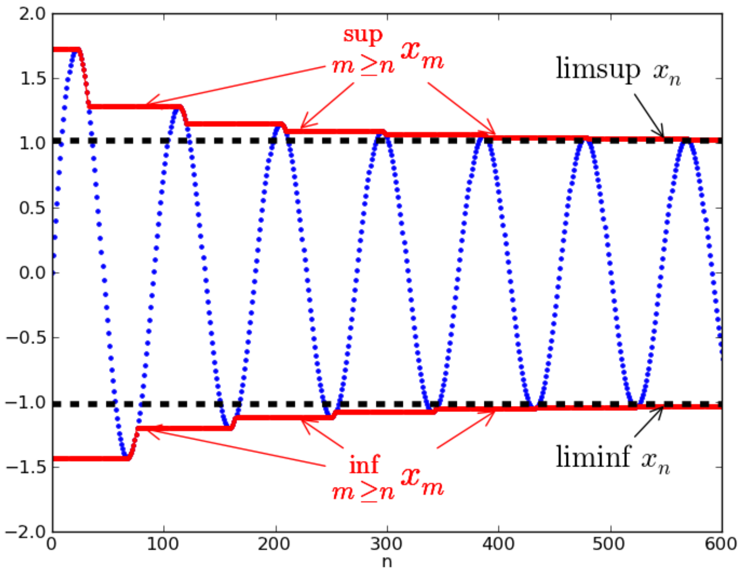
The next operations we'll show are measurable are sup, inf, lim sup and lim inf. This is perhaps one of the most important functions to be measurable since they are at the heart of many constructions in analysis. Recall that

$$\sup_i f_i(x) = \sup \{f_i(x) \mid i \in \mathbb{N}\}$$

and so this expression means we're taking the set $f_i(x)$ for each i and taking the sup of that set. In other words, for every point x we're taking the supremum of the set $\{f_1(x), f_2(x), ...\}$. Symmetrically the same for inf. For lim sup, recall that:

$$\limsup_i f_i(x) = \lim_{i\to\infty} \sup \{f_k(x) \mid k \geq i\}$$

that is, we are constantly shrinking the set over which we are doing the supremum, which in a sense means we're trying to "hug" the value from above as close as possible in the limit. Symmetrically the same for lim inf. If you're visual, I like to keep this image in mind: let's say the blue line is the value of $f_i(x_0)$ for a specified x_0. Then:



Comment here for future reference:

Proposition 2.1.11: Measurable functions and Limit Bounding

Let $f_i : X \to \overline{\mathbb{R}}$ be $\mathcal{M}$-measurable functions for each i and consider $\{f_i\}_{i=1}^{\infty}$. Then

$$g_1(x) = \sup_i f_i(x) \qquad g_2(x) = \inf_i f_i(x)$$

$$g_3(x) = \limsup_{i\to\infty} f_i(x) \qquad g_4(x) = \liminf_{i\to\infty} f_i(x)$$

The codomain $\overline{\mathbb{R}}$ was used so that the sup and inf is still considered defined if sup $= \pm\infty$ or inf $= \pm\infty$.

***Proof*:**
To show $\sup_i f_i$ and $\inf_i f_i$ are measurable, since the codomain domain is $\overline{\mathbb{R}}$, which is equivalent measure-theoretically to $\mathbb{R}$, we will show the pre-image of open rays is the union of measurable sets:

$$\begin{aligned} x \in g_1^{-1}((a,\infty]) &\Leftrightarrow g_1(x) > a \\ &\Leftrightarrow f_i(x) > a && \text{for some } i \\ &\Leftrightarrow x \in f_i^{-1}((a,\infty]) && \text{for some } i \\ &\Leftrightarrow x \in \bigcup_{i=1}^{n} f_i^{-1}((a,\infty]) \end{aligned}$$

and similarly for g_2, except considering $(-\infty, a]$, therefore:

$$g_1^{-1}((a,\infty]) = \bigcup_i^{\infty} f_i^{-1}((a,\infty]) \qquad g_2^{-1}(-\infty, a])) = \cup_i^{\infty} f_i^{-1}([-\infty, a))$$

since each f_i is measurable, we have a union of measurable sets, so g_1 and g_2 are measurable by proposition 1.1.11. Next, let

$$h_k(x) = \sup_{i>k} f_i(x) = \sup \{f_k(x) \mid k > i\}$$

then by what we've just established, h_k is measurable for each k. Then by of $\limsup$:

$$g_3 = \inf_k h_k$$

and so g_3 is measurable since we just established the inf of measurable functions is measurable. Similarly for g_4.

as an immediate corrollary:

Corollary 2.1.12: Measurability of Max and Min

Let f, g be measurable functions. Then so is $\max\{f, g\}$ and $\min\{f, g\}$

Corollary 2.1.13: Measurable Preserved under Pointwise limit

Let $\{f_i\}$ be a sequence of measurable function. If $\lim_{i\to\infty} f_i(x)$ exists for every x, and we define $f : X \to \overline{\mathbb{R}}$ to be

$$f(x) = \lim_{i\to\infty} f_i(x)$$

then f is measurable (i.e. point-wise convergence preserves measureability)

Recall that point-wise convergence *does not preserve continuity* (the steeple function's are the usual example). However, measureability is sufficiently general to be preserved under this weaker form of convergence! Naturally, if the sequence of functions $\{f_i\}$ are all $\mathcal{M}$-measurable and uniformally convergent, they are measurable

***Proof*:**
If f exists (i.e., every $f_n(x)$ converges pointwise to $f(x)$), then by proposition 2.1.11, $f = g_3 = g_4$, completing the proof.

2.1.14 Remark Note how important it is that *all* points converge! It is possible that a single point doesn't converge (ex. $\lim f_i(0)$ doesn't converge), and so the limit is *not* a measurable function, in fact it doesn't even need to be a function. Take for example the collection of functions $f_k : \mathbb{R} \to \mathbb{R}$ with the properties:

1. f_k is continuous (it can even be smooth)
2. $\int_{\mathbb{R}} f(x)dx = 1$ (In the Riemann sense), or really any constant $c \in \mathbb{R}_{>0}$
3. $\operatorname{supp} f \subseteq \left[\frac{-1}{k}, \frac{1}{k}\right]$

In other words, the collection of f_k are all smooth bump functions with volume 1, but the area in which it has support is shrinking, and so the "bump" of each f_k must be getting higher and higher (to define such functions, consider starting with $f_1 = e^{-\frac{1}{1-x^2}}$ where $x \in (-1,1)$ and 0 otherwise). Then $f_k \to 0$ at all but 1 point, namely $f_k(0)$ does *not* converge. As a consequecne, this limit is *not* a function!

This example shows the important of somehow "bounding" your area; we will get back to this observation in section 2.2. Some might think that this example is a bit cheeky, it still feels like the function is measurable if we just circumnavigate this problem by making the domain of the function $\mathbb{R} \cup \{\infty\}$, there is the trouble of defining a measure on the 1-point compactification of $\mathbb{R}$ while extending the measure of $\mathbb{R}$[a]. If the reader wonders if there is a way of defining some generlization of a function that will accept the fact that the f_k converge to something that still somehow preserves the fact that $\int f = 1$, then we will show in in chapter 9 that this converges to something called a *distribution*[b]

[a]In particular, $\mu(\{\infty\}) = \infty$, as can be shown using continuity form above, and so somehow we have "gained" an infinite amount of area. On the other-hand, if we just ignore this and consider that ∞ is just some point like any other point, then we have "lost" area

[b]The example named here is a distribution called the *delta distribution*, or sometimes *delta function* for reasons that is covered in chapter 9

Now that we proved that these operations preserving measurability, we will introduce some new notation to decompose real and complex functions which will soon become useful when defining integrable functions.

Definition 2.1.15: Real and Complex Decomposition

Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{C}$ be real and complex measurable functions respectively. Then

$$f^+ = \max\{f, 0\} \qquad f^- = \max\{-f, 0\}$$

and

$$\operatorname{sgn}(g(x)) = \begin{cases} \frac{g(x)}{|g(x)|} & g(x) \neq 0 \\ 0 & g(x) = 0 \end{cases} \qquad |g(x)| = |z| = |x+iy| = \sqrt{x^2+y^2}$$

which gives the decomposition of:

$$f = f^+ - f^-, \; g = (\operatorname{sgn} g)|g|$$

The functions f^+ and f^- are clearly well-defined and measurable. For the complex case, $|\cdot|$ is continuous everywhere and $z \mapsto \operatorname{sgn} z$ is continuous everywhere except 0. It is measurable since if $U \subseteq \mathbb{C}$ is open, then $\operatorname{sgn}^{-1}(U)$ is either V for some open set V, or $V \cup \{0\}$, meaning sgn is Borel measurable. In some textbooks, arg is used instead of sgn.

2.1.1 Standard Representation

Recall that every measurable function is the point-wise limit of a bunch of easy to work with function, in particular, linear combination of characteristic functions! As a reminder, if $\chi_E : X \to \mathbb{R}$ (or $\chi_E : X \to \{0, 1\}$) where $E \subseteq X$, then:

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

Then χ_E is called the characteristic (or indicator) function. It is sometimes also denoted as 1_E or I_E.

Definition 2.1.16: Simple function

Let $\{E_i\} \subseteq P(\mathbb{C})$ be some finite collection of sets. Then the finite linear combination of characteristic functions on E_i multiplied by a complex constant:

$$f = \sum_{i=1}^{n} z_i \chi_{E_i}$$

is called a *simple function*

Example 2.1.17: Simple Functions

1. Show that $|f(X)| < \infty$ then f is in fact a simple function (hint: let $E_i = z_i$ for each $z_i \in f(X)$). In fact, each coefficient can also be distinct (let it be the respective z_i from the range)
2. Show that if f and g are simple functions, so are $f + g$ and fg

Using these, we can now show that any measurable function is simply the limit of simple functions!

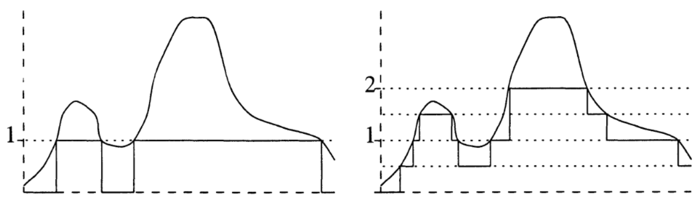
Theorem 2.1.18: Simple Function Approximation

Let $(X, \mathcal{M})$ be a measurable space. Then

1. If $f : X \to [0, \infty]$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that
$$0 \leq \varphi_1 \leq \varphi_2 \leq \cdots \leq f$$
where $\lim_{n\to\infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \to f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformally converges)

2. If $f : X \to \mathbb{C}$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that
$$0 \leq |\varphi_1| \leq |\varphi_2| \leq \cdots \leq |f|$$
where $\lim_{n\to\infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \to f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformally converges)

***Proof*:**
Essentially, you will get characteristic functions to converge to the value like so:



In particular, for $n \in \mathbb{N}_{\geq 0}$ and $0 \leq k \leq 2^{2n} - 1$, let
$$E_n^k = f^{-1}\left(\left(\frac{k}{2^n}, \frac{k+1}{2^n}\right)\right) \qquad F_n = f^{-1}((2^n, \infty])$$
and
$$\varphi_n = \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \chi_{E_n^k} 2^n \chi_{F_n}$$
then we get that $\varphi_n \leq \varphi_{n+1}$ for all n and $0 \leq f - \varphi_n \leq \frac{1}{2^n}$ where $f \leq 2^n$. The rest is left to check.

Generalizing to $\mathbb{C}$, decomposing to $f = g + ih = (g^+ + g^-) = i(h^+ + h^-)$, where the $+$ and $-$ functions are the positive and negative parts of f and g, then we can repeat the process for each of these and get $\psi_n^+, \psi_n^-, \zeta_n^+, \zeta_n^-$ and let $\varphi_n = (\psi_n^+ - \psi_n^-) + i(\zeta_n^+ - \zeta_n^-)$.

This shows that measurable functions are a very natural type of function to work with in analysis: they are all a pointwise-limit (and on a bounded domain a uniform limit) of simple functions. The particular sequence of simple functions will be called the *standard representation of f*. Using this result, we can define a new definition of equality up to zero-measure sets:

Proposition 2.1.19: Measurability up to Zero Set

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be two measure with ν being a complete measure, and let $f : X \to Y$ be $(\mathcal{M}, \mathcal{N})$-measurable. Then:

1. If $f = g$ up to a zero measure set with respect to μ (shorthanded to μ-a.e.), then g is measurable
2. If f_n is measurable for each $n \in \mathbb{N}$ and $f_n \to f$ μ-a.e, then f is measurable

***Proof*:**

1. Let $N = \{x \in X \mid f(x) \neq g(x)\}$. Let $B \in \mathcal{N}$ and consider $g^{-1}(B)$. Then by definition:

$$\begin{aligned} g^{-1}(B) &= \{x \in M \mid g(x) \in B\} \\ &= \{x \in \mathcal{M} \mid g(x) \in B,\ g(x) = f(x)\} \cup \{x \in \mathcal{M} \mid g \in B,\ g(x) \neq f(x)\} \end{aligned}$$

the latter set is a subset of N, and so is measurable by the completeness of f. The former set is of the form:

$$\begin{aligned} \{x \in \mathcal{M} \mid g(x) \in B,\ g(x) = f(x)\} &= \{x \in \mathcal{M} \mid f(x) \in B,\ g(x) = f(x)\} \\ &= f^{-1}(B) \cap \{x \mid f(x) = g(x)\} \\ &= f^{-1}(B) \cap (X - \{x \mid f(x) \neq g(x)\}) \\ &= f^{-1}(B) \cap (X - N) \end{aligned}$$

and since f measurable the pre-image of a measurable set is too, and since N is measuable, $X - N$ is too, so we have that $g^{-1}(B)$ is measurable.

2. Let's say N is the set on which $f_n \not\to f$. Define

$$g_n = f_n \chi_{X-N} + f \chi_N$$

Then notice that $g_n = f_n$ μ-a.e., and so by part 1 are measurable, and by construction $g_n \to f$ everywehre, and so f is measurable!

However, if μ is not complete, it turns out to make little difference since we can complete the measure, produce the result, and restrict back to the original measure without changing the result:

Proposition 2.1.20: Extending To Complete Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space and $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. if f is a $\overline{\mathcal{M}}$-measurable function on X, there is a $\mathcal{M}$-measurable function g such that $f = g$ $\overline{\mu}$-a.e.

***Proof*:**
If f is a simple function, then we see that the completion will at most remove finitely many points when we restrict every χ_E $(E \in \overline{\mathcal{M}})$ to g, and so $f = g$ $\overline{\mu}$-a.e.

Now let f be a $\overline{\mathcal{M}}$-measurable function, and by theorem 2.1.18 choose a sequence of $\overline{\mathcal{M}}$-measurable functions $\{\varphi_i\}_{i=1}^{\infty}$ that converge pointwise to f. The idea is that each $\varphi_i = \psi_i$ $\overline{\mu}$-a.e., that is, there exists a E_i where $\varphi_i(x) \neq \psi_i(x)$ for all $x \in E_i$ but $\overline{\mu}(E_i) = 0$, and the countable union of these will still be measure 0.

In particular Let $E = \bigcup_i^{\infty} E_i$. By definition of completion, there must exist some N such that $\mu(N) = 0$ and $E \subseteq N$. Let $g = \lim_{\chi_{X-N}} \psi_n$. Then g is measurable by corollary 2.1.13, and clearly $f = g$ on N^c, completing the proof.

2.2 Integration of non-negative Function

We are now in a position to define how to integrate functions (integrable functions coming soon)! Throughout this section, let $(X, \mathcal{M}, \mu)$ be a measure space

Definition 2.2.1: L^+

Let
$$L^+ = \{f : X \to [0, \infty] \mid f \text{ is a } \mathcal{M}\text{-measurable function}\}$$

Sometimes, L^+ is written as $L^+(X)$, $L^+(\mu)$, or $L^+(\mu, X)$ to emphasize which space of functions we are dealing with. If it is unambiguous, we use L^+.

Definition 2.2.2: Integral of Simple Function

Let $f = \sum_1^n a_i \chi_{E_i}$ be a simple function. Then define the *integral of f with respect to μ* to be

$$\int f d\mu = \sum_i^n a_i \mu(E_i)$$

In other words, we simply took the measure of the domain of χ_{E_i} for each i. As usual, we take the convention that $0 \cdot \infty = 0$. We will also allow $\int f = \infty$ to be a valid result. If there is no confusion, we will write $\int f$ instead of $\int f d\mu$. It should also be emphasized that it is the measure of the *domain* and the coefficient from the *codomain* that we are taking.

If we want to integrate over just some $A \subseteq X$ where $A \in \mathcal{M}$, then $\varphi|_A$ is also a simple function (simply take $\chi_{E \cap A}$ for each characteristic function). We usually denote this by

$$\int_A f d\mu = \int f \chi_A \, d\mu$$

Because of this notation, we can write $\int f d\mu$ as

$$\int f d\mu = \int_X f d\mu$$

Proposition 2.2.3: properties of Integrals of Simple Functions

Let φ and ψ be simple functions in L^+. Then

1. if $c \geq 0$, $\int c\varphi d\mu = c \int \varphi d\mu$
2. $\int (\varphi + \psi) d\mu = \int \varphi d\mu + \int \psi d\mu$
3. if $\varphi \leq \psi$ then $\int \varphi \leq \int \psi$
4. The map $A \mapsto \int_A 1 d\mu$ is a measure on $\mathcal{M}$ (more generally, $A \mapsto \int_A \varphi d\mu$ is a meaure on $\mathcal{M}$ for a simple function φ)

***Proof*:**
Do these as midterm exercises

With this definition, we can define integrable function more generally:

Definition 2.2.4: Integral of Measurable Function

Let $f \in L^+$. Then define

$$\int f d\mu := \sup \left\{ \int \varphi d\mu \,\middle|\, \varphi \text{ is a simple function, } 0 \leq \varphi \leq f \right\}$$

It's important to see why the integral of a simple function will give the same result as the "simple integral" (i.e. integral over simple functions from earlier), in particular since the set over which we are containing the supremum will contain the simple function in question, and so the maximum will be achieved by itself (and proposition2.2.3?). Furthermore, results 1 and 3 from proposition 2.2.3 holds for the definition of integration just given. The other results, in particular linearity, will be established soon. We can also see how the definition of the integral solidifies the idea that the integral is a sort of "weighted measure": the integral of a simple function is just a sum of measure of sets weighted by some constant, and we take the supremum of all such sets. We will in fact take a closer look at defining measure in terms of integration of functions in chapter 3.

Given the supremum definition of the integral, it could be rather hard to find the integral of a function in L^+. We essentially never check this, and treat the following theorem as the de-facto way of finding the integral:

Theorem 2.2.5: Monotone Convergence Theorem (MCT)

If $\{f_n\}_{i=1}^{\infty} \subseteq L^+$ is a sequence such that $f_i \leq f_{i+1}$ for all $i \in \mathbb{N}$ and $f = \lim_{n\to\infty} f_n$ $(= \sup_n f_n)$, then

$$\int f = \lim_{n\to\infty} \int f_n$$

or, since $f = \lim_{n\to\infty} f_n$:

$$\int \lim_{n\to\infty} f = \lim_{n\to\infty} \int f_n$$

i.e. the limit commutes

It might be tempting to then say that $\int$ is "continuous" in the appropriate domain since it commutes with limits. However, the f_i must be *increasging* for the monotone convergence theorem to apply, and hence it is better to say that $\int$ is continuous from bellow. We will later show that the limit need not always commute (and hence it will be innapropriate to say that the integral is continuous on the space of measurable functions).

2.2.6 Remark any times in Analysis, when the limit can commute with something (ex. integral, another limit, functions), then the proof that it commutes is usually given a name (ex. sequential continuity, MCT, DCT, Fatou lemma, Moore-Osgood theorem, etc.) see:

wikipedia: Interchanging Limits

***Proof*:**
We'll show $\leq$ and $\geq$. For $\leq$, notice that $\{\int f_n\}$ is an increasing sequence, and so converges (possibly to ∞). Furthermore, $f_n \leq f$ for all n, and so by proposition 2.2.3 updated for general integral functions, we have:

$$\int f_n \leq \int f$$

and so

$$\lim_{n\to\infty} \int f_n \leq f$$

Conversely, we'll show that if we "shrink" the left hand side by a factor of $\alpha \in (0, 1)$ so that we get $\geq$, then since this will be true for all α, we will get the result we want.

Fix some $\alpha \in (0, 1)$, and choose a simple function where $0 \leq \varphi \leq f$ (such a function exists by theorem 2.1.18). Let

$$E_n = \left\{x \in \overline{\mathbb{R}} \;\middle|\; f_n(x) \geq \alpha\varphi(x)\right\}$$

Then $\{E_n\}_{n=1}^{\infty}$ is an increasing sequence (via inclusion) of measurable sets who's union is X (since $\lim_{n\to\infty} f_n = f$ and the f_n's are increasing). Then

$$\int f_n \geq \int_{E_n} f_n \geq \alpha \int_{E_n} \varphi$$

Since the map $E_n \mapsto \int_{E_n} \varphi$ is measurable by proposition 2.2.3 and the E_n's are continuous from above, we have

$$\lim_{n\to\infty}\int_{E_n}\varphi=\int\varphi$$

and so

$$\lim_{n\to\infty}\int f_n\geq\alpha\int\varphi$$

Since this is true for all $\alpha<1$, it is true for $\alpha=1$, and so $\int f_n\geq\int\varphi$. Taking the supremum over all φ, we get that

$$\lim_{n\to\infty}\int f_n\geq\int f$$

And since we showed the $\leq$ direction, we in fact have:

$$\lim_{n\to\infty}\int f_n=\int f$$

as we sought to show

Thus, to find the value of $\int f$, it suffices to find some sequence of simple functions $\{\varphi_n\}$ that converges to f, which always exists since every measurable function has a standard representation (theorem 2.1.18).

An immediate consequence of the Monotone Convergence Theorem is that linearity also applies to integrals:

Proposition 2.2.7: Linearity of Integrals

Let $\{f_i\}_{i=1}^N\subseteq L^+$ where $N\in\mathbb{N}\cup\{\infty\}$ and let $f=\sum_i^N f_i$. Then

$$\int f=\sum_i^N\int f_n$$

***Proof*:**
Let's first consider the case where $n=2$ so that we have f_1 and f_2. Let $\{\varphi_i\}$ and $\{\psi_i\}$ be a sequence for the standard representation of f_1 and f_2. Then clearly $\{\varphi_i+\psi_i\}$ is a sequence for a standard representation of f_1+f_2. Then by the Monotone Convergence Theorem and linearity

of integrals with simple functions:

$$\begin{aligned}\int f_1 + f_2 &= \lim_{i\to\infty} \int \varphi_i + \psi_i \\ &\overset{\text{MCT}}{=} \int \lim_{i\to\infty} \varphi_i + \psi_i \\ &= \int \lim_{i\to\infty} \varphi_i + \lim_{i\to\infty} \psi_i \\ &= \int \lim_{i\to\infty} \varphi_i + \int \lim_{i\to\infty} \psi_i \\ &\overset{\text{MCT}}{=} \lim_{i\to\infty} \int \varphi_i + \int \lim_{i\to\infty} \psi_i \\ &= \int f_1 + \int f_2\end{aligned}$$

By induction, it holds that

$$\int \sum_i^n f_i = \sum_i^n \int f_i$$

Letting $n \to \infty$, we can once again apply the Monotone Convergence Theorem and get

$$\begin{aligned}\int \sum_i^\infty f_i &= \int \lim_{n\to\infty} \sum_i^n f_i \\ &\overset{\text{MCT}}{=} \lim_{n\to\infty} \int \sum_i^n f_i \\ &= \lim_{n\to\infty} \sum_i^n \int f_i \\ &= \sum_i^\infty \int f_n\end{aligned}$$

completing the proof

The next result is a generalization from the same result with Riemann integrals, except we replace the condition of "finitely many points" with "null-set amount of points". It follows the trend of integration "forgetting" up to a zero-measure amount of points:

Proposition 2.2.8: Zero Integral

Let $f \in L^+$. Then $\int f = 0$ if and only if $f = 0$ a.e.

***Proof*:**
Starting with simple functions, this is evident in both directions. If φ was our simple function and $\int \varphi = 0$, then $\sum_i^n a_i\mu(E_i) = 0$, so either each $a_i = 0$ or each $\mu(E_i) = 0$ which immediately implies φ is zero a.e. (not necessarily everywehre since it's possible that $E_i \neq \emptyset$ but $\mu(E_i) = 0$).

Conversely, if each $\mu(E_i) = 0$ a.e., then clearly $\int \varphi = 0$.

Moving onto to a measurable function f, the $\Leftarrow$ comes almost immediately. If $f = 0$ a.e., then for every simple function $\varphi \leq f$, then $\varphi = 0$ a.e., which we have established means $\int \varphi = 0$. Then by definition of $\int f$:

$$\int f = \sup_{\varphi \leq f} \int \varphi = 0$$

For the $\Rightarrow$ direction, assume that $\int f = 0$, and for the sake of contradiction let's say $f \neq 0$ a.e. The idea is that we can then construct a simple function φ where $\varphi \leq f$ whose integral is greater than zero, but $\int f = \sup_{\varphi \leq f} \int \varphi$. To that end take:

$$\bigcup_i^\infty E_n = \bigcup_i^\infty \left\{ x \,\middle|\, f(x) > \frac{1}{n} \right\} = \{x \mid f(x) > 0\} = E$$

Since $f \neq 0$ a.e., it must be that $\mu(E) > 0$, and so by construction for some n, $\mu(E_n) > 0$. But then

$$f \geq n^{-1}\chi_{E_n} \qquad \Leftrightarrow \qquad \int f \geq n^{-1}\mu(E_n) > 0$$

so the supremum of values in f contain simple functions of nonzero measure, showing that $\int f > 0$, a contradiction to our original assumption.

This theorem also tells us that if $f = g$ a.e., then $\int f = \int g$, since $f - g = 0$ a.e., so $\int f - g = 0$, so $\int f = \int g$. Using this, we can upgrade the Monotone Conversely Theorem by needing the convergence to happen up to a measure zero set:

Corollary 2.2.9: MCT a.e.

Let $\{f_n\}_{i=1}^\infty \subseteq L^+$ and $f \in L^+$. Then if $f_1 \leq f_2 \leq \cdots \leq f$ and $f_n \to f$ a.e., then

$$\int f = \lim_{n\to\infty} \int f_n$$

***Proof*:**
Let $f_n(x) \uparrow f(x)$ for all $x \in E$ such that $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ a.e. which implies $f_n - f_n\chi_E = 0$ a.e., so by the Monotone Conversely Theorem

$$\int f = \int f\chi_E \overset{\text{MCT}}{=} \lim_{n\to\infty} \int f_n\chi_E = \lim_{n\to\infty} \int f_n$$

as we sought to show.

We next address the fact that the sequence $\{f_i\}$ must be increasing (at least a.e.) for the MCT to work (recall we relied on the sequence being increasing in the construction of one of the direcitons of the inequality). If it were not, then it need not be the case that the limit passes through the integral!

Example 2.2.10: not increasing $\not\Rightarrow$ Limit commutes with $\int$

Let $X = \mathbb{R}$ and μ be the Lebesgue measure. Consider the two following examples:

$$\{n\chi_{(0,1/n)}\}_{n=1}^{\infty}$$

then it is clear that $n\chi_{(0,1/n)} \to 0$. However, notice that $\int n\chi_{(0,1/n)} = 1$ for each element in the sequence. Thus:

$$0 = \int \lim_{n\to\infty} n\chi_{(0,1/n)} \neq \lim_{n\to\infty} \int n\chi_{(0,1/n)} = 1$$

Even if the function is bounded, the problem remains: take $\chi_{(n,n+1)}$ which also converges pointwise to 0 but the sequence of the integrals of the functions is a sequence of constants 1 and so converges to 1 which is clearly not 0.

This is interesting if we interpret it from a sort of "physics" perspective: somehow, we have "lost our mass" in the process of taking our limit. There is in fact essentially 3 ways we can "lose mass', the first two begin the counter-examples above, the last one being a function that oscillates faster and faster (this last one requires a function that is also defined in the negative, and so we will get back to this)

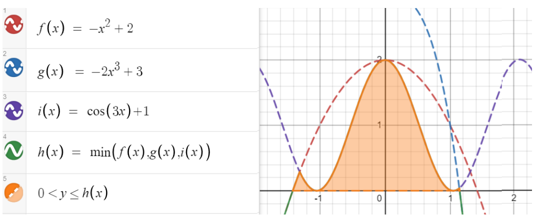
Notice that in both examples, there is some "escaping to infinity" going on. Perhaps if we have some overarching "bound" to the sequence, then we can pass the limit through, and that will indeed be the case. We address this when talking about the Dominated Convergence Theorem in the next section. However, not all hope is lost for the general case; there is weakening to only the lim inf and the equality to $\leq$ which works in the general case:

Theorem 2.2.11: Fatou's Lemma

If $\{f_n\} \subseteq L^+$ is *any* sequence, then

$$\int \liminf_n f_n \leq \liminf_n \int f_n$$

To visualize this, consider:



The integral of the orange region weill be less than or equal to the lim inf of those functions over that region. In this case, the integral would be equal, but in the case where the mass somehow "escapes", we will have an inequality.

***Proof*:**
This comes directly from properties of the $\liminf$. Let $k \geq 1$. Then

$$\inf_{i \geq k} f_i \leq f_j \quad \forall\ j \geq k$$

Then by theorem 2.2.3

$$\int \inf_{i \geq k} f_i \leq \int f_j \quad \forall j \geq k$$

and since this holds for all $j \geq k$, we get:

$$\int \inf_{i \geq k} f_i \leq \inf_{j \geq k} \int f_j$$

Since the $\liminf$ defines an increasing sequence of function, by letting $k \to \infty$ and applying the Monotone Conversely Theorem:

$$\int \liminf f_n = \lim_{k \to \infty} \int \inf_{n \geq k} f_n \leq \liminf \int f_n$$

as we sought to show

Like before, the result holds if we loosen the condition so that $f_n \to f$ a.e.

Corollary 2.2.12: Fatou's Lemma a.e.

let $\{f_n\} \subseteq L^+$ be any sequence and $f \in L^+$. If $f_n \to f$ a.e. Then

$$\int f \leq \liminf_n \int f_n$$

***Proof*:**
Using proposition 2.2.9, make $f_n \to f$ everywhere arguing that it is equivalent to do so, and then the rest is Fatou's Lemma.

Finally, we show that if $\int f < \infty$, we can guarantee that f cannot "explode" too much, and that even better, the set over which the value of f is non-zero is σ-finite, i.e., it is not "pathological"

Proposition 2.2.13

Let $f \in L^+$. If $\int f < \infty$, then $\{x \mid f(x) = \infty\}$ is a null set and $\{x \mid f(x) > 0\}$ is σ-finite

***Proof*:**
was left as an exercise to the reader in the book

2.3 Integration of Real and Complex Functions

We'll again let $(X, \mathcal{M}, \mu)$ be the fixed measure space for this section, except the codomain now will be either $\mathbb{R}$ or $\mathbb{C}$. For the case of $\mathbb{R}$, we can quickly extend our results from last section by splitting f into f^+ and f^- (which are both measurable if f is by corollary 2.1.12), and if at least one of f^+ or f^- is finite, then

$$\int f = \int f^+ - \int f^-$$

Notice that if they are both infinity, we have $\infty - \infty$ which is in general not well-defined. Thus, we will limit our attention to functions where $\int |f| < \infty$ since $|f| = f^+ + f^-$:

For the case of $\mathbb{C}$, using the triangle inequality we get that:

$$|f| = |\text{Re}f + \text{Im}f| \leq |\text{Re}f| + |\text{Im}f| \leq 2|f|$$

and so both $\text{Re}f$ and $\text{Im}f$ must be finite to get a well-defined result. In that case, we define

$$\int f := \int \text{Re}f + i \int \text{Im}f$$

We give a name for complex functions (the real functions being a special case) that satisfy this finiteness condition:

Definition 2.3.1: Integrable Function

Let $f : X \to \mathbb{C}$ be measurable. We say f is *integrable* if

$$\int |f| < \infty$$

Let L^1 represent the set of all integrable functions.

Like with L^+, we sometimes write $L^1(\mu)$, $L^1(X)$, or $L^1(X, \mu)$. If the codomain is $\mathbb{R}$, then we get that $\int |f^+| < \infty$ and $\int |f^-| < \infty$, as we claimed for our extension of the integral to $\mathbb{R}$. Furthermore, for any $E \in \mathcal{M}$, we will say that f is *integrable on* E if $\int_E |f| < \infty$.

The set of L^1 functions are well-behaved under addition and scalar multiplication in the sense that they form a vector-space:

Proposition 2.3.2: Vector Space on L^1

Let L^1 be the set of integrable functions. Then with $+$ and scalar multiplication, L^1 forms a vector-space (over $\mathbb{R}$ or $\mathbb{C}$ respective to the codomain). Furthermore, $\int$ is a linear functional on L^1 (i.e. $\int : L^1 \to \mathbb{R}$ is a linear functional)

***Proof*:**
the proof that $cf + dg$ is integrable comes from the triangle inequality ($|cf + dg| \leq |c||f| + |d||g| < \infty$). The fact that $cf \in L^1$ for any $c \in \mathbb{R}$ and $f \in L^1$ follows similarly.

$\int$ being a functional, we need to show linearity and $\int cf = c \int f$. For linearity, assume $h = f + g$

is integrable. Then $h^+ - h^- = f^+ - f^- + g^+ - g^-$, or, re-arranging, we get:

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

These are all functions in L^+, and so by linearity for of integrals for L^1 functions:

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+$$

re-arraning, we get:

$$\begin{aligned} &= \int f + g \\ &= \int h = \int h^+ - \int h^- = \int f^+ - \int f^- + \int g^+ - \int g^- \\ &= \int f + \int g \end{aligned}$$

showing linearity. Through a similar means, we can also show $\int cf = c \int f$, completing the proof.

2.3.3 Remark Since it is a vector-space, it has a basis and a dimension. What dimension is this vector-space? Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinal N (N may be finite, countable, uncountable, etc). If $N \geq \aleph_0$, that means that it is rather difficult to define a measure on the space L^1[a]. Trying to find ways to study "measure" notions infinite dimensional vector-spaces (ex. boundedness, σ-finiteness, etc.) is accomplished by defining various different concepts like *seminorms*, *norms*, or *inner products*, which we shall do in chapter 7.

[a]For an illustration of the oddities of infinite dimensional vector-spaces, consider the [hyper]-volume of a unit ball S^n as $n \to \infty$; you should find that it goes to 0!

Next, we would like to establish some properties of integrable functions. One result is a including a useful inequality including absolute values. Another is how the results still work as long as things happen "a.e.". Finally, the set of all "interesting" points (points where $f(x) \neq 0$) are still σ-finite:

Proposition 2.3.4: Properties of Integrable Functions

Let $f, g \in L^1$. Then:

1. $\left|\int f\right| \leq \int |f|$
2. $\{x \mid f(x) \neq 0\}$ is σ-finite
3. $\int_E f = \int_E g$ for all $E \in \mathcal{M}$ if and only if $\int |f - g| = 0$ if and only if $f = g$ a.e.

***Proof*:**

1. The result is essentially immediate in the $\mathbb{R}$ case. In the $\mathbb{C}$ case, let α where $|\alpha| = 1$ be such that it "rotates" the integral so that it's in the real case, and then apply a similar strategy to that of the $\mathbb{R}$ case.

2. Notice that $\{x \mid f(x) \neq 0\} = \{x \mid f^+(x) > 0\} \cup \{x \mid f^-(x) < 0\}$, which we've shown in proposition 2.2.13 to be σ-finite

3. The fact that $\int |f - g| = 0$ if and only if $f = g$ a.e. comes from proposition 2.2.8, (since $|f-g| \in L^+$) so we will prove $\int_E f = \int_E g$ if and only if $\int |f-g| = 0$. Let's say $\int |f-g| = 0$. Then to show that $\int_E f = \int_E g$, it is equivalent to show that

$$\left| \int_E f - \int_E g \right| = 0$$

by the epsilon principle (i.e. $a = b$ if and only if $|a-b| < \epsilon$ for all ϵ). Using proposition 2.3.4, we get that:

$$\left| \int_E f - \int_E g \right| \leq \int \chi_E |f - g| \leq \int |f - g| = 0$$

completing the $\Rightarrow$ direction.

For the other direction, let's argue contra-positively and assume $\int |f - g| \neq 0$ a.e. Let

$$u = \mathrm{Re}(f - g) \qquad v = \mathrm{Im}(f - g)$$

so that u^+, u^-, v^+, v^- are the corresponding functions. Since $f \neq g$ a.e., at least one of these must have a set with positive measure. Without loss of generality, let's say $E = \{x \mid u^+(x) > 0\}$ and $\mu(E) > 0$. Then

$$\mathrm{Re}\left(\int_E f - g \right) = \mathrm{Re}\left(\int_E f - \int_e g \right) = \int_E u^+ + \int_E u^- = \mu(E) + 0 > 0$$

since $u^- = 0$ on E. Thus, the measure will be positive.

As a consequence of this, we usually think of integrable functions *up to a zero set.* If we wanted to, we can limit our scope to of integration to functions f that are measurable on E and E^c has zero-measure to functions where $f = 0$ on E^c. This also means that if we are working with an $\overline{\mathbb{R}}$-value function that is finite a.e., then it is equivalent to treat it as a $\mathbb{R}$-value function.

Proposition 2.3.4 also allows us to re-define L^1 in terms of functions that are equivalent up to a measure zero set. That is, we form an equivalence relation on L^1 if and only if $f = g$ a.e., and label the collection of equivalence classes as L^1. We abuse notation and write "$f \in L^1$" to mean we are selecting a function that is integrable (or an a.e.-defined integrable function). This is because we will usually be working with functions instead of the equivalence classes, and so this is a common abuse of notation.

The reason to introduce this new definition of L^1 is so that we can define a new metric on L^1:

Definition 2.3.5: L^1 metric

Let L^1 be the set of equivalence classes as defined above. Then

$$\rho(f,g) = \int |f-g|$$

is a metric on L^1

Symmetry and triangle inequality come immediately without even needing to use the fact that these are equivalence classes. Where the new construction of L^1 becomes important is to say $\int |f-g| = 0$ if and only if $f = g$. Since this is true a.e., then we need these two to be in an equivalence class. Since we defined a metric, we can define a notion of convergence, in particular, f_n converges to f in L^1 if and only if $\int |f - f_n| \to 0$, meaning $f_n \to f$ almost everywhere (however, L^1 convergence does not imply a.e. point-wise convergence, and a.e. point-wise convergence does not imply L^1 convergence, as we'll show in section 2.4).

We will soon explore more properties of this metric space, in particular, we would want this metric space to be complete so that the limit of a sequence of L^1 function's is L^1 and find some nice dense set of L^1 (so far, we only have the [point-wise] limit of measurable function is measurable, but no such result yet for integrable functions). To do so, we need more tools to analyze swapping limits and integrals. The next result we'll cover allows us to generalize the Monotone Conversely Theorem to the general case of $f_n \to f$ a.e., with the extra condition that each $|f_n|$ is bounded by a single $|g|$ with finite area (in this way, we are getting rid of the "escaping to infinity" cases in example 2.2.10)

Theorem 2.3.6: Dominated Convergence Theorem (DCT)

Let $\{f_n\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of integrable functions such that

1. $f_n \to f$ a.e.
2. there exists a nonnegative $g \in L^1$ such that for all $n \in \mathbb{N}$, $|f_n| \leq g$ a.e.

Then

$$\int f = \lim_{n\to\infty} \int f_n$$

or equivalently:

$$\int \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int f_n$$

***Proof*:**
By proposition 2.1.19 and 2.1.20 , f is measurable (up to a null set, in which case we can appropriately redefine f on that null-set). Since $|f_n| < g$ for all $n \in \mathbb{N}$, $f < g$, and so $f \in L^1$.

Since all L^1 functions are broken down to real parts (complex functions into 2 real parts), it suffices to show the result for real value f_n and f. By assumption, we have $g + f_n \geq 0$ and $g - f_n \geq 0$. Thus, by Fatou's Lemma:

$$\int g + \int f \leq \liminf \int (g + f_n) = \int g + \liminf \int f_n$$

$$\int g - \int f \leq \liminf \int (g - f_n) = \int g - \limsup \int f_n$$

Therefore, $\limsup \int f_n \leq \int f \leq \liminf \int f_n$, and so the two value are actually equal and so

$$\int f = \lim_{n \to \infty} \int f_n$$

as we sought to show

Note that functions can still converge even if the hypothesis of the DCT fails:

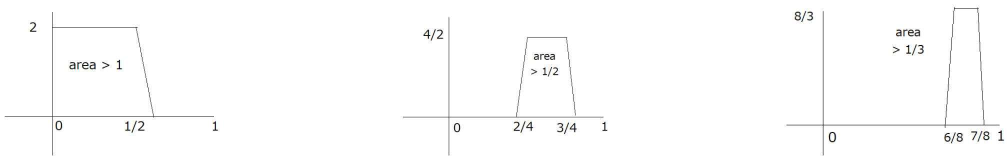
Example 2.3.7: Converges Without DCT

Construct a sequence of functions $\{f_n\}$ such that all the following hold:

1. $f_n : [0, 1] \to \mathbb{R}$ is continuous
2. $f_n \geq 0$
3. $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in [0, 1]$
4. $\lim_{n\to\infty} \int_0^1 f_n(x)\, dm = 0$
5. $\sup_n f_n(x)$ is *not* integrable on $[0, 1]$ with respect to the Lebesgue measure

(that is, just because the hypothesis of the dominated convergence theorem fails does not mean the conclusion does not hold: it is not an if and only if)

Proof. The proof essentially is a formalizing the following images:



(a) f_1 (b) f_2 (c) f_3

Figure 2.1: Visual intuition for question 3

We'll first define the indicator functions, then make these functions continuous. Let

$$g_1(x) = \begin{cases} 2^1/1 & x \in \left[0, \frac{1}{2}\right) \\ 0 & x \in \left[1 - \frac{1}{2}, 1\right] \end{cases} \quad g_2(x) = \begin{cases} 2^2/2 & x \in \left[\frac{1}{2}, \frac{1}{4}\right) \\ 0 & x \in \left[1 - \frac{1}{4}, 1\right] \end{cases} \quad g_3(x) = \begin{cases} 2^3/3 & x \in \left[\frac{3}{4}, \frac{7}{8}\right) \\ 0 & x \in \left[1 - \frac{7}{4}, 1\right] \end{cases}$$

and so on, defining:

$$g_n(x) = \begin{cases} 2^n/n & x \in [2^{n-1} - 2/2^{n-1}, 2^{n-1} - 1/2^{n-1}) \\ 0 & x \in [1 - 2^{n-1} - 1/2^{n-1}, 1] \end{cases}$$

Next, we can make each g_n continuous by simply adding a line to omit the jump:

$$f_1(x) = \begin{cases} 2^1/1 & x \in [0, 1/2) \\ -2^{10\cdot 1}\left(x - \left(\frac{1}{2} - 1/2^{10\cdot 1}\right)\right) & x \in (1/2, 1/2 + 1/2^{10\cdot 1}] \\ 0 & x \in [1 - (1/2 + 1/2^{10\cdot 1}), 1] \end{cases}$$

and defining each f_n similarly. Then it is clear by our construction that $f_n \to 0$: since the trapezoids are moving away, they will never cross our values again once they pass them (this is the opposite of the proof for question 2 where they constantly loop back, since the proof is similar the details were ommited for this proof). Furthermore,

$$\int f_n = 1/n + 1/2^{10n}$$

Thus, $\int f_n \xrightarrow{n\to\infty} 0$, showing that the integral also converges to 0. However,

$$\int \sup_n f_n \geq \int \sup_n g_n = \sum_{k=1}^{\infty} 1/k = \infty$$

showing that the $\sup_n f_n$ is *not* integrable, as we sought to show.

□

Therefore, the DCT is a sufficient, but not necessary condition for convergence. Sufficient and necessary conditions are a bit harder to come by, and we will not need to characterize limits commuting with integrals in these other ways, but for those who are curious:

https://journalofinequalitiesandapplications.springeropen.com/articles/10.1186/s13660-020-02502-w

This theorem will be key in finding the integral of many function constructed by limits, and will be used to prove completeness of L^1 (after defining Cauchy sequences in L^1 in section ref:HERE) A more immediate result is that the Dominated convergence theorem allows to define a form of countable linearity for L^1 functions:

Theorem 2.3.8: Linearity for Integrable Functions

Let $\{f_i\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of functions, and $f = \sum_{i=1}^{\infty} f_i$. If $\sum_{i=1}^{\infty} |f_i| < \infty$, then $\sum_{i=1}^{\infty} f_i$ converges a.e. to a function in L^1, and

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

***Proof*:**
First, since $\sum_{i=1}^{\infty} |f_i| < \infty$, by proposition 2.2.7,

$$\int \sum_{i=1}^{\infty} |f_i| = \sum_{i=1}^{\infty} \int |f_i|$$

Thus, $g = |\sum_{i=1}^{\infty} f_i|$ is a function in L^1. The function g is finite a.e. by proposition 2.3.4[2], so by the dominated convergence theorem, any partial sum of functions will commute, and so taking the limit, we get:

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

as we sought to show

The dominated convergence theorem also gives us a dense set in L^1 to work with, namely the simple functions:

Theorem 2.3.9: Simple Functions Dense in L^1

Let $f \in L^1$. Then for all $\epsilon > 0$, there exists an integrable simple function $\varphi = \sum a_i \chi_{E_i}$ such that

$$\int |f - \varphi| d\mu < \epsilon$$

Furthermore, if μ is a Lebesgue-Stieljes measure on $\mathbb{R}$, the sets E_i can be taken to be finite union of open intervals, and better, there is a continuous function g that vanishes outside a bounded interval (i.e. $g = 0$ outside the bounded interval) such

$$\int |f - g| d\mu < \epsilon$$

***Proof*:**
p. 55 in the book. Also proven in fuller generality that simple functions are dense in L^p spaces, which includes L^1.

For the purposes of Ordinary Differential Equations (ODE's), we want to establish is a quick result to relate differentiable functions and integral of functions:

Theorem 2.3.10: Relating Derivative and Integral

Let $(X, \mathcal{M}, \mu)$ be a measure space, and suppose $f : X \times [a, b] \to \mathbb{C}$ for $-\infty < a < b < \infty$ and that $f(\cdot, t) : X \to \mathbb{C}$ is integrable for every $t \in [a, b]$, so we can define $F(t) = \int_X f(x, t)\, d\mu$.

1. Suppose that there exists a $g \in L^1$ such that $|f(x, t)| \leq g(x)$ for all x, t. Then if $\lim_{t \to t_0} f(x, t) = f(x, t_0)$ for every x, $\lim_{t \to t_0} F(t) = F(t_0)$; that is, if $f(x, \cdot)$ is continuous at every x, so is F.

2. Suppose $\frac{\partial f}{\partial t}$ exists and there is a $g \in L^1(\mu)$ such that $\left|\frac{\partial f}{\partial t}(x, t)\right| \leq g(x)$ for all x, t. Then F is differentiable and
$$F'(x) = \int \frac{\partial f}{\partial t}(x, t)\, d\mu(x)$$

The way I like to think about is that there is a system X that at different times t acts/behaves differently. This difference in how the system acts at any given time t is captured by $f(x, t)$. The function F capture the "total value" of this action/behavior (note that f is non-negative too, so all behavior has a "non-negative" output).

Now, if the behavior varies continuously, then the total output varies continuously, and if the behavior is smooth smooth enough so that it is differentiable (at any slice), then F is also differentiable, and it is in fact even easy to compute the value of F'.

***Proof*:**

1. Since $|f(x,t)|$ is dominated by g, simply apply the Dominated Convergence Theorem:

$$\lim_{t\to t_0} F(t) = \lim_{t\to t_0} \int f(x,t)\, d\mu(x) = \int \lim_{t\to t_0} f(x,t)\, d\mu(x) = \int f(x,t_0)\, d\mu(x) = F(t_0)$$

where $\{t_n\}$ is any sequence converging to t (since any $f(x,t)$ satisfies the bound, no more restrictions are needed on the choice of sequence)

2. Since $\frac{\partial f}{\partial t}$ exists, the limit of the derivative exists, and so define

$$\frac{\partial f}{\partial t} = \lim_{t_n\to t_0} h_n(x) \qquad h_n(x) = \frac{f(x,t_n) - f(x,t_0)}{t_n - t_0}$$

$\{t_n\}$ is any sequence converging to t. Since h_n difference and division of measurable functions, by proposition 2.1.10 it is measurable, and by the Dominated Convergence Theorem $\frac{\partial f}{\partial t}$ is measurable. Furthermore, by the Mean Value Theorem:

$$|h_n(x)| \le \sup_{t\in[a,b]} \left|\frac{\partial f}{\partial t}(x,t)\right| \le g(x)$$

and so by the Dominated Convergence Theorem again:

$$\begin{aligned} F'(t_0) &= \lim_{t_n\to t} \left(\frac{F(t_n) - F(t_0)}{t_n - t_0}\right) \\ &= \lim_{t_n\to t} \left(\int h_n(x)\, d\mu(x)\right) \\ &\overset{\text{D.C.T.}}{=} \int \lim_{t_n\to t} h_n(x)\, d\mu(x) \\ &= \int \frac{\partial f}{\partial t}\, d\mu(x) \end{aligned}$$

completing the proof

The final fact we shall present is a continuation of proposition 2.2.13 in the sense that we aim to further establish connections between some subsets of the domain and the resulting integral value in the co-domain, and see that if $\int$ is limited to some finite area of integration E ($\int_E < \infty$), does it actually respect continuity? In fact, it does! Even better, it is in a sense uniformally continuous with respect to the measure of the set that is being integrated over.

Proposition 2.3.11: $\int$ Uniform Continuity

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f \in L^1$. Then for all $\epsilon > 0$, there exists a $\delta > 0$ such that
$$\mu(E) < \delta \Rightarrow \int_E |f| < \epsilon$$

***Proof*:**
Without loss of generality, let $f \geq 0$. Let $\epsilon > 0$, and choose φ such that $\int f < \int \varphi + \epsilon/2$, i.e. $\int f - \int \varphi < \epsilon/2$. Next, since φ is a simple function, we have that $\sup \varphi = M < \infty$ (being the finite sum of constant terms). Now, set $\delta = \frac{\epsilon}{2M}$ so that $\mu(E) < \delta$.
$$\int_E f - \varphi \leq \int f - \varphi$$
then re-arranging we get:
$$\int_E f \leq \int f - \varphi + \int_E \varphi < \frac{\epsilon}{2} + \frac{M\epsilon}{2M} = \epsilon$$
as we sought to show.

As a consequence of this, if we have a space of integrable functions $\mathcal{F}$ with the sup norm (convergence in the sup norm is equivalent to convergence), then for any $\mu(E) < \infty$, $\int_E$ is in fact *continuous*, that is, if $f_n \rightrightarrows f$, then:
$$\int_E \lim f_n = \lim \int_E f_n$$
let's fix some E where $\int_E 1 = M < \infty$. Take $f_n \rightrightarrows f$ so that $|f(x) - f_n(x)| < \epsilon$ for all $x \in E$ when $n > N$. Then:
$$\begin{aligned}\left|\int_E f(x)dx - \int_E f_n(x)\right| &= \left|\int_E f(x) - f_n(x)\right| \\ &\leq \int_E |f(x) - f_n(x)| \\ &\leq \int_E \epsilon \\ &= \epsilon M\end{aligned}$$

It quickly follows form here that the limit does indeed commute.

We close off this section by proving a useful result allowing for differentiation and integration to commute:

Theorem 2.3.12: Leibniz Integral Rule

Let $f(x,t)$ and its partial derivative $\frac{\partial f}{\partial x}(x,t)$ be continuous functions on a rectangle $R = [c,d] \times [a,b]$. If $x \in (c,d)$, then the function $F(x)$ defined by:

$$F(x) = \int_a^b f(x,t)\,dt$$

is differentiable, and its derivative is given by:

$$\frac{d}{dx}\left(\int_a^b f(x,t)\,dt\right) = \int_a^b \frac{\partial f}{\partial x}(x,t)\,dt$$

Namely, if the variables are independent, differentiation and integration commute.

***Proof*:**
By the definition of the derivative, we have:

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h}$$

Substitute the integral definition of F:

$$F'(x) = \lim_{h\to 0} \frac{1}{h}\left(\int_a^b f(x+h,t)\,dt - \int_a^b f(x,t)\,dt\right)$$

Because the integral of a sum/difference is the sum/difference of the integrals, we can combine them into a single integral:

$$F'(x) = \lim_{h\to 0} \int_a^b \frac{f(x+h,t) - f(x,t)}{h}\,dt$$

To prove the theorem, we must show that we can move the limit inside the integral. We do this by applying the Mean Value Theorem.

For a fixed t, consider the function f with respect to its first variable. By the Mean Value Theorem, there exists a number ξ_h strictly between x and $x+h$ such that:

$$\frac{f(x+h,t) - f(x,t)}{h} = \frac{\partial f}{\partial x}(\xi_h, t)$$

Substitute this back into our integral equation:

$$F'(x) = \lim_{h\to 0} \int_a^b \frac{\partial f}{\partial x}(\xi_h, t)\,dt$$

Now, we need to show that as $h \to 0$, the integral of $\frac{\partial f}{\partial x}(\xi_h, t)$ converges to the integral of $\frac{\partial f}{\partial x}(x,t)$.

Consider the absolute difference:

$$\left|\int_a^b \frac{\partial f}{\partial x}(\xi_h, t)\,dt - \int_a^b \frac{\partial f}{\partial x}(x,t)\,dt\right| \le \int_a^b \left|\frac{\partial f}{\partial x}(\xi_h, t) - \frac{\partial f}{\partial x}(x,t)\right| dt$$

Because $\frac{\partial f}{\partial x}$ is continuous on the closed and bounded (compact) rectangle $R = [c, d] \times [a, b]$, it is uniformly continuous on R.

Uniform continuity means that for any $\epsilon > 0$, there exists a $\delta > 0$ such that whenever the distance between two points in R is less than δ, the difference between the function values at those points is less than $\frac{\epsilon}{b-a}$.

Since ξ_h is between x and $x + h$, the distance between (ξ_h, t) and (x, t) is simply $|\xi_h - x|$, which is strictly less than $|h|$.

Therefore, if we choose $|h| < \delta$, we are guaranteed that:

$$\left|\frac{\partial f}{\partial x}(\xi_h, t) - \frac{\partial f}{\partial x}(x, t)\right| < \frac{\epsilon}{b-a}$$

for all $t \in [a, b]$.

Substituting this bound into our absolute difference gives:

$$\int_a^b \left|\frac{\partial f}{\partial x}(\xi_h, t) - \frac{\partial f}{\partial x}(x, t)\right| dt < \int_a^b \frac{\epsilon}{b-a}\, dt = \left(\frac{\epsilon}{b-a}\right)(b-a) = \epsilon$$

Since ϵ can be arbitrarily small, this proves that:

$$\lim_{h\to 0} \int_a^b \frac{\partial f}{\partial x}(\xi_h, t)\, dt = \int_a^b \frac{\partial f}{\partial x}(x, t)\, dt$$

as we sought to show.

2.3.1 Relating Lebesgue and Riemann Integral

Let's consider $f \in L^1(m)$ to be a Lebesgue integrable function on $\mathbb{R}$. As a preliminary to this book, Riemann integration of real-valued functions have been studied. So far, we have done nothing to show that the definitions of the integral will produce the same result. They surely must, since the Riemann integral correctly captures the area under surfaces. In this section, we show how the Lebesgue integral is in fact an extension of the Riemann integral, meaning if we restrict to the set of all Riemann integrable functions (which will all be Lebesgue integral), then the Lebesgue and Riemann integral give the same result.

As a quick reminder of Riemann integration, choosing to follow Darboux' characterization, let $[a, b]$ be a closed interval (or more generally a compact subset of $\mathbb{R}^n$) Then for any finite partition $P = \{t_0, ..., t_n\}$ where $a = t_0 < t_1 < \cdots < t_n = b$, define:

$$U_P(f) = \sum_{i=1}^n M_i(t_i - t_{i-1}) \qquad L_P(f) = \sum_{i=1}^n m_i(t_i - t_{i-1})$$

where M_i and m_i are the supremum and infimum of f on $[t_{i-1}, t_i]$. Then take:

$$\overline{I}_a^b(f) = \inf_P U_P(f) \qquad \underline{I}_a^b = \sup_P L_p(f)$$

over all possible partitions P. If $\overline{I}_a^b(f) = \underline{I}_a^b$, then their common value is denoted as $\int_a^b f(x)\, dx$ and f is called Riemann integrable on $[a, b]$.

We'll now compare this to the Lebesgue integral:

Theorem 2.3.13: Lebesgue-Vitali Theorem

Let f be a bounded real-valued function on $[a, b]$.

1. If f is Riemann integrable, then f is Lebesgue measurable (and hence integrable on $[a, b]$ since f is bounded) and
$$\int_a^b f(x)\, dx = \int_{[a,b]} f\, dm$$
2. f is Riemann integrable if and only if $\{x \in [a, b] \mid f \text{ is discontinuous at } x\}$ has Lebesgue measure 0.

Note that $\chi_{\mathbb{Q}}$ on $[0, 1]$ is discontinuous everywhere, with discontinuities having Lebesgue measure 1, was shown to not be Riemann integrable (this was in fact the example of a non Riemann-integrable function presented in the preliminary chapter). However, it is certainly Lebesgue integrable!

***Proof*:**

1. Let f be Riemann integrable. Let M_i and m_i are defined as the supremum and infimum as defined above, and for each partition P, let:
$$G_P = \sum_{i=1}^{n} M_i \chi_{(t_{i-1},t_i]} \qquad g_P = \sum_{i=1}^{n} m_i \chi_{(t_{i-1},t_i]}$$
so $U_p(f) = \int G_P\, dm$ and $L_p(f) = \int g_P\, dm$. Now, there exists a sequence of partitions P_n such that $\max_i(t_i - t_{i-1})$ tends to zero and P_{n+1} is a refinement of P_n (meaning g_{P_n} increase in value while G_{P_n} decreases) such that $U_{P_n}(f)$ and $L_{P_n}(f)$ converge to $\int_a^b f(x)\, dx$. Let $G = \lim G_{P_n}$ and $g = \lim g_{P_n}$. Since $g \le f \le G$, by the dominated convergence theorem:
$$\int G\, dm = \int g\, dm = \int_a^b f(x)\, dx$$
Therefore, $\int G - g = 0$ and so $G = g$ a.e. by proposition 2.2.8, and so $G = f$ a.e.
Finally, since G is measurable (being the point-wise limit of measurable functions)and m is complete, f is measurable and
$$\int_[a, b] f\, dm = \int G\, dm = \int_a^b f(x)\, dx$$
2. This was exercise 23 in the textbook

The big advantage we have with this is that we can now use the computational techniques that were developed for Riemann integrals to compute particular Lebesgue integrals! In most cases in analysis, functions are locally Riemann integrable, and so we might what does the generality give us. The main thing it gives us is *completeness*. So though computing Lebesgue integrals might no be so easy (or if they are not an elementary Riemann integrable function), they are easier to work with in a theoretical framework, and help us define concepts like L^p space (which are important for Fourier Analysis).

(Note that in the infinite case, the Riemann and Lebesgue integral need not match. For example, $\sin(x)/x$ is Riemann integrable and produce a finite value, but it is not Lebesgue integral, producing an infinite value))

(A quick word on the Gamma function)

2.4 Modes of Convergence

So far, we have accumulated the following forms of convergence of functions:

1. uniform convergence $f_n \rightrightarrows f$
2. point-wise convergence $f_n \to f$
3. point-wise convergence a.e. $f_n \to f$ a.e.
4. L^1 convergence $f_n \to f$ if and only if $\int |f_n - f| \to 0$

The first 3 are weaker than then the next: uniform convergence $\implies$ point-wise convergence $\implies$ a.e. convergence, but the converse is false in each case. Furthermore, L^1 convergence implies none of those 3, and none of those 3 imply L^1 convergence. The examples to keep in mind to compare are:

1. $f_n = n^{-1}\chi_{[0,n]}$
2. $f_n = \chi_{[n,n+1]}$
3. $f_n = n\chi_{[0,1/n]}$
4. $f_1 = \chi_{[0,1]}$, $f_2 = \chi_{[0,1/2]}$, $f_3 = \chi_{[1/2,1]}$, $f_4 = \chi_{[0,1/4]}$, $f_5 = \chi_{[1/4,2/4]}$, and so on

We now introduce another way of thinking about convergence: instead of saying that the $\int f_n$ approaches that of f (i.e., evaluate the integral and look at the limiting behavior of the integral of the function), we will want there to exist an N such that $n \geq N$, the measure of the difference between f_n and f are epsilon away from the function we are converging to f. This is not the same thing as L^1 convergence, since this (1), (3) and (4) all converge to 0 in this type of convergence. Before any further analysis let's define our terms:

Definition 2.4.1: Cauchy in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *cauchy in measure* if:

$$\forall \epsilon > 0,\ \exists N \text{ such that } \forall n, m \geq N,\ \mu(\{x \mid |f_m(x) - f_n(x)| \geq \epsilon\}) < \epsilon$$

Equivalently, we can write the definition as $\forall \epsilon > 0$,

$$\mu(\{x \mid |f_n(x), f_m(x)| \geq \epsilon\}) \to 0 \text{ as } n, m \to \infty$$

Definition 2.4.2: Convergence in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *convergent in measure* or *converge in measure* if:

$$\forall \epsilon > 0, \ \lim_{n \to \infty} \mu(\{x \mid |f(x) - f_n(x)| \geq \epsilon\}) = 0$$

With these definition, notice that (2) is in fact *not* Cauchy in measure: the difference between the two functions.

Proposition 2.4.3: L^1 convergence Then Measure Convergence

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f_n \to f$ in L^1. Then $f_n \to f$ in measure

***Proof*:**
Let ϵ be given, and define

$$E_{\epsilon,n} = \mu(\{x \mid |f_n(x) - f(x)| \geq \epsilon\})$$

Then:

$$\int |f - f_n| \geq \int_{E_{\epsilon,n}} |f - f_n| \geq \epsilon\mu(E_{\epsilon,n})$$

Thus:

$$\mu(E_{n,\epsilon}) \leq \epsilon^{-1} \int |f - f_n| \to 0$$

showing that it converges in measure.

Note that convergence in measure does not imply convergence in L^1 as we saw in example (1) and (3).

With this established, we can go onto estalbish why we care about convergence in measure: it gives us another way of finding point-wise convergent functions by establishing a condition weaker than L^1 convergence:

Theorem 2.4.4: Cauchy in Measure Then Pointwise a.e.

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ Cauchy in measure. Then there exists a f such that $f_n \to f$ in measure, and furthermore there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ a.e.
If $f_n \to g$ in measure, then $f = g$ a.e.

***Proof*:**

We can choose a subsequence $\{g_j\} = \{f_{n_j}\}$ of $\{f_n\}$ such that if $E_j = \{x : |g_j(x) - g_{j+1}(x)| \geq 2^{-j}\}$, then $\mu(E_j) \leq 2^{-j}$. If $F_k = \bigcup_{j=k}^{\infty} E_j$, then $\mu(F_k) \leq \sum_k^\infty 2^{-j} = 2^{1-k}$, and if $x \notin F_k$, for $i \geq j \geq k$ we have

$$|g_j(x) - g_i(x)| \leq \sum_{l=j}^{i-1} |g_{l+1}(x) - g_l(x)| \leq \sum_{l=j}^{i-1} 2^{-l} \leq 2^{1-j}. \tag{2.31}$$

Thus $\{g_j\}$ is pointwise Cauchy on F_k^c. Let $F = \bigcap_1^\infty F_k = \limsup E_j$. Then $\mu(F) = 0$, and if we set $f(x) = \lim g_j(x)$ for $x \notin F$ and $f(x) = 0$ for $x \in F$, then f is measurable (see Exercises 3 and 5) and $g_j \to f$ a.e. Also, (2.31) shows that $|g_j(x) - f(x)| \leq 2^{1-j}$ for $x \notin F_k$ and $j \geq k$. Since $\mu(F_k) \to 0$ as $k \to \infty$, it follows that $g_j \to f$ in measure. But then $f_n \to f$ in measure, because

$$\{x : |f_n(x) - f(x)| \geq \epsilon\} \subset \{x : |f_n(x) - g_j(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |g_j(x) - f(x)| \geq \tfrac{1}{2}\epsilon\},$$

and the sets on the right both have small measure when n and j are large. Likewise, if $f_n \to g$ in measure,

$$\{x : |f(x) - g(x)| \geq \epsilon\} \subset \{x : |f(x) - f_n(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |f_n(x) - g(x)| \geq \tfrac{1}{2}\epsilon\}$$

for all n, hence $\mu(\{x : |f(x) - g(x)| \geq \epsilon\}) = 0$ for all ϵ. Letting ϵ tend to zero through some sequence of values, we conclude that $f = g$ a.e.

Corollary 2.4.5: L^1 convergence and Finding Subsequence

Let $f_n \to f$ in L^1. Then there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ a.e.

***Proof*:**

By proposition 2.4.3, $f_n \to f$ in measure, and by theorem 2.4.4, there exists a subsequence that converges to f a.e.

As we saw, $f_n \to f$ a.e. does not imply $f_n \to f$ in measure, as we saw in example (2). This is due to the "escape to infinity" that is happening, which is why we care about the dominated convergence theorem to help us stop such "sneaky infinity problem". However, if μ was a finite measure, then $f_n \to f$ a.e. does imply $f_n \to f$ in measure. In fact, since the area is bounded, we can some even stronger convergent conditions:

Theorem 2.4.6: Ergoff's Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space and suppose that $\mu(X) < \infty$. If $\{f_i\}$ are measurable complex-valued functions such that $f_n \to f$ a.e., then for every $\epsilon > 0$, there exists a $E \subseteq X$ such that $\mu(E) < \epsilon$ and $f_n \to f$ uniformly on $\mu(E^c)$

***Proof*:**

Without loss of generality we may assume that $f_n \to f$ everywhere on X. For $k, n \in \mathbb{N}$ let

$$E_n(k) = \bigcup_{m=n}^{\infty} \{x : |f_m(x) - f(x)| \geq k^{-1}\}.$$

Then, for fixed k, $E_n(k)$ decreases as n increases, and $\bigcap_{n=1}^{\infty} E_n(k) = \varnothing$, so since $\mu(X) < \infty$ we conclude that $\mu(E_n(k)) \to 0$ as $n \to \infty$. Given $\epsilon > 0$ and $k \in \mathbb{N}$, choose n_k so large that $\mu(E_{n_k}(k)) < \epsilon 2^{-k}$ and let $E = \bigcup_{k=1}^{\infty} E_{n_k}(k)$. Then $\mu(E) < \epsilon$, and we have $|f_n(x) - f(x)| < k^{-1}$ for $n > n_k$ and $x \notin E$. Thus $f_n \to f$ uniformly on E^c.

The type of convergence in this theorem is sometimes called *almost uniform convergence*. Almost uniform convergence implies convergene a.e. and convergence in measure.

2.5 Product Measure

As we defined in definition 1.1.8, If $\mathcal{M}$ and $\mathcal{N}$ are σ-algebras, then we can define a new σ-algebra $\mathcal{M} \otimes \mathcal{N}$. If $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are measure spaces, we will show that we can define a new measure on $\mathcal{M} \otimes \mathcal{N}$ using μ and ν. We will use the tools we have done in constructing measure to construct this measure, and show that that there is a natural measure $\mu \times \nu$ with respect to the outer-measure given our space is σ-finite (this is just the uniqueness clause of theorem 1.3.7).

First, we'll give a name to sets in $\mathcal{M} \otimes \mathcal{N}$ that will be easier to measure (these will be our generating sets)

Definition 2.5.1: [Measurable] Rectangle

Let $A \in \mathcal{M}$ and $B \in \mathcal{N}$. Then $A \times B \in \mathcal{M} \otimes \mathcal{N}$ is called a *measurable rectangle* or just *rectangle* for short

Notice that if $A \times B$ and $C \times D$ are rectangles, they form an elementary family:

$$(A \times B) \cup (C \times D) = (A \cap C) \times (B \cap D) \qquad (A \times B)^c = (X \times B^c) \cup (A^c \times B)$$

and so by proposition 1.1.13, the collection of finite disjoint unions of rectangles forms an algebra $\mathcal{A}$, and they generate $\mathcal{M} \otimes \mathcal{N}$.

Furthermore, not all sets in $\mathcal{M} \otimes \mathcal{N}$ are measurable rectangles: if we take $(\mathbb{R}, \mathcal{L}(\mathbb{R}), m)$ and consider $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$, then the set of points $\Delta = \{(x, x) \mid x \in \mathbb{R}\}$ is in $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$ (being the countable intersection of ever smaller squares on the line), however it is very far from being the product of two sets in $\mathcal{L}(\mathbb{R})$.

We'll now define how to integrate simple functions on $X \times Y$ with the associated measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$: Let $A \times B$ be a rectangle that is the (finite or countable) disjoint union of rectangle $A_i \times B_i$. Then notice that:

$$\chi_{A\times B}(x, y) = \chi_A(x)\chi_B(y)$$

and so:

$$\chi_{A\times B}(x, y) = \sum_i \chi_{A_i\times B_i}(x, y) = \sum_i \chi_{A_i}(x)\chi_{B_i}(y)$$

If we integrate with respect to x, treating $\chi_{B_i}(y)$ as some constant, then by theorem 2.2.7

$$\int \chi_A(x)\chi_B(y)d\mu = \mu(A)\chi_B(y)d\mu = \mu(A)\chi_B(y) = \sum_i \mu(A_i)\chi_{B_i}(y)$$

It might be confusing which variable is associated to which measure, in which case it is common to write:

$$\int \chi_A(x)\chi_B(y)d\mu(x)$$

to emphasize the function with the x variable is the one being measured by μ. Integrating with respect to y yields:

$$\int \mu(A)\chi_B(y)d\nu = \mu(A)\nu(B) = \sum_i \mu(A_i)\nu(B_i)$$

Thus, we can define $\pi : \mathcal{A} \to \mathbb{R}$ where if $E \in \mathcal{A}$ is a finite disjoint union of rectangles $A_i \times B_i$, then:

$$\pi(E) = \sum_i \mu(A_i)\nu(B_i)$$

with the usual convention of $0 \times \infty = 0$ (i.e., if one of the dimensions is "0", then we are making the value 0). Notice that π is independent of representation, since any finite disjoint union of rectangles have a common refinement. Thus π is a pre-measure, and so by theorm 1.3.7, π generates an outer measure on $X \times Y$ whose restriction to $\mathcal{M} \otimes \mathcal{N}$ is a measure where restricted to $\mathcal{A}$ is π (or conversely, whose measure extends π). Such a measure is given a name:

Definition 2.5.2: Product Measure

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{M}, \nu)$ be measure spaces, and define π as in the previous construction. Then the induced outer-measure is called the *product measure* and is usually denoted $\mu \times \nu$.

Notice further that if μ and ν are σ-finite, so $\mu(X) = \sum_i^\infty \mu(A_i)$ and $\nu(Y) = \sum_i^\infty \nu(B_i)$, then $(\mu \times \nu)(X \times Y) = \sum_{i,j} \mu(A_i)\nu(B_j)$, and since all terms are finite, $\mu \times \nu$ is also σ-finite, in which case by the same theorem as invoked in defining $\mu \times \nu$, the product measure is the unique measure such that $\mu \times \nu(A \times B) = \mu(A)\nu(B)$.

Using induction, we can extend this construction to any finite product of measures, that is, for measure spaces $(X_i, \mathcal{M}_i, \mu_i)$, we can define $\mu_1 \times \cdots \mu_n$ on $\mathcal{M}_1 \otimes \mathcal{M}_2 \otimes \cdots \otimes \mathcal{M}_n$ such that

$$\mu_1 \times \mu_2 \times \cdots \mu_n(A_1 \times A_2 \times \cdots \times A_n) = \prod_i^n \mu_i(A_n)$$

and if every μ_i is σ-finite, then the result product measure is the unique extension from the premeasure on the rectangles of $\prod_i^n \mathcal{M}_i$.

Next, (should I prove associativity, it is exercise 45)

With what we've established so far, we can already measure sets using the outer-measure definition. However, this is very tedious, and ultimately comes down to some measure properties of the respective component measure. There is an easier way to use these component measure to get the resulting measure: notice in our construction of π that we integrated with respect to one variable first, and then the next. We will ostensibly generalizing this trick so that we can measure spaces, or more specifically integrate functions on product spaces, by fixing all variables except one, integrating on it, and then moving the result out (since it's a constant) and integrate the next variable, until we integrate over all variables.

Since we can generalise our results by induction, let's take $n = 2$ and consider the measure space $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$. The following definition tries to capture the idea of fixing everything except one variable:

Definition 2.5.3: n-section

Let $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$ be a product measure space, and consider $E \subseteq X \times Y$. Then for any $x \in X$, $y \in Y$, define the *x-section* E_x and *y-section* E^y to be:

$$E_x = \{y \in Y \mid (x, y) \in E\} \qquad E^y = \{x \in X \mid (x, y) \in E\}$$

If f is a function on $X \times Y$, we define the *x-section* f_x and *y-section* f^y to be:

$$f_x(y) = f(x, y) \qquad f^y(x) = f(x, y)$$

The most simple example of the section of a function would be:

$$(\chi_E)_x = \chi_{E_x} \qquad (\chi_E)^y = \chi_{E^y}$$

Proposition 2.5.4: Measure of Section

1. Let $E \in \mathcal{M} \otimes \mathcal{N}$. Then $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$ (i.e., all sets in a produce σ-algebra can be broken down into x-sections and y-section)
2. If f is $\mathcal{M} \otimes \mathcal{N}$-measurable, then f_x is $\mathcal{N}$-measurable and f^y is $\mathcal{M}$-measurable

***Proof*:**

1. The proof of this fact comes down to taking advantage of the fact that the pre-measure of rectangles generates $\mathcal{M} \otimes \mathcal{N}$. Let $\mathcal{R}$ be the collection of all subsets of $E \subseteq X \times Y$ such that $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$. This clearly contains the empty set,

and the set must contain all rectangles, since if $A \neq 0$ and $x \in A$, then $(A \times B)_x = B$ (if $x \notin A$, then $(A \times B)_x = \emptyset$). Now, since

$$\left(\bigcup_i^\infty E_i\right)_x = \bigcup_i^\infty (E_i)_x \qquad (E^c)_x = (E_x)^c$$

by some basic set manipulation, we see that $\mathcal{R}$ is a σ-algebra. Since it contains the generating set of $\mathcal{M} \otimes \mathcal{N}$, we have that $\mathcal{R} \supseteq \mathcal{M} \otimes \mathcal{N}$. Thus, all the sets in $\mathcal{M} \otimes \mathcal{N}$ can be decompose in this way (note that we have not proved equality: we have not proved that if every slice is measurable then E is measurable)

2. By some set-theory manipulation, and part (1):

$$f_x^{-1}(B) = (f^{-1}(B))_x \qquad f^y{-1}(B) = (f^{-1}(B))^y$$

and so f_x and f^y must be $\mathcal{N}$ and $\mathcal{M}$ measurable (respectively)

With this, we are almost ready to start proving we can integrate $A \times B \in \mathcal{M} \otimes \mathcal{N}$ with respect to 1 measure at a time. We first need to prove a technical result for the next theorem which gives (again) another way of constructing σ-algebra. This one is particularly useful, since it will let us prove certain sets are σ-algebras by using the MCT and DCT:

Definition 2.5.5: Monotone Class

Let X be a set and $P(X)$ the powerset of X. Then the subset $\mathcal{C} \subseteq P(X)$ is called a *monotone class* if

1. it is cosed under countable unions of increasing sets, that is, if $E_1 \subseteq E_2 \subseteq \cdots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\cup_i E_i \in \mathcal{C}$
2. it is cosed under intersection unions of decreasing sets, that is, if $E_1 \supseteq E_2 \supseteq \cdots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\cap_i E_i \in \mathcal{C}$

Clearly, every σ-algebra is a monotone class (we've even used this property in a couple of proofs). Furthermore, the intersection of any family of monotone class is again a monotone class, and so we can define the unique smallest monotone class that contains the set $\mathcal{E} \subseteq P(X)$, where $\mathcal{E}$ is called the monotone class generated set, and the monotone class is called the monotone class generated by $\mathcal{E}$, and is dentoed $\mathcal{C}(\mathcal{E})$

Lemma 2.5.6: Monotone Class Lemma

Let X be a set and $\mathcal{A} \subseteq X$ be an algebra on some subsets of X. Then the monotone class $\mathcal{C}$ generated by $\mathcal{A}$ is equal to the σ-algebra $\mathcal{M}$ generated by $\mathcal{A}$, that is:

$$\mathcal{C}(\mathcal{A}) = \mathcal{M}(\mathcal{A})$$

***Proof*:**
Since $\mathcal{M}$ is itself a monotone class, we automatically have $\mathcal{E} \subseteq \mathcal{M}$, so we'll prove $\mathcal{E} \supseteq \mathcal{M}$.

This is just a lot of set-theory manipulation, I'll leave it for another time

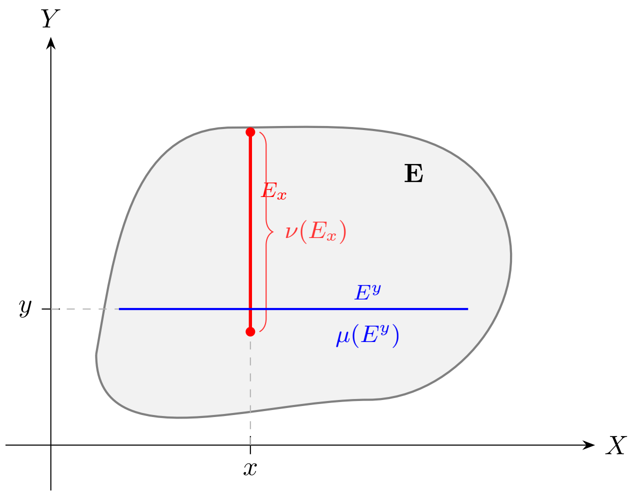
We now are able to start proving the main results:

Theorem 2.5.7: Fubini-Tonelli for Characteristic Functions

Let $(X, \mathcal{M}, \mu)$ and $(N, \mathcal{N}, \nu)$ be measure spaces with μ, ν being σ-finite. Then if $E \in \mathcal{M} \otimes \mathcal{N}$, the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y respectively, and

$$(\mu \times \nu)(E) = \int \nu(E_x) d\mu = \int \mu(E^y) d\nu$$

If you are visual, you can think of $\mu(E_x)$ as measuring the area under the slice (inside the appropriate measure so that it doesn't have measure 0), and see that are saying there exists a function that maps x to every slice of the graph:



In particular, it maps x the measure of that slice, so we get a new function which at every slice captures the effect of the X component. After that, we integrate the function to get the effect from the Y component.

Furthermore, it will turn out that we can measure the slices with respect to X, or with respect to Y, and get the same result, hence the second statement of this theorem! After proving the theorem, we'll go over why the assumption in the statement of the theorem are important.

***Proof*:**
We'll first prove it for when μ and ν are finite, and then generalize for when they are σ-finite. The way we'll do it is by starting with a dense subset of $\mathcal{M} \otimes \mathcal{N}$ for which we know the conclusion holds, and then generalize the result using the MCT and DCT. In particular, we will be taking advantage that rectangles already satisfy the property of the theorem, and since rectangles are a generating set for $\mathcal{M} \otimes \mathcal{N}$, we will be able to extend this result any measurable set.

Let $\mathcal{C}$ be the subset of $\mathcal{M} \otimes \mathcal{N}$ for which the result of the theorem is true, that is, the two maps are measurable and we can integrate one way or another. Then if $A \in \mathcal{M}$ and $B \in \mathcal{N}$, we have that $A \times B \in \mathcal{C}$. To see this, first compute the values $\nu(E_x)$ and $\mu(E^y)$:

$$\nu(E_x) = \chi_A(x)\nu(B) \qquad \mu(E^y) = \mu(A)\chi_B(y)$$

Then the functions is measurable since it's a slice (theorem 2.5.4), and:

$$\int \nu(E_x)d\mu = \int \chi_A(x)\nu(B)d\mu = \mu(A)\nu(B) = \int \mu(A)\chi_B(y)d\nu = \int \mu(E^y)d\nu$$

Showing the two integrals are equivalent, and since $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$, all equality hold, and so $E = A \times B \in \mathcal{C}$. By additivity of the integral, it follows that all finite disjoint unions of rectangles are also in $\mathcal{C}$. Thus, by lemma 2.5.6, it suffices to show that $\mathcal{C}$ is a monotone class, for then it is a σ-algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, meaning $\mathcal{C} \supseteq \mathcal{M} \otimes \mathcal{N}$, and so all measurable sets can be broken down in this way.

First, let $\{E_n\}_{n=1}^{\infty} \subseteq \mathcal{C}$ be an increasing subset and let $E = \bigcup_n^{\infty} E_n$. We want to show that $E \in \mathcal{C}$. Since $E_n \in \mathcal{C}$ for all $n \in \mathbb{N}$, the sequence of y-section functions f_n, $f_n(y) = \mu((E_n)^y)$ are measurable and increase pointwise to $f(y) = \mu(E^y)$. Since the limit of measurable functions is measurable (corollary 2.1.13), f is measurable. Thus, we can find the value of $\int \mu(E^y)d\nu$ by applying the Monotone Convergence Theorem:

$$\begin{aligned}
\int \mu(E^y)d\nu &= \int \lim_{n\to\infty} \mu((E_n)^y)d\nu \\
&= \lim_{n\to\infty} \int \mu((E_n)^y)d\nu && \text{MCT} \\
&= \lim_{n\to\infty} (\mu \times \nu)(E_n) && E_n \in \mathcal{C} \\
&= (\mu \times \nu)(E) && \cup_i E_i = E
\end{aligned}$$

Doing the same trick with respect to x, we can see that:

$$\int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

which combined together shows us that:

$$\int \mu(E^y)d\nu = \int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

and so $E \in \mathcal{C}$.

Next, let $\{E_n\}_{n=1}^{\infty} \subseteq \mathcal{C}$ be a decreasing subset and let $\bigcap_n E_n = E$. Then since $E_1 \in \mathcal{C}$, the function f_1, $f_1(y) = \mu((E_1)^y)$ is measurable. In fact, it's in L^1, since:

$$\mu((E_1)^y) \leq \mu(X) < \infty \qquad \nu(Y) < \infty$$

and so the resulting integral must have finite measure. Furthermore, f_1 will dominate the sequence of functions f_n, and so we can apply the Dominated Convergence Theorem to do the same limit trick we have just done and conclude that $E \in \mathcal{C}$. Therefore, $\mathcal{C}$ is a monotone class, hence a σ-algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, and the result is true for all measurable sets.

Now, suppose that μ and ν are σ-finite. Then we can write $X \times Y$ to be the union of an increasing sequence of rectangles $\{X_i \times Y_i\}$ of finite measure. Thus, for any $E \in \mathcal{M} \otimes \mathcal{N}$, take $E \cap (X_i \times Y_i)$, then this set now has finite measure, and so what we have just proved applies giving us:

$$(\mu \times \nu(E \cap (X_i \times Y_i)) = \int \chi_{X_i}(x)\nu(E_x \cap Y_i)d\mu = \int \mu(E^y \cap X_i)\chi_{Y_i}(y)d\nu$$

and so, a final application of the Monotone Convergence Theorem gives us our desired result

The hypothesis that $\mu \times \nu$ is σ-finite is necessary:

Example 2.5.8: Necessity of σ-finite

To compute the iterated integrals, we start by finding the x and y section. Using the definition of the diagonal we get that: $(\chi_D)_x = \chi_{\{x\}}$ and $(\chi_D)_y = \chi_{\{y\}}$. Thus, we get:

$$\int \left[\int \chi_{\{y\}} d\mu(x)\right] d\nu(y) = \int 0\nu(y) = 0$$

and

$$\int \left[\int \chi_{\{x\}} d\nu(y)\right] d\mu(x) = \int 1 d\mu(x) = 1$$

showing that the iterated iterated iterated integrals are not equal. For the integral $\int \chi_D \, d(\mu \times \nu)$, since χ_D is a simple function, we have shown in class that the integral is the measure of D with respect to the product measure $\mu \times \nu$ (namely, since χ_D will be in the supremum set over which we are computing the integral), which is:

$$\int \chi_D d(\mu \times \nu) = \mu \times \nu(D) = \inf \left\{ \sum_i^{\infty} \mu(A_i)\nu(B_i) \;\middle|\; A_i \times B_i \in \mathcal{A},\ D \subseteq \bigcup_i A_i \times B_i \right\}$$

where $\mathcal{A}$ is the collection of all rectangles of $\mathcal{M} \otimes \mathcal{N}$. We'll show $\mu \times \nu(D) = \infty$.

Take any countable cover of D by rectangle $\mathcal{C}$ (so $\{A_i \times B_i\}_{i=1}^{\infty} = \mathcal{C}$ for appropriate $A_i \times B_i$). Without loss of generality, we can make these covers disjoint; we can refine the rectangles. If we can show that there must be a non-zero measure set A_i and an infinite measure set B_i such that $A_i \times B_i \in \mathcal{C}$ then we are done. For the sake of contradiction, let's say no such set exists in any countable cover $\mathcal{C}$. We'll show that $\mathcal{C}$ doesn't cover D.

Split $\mathcal{C}$ into $F \subseteq \mathcal{C}$ and $I \subseteq \mathcal{C}$ where F is the collection of all sets where A_i has ***finite*** non-zero measure (the sets have finite measure since if $A_i \subseteq X$, $\mu(A_i) \leq \mu(X) = 1$), and I is the set where B_i has ***infinite*** measure. By our contradiction assumption, these set's are disjoint, since the first component of the sets in I must have zero measure.

The set F can only cover countably many points of D: Since each B_i has finite measure, any set $A_i \times B_i \in F$ can cover at most finitely many points of D (given appropriate A_i), and so $\bigcup_{A_i \times B_i \in F} A_i \times B_i$ can at most cover countably many points of D (since the 2nd component of D is covered by elements from B_i given appropriate A_i). Therefore, an uncountable number of points are not being covered by F.

The set I misses at least uncountably many points. Since each component A_i has zero-measure and the collection of x components of D (i.e. the set $[0, 1]$) has measure one, the union of the sets A_i cannot cover all of the x-components (given appropriate B_i components) or else

$0 = \mu(\bigcup_i A_i) \geq \mu([0,1]) = 1$, a contradiction. If the B_i component's don't line up correctly, then this set cover's even fewer points of D. Therefore, I can at most cover a measure zero amount of x-components points from D. Since any set two sets containing each other with a different measure (ex. $\alpha \subseteq \beta$, $0 = \mu(\alpha) \leq (\beta) = 1$) must have a difference with at least uncountable cardinality[a] ($|B \setminus A| \geq \aleph_1$), the set I must miss at least uncountably many points.

However, the cover F can only cover at most countably many points. Therefore, the cover $F \cup I = \mathcal{C}$ cannot cover all of D, a contradiction to the assumption that $\mathcal{C}$ is a cover for D. Hence, there is at least one set $A_i \times B_i$ with $\mu(A_i) > 0$ and $\mu(B_i) = \infty$ for any cover of D, meaning all covers of D must have a set with infinite measure, and so $\infty \leq \mu \times \nu(D)$, and hence

$$\int \chi_D \, d(\mu \times \nu) = \infty$$

and so the two iterated integrals, as well as the integral of χ_D has unequal measures, as we sought to show.

[a] If two sets containing each other had differed by countably many point, then they would have the same measure

Notice that the functions within the integral of the previous theorem are in fact the values of characteristic functions! Hence by the linearity of the integral, we have the result for simple functions. Thus, with the statement proven for simple functions on $\mathcal{M} \otimes \mathcal{N}$, we move on to proving it for integrable functions in general:

Theorem 2.5.9: Fubini-Tonelli Theorem

Let $(X, \mathcal{M}, \mu)$ and $(N, \mathcal{N}, \nu)$ be σ-finite measures. Then:

1. (Tonelli) If $f \in L^+(X \times Y)$, then the functions $g(x) = \int f_x d\nu$ and $h(y) \int f^y d\mu$ are in $L^+(X)$ and $L^+(Y)$ respectively, and
$$\int \left[\int f(x,y) d\nu(y) \right] d\mu(x) = \int f(x,y) d(\mu \times \nu) = \int \left[\int f(x,y) d\mu(x) \right] \nu(y)$$
2. (Fubini) If $f \in L^1(\mu \times \nu)$, then $f_x \in L^1(\nu)$ for a.e. $x \in X$ and $f^y \in L^1(\mu)$ for a.e. $y \in Y$. Furthermore, the a.e. defined function $g(x) = \int f_x d\nu$ and $h(x) = \int f^y d\mu$

The resulting integral is called the *iterated integral*

***Proof*:**

1. This essentially comes down to using MCT and the standard representation of measurable functions. By theorem 2.5.7, we have the simple functions satisfy the theorem. To extend it to integrable functions, take $f \in L^+(X \times Y)$, and let $\{f_n\}$ be some simple representation for it, that is, a sequence of simple functions converging upwards to f. From $\{f_n\}$, we have a corresponding $\{g_n\}$ and $\{h_n\}$ converging upwards to g and h: $g_n = \int (f_n)_x$ and $h_n = \int (f_n)^y$. By theorem 2.5.7, these functions are measurable, and so h and g are measurable. With this, we can put together all our information to form the following chain

of equalities:

$$\begin{aligned}
\int \left[\int f_x \, d\nu \right] d\mu = \int g d\mu &= \int \lim g_n \, d\mu \\
&= \lim \int g_n \, d\mu \\
&= \lim \int f_n \, d(\mu \times \nu) && \text{theorem 2.5.7} \\
&= \int f \, d(\mu \times \nu) && \text{MCT} \\
&= \lim \int f_n \, d(\mu \times \nu) \\
&= \lim \int h_n \, d\nu && \text{theorem 2.5.7} \\
\int \left[\int f^y \, d\mu \right] d\nu = \int h d\nu &= \int \lim h_n \, d\nu
\end{aligned}$$

Thus establishing the equality of Tonelli's Theorem. Furthermore, since $\int f \, d(\mu \times \nu) < \infty$, then $g < \infty$ a.e. and $h < \infty$ a.e., so $\int f_x < \infty$ and $\int f^y < \infty$, and therefore $f_x \in L^1(\nu)$ and $f^y \in L^1(\mu)$, which some of the preliminary results for Fubini's Theorem.

2. Let $f \in L^1(\mu \times \nu)$, so $f^+ \in L^+(\mu \times \nu)$ and $f^- \in L^+(\mu \times \nu)$ (or $\mathrm{Re} f$ and $\mathrm{sgn} f$). Thus, the result applies to each component function, and so by linearity applies to f

Often, the parenthesis are dropped, and we combine the integrals into a double integral:

$$\int \left[\int f(x, y) \, d\mu(x) \right] d\nu(y) = \int \int f(x, y) \, d\mu(x) \, d\nu(y) = \int \int f \, d\mu \, d\nu$$

Also, you might wonder if instead of assuming $f \in L^+(X \times Y)$ or $f \in L^1(X \times Y)$, is it possible to instead assume that each $f_x \in L^+(\nu), f^y \in L^+(\mu)$ (L^1 respectively) and that the iterated integrals $\int \int f \, d\mu \, d\nu$ and $\int \int f \, d\nu \, d\mu$ existed, then $f \in L^+(X \times Y)$ (L^1 respectively), that is, is the converse true? The answer is actually no:

Example 2.5.10: Converse not True

Let $X = Y = \mathbb{N}$, $\mathcal{M} = \mathcal{N} = P(\mathbb{N})$, $\mu = \nu$ be the counting measures. Define:

$$f(m, n) = \begin{cases} 1 & m = n \\ -1 & m = n + 1 \\ 0 & \text{otherwise} \end{cases}$$

Then $\int |f| d(\mu \times \nu) = \infty$, and $\int \int f d\mu d\nu$, $\int \int f d\nu d\mu$ exist and are unequal.

This is not quite a converse because the two iterated integrals are unequal, and I don't know of an example where they are equal, but $\int f d(\mu \times \nu)$ is not equal to it (i.e. infinity).

One practical application of Fubini-Tonelli's Theorem is to solve integrals by integrating over the simpler variable first. Usually, one first uses Tonelli's theorem to evaluate the integral $\int f \, d(\mu \times \nu)$ as an iterated integral $\int\int f\,d\mu\,d\nu$ and show that the result is finite, at which point Fubini's Theorem can be invoked to get $\int\int f\,d\mu\,d\nu = \int\int f\,d\nu\,d\mu$.

To close off the discussion on product measures, notice that if μ and ν are complete measure, $\mu\times\nu$ is almost never a complete measure. Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be measure spaces, take $A\in\mathcal{M}$ is a non-empty set with $\mu(A)=0$, and suppose $\mathcal{N}\neq P(Y)$ (For example, choose $(\mathbb{R},\mathcal{L},m)$ for both measure spaces). If $E\in P(Y)\setminus\mathcal{N}$, then $A\times E\notin\mathcal{M}\times\mathcal{N}$ since not all slices of $A\times E$ (in particular, the slice E) are in the respective measure spaces (proposition 2.5.4). However, $A\times E\subseteq A\times Y$, and $(\mu\times\nu)(A\times Y)=0$ (since the measure is finite, any out-measure is equal to the usual outer-measure on these sets, which is $\mu(A)\nu(Y)=0\cdot n=0$, since we take on the convention that $0\cdot\infty=0$). Thus, $\mu\times\nu$ is not complete. If we want, we can apply theorem 1.2.8 to complete the measure, however we can no longer directly apply Fubini-Tonelli's Theorem. (Put Why Here!). We thus have to slightly amend Fubini-Tonelli's theorem for the completion:

Theorem 2.5.11: Fubini-Tonelli's Theorem for Complete Measures

Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be complete σ-finite measures and let $(X\times Y,\mathcal{L},\lambda)$ be the completion of $(X\times Y,\mathcal{M}\otimes\mathcal{N},\mu\times\nu)$. If f is $\mathcal{L}$-measurable is either:

1. non-negative, that is $f\geq 0$
2. $f\in L^1(\lambda)$

Then f_x is $\mathcal{N}$-measurable for a.e. x and f^y is $\mathcal{M}$-measurable for a.e. y, and if (2) applies as well, then f_x and f^y are integrable for a.e. x and y. Furthermore, $x\mapsto\int f_x\,d\nu$ and $y\mapsto\int f^y\,d\mu$ are measurable (and in the case of (2), also integrable) and

$$\int f\,d\lambda = \int\int f(x,y)\,d\mu(x)\,d\nu(y) = \int\int f(x,y)\,d\nu(y)\,d\mu(x)$$

***Proof*:**
We'll do the following:

1. If $E\in\mathcal{M}\otimes\mathcal{N}$ and $\mu\times\nu(E)=0$, then $\nu(E_x)=\mu(E^y)=0$ for a.e. x and y
2. If f is $\mathcal{L}$-measurable and $f=0$ λ-a.e., then f_x and f^y are integrable for a.e. x and y, and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y (here, the completeness of μ and ν is needed)

lemma 1:

Proof. Note that $\chi_E\in L^+(X\times Y)$ and $(X,\mathcal{M},\mu)$ $(Y,\mathcal{N},\nu)$ are σ-finite, and so Tonelli's Theorem applies, meaning

$$\int\chi_E d(\mu\times\nu) = \int\left[\int(\chi_E)^y d\mu(x)\right]d\nu(y) = \int\left[\int(\chi_E)_x d\nu(y)\right]d\mu(x)$$

First, since χ_E is an indicator function, we see that $(\chi_E)_x=\chi_{E_x}$, since $\chi_{E_x}(y)=\chi_E(x,y)$.

Similarly, $(\chi_E)^y = \chi_{E^y}$. Therefore, we have:

$$\begin{aligned} 0 &= \int \chi_E d(\mu \times \nu) \\ &= \int \left[\int (\chi_E)^y d\mu(x) \right] d\nu(y) \\ &= \int \left[\int (\chi_E)_x d\nu(y) \right] d\mu(x) \end{aligned}$$

showing that the function $x \mapsto \int \chi_{E_x} d\nu$ and $y \mapsto \int \chi_{E^y} d\mu$ is 0 a.e., which implies $\nu(E_x) = 0$ and $\mu(E^y) = 0$ a.e. $\square$

lemma 2: Multiple parts of this proof take advantage of the fact that if μ is a measure and $\mu(A) = 0$, then $\int_A f d\mu = 0$ (since all simple functions φ approximating f with the property of $\varphi \leq f$ will have their integral be of the form $\sum_i^n c_i \chi_{A_i} = \sum_i^n c_i \cdot 0 = 0$ for $A_i \subseteq A$, and so the supremum of the integral of such simple functions must also be 0)

Proof. Let f be $\mathcal{L}$-measurable and $f = 0$ λ-a.e. We want to show that f_x and f^y is integrable for a.e. x and y and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y. What we'll do is take advantage of Fubini-Tonelli and theorem 2.12 (which states that there exists a $\mathcal{M} \otimes \mathcal{N}$-measurable function g such that $f = g$ λ-a.e.)

First, since $f = g$ λ-a.e., $f - g = 0$ λ-a.e., meaning $(f - g)_x = f_x - g_x = 0$ for a.e. x (assuming completeness) and $f^y - g^y = 0$ for a.e. y (assuming completeness), and so $f_x = g_x$ and $f^y = g^y$ for a.e. x and y. This will be useful to us later.

Next, Consider $\int f d\lambda$. We want to take advantage of Fubini's theorem, which is true for g, and so we must re-write this expression in such a way that we get it equal to g. Let's say $S = \{(x, y) \in X \times Y \mid f(x, y) \neq g(x, y)\}$. We know that $\lambda(S) = 0$, and so:

$$0 = \int f \, d\lambda = \int_S f \, d\lambda + \int_{S^c} f \, d\lambda = \int_{S^c} f \, d\lambda \tag{2.1}$$

Since $S^c \in \mathcal{L}$, we have that $S^c = A \coprod N$ where $A \in \mathcal{M} \otimes \mathcal{N}$ and $N \subseteq B$ where $\mu \times \nu(B) = 0$. Thus, we get:

$$\int_{S^c} f d\lambda = \int_A f d\lambda + \int_N f d\lambda = \int_A f d\lambda$$

since $f = g$ on S^c, in particular $f = g$ on $A \in \mathcal{M} \otimes \mathcal{N}$, and λ is the extension of $\mu \times \nu$, we can re-write this as:

$$\int_A f d\lambda = \int_A g d(\mu \times \nu)$$

and so Fubini-Tonelli applies, giving us the iterated integrals are equal to 0 (carrying the 0 from equation (2.1)):

$$\int \left[\int_{A_x} g_x d\nu(y) \right] d\mu(x) = 0$$

and similarly for the y-section. Since $f = g$ on S^c, by what we've shown earlier, $f_x = g_x$ on A_x, (and similarly $f^y = g^y$ on A^y) and so:

$$\int_{A_x} f_x d\nu(y) = 0 \qquad \text{for a.e. } x$$

and similarly for the y-sections, which brings us close to our goal (it is possible that $A_x \subsetneq X$ or $A^y \subsetneq Y$). We now have to add back in points which we eliminated since the integral was zero on S and N:

$$\int_S f d\lambda = 0 \qquad \int_N f d\lambda = 0$$

Since they are measure zero sets in $\mathcal{L}$, there exists a $C, D \in \mathcal{M} \otimes \mathcal{N}$ such that $\mu \times \nu(C) = \mu \times \nu(D) = 0$ and $S \subseteq C$, $N \subseteq D$. Let $E = C \cup D$ so that $S \cup N \subseteq E$ and $\mu \times \nu(E) = 0$. Thus, we have:

$$\int_E f d\lambda = 0$$

since $\mu \times \nu(E) = 0$ and $E \in \mathcal{M} \otimes \mathcal{N}$, by the first lemma, $\mu(E^y) = \nu(E_x) = 0$ for a.e. x and y. But then, for a.e. x and y, $\int_{E_x} f_x d\nu(y) = \int_{E^y} f^y d\mu(x) = 0$. Since $X \times Y = S^c \cup S = A \cup E$ (by how we've broken the set down), then $X = A_x \cup E_x$ and $Y = A^y \cup E^y$, and so

$$0 \leq \int f_x d\nu(y) \leq \int_{A_x} f_x d\nu(y) + \int_{E_x} f_x d\nu(y) = 0$$

for a.e. x, implying $\int f_x d\nu(y) = 0$ for a.e. x (and similarly for y). Hence, they are integrable for a.e. x and y, completing the proof. □

With this, we can prove Fubini-Tonelli's Theorem for complete measures. First, let's build-up some result and notation. Let $f \in L^+(\lambda)$ (resp. $f \in L^1(\lambda)$). Then by theorem 2.12, there exists a g such that $f = g$ λ-a.e., meaning that $f - g = 0$ λ-a.e. Thus, by the 2nd lemma, $\int (f-g)_x = 0$ and $\int (f-g)^y = 0$ are integrable for a.e. x and y. Therefore, $\int f_x - g_x = 0$ and $\int f^y - g^y = 0$ for a.e. x and y, and so $\int f_x = \int g_x$ and $\int f^y = \int g^y$ for a.e. x and y.

We can now start concluding the proof. Since μ and ν are complete and $f_x = g_x$ ($f^y = g^y$) for a.e. x and y, by proposition 2.11, f_x is $\mathcal{N}$-measurable and f^y is $\mathcal{M}$-measurable for a.e. x and y, completing the 1st assertion of the theorem. In the $f \in L^1(\lambda)$ case, the proof we've just done in the previous paragraph shows that each f_x and f^y are integrable for a.e. x and y.

Next, since $G_x : x \mapsto \int g_x d\nu$ and $G^y : y \mapsto \int g^y d\mu$ is measurable (resp. integrable), by theorem 2.11, $F_x : x \mapsto \int f_x d\nu$ and $F^y : y \mapsto \int f^y d\mu$ is measurable (resp. integrable). Finally, since $F_x = G_x$ for a.e. x and $F^y = G^y$ for a.e. y, their integrals match, and since g is $\mathcal{M} \otimes \mathcal{N}$ measurable, Tonelli's (resp. Fubini's) theorem applies, and so by one more application of theorem 2.11 :

$$\int f \, d\lambda = \int \int f(x,y) \, d\mu(x) \, d\nu(y) = \int \int f(x,y) \, d\nu(y) \, d\mu(x)$$

Completing the proof

2.6 The n-dimensional Lebesgue Integral

In this section, we focus on $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ with the completed product measure $m \times m \times \cdots \times m = m^n$ on $\mathcal{L} \otimes \mathcal{L} \otimes \cdots \otimes \mathcal{L} = \mathcal{L}^n$. The domain $\mathcal{L}^n$ of m^n is called the set of *Lebesgue measurable sets in* $\mathbb{R}^n$: $(\mathbb{R}^n, \mathcal{L}^n, m^n)$ (sometimes, we will restrict m^n to have domain $(\mathcal{B}_\mathbb{R})^n = \mathcal{B}_{\mathbb{R}^n}$). If it is unambiguous, we will usually write $(\mathbb{R}^n, \mathcal{L}^n, m)$, dropping the power on the m, and for the $n = 1$ case, we will often write $\int f(x)\,dx := \int f\,dm$.

The following 3 theorems are simply generalizations of previous properties of the Lebesgue measure. In the following, let $E = \prod_{i=1}^n E_i$ where $E_i \subseteq \mathbb{R}$. We will call each E_i the *sides* of E

Theorem 2.6.1: Simplifying outer-measure

Suppose $E \in \mathcal{L}^n$

1. $m(E) = \inf\{m(U) \mid E \supseteq U,\ U \text{ is open}\} = \sup\{m(K) \mid K \subseteq E,\ K \text{ is compact}\}$
2. $E = A_1 \cup N_1 = A_2 \setminus N_2$ where $A_1 \in F_\sigma$, $A_2 \in G_\delta$, and $m(N_1) = m(N_2) = 0$
3. If $m(E) < \infty$, for all $\epsilon > 0$, there is a finite disjoint collection of rectangles $\{R_i\}_{i=1}^n$ whose sides are intervals such that $m(E\Delta \cup_i^n R_i) < \epsilon$

***Proof*:**
p. 70 in folland

Theorem 2.6.2: Completeness of $L^1(m)$

Let $f \in L^1(m)$ and let $\epsilon > 0$. Then there exists a simple function $\varphi = \sum_{i=1}^n a_i \chi_R$ where each R_i is a product of intervals such that

$$\int |f - \varphi| < \epsilon$$

and there is a continuous function that vanishes outside a bounded set such that

$$\int |f - g| < \epsilon$$

***Proof*:**
Just like in theorem 2.3.9, approximate f by a simple function, then apply proposition 2.6.1(3) to approximate φ appropriately. Finally, approximate φ by continuous functions by applying the natural generalization of the argument in theorem 2.3.9

Theorem 2.6.3: Translation Invariance in $\mathbb{R}^n$

Let $(\mathbb{R}^n, \mathcal{L}^n, m)$ be a measure space. Then m is translation invariant, that is, for any $a \in \mathbb{R}^n$, if we define $\tau_a : \mathbb{R}^n \to \mathbb{R}^n$, $\tau_a(x) = a + x$, then

1. If $E \in \mathcal{L}^n$, then $\tau_a(E) \in \mathcal{L}^n$ and $m(E) = m(\tau_a(E))$
2. If $f : \mathbb{R}^n \to \mathbb{C}$ is Lebesgue measurable, then so is $f \circ \tau_a$. Furthermore, if either $f \geq 0$ or $f \in L^1(m)$, then $\int (f \circ \tau_a)\,dm = \int f\,dm$

Notice that we did not mention scaling. This is because scaling is not abstracted to linear transformations which requires more work to show how the scaling of a linear transformation affects the value, and so we will prove that later in this section.

> ***Proof*:**
> Folland p.71

(here, some on cube approximation and Jordan content)

I'll for now simply state the effect of linear transformations and come back to this:

Theorem 2.6.4: Linear Transformation on Integrability

Suppose $T \in GL_n(\mathbb{R})$. Then:

1. If f is a Lebesgue measurable function on $\mathbb{R}^n$, so is $f \circ T$. If $f \geq 0$ or $f \in L^1(m)$, then:

$$\int f\, dm = |\det(T)| \int f \circ T\, dm$$

2. If $E \in \mathcal{L}^n$, then so is $T(E)$, and $m(T(E)) = |\det(T)|m(E)|$

> ***Proof*:**
> here

There is also the statement on how diffeomorphisms affect the integral.

3

Signed Measure and Differentiation

In this chapter, we will take a close look at how to define measure's with respect to integrals. In particular, given the appropriate integrable function f, we will want to define a measure μ such that

$$\mu(A) = \int_A f$$

After defining some terms in section 3.1, we will focus in section 3.2 and 3.3 on how to find a function f given a measure μ so that $\mu(A) = \int_A f$. The last 2 sections then focus on the special case where the function f that we found can actually be the *derivative*. In the simple case where $X = \mathbb{R}$, then we can define:

$$F(x) = \int_{(-\infty,x]} f dx = \int_{-\infty}^{x} f dx$$

and ask ourselves what conditions does f or F requires so that $f = F'$ (or, $f = F'$ a.e.). This line of reasoning will ultimately lead to a generalisation of the Fundamental Theorem of Calculus, which will tell us when can we recover the function f given we know $\int_A f$ for all A.

3.1 Signed Measures

Definition 3.1.1: Signed Measure

Let $(X, \mathcal{M})$ be a measurable space. A *signed measure* on $(X, \mathcal{M})$ is a function: $\nu : \mathcal{M} \to [-\infty, \infty]$ such that:

1. $\nu(\emptyset) = 0$
2. ν has at most one value of $\pm\infty$
3. If $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ is a disjoint collection, then

$$\nu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \nu(E_i)$$

   where if $\nu\left(\bigcup_{i=1}^{\infty} E_i\right) < \infty$, then $\sum_{i=1}^{\infty} \nu(E_i)$ converges absolutely.

Example 3.1.2: Signed Measures

Let $(X, \mathcal{M})$ be a measurable space.

1. Clearly, every measure on $\mathcal{M}$ is a signed measure. To distinguish between the two, we will often call measures *positive measures*.
2. Let μ_1 and μ_2 be two measure on $\mathcal{M}$ with at least one being finite. Then $\nu = \mu_1 - \mu_2$ is a signed measure.
3. If f is a $\mathcal{M}$-measurable function, then if at least one of $\int f^+$ or $\int f^-$ is finite,

$$\nu(E) = \int_E f$$

   is a signed measure. In this case, we call f the *extended μ-integrable function* (in contrast to $f \in L^1$ where both terms need to be finite)

The fact that it converges absolutely is to make sure that the order in which we sum the elements does not matter: If it did not converge absolutely, then it is possible to make the sum approach any value we want. Since $\mathcal{M}$ doesn't necessarily have an inheret order (in fact it almost never does), then having this dependence on the order of summing the elements is inconsistent with our notion of area (this is also why we only allow one of the direction to be infinite). It might seem that we must thus be careful on how we define our signed measure to make sure this condition holds, however we will soon show that every signed measrue can be decomposed into 2 measures. If the measure of all sets with positive measure and negative measure are both infinite ($\pm\infty$), then we have contradicted the definition of a signed measure. As just mentioned, all signed measures can be decomposed into two positive measures. We will spend this section proving that fact. In the next section, we will show that under appropriate conditions and choices of signed measure ν and positive meaure μ, we can find a measruable function f such that $\nu(E) = \int_E f d\mu$. This will essentially be the generalization of the derivative!

We start by building up to decomposing a signed measure into two positive meaures:

Proposition 3.1.3: Continuity from Above/Bellow for Signed Meaures

Let ν be a signed measure on $(X, \mathcal{M})$.

1. If $\{E_i\}$ is an increasing sequence in $\mathcal{M}$, then $\nu\left(\bigcup_i E_i\right) = \lim_{i\to\infty} \nu(E_i)$
2. If $\{E_i\}$ is an decreasing sequence in $\mathcal{M}$ where $\nu(E_1)$ is finite, then $\nu\left(\bigcap_i E_i\right) = \lim_{i\to\infty} \nu(E_i)$

***Proof*:**
Very similar to proposition 1.2.5

For the next proposition, we introduce a quick new definition: a set $P \in \mathcal{M}$ is called *positive* (resp. *negative*) if for all $S \subseteq P$ such that $S \in \mathcal{M}$, then $\nu(S) \geq 0$ (resp. $\nu(S) \leq 0$)

Lemma 3.1.4

Any countable union of positive sets is a positive set

***Proof*:**
Use proposition 3.1.3

Theorem 3.1.5: Hahn Decomposition Theorem

If ν is a signed measure on $(X, \mathcal{M})$, there exists a positive set P and a negative set N for ν $P \coprod N = X$ and $P \cap N = \emptyset$. If P' and N' is another such pair, then $P \Delta P$ and $N \Delta N'$ is a null set for ν

***Proof*:**
Let ν be a signed measure, so only ∞ or $-\infty$ can be achieved. If ∞ is achieved, take $-\nu$ so that ∞ is not achieved. Now, let m be the supremum of $\nu(E)$ where E ranges over all positive sets. Thus, there is a sequence of positive sets $\{P_i\}$ such that $\nu(P_i) \to m$. Let $P = \bigcup_i P_i$. Then by the previous lemma and proposition, P is a positive set and $\nu(P) = m$. Since ν does not achieve ∞, $m < \infty$. I claim that $N = X \setminus P$ is negative (thus completing the first assertion of the proof).

First, N cannot contain non-null positive sets. If $E \subseteq N$ was a positive set (i.e. $m(E) > 0$), then $E \cap P = \emptyset$ and $E \cup P$ is a positive set, so $m(E \cup P) = m(E) + m(P) > m$, which is a contradiction to m being the supremum.

Now, let's say there exists a $A \subseteq N$ such that $\nu(A) > 0$. We will reach a contradiction in using the following fact: since A cannot be positive, then there must exist a $B \subseteq A$ such that $\nu(B) < 0$. Then let $C = A \setminus B$ so that $\nu(C) > \nu(A)$ and $C \subseteq A$.

With this, we proceed in forming a contradiction: Define the following sequence $\{A_i\}$ and $\{n_i\}$ as follows:

1. Let n_1 be the smallest positive integer such that there exists a $A_1 \subseteq N$ where $\nu(A_1) > n_1^{-1}$ (this set exists since there is a $A \subseteq N$ where $\nu(A) > 0$
2. Let n_k be the smallest positive integer such that there exists a $A_k \subseteq A_{k-1}$ such that $\nu(A_{k-1}) > \nu(A_k) + n_k^{-1}$ (since A_{k-1} is positive, we use the earlier remark fact to find such a set)

From this sequence of positive decreasing sets, $A_1 \supseteq A_2 \supseteq \cdots$, let $A = \bigcap_i A_i$. Then by the fact that ν does not attain ∞ and the previous proposition:

$$\infty > \nu(A) = \lim \nu(A_i) > \sum_{i=1}^{\infty} n_i^{-1}$$

since the series converges, it must be that $n_i \to \infty$. Finally, once again take $B \subseteq A$ such that $\nu(A) > \nu(B) + n^{-1}$ for some positive integer n. Since $n_i \to \infty$, there exists an i sufficiently large so that $n < n_i$ and $B \subseteq A_i$, meaning we can replace n_i by n in the above construction. However, by construction, we picked the *minimum* integer every-time, and so this contradicts minimality! Therefore, it cannot be that there exists a set $A \subseteq N$ such that $\nu(A) > 0$, and so N is indeed negative.

If P' and N' where two other sets satisfying the decomposition condition, then we have that $P \setminus P' \subseteq P$ (since it's a subset) and $P \setminus P' \subseteq N'$ (since it is P minus all the positive set P') , so it is both positive and negative, which can only be possible if it is a null-set (similarly for $N \setminus N'$)

Using the Hahn decomposition, we can define a positive measure on P and N. Before proceeding to that theorem, we quickly give a name to the relation between the two measures that will be defined:

Definition 3.1.6: Mutually Singular Signed Measures

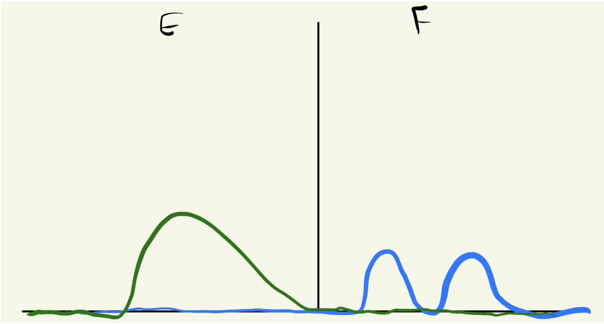
Let ν and μ be be two signed measure on $(X, \mathcal{M})$. Then they are called *mutually singular* or μ *is singular with respect to* ν if there exists a $E, F \in \mathcal{M}$ such that $E \cup F = X$, $E \cap F = \emptyset$, and:

1. $\mu(E) = 0$
2. $\nu(F) = 0$

and we write mutually singular signed measures as:

$$\mu \perp \nu$$

In a sense, this means that these two sets are are meaningful on disjoint sets, in that sense they are "perpendicular" to each other. I like to keep an image like this in mind



Notice that it is possible that μ have zero values on F, and ν have zero values on E. They key is that they *must* have zero values on their respective sets.

Mutually singular measures can be applied more generally than in the context of decomposing a signed measure. For example, of δ_0 is the dirac-delta measure at 0 (i.e. $\delta(\{0\}) = 1$ and zero everywehre else), then $\delta_0 \perp \lambda$.

Theorem 3.1.7: Jordan Decomposition Theorem

If ν is a signed measure, there exists unique positive measure ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$

***Proof*:**
By theorem 3.1.5, let $X = P \cup N$ be the Hahn decomposition of X with respect to ν. Define $\nu^+(E) = \nu(E \cap P)$ and $\nu^-(E) = -\nu(E \cap N)$. Then by definition, $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.

If there was another decomposition $\nu = \mu^+ - \mu^-$ where $\mu^+ \perp \mu^-$, then we have $E, F \in \mathcal{M}$ such that $E \coprod F = X$ and $E \cap F = \emptyset$, $\mu^+(F) = \mu^-(E) = 0$, so E, F is another Hahn decomposition for ν, and so $P \Delta E$ is null. Thus, for any $A \in \mathcal{M}$

$$\mu^+(A) = \mu^+(A \cap E) = \nu(A \cap E) = \nu(A \cap P) = \nu^+(A)$$

thus, $\mu^+ = \nu^+$, and similarly $\mu^- = \nu^-$, completing the proof.

Definition 3.1.8: Jordan Decomposition

Let ν be a signed measure on $(X, \mathcal{M})$. THen ν^+ and ν^- are caleld the *positive* and *negative variation* of ν, and $\nu = \nu^+ - \nu^-$ is called the *Jordan decomposition of* ν. The *total variation of* ν, denoted $|\nu|$ is defined as $|\nu| = \nu^+ + \nu^-$.

As an exercise, you should verify that $E \in \mathcal{M}$ is ν-null if and only if $|\nu|(E) = 0$, and $\nu \perp \mu$ if and only if $|\nu| \perp \mu$ if and only if $\nu^+ \perp \mu$ and $\nu^- \perp \mu$. With this definition, we will say that ν is finite (resp. σ-finite) if $|\nu|$ is finite (resp. σ-finite)

Finally, we can also get back to showing that any signed measure is of the form $\nu(E) = \int_E d\mu$ for appropriate μ, namely $\mu = |\nu|$ and $f = \chi_P - \chi_N$ for appropriate P and N from a Hahn decomposition. From this, we can define the integration with respect to a signed measure ν as follows:

Definition 3.1.9: Integration w.r.t. Signed Measures

Let ν be a signed measure, $\nu^+ - \nu^-$ be it's Hahn decomposition, and $f : X \to \mathbb{C}$ be a real function. Then we will say that f is *integrable with respect to the signed measure* ν if:

$$f \in L^1(\nu^+) \cap L^1(\nu^-) = L^1(\nu)$$

and define it's integral to be:

$$\int f d\nu = \int f d\nu^+ - \int f d\nu^- \qquad (f \in L^1(\nu))$$

3.2 Lebesgue-Radon-Nikodym Theorem

In this section, we show under what circumstances we can decompose a signed measure μ into $\nu + \rho$ where $\nu(A) = \int_A f$ and ρ is the "left-over" (also known as the singular part). We first define the "opposite" notion of mutually singular signed measures. The motivation for the vocabulary will come in section ref:HERE

Definition 3.2.1: Absolutely Continuous w.r.t. μ

Let ν be a signed measure on $(X, \mathcal{M})$ and μ is a positive measure[a] on $(X, \mathcal{M})$. We say that ν is *absolutely continuous with respect to* μ when if $\mu(E) = 0$ (given $E \in \mathcal{M}$) then $\nu(E) = 0$, and we write is as:

$$\nu \ll \mu$$

[a]this can be extended to a signed measure by taking $\nu \ll \mu$ if and only if $\nu \ll |\mu|$, but we don't need this for this document

I like to remember the direction of the arrow by remembering $\mu(E) = 0 \Rightarrow \nu(E) = 0$, so the arrow is pointing towards ν, so we write the arrow's pointing towards ν: $\nu \ll \mu$. The ν is absolutely continuous with respect to μ, and so it's ν "w.r.t." μ: ν is on the left, the μ on the right. If μ is any measure and 0 is the zero measure, it is always the case that $\mu \ll \mu$ and $0 \ll \mu$. The most important example of an absolutely continuous measures is given by meaures defined via integration:

Definition 3.2.2: Measures Defined By Densities

Let μ be a measure on x and let $f \in L^1(X)$. Then:

$$\nu(E) = \int_E f d\mu$$

is a measure with respect to f.

Clearly $\nu \ll \mu$, since if $\mu(E) = 0$, then $\int_E f d\mu = 0 = \nu(E)$.

It should be verified that $\nu \ll \mu$ if and only if $|\nu| \ll \mu$ if and only if $\nu^+ \ll \mu$ and $\mu^- \ll \mu$. Notice that absolute continuity is sort of the opposite of modular singularity in the sense that if $\nu \perp \mu$ and $\nu \ll \mu$, then it must be that $\nu = 0$. Note that $\mu \perp \nu$ does not imply $\mu(E) = 0 \Rightarrow \nu(E^c) = 0$: take

ν to be the Lebesgue signed measure on $\mathbb{R}$, so that the positive and negative sets are $[0, \infty)$ and $(-\infty, 0)$ and experiment.

The following theorem should give you a taste on why the term "continuity" comes around when ν is finite. In particular, it should give you a feeling of uniformally continuous. Later on, we'll show that absolute continuity is stronger than uniform continuity, and so this feeling should make sense:

Theorem 3.2.3: Epsilon-Delta Absolute Continuity

Let ν be a finite signed measure on $(X, \mathcal{M})$ and μ a positive measure on $(X, \mathcal{M})$. Then $\nu \ll \mu$ if and only if for every $\epsilon > 0$, there exists a $\delta > 0$ such that $\mu(E) < \delta \Rightarrow |\nu(E)| < \epsilon$.

***Proof*:**
Since $\nu \ll \mu$ if and only if $|\nu| \ll \mu$, and moreover $|\nu(E)| \leq |\nu|(E)$, it suffices to show it in the case where $\nu = |\nu|$ is positive.

- ($\Leftarrow$) Let the ϵ-δ definition hold so that for any $\epsilon > 0$, there exists a $\delta > 0$ such that for all $E \in \mathcal{M}$ that satisfy $\mu(E) < \delta$, then $\nu(E) < \epsilon$. If we pick all E such that $\mu(E) = 0$, then for any ϵ arbitrarily small, any δ will work, and so $\nu(E) < \epsilon$ for all ϵ, and so it must be that $\nu(E) = 0$

- ($\Rightarrow$) assume the contra-positive; that the ϵ-δ condition is not satisfied, meaning there exists a $\epsilon > 0$ such that for all $\delta > 0$, there is a E_δ such that $\mu(E_\delta) < \delta \Rightarrow \nu(E_\delta) \geq \epsilon$. In particular, for every $n \in \mathbb{N}$, $\mu(E_n) < 2^{-n} \Rightarrow \nu(E_n) \geq \epsilon$. let $F_k = \bigcup_{n=k}^{\infty} E_n$ and $F = \bigcap_k F_k$. Then

$$\mu(F_k) < \sum_{k=n}^{\infty} \frac{1}{2^n} = 2^{1-k}$$

  and since $F \subseteq F_k$ for all F_k, it must be that $\mu(F) = 0$. However, by assumpiton $\nu(F_k) \geq \epsilon$ for all k. Since ν is finite, $\nu(F_1)$ is finite, so continuity from above implies

$$\nu(F) = \lim \nu(F_k) \geq \epsilon$$

  and so $\nu \not\ll \mu$

An easy way to find a ν that is absolutey continuous with respect to μ is if we have a measure μ and a extended μ-integrable function f and define $\nu(E) = \int_E f \, d\mu$. If $f \in L^1$, then ν is finite, in which case we have the following corrollary:

Corollary 3.2.4: Absolute Continuity And L^1

If $f \in L^1(\mu)$, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\forall E \in \mathcal{M},\ \mu(E) < \delta \ \Rightarrow \ \left| \int_E f \right| < \epsilon$$

***Proof*:**
apply the previous theorem on Ref and Imf

The definition of ν with respect to $\int_E f d\mu$ is common enough that we define the following notation for it:

Definition 3.2.5: Density Notation

$$d\nu = f d\mu$$

This should give you the feeling of change of variables, and indeed the two concepts will be related. I like to remember this notation by thinking that if we add an integral with respect to E, $\int_E$ to both sides, and imagine there is the constant function in-front of the $d\nu$ on the left hand side, we get back our previous notation of $\nu(E) = \int_E f d\mu$.

We are now in a position to fully classify signed measures given positive measures. The way we will generlize this is to start by remember that an extended μ integrable function f has one part that allow's for an infinite value. Thus, we can think of the domain of f as being split between a signed measure and a positive measure. To avoid pathologies, we will work over σ-finite measures (as usual). We first prove some technical lemma before continuing to the main result:

Lemma 3.2.6

Let μ and ν be finite measures on $(X, \mathcal{M})$. Then either $\mu \perp \nu$, or there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\nu \geq \epsilon\mu$ on E (that is, ν and μ have domain $M_E = \{E \cap F \mid F \in \mathcal{M}\}$, and E is a positive set for $\nu - \epsilon\mu$)

***Proof*:**
Let $X = P_n \coprod N_n$ be the Hahn decomposition for $\nu - n^{-1}\mu$, and let $P = \bigcup_n^\infty P_n$ and $N = \bigcap_n^\infty N_n = P^c$. Then N is a negative set on $\nu - n^{-1}\mu$ for all n a.e., and

$$0 \leq \nu(N) \leq n^{-1}\mu(N)$$

for all n, so $\nu(N) = 0$. If $\mu(P) = 0$, then $\nu \perp \mu$. If $\mu(P) > 0$, then $\mu(P_n) > 0$ for some n, and P_n is a positive set for $\nu - n^{-1}\mu$

Lemma 3.2.7: Sum and Absolute Continuity

Suppose $\{\nu_i\}$ is a sequence of positive meaures. If $\nu_i \perp \mu$ for all i, then $\sum_i^\infty \nu_i \perp \mu$ and if $\nu_i \ll \mu$ for all i then $\sum_i^\infty \nu_i \ll \mu$

***Proof*:**
Let $\nu_i \perp \mu$ for all i, so that $X = A_i \coprod B_i$ where $\nu_i(A_i) = 0$ and $\mu(B_i) = 0$ for all i. Since $\mu(B_i) = 0$ for all i, then $\mu\left(\bigcup_i B_i\right) = 0$. Similarly, $\sum_i^n \nu_i\left(\bigcap_i A_i\right) = 0$ since $\cap_i A_i \subseteq A_i$ for each i. By some basic set theory

$$X = \left(\bigcap_i A_i\right) \coprod \left(\bigcup_i B_i\right) \qquad \left(\bigcap_i A_i\right) \cap \left(\bigcup_i B_i\right) = 0$$

since each x is in either A_i or B_i and $A_i \coprod B_i = X$ for each i. Therefore, we have:

$$\sum_i \nu_i \perp \mu$$

For the second statement, let's say $\nu_i \ll \mu$ for all i. This implies that:

$$\begin{aligned} \mu(E) = 0 &\Rightarrow \nu_1(E) = 0 \\ &\Rightarrow \nu_2(E) = 0 \\ &\vdots \\ &\Rightarrow \nu_n(E) = 0 \\ &\vdots \end{aligned}$$

meaning if $\mu(E) = 0$, then each $\nu_i(E) = 0$. But then $\sum_i \nu_i(E) = 0$, and so

$$\sum_i^\infty \nu_i \ll \mu$$

as we sought to show.

Theorem 3.2.8: Lebesgue-Radon-Nikodym Theorem

Let ν be a signed σ-finite measure on $(X, \mathcal{M})$ and μ be a positive σ-finite measure on $(X, \mathcal{M})$. Then, there exists unique σ-finite measure λ and ρ on $(X, \mathcal{M})$ such that

$$\lambda \perp \mu, \qquad \rho \ll \mu \quad \text{and} \quad \nu = \lambda + \rho$$

Furthermore, there is an extended μ-integrable function f such that $d\rho = f d\mu$, and any two such functions are equal μ-a.e. and so we get:

$$\nu(A) = \int_A f d\mu + \lambda(A)$$

In other words, λ and ρ "split" the measure μ based on where μ is zero or not , and together they can be combined to make ν. Notice that there is no prior relation between the signed measure ν and the positive measure μ. This means that we can replace μ with μ', and find $\lambda' \perp \mu'$ and $\rho' \ll \mu'$, and still get $\nu = \lambda' + \rho'$, and so this decomposition of a signed measure with respect to a measure μ heavily depends on μ. This decomposition might make more sense if we conider the special case where $\nu \ll \mu$, in which case we ar left with $\nu = \rho$. The last statement of the theorem tell's us that:

$$d\rho = f d\mu \qquad \text{which is shorthand for} \qquad \rho(E) = \int_E f d\mu$$

Notice that this looks like the change of varaibles we were talking about earlier. Thus, in this case, this theorem is ultimately about a "change of variables" into the μ measure!

Finally, notice that this "change of variables" show's that the measure ν is sort of the "integral" of the measure of f, and so in a sense f is thus the "derivative" of ν since you "get back" ν by integrating. In other words, this should start giving you Fundamental Theorem of Calculus where $d/dx(\int f)) = f$. However, be weary that the function f is not [always] obtained by taking the limit of some function F, i.e. does does not always exist a function F such that $f(a) = F'(a) = \lim_{x\to a} \frac{F(x+a)-F(a)}{x-a}$. However, in the next section, we will see that in some special cases we *can* define a function F (which we obtain through ν) where $F' = f$. Furthermore, this does not imply that we can compute real measures by defining a distribution F like in the case of Lebesgue measures (which is the other result of the FTC). Thus, the function f can be thought of as a sort of generalization of the notion of derivative.

***Proof*:**
We will break down this proof into 3 cases: when ν and μ are finite positive measures, when ν and μ are σ-finite positive meaures, and finally when ν is a signed measure.

For the first case, let:

$$\mathcal{F} = \left\{ f : X \to [0,\infty] \;\middle|\; \int_E f\,d\mu \leq \nu(E) = \int_E 1 d\nu \text{ for all } E \in \mathcal{M} \right\}$$

Remember that we can define a measure through the integral, so what we are asking for is if we can find all possible functions which can define us a measure that will be smaller than ν. We will end up taking the largest such function f, and labelling $d\rho = f d\mu$, and $d\lambda = d\nu - d\rho$. We can already see that $\rho \ll \mu$. What's left to show is that such an f exists, that $\lambda \perp \mu$ and that the result is unique.

Notice also how the construction can make you think of change of variables. Let's say for a moment that $\int_E f d\mu = \int_E 1 d\nu$. Then it is as though f is correcting the value so that the values are equal. This is actually what we are looking for, so we look for all f such that $\int_E f d\mu \leq \int_E 1 d\nu$ and take the supremum.

First, $\mathcal{F}$ is nonempty since $0 \in \mathcal{F}$. Thus, let $a = \sup \left\{ \int f\,d\mu \;\middle|\; f \in \mathcal{F} \right\}$ (notice that $a \leq \nu(X) < \infty$ since ν is finite), and choose a sequence $\{f_n\} \subseteq \mathcal{F}$ such that $\int f_n d\mu \to a$ and let $f = \sup_n f_n$. We would like to show that $a = \int f$. To that end, notice that if $f, g \in \mathcal{F}$, then $\max(f,g) \in \mathcal{F}$: if $A = \{x \mid f(x) > g(x)\}$, for any $E \in \mathcal{M}$:

$$\int_E \max(f,g) d\mu = \int_{E\cap A} f d\mu + \int_{E\setminus A} g d\mu \leq \nu(E\cap A) + \nu(E \setminus A) = \nu(E)$$

Let $g_n = \max(f_1, f_2, ..., f_n)$. Then $g_n \in \mathcal{F}$, g_n increases pointwise to f, and $\int g_n\,d\mu \geq \int f_n\,d\mu$ by monotonicity. Thus, since a is the supremum, it must be that $\lim \int g_n\,d\mu = a$, and so by the MCT

$$a = \lim \int g_n d\mu = \int \lim g_n d\mu = \int f$$

and so $f \in \mathcal{F}$ (i.e. the supremum is "achieved") and $\int f\,d\mu = a$. In particular, $f < \infty$ a.e., so we may take f to be real-valued everywhere. Let $d\rho = f d\mu$ (we will soon show this $d\rho$ satisfies the conditions of the theorem).

We know define the measure λ which is singular with respect to μ. Let $d\lambda = d\nu - f d\mu$ (i.e. $\lambda(E) = \nu(E) - \int_E f d\mu$). Then $d\lambda$ is positive, since $f \in \mathcal{F}$, and is singular with respect to

μ. It it weren't, then by lemma 3.2.6, there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\lambda \geq \epsilon\mu$ on E. Then $\epsilon\chi_E d\mu \leq d\lambda = d\nu = f d\mu$, or rearranging $(f + \epsilon\chi_E)d\mu \leq d\nu$. Since $f \in \mathcal{F}$ and the $\epsilon\chi_E$ only effects the value on E, we just showed that $f + \epsilon\chi_E \in \mathcal{F}$. However, $\int f + \epsilon\chi_E d\mu = a + \epsilon\mu(E) > a$, contradicting the maximality of a. Hence it must be that $\lambda \perp \mu$.

Therfore, we have that $d\nu = f d\mu + d\lambda$ where $\lambda \perp \mu$. Recalling we set $d\rho = f d\mu$, we get that $d\nu = d\rho + d\lambda$, and by definition of ρ we have $\rho \ll \mu$ (see definition 3.2.2). Converting into our more usual notation, we see that $\nu(E) = \rho(E) + \lambda(E)$ for all E, and so $\nu = \rho + \lambda$ with $\lambda \perp \mu$ and $\rho \ll \mu$. It remains to show that this decomposition is unique. To that end, let's say $d\nu = d\lambda' + f' d\mu$. Then $d\lambda - d\lambda' = (f - f')d\mu$. Then (show that $d\lambda - d\lambda' \perp \mu$) and $(f' - f)d\mu \ll d\mu$, and since both sides must be equal, this forces that $d\lambda - d\lambda' = (f' - f)d\mu = 0$. Thus, $\lambda = \lambda'$, and by proposition 2.3.4, $f = f'$ μ-a.e. Thus, if ν and μ are of finite measure, the theorem is done.

Now let ν and μ be σ-finite measures so that X is a countable union of μ-finite sets and a countable union of ν-finite sets. By refining the two by taking intersections, we get a new collection $\{A_i\} \subseteq \mathcal{M}$ $\mu(A_i)$ and $\nu(A_i)$ is finite for all i and $X = \bigcup_i^\infty A_i$. We can define

$$\mu_i(E) = \mu(E \cap A_i) \qquad \nu_i(E) = \nu(E \cap A_i)$$

Then by what we've just shown, we have the decomposition:

$$d\nu_i = d\lambda_i - f_i d\mu$$

where $\lambda_i \perp \mu_i$. Now $\mu_i(A_i^c) = \nu_i(A_i^c) = 0$, so $\int_{A_i^c} f d\mu = 0$ and

$$\lambda_i(A_i^c) = \nu_i(A_i^c) - \int_{A_i^c} f d\mu = 0$$

for our construciton, let f_i be 0 on A_i^c. Now, let $\lambda = \sum_{i=1}^\infty \lambda_i$ and $f = \sum_{i=1}^\infty f_i$. Then by lemma 3.2.7:

$$d\nu = d\lambda - f d\mu$$

and $d\lambda$ and $f d\mu$ are σ-finite, as desired.

Finally, taking the case of ν being a signed measure, using the Jordan decomposition, apply the preceeding argument to ν^+ and $\nu-$, compelting the proof

Since the decomposition exists, it is given a name:

Definition 3.2.9: Lebesgue Decomposition

Let ν be a signed σ-finite measure and μ a positive σ-finite measure. Then the decomposition

$$\nu = \lambda + \rho \qquad \lambda \perp \mu \quad \rho \ll \mu$$

is called the *Lebesgue decomposition of ν with respect to μ.*

if $\nu \ll \mu$, then the Lebesgue-Radon-Nikodym theorem tell's us that $d\nu = f d\mu$. If the extra condition that $\nu \ll \mu$ is added, the theorem is often called *Radon-Nikodym's Theorem*, and f is given a special name:

Definition 3.2.10: Radon-Nikodym Derivative

Let $\nu \ll \mu$ and f be the function obtain from the Lebesgue Decomposition such that $d\nu = f d\mu$. Then f is called the *Radon-Nikodym derivative of ν with respect to μ*. We often denote f by:

$$\frac{d\nu}{d\mu} := f$$

As a consequence of this notation, we get the following representation of $\nu(E) = \rho(E)$ that looks like a fraction cancellation:

$$d\nu = \frac{d\nu}{d\mu} d\nu$$

Strictly speaking, $\frac{d\nu}{d\mu}$ should really be the class of function's that are equal to f μ-a.e., however just like with L^1, we often abuse notation and pick some function in this class when we use this notation). Most of the time, we will be working with the case of $\nu \ll \mu$. In Distribution theory, we will be studying the case where $\nu \not\ll \mu$.

Writing f in this form is very fitting, as in f respects the same properties as the derivative! Once we reach the special case of the measures being Lebesgue measures and the functions being differentiable functions, then f will in fact be the derivative of f.

Without yet doing concrete computations of f, if we take the Radon-Nikodym of $\nu_1 + \nu_2$ where $\nu_1 + \nu_1 \ll \mu$ then we have

$$\frac{d(\nu_1 + \nu_2)}{d\mu} = \frac{d\nu_1}{d\mu} + \frac{d\nu_1}{d\mu}$$

to see this, let $d(\nu_1 + \nu_2)/d\mu = f$. Since $\nu_1 + \nu_2 \ll \mu$, $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$. Then taking $d\nu_1 = f_1 d\mu$ and $d\nu_2 = f_2 d\mu$, then:

$$\begin{aligned}
&= \int \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} d\mu \\
&= \int \frac{d\nu_1}{d\mu} d\mu + \int \frac{d\nu_2}{d\mu} d\mu \\
&= \int 1 d\nu_1 + \int 1 d\nu_2 \\
&= \nu_1(E) + \nu_2(E) \\
&= (\nu_1 + \nu_2)(E) \\
&= \int 1 d(\nu_1 + \nu_2) \\
&= \int_E \frac{d(\nu_1 + \nu_2)}{d\mu} d\mu
\end{aligned}$$

and since the two integrals are equal over every measurable set, we get:

$$\frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} = \frac{d(\nu_1 + \nu_2)}{d\mu}$$

which is the same property of differentiable functions. In particular, notice that similarly we have the following chain-rule property:

Proposition 3.2.11: Chain Rule For Measures

Let ν be a signed σ-finite measure on $(X, \mathcal{M})$, μ and λ positive σ-finite measures on $(X, \mathcal{M})$ such that $\nu \ll \mu$ and $\mu \ll \lambda$. Then:

1. If $g \in L^1(\nu)$, then $g(d\nu/d\mu) \in L^1(\mu)$ and
$$\int g d\nu = \int g(d\nu/d\mu) d\mu$$
2. Absolute continuity, $\ll$, is transitive: $\nu \ll \lambda$ and
$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda}$$

***Proof*:**

1. This proof uses (what now should be) the standard measure-theory way to generalize results. First, we can consider ν^+ and ν^- seperately, so without loss of generality let's say $\nu \geq 0$. If we let $g = \chi_E$, then:
$$\int g \frac{d\nu}{d\mu} d\mu = \int \int \chi_E f d\mu = \int_E f d\mu$$
which is true by definition of f. Thus, it is true for any simple functions since the integral and the construction is linear, and so it is also true for non-negative function's by the MCT, and so it is true for functions in $L^1(\nu)$ by linearity again.
2. If we take ν to be μ and μ to be λ and set $g = \chi_E \frac{d\nu}{d\mu}$, we get:
$$\nu(E) = \int_E \frac{d\nu}{d\mu} d\mu = \int_E \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} d\lambda$$
for all $E \in \mathcal{M}$, including X, and so by proposition 2.3.4
$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} \quad \lambda\text{-a.e.}$$
as we sought to show

As a consequence, if $\mu \ll \lambda$ and $\lambda \ll \mu$, then $(d\lambda/d\mu)(d\mu/d\lambda) = 1$ λ-a.e. and μ-a.e. Note that all of this is for the special case of $\nu \ll \mu$. If this is not the case (say $\nu \ll \mu$) we get a different notion of "$d\nu/d\mu$ (the example of $\delta_0 \perp \lambda$ has a "Radon-Nikodym derivative" of the dirac delta function).

Note that σ-finiteness is necessary:

Example 3.2.12: Withiout σ-Finiteness

Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, m the Lebesgue measure and μ the counting measure on $\mathcal{M}$

1. $m \ll \mu$ but $dm \neq f d\mu$ for any f

Proof. First, if $\mu(E) = 0$, then $|E| = 0$, which we have shown that this implies $m(E) = 0$, and so $m \ll \mu$.

However, there does not exist an f such that $m(E) = \int_E f d\mu$. To see this, consider any $p \in [0,1]$. Then

$$0 = m(\{p\}) = \int_{\{p\}} f d\mu = \int_{[0,1]} \chi_{\{p\}} f d\mu \stackrel{!}{=} f(p)\mu(\{p\}) = f(p)$$

where $\stackrel{!}{=}$ comes from the fact that $\chi_{\{p\}} f$ is in fact a simple funciton (being of the form $\sum_i c_i \chi_{E_i}$ with $i = 1$, $E_i = \{p\}$ and $c_i = f(p)$), and so matches the definition of the integral for simple functions (which we have shown in class is the value of the usual integl). Hence, we have that $f(p) = 0$ for all $p \in [0,1]$. However, this implies that $f \equiv 0$. This leads to a contardiction, since:

$$1 = m([0,1]) = \int_{[0,1]} f d\mu = \int_{[0,1]} 0 d\mu = 0$$

but $1 \neq 0$ □

Next, show that μ has no Lebesgue decomposition with respect to m

Proof. For the sake of contradiction, let $\mu = \lambda + \rho$ where $\lambda \perp m$ and $\rho \ll m$. Then for any $c \in [0,1]$

$$\mu(\{c\}) = \lambda(\{c\}) = 1$$

since $m(\{c\}) = 0$ and $\rho \ll m$, $\rho(\{c\}) = 0$. But then, λ is never null on non-zero sets by monotonicity, implying that m must *always* be null. However, this is not true since m is the Lebesgue measure – a contradiction. Thus, μ does not have such a decomposition with respect to m. □

Before we move on to our last result of the section, I want to take a moment to talk about generalizations of the Radon-Nikodym theorem. In the theorem, we assumed that the measures must be σ-finite. However, in contrast to previous example where without σ-finiteness the theorem fails badly, we seem to have these assumption because the generalization requires more build-up. In particular, to fix the lack of control of σ-finiteness, we would have to define the notion of being *truly continuous*. If μ is σ-finite, being absolutely continuous and truly continuous are equivalent. The statement of the Radon-Nikodym can be reformulated in terms of truly continuous measures, and I found a stack-exchange post on this:

https://mathoverflow.net/questions/258936/radon-nikodym-theorem-for-non-sigma-finite-measures

Finally, we conclude with this final proposition we'll use later on:

Proposition 3.2.13

Let $\mu_1, ..., \mu_n$ be measures on $(X, \mathcal{M})$. Then there exists a measure μ (namely $\sum_i \mu_i$) such that $\mu_i \ll \mu$

***Proof*:**
exercise

3.3 Complex Measure

We know show how we can generalize the results when the codomain is $\mathbb{C}$. The reason to define complex measure is generalize the notion of the Radon-Nikodym derivative to the complex case so that we may define complex results. Most of the work we have done before translates one-to-one for the complex measures, with the exception of defining the absolute variation which we will spend most of this section covering.

Definition 3.3.1: Complex Measure

Let $(X, \mathcal{M})$ be a measure space. A function $\nu : \mathcal{M} \to \mathbb{C}$ is called a *complex measure* if

1. $\nu(\emptyset) = 0$
2. If $\{E_i\} \subseteq \mathcal{M}$ is a pair-wise disjoint sequence, then if $\sum_i^\infty \nu(E_i)$ converges absolutely, then

$$\nu\left(\bigcup_i^\infty E_i\right) = \sum_i^\infty \nu(E_i)$$

Just like for signed measures, the absolute convergence is so that the order in which we measure the sets does not matter.

Notice that the sequence $X, \emptyset, \emptyset, \ldots$ is a valid disjoint sequence, and so $\sum_i \nu(X_i) = \nu(X)$ must converge absolutely, and so ν must be a finite measure. Thus, a measure is a complex measure if it is finite. Another way of saying it is that if μ is a positive measure and $f \in L^1(\mu)$ (recall f is a complex function), then $fd\mu$ is also a complex measure. As is usually done with complex valued function, we define the signed measures $\nu_r(E) = \pi_r(\nu(E))$ and $\nu_i(E) = \frac{\pi_i(\nu(E))}{i}$ to take on the real and complex values of ν, so ν_i and ν_r are *finite* signed measures and $\nu = \nu_r + i\nu_i$. As a consequence, the range of ν is a bounded subset of $\mathbb{C}$.

Many of the previous concepts we've seen naturally generalize to the complex case:

1. We define $L^1(\nu)$ to be $L^1(\nu_r) \cap L^1(\nu_i)$
2. For $f \in L^1(\nu)$, set $\int f d\nu := \int f d\nu_r + i \int f d\nu_i$
3. If ν and μ are complex measures, then we that $\mu \perp \nu$ if $\nu_a \perp \nu_b$ for all $a, b \in \{r, i\}$.
4. If λ is a positive measure, we say $\nu \ll \lambda$ if $\nu_r \ll \lambda$ and $\nu_i \ll \lambda$
5. The Lebesgue-Radon-Nikodym Theroem generlizes to complex measures nicely; you would apply theorem 3.2.8 to the real and imaginary part seperately. To make sure there is no ambiguity in understanding the generalization:

   Let ν be a complex measure and μ a σ-finite measure on $(X, \mathcal{M})$. Then there exists a complex measure λ and a $f \in L^1(\mu)$ such that

$$d\nu = d\lambda + fd\mu \qquad \lambda \perp \mu$$

Furthermore, if there existed another λ' such that $\lambda' \perp \mu$ and $d\nu = d\lambda' + f' d\mu$, then $\lambda = \lambda'$ μ-a.e. and $f = f'$ μ-a.e.
If $\nu \ll \mu$, then we denote f a $\frac{d\nu}{d\mu}$

We next want to generalize the notion of *total variation* to the complex case. In the signed measure case, this was an easy task since $\mathbb{R}$ has an inherent order. In the complex case, this is not so straight forward: Given that a surface $S \subseteq \mathbb{C}$ is the image of some complex measure $\nu(E) = S$, what does it mean to capture the "total-variation" on that surface? There is more than one sensible answer to this question, namely since there is no "natural order" to $\mathbb{C}$, and so it would be nice to have succint way to capture it's *total variation*. There is in fact a nice and concise way of defining this which also generalizes the previous way of looking at total variation that captures the idea in 1 formula, and so we will build-up to it.

First, let ν be a complex measure and let $\mu = |\nu_r| + |\nu_i|$. Next, notice that $\nu \ll \mu$, since if $\mu(E) = 0$, then it must be that both $|\nu_r|$ and $|\nu_i|$ are zero, and so $\nu_r(E) = \nu_i(E) = 0$, and so $\nu(E) = 0$. Therefore, the Lebesgue decomposition of ν with respect to μ gives us a f such that $d\nu = f d\mu$. As an example of such an f, let's say that ν was simply a finite signed measure (which is a special case of a complex measure), so $\mu = |\nu|$. Then $f = (\chi_P - \chi_N)$ and:

$$d\nu = (\chi_P - \chi_N)d\mu = (\chi_P - \chi_N)d|\nu|$$

Which is the Jordan decomposition theorem we have shown earlier! Now, notice that $|\chi_P - \chi_N| = 1$ so that we can write:

$$d|\nu| = |\chi_P - \chi_N| d\mu \quad \text{which is shorthand for} \quad |\nu|(E) = \int_E |\chi_P - \chi_N| d\mu$$

If we re-write the last equation as $d|\nu| = |f|d\mu$, then we have that the measure $|\nu|$ is defiend by the absolute value of the measure defined by the derivative with respect to $|\nu|$. If we come back to the case of ν being a general complex measure, then we would get $\mu = |\nu_r| + |\nu_i|$, giving us a less obvious, but certainly existent decomposition $d|\nu| = |f|d\mu$ (i.e. $|\nu|(E) = \int_E |f|d\mu$). Thus, it seems that this property somehow "captures" the idea of total variation. We chose a particular μ for both decomposition, however it turns out that this property we have stated is independent of choice of μ up to an appropriate zero-measure set, and so we will define total-variation with respect to the following property:

Definition 3.3.2: Total Variation And Complex Differentiation

Let ν be a complex measure. Then $|\nu|$ is defined such that when $d\nu = f d\mu$ where μ is a positive measure, then $d|\nu| = |f|d\mu$

What we have shown with the μ we saw earlier is that $d\nu$ is always decomposeable into $d\nu = f d\mu$. We must now show that $|\nu|$ is in-fact independent of choice of μ (i.e. any choice will get the same result up zero an appropriate zero-measure set) given μ has the property of the definition. Let's say $d\nu = f_1 d\mu_1 = f_2 d\mu_2$ satisfy the total-variation property. Let $\rho = \mu_1 + \mu_2$. By proposition 3.2.11, we have that:

$$f_1 \frac{d\mu_1}{d\rho} d\rho = d\nu = f_2 \frac{d\mu_2}{d\rho} d\rho$$

Thus $f_1 \frac{d\mu_1}{d\rho} = f_2 \frac{d\mu_2}{d\rho}$ ρ-a.e. Since $\frac{d\mu_1}{d\rho}$ and $\frac{d\mu_2}{d\rho}$ are both non-negative, we have:

$$|f_1| \frac{d\mu_1}{d\rho} d\rho = \left| f_1 \frac{d\mu_1}{d\rho} d\rho \right| = \left| f_2 \frac{d\mu_2}{d\rho} d\rho \right| = |f_2| \frac{d\mu_2}{d\rho} d\rho \quad \rho\text{-a.e.}$$

Thus:

$$|f_1|d\mu_1 = |f_1|\frac{d\mu_1}{d\rho}d\rho = |f_2|\frac{d\mu_2}{d\rho}d\rho = |f_2|d\mu_2$$

and so, ν is independent of choice of μ. We next give some useful properties of total variation:

Proposition 3.3.3: Properties Of Total Variation

Let $(X, \mathcal{M}, \nu)$ be a complex measure space.

1. $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$
2. $\nu \ll |\nu|$ and $\frac{d\nu}{d|\nu|}$ has absolute value 1 $|\nu|$-a.e.
3. $L^1(\nu) = L^1(|\nu|)$ and if $f \in L^1(\nu)$, then $|\int f d\nu| \leq \int |f| d|\nu|$

***Proof*:**
Pick f such that $d\nu = f d\mu$ (as we've shown exists when defining $|\nu|$). Then:

$$|\nu(E)| = \left|\int_E f d\mu\right| \leq \int_E |f| d\mu = |\nu|(E)$$

showing the first claim, and also showing that $\nu \ll |\nu|$. For the next claim, let $g = \frac{d\nu}{d|\nu|}$ so

$$f d\mu = d\nu = g d|\nu|$$

so $g|f| = f$ μ-a.e. and so $|\nu|$-a.e. Since $|f| > 0$ $|\nu|$-a.e. as well, it must be that $g = 1$ $|\nu|$-a.e.
Next, (left as an exercise by Folland)

The next property we introduce shows that it interacts well with the triangle inequality

Proposition 3.3.4: Triangle Inequality Of Total Variation

Let ν_1 and ν_2 be complex measures on $(X, \mathcal{M})$. Then

$$|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$$

***Proof*:**
By proposition 3.2.13, we can write $\nu_1 = f_1 d\mu$ and $\nu_2 = f_2 d\mu$ for the same μ. Then:

$$d|\nu_1 + \nu_2| = |f_1 + f_2| d\mu \leq |f_1| d\mu + |f_2| d\mu = d|\nu_1| + d|\nu_2|$$

as we sought to show

Finally, though this characterization of total variation is a good definition, it can be hard to work with. The following gives an equivalent definition of total variation that is easier computationally:

Proposition 3.3.5: Equivalence Of Total Variation

Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$, define

1. $\mu_1(E) = \sup\{\sum_{i=1}^{n} |\nu(E_i)| \mid n \in \mathbb{N},\ E_1, ..., E_n \text{ disjiont},\ E = \coprod_{i=1}^{n} E_i\}$
2. $\mu_2(E) = \sup\{\sum_{i=1}^{\infty} |\nu(E_i)| \mid E_1, ... \text{ disjiont},\ E = \coprod_{i=1}^{\infty} E_i\}$
3. $\mu_3(E) = \sup\left\{\left|\int_E f d\nu\right| \mid |f| \leq 1\right\}$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$

***Proof*:**
Throughout this proof, let:

$$S_1^E = \left\{\sum_{i=1}^{n} |\nu(E_i)| \;\middle|\; n \in \mathbb{N},\ E_1, ..., E_n \text{ disjiont},\ E = \coprod_{i=1}^{n} E_i\right\}$$

and

$$S_2^E = \left\{\sum_{i=1}^{\infty} |\nu(E_i)| \;\middle|\; E_1, ... \text{ disjiont},\ E = \coprod_{i=1}^{\infty} E_i\right\}$$

($\mu_1 \leq \mu_2$) Notice that $S_1^E \subseteq S_2^E$ (since for any finite disjoint sequence, we can extend it to an infinite disjoint sequence by letting $E_i = \emptyset$ for the rest of the terms), we get $\sup S_1^E \leq \sup S_2^E$, and since this is true for all E, $\mu_1 \leq \mu_2$

($\mu_2 \leq \mu_3$) In the process we will use the fact that there exists an f such that:

$$d\nu = f d|\nu|$$

By proposition 3.3.3, $\left|\frac{d\nu}{d|\nu|}\right| = |f| = 1$ $|\nu|$-a,e, and so $|f| \leq 1$ $|\nu|$-a.e. For the $|\nu|$-zero-measure points that are not less than or equal to 1, we can simply choose them to be equal to 0. This also implies that $|\overline{f}| \leq 1$. With this information we get that:

$$\overline{f} d\nu = \overline{f} f d|\nu| = |f|^2 d|\nu| = 1 d|\nu| = d|\nu| \quad |\nu|\text{-a.e.}$$

and so:

$$\begin{aligned}\sum_{i}^{\infty} |\nu(E_i)| &\leq \sum_{i}^{\infty} |\nu|(E_i) \\ &= \int_E d|\nu| \\ &= \int_E \overline{f} d\nu \\ &\leq \left|\int_E \overline{f} d\nu\right|\end{aligned}$$

and since $|f| \leq 1$, this is an element of $\left\{\left|\int_E f d\nu\right| \mid |f| \leq 1\right\}$, and since this was true for arbitrary E, we get: $\mu_2 \leq \mu_3$

($\mu_3 = |\nu|$) We'll first show $\mu_3 \leq |\nu|$. By the definition of the supremum, for all $\epsilon > 0$, there exists an f where $|f| \leq 1$ such that

$$\begin{aligned}
\mu_3 &\leq \left|\int_E f d\nu\right| + \epsilon \\
&\leq \int_E |f| d|\nu| + \epsilon && \text{prop. 3.13[c]} \\
&\leq \int_E d|\nu| + \epsilon && |f| \leq 1 \\
&\leq |\nu|(E) + \epsilon
\end{aligned}$$

and as $\epsilon \to 0$, we have that $\mu_3 \leq |\nu|(E)$. Going the other way, we can again define $\frac{d\nu}{d|\nu|}$, which by proposition 3.13 its absolute value is 1 $|\nu|$-a.e. Let $\overline{f} = \frac{d\nu}{d|\nu|}$ so that $1 = |f|^2 = f\overline{f}$ $|\nu|$-a.e. If f ever has value greater than 1, we can simply redefine them to be 0 since those value have measure 0. Then:

$$\begin{aligned}
|\nu|(E) &= \int_E d|\nu| \\
&= \int_E 1 d|\nu| \\
&= \int_E |f|^2 d|\nu| \\
&= \int_E |f| d|\nu| && \text{since } |f| = 1 \ |\nu|\text{-a.e.} \\
&= \int_E f\overline{f} d|\nu| \\
&= \int_E f \frac{d\nu}{d|\nu|} d|\nu| \\
&= \int_E f d\nu && \text{prop. 3.9} \\
&\leq \left|\int_E f d\nu\right|
\end{aligned}$$

and since $|f| \leq 1$,

$$\left|\int_E f d\nu\right| \in \left\{ \left|\int_E f d\nu\right| \;\middle|\; |f| \leq 1 \right\}$$

and since this is true for arbitrary E:

$$|\nu| \leq \mu_3$$

and so, combining the two inequalities, we get $\mu_3 = |\nu|$

($\mu_3 \leq \mu_1$) We will do as the hint suggets and use simple functions to approximate f, in particular we will use the standard representation from theorem 2.10, let's label the increasing sequence $\{\varphi_k\}_{k=1}^{\infty}$. To take advantage of these functions, first break up ν into positive meaures, $\nu_r^+, \nu_i^+, \nu_r^-, \nu_i^-$ so that:

$$\left|\int_E f d\nu\right| = \left|\int_E f d\nu_r^+ - \int_E d\nu_r^- + i\left(\int_E f d\nu_i^+ - \int_E f d\nu_i^-\right)\right|$$

Then by the definition of integral, for all $\epsilon/4 > 0$, there must exists an $n \in \mathbb{N}$ such that

$$\begin{aligned}
&= \left|\int_E f d\nu_r^+ - \int_E f d\nu_r^- + i\left(\int_E f d\nu_i^+ - \int_e f d\nu_i^-\right)\right| \\
&\leq \left|\int_E \varphi_n d\nu_r^+ + \frac{\epsilon}{4} - \int_E \varphi_n d\nu_r^- - \frac{\epsilon}{4} + i\left(\int_E \varphi_n d\nu_i^+ + \frac{\epsilon}{4} - \int_e \varphi_n d\nu_i^- - \frac{\epsilon}{4}\right)\right| \\
&\leq \left|\int_E \varphi_n d\nu_r^+ - \int_E \varphi_n d\nu_r^- + i\left(\int_E \varphi_n d\nu_i^+ - \int_E \varphi_n d\nu_i^-\right)\right| + \epsilon \\
&= \left|\int_E \varphi_n d\nu\right| + \epsilon
\end{aligned}$$

Then by the definition of a simple function, $\varphi_n = \sum_{i=1}^n c_i \chi_{F_i}$ for appropriate F_i and c_i. Note that $c_i \leq 1$ since $|\varphi_n| \leq |f| \leq 1$. Also, note that $\chi_E \chi_{F_i} = \chi_{E \cap F_i}$. Then by definition of an integral:

$$\left|\int_E \varphi_n d\nu\right| + \epsilon = \left|\sum_{i=1}^n c_i \nu(F_i \cap E)\right| + \epsilon \leq \sum_{i=1}^n |c_i| \, |\nu(F_i \cap E)| + \epsilon$$

This is almost the result we need. The problem is that the collection $\{F_i \cap E\}$ is not guaranteed to cover E since there might be portions that are yet to be "precise" enough to bring it into the approximation. Since we only need to get bigger, we can solve this by adding $\bigcap_{i=1}^n F_i^c \cap E$, which is disjoint of every $F_i \cap E$, and so:

$$\begin{aligned}
\sum_{i=1}^n |c_i| \, |\nu(F_i \cap E)| + \epsilon &\leq \sum_{i=1}^n |c_i| \, |\nu(F_i \cap E)| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon \\
&\leq \sum_{i=1}^n |\nu(F_i \cap E)| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon
\end{aligned}$$

and so, combining all the inqualities, we get:

$$\left|\int_E f d\nu\right| \leq \sum_{i=1}^n |\nu(F_i \cap E)| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon$$

and so, by the definition of the supremum:

$$\left|\int_E f d\nu\right| \leq \sup S_1^E$$

and since this is true for arbitrary $\left|\int_E f d\nu\right|$:

$$\mu_3 \leq \mu_1$$

and so, we've established $\mu_1 \leq \mu_2 \leq \mu_3 [= |\nu|] \leq \mu_1$, and so they are all equal, as we sought to show

3.4 Differentiation of Euclidean Space

We now analyze the special case where $\mu = m$ is the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Since we are working over the Borel sets of $\mathbb{R}^n$, we have a metric available to us, and so we can define the pointwise derivative of ν with respect to m at x:

Definition 3.4.1: Pointwise Derivative

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, m)$ be our measure space, ν a signed or complex measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$, and $x \in \mathbb{R}^n$. Then if $B_r(x)$ is a ball of radius r around x, then if the following limit exists:

$$\lim_{r\to 0} \frac{\nu(B_r(x))}{m(B_r(x))} = F(x)$$

then we say that $F(x)$ is the *pointwise derivative of ν with respect to m at x*

We can replace balls of radius x with sets that shrink suitably quickly in some "regular" way. This will be advantageous to us later, since it will turn out we can make these "nicely shrinking" sets to be $(x, x+h)$ (or the appropriate generalisation in the $\mathbb{R}^n$ case) which will give us

$$\lim_{r\to 0} \frac{\nu(x+h)}{h}$$

notice that if we define ν in terms of a distribution as we've done for the outer measure in section 1.3, say G is our distribution, then our numerator becomes $G(x+h) - G(x)$, so in this case the definition *exactly* matches be the definition of the derivative in one dimension:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \overset{\text{F.t.C.}}{=} \frac{1}{h}\int_x^{x+h} f'(x)dx$$

We will get back to this important case after some more examination of the general case.

If $\nu \ll m$ so that $d\nu = f dm$, then $\nu(B_r(x))/m(B_r(x)) = \frac{1}{m(B_r(x))}\int_{B_r(x)} f d\mu$ is the average value of f on $B_r(x)$ (recall the 1-dimensional intuition of $\frac{1}{b-a}\int_a^b f(x)dx$). Since $r \to 0$, we can hope that $F = f$ m-a.e. (since the averaging is getting more precise with the smaller range), telling us that we can recover a density F with information about f. This will indeed be the case if $\nu(B_r(x))$ is finite for all r, x; this will be what we explore in this section. Essentially, from the point of view of f, what we're showing is that the derivative of the integral of f is f, i.e., the Fundamental Theorem of Calculus (FTC). For the rest of this section, the term integrable and a.e. is with respect to the Lebesgue measure m.

We start with an important technical lemma:

Lemma 3.4.2: build-up lemma I

Let $\mathcal{C}$ be a collection of open balls in $\mathbb{R}^n$ and let $U = \bigcup_{B\in\mathcal{C}} B$. Then for any $c \in \mathbb{R}$ such that $c < m(U)$, there exists a fininte disjoint subset of $\mathcal{C}$, $B_1, ..., B_k$, such that

$$\sum_{i=1}^k m(B_i) > \frac{c}{3^n}$$

In other words, for any value c less than $m(U)$, we can show that we only need to pick a finite number of balls from $\mathcal{C}$ and still keep the inequality, up to a correction factor of 3^{-n} where n is the dimension of our space, $\mathbb{R}^n$, and 3 comes form making sure the radius of the balls will be big enough to cover what we need.

***Proof*:**
We essentially sneakily get compactness by using theorem 2.6.1[1]. Using this theorem, we know that there exists a compact set $K \subseteq U$ such that $c < \mu(K)$, so there exists a finite subcover of balls from $\mathcal{C}$, say $A = \{A_1, ..., A_m\}$, that cover K. Since balls have finite radius and there are finitely many balls, let B_1 be the ball with largest radius from A. Then let B_2 be the largest ball from A that are disjoint from B_1. Continuing until we have exhausted all possibilites, we get a collection of balls $B = \{B_1, ..., B_k\}$. If there is an A_i that is not in B, then there must be a B_j such that $A_i \cap B_j \neq \emptyset$, and if j is the smallest integer with this property, then by our construction A_i can have radius that is at most that of B_j (or else we would have picked A_i instead of B_j in our construction). Thus, we can take a ball B_j^* whose radius is 3 times that of B_j and have the same center as B_i (i.e., they are concentric) so that $A_i \subseteq B_j^*$. Then by this construction, $K \subseteq \bigcup_j^k B_j^*$, and so

$$c < \mu(K) \leq \sum_j^k m(B_j^*) = 3^n \sum_j^k m(B_j)$$

and so $\frac{c}{c^n} \leq \sum_j^k m(B_j)$, as we sought to show

With this lemma ready to be referenced, we define another important concept that we will use

Definition 3.4.3: Locally Integrable

Let $f : \mathbb{R}^n \to \mathbb{C}$ be a function. Then we say that f is *locally integrable* (with respect to the Lebesgue measure) if:

$$\forall K \in \mathcal{B}_{\mathbb{R}^n}, m(K) < \infty, \int_K |f(x)|dx < \infty$$

and is denoted L^1_{loc}

Definition 3.4.4: Average

Let $f \in L^1_{\text{loc}}$. Then we define the average $A_r f(x)$ on $B_r(x)$ to be

$$A_r f(x) = \frac{1}{B_r(x)} \int_{B_r(x)} f(y)dy$$

This is simply the observation we made in the previous paragraph but made into a definition. We next prove that varying r or x by a bit will only change $A_r f(x)$ by a bit, i.e., $A_r f(x)$ is continuous:

Lemma 3.4.5: Build-Up Lemma II

Let $f \in L^1_{\text{loc}}$. Then $A_r f(x)$ is continuous on both r $(r > 0)$ and x $(x \in \mathbb{R}^n)$

Notice that f certainly need not be continuous, while $1/m(B_r(x))$ is clearly continuous. It is the $\int$ operation that is forcing this result to be continuous.

***Proof*:**
First, taking advantage that we're working over the Lebesgue measure in $\mathbb{R}^n$, so that if $c = B_1(0)$, then $cr^n = B_r(x)$ and $m(\partial B_r(x)) = 0$ (where $\partial B_r(x) = \{y \mid \|x - y\| = r\}$). This will come back soon.

To actually prove continuity, we'll show that $A_{\lim_{r\to r_0}} f(x) = \lim_{r\to r_0} A_r f(x)$ and $A_r f(\lim_{x\to x_0}) = \lim_{x\to x_0} A_r(x)$. To that end, take a sequene $r \to r_0$ and $x \to x_0$, and notice that the $\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ pointwise on $\mathbb{R}^n \setminus \partial B_{r_0}(x_0)$ (imagine $x = x_0$ is fixed and the radius goes to r_0 by oscilating around r_0 so that the pointwise limit doesn't exist for the border). Thus, $\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ a.e., and so we have that $\chi_{B_{r_0+1}(x_0)}$ dominates the $\chi_{B_r(x)}$ if $r < r_0 + 1/2$ and $|x - x_0| < 1/2$. Thus, by the Dominated Converting Theorem:

$$\begin{aligned}
A_{\lim\limits_{r\to r_0} r} f(\lim_{x_0\to x} x) &= \int_{B_{\lim\limits_{r\to r_0} r}(\lim\limits_{x\to x_0} x)} f(y)dy \\
&= \int_{B_r(x)} \lim_{r,x} f(y)dy \\
&= \lim_{r,x} \int_{B_r(x)} f(y)dy && \text{DCT}
\end{aligned}$$

and so $\int_{B_r(x)} f(y)dy$ is continuous. Since $B_r(x) = cr^n$ and r^n is a continuous funciton, and the product of two continuous function is continuous, we have that

$$A_r f(x) = \frac{1}{cr^n} \int_{B_r(x)} f(y)dy$$

is also continuous, as we sought to show

Since $A_r f$ is continuous, it is measurable. Next, we can define the largest possible average given a point x:

Definition 3.4.6: Hardy-Littlewood Maximal Function

Let $f \in L^1_{\text{loc}}$. Then the *Hardy-Littlewood Maximal function* is defined as

$$Hf(x) := \sup_{r>0} A_r|f|(x) = \sup_{r>0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y)|dy$$

Note that Hf is measurable since $(Hf)^{-1} = \bigcup_{r>0}(A - r|f|)^{-1}((a, \infty))$ is open for any $a \in \mathbb{R}$ by the previous lemma. What the next theorem shows us is that if we fix some number $\alpha \in \mathbb{R}_{>0}$, and try to take the measure of all points where $Hf(x) > \alpha$ (i.e., we are measuring the area of the largest possible average over an area of points with that larger average), then this value is in fact bounded by $\int |f(x)|$ provided an appropraite constant C/α:

Theorem 3.4.7: The Maximal Theorem

Let $(\mathbb{R}^n, \mathcal{L}, m)$ be a measure space. There exists a constant C such that for all $f \in L^1$ and all $\alpha > 0$

$$m(\{x \mid Hf(x) > \alpha\}) \leq \frac{C}{\alpha} \int |f(x)| dx$$

***Proof*:**
Let $E_\alpha = \{x \mid Hf(x) > \alpha\}$. By suprmum property of $Hf(x)$, there exists a $r_x > 0$ such that

$$\frac{1}{m(B_{r_x}(x))} \int_{B_{r_x}(x)} |f(y)|dy = A_{r_x}|f|(x) > \alpha \quad \Leftrightarrow \quad m(B_{r_x}(x)) < \frac{1}{\alpha} \int_{B_{r_x}(x)} |f(y)|dy \qquad (3.1)$$

Clearly, the balls $B_{r_x}(x)$ cover E_α. So for any $c < m(E_\alpha)$, by lemma 3.4.2, there exists a finite collection $r_{x_1}, ..., r_{x_k} \in E_\alpha$ such that

$$\sum_{i=1}^{n} m(B_{r_{x_j}}(x_j)) > \frac{c}{3^n}$$

and so, combining the inequalities we have and moving values around, we get:

$$c < 3^n \sum_{i=1}^{n} m(B_{r_{x_j}}(x_j)) \overset{\text{eq. } 3.1}{\leq} \frac{3^n}{\alpha} \int_{B_{r_{x_j}}} |f(y)|dy \leq \frac{3^n}{\alpha} \int_{\mathbb{R}^n} |f(y)dy|$$

Since this is true for all c, letting $c \to m(E_\alpha)$, we get the desired result, as we sought to show.

With all this build-up, we are ready to show the result we stated at the beginning, that $A_r f(x) \to f(x)$ a.e.

Before stating the theorem, it is useful to define the notion of limit superior for real-valued functions:

Definition 3.4.8: Limit Superior

Let $\varphi : \mathbb{R} \to \mathbb{R}$. Then:

$$\limsup_{r \to R} \varphi(r) := \lim_{\epsilon \to 0} \sup_{0 < |r-R| < \epsilon} \varphi(r) = \inf_{\epsilon > 0} \sup_{0|r-R| < \epsilon} \varphi(r)$$

As an exercise, show that:

$$\lim_{r \to R} \varphi(r) = c \quad \text{iff} \quad \limsup_{r \to R} |\varphi(r) - c| = 0$$

Theorem 3.4.9: Almost Lebesgue Differentiation Thoerem

If $f \in L^1_{\text{loc}}$. Then

$$\lim_{r \to 0} A_r f(x) = f(x)$$

for a.e. $x \in \mathbb{R}^n$. Phrased another way:

$$\lim_{r \to 0} \frac{1}{mB_r(x)} \int_{B_r(x)} f(x) - f(y)dy = 0$$

for a.e. $x \in \mathbb{R}^n$.

***Proof*:**
First, let's show we can limit our attention to function in L^1. In particular, If $A_r f(x) \to f(x)$ for a.e. x, for any $N \in \mathbb{N}$, we can bound what x's we choose by $|x| \leq N$. Since for any $r \leq 1$, the values $A_r f(x)$ depend only on $f(y)$ for $|y| \leq N+1$, we can replace f with $f\chi_{B_{N+1}(0)}$, so $f \in L^1$

By theorem 2.6.2, for every $\epsilon > 0$, there exists a continuous integrable function g such that $\int |g(y) - f(y)|dy < \epsilon$. Since g is continuous, for every $x \in \mathbb{R}^n$, for all $\delta > 0$, there exists an $r > 0$ such that $|y - x| < r$ implies $|g(y) - g(x)| < \delta$. From this, we get the result to work for g:

$$\begin{aligned}
|A_r g(x) - g(x)| &= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y)dy - g(x) \right| \\
&= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y)dy - \frac{m(B_r(x))}{m(B_r(x))} g(x) \right| \\
&= \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} g(y) - g(x)dy \right| \\
&< \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} \delta dy \right| \\
&= \frac{1}{m(B_r(x))} |m(B_r(x))\delta| \\
&= \delta
\end{aligned}$$

and so, since δ can be anything, we hve that $A_r g(x) \to g(x)$ for a.e. x.

Now we use this result to prove it for f. We will use the Maximal theorem to bound the result. Let:

$$\begin{aligned}
\limsup_{r \to 0} |A_r f(x) - f(x)| &= \limsup_{r \to 0} |A_r f(x) - A_r g(x) + A_r g(x) - g(x) + g(x) - f(x)| \\
&= \limsup_{r \to 0} |A_r(f - g)(x) + (A_r g - g)(x) + (g - f)(x)| \\
&\leq H(f - g)(x) + 0 + |f - g|(x)
\end{aligned}$$

Or, in terms of sets, if

$$E_\alpha = \left\{ x \;\middle|\; \limsup_{r \to 0} |A_r f(x) - f(x)| > \alpha \right\} \qquad F_\alpha = \{x \mid |f - g| > \alpha\}$$

then

$$E_\alpha \subseteq F_{\alpha/2} \cup \{x \mid H(f-g)(x) > \alpha/2\}$$

Furthermore, we have that

$$(\alpha/2)m(F_{\alpha/2}) \leq \int_{F_{\alpha/2}} |f(x) - g(x)|dx < \epsilon$$

Thus, by the maximal theroem:

$$m(E_\alpha) \leq \frac{2\epsilon}{\alpha} + \frac{2C\epsilon}{\alpha}$$

and since ϵ is arbitrary, $m(E_\alpha) = 0$ for all $\alpha > 0$. However, $\lim_{r\to 0} A_r f(x)$ for all $x \notin \bigcup_{i=1}^\infty E_{1/n}$, finishing the proof.

In the theorem, we proved it for $f(x)$; it can in fact be shown that it can be proven for $|f|$. To show this, we define the following concept for reference:

Definition 3.4.10: Lebesgue Set

Let f be a measurable function. Then the *Lebesgue set* is :

$$L_f := \left\{ x \;\middle|\; \lim_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(x) - f(y)|dy = 0 \right\}$$

The following theorem shows that the points on which the Lebesgue set fails on is measure 0, and so the result is true for L_f as well

Theorem 3.4.11: Upgrade To Lebesgue Set

Let $f \in L^1_{\text{loc}}$. Then $m((L_f)^c) = 0$

***Proof*:**

This proof will take advantage of the zero measure set in theorem 3.4.9, and that we can form a dense set of $\mathbb{C}$ which will correspond the zero-measure set from this previous theorem, meaning we can make a limit argument that will work for the proof.

First, for each $c \in \mathbb{C}$, we can apply theorem 3.4.9 to $g_c = |f(x) - c|$ to get that:

$$\lim_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - c|dy = |f(x) - c|$$

up to some Lebesgue zero measure set, say N_c. Now, Let D be any countable dense subset of $\mathbb{C}$ and let $N = \bigcup_{c\in D} N_c$ (so $m(N) = 0$). If $x \notin N$, then for any $\epsilon > 0$, we can choose some $c \in D$ such that $|f(x) - c| < \epsilon$. Thus:

$$|f(y) - f(x)| < |f(y) - c| + \epsilon$$

Thus, we have that:

$$\limsup_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)|dy \leq |f(x) - c| + \epsilon \leq 2\epsilon$$

and since ϵ can be arbitrary small, we have that $\limsup_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)|dy = 0$, and so if $x \notin N$, the theorem applies. But then $0 = m(N) = m(L_f)^c)$, as we sought to show.

We have essentially proven the main results of this section; the last thing we want to do is generalize the types of sets that shrink to x. So far, we have been using balls, however that is not necessary. Thus, we will generalize the types of sets and state the most general version of this "averaging" theorem

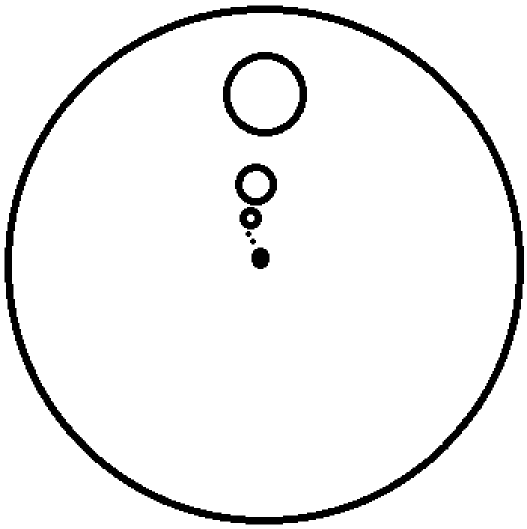
Definition 3.4.12: Sets That Shrink Nicely

A family of set $\{E_r\}_{r>0}$ of Borel subsets of $\mathbb{R}^n$ is said to *shrink nicely* if :

1. $E_r \subseteq B_r(x)$ for each r (so $m(E_r) \leq m(B_r(x))$)
2. There is a constant $\alpha > 0$ that is indepenent of r such that $m(E_r) > \alpha m(B_r(x))$

Example 3.4.13: Nicely Shrinking Sets

1. Let $U \subseteq B_1(0)$ be any borel subset. such that $m(U) > 0$, and consider $E_r = \{x + ry \mid y \in U\}$. You can imagine that U is some circle of $B_1(0)$ and see that, if $x = 0$:



   Then we see that E_r shrinks to to x (even if $x \notin B_1(0)$)
2. in $\mathbb{R}$, the set $E_r = (x, x + r]$ also shrink nicely.

With this, we can finally state the result we've building up to in full generality:

Theorem 3.4.14: The Lebesgue Differentiation Theorem

Let $f \in L^1_{\text{loc}}$. Then for every x in the Lebesgue set, that is almost every x:

$$\lim_{r \to 0} \frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| dy = 0$$

and

$$\lim_{r \to 0} \frac{1}{m(E_r)} \int_{E_r} f(y) dy = f(x)$$

for every family of nicely shrinking sets $\{E_r\}_{r>0}$ that shrink nicely to x

Written in terms of measure's this tells us that if $d\nu = f d\mu$, then:

$$\lim_{r \to 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

***Proof*:**
To prove the first equality, we simply use the definition of nicely shrinking sets:

$$\begin{aligned} \frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| dy &\leq \frac{1}{m(E_r)} \int_{B_r(x)} |f(y) - f(x)| dy \\ &\leq \frac{1}{\alpha m(B_r(x))} \int_{E_r} |f(y) - f(x)| dy \end{aligned}$$

so we can use theorem 3.4.11.

For the second equality, intead of putting absolute value, can start without them, in which case we would do the same think and then use theorem 3.4.9, and move the $f(x)$ to the other side.

3.4.15 Points As Average Notes that this also means that points can be interpreted as the *limit of averages*; in a sense they are inherently "singular". In a sense, it is more natural to work not by evaluating at points but by integrating against function in a small neighbourhood of a point (so that it capture local information). This is exactly what will be done in chapter 9 and the Lebesgue Differentiation theorem will insure functions in the traditional sense will be a special case of distributions.

3.4.1 Regular Measures

We will apply these results to a special class of measures which, as is common with such definitions, we take some aspects that we like from the Lebesgue measure on $\mathbb{R}^n$, and want to study measure's more generally (in this case, borel measures on $\mathbb{R}^n$) that have the same property:

Definition 3.4.16: Regular Measure

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \nu)$ be a borel measure space. Then ν is called *regular* if:

1. $\nu(K) < \infty$ for all compact subset $K \subseteq \mathbb{R}^n$
2. $\nu(E) = \inf \{\nu(U) \mid U \text{ open}, E \subseteq U\}$ for every $E \in \mathcal{B}_{\mathbb{R}^n}$

It is in fact provable that the first condition implies the second. We have shown this for $\mathbb{R}$ (i.e. $n = 1$) in theorem 1.4.2 and 1.4.4. In the study of Radon measures, this proof can be done for arbitrary n (Folland chapter 7.2).

Notice that a regular measure is automatically σ-finite, since $\mathbb{R}^n$ is σ-compact. As usual, a signed or complex measure ν is regular if $|\nu|$ is regular.

If $f \in L^+(\mathbb{R}^n)$, then the measure fdm is regular if and only if $f \in L^1_{\text{loc}}$. This means that we can generalize the Lebesgue differentiation theorem to regular measures in general instead of measures defined on densities which are locally integrable. Let's first check that that assertion is indeed the case. First, it should be clear that $f \in L^1_{\text{loc}}$ is equivalent to $\nu(K) < \infty$ for all compact sets. To show 2, we'll first show it works for bounded borel sets E. By theorem 2.6.1, for any $\delta > 0$, there exists an open set $U \supseteq E$ such that $\mu(U) < \mu(E) + \delta$, thus $\mu(U \setminus E) < \delta$. By theorem 3.2.3, for all $\epsilon > 0$, there exists a $U \supseteq E$ such that $\int_{U \setminus E} fdm < \epsilon$, so $\int_U fdm < \int_E fdm + \epsilon$, thus $\nu(U) < \nu(E) + \epsilon$. This completes the requirement of the infimum. For the unbounded case, take $E = \cup_i E_i$ where each E_i is finite, finding it in each case while choosing $\int_{U_i \setminus E_i} fdm < \epsilon 2^{-i}$.

With this equivalence, it becomes worthwile exploring Lebesgue Differentiation Theorem in the case of regular measures:

Theorem 3.4.17: Lebesgue Differentiation With Regular Measure

Let ν be a regular signed or complex Borel measure on $\mathbb{R}^n$, and let $d\nu = d\lambda + fdm$ be its Lebesgue decomposition. Then for m-almost every $x \in \mathbb{R}^n$:

$$\lim_{r \to 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

for every nicely shrinking family $\{E_r\}_{r>0}$ that nicely shrinks to x

***Proof*:**
We have essentially already done this proof, except now we have to consider we are adding $d\lambda$, $d\nu = d\lambda + fdm$, and so we would like to show that $\frac{\lambda(E_r)}{m(E_r)} \to 0$.

First, notice that $d|\nu| = d|\lambda| + |f|d\mu$ (or $|f|dm$?). Thus, regularity in ν implies regularity of λ and fdm (this should be proven, though it is a bit tedious[a]). Since $f \in L^1_{\text{loc}}$, it is enough to show that if $\lambda \perp m$, then for all x, $\lambda(E_r)/m(E_r) \to 0$ m-a.e as $r \to 0$. Without loss of generality, we can let $E_r = B_r(x)$ and assume λ is positive since:

$$\left|\frac{\lambda(E_r)}{m(E_r)}\right| \leq \frac{|\lambda|(E_r)}{m(E_r)} \leq \frac{|\lambda|(B_r(x))}{m(E_r)} \leq \frac{|\lambda|(B_r(x))}{\alpha m(B_r(x))}$$

for some $\alpha > 0$. Now, let's say $\lambda \geq 0$ (if $\lambda = 0$, then we'd be done), and take $A \in \mathcal{B}_{\mathbb{R}^n}$ such that $\lambda(A) = m(A^c) = 0$ (considering that $\lambda \perp m$). We will do the usual trick when it comes to showing that something has measure zero by assuming it doesn't, havign some sets define define in terms of a " $1/k$ sequence", so to speak, and show we reach a contradiction. Consider:

$$F_k = \left\{ x \in A \;\middle|\; \limsup_{r \to 0} \frac{\lambda(B_r(x))}{m(B_r(x))} > \frac{1}{k} \right\}$$

We will show that $m(F_k) = 0$ for all k, meaning the set of points for which $\lambda(E_r)/m(E_r) \to 0$ is 0 m-a.e. First, since λ is regular, for all $\epsilon > 0$, there exists a $U_\epsilon \supseteq A$ such that $\lambda(U_\epsilon) < \epsilon$. Since U_ϵ is open, for each point $x \in U_\epsilon$, there exists a ball such that $B(x) \subseteq U_\epsilon$. By definition of F_k, $\lambda(B(x)) > \frac{m(B(x))}{k}$. or $k\lambda(B(x)) > m(B(x))$. Next, take $V_\delta = \cup_x B(x)$. Then by construction $F_k \subseteq V_\epsilon$. We'll show that $m(V_\epsilon) \leq 3^n k\epsilon$ for all ϵ, and so $m(F_k) = 0$ or all k. To do this, notice that by assumption there is a constant c such that $c < m(V_\epsilon)$. Thus, by lemma 3.4.2, there exist disjoint balls around points $x_1, ..., x_n$ such that

$$c < 3^n \sum_1^n m(B_{x_i}) \leq 3^n k \sum_1^n \lambda(B_{x_i}) \leq 3^n k\lambda(V_\epsilon) \leq 3^n k\lambda(U_\epsilon) \leq 3^n k\epsilon$$

showing that $m(F_k) \leq m(V_\epsilon) < 3^n k\epsilon$, and since ϵ is arbitrary, we have that $m(F_k) = 0$ for all k, as we sought to show.

[a] It was homework 8, q26

3.5 Functions of Bounded Variation

We will now work towards the Fundamental Theorem of Calculus in its generalized form. This generalization is not in the direction of higher dimensions, as is the case with Stokes Theorem which is a generalization into higher dimensions, but requires much nicer conditions like working over smooth manifolds and differential forms. Rather, we want to stay in $\mathbb{R}$ (or $\mathbb{C}$) and see if we can weaken the conditions (i.e. Weaker than differentiable) and see when we can still apply FTC. Essentially, we want to see if we can define a *density* from probability using the closest notion of a derivative of a function.

In the case of $\mathbb{R}$, we have alraedy explored in section 1.4 that Borel measures defined on $\mathbb{R}$ are defined through an associated distribution F, which is a right-continuous increasing function. In particular, we defined μ_F based on a increasing right-continuous function F (i.e. a distribution) to be $\mu([a, b]) = F(b) - F(a)$. We will use the tools we build up in the last section to finally prove the Fundamental theorem of Calculus in full generality. We will then explore when we can define distributions when we have a complex function, and substitute the notion of increasing function with a function of *bounded variation.*

The first thing we will do is show how increasing functions are a.e. differentiable (which to me is sort of crazy)!!

Proposition 3.5.1: Increasing Then a.e. Differentiable

Let $F : \mathbb{R} \to \mathbb{R}$ be an increasing function. Then the set of points where F is discontinuous is countable.
Furthermore, if $G(x) = \lim_{y \to x^+} F(y) = F(x+)$, then F and G are differentiable a.e. and $F' = G'$ a.e.

***Proof*:**
First, if x is continuous at x, then $F(x+) = F(x-)$, so the interval $(F(x-), F(x+))$ is empty. If it is not empty, then since F is increasing, each $(F(x-), F(x+))$ must be disjoint. So, for any $N \in \mathbb{N}$, for all $|x| \leq N$, we have that $(F(x-), F(x+)) \subseteq (F(-N), F(N))$. Now, here is the trick of this proof: notice that

$$\sum_{|x|<N} [F(x+) - F(x-)] \leq F(N) - F(-N) < \infty$$

since the $F(x-)$ is equal to $F(x_+)$ at appropriate locations, and so the sum cancels out (note that this sum is absolutely continuous since $F(x+) \geq F(x-)$). To see more clearly what's going on ,if we label all the points in $|x| < N$ through an indexing set A, and let $x_\alpha = (F(x_\alpha -) - F(x_\alpha -))$, then:

$$\sum_{\alpha < A} x_\alpha < \infty$$

From here, we use a the fact that this the sum of uncountably many positive x_α, and use the fact that the sum of uncountably many positive elements must be infinite if more than countably elements are non-zero, but since the sum *must* be finite $\{x \in (-N, N) \mid f(x+) \neq F(x-)\}$ is countable. Since this is true for all N, the set of discontinuities of F must be countable, proving the first assertion.

For the second assertion, first note that G is increasing and right continuous and $G = F$ at all points excepts where F is discontinuous. To show G is differentaible a.e, by proposition 1.4.2

$$G(x+h) - G(x) = \begin{cases} \mu_G((x, x+h]) & \text{if } h > 0 \\ \mu_G((x+h, x]) & \text{if } h < 0 \end{cases}$$

Notice that $\{(x-r, x]\}$ and $\{(x, x+r]\}$ shrink nicely to x as $r = |h| \to 0$. Furhtermore μ_G is regual by theorem 1.4.2. Thus, by theorem 3.4.17, we have that:

$$G'(x) = \lim_{r \to 0} \frac{G(x+h) - G(x)}{h} = \lim_{r \to 0} \frac{\mu_G(E_r)}{m(E_r)}$$

exists for a.e. x. The final part of this proof is to show that if $H = G - F$, then H' exists and is equal to zero a.e.

Let $\{x_i\}$ be an enumeration of the points where $H \neq 0$. Since $H = G - F$, $H(x_i) > 0$. Let's say $\delta_i : \mathbb{R} \to \mathbb{R}$ is a function where $\delta_i(x_i) = 1$ and zero eveywehre else. Then for any $N \in \mathbb{N}$, since H is increasing we have $\sum_{\{i \mid |x_i| < N\}} H(x_i) < \infty$. Thus, we can define a new measure:

$$\mu = \sum_i H(x_i)\delta_i$$

By what we've shown, μ is finite on all compact sets by the previous lines reasoning, and so by the theorem's we've been using μ is regular. Furthermore, $\mu \perp m$ since $m(E) = \mu(E^c) = 0$ where $E = \{x_i\}_{i=1}^{\infty}$. Thus:

$$\left|\frac{H(x+h) - H(x)}{h}\right| \leq \frac{H(x+h) + H(x)}{|h|} \overset{!}{\leq} \frac{\mu\left(x - 2|h|, x + 2|h|\right)}{|h|}$$

where $\overset{!}{\leq}$ comes from trying to ingulf $H(x+h)$ and $H(x)$ into an interval.

Since all measure are regular, by theorem 3.4.17 this limit exists and $H' = 0$ a.e., completing the proof.

With this, we now have the tools to try and explore how to think about "increasing function" of the form $F : \mathbb{R} \to \mathbb{C}$. Since $\mathbb{C}$ has no inheret order, we must be more precise on how we define "increasing" and how do we make sure it doesn't explode (just like how when defining the borel measures through a distribution)

To understand the motivation behind the following definitions, imagine the following scenario: let F be the function representing the movement of a particle as it travels accross in time with respect to t (the graph being the position of the particle relative to, say, where it started). The *total variation* from time a to time b of the particle would be the total distance the particle has travelled in that time frame. if F were differentiable, this would simply be:

$$\int_a^b |F'(x)| dx$$

However, what if F is not differentiable? In that case, we will take the following approach: partition $[a, b]$ into $[a, t_1], [t_1, t_2], ..., [t_{n-1}, b]$. Then, linearly approximate F by a straight line from $(t_{i-1}, F(t_{i-1}))$ to $(t_i, F(t_i))$. Then, take the limit over all partitions (in particular, as partitions get finer). If you are worried that such a limit is not well-defined, remember that polynomials can arbitrarily well approximate continuous functions on $\mathbb{R}$, and in most cases F is right-continuous. This heuristic might be more rigorous, but it at least make me feel comfortable with doing this idea

We will do a similar idea for the complex case, except we will take the norm (which we'll also call absolute value) of the distances:

Definition 3.5.2: Total Variation

Let $f : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$. Then define the *total variation of* F to be:

$$T_F(x) = \sup\left\{\sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| \;\middle|\; n \in \mathbb{N},\ -\infty < x_0 < x_1 < \cdots < x_n = x\right\}$$

I should be clear by construction that T_F is an increasing function. Since we put absolute values inside the sum, then as we refine the partition, the sum will get bigger. Therefore, If we have $a < b$ and consider $T_F(b)$, it is harmless to assume a is one of the subdivision points (since we can refine and get a partition with a being part of it). It follows that we have:

$$T_F(b) - T_F(a) = \sup\left\{\sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| \;\middle|\; n \in \mathbb{N},\ a = x_0 < x_1 < \cdots < x_n = x\right\}$$

Using this fact, we can define the following:

Definition 3.5.3: Bounded Variation

Let T_F be the total variation of the function F. Then if:

$$\lim_{t\to\infty} T_F(t) = T_F(\infty) < \infty$$

Then F is said to be of *bounded variation* (on $\mathbb{R}$). We denote the set of of all functions of bounded variation as BV. If we are concerned with all function of bounded variation on an interval $[a,b]$, we will denote this as $BV([a,b])$.

Note that if $F \in BV$, then $F \in BV([a,b])$, since it's total variation on $[a,b]$ is $T_F(b) - T_F(a)$ which is less thatn the total vriation of T_F. If $F \in BV([a,b])$, then we can extend F to be in BV by letting $F(x) = F(a)$ for all $x \leq a$ and $F(b) = F(x)$ for all $b \leq x$. Further note that we rarely actually compute $T_F(b)$, but we use the fact that a function is in BV to get many theoretical results:

Example 3.5.4: Bounded Variation

1. If $F : \mathbb{R} \to \mathbb{R}$ is bounded and increasing, then $F \in BV$ $(T_f(x) = F(x) - F(-\infty))$
2. If $F, G \in BV$ and $a, b \in \mathbb{C}$ then $aF + bG \in BV$
3. If F is differentiable on $\mathbb{R}$ and F' is bounded, then $F \in BV([a,b])$ for $-\infty < a < b < \infty$ (by the mean value theorem)
4. If $F(x) = \sin(x)$, then $F \in BV([a,b])$ for all $-\infty < a < b < \infty$, but $F \in BV$
5. if $F = x \sin\left(\frac{1}{x}\right)$ for $x \neq 0$ and $F(0) = 0$, then $F \notin BV([a,b])$ for $a \leq 0 < b$ and $a < 0 \leq b$

Before we prove stuff about BV functions, let's first prove a lemma:

Lemma 3.5.5: Sum Is Increasing

Let $F \in BV$ be a real valued function. Then $T_F + F$ and $T_F - F$ are increasing functions.

***Proof*:**
Take $x < y$ and $\epsilon > 0$. Since T_F is a supremum, take appropriat $x_0 < \cdots x_n = x$ such that

$$\sum_i^n |F(x_i) - F(x_{i-1})| \geq T_F(x) - \epsilon$$

Next, notice that we can make this into an element of the supremum set of $T_f(y)$ (also known as an approximating sum) by doing

$$\sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)|$$

With this sum, take $F(y) = [F(y) - F(x)] + F(x)$. With some manipulation, we get:

$$T_f(y) \pm F(y) \geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x)$$

Since F is a real valued function, we cna simply take cases to notice that:

+	F(x) - F(y)	+	F(x) - F(y)	$2\lvert F(y) - F(x)\rvert$
+	F(x) - F(y)	-	F(x) - F(y)	0
-	F(y) - F(x)	+	F(x) - F(y)	0
-	F(y) - F(x)	-	F(x) - F(y)	$2\lvert F(y) - F(x)\rvert$

$$\begin{aligned} T_f(y) \pm F(y) &\geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x) \\ &\geq T_f(x) - \epsilon \pm F(x) \end{aligned}$$

letting $\epsilon \to 0$, we see that:

$$T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$$

and since x and y was arbitrary, we have establish that $T_F + F$ and $T_F - F$ are both increasing functions, we as sought to show

Theorem 3.5.6: Properties Of BV

1. $T_F \in BV$ is an increasing function
2. $F \in BV$ if and only if $\text{Re}F \in BV$ and $ImF \in BV$
3. If $F : \mathbb{R} \to \mathbb{R}$ then $F \in BV$ if and only if F is the difference of two bounded increasing functions; for $F \in BV$ these functions may be taken to be $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$
4. If $F \in BV$, then $F(x+) = \lim_{y \to x^-} F(y)$ and $F(x-) = \lim_{y \to x^+} F(y)$ exist for all $x \in \mathbb{R}$, as do $F(\pm\infty) = \lim_{y \to \pm\infty} F(y)$
5. If $F \in BV$, the set of points at which F is discontinuous is countable
6. If $F \in BV$ and $G(x) = F(x+)$, then F' and G' are exist and are equal a.e.

***Proof*:**

1. This essentially comes from the definition
2. This is essentially immediate, and so is left as exam review
3. Look at example 3.5.4 to see how the $\Rightarrow$ direction goes. For the $\Leftarrow$ direction, by lemma 3.5.5, the equation $F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$ express F as the difference of two increasing

functions. To show boundedness, we showed that $T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$ for $y > x$. Re-arranging and taking advantage of the fact that $F \in BV$, we get:

$$|F(y) - F(x)| \leq T_F(y) - T_F(x) \leq T_F(\infty) - T_F(-\infty) < \infty$$

showing that F (and hence T_F) is bounded

4,5,6 follow from 1,2,3 along with proposition 3.5.1.

Note that (6) is particularly powerful since we now have that a function simply needs to be of bounded variation to have a derivative a.e. (not necessarily increasing anymore). The converse is false: being differentiable a.e. (even everywhere) does not imply BV (think of $\sin(1/x)$ on $(0, 1]$).

The equation in part 2 are important enough to be given a name:

Definition 3.5.7: Jordan Decomposition

Let $F \in BV$ be a real-valued function. Then the *jordan decomposition* of F is:

$$F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$$

where $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$ are called the *positive variation of* F and *negative variation of* F respectively.

Since $x^+ = \max(x, 0) = \frac{1}{2}(|x| + x)$ and $x^- = \max(-x, 0) = \frac{1}{2}(|x| - x)$ for $x \in \mathbb{R}$, we get that:

$$\frac{1}{2}(T_F \pm F)(x) = \sum \left\{ \sum_i^n [F(x_i)F(x_{i-1})]^\pm \;\middle|\; x_0 < \cdots < x_n = x \right\} \pm \frac{1}{2}F(-\infty)$$

We are almost at a stage to connect borel measures to BV functions. There is one more peice of the puzzle that we need:

Definition 3.5.8: Normalized Bounded Variation (NBV)

Let $F \in BV$ be a function of bounded variation. Then if F is right-continuous and $F(-\infty) = 0$, then F is said to be a function of *normalized bounded variation*. We will denote the collection of such functions as:

$$NBV := \{F \in BV \mid F \text{ is right continuous and } F(-\infty) = 0\}$$

Example 3.5.9: NBV

Let $F \in BV$. Define $G = F(x+) - F(-\infty)$. Then $G \in NBV$, and $F' = G'$ a.e.

First, to show that $G \in BV$, by theorem 3.5.6[2,3], if F is real and $F = F_1 - F_2$ where F_1, F_2 are increasing, then $G(x) = F_1(x+) - [F_2(x+) - F(-\infty)]$, which is the difference of two increasing functions. That $G(-\infty) = 0$ comes by definition. Hence $G \in NBV$.

In fact, every $F \in BV$ almost defines an NBV function through T_F:

Lemma 3.5.10: BV, then T_F almost NBV

If $F \in BV$, then $T_F(-\infty) = 0$. If F is also right continuous, then T_F is also right-continuous (making $T_F \in NBV$)

***Proof*:**
Let $\epsilon > 0$ so that for any x and $x_0 < \cdots < x$ we ahve:

$$\sum F(x_{i+1}) - F(x_i) = T_f(x) - T_f(x_0) > T_F(x) - \epsilon$$

So $T_f(x_0) < \epsilon$ or more generally for any $y \leq x_0$ $T_f(y) < \epsilon$. Since epislon is arbitray, as we let $y \to 0$ we get $T_f(-\infty) = 0$

For the next assertion, suppose F is right-continuous. I'll finish this proof later:

Now suppose that F is right continuous. Given $x \in \mathbb{R}$ and $\epsilon > 0$, let $\alpha = T_F(x+) - T_F(x)$, and choose $\delta > 0$ so that $|F(x+h) - F(x)| < \epsilon$ and $T_F(x+h) - T_F(x+) < \epsilon$ whenever $0 < h < \delta$. For any such h, by (3.24) there exist $x_0 < \cdots < x_n = x + h$ such that

$$\sum_1^n |F(x_j) - F(x_{j-1})| \geq \tfrac{3}{4}[T_F(x+h) - T_F(x)] \geq \tfrac{3}{4}\alpha,$$

and hence

$$\sum_2^n |F(x_j) - F(x_{j-1})| \geq \tfrac{3}{4}\alpha - |F(x_1) - F(x_0)| \geq \tfrac{3}{4}\alpha - \epsilon.$$

Likewise, there exist $x = t_0 < \cdots < t_m = x_1$ such that $\sum_1^n |F(t_j) - F(t_{j-1})| \geq \frac{3}{4}\alpha$, and hence

$$\begin{aligned}
\alpha + \epsilon &> T_F(x+h) - T_F(x) \\
&\geq \sum_1^m |F(t_j) - F(t_{j-1})| + \sum_2^n |F(x_j) - F(x_{j-1})| \\
&\geq \tfrac{3}{2}\alpha - \epsilon.
\end{aligned}$$

Thus $\alpha < 4\epsilon$, and since ϵ is arbitrary, $\alpha = 0$. ∎

We finally get to establish the connection between complex meaures and functions in BV:

Theorem 3.5.11: Complex Meaures And BV

If μ is a complex Borel meaure on $\mathbb{R}$ (i.e. a finite signed measure) and $F(x) = \mu((-\infty, x])$, then $F \in NBV$. Conversely, if $F \in NBV$, then there is a unique complex Borel measure μ_F such that $F(x) = \mu_F((-\infty, x])$

***Proof*:**
Let μ a complex measure. Then we can decompose it into $\mu = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$ where $\mu_i^\pm$ are positive finite measures. If $F_i^\pm(x) = \mu_i^\pm((-\infty, x])$, then $F_i^\pm$ is increasing, right continuous, $F_i^\pm(-\infty) = 0$, and $F_i^\pm(\infty) = \mu_i^\pm(\mathbb{R}) < \infty$. Thus, by theorem 3.5.6[2,3], the function

$$F = F_1 + -F_1^- + i(F_2^+ - F_2^-) \in NBV \tag{3.2}$$

Conversely, by theorem 3.5.6 and lemma 3.5.10, any $F \in NBV$ can be written in the from of equation (3.2) with the $F_i^\pm$ increasing and in NBV. By Theorem 1.4.2, each $F_i^\pm$ gives a measure $\mu_i^\pm$, so $F(x) = \mu_F((-\infty, x])$ where $\mu_F = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$.

Finally, to show $|\mu_F| = \mu_{T_F}$ (this was left as an outlined exercise 28)

Corollary 3.5.12: Total Variation Of μ_F

Let $F \in NBV$, μ_F be the associated complex measure on $\mathbb{R}$, and $G(x) = |\mu_F|((-\infty, x])$. Then:

$$|\mu_F| = \mu_{T_F}$$

***Proof*:**
We break down the proof in the following steps:

1. From the definition of T_F, $T_F \leq G$

   Proof. We need to show $T_F(x) \leq G(x)$ In the previous homework, we showed that:

   $$|\mu_F|((-\infty, x]) = \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; \coprod_i E_i = (-\infty, x]\right\}$$

   In particular, we can choose are disjoint sets to be h-intervals and get:

   $$\begin{aligned} |\mu_F|((-\infty, x]) &= \sup\left\{\sum_i^\infty |\mu_F((x_i, x_{i+1}])| \;\middle|\; \coprod_i (x_i, x_{i+1}] = (-\infty, x]\right\} \\ &= \sup\left\{\sum_i^\infty |F(x_{i+1}) - F(x_i)| \;\middle|\; \coprod_i (x_i, x_{i+1}] = (-\infty, x]\right\} \end{aligned}$$

   and by definition of T_F

   $$T_F(x) = \sup\left\{\sum_i^N |F(x_{i+1}) - F(x_i)| \;\middle|\; N \in \mathbb{N},\ x_0 < \cdots < x_N = x\right\}$$

Notice that the set for $|\mu_F|$ contains that of $T_F(x)$ since we can set the rest of the infinite value for T_F to 0, and so $T_F \le G = |\mu_F|$ □

2. $|\mu_F(E)| \le \mu_{T_F}(E)$ when E is an interval, and hence E is a borel set.

Proof. Without loss of generality, we will work with intervals of the form $(a, b]$. We only require these type of intervals for the σ-algebra that will be generatedi n part (c).

First, since $F \in NBV$, then it is right-continuous, and so by lemma 3.28 T_F is right-continuous, and so by lemma 3.28 again $T_F \in NBV$. Thus, by theorem 3.29, there is an assocaited complex borel meaure such that $T_F(x) = \mu_{T_F}((-\infty, x])$. So $\mu_{T_F}((a, b]) = T_F(b) - T_F(a)$. With that, we proceed with the proof.

First, the emptyset (the trivial interval) is immediate since $|\mu_F(\emptyset)| = 0 = \mu_{T_F}(\emptyset)$. Next, to show the inequality is true for bounded intervals, notice that:

$$|\mu_F((a, b])| = |\mu_F((-\infty, b]) + \mu_F((-\infty, a]) = |F(b) - F(a)|$$

Next, recall that:

$$\mu_{T_F}((a, b]) = T_F(b) - T_F(a) = \sup \left\{ \sum_i^n |F(x_{i+1}) - F(x_i)| \;\middle|\; n \in \mathbb{N}\ a = x_0 < \cdots < x_n = b \right\}$$

So, $|F(b) - F(a)|$ is inside the supremum set, and so:

$$= |F(b) - F(a)| \le T_F(b) - T_F(a) = \mu_{T_F}((a, b])$$

Next, for rays, notice we can split a ray into the countable union of h-intervals:

$$|\mu_F((-\infty, b])| = \left| \sum_{i=0}^{\infty} \mu_F((b - i - 1, i - k]) \right| \le \sum_{i=0}^{\infty} |\mu_F((b - i - 1, i - k])|$$

where we can now apply the results for bounded intervals to get that:

$$\le \sum_{i=0}^{\infty} |\mu_F((b - i - 1, i - k])| \le \sum_{i=0}^{\infty} \mu_{T_F}((a, b]) = \mu_{T_F}((-\infty, b])$$

Thus $|\mu_F((-\infty, b])| \le \mu_{T_F}((-\infty, b])$, and similarly for rays of the form (a, ∞).

To show it's true for all Borel sets, first recall that h-intervals form an algebra (their disjoint union forms an elementary family, and hence by proposition 1.7). Next, let $\mathcal{C} = \{E \in \mathcal{B}_{\mathbb{R}} \mid |\mu_F(E)| \le \mu_{T_F}(E)\}$. This set clearly contains all h-intervals by what we've just shown. We'll show that this set is in fact a monotone class, and hence by the monotone class lemma contains all Borel sets.

This is simply a matter of using the appropraite definitions. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{C}$ be an increasing sequence. Then:

$$\begin{aligned}
\left|\mu_F\left(\bigcup_i^\infty E_i\right)\right| &= \left|\lim_{i\to\infty}\mu_F\left(\bigcup_i^\infty E_i\right)\right| && \text{definition}\\
&= \lim_{i\to\infty}\left|\mu_F\left(\bigcup_i^\infty E_i\right)\right| && \text{limit property}\\
&\le \lim_{i\to\infty}\mu_{T_F}(E_i) && \text{all } E_i \in \mathcal{C}\\
&= \mu_{T_F}\left(\bigcup_i^\infty E_i\right) && \text{definition}
\end{aligned}$$

and hence $\mathcal{C}$ is closed under increasing sequences. A similar argument works for decreasing sequences (in particular since μ_{T_F} is finite, it will hold true for any decreasing sequence). Hence, by the monotone class lemma, $\mathcal{C}$ contains the Borel set, showing that $|\mu_F(E)| \le \mu_{T_F}(E)$ for all $E \in \mathcal{B}_{\mathbb{R}}$, completing the proof.

□

3. $|\mu_F| \le \mu_{T_F}$, and hence $G \le T_F$ (use exercise 21.)

   Proof. Recall that

$$|\mu_F|(E) = \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}$$

where I modified what's inside the parenthesis to make it more explicit that the sets are inside $\mathcal{B}_{\mathbb{R}}$. Using what we have just proven:

$$\begin{aligned}
|\mu_F|(E) &= \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}\\
&\le \sup\left\{\sum_i^\infty \mu_{T_F}(E_i) \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}\\
&\le \sup\left\{\mu_{T_F}(\coprod_i E_i) \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\} && \text{disjoint}\\
&\le \sup\left\{\sum_i^\infty \mu_{T_F}(E) \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}\\
&= \mu_{T_F}(E)
\end{aligned}$$

Thus, $|\mu_F|(E) \le \mu_{T_F}(E)$. If we take $E = (-\infty, x]$, then we get $|\mu_F|(E) = G(E) \le \mu_{T_F}(E)F$, completing the proof.

In particular, since we've shown the other direction, we have that $G = \mu_{T_F}$. □

With the connection established for between NBV functions and complex measures, it is natural to ask which function in NBV correspond to measure μ such that $\mu \perp m$ or $\mu \ll m$? The following proposition answers that:

Proposition 3.5.13: NBV And Radon-Nikodym with respect to m

If $F \in NBV$, then $F' \in L^1(m)$. Moreover:

1. $\mu_F \perp m$ if and only if $F' = 0$ a.e.
2. $\mu_F \ll m$ if and only if $F(x) = \int_{-\infty}^{x} F'(t)dt$

***Proof*:**
First, note that μ_F is regular by theorem 1.4.4. Next, observe that:

$$F'(x) = \lim_{r \to 0} \frac{\mu_F(E_r)}{m(E_r)}$$

where $E_r = (x, x+r]$ or $(x-r, x]$ by applying theorem 3.4.17

Since we have now characterized μ_F in terms of F, we can characterize $\mu_F \ll m$ in terms of F:

Definition 3.5.14: Absolute Continuity w.r.t. F

A function $F : \mathbb{R} \to \mathbb{C}$ is said to be *absolutely continuous* if

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that}$$

$$\text{for any set of disjoint intervals } (a_1, b_1), ..., (a_N, b_N),$$

$$\sum_{1}^{N} (b_i - a_i) < \delta \Rightarrow \sum_{i}^{N} |F(b_i) - F(a_i)| < \epsilon$$

More generally, F is absolutely continuous on $[a, b]$ if all the open intervals lie in $[a, b]$.

Notice that if F is absolutely continuous, then F is automatically uniformly continuous (pick $N = 1$). To understand why this is, and how it relates to other forms of continuity, it is helpful to use a driving analogy. Imagine driving a car, where the function $F(x)$ represents your distance traveled at time x:

- *Continuous:* You cannot teleport. You must drive through every point.
- *Uniformly Continuous:* You cannot have a vertical asymptote (you can't travel an infinite distance in finite time). It places a "global" rule: for a given time window δ, your distance will never exceed ϵ. However, it only looks at *one continuous interval at a time.*
- *Lipschitz Continuous:* You have a *strict, hard speed limit.* Your speed (the derivative $F'(x)$) can never exceed M.
- *Absolutely Continuous:* You *do not have a strict speed limit*, but your total fuel (or total movement) is restricted by the time allowed. You are allowed to momentarily hit "infinite" speed, but only for such a fleeting instant that the total distance gained during that spike is minuscule.

Why do we need absolute continuity if we have uniform continuity? Uniform continuity has a blind spot: it allows a function to concentrate all of its growth into a scattered collection of tiny, microscopic intervals whose total length is practically zero.

Example 3.5.15: The Devil's Staircase

A prime example is the *Cantor Function* (the "Devil's Staircase"). It is uniformly continuous on $[0, 1]$. It starts at 0 and ends at 1. However, it does 100% of its climbing on the Cantor set—a fractal set that has a total length (measure) of *exactly zero*.

Absolute continuity was invented to prevent this "Devil's Staircase" loophole. By checking a *collection* of N disjoint intervals whose *combined* total width is $< \delta$, absolute continuity ensures you cannot "sneak" a large amount of vertical growth into a small amount of horizontal space, even if you scatter that space into a million tiny pieces.

Proof. Recall the definition of absolute continuity: for every $\epsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of disjoint open intervals (a_i, b_i) in $[0, 1]$,

$$\sum_{i=1}^{N}(b_i - a_i) < \delta \Rightarrow \sum_{i=1}^{N}|F(b_i) - F(a_i)| < \epsilon$$

We will negate this statement. Let $\epsilon = 1/2$. We must show that for any $\delta > 0$, we can find a collection of disjoint intervals whose total length is less than δ, but over which the sum of the variations of F is at least $1/2$.

Recall the construction of the Cantor set C. At the n-th stage of the construction, we have a set C_n consisting of 2^n disjoint closed intervals, each of length 3^{-n}. The total length of the intervals making up C_n is $(2/3)^n$.

Because $\lim_{n\to\infty}(2/3)^n = 0$, given any $\delta > 0$, we can choose an n large enough such that $(2/3)^n < \delta$. Let these 2^n intervals be denoted by $[a_k, b_k]$ for $k = 1, \ldots, 2^n$. Their interiors (a_k, b_k) form a disjoint collection of open intervals whose total length is:

$$\sum_{k=1}^{2^n}(b_k - a_k) = \left(\frac{2}{3}\right)^n < \delta$$

By the definition of the Cantor function, F is constant on the intervals removed during the construction of the Cantor set. Therefore, all of the increase of F from $F(0) = 0$ to $F(1) = 1$ happens precisely over the intervals of C_n. Across each interval $[a_k, b_k]$ in C_n, F increases by exactly 2^{-n}.

Summing the variation of F over these open intervals gives:

$$\sum_{k=1}^{2^n}|F(b_k) - F(a_k)| = \sum_{k=1}^{2^n}2^{-n} = 2^n \cdot 2^{-n} = 1$$

Since $1 \geq \epsilon = 1/2$, the condition for absolute continuity fails. Thus, F is not absolutely continuous. □

To see why absolute continuity is weaker than Lipschitz, consider $F(x) = \sqrt{x}$ on $[0, 1]$. It is not Lipschitz because as $x \to 0$, the slope approaches infinity (no maximum speed limit). However, it *is* absolutely continuous because the "infinite speed" lasts for such a short duration that the accumulated distance remains bounded.

If F is everywhere differentiable and F' is bounded, then F is absolutely continuous, since $|F(b_i) - F(a_i)| \leq (\max |F'|)(b_i - a_i)$ by the Mean Value Theorem. More generally, on a compact set, we have that:

$$\text{absolutely continuous} \subseteq \text{uniformly continuous} \subseteq \text{continuous}$$

The rules change on a compact interval because a closed and bounded domain forces "nice" behavior. The function cannot escape to infinity, preventing unexpected asymptotic stretching. For example, $F(x) = x^2$ is continuous but not uniformly continuous on $\mathbb{R}$. But on a compact interval like $[0, 5]$, it is uniformly continuous (by the Heine-Cantor Theorem) because the domain is capped, and there is a "steepest point."

Because of this bounded nature, on a compact interval, the following hierarchy holds beautifully:

$$\begin{gathered} \text{continuously differentiable} \\ \subseteq \\ \text{Lipschitz continuous} \\ \subseteq \\ \text{absolutely continuous} \\ \subseteq \\ \text{Bounded Variation} \\ \subseteq \\ \text{differentiable a.e.} \end{gathered}$$

where Lipschitz continuous means for all $x, y \in \mathbb{R}$, there exists an M such that $|F(x)-F(y)| \leq M|x-y|$. After going over the FTC for Lebesgue integrals, you can prove as an exercise that Lipschitz continuous implies absolutely continuous. Usually, Lipschitz continuity is used as a simple way of finding absolutely continuous functions.

In modern mathematics, absolute continuity is the right condition for the question: *"When does $F(x) = F(a) + \int_a^x F'(t)dt$ hold true?"* For the Cantor function, the derivative is 0 almost everywhere, so integrating it gives 0, even though the function climbed from 0 to 1. Absolute continuity strictly outlaws this unmeasurable growth, ensuring the derivative accounts for 100% of the function's growth.

To make sure this new notion links with the absolute continuity of the measure:

Proposition 3.5.16: Linking Absolute Continuity Of F And Measures

Let $F \in NBV$. Then F is absolutely continuous if and only if $\mu_F \ll m$

***Proof*:**

($\Leftarrow$) Let $\mu_F \ll m$. Then since μ_F is finite, we can apply theorem 3.2.3 to the sets $E = \bigcup_i^N (a_i, b_i]$, and get that F is absolutely continuous

($\Rightarrow$) Let E be a borel set such that $m(E) = 0$. Let $\epsilon > 0$ so that we can choose a δ that satisfies the definition of absolute continuity of F. By theorem 1.4.4, we can find open sets $U \supseteq U_1 \cdots \supseteq E$ such that $m(U_1) < \delta$ (which also implies $m(U_i) < \delta$ by monotonicity) and

$\mu_F(U_i) \to \mu_F(E)$. Since each U_i is the countable union of disjoint intervals $(a_{i,j}, b_{i,j})$, for all $N \in \mathbb{N}$:

$$\sum_k^N |\mu_F((a_{i,j}, b_{i,j}))| \leq \sum_k^N |F(b_{i,j}) - F(a_{i,j})| < \epsilon$$

by absolute continuity. Letting $N \to \infty$, we get that $|\mu_F(U_i)| < \epsilon$, and so $|\mu_F(E)| < \epsilon$. Since this is true for all epsilon, it must be that $|\mu_F(E)| = 0$, showing that $\mu_F \ll m$, as we sought to show

With this, we can relate L^1 functions to NBV functions:

Corollary 3.5.17: Relating L^1 And NBV Functions

Let $f \in L^1$. Then the function $F(x) = \int_{-\infty}^x f(t)dt$ is in NBV, is absolutely continuous, and $f = F'$ a.e. Conversely, if $F \in NBV$, is absolutely continuous, then $F' \in L^1(m)$ and $F(x) = \int_{-\infty}^x F'(t)dt$

***Proof*:**
This follows from proposition 3.5.13 and 3.5.16.

If the function is bounded, then we get a slightly stronger result:

Lemma 3.5.18

If F is absolutely continuous on $[a, b]$, then $F \in BV([a, b])$

***Proof*:**
We will find an upper bound for the total variation of F, that is, we'll find an N such that $T_f(b) < N$. Choose $\epsilon = 1$ and let δ be the appropriate choice to satisfy absolute continuity: for any finite set of disjoint intervals (a_i, b_i):

$$\sum (b_i - a_i) < \delta \Rightarrow \sum |F(b_i) - F(a_i)| < \epsilon$$

Choose N to be the greatest integer that is smaller than $\delta^{-1}(b-a)+1$. Then, for any subdivision $a = x_0 < x_1 < \cdots < x_n = b$, we can collect the subintervals (x_i, x_{i+1}) into at most N groups such that the sum of each group is less than δ (adding subdivision's if necessary). Then the sum $\sum |F(x_{i+1}) - F(x_i)|$ over each group is 1, and so the total variation of F on $[a, b]$ is at most N

Is the converse true: If $F \in BV$, then does it imply that F is aboslutely continuous? The answer is no. Even better, F can even be NBV, uniformally continuous, and increasing (so differentiable a.e.), but still not absolutely continuous:

Example 3.5.19: NBV, Continuous, But Not Absolutely Continuous
Let F be the cantor function. Note that F is increasing, $F(-\infty) = 0$, and $T_F(\infty) = 1/2$ (since it's continuous, it is integrable, and integarting give $1/2$). Thus μ_F is defined. However, it is not absolutely continuous with respect to m. However, F is not absolutely continuous (you should prove this).

Theorem 3.5.20: Fundamental Theorem Of Calculus For Lebesgue Integrals

Let $-\infty < a < b < \infty$ and $f : [a, b] \to \mathbb{C}$. Then the following are equivalent:

(a) F is absolutely continuous

(b) $F(x) - F(a) = \int_a^x f(t)dt$ for some $f \in L^1([a, b], m)$

(c) F is differentiable a.e. on $[a, b]$, $F' \in L^1([a, b], m)$ and $F(x) - F(a) = \int_a^x F'(t)dt$

***Proof*:**
First, to show $(a) \Rightarrow (c)$, we can subtract some constant from F so that $F(a) = 0$. Doing the usual extension of setting $F(x) = 0$ for $x < a$ and $F(x) = F(b)$ for $x > b$, then $F \in NBV$. Thus, by lemma 3.5.18 using corrollary 3.5.17. Then, that $(c) \Rightarrow (b)$ comes immetidaly since (c) is a stronger condition than (b). For $(b) \Rightarrow (a)$, if we set $f(t) = 0$ for $t \notin [a, b]$ then $f \in L^1(m)$ and so by corrollary 3.5.17 we get the final result.

If $F \in NBV$, then we usually write an integral of a function g with respect to the associated μ_F as $\int gdF$ or $\int g(x)dF(x)$. These types of integrals are called *Lebesgue-Stieltjes integrals.* I'm not sure why they were introduced, but this theorem was presented for generalizing integration by parts to these types of integrals

Theorem 3.5.21: Integration By Parts For Lebesgue-Stieltjes Integrals

Let $F, G \in NBV$ where at least one of them is continuous. Then for $-\infty < a < b < \infty$

$$\int_{(a,b]} FdG + \int_{(a,b]} GdF = F(b)G(b) - F(a)G(a)$$

***Proof*:**
F and G are linear combinations of increasing functions in NBV (by Theorem 3.27(a,b) in Folland), so a simple calculation shows that it suffices to assume F and G increasing. Suppose for the sake of definiteness that G is continuous, and let $\Omega = \{(x, y) : a < x \le y \le b\}$. We use Fubini's theorem to compute $\mu_F \times \mu_G(\Omega)$ in two ways:

$$\begin{aligned} \mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{(a,y]} dF(x)\, dG(x) = \int_{(a,b]} [F(y) - F(a)]\, dG(y) \\ &= \int_{(a,b]} F\, dG - F(a)[G(b) - G(a)], \end{aligned}$$

and since $G(x) = G(x-)$,

$$\begin{aligned} \mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{[x,b]} dG(y)\, dF(x) = \int_{(a,b]} [G(b) - G(x)]\, dF(x) \\ &= G(b)[F(b) - F(a)] - \int_{(a,b]} G\, dF. \end{aligned}$$

Subtracting these two equations, we get the desired result.

Finally, we end this chapter with a quick discussion about different terminology when it comes down to decomposing measures.

Definition 3.5.22: Discrete And Continuous Measures

Let μ be a complex Borel measure on $\mathbb{R}^n$

1. μ is called *discrete* if if there is a coutnable subset $\{x_i\}_i \subseteq \mathbb{R}^n$ such that $\sum_i c_i < \infty$ and $\mu = \sum c_i \delta_{x_i}$ where $\delta_{x_i}(x_i) = 1$ and is zero everywhere else (i.e. the pointmass at x).

2. μ is called *continuous* if for all $x \in \mathbb{R}^n$, $\mu(\{x\}) = 0$

It is easy to see that any measure can be broken down into $\mu = \mu_d + \mu_c$ where μ_d is discrete and μ_c is continuous. The main thing is to insure that μ_d is over a countable set: let $E = \{x \mid \mu(\{x\}) \neq 0\}$. Taking any countable subset $F \subseteq E$, then $\sum_{x \in F} \mu(\{x\})$ converges absolutely to $\mu(F)$ so $\left\{x \in E \;\middle|\; \mu(\{x\}) > k^{-1}\right\}$ is finite, and so it must be that E is countable, so $\mu_d(A) = \mu(A \cap E)$ is discrete and $\mu_c(A) = \mu(A \setminus E)$ is continuous.

If μ is discrete, then $\mu \perp m$, and if μ is continuous then $\mu \ll m$. Thus, we can further break down the decomposition into:

$$\mu = \mu_d + \mu_{ac} + \mu_{sc}$$

where μ_d is the discrete part, μ_{ac} is the absolutely continuous part, and μ_{sc} is the *singularly continuous* part. So far, we have mainly worked with measures with zero singularly continuous parts. There do exist measures with nonzero singularly continuous parts (Folland says the surface measure of the sphere for $n > 1$ is an example). An example for $n = 1$ is rarer, but they do exist. By proposition 3.5.13, we know that we are looking for a corresponding $F \in NBV$ such that $F' = 0$ a.e. In fact, the cantor function is such an example (when extended in the way we keep doing to make sure functions are in NBV). In fact, using the cantor function, we can construct an increasing function F that is continuous, $F' = 0$ a.e., and that is *strictly increasing*:

Example 3.5.23: Interesting Singularly Continuous Function

1. If $\{F_i\}_{i=1}^{\infty}$ is a sequence on nonegative increasing functions on $[a, b]$ such that $F(x) = \sum_1^{\infty} F_i(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_1^{\infty} F_i'(x)$ for a.e. $x \in [a, b]$ (It suffices to assume $F_i \in NBV$. Consider the measure μ_{F_i})

***Proof*:**
First, notice that since the F_i are nonnegative, by letting $F(x) = 0$ for $x < a$ and $F(b) = F(y)$ for $b < y$, we have that $F_i(-\infty) = 0$. Furthermore, the the function $G(x) = F(x+)$ and $G_i(x) = F_i(x+)$ is equal a.e. with the same derivative wherever it is defined (which it is a.e.), and so if G and G_i has a particular derivative, so do F and F_i. Thus, Without loss of generality, we'll assume $F, F_i \in NBV$.

Since they are in NBV, consider the induced complex borel measure μ_F and μ_{F_i} given by theorem 3.29. Since these measures are finite, they are σ-finite, and so admit a Radon-Nykodym derivative:

$$d\mu_F = f dm + d\lambda \qquad d\mu_{F_i} = f_i dm + d\lambda_i$$

where $f dm \ll m$, $f_i dm \ll m$ and $d\lambda \perp m$, $d\lambda_i \perp m$. We will show that $f dm = \sum f_i dm$. First, we need to show that $\mu_f = \sum \mu_{F_i}$, and then use the previous homework exercise (question 8) to conclude the equality.

By definition of F, $\mu_F((-\infty, b]) = \sum_i^\infty \mu_{F_i}((-\infty, b])$. Since the two measures agree on a generating set (closed rays form a generated set), they agree everywehre, meaning $\mu_F = \sum_i^\infty \mu_{F_i}$. Thus, we have that (using absolute convergence to re-arrange the sum)

$$f dm + d\lambda = d\mu_F = \sum_{i=1}^{\infty} d\mu_{F_i} = \sum f_i dm + \sum \lambda_i$$

Since $(\sum d\lambda_i) \perp m$ and $(\sum f_i dm) \ll m$, by question 8:

$$d\lambda = \sum d\lambda_i \qquad f dm = \sum f_i dm$$

Finally, we can invoke theorem 3.22 to connect the information about f and f_i to derivative information. By letting $E_r = (x, x+r]$ (since the F and F_i's are always positive), we get:

$$\begin{aligned}
F'(x) &= \lim_{r \to 0} \frac{F(x+r) - F(x)}{r} \\
&= \lim_{r \to 0} \frac{\mu_F(E_r)}{m(E_r)} \\
&= f(x) && \text{thm. 3.22} \\
&= \sum f_i(x) && m\text{-a.e.} \\
&= \sum \left(\lim_{r \to 0} \frac{\mu_{F_i}(E_r)}{m(E_r)} \right) && \text{thm 3.22} \\
&= \sum \left(\lim_{r \to 0} \frac{F_i(x+r) - F_i(x)}{r} \right) \\
&= \sum F'(x)
\end{aligned}$$

as we sought to show.

2. Let F denote the cantor function on $[0,1]$ (see 1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0,1]$ with rational endpoints, and let $F_n(x) = F\left(\frac{x - a_n}{b_n - a_n}\right)$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly

increasing on $[0,1]$ and $G' = 0$ a.e. (use exercise 39)

***Proof*:**
We break this proof down in many steps.

First, F is differentiable a.e. Notice that F is increasing and bounded, and so by theorem 3.27(e) F is a.e. differentiable. Notice that we can extend F as $F(x) = F(0)$ $x \leq 0$ and $F(1) = F(y)$ $1 \leq y$. In this way, $F \in NBV$.

Next, $F' = 0$ a.e. To see this, notice that by definition of F, the function is constant on C^c where C is the cantor set. Since the Cantor set is closed, C^c is open. On any open interval of C^c, F is constant, and so $F' = 0$ on C^c.

As a consequence, every F_n has zero derivative almost everywhere. At any point where F is differentiable:

$$F_n' = F'\left(\frac{x-a_n}{b_n-a_n}\right)\left(\frac{x-a_n}{b_n-a_n}\right)' = 0\left(\frac{x-a_n}{b_n-a_n}\right)' = 0$$

Showing that F_n' is a.e. 0.

Using this result, we can apply exercise 39 to get $G' = \sum_n 2^{-n}F_n$. Since each $2^{-n}F_n$ is a non-negative increasing function has derivative 0 a.e.:

$$G' = \sum_{n=1}^{\infty} 2^{-n}F_n' = 0 \qquad \text{a.e.}$$

Which gives us our first result.

Next, let's prove G is continuous. To see this, let $G_n = \sum_{k=1}^{n} 2^{-k}F_k$. Notice that $2^{-k}F_k$ is continuous being the product of two continuous functions (F_k is continuous being the composition of two continuous functions). By definition of F_k, the largest value if 1, and so $\left|\sum_1^n 2^{-k}F_n\right| \leq \sum_1^n 2^{-k} = 1 - 2^{-n}$. Then we can combine these two pieces of information to show that G_n uniformally converges to G. In particular, for all $\epsilon > 0$, chosoe $n > \frac{1}{2^\epsilon}$ so that $|G - G_n| < \epsilon$ for all $x \in [0,1]$. Thus, G is continuous.

Finally, to show G is strictly increasing, choose $x, y \in [a,b]$ such that $x < y$. Since x is strictly less than y, there exists an ϵ such that $x \notin [y-\epsilon, y]$. Since the rational's are dense, we can choose $[a_n, b_n]$ from our enumeration such that $x < a_n < b_n < y$ so that $x, y \notin [a_n, b_n]$. As a consequence, notice that $G(x)$ must be at least 2^{-n} away from $G(y)$, since:

$$\sum_n^{\infty} 2^{-n}F_n(x) < \sum_n^{\infty} 2^{-n}F_n(y)$$

and so G is strictly increasing.

I also want to give a quick word on convexity. I'd like to write more about convex functions, like their definition and some properties, but for now I'll just write down in bullet form what I found interesting:

1. Convex functions on compact intervals are absolutely continuous
2. As a consequence, FTC applies. In fact convex functions can be characterized like so : f is convex if and only if f' is strictly increasing. If f is twice differentiable, this leads to the

common notion of $f''(x) \geq 0$

3. In a sense, linear functions are a sepcial type of convex function. Convex functions are closed under addition and sclara multiplication. Thus, linear function being both convex and concave mean they are closd under subraction, which is why they form a vector-space, but convex functions do not. The inequalities of linear function are also all “trivial”, preciesly because of linearity, while those of convex functions are more complex.

4

Radon Measures and the Riesz Representation Theorem

Measures on the real line were built in chapter 2 from below, out of their values on intervals: a distribution function determines a premeasure, and theorem 1.4.2 extends it to a Borel measure. That route is tied to the order structure of $\mathbb{R}$ and does not survive the passage to a general space. On a locally compact Hausdorff space there is no distribution function to start from, and yet the measures one cares about there - the ones that integrate continuous functions of compact support - are pinned down by a single piece of data: the linear functional $f \mapsto \int f \, d\mu$ they induce on $C_c(X)$. This chapter shows that this functional loses nothing. Every positive linear functional on $C_c(X)$ is integration against a measure, and against essentially only one; this is the Riesz-Markov representation theorem, the second of the two bridges between set functions and functionals that this book builds (the first being the Lebesgue-Radon-Nikodym theorem).

The theorem turns the construction of a measure into the construction of an invariant, or otherwise constrained, functional - often a far easier task, since a functional can be built by averaging where a measure cannot. That is exactly the leverage the next chapter needs: on a locally compact group one writes down a left-invariant functional on $C_c(G)$ almost by hand, and Riesz-Markov converts it into the left-invariant measure - Haar measure (chapter 6) - on which all of harmonic analysis rests. The first section assembles the topological tools the representation theorem needs on a locally compact Hausdorff space; the second proves the theorem.

4.1 Radon Measures and $C_c(X)$ on Locally Compact Hausdorff Spaces

This section assembles the three ingredients the representation theorem needs: the spaces it lives on, the test functions that carry the functional, and the measures it produces. The spaces are the locally compact Hausdorff ones, general enough to include $\mathbb{R}^n$, every compact space, and every discrete space, yet rigid enough to supply the two technical tools it needs: bump functions and partitions of unity. The test functions are the continuous functions of compact support; the target measures are the Radon measures, singled out by a regularity package that couples a measure to the topology tightly enough for it to be recovered from its values on continuous functions.

The setting is not $\mathbb{R}^n$ but any space in which points can be separated and every point sits inside a compact neighbourhood. Both conditions are needed, and each does distinct work: the Hausdorff axiom makes compact sets closed and lets Urysohn's lemma run, while local compactness supplies the compact neighbourhoods that shrink test functions down to compact support.

Definition 4.1.1: Locally Compact Hausdorff Space

A topological space X is *locally compact* if every point has a compact neighbourhood, that is, for each $x \in X$ there is a compact set whose interior contains x. It is *Hausdorff* if any two distinct points have disjoint open neighbourhoods. A *locally compact Hausdorff space* (LCH space) is a space that is both.

The single structural fact drawn from these axioms, used at every turn below, is that compact sets can be surrounded by open sets with compact closure. An open set V is *precompact* if its closure $\overline{V}$ is compact. In a general space precompact opens need not exist in any useful supply; in an LCH space they are abundant, and abundant precisely where they are needed.

Proposition 4.1.2: Separating a Compact Set from the Boundary

Let X be an LCH space, $K \subseteq X$ compact, and $U \subseteq X$ open with $K \subseteq U$. Then there is a precompact open set V with

$$K \subseteq V \subseteq \overline{V} \subseteq U$$

***Proof*:**
Every point of K has a compact neighbourhood, so it has an open neighbourhood with compact closure; in a Hausdorff space, shrinking within that closure, one can arrange this precompact neighbourhood to sit inside U. Concretely, fix $x \in K$. Local compactness gives an open $W_x \ni x$ with $\overline{W_x}$ compact. The set $\overline{W_x}$ is a compact Hausdorff space, hence normal, and $\overline{W_x} \cap (X \setminus U)$ is a closed subset of it not containing x; normality separates x from it inside $\overline{W_x}$, producing an open $V_x \ni x$ with $\overline{V_x}$ compact (a closed subset of $\overline{W_x}$) and $\overline{V_x} \subseteq U$. As x ranges over K the sets V_x form an open cover; extract a finite subcover $V_{x_1}, \ldots, V_{x_m}$ and set $V = V_{x_1} \cup \cdots \cup V_{x_m}$. Then V is open, $K \subseteq V \subseteq U$, and $\overline{V} = \overline{V_{x_1}} \cup \cdots \cup \overline{V_{x_m}}$ is a finite union of compact sets, hence compact and contained in U, completing the proof.

The point-set groundwork here is exactly the local-compactness layer developed for topological groups in [7]. What that discussion supplies, and what proposition 4.1.2 repackages for the present purpose, is the capacity to insert a compact buffer $\overline{V}$ between an arbitrary compact set and any open set containing it. Every construction below threads a function through that buffer.

With the spaces fixed, the test functions come next. The functional in the representation theorem acts on continuous functions, but not all of them: a continuous function on $\mathbb{R}$ can fail to be integrable, and worse, the sup defining the eventual measure would diverge. Restricting to compact support removes both problems at once and keeps the space closed under the operations the proof needs.

Definition 4.1.3: $C_c(X)$ and Support

Let X be a topological space and $f\colon X \to \mathbb{R}$ continuous. The *support* of f is the closure of the set where f is nonzero,

$$\operatorname{supp} f = \overline{\{x \in X : f(x) \neq 0\}}$$

The space $C_c(X)$ consists of all continuous $f\colon X \to \mathbb{R}$ whose support is compact.

Two pieces of notation compress the constraints that recur in every argument below. They record, respectively, that a function caps a compact set from above and that a function is squeezed inside an open set.

Definition 4.1.4: The Relations $K \prec f$ and $f \prec U$

Let X be an LCH space, $K \subseteq X$ compact, and $U \subseteq X$ open. For $f \in C_c(X)$:

- write $K \prec f$ if $0 \leq f \leq 1$ on X and $f \equiv 1$ on K;
- write $f \prec U$ if $0 \leq f \leq 1$ on X and $\operatorname{supp} f \subseteq U$.

The combined assertion $K \prec f \prec U$ means both hold: $f \in C_c(X)$ takes values in $[0, 1]$, equals 1 on K, and is supported in U.

The relation $K \prec f \prec U$ is the exact profile of a *bump function*: a continuous plateau of height 1 over K, tapering to 0 before leaving U, and vanishing outside a compact set. The next lemma asserts that such bumps always exist. On $\mathbb{R}^n$ one can write a bump down by hand, mollifying the indicator of a slightly enlarged K; the content of the lemma is that the topology of an LCH space alone, with no smooth structure and no metric, already forces enough continuous functions into existence.

Lemma 4.1.5: Locally Compact Urysohn Lemma

Let X be an LCH space, $K \subseteq X$ compact, and $U \subseteq X$ open with $K \subseteq U$. Then there exists $f \in C_c(X)$ with

$$K \prec f \prec U$$

***Proof*:**
The strategy is to trap the whole construction inside a single compact Hausdorff space, where the classical Urysohn lemma applies, and then to check that the resulting function extends by zero to a genuine element of $C_c(X)$.

Step 1: Trap K in a compact buffer. By proposition 4.1.2 choose a precompact open V with

$$K \subseteq V \subseteq \overline{V} \subseteq U$$

The closure $\overline{V}$ is compact, and a compact Hausdorff space is normal; this normality is what the next step consumes.

Step 2: Apply the normal-space Urysohn lemma on $\overline{V}$. Inside $\overline{V}$ consider the two sets

$$A = K, \qquad B = \overline{V} \setminus V$$

Both are closed in $\overline{V}$: the set K is compact, hence closed in the Hausdorff space $\overline{V}$, and B is the intersection of the closed set $X \setminus V$ with $\overline{V}$. They are disjoint, since $K \subseteq V$ misses $B = \overline{V} \setminus V$. By the normal-space Urysohn lemma [7], there is a continuous $g \colon \overline{V} \to [0, 1]$ with $g \equiv 1$ on $A = K$ and $g \equiv 0$ on $B = \overline{V} \setminus V$.

Step 3: Extend by zero. Define $f \colon X \to \mathbb{R}$ by

$$f(x) = \begin{cases} g(x) & x \in \overline{V} \\ 0 & x \in X \setminus \overline{V} \end{cases}$$

This is well-defined: the two clauses are prescribed on the disjoint sets $\overline{V}$ and $X \setminus \overline{V}$, which partition X, so exactly one clause applies at each point. It remains to check that f is continuous, and that $\operatorname{supp} f$ is compact and contained in U.

Continuity. Because g vanishes on $\overline{V} \setminus V$, the function f vanishes on the open set $X \setminus \overline{V}$ as well as on $\overline{V} \setminus V$, hence on all of $X \setminus V$. Thus X is covered by two *closed* sets on each of which f is continuous: on $\overline{V}$, where $f = g$ is continuous, and on $X \setminus V$, where $f \equiv 0$. These two closed sets cover X, since $\overline{V} \cup (X \setminus V) = X$, and f is continuous on each; by the closed pasting lemma (a finite closed cover on each member of which a function is continuous glues to a continuous function) f is continuous on X.

Support. The set $\{f \neq 0\}$ is contained in V, so

$$\operatorname{supp} f = \overline{\{f \neq 0\}} \subseteq \overline{V} \subseteq U$$

As $\operatorname{supp} f$ is a closed subset of the compact set $\overline{V}$, it is compact, so $f \in C_c(X)$. By construction $0 \leq f \leq 1$ and $f \equiv 1$ on K, so $K \prec f$, and $\operatorname{supp} f \subseteq U$ gives $f \prec U$. Hence $K \prec f \prec U$, as we sought to show.

The lemma is Urysohn's separation theorem transported from normal spaces to LCH spaces, with compact support thrown in for free. The mechanism deserves a second look: local compactness is used exactly once, in proposition 4.1.2, to manufacture the compact buffer $\overline{V}$; once inside that buffer, the argument is purely the theory of normal spaces. This is the reason LCH is the natural home for the representation theorem. An LCH space need not be normal globally, but every compact set sits inside a compact, hence normal, buffer, and that local normality is all the separation the theory ever calls on.

Example 4.1.6: A Bump on the Real Line

Take $X = \mathbb{R}$, $K = [-1, 1]$, and $U = (-2, 2)$. The recipe of lemma 4.1.5 produces a continuous $f \in C_c(\mathbb{R})$ with $[-1, 1] \prec f \prec (-2, 2)$; an explicit representative is the piecewise-linear tent that ramps down to 0 over $[1, \frac{3}{2}]$, well before the endpoints of U,

$$f(x) = \begin{cases} 1 & |x| \leq 1 \\ 3 - 2|x| & 1 \leq |x| \leq \frac{3}{2} \\ 0 & |x| \geq \frac{3}{2} \end{cases}$$

Direct inspection confirms the profile: f is continuous, $0 \leq f \leq 1$, $f \equiv 1$ on $[-1, 1]$, and $\mathrm{supp} f = [-\frac{3}{2}, \frac{3}{2}] \subseteq (-2, 2)$, so indeed $\mathrm{supp} f$ is compact and contained in U. Ramping to 0 strictly inside U is what makes $\mathrm{supp} f \subseteq U$, hence $f \prec U$, hold. Here the precompact buffer of proposition 4.1.2 may be taken as $V = (-\frac{3}{2}, \frac{3}{2})$, with $\overline{V} = [-\frac{3}{2}, \frac{3}{2}]$ the compact interval in which the classical Urysohn lemma operates.

A single bump handles one compact set inside one open set. To integrate a function against a functional, or to compare a functional's values across an open cover, the bump must be replicated across several opens at once and the pieces made to sum to 1. This is the finite partition of unity, and it is the second and last topological tool the chapter needs. Its proof is a bootstrap: shrink the cover so that the closures stay inside the original opens, bump each shrunken piece, and normalise.

Lemma 4.1.7: Finite Partition of Unity

Let X be an LCH space and $K \subseteq X$ compact, covered by finitely many open sets $U_1, \ldots, U_n$. Then there exist $h_1, \ldots, h_n \in C_c(X)$ with $h_i \prec U_i$ for each i, such that

$$\sum_{i=1}^{n} h_i \equiv 1 \quad \text{on } K, \qquad 0 \leq \sum_{i=1}^{n} h_i \leq 1 \quad \text{on } X$$

The collection $\{h_i\}$ is called a *partition of unity on K subordinate to the cover $\{U_i\}$*.

***Proof*:**
Step 1: Shrink the cover. Each point $x \in K$ lies in some U_i, and by proposition 4.1.2 applied to the compact set $\{x\}$ and the open set U_i, it has a precompact open neighbourhood W_x with $\overline{W_x}$ compact and $\overline{W_x} \subseteq U_i$. The sets $\{W_x\}_{x \in K}$ cover the compact set K, so finitely many $W_{x_1}, \ldots, W_{x_m}$ suffice. For each index i let

$$K_i = \bigcup \{ \overline{W_{x_j}} : \overline{W_{x_j}} \subseteq U_i \}$$

a finite union of compact sets, hence compact, with $K_i \subseteq U_i$. Every $\overline{W_{x_j}}$ lands in at least one U_i, so $K \subseteq \bigcup_j \overline{W_{x_j}} \subseteq \bigcup_i K_i$; thus $\{K_i\}$ is a compact refinement with $K_i \subseteq U_i$ covering K.

Step 2: Bump each piece. By the locally compact Urysohn lemma lemma 4.1.5, for each i choose $g_i \in C_c(X)$ with

$$K_i \prec g_i \prec U_i$$

Then $g_i \equiv 1$ on K_i, $0 \leq g_i \leq 1$, and $\mathrm{supp} g_i \subseteq U_i$. Since $K \subseteq \bigcup_i K_i$, at every point of K some g_i equals 1, so the sum $\sum_i g_i \geq 1$ on K.

Step 3: Normalise, carefully at the boundary. The naive normalisation $g_i / \sum_j g_j$ fails where the denominator vanishes, so first separate K from that danger. Set $g = \sum_{i=1}^{n} g_i$; it is continuous, and $g \geq 1$ on K, so K is contained in the open set $\{g > 0\}$. Apply lemma 4.1.5 once more to obtain $g_0 \in C_c(X)$ with

$$K \prec g_0 \prec \{g > 0\}$$

Now $g + (1 - g_0)$ is strictly positive everywhere. Fix $x \in X$. If $g(x) > 0$ the sum is positive because $1 - g_0 \geq 0$. If instead $g(x) = 0$, then $x \notin \{g > 0\}$, and since $\operatorname{supp} g_0 \subseteq \{g > 0\}$ this forces $g_0(x) = 0$, so $1 - g_0(x) = 1 > 0$. Either way $g(x) + (1 - g_0(x)) > 0$, so the denominator below never vanishes. Define

$$h_i = \frac{g_i}{g + (1 - g_0)} \qquad (i = 1, \ldots, n)$$

Each h_i is continuous, since the denominator never vanishes, and $\operatorname{supp} h_i \subseteq \operatorname{supp} g_i \subseteq U_i$ with $\operatorname{supp} h_i$ compact, so $h_i \prec U_i$. On K one has $g_0 = 1$, whence the denominator equals g, and therefore

$$\sum_{i=1}^{n} h_i = \frac{\sum_i g_i}{g} = \frac{g}{g} = 1 \quad \text{on } K$$

On all of X, $\sum_i h_i = g/(g + (1 - g_0)) \leq 1$ because $1 - g_0 \geq 0$, and $\sum_i h_i \geq 0$ since each $h_i \geq 0$. Thus $0 \leq \sum_i h_i \leq 1$ throughout and $\sum_i h_i \equiv 1$ on K, completing the proof.

The partition of unity converts a global statement about K into a sum of local statements, one per open U_i. Writing $\sum_i h_i \equiv 1$ on K decomposes the constant function 1 (restricted to K) into pieces h_i each confined to a single U_i, so any object that behaves well on each U_i separately can be reassembled over K by summing the h_i against it. In the representation theorem this is precisely how a functional's local behaviour, controlled by a single bump at a time, is stitched into a global integral; the partition is the instrument that turns lemma 4.1.5 from a pointwise into a global tool.

Example 4.1.8: Splitting the Interval $[0, 1]$

Cover the compact set $K = [0, 1] \subseteq \mathbb{R}$ by the two opens $U_1 = (-1, \frac{2}{3})$ and $U_2 = (\frac{1}{3}, 2)$. Confining the ramps to the transition window $[\frac{5}{12}, \frac{7}{12}]$, which sits strictly inside the overlap $U_1 \cap U_2 = (\frac{1}{3}, \frac{2}{3})$, keeps each support clear of the boundaries of its U_i. An explicit subordinate partition of unity is

$$h_1(x) = \begin{cases} 1 & 0 \leq x \leq \frac{5}{12} \\ 6(\frac{7}{12} - x) & \frac{5}{12} \leq x \leq \frac{7}{12} \\ 0 & \text{otherwise} \end{cases} \qquad h_2(x) = \begin{cases} 6(x - \frac{5}{12}) & \frac{5}{12} \leq x \leq \frac{7}{12} \\ 1 & \frac{7}{12} \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Both are continuous with $\operatorname{supp} h_1 = [0, \frac{7}{12}] \subseteq U_1$ and $\operatorname{supp} h_2 = [\frac{5}{12}, 1] \subseteq U_2$ (each support lands strictly inside its open, since $\frac{7}{12} < \frac{2}{3}$ and $\frac{5}{12} > \frac{1}{3}$), so $h_1 \prec U_1$ and $h_2 \prec U_2$, and both lie in $C_c(\mathbb{R})$. On the transition window $[\frac{5}{12}, \frac{7}{12}]$ one computes $h_1(x) + h_2(x) = 6(\frac{7}{12} - x) + 6(x - \frac{5}{12}) = 1$, while on $[0, \frac{5}{12}]$ the sum is $1 + 0$ and on $[\frac{7}{12}, 1]$ it is $0 + 1$; thus $h_1 + h_2 \equiv 1$ on all of $K = [0, 1]$. Off $[0, 1]$ neither bump exceeds 1 and both taper to 0, so $0 \leq h_1 + h_2 \leq 1$ everywhere.

The last ingredient is the class of measures the representation theorem produces. A positive functional on $C_c(X)$ carries only as much information as continuous functions can see, so the measure it represents cannot be an arbitrary Borel measure; it must be one that its values on open and compact sets already determine. The regularity conditions below encode exactly that determinacy. Outer regularity computes any Borel set's measure from the open sets above it, inner regularity computes an open set's measure from the compact sets below it, and finiteness on compacts keeps the functional's values finite. Together they define a Radon measure.

Definition 4.1.9: Radon Measure

Let X be an LCH space and μ a Borel measure on X, that is, a measure defined on the Borel σ-algebra of X. Call μ a *Radon measure* if it satisfies:

1. *Finite on compacts:* $\mu(K) < \infty$ for every compact $K \subseteq X$;
2. *Outer regular on Borel sets:* for every Borel set E,
$$\mu(E) = \inf \{\mu(U) \mid E \subseteq U,\ U \text{ open}\}$$
3. *Inner regular on open sets:* for every open set U,
$$\mu(U) = \sup \{\mu(K) \mid K \subseteq U,\ K \text{ compact}\}$$

The asymmetry between conditions (2) and (3) is deliberate and worth dwelling on. Outer regularity is demanded on all Borel sets, but inner regularity only on open sets; requiring inner regularity everywhere would exclude some legitimate measures on pathological non-σ-finite spaces, whereas the open-set version is exactly what the representation theorem delivers and exactly what its applications use. On well-behaved spaces the gap closes: when X is σ-compact, a Radon measure is automatically inner regular on all Borel sets of finite measure, so the distinction is invisible on $\mathbb{R}^n$ and every space encountered in practice. The definition is stated at its most general so that the representation theorem can assert uniqueness without a side hypothesis.

The measures built earlier in this book from distribution functions are the prototype, and they are already Radon; nothing new need be checked, since the two regularity axioms were established for them in the chapters preceding chapter 4.

Example 4.1.10: Lebesgue-Stieltjes Measures Are Radon

Let $F\colon \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and let μ_F be the Borel measure it defines by theorem 1.4.2, characterised by $\mu_F((a, b]) = F(b) - F(a)$. Then μ_F is a Radon measure on $\mathbb{R}$.

Each Radon axiom is a result already in hand. *Finiteness on compacts:* a compact $K \subseteq \mathbb{R}$ lies in some bounded interval $(a, b]$, so $\mu_F(K) \le \mu_F((a, b]) = F(b) - F(a) < \infty$. *Outer and inner regularity:* these are precisely the two displayed identities of theorem 1.4.4, which computes $\mu_F(E) = \inf\{\mu_F(U) : U \supseteq E \text{ open}\}$ for every measurable E and $\mu_F(E) = \sup\{\mu_F(K) : K \subseteq E \text{ compact}\}$; the first is outer regularity on Borel sets, and the second gives inner regularity in particular on open sets. Thus every Lebesgue-Stieltjes measure is Radon, and taking $F(x) = x$ recovers Lebesgue measure on $\mathbb{R}$ as the archetypal Radon measure.

The example closes the loop with which chapter 4 opened. The question posed there, which set functions on a space deserve to be called measures, was answered on $\mathbb{R}$ by the distribution-function construction; the Radon axioms lift that answer to any LCH space by isolating the regularity that the $\mathbb{R}$ construction enjoyed automatically. What remains is to show that these axioms are not merely a convenient bookkeeping class but the exact image of the positive functionals on $C_c(X)$, which is the content of section 4.2.

A Radon measure was required to be inner regular only on open sets. On the σ-finite part of the space that extends to every Borel set, and it is this stronger regularity that the applications of the representation theorem draw on.

Proposition 4.1.11: Inner Regularity on σ-Finite Sets

Every Radon measure μ on an LCH space X is inner regular on each of its σ-finite Borel sets: if E is a Borel set that is σ-finite for μ, then

$$\mu(E) = \sup\{\mu(K) \mid K \subseteq E,\ K \text{ compact}\}$$

***Proof*:**
Consider first the case $\mu(E) < \infty$. Fix $\varepsilon > 0$. Outer regularity gives an open $U \supseteq E$ with $\mu(U) < \mu(E) + \varepsilon$, and inner regularity on the open set U gives a compact $F \subseteq U$ with $\mu(F) > \mu(U) - \varepsilon$. The set F need not lie in E, but its overspill sits in $U \setminus E$, a Borel set of measure $\mu(U \setminus E) = \mu(U) - \mu(E) < \varepsilon$. Outer regularity applied to $U \setminus E$ produces an open $V \supseteq U \setminus E$ with $\mu(V) < \varepsilon$. Put $K = F \setminus V$: it is a closed subset of the compact F, hence compact, and $K \subseteq E$ because everything of F outside E was covered by V. Finally, using $\mu(F) > \mu(U) - \varepsilon \geq \mu(E) - \varepsilon$,

$$\mu(K) = \mu(F) - \mu(F \cap V) \geq \mu(F) - \mu(V) > (\mu(E) - \varepsilon) - \varepsilon = \mu(E) - 2\varepsilon$$

As $K \subseteq E$ is compact and ε is arbitrary, the supremum of $\mu(K)$ over compact $K \subseteq E$ equals $\mu(E)$.

If instead $\mu(E) = \infty$, write $E = \bigcup_n E_n$ as an increasing union of Borel sets with $\mu(E_n) < \infty$; then $\mu(E_n) \to \infty$ by continuity from below. Given any N, choose n with $\mu(E_n) > N$; the finite case supplies a compact $K \subseteq E_n \subseteq E$ with $\mu(K) > N$. Hence the supremum is infinite, matching $\mu(E)$.

Corollary 4.1.12: Regularity of σ-Finite Radon Measures

Every σ-finite Radon measure is regular, that is, outer and inner regular on all Borel sets. In particular, if X is σ-compact then every Radon measure on X is regular.

***Proof*:**
Outer regularity on all Borel sets is part of definition 4.1.9, and when μ is σ-finite every Borel set is σ-finite, so proposition 4.1.11 supplies inner regularity on all Borel sets. If $X = \bigcup_n K_n$ is σ-compact with each K_n compact, then $\mu(K_n) < \infty$ by finiteness on compacts, so μ is σ-finite and the first statement applies.

4.2 The Riesz-Markov Representation Theorem

The bump functions and partitions of unity of section 4.1 are the tools; the measure they build is the payoff. A positive linear functional I on $C_c(X)$ assigns a number to each compactly supported continuous function, monotonically in the sense that $f \leq g$ forces $I(f) \leq I(g)$. Integration against a Radon measure is the obvious example: $f \mapsto \int_X f\, d\mu$ is linear, and it is positive because $f \geq 0$ makes the integrand nonnegative. The content of this section is that there are no other examples. Every positive functional on $C_c(X)$ integrates against one Radon measure, and against exactly one.

This converts the analytic datum I, a rule for averaging functions, into the geometric datum μ, a rule for sizing sets, and it is the mechanism by which the construction of Haar measure in chapter 6 will produce a measure out of the invariance of an averaging operation.

The functional records only what the measure does to continuous functions, so the measure must be reconstructed from that restriction. The reconstruction runs through an outer measure: define the size of an open set as the largest mass I can place inside it, extend to all sets by outer approximation, and appeal to the Carathéodory machine of theorem 1.3.4 to cut down to a σ-algebra on which the result is a genuine measure. The work is then to check that this measure is Radon, that it represents I, and that it is forced.

Theorem 4.2.1: Riesz-Markov Representation

Let X be a locally compact Hausdorff space and let $I\colon C_c(X) \to \mathbb{R}$ be a positive linear functional, so that $I(f) \geq 0$ whenever $f \geq 0$. Then there is a unique Radon measure μ on X such that

$$I(f) = \int_X f\,d\mu$$

for every $f \in C_c(X)$.

The proof occupies the remainder of the section. Existence is a construction in five stages: build a candidate set function, promote it to an outer measure, verify Carathéodory measurability of the open sets, extract the regularity that makes it Radon, and finally match I against integration. Uniqueness is a short argument once the regularity is in hand.

***Proof*:**
Write $C_c^+(X)$ for the nonnegative functions in $C_c(X)$. Throughout, $f \prec U$ means $f \in C_c^+(X)$ with $0 \leq f \leq 1$ and $\operatorname{supp} f \subseteq U$, and $K \prec f$ means $f \in C_c^+(X)$ with $0 \leq f \leq 1$ and $f \equiv 1$ on K, as fixed in section 4.1.

Step 1: The candidate set function. For $U \subseteq X$ open define

$$\mu(U) = \sup\{I(f) \mid f \prec U\} \tag{4.1}$$

and for an arbitrary $E \subseteq X$ define

$$\mu^*(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\} \tag{4.2}$$

Two consistency points must be settled before μ^* can be used. First, the supremum in (4.1) is over a nonempty set: the zero function satisfies $0 \prec U$, so $\mu(U) \geq I(0) = 0$, and $\mu(\emptyset) = 0$ because $f \prec \emptyset$ forces $f = 0$. Second, if U is itself open then the two definitions agree. Indeed $U \subseteq U$ is an admissible cover in (4.2), so $\mu^*(U) \leq \mu(U)$; and if $U \subseteq V$ with V open, then any $f \prec U$ also satisfies $f \prec V$, whence $\mu(U) \leq \mu(V)$, so $\mu(U) \leq \inf_V \mu(V) = \mu^*(U)$. Thus $\mu^*(U) = \mu(U)$ for open U, and no ambiguity arises from writing μ for both. The same monotonicity of (4.1) in the open set will be used repeatedly: $U \subseteq V$ open implies $\mu(U) \leq \mu(V)$.

Step 2: μ^* is an outer measure. The three axioms of definition 1.3.1 must be verified. That $\mu^*(\emptyset) = 0$ is immediate from $\mu(\emptyset) = 0$. Monotonicity is equally direct: if $A \subseteq B$ then every open

$U \supseteq B$ also contains A, so the infimum in (4.2) defining $\mu^*(A)$ is over a larger collection and hence no larger, giving $\mu^*(A) \leq \mu^*(B)$.

The one substantial axiom is countable subadditivity. The heart of it is the open case, where the partition of unity enters. Let $U = \bigcup_{j=1}^{\infty} U_j$ with each U_j open. The claim is

$$\mu(U) \leq \sum_{j=1}^{\infty} \mu(U_j) \tag{4.3}$$

To see this, take any $f \prec U$. Its support $K = \operatorname{supp} f$ is compact and contained in $\bigcup_j U_j$, so by compactness finitely many of the U_j already cover K, say $K \subseteq U_1 \cup \cdots \cup U_n$ after relabelling. By lemma 4.1.7 there are functions $h_i \prec U_i$ for $i = 1, \ldots, n$ with $\sum_{i=1}^{n} h_i \equiv 1$ on K. Since f is supported in K, on all of X one has $f = \sum_{i=1}^{n} f h_i$: at a point of K this is $f \cdot 1$, and off K every term vanishes because f does. Each product $f h_i$ lies in $C_c^+(X)$, is bounded by $h_i \leq 1$, and is supported in $\operatorname{supp} h_i \subseteq U_i$, so $f h_i \prec U_i$. Linearity of I then gives

$$I(f) = \sum_{i=1}^{n} I(f h_i) \leq \sum_{i=1}^{n} \mu(U_i) \leq \sum_{j=1}^{\infty} \mu(U_j)$$

where the first inequality is (4.1) applied to each $f h_i \prec U_i$. Taking the supremum over $f \prec U$ yields (4.3).

The passage to arbitrary sets is now formal. Let $\{E_j\}_{j=1}^{\infty}$ be any sets and put $E = \bigcup_j E_j$. If some $\mu^*(E_j) = \infty$ the inequality $\mu^*(E) \leq \sum_j \mu^*(E_j)$ is trivial, so assume every $\mu^*(E_j)$ is finite. Fix $\varepsilon > 0$. By (4.2) choose for each j an open $U_j \supseteq E_j$ with

$$\mu(U_j) \leq \mu^*(E_j) + \varepsilon 2^{-j}$$

Then $U = \bigcup_j U_j$ is open and contains E, so by (4.2), (4.3), and the choice of the U_j,

$$\mu^*(E) \leq \mu(U) \leq \sum_{j=1}^{\infty} \mu(U_j) \leq \sum_{j=1}^{\infty} \mu^*(E_j) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $\mu^*(E) \leq \sum_j \mu^*(E_j)$. Thus μ^* is an outer measure.

Step 3: Open sets are Carathéodory-measurable. By definition 1.3.3 a set E is μ^*-measurable exactly when

$$\mu^*(S) \geq \mu^*(S \cap E) + \mu^*(S \setminus E) \quad \text{for every } S \subseteq X \tag{4.4}$$

the reverse inequality being automatic from subadditivity. It suffices to establish (4.4) for every open E, and then only for test sets S with $\mu^*(S) < \infty$, since (4.4) is trivial when $\mu^*(S) = \infty$.

Fix an open E and a test set S with $\mu^*(S) < \infty$. Let $\varepsilon > 0$. By (4.2) choose an open $U \supseteq S$ with $\mu(U) \leq \mu^*(S) + \varepsilon$. It is enough to bound $\mu^*(U \cap E) + \mu^*(U \setminus E)$, because $S \cap E \subseteq U \cap E$ and $S \setminus E \subseteq U \setminus E$ and μ^* is monotone. The set $U \cap E$ is open, so by (4.1) there is $f \prec U \cap E$ with

$$I(f) > \mu(U \cap E) - \varepsilon \tag{4.5}$$

Now $K = \operatorname{supp} f$ is a compact subset of $U \cap E$, so $U \setminus K$ is open and contains $U \setminus E$. Apply (4.1) again on this open set: there is $g \prec U \setminus K$ with

$$I(g) > \mu(U \setminus K) - \varepsilon \geq \mu^*(U \setminus E) - \varepsilon \tag{4.6}$$

the last inequality by monotonicity, since $U \setminus E \subseteq U \setminus K$ (which holds because $K \subseteq U \cap E \subseteq E$). The two bump functions have disjoint supports: $\operatorname{supp} f \subseteq K$ while $\operatorname{supp} g \subseteq U \setminus K$. Consequently $f + g$ takes values in $[0, 1]$, because at any point at most one summand is nonzero, and $\operatorname{supp}(f + g) \subseteq U$, so $f + g \prec U$. Therefore $I(f) + I(g) = I(f + g) \leq \mu(U)$ by (4.1). Using $\mu^*(U \cap E) = \mu(U \cap E)$ from Step 1 and then (4.5), (4.6),

$$\mu^*(U \cap E) + \mu^*(U \setminus E) = \mu(U \cap E) + \mu^*(U \setminus E) < \big(I(f) + \varepsilon\big) + \big(I(g) + \varepsilon\big) \leq \mu(U) + 2\varepsilon$$

Since $\mu(U) \leq \mu^*(S) + \varepsilon$ and $S \cap E \subseteq U \cap E$, $S \setminus E \subseteq U \setminus E$, monotonicity gives

$$\mu^*(S \cap E) + \mu^*(S \setminus E) \leq \mu(U) + 2\varepsilon \leq \mu^*(S) + 3\varepsilon$$

Letting $\varepsilon \to 0$ yields (4.4). Hence every open set is μ^*-measurable.

The μ^*-measurable sets form a σ-algebra containing every open set, so they contain the σ-algebra generated by the open sets, that is, the Borel sets. By theorem 1.3.4 the collection $\mathcal{M}$ of μ^*-measurable sets is a σ-algebra and $\mu = \mu^*|_{\mathcal{M}}$ is a complete measure. Restricting attention to the Borel σ-algebra $\mathcal{B}_X \subseteq \mathcal{M}$, the resulting μ is a Borel measure. From now on μ denotes this measure; (4.1) continues to compute it on open sets by Step 1.

Step 4: μ is a Radon measure. Three properties from definition 4.1.9 require checking: finiteness on compact sets, outer regularity on Borel sets, and inner regularity on open sets.

For finiteness on compacts, the following bound does the work: for any compact K and any $\varphi \in C_c^+(X)$ with $\varphi \geq 1$ on K,

$$\mu(K) \leq I(\varphi) \tag{4.7}$$

To prove (4.7), put $U_\alpha = \{x \in X \mid \varphi(x) > \alpha\}$ for $0 < \alpha < 1$, an open set containing K because $\varphi \geq 1 > \alpha$ there. If $g \prec U_\alpha$, then $g \leq \alpha^{-1}\varphi$ everywhere: on U_α one has $\alpha^{-1}\varphi > 1 \geq g$, while off U_α the function g vanishes because $\operatorname{supp} g \subseteq U_\alpha$. Positivity of I then gives $I(g) \leq \alpha^{-1} I(\varphi)$. Taking the supremum over $g \prec U_\alpha$ and using monotonicity of μ,

$$\mu(K) \leq \mu(U_\alpha) \leq \alpha^{-1} I(\varphi)$$

Letting $\alpha \to 1$ yields (4.7). In particular, for a given compact K, lemma 4.1.5 supplies some f with $K \prec f$, and then $f \geq 1$ on K gives $\mu(K) \leq I(f) < \infty$. Thus μ is finite on every compact set.

Outer regularity is built into the construction. For a Borel set E, definition (4.2) is exactly $\mu(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\}$. Thus μ is outer regular on all Borel sets.

Inner regularity on open sets is the content of (4.1). Let U be open and let $c < \mu(U)$. By (4.1) there is $f \prec U$ with $I(f) > c$. Set $K = \operatorname{supp} f$, a compact subset of U. For any open V with $K \subseteq V$ one has $f \prec V$, so $I(f) \leq \mu(V)$; taking the infimum over such V and using $\mu(K) = \inf\{\mu(V) \mid K \subseteq V,\ V \text{ open}\}$ from (4.2) gives $I(f) \leq \mu(K)$. Hence $\mu(K) \geq I(f) > c$ with $K \subseteq U$ compact. Since $c < \mu(U)$ was arbitrary,

$$\mu(U) = \sup\{\mu(K) \mid K \subseteq U,\ K \text{ compact}\}$$

so μ is inner regular on open sets. Therefore μ is a Radon measure.

Step 5: $I(f) = \int_X f\,d\mu$ for all $f \in C_c(X)$. It suffices to prove

$$I(f) \leq \int_X f\,d\mu \quad \text{for every real } f \in C_c(X) \tag{4.8}$$

for then applying (4.8) to $-f$ gives $-I(f) \leq -\int_X f\,d\mu$, hence the reverse inequality, and the two together give equality. The complex case follows by taking real and imaginary parts.

Fix a real $f \in C_c(X)$ and let $K = \operatorname{supp} f$. Since f is continuous with compact support it is bounded, say $f(X) \subseteq [a, b]$ with $a < b$. Fix $\varepsilon > 0$ and choose real numbers

$$y_0 < a < y_1 < y_2 < \cdots < y_n = b$$

with $y_i - y_{i-1} < \varepsilon$ for each i and with $y_0 < a$ so that $f(x) > y_0$ for all x. Define the Borel sets

$$E_i = \{x \in K \mid y_{i-1} < f(x) \leq y_i\}, \qquad i = 1, \ldots, n$$

These are disjoint and their union is K, since every $x \in K$ has $a < f(x) \leq b$ and so falls in exactly one band. Each E_i is the intersection of K with the preimage of a half-open interval under the continuous f, hence Borel.

The plan is to trap f between step functions on the E_i and to compare I and the integral band by band. By outer regularity choose open sets $U_i \supseteq E_i$ with

$$\mu(U_i) < \mu(E_i) + \frac{\varepsilon}{n} \tag{4.9}$$

and, by shrinking, arrange in addition that $f(x) < y_i + \varepsilon$ for all $x \in U_i$: this is possible because $f < y_i + \varepsilon$ on the open set $\{f < y_i + \varepsilon\} \supseteq E_i$, and intersecting the chosen U_i with this open set preserves both $E_i \subseteq U_i$ and (4.9). The sets $U_1, \ldots, U_n$ cover the compact set K, so by lemma 4.1.7 there are $h_i \prec U_i$ with $\sum_{i=1}^n h_i \equiv 1$ on K. Since $\operatorname{supp} f = K$, exactly as in Step 2 one has $f = \sum_{i=1}^n f h_i$ on all of X.

Now estimate $I(f)$ from above. The function $\varphi = \sum_{i=1}^n h_i$ lies in $C_c^+(X)$ and equals 1 on K, so (4.7) applies to it. By linearity $I(\varphi) = \sum_i I(h_i)$, hence

$$\mu(K) \le \sum_{i=1}^n I(h_i) \tag{4.10}$$

On U_i the function f is bounded above by $y_i + \varepsilon$, and h_i is supported in U_i with $0 \le h_i \le 1$, so $fh_i \le (y_i + \varepsilon)h_i$ pointwise. Monotonicity and linearity of I then give $I(fh_i) \le (y_i + \varepsilon)I(h_i)$, for either sign of $y_i + \varepsilon$. Summing over i,

$$I(f) = \sum_{i=1}^n I(fh_i) \le \sum_{i=1}^n (y_i + \varepsilon) I(h_i)$$

The bound $I(h_i) \le \mu(U_i)$ from $h_i \prec U_i$ can only be inserted where the coefficient of $I(h_i)$ is nonnegative, so shift by the constant $|a| + \varepsilon$: write $y_i + \varepsilon = (y_i + \varepsilon + |a| + \varepsilon) - (|a| + \varepsilon)$, where the first factor is nonnegative because $y_i > a \ge -|a|$. The subtracted constant contributes $-(|a| + \varepsilon)\sum_i I(h_i) \le -(|a| + \varepsilon)\mu(K)$ by (4.10), since $|a| + \varepsilon \ge 0$. The nonnegative first factor multiplies $I(h_i) \le \mu(U_i)$ termwise. Carrying this out,

$$\begin{aligned} I(f) &\le \sum_{i=1}^n (y_i + \varepsilon + |a| + \varepsilon)\, \mu(U_i) - (|a| + \varepsilon)\, \mu(K) \\ &< \sum_{i=1}^n (y_i + \varepsilon + |a| + \varepsilon) \left(\mu(E_i) + \frac{\varepsilon}{n}\right) - (|a| + \varepsilon)\, \mu(K) \end{aligned}$$

Since $\bigsqcup_i E_i = K$, the terms carrying the constant $|a| + \varepsilon$ telescope: $\sum_i (|a| + \varepsilon)\mu(E_i) = (|a| + \varepsilon)\mu(K)$, cancelling the final term up to the ε/n remainders. Expanding and regrouping,

$$I(f) \le \sum_{i=1}^n (y_i + \varepsilon)\, \mu(E_i) + \frac{\varepsilon}{n} \sum_{i=1}^n (y_i + \varepsilon + |a| + \varepsilon)$$

The second sum is a remainder. Assuming $\varepsilon \le 1$, each term satisfies $y_i + \varepsilon + |a| + \varepsilon \le b + |a| + 2$, so summing n terms and multiplying by ε/n bounds the remainder by $C\varepsilon$ with $C = b + |a| + 2$, a constant depending only on f through the fixed interval $[a, b]$. For the main term, $x \in E_i$ gives $y_i < y_{i-1} + \varepsilon < f(x) + \varepsilon$, hence $y_i + \varepsilon < f(x) + 2\varepsilon$ on E_i. Since $y_i + \varepsilon$ is constant on E_i, integrating this pointwise bound gives

$$\sum_{i=1}^n (y_i + \varepsilon)\, \mu(E_i) \le \sum_{i=1}^n \int_{E_i} (f + 2\varepsilon)\, d\mu = \int_K f\, d\mu + 2\varepsilon\, \mu(K) = \int_X f\, d\mu + 2\varepsilon\, \mu(K)$$

where $\int_K f\, d\mu = \int_X f\, d\mu$ because f vanishes off K. Collecting the two bounds,

$$I(f) \le \int_X f\, d\mu + 2\varepsilon\, \mu(K) + C\varepsilon$$

Both $\mu(K) < \infty$ and C are finite and independent of ε, so letting $\varepsilon \to 0$ gives (4.8). This establishes $I(f) = \int_X f\, d\mu$ for all $f \in C_c(X)$, completing the existence half.

Step 6: Uniqueness. Suppose μ and ν are Radon measures with $\int_X f\,d\mu = I(f) = \int_X f\,d\nu$ for all $f \in C_c(X)$. It suffices to show $\mu(K) = \nu(K)$ for every compact K. Indeed, each open U is inner regular, $\mu(U) = \sup\{\mu(K) \mid K \subseteq U,\ K \text{ compact}\}$, so agreement on compacts forces $\mu(U) = \nu(U)$ on open sets; and each Borel E is outer regular, $\mu(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\}$, so agreement on open sets forces $\mu(E) = \nu(E)$ on all Borel sets.

Fix a compact K and $\varepsilon > 0$. By outer regularity of ν choose an open $U \supseteq K$ with $\nu(U) < \nu(K)+\varepsilon$; shrinking U if necessary, arrange $\overline{U}$ compact using local compactness. By lemma 4.1.5 take $f \in C_c(X)$ with $K \prec f \prec U$, so $\chi_K \leq f \leq \chi_U$ pointwise. Then

$$\mu(K) = \int_X \chi_K\,d\mu \leq \int_X f\,d\mu = \int_X f\,d\nu \leq \int_X \chi_U\,d\nu = \nu(U) < \nu(K) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $\mu(K) \leq \nu(K)$. Interchanging the roles of μ and ν gives $\nu(K) \leq \mu(K)$, hence $\mu(K) = \nu(K)$. By the reduction above, μ and ν agree on every open set and therefore on every Borel set. The Radon measure representing I is unique, completing the proof.

The theorem translates between two descriptions of the same data. A positive functional is an averaging rule with no set-theoretic content; a Radon measure is an assignment of size to sets. The construction shows that the functional already encodes the measure: the value $\mu(U) = \sup\{I(f) \mid f \prec U\}$ reads off the largest averaged mass I can concentrate inside U, and outer approximation then sizes every set. Regularity is not an extra hypothesis imposed on the output but the exact structure the construction produces, which is why the representing measure comes out Radon rather than merely Borel. Uniqueness records the converse, that a Radon measure is determined by its action on $C_c(X)$: two Radon measures agreeing on all continuous compactly supported functions agree on compact sets by squeezing χ_K between bump functions, and regularity carries that agreement to the whole Borel σ-algebra.

The identification is only as sharp as the class of measures it ranges over. Positivity of I is what forces the candidate to be a measure at all, since it is what makes (4.1) monotone in the open set. Restricting to Radon measures is what makes the correspondence a bijection: without regularity, distinct Borel measures can integrate every $f \in C_c(X)$ to the same value, so the measure is recovered only up to its Radon regularization. On a space where the topology already tames measures, this subtlety collapses. When X is second countable and locally compact, every Borel measure finite on compacts is automatically outer and inner regular, so the Radon qualifier is redundant and $I \mapsto \mu$ is a bijection onto all locally finite Borel measures.

Example 4.2.2: Lebesgue Measure from the Riemann Integral

Take $X = \mathbb{R}$ with its usual topology, a locally compact Hausdorff space, and let I be the Riemann integral,

$$I(f) = \int_{-\infty}^{\infty} f(x)\,dx$$

which is defined on every $f \in C_c(\mathbb{R})$ because such an f is continuous and supported in a compact interval, hence Riemann integrable there. Linearity is standard, and positivity holds because a nonnegative continuous function has nonnegative Riemann integral. Theorem 4.2.1 produces a unique Radon measure μ on $\mathbb{R}$ with $\int_{\mathbb{R}} f\,d\mu = \int_{-\infty}^{\infty} f(x)\,dx$ for all $f \in C_c(\mathbb{R})$.

This μ is Lebesgue measure. To read off its value on an interval $[a,b]$, sandwich the indicator $\chi_{[a,b]}$ between two continuous trapezoids. For small $\delta > 0$ with $2\delta < b-a$, let $f_\delta \in C_c(\mathbb{R})$ equal 1 on $[a,b]$, equal 0 outside $(a-\delta, b+\delta)$, and interpolate linearly on the two flanking intervals of width δ, and let $g_\delta \in C_c(\mathbb{R})$ equal 1 on $[a+\delta, b-\delta]$, equal 0 outside (a,b), and interpolate linearly on the two intervals of width δ. Then $g_\delta \le \chi_{[a,b]} \le f_\delta$ pointwise, so monotonicity of the integral gives

$$\int_{\mathbb{R}} g_\delta \, d\mu \le \mu([a,b]) \le \int_{\mathbb{R}} f_\delta \, d\mu$$

The two integrals are Riemann integrals by the representation. Computing each trapezoid's area as a central rectangle plus two triangles,

$$\int_{-\infty}^{\infty} g_\delta(x)\,dx = (b-a) - \delta, \qquad \int_{-\infty}^{\infty} f_\delta(x)\,dx = (b-a) + \delta$$

since f_δ adds two triangles of combined area δ to the rectangle $b-a$, while g_δ removes the same. Hence $(b-a)-\delta \le \mu([a,b]) \le (b-a)+\delta$, and letting $\delta \to 0$ gives $\mu([a,b]) = b-a$. Thus μ agrees with the Lebesgue measure built in theorem 1.4.2 from the distribution function $F(x) = x$. The Riesz-Markov route recovers Lebesgue measure from the bare positivity of the Riemann integral, with no explicit outer-measure construction on intervals: the general theorem does that work once and for all.

The same regularity that pins down the correspondence also lets an arbitrary measurable function be edited, off a set of small measure, into a continuous one. This is Lusin's theorem, the precise form of the slogan that "measurable functions are nearly continuous."

Theorem 4.2.3: Lusin's Theorem

Let X be a locally compact Hausdorff space, μ a Radon measure on X, and $f\colon X \to \mathbb{C}$ a measurable function vanishing outside a set of finite measure. For every $\varepsilon > 0$ there is $g \in C_c(X)$ with

$$\mu\big(\{x \in X \mid f(x) \ne g(x)\}\big) < \varepsilon,$$

and g may be taken with $\|g\|_\infty \le \|f\|_\infty$.

***Proof*:**
Let $A = \{x \mid f(x) \ne 0\}$, so $\mu(A) < \infty$. By inner regularity (corollary 4.1.12) fix a compact K_0 with $\mu(A \setminus K_0) < \varepsilon/2$ and a relatively compact open $U \supseteq K_0$; it suffices to produce $g \in C_c(X)$ supported in $\overline{U}$ agreeing with f off a subset of K_0 of measure $< \varepsilon/2$. Decomposing f into real and imaginary, positive and negative parts and rescaling, we may assume $0 \le f < 1$ on K_0 and write the binary expansion

$$f = \sum_{n=1}^{\infty} 2^{-n} \mathbf{1}_{T_n}, \qquad T_n \subseteq K_0 \text{ measurable.}$$

For each n, regularity (corollary 4.1.12) gives a compact K_n and open V_n with $K_n \subseteq T_n \subseteq V_n \subseteq U$ and $\mu(V_n \setminus K_n) < \varepsilon\, 2^{-n-1}$, and the locally compact Urysohn lemma (lemma 4.1.5) gives $h_n \in C_c(X)$ with $\mathbf{1}_{K_n} \le h_n \le \mathbf{1}_{V_n}$. The series $g = \sum_n 2^{-n} h_n$ converges uniformly, so $g \in C_c(X)$ with support in $\overline{U}$, and on $K_0 \setminus \bigcup_n (V_n \setminus K_n)$ each h_n equals $\mathbf{1}_{T_n}$, so $g = f$ there. Hence $f \ne g$

only on $(A \setminus K_0) \cup \bigcup_n (V_n \setminus K_n)$, of measure $< \varepsilon/2 + \sum_n \varepsilon\, 2^{-n-1} = \varepsilon$. Finally, composing g with the radial retraction $z \mapsto z\, \min(1, \|f\|_\infty/|z|)$ of $\mathbb{C}$ onto the closed disc of radius $\|f\|_\infty$ leaves g in $C_c(X)$, does not enlarge the disagreement set, and enforces $\|g\|_\infty \le \|f\|_\infty$.

5

Hausdorff Dimension and Kakeya Sets

The Lebesgue measure m assigns a notion of "volume" to measurable subsets of $\mathbb{R}^n$, and throughout the preceding chapters this has been the principal tool for quantifying the size of sets. But m is a blunt instrument for the many sets it assigns measure zero: a single point, a line segment in $\mathbb{R}^2$, and the classical Cantor set C all satisfy $m = 0$, yet they differ in every geometric sense. A single point is 0-dimensional, a line segment is 1-dimensional, and the Cantor set—uncountable, perfect, totally disconnected (proposition 1.4.9)—seems to occupy some intermediate regime. To make this distinction precise requires a family of measures that probes sets at every "dimensional scale," and the critical scale at which the measure transitions from infinity to zero defines a notion of dimension that applies to arbitrary subsets of $\mathbb{R}^n$.

This chapter develops this notion—the *Hausdorff dimension*—and then applies it to a geometric problem whose simplicity belies its difficulty: if a subset of $\mathbb{R}^n$ contains a unit line segment in every direction, how small can it be? Such sets, called *Besicovitch sets* or *Kakeya sets*, can have Lebesgue measure zero. The Kakeya conjecture asserts that despite having zero volume, they must still have full Hausdorff dimension n. The conjecture is resolved for $n = 2$ (Davies, 1971) and $n = 3$ (Wang–Zahl, 2025), but remains open in all higher dimensions. The tools developed in this chapter—Hausdorff measure, Minkowski dimension, and the geometry of directional sets—form the foundation for this problem and its connections to harmonic analysis.

5.1 Hausdorff Measure and Hausdorff Dimension

The construction of Hausdorff measure follows the same outer-measure paradigm used to build the Lebesgue measure in proposition 1.3.2: cover the set by countably many pieces, assign a cost to each piece, and take the infimum over all such covers. For Lebesgue measure the cost of a piece is its volume; for Hausdorff measure the cost is the s-th power of its diameter, where $s \geq 0$ is a free parameter. By varying s, one obtains a one-parameter family of measures: when s is large, coverings are cheap and the measure collapses to zero; when s is small, no covering is efficient and the measure blows up. The transition point between these two regimes is the Hausdorff dimension.

Definition 5.1.1: Hausdorff Measure

Let $E \subseteq \mathbb{R}^n$ and $s \geq 0$. For each $\delta > 0$, define the *δ-approximate s-dimensional Hausdorff measure* of E by

$$\mathcal{H}^s_\delta(E) = \inf \left\{ \sum_{j=1}^{\infty} (\operatorname{diam} U_j)^s \;\middle|\; E \subseteq \bigcup_{j=1}^{\infty} U_j, \quad \operatorname{diam} U_j < \delta \right\}$$

where the infimum is taken over all countable covers of E by sets $U_j \subseteq \mathbb{R}^n$ with $\operatorname{diam} U_j < \delta$. The convention $0^0 = 1$ is adopted.
As δ decreases, fewer covers are admissible, so $\mathcal{H}^s_\delta(E)$ is non-decreasing in $1/\delta$. Define the *s-dimensional Hausdorff measure* of E by

$$\mathcal{H}^s(E) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(E) = \sup_{\delta > 0} \mathcal{H}^s_\delta(E) \in [0, \infty]$$

The parameter δ forces the cover to use "fine" pieces: one cannot cheat by covering E with a single large set and paying a cost proportional only to the diameter of that set. As $\delta \to 0$, only the local geometry of E contributes. The parameter s controls the "dimensional weight" assigned to each covering set: a set of diameter r is penalized as r^s, so higher s makes small-diameter covers cheaper. This interplay between δ (resolution) and s (dimensional weight) is what allows $\mathcal{H}^s$ to detect dimension.

Proposition 5.1.2: $\mathcal{H}^s$ Is a Metric Outer Measure

For each $s \geq 0$, the function $\mathcal{H}^s : P(\mathbb{R}^n) \to [0, \infty]$ is an outer measure. Furthermore, $\mathcal{H}^s$ is a *metric outer measure*: if $A, B \subseteq \mathbb{R}^n$ satisfy $\operatorname{dist}(A, B) > 0$, then

$$\mathcal{H}^s(A \cup B) = \mathcal{H}^s(A) + \mathcal{H}^s(B)$$

As a consequence, by theorem 1.3.4, all Borel sets in $\mathbb{R}^n$ are $\mathcal{H}^s$-measurable.

***Proof*:**
Step 1: $\mathcal{H}^s$ is an outer measure. The argument follows the same template as proposition 1.3.2. First, $\mathcal{H}^s(\emptyset) = 0$: the empty cover (no sets) has cost 0. Monotonicity is immediate: if $A \subseteq B$, every cover of B is a cover of A, so $\mathcal{H}^s_\delta(A) \leq \mathcal{H}^s_\delta(B)$ for each δ, and the inequality persists in the limit.

For countable subadditivity, let $\{E_i\}_{i=1}^{\infty}$ be a countable collection and fix $\delta > 0$ and $\epsilon > 0$. For each i, choose a cover $\{U_j^i\}_{j=1}^\infty$ of E_i with diam $U_j^i < \delta$ and

$$\sum_{j=1}^{\infty} (\text{diam } U_j^i)^s \leq \mathcal{H}_\delta^s(E_i) + \frac{\epsilon}{2^i}$$

Then $\{U_j^i\}_{i,j}$ covers $\bigcup_i E_i$, so

$$\mathcal{H}_\delta^s\left(\bigcup_i E_i\right) \leq \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} (\text{diam } U_j^i)^s \leq \sum_{i=1}^{\infty} \mathcal{H}_\delta^s(E_i) + \epsilon$$

Since ϵ was arbitrary, $\mathcal{H}_\delta^s(\bigcup_i E_i) \leq \sum_i \mathcal{H}_\delta^s(E_i)$. Taking $\delta \to 0$, countable subadditivity of $\mathcal{H}^s$ follows.

Step 2: $\mathcal{H}^s$ is a metric outer measure. Let $d = \text{dist}(A, B) > 0$ and choose any $\delta < d$. If $\{U_j\}$ is a cover of $A \cup B$ with diam $U_j < \delta < d$, then each U_j satisfies $U_j \cap A = \emptyset$ or $U_j \cap B = \emptyset$: if some $x \in U_j \cap A$ and $y \in U_j \cap B$ existed, then $|x - y| \leq \text{diam } U_j < d = \text{dist}(A, B)$, a contradiction. Partitioning the cover into $\mathcal{A} = \{U_j \mid U_j \cap A \neq \emptyset\}$ and $\mathcal{B} = \{U_j \mid U_j \cap B \neq \emptyset\}$ gives

$$\sum_j (\text{diam } U_j)^s = \sum_{U_j \in \mathcal{A}} (\text{diam } U_j)^s + \sum_{U_j \in \mathcal{B}} (\text{diam } U_j)^s \geq \mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B)$$

Taking the infimum over all such covers: $\mathcal{H}_\delta^s(A \cup B) \geq \mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B)$. The reverse inequality holds by subadditivity, so equality holds for each $\delta < d$, and hence in the limit $\delta \to 0$.

Step 3: Borel measurability. The fact that every metric outer measure on $\mathbb{R}^n$ makes all Borel sets measurable (in the sense of definition 1.3.3) follows from a standard argument: the metric property implies that every open set G satisfies the Carathéodory condition $\mathcal{H}^s(S) = \mathcal{H}^s(S \cap G) + \mathcal{H}^s(S \cap G^c)$ for all test sets S, and since open sets generate the Borel σ-algebra $\mathcal{B}_{\mathbb{R}^n}$, the claim follows from theorem 1.3.4. The verification that open sets satisfy the Carathéodory condition uses a telescoping decomposition of $S \cap G$ into annular shells $A_k = \left\{x \in S \cap G \;\middle|\; \frac{1}{k+1} \leq d(x, G^c) < \frac{1}{k}\right\}$ with pairwise positive separation, and is left as an exercise.

This gives a second major application of theorem 1.3.4: the Carathéodory machinery, applied to the Hausdorff outer measure, produces a Borel measure for each dimensional parameter s. The resulting measure space $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \mathcal{H}^s)$ is the arena in which Hausdorff dimension is defined. Note that the σ-algebra of $\mathcal{H}^s$-measurable sets may be strictly larger than $\mathcal{B}_{\mathbb{R}^n}$ (just as the Lebesgue σ-algebra $\mathcal{L}$ is strictly larger than $\mathcal{B}_{\mathbb{R}}$), but for the purposes of dimension theory the Borel sets suffice.

A point of contrast with the Lebesgue outer measure from proposition 1.3.2 is worth emphasizing. The Lebesgue outer measure is constructed from a premeasure on intervals (or rectangles), and its restriction to Borel sets is the unique extension of that premeasure (theorem 1.3.7). The Hausdorff outer measure has no such premeasure origin—it is defined directly on all subsets of $\mathbb{R}^n$ using coverings by *arbitrary* sets, not just intervals. The cost functional $(\text{diam})^s$ plays the role that the volume functional played for the Lebesgue construction. This generality is what allows $\mathcal{H}^s$ to detect fractal behavior: by optimizing over all possible coverings (not just grids), the measure can adapt to the local geometry of irregular sets.

Before defining dimension, it is useful to ground the new construction in the familiar. The case $s = n$ recovers the Lebesgue measure up to a normalizing constant:

Proposition 5.1.3: Hausdorff Measure and Lebesgue Measure

Let m denote the Lebesgue measure on $\mathbb{R}^n$. Then for every Lebesgue measurable set $E \subseteq \mathbb{R}^n$:

$$\mathcal{H}^n(E) = \frac{2^n}{\omega_n}\, m(E)$$

where $\omega_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$ is the volume of the unit ball in $\mathbb{R}^n$. The constant $c_n = 2^n/\omega_n$ arises because $\mathcal{H}^n$ measures sets using $(\text{diam})^n$ while m uses side-lengths (or ball volumes).

***Proof Sketch*:**
For the inequality $\mathcal{H}^n(E) \leq c_n\, m(E)$: fix $\delta > 0$. By the definition of Lebesgue measure (or by theorem 2.6.1), E can be covered by countably many cubes $\{Q_j\}$ of side length $\delta/\sqrt{n}$, each having diameter $\text{diam}(Q_j) = \delta/\sqrt{n} \cdot \sqrt{n} = \delta$. The s-cost of each cube is $(\text{diam } Q_j)^n = \delta^n$, while its Lebesgue measure is $(\delta/\sqrt{n})^n$. Since $\sum_j m(Q_j) \geq m(E)$, the number of cubes needed is at least $m(E)/(\delta/\sqrt{n})^n = m(E) n^{n/2}/\delta^n$. Then

$$\mathcal{H}^n_\delta(E) \leq \sum_j \delta^n = N \cdot \delta^n$$

where N is the number of cubes. By choosing the cover to have $\sum_j m(Q_j) \leq m(E) + \epsilon$, the bound becomes $\mathcal{H}^n_\delta(E) \leq (m(E)+\epsilon) n^{n/2}$. Taking $\delta \to 0$ and $\epsilon \to 0$: $\mathcal{H}^n(E) \leq n^{n/2} m(E)$. (The sharper constant $c_n = 2^n/\omega_n$ requires a more careful argument using ball covers instead of cube covers.)

The reverse inequality $\mathcal{H}^n(E) \geq c_n\, m(E)$ uses the *isodiametric inequality*: among all sets of diameter d in $\mathbb{R}^n$, the ball of diameter d has the largest volume, namely $\omega_n(d/2)^n$. Hence for any cover $\{U_j\}$ of E with $\text{diam } U_j < \delta$:

$$m(E) \leq \sum_j m(U_j) \leq \sum_j \omega_n \left(\frac{\text{diam } U_j}{2}\right)^n = \frac{\omega_n}{2^n} \sum_j (\text{diam } U_j)^n$$

Taking the infimum over all such covers: $m(E) \leq \frac{\omega_n}{2^n} \mathcal{H}^n_\delta(E)$. Taking $\delta \to 0$: $m(E) \leq \frac{\omega_n}{2^n} \mathcal{H}^n(E)$, which rearranges to $\mathcal{H}^n(E) \geq \frac{2^n}{\omega_n} m(E) = c_n\, m(E)$.

5.1.4 Remark: The Isodiametric Inequality The isodiametric inequality states that among all subsets of $\mathbb{R}^n$ with a given diameter, the ball achieves the maximal volume. A proof using Steiner symmetrization can be found in Evans–Gariepy *Measure Theory and Fine Properties of Functions*. The inequality is logically independent of the rest of this chapter and is used only in the proof of proposition 5.1.3.

The normalization means $\mathcal{H}^n$ and m carry the same information on $\mathbb{R}^n$. Some authors define $\mathcal{H}^n$ with the normalizing constant built in; the convention used here (no constant in the definition, constant appearing in the comparison) is more common in fractal geometry.

Example 5.1.5: Hausdorff Measure at Integer Dimensions

1. $\mathcal{H}^0$ is the counting measure. If $|E| = k$ is finite, then $\mathcal{H}^0(E) = k$: cover each point by a set of diameter $< \delta$, and each contributes $(\mathrm{diam})^0 = 1$. If E is infinite, $\mathcal{H}^0(E) = \infty$. This is consistent with the convention $0^0 = 1$.

2. If $\gamma : [a, b] \to \mathbb{R}^2$ is a C^1 curve with $|\gamma'(t)| > 0$, then $\mathcal{H}^1(\gamma([a, b]))$ equals the arc length $\int_a^b |\gamma'(t)|\, dt$. The Hausdorff 1-measure thus recovers the classical notion of length for smooth curves. More generally, $\mathcal{H}^k$ restricted to a smooth k-dimensional submanifold of $\mathbb{R}^n$ agrees (up to a normalizing constant) with the k-dimensional volume form induced by the ambient metric.

3. For non-integer values of s, the measure $\mathcal{H}^s$ does not correspond to any classical geometric quantity. The set E at which $\mathcal{H}^s(E)$ transitions from infinity to zero has "effective dimension" s—it is "too large" to be measured by any $\mathcal{H}^t$ with $t < s$ (giving infinity) and "too small" for $\mathcal{H}^t$ with $t > s$ (giving zero). The next proposition makes this transition precise.

The central observation about $\mathcal{H}^s$ is that for a given set E, the value $\mathcal{H}^s(E)$ jumps from ∞ to 0 at a single critical exponent. This jump is a consequence of the power-law relationship between the cost functions at different exponents:

Proposition 5.1.6: Critical Exponent

Let $E \subseteq \mathbb{R}^n$. If $\mathcal{H}^s(E) < \infty$ for some $s \geq 0$, then $\mathcal{H}^t(E) = 0$ for all $t > s$. Equivalently, if $\mathcal{H}^t(E) > 0$ for some t, then $\mathcal{H}^s(E) = \infty$ for all $s < t$.

***Proof*:**
Fix $\delta > 0$ and let $\{U_j\}$ be a cover of E with diam $U_j < \delta$. For $t > s$:

$$\sum_j (\text{diam } U_j)^t = \sum_j (\text{diam } U_j)^{t-s} (\text{diam } U_j)^s \leq \delta^{t-s} \sum_j (\text{diam } U_j)^s$$

Taking the infimum: $\mathcal{H}^t_\delta(E) \leq \delta^{t-s} \mathcal{H}^s_\delta(E)$. If $\mathcal{H}^s(E) < \infty$, then $\mathcal{H}^s_\delta(E) \leq \mathcal{H}^s(E) + 1$ for small δ, so $\mathcal{H}^t_\delta(E) \leq \delta^{t-s}(\mathcal{H}^s(E) + 1) \to 0$ as $\delta \to 0$. Thus $\mathcal{H}^t(E) = 0$.

This jump behavior defines a unique threshold:

Definition 5.1.7: Hausdorff Dimension

Let $E \subseteq \mathbb{R}^n$. The *Hausdorff dimension* of E is

$$\dim_H(E) = \inf \{s \geq 0 \mid \mathcal{H}^s(E) = 0\} = \sup \{s \geq 0 \mid \mathcal{H}^s(E) = \infty\}$$

where the two expressions are equal by proposition 5.1.6. At the critical value $s_0 = \dim_H(E)$, the measure $\mathcal{H}^{s_0}(E)$ can be 0, ∞, or any value in between.

By proposition 5.1.3, $\dim_H(E) = n$ whenever $m(E) > 0$, and $\dim_H(E) \leq n$ for all $E \subseteq \mathbb{R}^n$. The utility of $\dim_H$ is precisely for sets where $m(E) = 0$: it provides a finer stratification of null sets.

A countable set has dimension 0; a smooth curve in $\mathbb{R}^2$ has dimension 1; a "thick dust" like the Cantor set has some intermediate fractional dimension. The value $\mathcal{H}^{s_0}(E)$ at the critical exponent $s_0 = \dim_H(E)$ itself is often difficult to compute and can be 0, positive and finite, or ∞. For many problems in geometric measure theory (including the Kakeya conjecture), only the dimension matters, and the precise value of $\mathcal{H}^{s_0}(E)$ is secondary.

Proposition 5.1.8: Properties of Hausdorff Dimension

Let $E, F \subseteq \mathbb{R}^n$ and let $\{E_i\}_{i=1}^{\infty} \subseteq \mathbb{R}^n$ be a countable collection.

1. **(Monotonicity)** If $E \subseteq F$, then $\dim_H(E) \le \dim_H(F)$.
2. **(Countable stability)** $\dim_H\left(\bigcup_{i=1}^{\infty} E_i\right) = \sup_i \dim_H(E_i)$.
3. **(Countable sets)** If E is countable, then $\dim_H(E) = 0$.
4. **(Lipschitz maps)** If $f : E \to \mathbb{R}^m$ satisfies $|f(x) - f(y)| \le L|x - y|$ for all $x, y \in E$, then $\dim_H(f(E)) \le \dim_H(E)$. In particular, $\dim_H$ is invariant under bi-Lipschitz maps.

***Proof*:**

1. If $E \subseteq F$, then $\mathcal{H}^s(E) \le \mathcal{H}^s(F)$ for all s, so $\mathcal{H}^s(F) = 0$ implies $\mathcal{H}^s(E) = 0$, giving $\dim_H(E) \le \dim_H(F)$.
2. Let $d = \sup_i \dim_H(E_i)$. By monotonicity, $\dim_H(\bigcup_i E_i) \ge d$. For the reverse, if $s > d$, then $\mathcal{H}^s(E_i) = 0$ for all i, so by countable subadditivity $\mathcal{H}^s(\bigcup_i E_i) \le \sum_i \mathcal{H}^s(E_i) = 0$, hence $\dim_H(\bigcup_i E_i) \le s$. Since $s > d$ was arbitrary, $\dim_H(\bigcup_i E_i) \le d$.
3. A single point $\{x\}$ has $\dim_H(\{x\}) = 0$ since $\mathcal{H}^s(\{x\}) = 0$ for all $s > 0$ (cover by a ball of diameter δ: cost $= \delta^s \to 0$). By countable stability, any countable set has dimension 0.
4. If $\{U_j\}$ covers E with diam $U_j < \delta$, then $\{f(U_j \cap E)\}$ covers $f(E)$ with diam $f(U_j \cap E) \le L \operatorname{diam} U_j < L\delta$. Thus

$$\mathcal{H}^s_{L\delta}(f(E)) \le \sum_j (L \operatorname{diam} U_j)^s = L^s \sum_j (\operatorname{diam} U_j)^s$$

   Taking the infimum and the limit $\delta \to 0$: $\mathcal{H}^s(f(E)) \le L^s \mathcal{H}^s(E)$. If $\mathcal{H}^s(E) = 0$ then $\mathcal{H}^s(f(E)) = 0$, giving $\dim_H(f(E)) \le \dim_H(E)$. If f is bi-Lipschitz, apply the same argument to f^{-1} to get the reverse inequality.

Property (4) implies that $\dim_H$ is a bi-Lipschitz invariant but *not* a topological invariant. The fat Cantor set C_α of measure $\alpha > 0$ from lemma 1.4.10 satisfies $\dim_H(C_\alpha) = 1$ (since $m(C_\alpha) > 0$ forces $\mathcal{H}^1(C_\alpha) > 0$), yet C_α is homeomorphic to the standard Cantor set C by corollary 1.4.16. Since the homeomorphism need not be Lipschitz, the dimensions are free to differ. This distinction matters: the Kakeya conjecture (which asserts that certain sets have full Hausdorff dimension) would be trivially false if dimension were a topological invariant, since any compact set is homeomorphic to a set of dimension 0 in a sufficiently large ambient space.

Property (2)—countable stability—is the feature that most sharply distinguishes Hausdorff dimension from the Minkowski dimension studied in the next section. It ensures that the dimension of a countable union is controlled by the dimensions of the individual pieces, a property that is indispensable for measure-theoretic arguments but that the Minkowski dimension fails to satisfy.

The upper bound on dimension is typically obtained by exhibiting an explicit covering and computing its cost, as in the standard representation of the Cantor set by stage-k intervals. Lower bounds are harder: one must show that *no* covering can be too cheap. The most effective general technique is to "distribute mass" on the set and use the following duality:

Lemma 5.1.9: Mass Distribution Principle

Let $E \subseteq \mathbb{R}^n$ and suppose there exists a Borel measure μ with $\mu(E) > 0$ and a constant $C > 0$ such that

$$\mu(B(x,r)) \leq Cr^s \qquad \text{for all } x \in \mathbb{R}^n \text{ and all } r > 0$$

Then $\dim_H(E) \geq s$.

***Proof*:**
Let $\{U_j\}$ be any cover of E with diam $U_j < \delta$. Each U_j is contained in a ball of radius diam U_j, so $\mu(U_j) \leq C(\text{diam } U_j)^s$. Then

$$0 < \mu(E) \leq \mu\left(\bigcup_j U_j\right) \leq \sum_j \mu(U_j) \leq C \sum_j (\text{diam } U_j)^s$$

Hence $\sum_j (\text{diam } U_j)^s \geq \mu(E)/C > 0$ for every admissible cover, giving $\mathcal{H}^s_\delta(E) \geq \mu(E)/C > 0$ and thus $\mathcal{H}^s(E) > 0$, so $\dim_H(E) \geq s$.

The lemma reduces dimension lower bounds to the construction of a measure with controlled local growth. The idea is a duality: to show $\dim_H(E) \geq s$, one produces a measure supported on E that "sees" the set as s-dimensional. The condition $\mu(B(x,r)) \leq Cr^s$ says that μ distributes its mass at rate r^s per ball—matching the growth rate of an s-dimensional object. If such a measure exists, no covering can be too efficient, and the Hausdorff measure at exponent s must be positive.

For self-similar sets, the natural "equidistributed" measure typically satisfies the required ball condition. For the Cantor set, this is the Cantor measure, which assigns equal mass to each interval at every stage of the construction.

Example 5.1.10: Hausdorff Dimension of the Middle-Thirds Cantor Set

Let C denote the standard middle-thirds Cantor set (definition 1.4.8). Then $\dim_H(C) = \frac{\log 2}{\log 3} \approx 0.631$.

Upper bound. At stage k of the construction, C is covered by 2^k closed intervals, each of length 3^{-k}. These intervals have diameter 3^{-k}, so for $s = \frac{\log 2}{\log 3}$:

$$\mathcal{H}^s_{3^{-k}}(C) \leq 2^k \cdot (3^{-k})^s = 2^k \cdot 3^{-k\log 2/\log 3} = 2^k \cdot 2^{-k} = 1$$

The computation uses $3^{-\log 2/\log 3} = 2^{-1}$, which follows from exponentiating $-\frac{\log 2}{\log 3} \cdot \log 3 = -\log 2$. Thus $\mathcal{H}^s(C) \leq 1 < \infty$, giving $\dim_H(C) \leq s$. Note that for any $s' < s$, the same covering gives $\mathcal{H}^{s'}_{3^{-k}}(C) \leq 2^k \cdot 3^{-ks'} = (2 \cdot 3^{-s'})^k$, and since $s' < s$ implies $2 \cdot 3^{-s'} > 2 \cdot 3^{-s} = 1$, this diverges as $k \to \infty$. The covering at exponent s is thus "perfectly balanced"—the cost neither diverges nor converges to zero.

Lower bound. Define the Cantor measure μ on C as follows: at stage k, assign mass 2^{-k} to each of the 2^k intervals I_k^j. Concretely, if E is a union of stage-k intervals, then $\mu(E) =$ (number of stage-k intervals in E) $\cdot 2^{-k}$. This definition is consistent across stages (each stage-k interval contains exactly two stage-$(k+1)$ intervals, and $2 \cdot 2^{-(k+1)} = 2^{-k}$), so it defines a premeasure on the algebra of finite unions of stage-k intervals, and extends to a Borel measure μ on C with $\mu(C) = 1$ by theorem 1.3.7. The claim is that $\mu(B(x,r)) \leq Cr^s$ for a constant C and $s = \frac{\log 2}{\log 3}$.

For any $x \in C$ and $r > 0$, choose k so that $3^{-(k+1)} \leq r < 3^{-k}$. The ball $B(x,r)$ intersects at most 2 stage-k intervals (since consecutive stage-k intervals are separated by a gap of length $\geq 3^{-k}/2$). Therefore

$$\mu(B(x,r)) \leq 2 \cdot 2^{-k} = 2^{-(k-1)}$$

Since $r \geq 3^{-(k+1)}$, raising to the power s:

$$r^s \geq 3^{-(k+1)s} = 3^{-s} \cdot 3^{-ks} = 3^{-s} \cdot 2^{-k}$$

Combining: $\mu(B(x,r)) \leq 2^{-(k-1)} = 2 \cdot 2^{-k} \leq 2 \cdot 3^s \cdot r^s$. By the mass distribution principle, $\dim_H(C) \geq s = \frac{\log 2}{\log 3}$.

The same technique applies to any self-similar Cantor set where a fixed fraction is removed at each step. The formula $\dim_H = \log(\text{pieces})/\log(\text{scaling factor})$ is a general phenomenon for self-similar sets satisfying an "open set condition," though the proof of this general statement requires additional machinery beyond the mass distribution principle.

Example 5.1.11: Further Dimension Computations

1. **Fat Cantor sets.** The fat Cantor set C_α with $m(C_\alpha) = \alpha > 0$ satisfies $\dim_H(C_\alpha) = 1$. The lower bound follows from $\mathcal{H}^1(C_\alpha) \geq c_1\, m(C_\alpha) > 0$ by proposition 5.1.3, and $\dim_H(C_\alpha) \leq 1$ since $C_\alpha \subseteq \mathbb{R}$. Despite being homeomorphic to the standard Cantor set by corollary 1.4.16, the two sets have different Hausdorff dimensions ($\log 2/\log 3 < 1$). Hausdorff dimension is thus a metric—not topological—invariant.

2. **The rationals and irrationals.** $\dim_H(\mathbb{Q}) = 0$ since $\mathbb{Q}$ is countable. By proposition 5.1.8(2), $1 = \dim_H(\mathbb{R}) = \max\{\dim_H(\mathbb{Q}), \dim_H(\mathbb{R} \setminus \mathbb{Q})\}$, so $\dim_H(\mathbb{R} \setminus \mathbb{Q}) = 1$.

3. **Smooth submanifolds.** If $M \subseteq \mathbb{R}^n$ is a smooth k-dimensional submanifold (for instance, a smooth curve in $\mathbb{R}^2$ or a smooth surface in $\mathbb{R}^3$), then $\dim_H(M) = k$. The upper bound follows by covering M with coordinate patches, each of which is bi-Lipschitz equivalent to an open subset of $\mathbb{R}^k$. The lower bound follows from proposition 5.1.3 applied within each coordinate chart: the k-dimensional Lebesgue measure of the chart is positive, so $\mathcal{H}^k(M) > 0$. This confirms that Hausdorff dimension extends the classical notion of dimension for smooth objects.

The computations above illustrate a general principle: for "tame" sets (smooth manifolds, self-similar fractals satisfying the open set condition), Hausdorff dimension is computable and agrees with geometric intuition. The subtlety arises for sets constructed through limiting processes without self-similar structure—such as the Besicovitch sets studied in the next section, where even establishing a lower bound on $\dim_H$ requires fundamentally new ideas.

Exercise 5.1.1

1. Let C_r be the Cantor set obtained by removing a central interval of length $r \in (0,1)$ at each stage (the middle-thirds Cantor set corresponds to $r = 1/3$). Show that $\dim_H(C_r) = \frac{\log 2}{\log(2/(1-r))}$.
2. Let $f : [0,1] \to \mathbb{R}$ be α-Hölder continuous, that is, $|f(x)-f(y)| \le C|x-y|^\alpha$ for some $C > 0$ and $\alpha \in (0,1]$. Show that $\dim_H(\operatorname{graph}(f)) \le 2-\alpha$, where $\operatorname{graph}(f) = \{(x, f(x)) \mid x \in [0,1]\} \subseteq \mathbb{R}^2$. (*Hint*: cover $[0,1]$ by N intervals of length $1/N$; each piece of the graph is contained in a rectangle of width $1/N$ and height C/N^α.)
3. Show that every metric outer measure on $\mathbb{R}^n$ makes all Borel sets measurable. (*Hint*: it suffices to show that every open set G satisfies the Carathéodory condition. For a test set S, write $S \cap G = \bigcup_{k=1}^\infty A_k$ where $A_k = \left\{ x \in S \cap G \;\middle|\; \frac{1}{k+1} \le d(x, G^c) < \frac{1}{k} \right\}$, and use the metric property on A_k's with indices differing by ≥ 2.)
4. Show that $\dim_H(rE) = \dim_H(E)$ for all $r \neq 0$ and $\dim_H(E + a) = \dim_H(E)$ for all $a \in \mathbb{R}^n$ (compare with proposition 1.4.7).
5. Construct an uncountable set $E \subseteq \mathbb{R}$ with $\dim_H(E) = 0$. (*Hint*: consider a Cantor-type construction where the pieces shrink faster than geometrically.)

5.2 Minkowski Dimension

The Hausdorff dimension has excellent theoretical properties—countable stability, Lipschitz invariance, and a clean relationship to Hausdorff measure—but computing it can be demanding, since the definition requires optimizing over *all* countable covers. In practice, a coarser notion that counts δ-balls rather than arbitrary covers is often more tractable. This leads to the *Minkowski dimension* (also called the box-counting dimension), which uses only one type of covering set (balls or cubes of a fixed radius) and extracts the dimension from the growth rate of the covering number.

Definition 5.2.1: Minkowski Dimension

Let $E \subseteq \mathbb{R}^n$ be a bounded set. For each $\delta > 0$, let $N(E, \delta)$ denote the smallest number of sets of diameter at most δ needed to cover E. Define the *upper Minkowski dimension* and *lower Minkowski dimension* of E by

$$\overline{\dim}_M(E) = \limsup_{\delta \to 0} \frac{\log N(E,\delta)}{-\log \delta} \qquad \underline{\dim}_M(E) = \liminf_{\delta \to 0} \frac{\log N(E,\delta)}{-\log \delta}$$

If the two coincide, their common value is the *Minkowski dimension* $\dim_M(E)$.

The ratio $\log N(E,\delta)/(-\log \delta)$ measures the exponent in the scaling $N(E,\delta) \sim \delta^{-d}$: if covering E at resolution δ requires $\sim \delta^{-d}$ balls, then E "looks d-dimensional" at that scale. The $\limsup$ and $\liminf$ capture the worst-case and best-case scaling behavior as $\delta \to 0$.

There is an equivalent volume-based formulation that is sometimes more convenient. Let $E_\delta = \{x \in \mathbb{R}^n \mid d(x, E) < \delta\}$ denote the open δ-neighborhood of E. The volume $m(E_\delta)$ grows as E is "thickened" by δ, and the rate of growth encodes the dimension:

$$\overline{\dim}_M(E) = n - \liminf_{\delta \to 0} \frac{\log m(E_\delta)}{\log \delta}$$

To see the connection: E_δ is approximately the union of the $N(E, \delta)$ balls of radius δ in an optimal cover, so $m(E_\delta) \approx N(E, \delta) \cdot \omega_n \delta^n$. Taking logarithms and dividing by $\log \delta$ recovers the covering-number formulation. This perspective is particularly useful for relating Minkowski dimension to Lebesgue measure estimates, which is the context in which it appears in the Kakeya conjecture.

The covering number $N(E, \delta)$ admits several equivalent formulations, and these equivalences are useful in different computational contexts:

Proposition 5.2.2: Equivalent Formulations

Let $E \subseteq \mathbb{R}^n$ be bounded. The following quantities all have the same lim sup and lim inf as $\delta \to 0$ when divided by $-\log \delta$:

1. $N(E, \delta)$: smallest number of sets of diameter $\leq \delta$ covering E.
2. $N'(E, \delta)$: smallest number of closed balls of radius δ covering E.
3. $M(E, \delta)$: largest number of disjoint balls of radius δ centered at points of E.
4. $N_{\text{cube}}(E, \delta)$: number of cubes in a δ-grid of $\mathbb{R}^n$ that intersect E.

***Proof*:**
The equivalences follow from volume comparison. By definition, $N(E, \delta) \leq N'(E, \delta)$ since balls of radius δ have diameter 2δ, but a cover by sets of diameter $\leq \delta$ can be replaced by enclosing balls, giving $N'(E, \delta) \leq N(E, \delta)$. For $M(E, \delta)$ and $N'(E, \delta)$: any ball of radius δ covers at most one center of a disjoint packing, so $M(E, \delta) \leq N'(E, \delta)$. Conversely, the balls of radius 2δ centered at packing points cover E, so $N'(E, 2\delta) \leq M(E, \delta)$.

For N_{cube}: any cube of side δ has diameter $\delta\sqrt{n}$, and any ball of radius δ is contained in at most 3^n cubes of side δ, giving $N_{\text{cube}}(E, \delta) \leq 3^n N'(E, \delta)$ and $N'(E, \delta\sqrt{n}) \leq N_{\text{cube}}(E, \delta)$.

In each case, passing from δ to $c\delta$ for a fixed constant c shifts $\log N$ by at most $\log(c^n)$, which is $O(1)$ and hence negligible compared to $-\log \delta$ as $\delta \to 0$. Therefore all formulations yield the same lim sup and lim inf.

The relationship between Minkowski and Hausdorff dimensions is governed by a one-sided inequality:

Proposition 5.2.3: Hausdorff Versus Minkowski

For any bounded $E \subseteq \mathbb{R}^n$:

$$\dim_H(E) \leq \underline{\dim}_M(E) \leq \overline{\dim}_M(E)$$

***Proof*:**
Fix $s > \overline{\dim}_M(E)$ and $\delta > 0$. By definition of the upper Minkowski dimension, for all sufficiently

small δ there holds $N(E,\delta) \le \delta^{-s}$. Cover E by $N(E,\delta)$ sets of diameter $\le \delta$:

$$\mathcal{H}^s_\delta(E) \le N(E,\delta) \cdot \delta^s \le \delta^{-s} \cdot \delta^s = 1$$

Hence $\mathcal{H}^s(E) = \lim_{\delta\to 0} \mathcal{H}^s_\delta(E) \le 1 < \infty$, so $\dim_H(E) \le s$. Since $s > \overline{\dim}_M(E)$ was arbitrary, $\dim_H(E) \le \overline{\dim}_M(E)$. The inequality $\underline{\dim}_M(E) \le \overline{\dim}_M(E)$ is immediate from $\liminf \le \limsup$.

The inequality $\dim_H \le \underline{\dim}_M$ can be strict: the Hausdorff dimension optimizes over all covers (including covers with variable-diameter sets), while the Minkowski dimension uses only covers at a single scale δ. This constraint prevents Minkowski dimension from "adapting" to sets with non-uniform local structure. The proof of proposition 5.2.3 makes the source of the inequality transparent: the uniform cover at scale δ used to bound $\mathcal{H}^s_\delta(E)$ is only one of the many possible covers over which $\mathcal{H}^s_\delta$ takes an infimum. The Hausdorff measure can find cheaper covers, which is why $\dim_H$ can be strictly smaller.

Despite this deficiency, Minkowski dimension has a significant practical advantage: the covering number $N(E,\delta)$ can often be computed or estimated directly from the geometry of E, without the need to optimize over all covers. For this reason, many results in geometric measure theory (including the Kakeya conjecture) are first formulated in terms of Minkowski dimension, and the stronger Hausdorff statement is established separately. The volume-based formulation of Minkowski dimension also connects it directly to the "δ-tube" estimates that appear in the discretized Kakeya problem.

Example 5.2.4: The Set $\{1/n\}$: Separation of Dimensions

Let $E = \{1/n : n \ge 1\} \subseteq \mathbb{R}$. Since E is countable, $\dim_H(E) = 0$ by proposition 5.1.8(3). The Minkowski dimension is positive: the clustering of points near the origin inflates the box count.

To compute $N(E,\delta)$, partition E into a "sparse" part and a "dense" part. Fix $\delta > 0$ small and let $n_0 = \lfloor 1/\sqrt{\delta} \rfloor$.

Sparse part. For $n \le n_0$, consecutive points $1/n$ and $1/(n+1)$ have spacing $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \ge \frac{1}{n_0(n_0+1)} \ge \frac{\delta}{2}$ for δ small. Each such point requires its own δ-interval, contributing $n_0 \sim \delta^{-1/2}$ intervals.

Dense part. For $n > n_0$, all points lie in $[0, 1/n_0] \subseteq [0, 2\sqrt{\delta}]$. This interval is covered by at most $\lceil 2\sqrt{\delta}/\delta \rceil = \lceil 2/\sqrt{\delta} \rceil \sim \delta^{-1/2}$ intervals of length δ.

Combining both parts: $N(E,\delta) \sim \delta^{-1/2}$. Therefore

$$\dim_M(E) = \lim_{\delta\to 0} \frac{\log(\delta^{-1/2})}{-\log\delta} = \frac{1}{2}$$

The gap $\dim_H(E) = 0 < 1/2 = \dim_M(E)$ arises because the points cluster near 0, inflating the box count at small scales. The Hausdorff dimension ignores this clustering by using a cover with variable-diameter sets (large balls for the sparse region, small balls near 0). The Minkowski dimension is constrained to a single scale δ and pays a penalty for the dense cluster.

For well-behaved sets such as self-similar fractals, the two dimensions coincide:

Example 5.2.5: Minkowski Dimension of the Cantor Set

The middle-thirds Cantor set C satisfies $\dim_M(C) = \log 2/\log 3$.

At stage k, C is covered by 2^k intervals of length 3^{-k}, so $N(C, 3^{-k}) \le 2^k$. Conversely, each of the 2^k stage-k intervals contains a point of C at distance $\ge 3^{-k}/2$ from the others, so $N(C, 3^{-k}) \ge 2^k$. Thus $N(C, 3^{-k}) = 2^k$ for each k, and

$$\dim_M(C) = \lim_{k\to\infty} \frac{\log 2^k}{\log 3^k} = \frac{\log 2}{\log 3}$$

This confirms $\dim_M(C) = \dim_H(C)$.

Products of sets interact predictably with Minkowski dimension, which is useful for constructing higher-dimensional examples:

Proposition 5.2.6: Minkowski Dimension of Products

Let $A \subseteq \mathbb{R}^m$ and $B \subseteq \mathbb{R}^n$ be bounded. Then

$$\overline{\dim}_M(A \times B) \le \overline{\dim}_M(A) + \overline{\dim}_M(B)$$

If $\dim_M(A)$ and $\dim_M(B)$ both exist (upper equals lower), then $\dim_M(A \times B) = \dim_M(A) + \dim_M(B)$.

***Proof*:**

Any cover of A by N_A sets of diameter $\le \delta$ and cover of B by N_B sets of diameter $\le \delta$ induces a cover of $A \times B$ by $N_A \cdot N_B$ sets of diameter $\le 2\delta$ (using the product metric). Therefore $N(A \times B, 2\delta) \le N(A, \delta) \cdot N(B, \delta)$, giving

$$\frac{\log N(A \times B, 2\delta)}{-\log(2\delta)} \le \frac{\log N(A, \delta) + \log N(B, \delta)}{-\log(2\delta)}$$

Taking $\limsup$ and using $-\log(2\delta) \sim -\log\delta$ establishes the upper Minkowski bound. The lower bound $\underline{\dim}_M(A \times B) \ge \underline{\dim}_M(A) + \underline{\dim}_M(B)$ follows from a packing argument (a maximal δ-separated set in $A \times B$ projects to δ-separated sets in A and B), and when the limits exist, the two bounds combine to give equality.

The product formula confirms the intuition that taking a Cartesian product "adds dimensions": the product of a d_1-dimensional set and a d_2-dimensional set is $(d_1 + d_2)$-dimensional. For Hausdorff dimension, the analogous inequality $\dim_H(A \times B) \ge \dim_H(A) + \dim_H(B)$ holds, but equality can fail without additional regularity assumptions (though equality does hold when one of the factors has equal upper and lower Minkowski dimensions).

5.2.7 Remark: Countable Stability Fails for Minkowski Dimension Unlike Hausdorff dimension, Minkowski dimension is *not* countably stable. The set $E = \{1/n\}$ is a countable union of points, each of Minkowski dimension 0, yet $\dim_M(E) = 1/2$. This failure is the principal theoretical deficiency of Minkowski dimension. For compact sets arising from geometric constructions—including the Besicovitch sets studied in the next section—the two dimensions often coincide, and the Kakeya conjecture asserts that both equal n.

The tools developed in this section and the previous one—Hausdorff dimension for sharp theoretical bounds, Minkowski dimension for computationally tractable estimates, and the inequality $\dim_H \leq \dim_M$ relating the two—form the dimensional framework for studying the Kakeya conjecture. The next section constructs the sets to which this framework will be applied: the Besicovitch sets, which contain a line segment in every direction yet have zero Lebesgue measure.

Exercise 5.2.1

1. Let $E_\alpha = \{n^{-\alpha} : n \geq 1\}$ for $\alpha > 0$. Show that $\dim_M(E_\alpha) = \frac{1}{1+\alpha}$.
2. Show that $\dim_M(\mathbb{Q} \cap [0,1]) = 1$ while $\dim_H(\mathbb{Q} \cap [0,1]) = 0$. This gives a maximal gap between the two dimensions.
3. Verify proposition 5.2.6 explicitly for $A = B = C$ (the middle-thirds Cantor set). Compute $\dim_M(C \times C)$ and confirm it equals $2\log 2/\log 3$.
4. Show that if $E \subseteq \mathbb{R}^n$ is bounded and $\dim_M(E)$ exists, then $\dim_M(\overline{E}) = \dim_M(E)$.
5. Show that for any bounded $E \subseteq \mathbb{R}^n$ and $\lambda > 0$, $\overline{\dim}_M(\lambda E) = \overline{\dim}_M(E)$ where $\lambda E = \{\lambda x \mid x \in E\}$. Contrast this with the behavior of Lebesgue measure under dilation (proposition 1.4.7).

5.3 Besicovitch Sets

In 1917, Sōichi Kakeya posed the following question: what is the smallest area of a region in the plane within which a unit needle can be continuously rotated through 180 degrees? If the needle is rotated about its center, the region is a circle of area $\pi/4$. A more efficient motion traces a *deltoid* (a three-cusped hypocycloid) of area $\pi/8$. In 1928, Pál proved that among *convex* regions, the equilateral triangle of unit height has the smallest area. One might expect that the answer for general (non-convex) regions is some small but positive number.

Two distinct questions emerge from Kakeya's original problem. The first, called the *Kakeya needle problem*, asks for the infimum of areas of regions in which a needle can be *continuously* rotated. Besicovitch showed in 1928 that this infimum is zero: the needle can be rotated continuously through all directions within a set of arbitrarily small area, using a "shifting" motion that translates the needle to avoid sweeping large regions. The second question, which is the focus of this section, drops the continuity requirement entirely and asks: what is the smallest measure of a set that *contains* a unit segment in every direction (without any requirement that the segments be related by a continuous motion)? Besicovitch answered this in 1919: such sets can have Lebesgue measure *zero*. This result predates his solution to the needle problem and is the more fundamental of the two—the set problem concerns the static geometry of directional containment, stripped of all topological or kinematic constraints.

The existence of measure-zero sets containing a segment in every direction is one of the most counterintuitive facts in geometric measure theory. Such a set carries full directional information—it "sees" every direction in S^{n-1}—yet is negligible from the perspective of Lebesgue measure. The connection to the Cantor set constructions in proposition 1.4.9 is real: both phenomena exploit the gap between geometric richness and measure-theoretic size.

Definition 5.3.1: Besicovitch Set

A set $K \subseteq \mathbb{R}^n$ is called a *Besicovitch set* (or *Kakeya set*) if for every direction $e \in S^{n-1}$, there exists a point $a \in \mathbb{R}^n$ such that the unit segment $\{a + te \mid 0 \le t \le 1\}$ is contained in K.

A Besicovitch set need not be measurable *a priori*, but all known constructions produce Borel (in fact, compact) sets. Any closed disk of radius $1/2$ or larger is a trivial Besicovitch set—every unit segment can be translated to fit inside it—and has positive measure. The content of Besicovitch's theorem is that the measure can be driven to zero while retaining all directions. The existence question thus splits into two parts: first, construct sets of arbitrarily small measure containing all directions; then, pass to a set of measure exactly zero.

The first part uses the *Perron tree* construction (named after Oskar Perron, who gave an early version of this argument in 1928). The key geometric mechanism is that thin triangles pointing in nearly parallel directions can be translated to overlap heavily: the overlap region is large near the apex (where the triangles are narrow and close together) and small near the base (where they spread apart). This produces a net reduction in area while preserving all directional content.

Lemma 5.3.2: Perron Tree

Let $T \subseteq \mathbb{R}^2$ be a triangle with base of length b on a horizontal line and apex at height h above the base. For each point P on the base, the segment from P to the apex defines a direction $\theta(P) \in (0, \pi)$, and as P ranges over the base, $\theta(P)$ sweeps an interval $[\theta_{\min}, \theta_{\max}]$. Then for any $\epsilon > 0$, there exists a compact set $\mathcal{P} \subseteq \mathbb{R}^2$ with $m(\mathcal{P}) < \epsilon$ containing, for each $\theta \in [\theta_{\min}, \theta_{\max}]$, a line segment of length h in direction θ.

***Proof*:**
The construction proceeds by iterated splitting and sliding.

Step 1: One splitting. Divide the base of T into 2 equal segments, producing two sub-triangles T_L and T_R sharing the same apex. Each sub-triangle contains segments in half of the directions: T_L handles $[\theta_{\min}, \theta_{\text{mid}}]$ and T_R handles $[\theta_{\text{mid}}, \theta_{\max}]$. Translate T_R horizontally so that its apex coincides with that of T_L. In the resulting figure $\mathcal{P}_1 = T_L \cup T_R'$, all original directions are present, but the two sub-triangles overlap near the apex, reducing the total area.

To compute the overlap, observe that each sub-triangle narrows linearly from base to apex. At height y above the base ($0 \le y \le h$), each sub-triangle occupies a horizontal interval of width $\frac{b}{2}(1 - y/h)$: the factor $(1 - y/h)$ reflects the linear taper from base width $b/2$ to apex width 0. Before the translation, T_L and T_R sit side-by-side with no overlap. After translating T_R so that the two apexes coincide, the horizontal offset between the sub-triangles at height y is $\frac{b}{2}(1 - y/h)$ (the same as the width, since the apex shift equals the base half-width). The overlapping region at height y therefore has width

$$2 \cdot \frac{b}{2}(1 - y/h) - \frac{b}{2} = \frac{b}{2}\left(1 - 2\frac{y}{h} + \frac{y}{h}\right) = \frac{b}{2} \cdot \frac{y}{h}$$

(valid for y where the sub-triangles genuinely overlap). Integrating the area of the overlap:

$$m(T_L \cap T_R') = \int_0^h \frac{b}{2} \cdot \frac{y}{h}\, dy = \frac{bh}{4}$$

Since $m(T_L) + m(T_R) = m(T) = bh/2$, the area of the union is $m(\mathcal{P}_1) = bh/2 - bh/4 = bh/4 = m(T)/2$. The overlap saved exactly half the original area.

Step 2: Iteration. Apply the same splitting-and-sliding procedure to each of the two sub-triangles in $\mathcal{P}_1$, obtaining 4 sub-sub-triangles with total area $m(T)/4$. After k iterations, the construction produces 2^k thin triangles with total area $\leq m(T)/2^k$. Choosing k large enough that $m(T)/2^k < \epsilon$ completes the construction.

Step 3: Directional preservation. At each stage, every sub-triangle retains segments in its assigned sub-interval of directions, and the translation preserves these segments (it shifts their starting points but does not rotate them). The union $\mathcal{P}_k$ therefore contains a segment of length $h(1-2^{-k}) \geq h/2$ in each direction $\theta \in [\theta_{\min}, \theta_{\max}]$. A final rescaling by a factor of $h/(h\cdot(1-2^{-k}))$ adjusts the segment lengths to exactly h, at the cost of increasing the area by a bounded factor. Since k can be chosen arbitrarily large, the total area can still be made $< \epsilon$.

The area estimate $m(\mathcal{P}_k) = m(T)/2^k$ computed above is for a specific splitting scheme. In fact, with a more aggressive sliding strategy (translating the sub-triangles further before each iteration), the area can be reduced even faster. The precise optimum is not needed for the existence result; what matters is that the area can be made arbitrarily small.

The construction should be visualized as follows. At stage 0, the set is a single triangle—a solid wedge containing segments in all directions in a fixed angular interval. At stage 1, the triangle is split into two thinner wedges that are slid together so their tips overlap, producing a figure shaped like an elongated "V" or a narrow tree with two branches. At stage 2, each branch is split and slid again, producing four branches. The process continues: at stage k, the set consists of 2^k very thin triangular "branches" that overlap extensively near their tips. The total area shrinks exponentially with k, yet every original direction is still represented by some branch. The visual resemblance to a tree with an increasingly dense canopy and decreasing total leaf area explains the name "Perron tree."

Example 5.3.3: Perron Tree: First Two Stages

Start with an isosceles triangle T with base $[-1, 1] \times \{0\}$ and apex $(0, 1)$. This triangle has area 1 and contains segments from the base to the apex sweeping directions in the interval $[\pi/4, 3\pi/4]$ (approximately, for this aspect ratio). A segment from the base point $(-1, 0)$ to the apex $(0, 1)$ points in direction $\arctan(1/1) = \pi/4$, and a segment from $(1, 0)$ to $(0, 1)$ points in direction $\pi - \pi/4 = 3\pi/4$. As the base point sweeps from -1 to 1, the direction sweeps from $\pi/4$ to $3\pi/4$.

Stage 1. Split the base at 0. The left sub-triangle T_L has vertices $(-1, 0)$, $(0, 0)$, $(0, 1)$; the right sub-triangle T_R has vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. Translate T_R left by 1 unit so its apex coincides with T_L's. The translated T_R' has vertices $(-1, 0)$, $(0, 0)$, $(-1, 1)$. The union $T_L \cup T_R'$ is a parallelogram-like figure of area 1/2: the overlap occurs in the upper portion where both triangles taper to width zero. T_L retains segments in directions $[\pi/4, \pi/2]$ and T_R' retains directions in

$[\pi/2, 3\pi/4]$, so all original directions are present.

Stage 2. Split each of T_L and T'_R at their base midpoints, obtaining 4 thin triangles. Slide each pair together as before. The resulting figure has area $1/4$ and still contains segments in all directions in $[\pi/4, 3\pi/4]$. The figure now consists of 4 thin triangles that share significant overlap in their upper portions, creating a branching pattern.

After k stages, the area is 2^{-k}, the set consists of 2^k thin triangles of base width 2^{1-k}, and the figure visually resembles a fractal "tree"—hence the name Perron tree. The tree structure arises because at each stage, the splitting doubles the number of branches while the sliding compresses them together.

With the Perron tree in hand, the passage to a measure-zero Besicovitch set proceeds in two steps: first cover all directions, then iterate the compression to reduce the area to zero.

Theorem 5.3.4: Existence of Measure-Zero Besicovitch Sets

There exists a compact Besicovitch set $K \subseteq \mathbb{R}^2$ with $m(K) = 0$.

***Proof*:**
Step 1: Covering all directions. The Perron tree from lemma 5.3.2 produces a set containing segments in all directions within a single angular interval. Since $[0, \pi)$ is compact, finitely many such intervals cover it: choose N triangles $T^{(1)}, \dots, T^{(N)}$, suitably rotated, so that their angular intervals cover $[0, \pi)$. For any $\epsilon > 0$, applying lemma 5.3.2 to each $T^{(j)}$ produces compact sets $\mathcal{P}^{(j)}$ with $m(\mathcal{P}^{(j)}) < \epsilon/N$. Their union $K_\epsilon = \bigcup_{j=1}^{N} \mathcal{P}^{(j)}$ is compact, satisfies $m(K_\epsilon) < \epsilon$, and contains a segment (of some fixed length $\ell > 0$, independent of ϵ) in every direction. By rescaling, a segment of length ℓ in every direction can be promoted to a *unit* segment in every direction, at the cost of multiplying the area by $(1/\ell)^2$; since ϵ was arbitrary, the area of the rescaled set can still be made $< \epsilon$.

Step 2: Passage to measure zero. The sets K_ϵ from Step 1 are not nested (different values of ϵ may produce geometrically unrelated configurations), so taking $K = \bigcap_n K_{1/n}$ does not yield a set of measure zero—the intersection may even be empty. Instead, the construction must be refined to produce a nested sequence.

Choose a compact set K_0 containing unit segments in all directions (for instance, a sufficiently large closed disk). The strategy is to apply the Perron tree compression *within* the existing set K_0, producing a subset $K_1 \subseteq K_0$ of smaller measure that still contains all directions, and then iterate.

Concretely, partition the directions $[0, \pi)$ into N sub-intervals $I_1, \dots, I_N$ and identify the corresponding "sectors" of K_0: for each I_j, the set of points in K_0 lying on unit segments with directions in I_j forms a roughly triangular region Σ_j. Within each sector Σ_j, apply the splitting-and-sliding procedure of lemma 5.3.2, translating sub-triangles *inward* (toward the interior of Σ_j). The resulting set $K_1 = \bigcup_{j=1}^{N} \mathcal{P}_j$ consists of thin sub-triangles that are contained in K_0 (the inward sliding ensures this), and satisfies $m(K_1) \leq m(K_0)/2$. Each direction in $[0, \pi)$ is still represented, since the Perron tree within each sector preserves all directions in I_j.

Iterating this procedure produces a nested sequence $K_0 \supseteq K_1 \supseteq K_2 \supseteq \cdots$ with $m(K_k) \leq m(K_0)/2^k$.

Define $K = \bigcap_{k=0}^{\infty} K_k$. Since $m(K_0) < \infty$ and the sequence is nested, continuity from above (proposition 1.2.5) gives $m(K) = \lim_{k\to\infty} m(K_k) = 0$.

Step 3: K contains all directions. For each direction $\theta \in [0, \pi)$ and each k, the set of starting points a such that $\{a + te_\theta : 0 \le t \le 1\} \subseteq K_k$ is a nonempty compact set $S_k(\theta) \subseteq K_k$ (nonempty by construction, compact as a closed subset of K_k). The sequence $S_0(\theta) \supseteq S_1(\theta) \supseteq \cdots$ is nested and nonempty, so $\bigcap_k S_k(\theta) \ne \emptyset$ by the finite intersection property. Any point $a \in \bigcap_k S_k(\theta)$ satisfies $\{a + te_\theta : 0 \le t \le 1\} \subseteq K_k$ for all k, hence $\{a + te_\theta : 0 \le t \le 1\} \subseteq K$. Therefore K is a Besicovitch set.

The passage from "small area" to "zero area" in Step 2 is the most delicate part of the construction. The argument that the Perron tree compression can be made to produce nested sets requires geometric care—in particular, the sub-triangles must be slid toward the interior of the parent triangle, not away from it. A complete treatment of this nesting, with explicit coordinates at each stage, can be found in Falconer's *The Geometry of Fractal Sets* or Mattila's *Geometry of Sets and Measures in Euclidean Spaces*.

The comparison to the Vitali set construction (theorem 1.0.1) is instructive. Vitali sets are *unmeasurable*—they cannot be assigned any consistent measure. Besicovitch sets are measurable (in fact, compact and Borel) with measure *exactly zero*. Both constructions reveal fundamental limitations: Vitali sets show that not all subsets of $\mathbb{R}$ can be measured; Besicovitch sets show that measure zero does not preclude rich geometric structure. However, Vitali sets require the axiom of choice, while Besicovitch sets are constructed explicitly (within ZF). This constructivity makes the Besicovitch set a more "concrete" pathology—it is not an artifact of set-theoretic axioms but a genuine feature of Euclidean geometry.

The existence of measure-zero Besicovitch sets shows that the Lebesgue measure is too coarse to capture the directional complexity of subsets of $\mathbb{R}^n$. A finer invariant is needed, and Hausdorff dimension from section 5.1 provides exactly this: while $m(K) = 0$ for all Besicovitch sets K, the Hausdorff dimension $\dim_H(K)$ can still be positive and potentially as large as n. The question of how large $\dim_H(K)$ must be is the Kakeya conjecture.

The construction generalizes to higher dimensions via a product argument:

Proposition 5.3.5: Besicovitch Sets in $\mathbb{R}^n$

For every $n \ge 2$, there exists a compact Besicovitch set $K \subseteq \mathbb{R}^n$ with $m(K) = 0$.

***Proof*:**
Let $K_2 \subseteq \mathbb{R}^2$ be a compact Besicovitch set with $m_2(K_2) = 0$ (where m_2 denotes the 2-dimensional Lebesgue measure). The product $K_2 \times [0,1]^{n-2} \subseteq \mathbb{R}^n$ has n-dimensional measure $m_n(K_2 \times [0,1]^{n-2}) = m_2(K_2) \cdot 1 = 0$ by theorem 2.5.9. However, this product set only contains segments whose directions lie in the 2-plane $\mathbb{R}^2 \times \{0\}^{n-2}$: a unit segment in $K_2 \times [0,1]^{n-2}$ has the form $(a + te_1, a_3, \ldots, a_n)$ where $e_1 \in S^1 \subseteq \mathbb{R}^2$ and $a_3, \ldots, a_n$ are fixed. Any direction $e \in S^{n-1}$ with a nonzero component outside the first two coordinates is missing.

To capture all directions in S^{n-1}, take a finite collection of orthogonal transformations $R_1, \ldots, R_M \in O(n)$ such that

$$\bigcup_{j=1}^{M} R_j(\mathbb{R}^2 \times \{0\}^{n-2})$$

covers a neighborhood of every direction in S^{n-1}. Such a finite collection exists by a compactness argument: for each direction $e \in S^{n-1}$, there is a rotation R mapping e into $\mathbb{R}^2 \times \{0\}^{n-2}$ (and in fact into a neighborhood of directions), and the resulting open cover of S^{n-1} has a finite subcover. For each R_j, the set $R_j(K_2 \times [0,1]^{n-2})$ is compact and has $m_n = 0$ (orthogonal transformations preserve Lebesgue measure by theorem 2.6.4 since $|\det R_j| = 1$). The union

$$K = \bigcup_{j=1}^{M} R_j(K_2 \times [0,1]^{n-2})$$

is a finite union of measure-zero compact sets, hence compact and measure zero. For any direction $e \in S^{n-1}$, some R_j maps e into $\mathbb{R}^2 \times \{0\}^{n-2}$, and the corresponding rotated copy $R_j(K_2 \times [0,1]^{n-2})$ contains a unit segment in direction e. Therefore K is a Besicovitch set.

This product-and-rotate construction is crude—it gives a Besicovitch set but reveals nothing about its dimension. To understand why, note that $K_2 \times [0,1]^{n-2}$ has Hausdorff dimension at most $\dim_H(K_2) + (n-2) \le 2 + (n-2) = n$ by proposition 5.2.6 (applied to Hausdorff dimension, where the analogous inequality holds). Rotating and taking a finite union does not change the dimension, so the construction gives $\dim_H(K) \le n$—which is already known from $K \subseteq \mathbb{R}^n$. The construction provides no *lower* bound on $\dim_H(K)$ beyond the trivial $\dim_H(K) \ge 1$ (from the presence of line segments).

The dimension of a Besicovitch set is the central open question. By proposition 5.1.8(1), every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) \le n$ (since $K \subseteq \mathbb{R}^n$). The question is whether $\dim_H(K) = n$ always holds—can the dimension be strictly less than n? In $\mathbb{R}^2$, the answer is no:

Theorem 5.3.6: Davies' Theorem

Every Besicovitch set in $\mathbb{R}^2$ has Hausdorff dimension 2.

Davies' proof (1971) uses a projection argument that is specific to $\mathbb{R}^2$. The key idea is as follows. If $K \subseteq \mathbb{R}^2$ is a Besicovitch set, consider the projection $\pi_\theta : \mathbb{R}^2 \to \mathbb{R}$ onto the line perpendicular to direction θ. Since K contains a unit segment in direction θ, the projection $\pi_\theta(K)$ contains an interval of length $\sin(\alpha)$ for some angle α depending on θ. By a result of Marstrand (1954) relating the Hausdorff dimension of a set to the dimensions of its projections, the fact that $\pi_\theta(K)$ has positive length for *every* direction θ forces $\dim_H(K) \ge 2$. (In precise terms: if $\dim_H(K) < 2$, then $\pi_\theta(K)$ would have zero length for *almost every* θ by Marstrand's projection theorem, contradicting the Besicovitch property.)

This argument fails in $\mathbb{R}^n$ for $n \ge 3$ because the relationship between a set and its projections becomes weaker in higher dimensions. A set of Hausdorff dimension $d < n$ in $\mathbb{R}^n$ projects to a set of positive measure in *some* directions but not necessarily in enough directions to force $d = n$. The combinatorial structure of how lines in $\mathbb{R}^n$ interact with lower-dimensional projections is far richer and less constrained than in $\mathbb{R}^2$, and exploiting the Besicovitch property (a segment in *every* direction) requires fundamentally different techniques. The natural generalization of Davies' theorem—that every Besicovitch set in $\mathbb{R}^n$ has Hausdorff dimension n—is the Kakeya conjecture, which forms the subject of the remaining sections.

To summarize the situation: Besicovitch sets of measure zero exist in every $\mathbb{R}^n$ with $n \ge 2$, so the Lebesgue measure cannot distinguish between a "large" Besicovitch set (like a disk) and a "small" one (measure zero). The Hausdorff dimension provides the finer distinction. In $\mathbb{R}^2$, Davies' theorem settles the matter: $\dim_H(K) = 2$ for every Besicovitch set. In $\mathbb{R}^3$, the Kakeya conjecture was open for over 50 years until Wang and Zahl resolved it in 2025. In $\mathbb{R}^n$ for $n \geq 4$, it remains one of the central open problems in geometric measure theory.

Exercise 5.3.1

1. Verify that at stage k of the Perron tree construction in lemma 5.3.2, the total area of the 2^k sub-triangles (before accounting for overlaps) is $m(T)$, and that the overlap estimate $m(\mathcal{P}_k) \leq m(T)/2^k$ holds by induction.
2. Show that if $K \subseteq \mathbb{R}^n$ is a Besicovitch set and $T \in GL_n(\mathbb{R})$ is an invertible linear map, then $T(K)$ is also a Besicovitch set. Does this remain true if T is injective but not surjective?
3. Show that a Besicovitch set $K \subseteq \mathbb{R}^n$ cannot be contained in any affine hyperplane. Conclude that $\dim_H(K) \geq 1$ for all $n \geq 2$. (This lower bound is far from optimal.)
4. Explain why the product construction $K_2 \times [0,1]^{n-2}$ does not directly contain unit segments in all directions of S^{n-1} (identify the missing directions), and verify that the finite rotation argument in proposition 5.3.5 resolves this.
5. (Nikodym set) A *Nikodym set* $N \subseteq \mathbb{R}^2$ is a set such that for every point $x \in \mathbb{R}^2$, there exists a line ℓ_x through x with $\ell_x \setminus \{x\} \subseteq N$. Show that Nikodym sets of measure zero exist. (*Hint*: the construction is closely related to the Besicovitch construction.)
6. Show that the Besicovitch set K constructed in theorem 5.3.4 is nowhere dense (definition 1.4.12). Conclude that K is topologically "small" (meager, being a compact set with empty interior) yet geometrically "rich" (containing a unit segment in every direction). Compare this with the discussion of the Cantor set in proposition 1.4.13.

5.4 The Kakeya Conjecture and Maximal Functions

The preceding section established that Besicovitch sets—compact sets of Lebesgue measure zero containing a unit segment in every direction—exist in every $\mathbb{R}^n$ with $n \geq 2$. Davies' theorem (theorem 5.3.6) showed that in $\mathbb{R}^2$, such sets must have full Hausdorff dimension despite having zero measure. The central open problem is whether this remains true in all dimensions:

Conjecture 5.4.1: Kakeya Set Conjecture

Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) = n$. Equivalently (since $\dim_H \leq \dim_M$ by proposition 5.2.3), every Besicovitch set satisfies $\dim_M(K) = n$.

The equivalence stated above is one-directional: the Hausdorff statement is strictly stronger than the Minkowski statement (since $\dim_H \leq \dim_M$, full Hausdorff dimension implies full Minkowski dimension, but not conversely). Both versions are commonly referred to as "the Kakeya conjecture," and both are open for $n \geq 4$. In $n = 2$, both are settled by theorem 5.3.6. In $n = 3$, both were resolved by Wang and Zahl in 2025, as discussed at the end of this chapter.

The conjecture admits a natural discretization in terms of "δ-tubes" that makes it more amenable to combinatorial and analytic methods. The passage from continuous sets to discrete collections of tubes is a standard technique in geometric measure theory: it replaces the measure-zero set K with a family of small-but-positive-volume objects, converting a dimension problem into a volume problem.

Definition 5.4.2: δ-Tube

Let $e \in S^{n-1}$ be a direction, $a \in \mathbb{R}^n$ a center point, and $\delta > 0$. The *δ-tube* centered at a in direction e is the set

$$T_e^\delta(a) = \{x \in \mathbb{R}^n \mid |(x-a) - ((x-a)\cdot e)e| \le \delta, \quad 0 \le (x-a)\cdot e \le 1\}$$

This is a cylinder of length 1, radius δ, with its long axis in direction e and centered at the point $a + \frac{1}{2}e$. Its volume is $|T_e^\delta(a)| = \omega_{n-1}\delta^{n-1}$, where ω_{n-1} is the volume of the unit ball in $\mathbb{R}^{n-1}$.

A δ-tube should be thought of as a "thickened" line segment: as $\delta \to 0$, it shrinks to the unit segment $\{a + te : 0 \le t \le 1\}$. A Besicovitch set K containing a segment in every direction can be "thickened" by replacing each segment with a δ-tube, and the Kakeya conjecture can be rephrased in terms of the volume of the resulting union of tubes.

A family of directions is called *δ-separated* if any two directions $e, e' \in S^{n-1}$ satisfy $|e - e'| \ge \delta$. A maximal δ-separated subset of S^{n-1} has cardinality $\sim \delta^{-(n-1)}$ (since S^{n-1} has $(n-1)$-dimensional surface area $n\omega_n$ and each δ-cap has area $\sim \delta^{n-1}$). The discretized Kakeya conjecture asserts that any such family of tubes has nearly maximal union volume:

Proposition 5.4.3: Discretized Kakeya Conjecture

The Kakeya set conjecture (**??** 5.4.1) is equivalent to the following statement: for every $\epsilon > 0$, there exists a constant $C_\epsilon > 0$ such that if $\mathbb{T}$ is a collection of δ-tubes $\{T_{e_j}^\delta(a_j)\}_{j=1}^N$ with δ-separated directions and $N \ge c\delta^{-(n-1)}$, then

$$\left| \bigcup_{j=1}^N T_{e_j}^\delta(a_j) \right| \ge C_\epsilon\, \delta^\epsilon$$

The bound $|\bigcup T_j| \ge C_\epsilon \delta^\epsilon$ is a weak lower bound: it says the volume cannot decay faster than any power of δ. For comparison, the total volume of all N tubes (if disjoint) would be $N \cdot \omega_{n-1}\delta^{n-1} \sim 1$, while the trivial lower bound from a single tube is $\omega_{n-1}\delta^{n-1}$. The conjecture sits between these extremes, asserting that the union cannot be much smaller than a fixed positive constant (independent of δ).

***Proof*:**
Step 1: Kakeya conjecture implies the discretized statement. Assume the Kakeya set conjecture holds, and let $\mathbb{T} = \{T_{e_j}^\delta(a_j)\}_{j=1}^N$ be a collection of δ-tubes with δ-separated directions and $N \ge c\delta^{-(n-1)}$. Define $K = \bigcap_{k=1}^\infty \bigcup_{j=1}^N \overline{T_{e_j}^{1/k}(a_j)}$, which is a compact set containing a unit segment in each direction e_j. By extending to a full Besicovitch set (adding segments in the finitely many missing directions), the Kakeya conjecture gives $\dim_M(K) = n$.

The δ-neighborhood $K_\delta = \{x \mid d(x, K) < \delta\}$ contains all the tubes $T_{e_j}^\delta(a_j)$. By the volume formulation of Minkowski dimension (section 5.2), $\dim_M(K) = n$ implies that for any $\epsilon > 0$ and all sufficiently small δ:

$$|K_\delta| \ge c_\epsilon\, \delta^{n-n+\epsilon} = c_\epsilon\, \delta^\epsilon$$

More precisely, the definition $\dim_M(K) = n$ means $\lim_{\delta \to 0} \frac{\log |K_\delta|}{\log \delta} = n - n = 0$ (using the n-dimensional volume formulation), so $|K_\delta|$ decays slower than any power of δ. Since $|\bigcup T_j| \leq |K_\delta|$, the discretized bound follows.

Step 2: Discretized statement implies the Kakeya conjecture. Conversely, let K be a Besicovitch set and fix $\epsilon > 0$. For each $\delta > 0$, choose a maximal δ-separated set of directions $\{e_j\}_{j=1}^N$ on S^{n-1} (so $N \sim \delta^{-(n-1)}$). For each e_j, the set K contains a unit segment in direction e_j, so there exist center points a_j such that the δ-tube $T^\delta_{e_j}(a_j)$ is contained in the δ-neighborhood K_δ. In particular, $|\bigcup_j T^\delta_{e_j}(a_j)| \leq |K_\delta|$.

By the discretized statement, $|K_\delta| \geq |\bigcup T_j| \geq C_\epsilon \delta^\epsilon$. Taking logarithms and dividing by $\log \delta$ (which is negative for $\delta < 1$):

$$\frac{\log |K_\delta|}{\log \delta} \leq \frac{\log C_\epsilon + \epsilon \log \delta}{\log \delta} = \epsilon + \frac{\log C_\epsilon}{\log \delta} \to \epsilon \quad \text{as } \delta \to 0$$

By the volume formulation of Minkowski dimension, $n - \dim_M(K) = \lim_{\delta \to 0} \frac{\log |K_\delta|}{\log \delta} \leq \epsilon$, giving $\dim_M(K) \geq n - \epsilon$. Since $\epsilon > 0$ was arbitrary, $\dim_M(K) = n$.

The passage from $\dim_M(K) = n$ to $\dim_H(K) = n$ requires an additional argument: the discretized statement must be applied at multiple scales simultaneously (not just a single δ) to produce efficient covers for the Hausdorff measure. The details of this "multi-scale" refinement are standard and can be found in Wolff's lecture notes or Mattila's textbook.

The discretized formulation transforms the Kakeya conjecture from a question about abstract Hausdorff dimension into a concrete volume estimate for unions of tubes. This reformulation is the form in which the conjecture is typically attacked, since tubes are rigid geometric objects whose intersections can be analyzed explicitly. The ϵ-loss in the bound $|\bigcup T_j| \geq C_\epsilon \delta^\epsilon$ may seem artificial, but it is essential: a bound of the form $|\bigcup T_j| \geq c$ (with no δ-dependence) is equivalent to the stronger statement that K has positive Lebesgue measure, which is false for Besicovitch sets. The δ^ϵ factor allows the volume to decay, but only sub-polynomially.

The intersection structure of tubes is governed by a basic geometric estimate: two tubes at angle α intersect in a region of controlled volume. This estimate is the foundation for all results on the Kakeya problem:

Lemma 5.4.4: Tube Intersection Estimate

Let $T^\delta_e(a)$ and $T^\delta_{e'}(a')$ be two δ-tubes in $\mathbb{R}^n$ with $|e - e'| = \alpha \geq \delta$. Then

$$|T^\delta_e(a) \cap T^\delta_{e'}(a')| \lesssim \frac{\delta^{n-1}}{\alpha}$$

where the implicit constant depends only on n. In particular, when $\alpha \sim 1$ (nearly perpendicular tubes), the intersection has volume $\sim \delta^{n-1}$, while when $\alpha \sim \delta$ (nearly parallel tubes), the intersection can be as large as $\sim \delta^{n-2}$.

***Proof*:**
The intersection of two cylinders of radius δ and length 1 whose axes meet at angle α is contained in a "slab" of thickness $\delta/\sin\alpha$ in the direction bisecting the two axes, and of cross-section $\sim \delta^{n-2}$ in the remaining directions. Since $\sin\alpha \geq c\alpha$ for $\alpha \in [0, \pi/2]$, the volume of the intersection is bounded by

$$|T_e^\delta(a) \cap T_{e'}^\delta(a')| \lesssim \delta^{n-2} \cdot \frac{\delta}{\alpha} = \frac{\delta^{n-1}}{\alpha}$$

completing the estimate.

The tube intersection estimate controls the degree to which two tubes can overlap as a function of their angular separation. When $\alpha \sim 1$, the tubes cross nearly perpendicularly, and their intersection is a small "dot" of volume $\sim \delta^{n-1}$. When $\alpha \sim \delta$, the tubes are nearly parallel and can overlap along their entire length, giving an intersection of volume $\sim \delta^{n-2}$—much larger. The worst case for the Kakeya problem is when many tubes are nearly parallel, since their overlaps waste volume. The Kakeya conjecture asserts that no arrangement of tubes can exploit this overlap efficiently enough to make the total union volume small.

The total overlap of a collection of N tubes with δ-separated directions can be quantified. Denoting $F = \sum_{j=1}^N \chi_{T_j}$, the L^2 norm squared of F counts pairwise intersections:

$$\|F\|_{L^2}^2 = \sum_{i,j} |T_i \cap T_j|$$

The diagonal terms contribute $N \cdot |T_j| \sim N\delta^{n-1}$. The off-diagonal terms are controlled by the tube intersection estimate and depend on the distribution of angles between pairs of tubes. Bounding this L^2 norm is the key to Córdoba's argument and the starting point for all higher-dimensional approaches.

With the tube intersection estimate in hand, the Kakeya problem can be approached through a maximal function that encodes the "worst-case" averaging behavior of a function along tubes. The connection to the Hardy–Littlewood maximal function (definition 3.4.6) is direct: the Hardy–Littlewood function takes suprema over balls centered at a point, while the Kakeya maximal function takes suprema over tubes in a given direction. The directional constraint is what makes the Kakeya problem harder: the Hardy–Littlewood maximal theorem (theorem 3.4.7) gives L^p bounds for $p > 1$ with constants independent of the ball radius, while the Kakeya maximal function incurs a loss of $\delta^{-\epsilon}$ (and the conjecture asserts this loss is the worst possible).

Definition 5.4.5: Kakeya Maximal Function

Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $\delta > 0$. The *Kakeya maximal function of* f is the function $f_\delta^* : S^{n-1} \to [0, \infty]$ defined by

$$f_\delta^*(e) = \sup_{a \in \mathbb{R}^n} \frac{1}{|T_e^\delta(a)|} \int_{T_e^\delta(a)} |f(x)|\, dx$$

where the supremum is over all positions $a \in \mathbb{R}^n$ of the δ-tube in direction e.

For each direction e, $f_\delta^*(e)$ captures the maximum average value of $|f|$ over any tube pointing in direction e. If $f = \chi_K$ for a Besicovitch set K, then $f_\delta^*(e) \geq c > 0$ for every direction (since K contains a segment, hence a substantial fraction of a tube, in every direction), and bounding f_δ^* from above constrains how "concentrated" the set K can be. The Kakeya maximal conjecture asserts a quantitative L^p bound:

Conjecture 5.4.6: Kakeya Maximal Conjecture

For every $\epsilon > 0$, there exists $C_\epsilon > 0$ such that for all $\delta > 0$ and $f \in L^n(\mathbb{R}^n)$:

$$\|f_\delta^*\|_{L^n(S^{n-1})} \leq C_\epsilon \, \delta^{-\epsilon} \|f\|_{L^n(\mathbb{R}^n)}$$

The exponent $p = n$ is critical: for $p < n$, the bound fails (as can be shown using a bush-like configuration), and for $p > n$, the bound follows from simpler covering arguments. Testing the maximal conjecture on $f = \chi_K$ recovers the set conjecture: if $\|f_\delta^*\|_{L^n} \leq C_\epsilon \delta^{-\epsilon}$ and $f_\delta^*(e) \geq c$ for every e, then the surface area of S^{n-1} gives $c^n \cdot n\omega_n \leq C_\epsilon^n \delta^{-n\epsilon} \|f\|_{L^n}^n$, forcing $\|f\|_{L^n}^n \geq c'\delta^{n\epsilon}$, hence $|K_\delta| \geq c'\delta^{n\epsilon}$ for any ϵ, establishing $\dim_M(K) = n$. The maximal conjecture is thus strictly stronger than the set conjecture—it provides quantitative control, not just a dimension bound.

Example 5.4.7: The Maximal Function for Simple Configurations

Consider $n = 2$ and $f = \chi_R$ where R is a rectangle of width δ and length 1 aligned with the x-axis. Then $f_\delta^*(e_1) = 1$ (the tube in direction $e_1 = (1, 0)$ can be chosen to coincide with R), while $f_\delta^*(e) \lesssim \delta/|e - e_1|$ for $|e - e_1| \gg \delta$ (a tube at angle α to R intersects R in a set of area $\sim \delta^2/\alpha$, and dividing by the tube area δ gives the average $\sim \delta/\alpha$). Integrating over S^1: $\|f_\delta^*\|_{L^2}^2 \lesssim 1 + \delta^2 \sum_{k=1}^{1/\delta} 1/(k\delta)^2 \cdot \delta \sim \log(1/\delta)$. This shows the $\log(1/\delta)$ factor in Córdoba's estimate is achieved (up to constants) by a single rectangle, confirming the bound is sharp.

In $\mathbb{R}^2$, the maximal conjecture is a theorem, proved by Córdoba in 1977 using a technique known as the "bush argument." The proof is a model for higher-dimensional approaches and uses only the tube intersection estimate and the Cauchy–Schwarz inequality:

Theorem 5.4.8: Córdoba's Estimate

In $\mathbb{R}^2$, for every $\delta > 0$ and $f \in L^2(\mathbb{R}^2)$:

$$\|f_\delta^*\|_{L^2(S^1)} \leq C \left(\log \frac{1}{\delta} \right)^{1/2} \|f\|_{L^2(\mathbb{R}^2)}$$

In particular, the Kakeya maximal conjecture holds in $\mathbb{R}^2$ (since $(\log 1/\delta)^{1/2} \leq C_\epsilon \delta^{-\epsilon}$ for any $\epsilon > 0$).

***Proof*:**

By a standard linearization argument, it suffices to prove the dual estimate: if $\{T_j\}_{j=1}^N$ is a collection of δ-tubes with δ-separated directions (so $N \sim 1/\delta$), and $\{a_j\}_{j=1}^N$ are non-negative coefficients, then

$$\left\| \sum_{j=1}^N a_j \chi_{T_j} \right\|_{L^2(\mathbb{R}^2)}^2 \leq C \log \frac{1}{\delta} \sum_{j=1}^N a_j^2 |T_j|$$

Taking $a_j = 1$ for all j gives a bound on the L^2 norm of the sum of characteristic functions, which by duality controls the maximal function.

Step 1: Expanding the L^2 norm. Compute

$$\left\|\sum_j a_j \chi_{T_j}\right\|_{L^2}^2 = \sum_{i,j} a_i a_j |T_i \cap T_j|$$

Split this into diagonal terms ($i = j$) and off-diagonal terms ($i \neq j$).

Step 2: Diagonal terms. The diagonal contributes $\sum_j a_j^2 |T_j| = \omega_1 \delta \sum_j a_j^2$, where $|T_j| = \omega_1 \delta$ is the area of a δ-tube in $\mathbb{R}^2$ (a rectangle of width 2δ and length 1).

Step 3: Off-diagonal terms. For $i \neq j$, the tube intersection estimate (lemma 5.4.4) gives $|T_i \cap T_j| \lesssim \delta^2/|e_i - e_j|$ in $\mathbb{R}^2$ (since $n = 2$, the bound is $\delta^{n-1}/\alpha = \delta/\alpha$; including the width of the tubes gives δ^2/α after a more careful accounting). Order the directions so that $e_1, e_2, \ldots, e_N$ are arranged consecutively around S^1 with $|e_j - e_{j+1}| \sim \delta$. Then $|e_i - e_j| \gtrsim |i - j|\delta$, and by Cauchy–Schwarz ($a_i a_j \leq \frac{1}{2}(a_i^2 + a_j^2)$):

$$\begin{aligned}
\sum_{i \neq j} a_i a_j |T_i \cap T_j| &\lesssim \sum_{i \neq j} a_i a_j \frac{\delta^2}{|i - j|\delta} \\
&= \delta \sum_{i \neq j} \frac{a_i a_j}{|i - j|} \\
&\leq \delta \sum_j a_j^2 \sum_{k=1}^{N} \frac{1}{k} \\
&\lesssim \delta \log N \sum_j a_j^2 \\
&\sim \delta \log \frac{1}{\delta} \sum_j a_j^2
\end{aligned}$$

where the second inequality uses the Cauchy–Schwarz step and the symmetry of the sum. The harmonic sum $\sum_{k=1}^{N} 1/k \sim \log N \sim \log(1/\delta)$ produces the logarithmic factor.

Step 4: Combining. Adding the diagonal and off-diagonal contributions:

$$\left\|\sum_j a_j \chi_{T_j}\right\|_{L^2}^2 \lesssim \delta \log \frac{1}{\delta} \sum_j a_j^2 = \log \frac{1}{\delta} \sum_j a_j^2 |T_j|$$

The duality between $\sum a_j \chi_{T_j}$ and the maximal function f_δ^* (via testing against f) then yields the claimed bound $\|f_\delta^*\|_{L^2} \leq C(\log 1/\delta)^{1/2} \|f\|_{L^2}$.

The factor $(\log 1/\delta)^{1/2}$ in Córdoba's bound is an artifact of the harmonic sum over tube intersections. It is sub-polynomial in $1/\delta$ (growing slower than any power $\delta^{-\epsilon}$), so the Kakeya maximal conjecture follows immediately. As a corollary, the Kakeya set conjecture in $\mathbb{R}^2$ follows from the maximal conjecture, giving a second proof of theorem 5.3.6.

The structure of the proof reveals why $n = 2$ is tractable. In $\mathbb{R}^2$, the directions live on S^1 and can be linearly ordered. The harmonic sum $\sum_{k=1}^{N} 1/k \sim \log N$ captures the total pairwise overlap, and this sum grows slowly enough that the Cauchy–Schwarz step produces a useful bound. In $\mathbb{R}^n$ for $n \geq 3$, the directions live on S^{n-1} and cannot be linearly ordered. The pairwise intersection sum becomes a double sum over an $(n-1)$-dimensional index set, and the geometric structure of the intersections depends not just on the angle between two tubes but also on their relative positions in space. A direct repetition of Córdoba's approach in $\mathbb{R}^n$ yields the bound $\dim_H(K) \geq (n+1)/2$, which for $n = 3$ gives $\dim_H \geq 2$—weaker than Wolff's 5/2. The additional gain in Wolff's argument comes from exploiting the *linear* structure of tubes (not just their pointwise intersections):

Theorem 5.4.9: Wolff's Bound

Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) \geq (n+2)/2$.

***Proof Key ideas*:**
The proof introduces the "hairbrush" argument, which improves on Córdoba's "bush" argument by exploiting the linear structure of tubes (rather than just their pointwise intersections).

Step 1: Bush versus hairbrush. In Córdoba's argument, the worst case is a "bush": many tubes passing through a common δ-ball. In a bush, the tubes spread out radially, and their union has volume $\sim N\delta^{n-1}$ (each tube contributes its full volume beyond the common ball). This gives $|\bigcup T_j| \geq c$ when $N \sim \delta^{-(n-1)}$, which is the full conjecture—but only if a bush actually occurs. When tubes are spread out and no bush exists, Córdoba's argument loses efficiency.

Step 2: Wolff's observation. Instead of looking for a point through which many tubes pass, look for a *tube* T_0 that is intersected by many other tubes. The tubes intersecting T_0 form a "hairbrush" rooted at T_0. The intersections are spread *along* T_0 (not concentrated at a point), and the tubes emanating from these intersection points spread into the ambient space.

Step 3: Volume estimate. The hairbrush structure produces a volume gain over the naive estimate. By a pigeonhole argument, there exists a tube T_0 intersected by $\gtrsim N^{1/2}$ other tubes (this uses the total intersection count). The $N^{1/2}$ tubes of the hairbrush have δ-separated intersection points along T_0 and δ-separated directions, so their union has volume $\gtrsim N^{1/2} \cdot \delta^{n-1} \sim \delta^{(n-2)/2}$. This leads to the bound $\dim_M(K) \geq (n+2)/2$ (and with more work, $\dim_H(K) \geq (n+2)/2$).

The pigeonhole step is the heart of the argument. The total number of "incidences" (pairs (T_i, T_j) with $T_i \cap T_j \neq \emptyset$ and $i \neq j$) is at least $cN^2\delta$ (from a counting argument using the volume of the union). Distributing these incidences among N tubes, at least one tube T_0 must have $\geq cN\delta \sim N^{1/2} \cdot N^{1/2}\delta$ incidences. The $N^{1/2}\delta$ factor, combined with the geometry of tube intersections along T_0, gives the improved volume bound.

For $n = 3$, Wolff's bound gives $\dim_H(K) \geq 5/2$. The gap between 5/2 and the conjectured value of 3 resisted all attempts for nearly three decades, until Wang and Zahl closed it in 2025 (see the end of this chapter). For $n \geq 4$, Wolff's bound of $(n+2)/2$ remains the best known general result, though incremental improvements have been obtained using arithmetic combinatorics (Bourgain, Katz–Tao) and polynomial methods (Guth).

The following table summarizes the state of knowledge for the Kakeya set conjecture in low dimensions, illustrating the progression from Wolff's bound to the resolution in $\mathbb{R}^3$:

n	Conjectured $\dim_H$	Best lower bound	Reference
2	2	2 (resolved)	Davies (1971)
3	3	3 (resolved)	Wang–Zahl (2025)
4	4	3	Wolff (1995)
5	5	7/2	Wolff (1995)
n	n	$(n+2)/2$	Wolff (1995)

Exercise 5.4.1

1. Verify the tube intersection estimate (lemma 5.4.4) explicitly in $\mathbb{R}^2$: show that two rectangles of width 2δ and length 1 whose long axes meet at angle α have intersection area $\lesssim \delta^2/\alpha$.
2. Show that the Kakeya maximal conjecture implies the Kakeya set conjecture. Specifically, if K is a Besicovitch set and $f = \chi_{K_\delta}$ (the δ-neighborhood of K), show that the maximal conjecture gives $|K_\delta| \geq c_\epsilon \delta^{n\epsilon}$, and conclude $\dim_M(K) = n$.
3. In $\mathbb{R}^2$, use Córdoba's estimate directly (not through the maximal function) to show that a union of $N = 1/\delta$ tubes with δ-separated directions has area $\gtrsim 1/\log(1/\delta)$. (*Hint*: let $F = \sum_j \chi_{T_j}$. Compute $\|F\|_{L^1}$ and $\|F\|_{L^2}$, then apply Cauchy–Schwarz: $\|F\|_{L^1}^2 \leq |\operatorname{supp}(F)| \cdot \|F\|_{L^2}^2$.)
4. Compute the volume of the intersection of two δ-tubes in $\mathbb{R}^3$ at angle α, and verify the estimate $\lesssim \delta^2/\alpha$.
5. Show that the "bush" configuration—N tubes all passing through a common δ-ball—achieves the volume lower bound $|\bigcup T_j| \gtrsim 1$ when the directions are δ-separated. This shows the Kakeya conjecture would be trivially true if all Besicovitch sets had a "bush" structure.
6. Explain why the Kakeya maximal conjecture remains open in $\mathbb{R}^3$ even after the Wang–Zahl resolution of the set conjecture. What additional information would the maximal conjecture provide?

5.5 The Finite Field Kakeya Conjecture

In 1999, Wolff proposed studying the Kakeya problem over finite fields as a model for the Euclidean case. The finite field setting strips away all analytic structure—there are no limits, no measures, no covering arguments—and reduces the problem to pure combinatorics. A "line" in $\mathbb{F}_q^n$ is a set of q points $\{x + ty : t \in \mathbb{F}_q\}$ for $x, y \in \mathbb{F}_q^n$ with $y \neq 0$, and a "Kakeya set" is a set containing a line in every direction. The question is: how large must such a set be?

Wolff hoped that techniques for the finite field problem would transfer to $\mathbb{R}^n$. The finite field analogue was resolved in 2008 by Dvir, using an algebraic method—the "polynomial method"—that is entirely different from the combinatorial and analytic approaches used in the Euclidean setting. The proof is short and self-contained, making it a compelling application of algebraic techniques to a combinatorial problem.

Definition 5.5.1: Finite Field Kakeya Set

Let $\mathbb{F}_q$ be a finite field with q elements and $\mathbb{F}_q^n$ the n-dimensional vector space over $\mathbb{F}_q$. A set $K \subseteq \mathbb{F}_q^n$ is called a *Kakeya set* if for every direction $y \in \mathbb{F}_q^n \setminus \{0\}$, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subseteq K$.

Since $\mathbb{F}_q^n$ is a finite set of cardinality q^n, the question is quantitative: what is the minimum cardinality of a Kakeya set? There are $(q^n - 1)/(q - 1)$ directions (points of projective space $\mathbb{P}^{n-1}(\mathbb{F}_q)$), and each line has q points, so a Kakeya set contains at least $(q^n - 1)/(q - 1)$ lines. If these lines were disjoint, the minimum size would be $q \cdot (q^n - 1)/(q - 1) \sim q^n$. The finite field Kakeya conjecture asserts that even with maximal overlap, the size is still $\sim q^n$.

The analogy with the Euclidean case is as follows. A "Besicovitch set" in $\mathbb{F}_q^n$ would be a Kakeya set of size $o(q^n)$—much smaller than the ambient space. The finite field conjecture states that no such set exists: every Kakeya set has size $\geq cq^n$, meaning it must occupy a positive fraction of $\mathbb{F}_q^n$. In the Euclidean setting, Besicovitch sets of Lebesgue measure zero *do* exist, so the analogy is not perfect. The finite field conjecture is closer in spirit to the Minkowski dimension statement: the "dimension" of a Kakeya set (measured by the exponent in $|K| \sim q^d$) must equal n.

Before Dvir's work, partial results on the finite field conjecture were obtained by Wolff (1999), Bourgain–Katz–Tao (2004), and others, using combinatorial methods that mirrored the Euclidean techniques (bush and hairbrush arguments). These gave bounds of the form $|K| \geq cq^{\alpha n}$ for various $\alpha < 1$. Dvir's proof, by contrast, gives the optimal exponent $\alpha = 1$ in a single stroke, using an entirely algebraic approach.

The key to Dvir's proof is a vanishing lemma that connects the geometry of lines to the algebra of polynomials:

Lemma 5.5.2: Vanishing on Kakeya Sets

Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set and let $P \in \mathbb{F}_q[x_1, \ldots, x_n]$ be a polynomial of degree $\deg P < q$. If P vanishes on K (i.e., $P(x) = 0$ for all $x \in K$), then P is the zero polynomial.

***Proof*:**
Suppose $P \neq 0$ and $\deg P = d < q$. Write $P = P_d + P_{d-1} + \cdots + P_0$ where P_j is the homogeneous component of degree j and $P_d \neq 0$.

Let $y \in \mathbb{F}_q^n \setminus \{0\}$ be any direction. Since K is a Kakeya set, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subseteq K$. Consider the univariate polynomial $Q(t) = P(x + ty)$. Since P vanishes on K and $x + ty \in K$ for all $t \in \mathbb{F}_q$, the polynomial Q vanishes at all q elements of $\mathbb{F}_q$. The degree of Q as a polynomial in t is at most $d < q$. A univariate polynomial of degree $< q$ over $\mathbb{F}_q$ with q roots must be the zero polynomial. Therefore $Q \equiv 0$.

The leading coefficient of $Q(t) = P(x + ty)$ (as a polynomial in t) is $P_d(y)$: expanding $P(x + ty)$, the terms of degree d in t come exclusively from the homogeneous part P_d, giving the leading term $t^d P_d(y)$. Since $Q \equiv 0$, all coefficients vanish, and in particular $P_d(y) = 0$.

This holds for every $y \in \mathbb{F}_q^n \setminus \{0\}$. Since P_d is homogeneous, $P_d(0) = 0$ as well (every monomial in P_d has positive degree). Thus P_d vanishes on all of $\mathbb{F}_q^n$. A nonzero polynomial of degree d in n variables over $\mathbb{F}_q$ can vanish on at most $d \cdot q^{n-1}$ points of $\mathbb{F}_q^n$ (by the Schwartz–Zippel lemma). Since $d < q$, this bound is $< q^n = |\mathbb{F}_q^n|$, so P_d cannot vanish on all of $\mathbb{F}_q^n$ unless $P_d = 0$. This contradicts $\deg P = d$, completing the proof.

The vanishing lemma has a clean algebraic content: a Kakeya set is "algebraically large" in the sense that no low-degree polynomial can vanish on it. The theorem then follows by a dimension count:

Theorem 5.5.3: Dvir's Theorem

Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set. Then $|K| \geq \binom{q+n-1}{n} \geq \frac{q^n}{n!}$.

***Proof*:**
The vector space $V = \{P \in \mathbb{F}_q[x_1, \dots, x_n] \mid \deg P < q\}$ of polynomials of degree $< q$ in n variables has dimension

$$\dim V = \binom{q+n-1}{n}$$

over $\mathbb{F}_q$ (the number of monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n < q$). Consider the evaluation map $\mathrm{ev} : V \to \mathbb{F}_q^K$ defined by $\mathrm{ev}(P) = (P(x))_{x \in K}$. By lemma 5.5.2, $\ker(\mathrm{ev}) = \{0\}$, so ev is injective. Since the codomain has dimension $|K|$ over $\mathbb{F}_q$:

$$|K| \geq \dim V = \binom{q+n-1}{n} \geq \frac{q^n}{n!}$$

The last inequality follows from $\binom{q+n-1}{n} = \frac{(q+n-1)(q+n-2)\cdots q}{n!} \geq \frac{q^n}{n!}$.

The proof is three lines once the vanishing lemma is established: the polynomial space is too large to be annihilated by evaluation on a small set. The logic is: (1) a Kakeya set cannot support the vanishing of any nonzero low-degree polynomial (by the vanishing lemma), so (2) the evaluation map from the space of low-degree polynomials to $\mathbb{F}_q^K$ must be injective, and (3) injectivity forces $|K| \geq \dim V$. The proof thus converts a geometric property (containing a line in every direction) into an algebraic property (no low-degree polynomial vanishes on K), and then uses linear algebra to extract a size bound.

The bound $|K| \geq q^n/n!$ is sharp up to the constant $1/n!$. The factor $n!$ comes from the binomial coefficient and can be improved: Dvir–Kopparty–Saraf–Sudan (2013) showed $|K| \geq q^n/2^n$ by a refinement involving the multiplicity of vanishing rather than just vanishing itself. The optimal constant remains open, though for large q the bound $|K| \geq (1 + o(1))q^n/2^n$ is expected to be close to sharp.

A natural question is whether the polynomial method gives any information about the *structure* of Kakeya sets, not just their size. Over finite fields, the answer is partially yes: the method shows that Kakeya sets must be "algebraically spread" (they cannot be contained in a low-degree variety), and subsequent work has used this to study the geometric distribution of points in Kakeya sets. Over $\mathbb{R}^n$, as the following remark explains, the method requires substantial modification.

Example 5.5.4: Dvir's Bound in Small Cases

1. Let $q = 3$ and $n = 2$, so $\mathbb{F}_3^2$ has 9 points. The theorem gives $|K| \geq \binom{4}{2} = 6$. A Kakeya set in $\mathbb{F}_3^2$ must contain a line (of 3 points) in each of the 4 directions of $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0), (0,1), (1,1), (1,2)\}$. The 4 lines have 12 points in total, but with overlaps. One can verify that the minimum Kakeya set has $|K| = 7$ (achievable by arranging the 4 lines to

share a common point and one additional overlap), consistent with the lower bound of 6.

2. Let $q = 2$ and $n = 3$, so $\mathbb{F}_2^3$ has 8 points. The theorem gives $|K| \geq \binom{4}{3} = 4$. There are $(2^3 - 1)/(2 - 1) = 7$ directions in $\mathbb{P}^2(\mathbb{F}_2)$. Each "line" has only 2 points ($\{x, x + y\}$ for $t \in \{0, 1\} = \mathbb{F}_2$). With 7 lines of 2 points each, a total of 14 point-memberships must be distributed among at most 8 points. The complement of any single point in $\mathbb{F}_2^3$ has 7 points and can be checked to be a Kakeya set (every direction is represented), so $|K| = 7$ is achievable. The lower bound of 4 is far from tight in this case, reflecting the looseness of the $1/n!$ constant for small q.

5.5.5 Remark: Why the Polynomial Method Does Not Transfer to $\mathbb{R}^n$ The polynomial method is effective over $\mathbb{F}_q$ because a nonzero polynomial of degree d can vanish on at most $d \cdot q^{n-1}$ points of $\mathbb{F}_q^n$ (the Schwartz–Zippel lemma)—a quantitative bound that makes the dimension count work. Over $\mathbb{R}$, no such bound exists: a nonzero polynomial of degree d can vanish on an uncountable set (e.g., the zero set of $x_1^2 + x_2^2 - 1$ is the unit circle, which has the cardinality of $\mathbb{R}$). The discrete counting argument that powers Dvir's proof has no continuous analogue.

The conceptual barrier is that over finite fields, "small" sets are sets with few elements, and "large" sets have many elements—a distinction captured by cardinality. Over $\mathbb{R}^n$, "small" sets are sets of low Hausdorff dimension, and "large" sets have high dimension—a distinction that cardinality cannot detect ($\mathbb{R}$ and $\mathbb{R}^2$ have the same cardinality but different dimensions). The polynomial method over finite fields exploits cardinality, not dimension, which is why it does not transfer.

A partial descendant of the polynomial method in the Euclidean setting is the *polynomial partitioning* technique introduced by Guth and Katz (2010), which uses a polynomial to partition $\mathbb{R}^n$ into cells of controlled geometry rather than to vanish on a set. The idea is to choose a polynomial P of degree d such that $\mathbb{R}^n \setminus Z(P)$ (the complement of the zero set) splits into $\sim d^n$ cells, each containing a controlled number of tubes. The zero set $Z(P)$ itself captures the tubes that lie on an algebraic variety, which can be handled separately. This technique has led to progress on related problems (such as the Erdős distinct distances problem) but has not yet resolved the Euclidean Kakeya conjecture.

The contrast between the finite field and Euclidean settings encapsulates a broader theme in modern combinatorial geometry: problems that are "purely combinatorial" over finite fields often require analytic and multi-scale techniques over $\mathbb{R}$. The Kakeya problem is a paradigmatic example of this phenomenon, and the interplay between algebraic, combinatorial, and analytic methods remains an active area of research.

Exercise 5.5.1

1. Verify Dvir's bound for $q = 2$, $n = 3$: compute the number of directions in $\mathbb{P}^2(\mathbb{F}_2)$, construct a Kakeya set in $\mathbb{F}_2^3$ explicitly, and check $|K| \geq \binom{4}{3} = 4$.

2. Show that lemma 5.5.2 fails if the degree condition is relaxed to $\deg P \leq q$ (rather than $\deg P < q$). (*Hint*: consider the polynomial $P(x) = \prod_{a \in \mathbb{F}_q}(x_1 - a)$, which has degree q and vanishes on $\mathbb{F}_q^n$ regardless of whether K is a Kakeya set.)

3. (Schwartz–Zippel lemma) Prove that a nonzero polynomial $P \in \mathbb{F}_q[x_1, \ldots, x_n]$ of total degree d has at most $d \cdot q^{n-1}$ zeros in $\mathbb{F}_q^n$. (*Hint*: induct on n. For $n = 1$ this is the fundamental

theorem of algebra over $\mathbb{F}_q$. For $n > 1$, write $P = \sum_{j=0}^{d} P_j(x_1, \ldots, x_{n-1})x_n^j$ and consider $a \in \mathbb{F}_q$ such that $P(\cdot, a)$ is nonzero.)

4. Show that a random subset $K \subseteq \mathbb{F}_q^n$ of size $|K| = q^{n-1}/2$ is, with high probability, *not* a Kakeya set when q is large. (*Hint*: for a fixed direction, the probability that a specific line lies in K is $(|K|/q^n)^q$. Apply a union bound over all directions.)

5. The *joints problem* asks: given L lines in $\mathbb{R}^3$, how many "joints" (points where three non-coplanar lines meet) can there be? Guth and Katz (2010) proved the sharp bound $O(L^{3/2})$ using the polynomial method. Formulate and prove the analogous result over $\mathbb{F}_q^3$: if $\mathcal{L}$ is a set of lines in $\mathbb{F}_q^3$ and J is the set of joints, show $|J| \le C|\mathcal{L}|^{3/2}$ using a vanishing argument analogous to lemma 5.5.2. (*Hint*: if $|J| > c|\mathcal{L}|^{3/2}$, find a polynomial of degree $< (|\mathcal{L}|/c)^{1/2}$ vanishing on J, then show its gradient vanishes at every joint, contradicting non-coplanarity.)

5.6 The Wang–Zahl Theorem and Open Frontiers

The results presented so far establish the Kakeya set conjecture in $\mathbb{R}^2$ (Davies, 1971) and provide the lower bound $\dim_H(K) \ge (n+2)/2$ in $\mathbb{R}^n$ (Wolff, 1995). In $\mathbb{R}^3$, Wolff's bound gives $\dim_H(K) \ge 5/2$, leaving a gap of $1/2$ below the conjectured value of 3. This gap was narrowed incrementally over three decades through the introduction of techniques from arithmetic combinatorics, additive number theory, and polynomial methods. The following is a summary of the main advances:

1. Wolff (1995): $\dim_H(K) \ge (n+2)/2$. (Hairbrush argument.)
2. Katz–Łaba–Tao (2000): $\dim_M(K) > 5/2 + \epsilon_0$ in $\mathbb{R}^3$ for an explicit small $\epsilon_0 > 0$. (Additive combinatorics, sticky Kakeya configurations.)
3. Katz–Tao (2002): $\dim_H(K) \ge (2 - \sqrt{2})(n-4) + 3$ for $n \ge 4$. (Sum-product estimates from arithmetic combinatorics.)
4. Katz–Zahl (2019): $\dim_H(K) \ge 5/2 + \epsilon_1$ in $\mathbb{R}^3$ for a small $\epsilon_1 > 0$. (Polynomial partitioning and degree reduction.)
5. Wang–Zahl (2022): Structural classification of near-extremal Kakeya sets in $\mathbb{R}^3$ into "sticky" and "non-sticky" regimes.
6. Wang–Zahl (2025): $\dim_H(K) = 3$ and $\dim_M(K) = 3$ for all Besicovitch sets in $\mathbb{R}^3$.

Each step introduced new ideas that reshaped the problem.

The Katz–Łaba–Tao result (2000) was the first to break Wolff's $5/2$ barrier in $\mathbb{R}^3$. Their approach introduced the notion of "sticky" Kakeya sets: configurations where, at every intermediate scale $\delta < r < 1$, the tubes cluster into thin slabs. They showed that sticky configurations are in some sense the "hardest" case, and handled non-sticky configurations by a refined hairbrush argument. The improvement $5/2 + \epsilon_0$ was small but conceptually significant: it demonstrated that the barrier at $5/2$ was not fundamental.

The Katz–Tao papers (2002) brought arithmetic combinatorics—specifically, sum-product estimates over $\mathbb{R}$—into the Kakeya problem. The connection is through the "arithmetic Kakeya conjecture":

if the set of slopes in a Kakeya configuration has additive structure (e.g., is close to an arithmetic progression), then sum-product estimates force the set to be large. This approach established the first bounds that improve over $(n+2)/2$ in high dimensions.

The Katz–Zahl paper (2019) adapted the polynomial partitioning technique from incidence geometry to the Kakeya setting. By partitioning $\mathbb{R}^3$ using the zero set of a polynomial, they decomposed the Kakeya problem into "cellular" and "algebraic" components, handling each with different methods. This gave a second independent proof that the 5/2 barrier is not sharp.

The Wang–Zahl papers (2022, 2025) unified and extended these ideas into a comprehensive structural theory of Kakeya sets at multiple scales, culminating in the resolution of the three-dimensional case.

Theorem 5.6.1: Wang–Zahl Theorem

Every Besicovitch set $K \subseteq \mathbb{R}^3$ has Hausdorff dimension 3 and Minkowski dimension 3.

The proof, contained in the preprint "Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions" (arXiv:2502.17655, February 2025), is approximately 130 pages. The core argument is an induction on scales, and the key innovation is a reformulation of the Kakeya problem in terms of a more general "super-Kakeya" condition involving convex sets (called "grains") rather than tubes. The term "super-Kakeya" refers to the fact that the reformulated problem is *stronger* than the original Kakeya conjecture: it asserts a volume bound for unions of convex sets satisfying a condition that includes the Kakeya condition as a special case. The stronger formulation is paradoxically easier to prove, because it is preserved under the induction on scales (the original Kakeya condition is not, since passing from scale δ to scale $\sqrt{\delta}$ distorts tubes into more general convex sets).

5.6.2 Remark: Key Ideas of the Proof The proof of theorem 5.6.1 has three main components.
1. Induction on scales. The problem at scale δ (tubes of radius δ) is reduced to a problem at scale $\sqrt{\delta}$ (tubes of radius $\sqrt{\delta}$). At the coarser scale, the tubes are replaced by "grains"—convex sets of controlled dimensions that capture the large-scale geometry. The inductive step shows that if the volume bound fails at scale δ, it must also fail at scale $\sqrt{\delta}$ with a quantitatively worse failure, which eventually contradicts the trivial bound at scale ~ 1.
2. Sticky versus non-sticky configurations. In their 2022 paper, Wang and Zahl classified near-extremal Kakeya configurations into two regimes. In the "sticky" regime, tubes cluster into thin slabs at every intermediate scale, creating a self-similar structure. In the "non-sticky" regime, tubes spread out at intermediate scales. The two regimes require different arguments: the non-sticky case can be handled by a refined version of Wolff's hairbrush argument (combined with polynomial methods), while the sticky case requires a new inductive structure using the grain decomposition.
3. The role of dimension 3. When a grain in the inductive step degenerates (becomes flat or thin), the problem reduces to a lower-dimensional Kakeya problem: a 2-dimensional problem for flat grains, or a 1-dimensional problem for thin grains. In $\mathbb{R}^3$, these lower-dimensional problems are already solved (by Davies' theorem and trivial estimates). This "dimension reduction" is the mechanism that closes the induction. In $\mathbb{R}^n$ for $n \geq 4$, the degenerate cases reduce to Kakeya problems in $\mathbb{R}^3, \mathbb{R}^4, \ldots$, which remain open, blocking the induction.

The Wang–Zahl theorem settles the Kakeya set conjecture in $\mathbb{R}^3$, but several related problems remain open. The state of the art and the main open directions are summarized below.

5.6.3 Remark: Open Problems and Directions *The conjecture in higher dimensions.* The Kakeya set conjecture remains open for all $n \geq 4$. The barrier is the dimension reduction step in the Wang–Zahl argument: degenerate configurations in $\mathbb{R}^n$ reduce to Kakeya problems in $\mathbb{R}^{n-1}$ (and lower), which are not yet solved for $n-1 \geq 4$. Resolving the conjecture in $\mathbb{R}^4$ would likely require either a different inductive scheme or new ideas to handle the degenerate cases without reducing to lower dimensions.
The Kakeya maximal conjecture. Even in $\mathbb{R}^3$, the Kakeya maximal conjecture (**??** 5.4.6) remains open. The Wang–Zahl proof establishes the set conjecture but does not produce the quantitative L^p bounds needed for the maximal statement. Strengthening the result to a maximal function estimate would have immediate consequences for the restriction and Bochner–Riesz conjectures in harmonic analysis.
Furstenberg sets. A *Furstenberg set* is a variant of a Besicovitch set where, instead of requiring a full unit segment in every direction, one requires a subset of Hausdorff dimension $\geq s$ inside a unit segment in every direction. The Furstenberg set conjecture predicts the minimum Hausdorff dimension of such sets as a function of s and n. Wang and Zahl's techniques yield partial results on Furstenberg sets, but the full conjecture remains open.
The (n,k)-Besicovitch conjecture. An (n,k)-Besicovitch set is a set of measure zero in $\mathbb{R}^n$ containing a translate of every k-dimensional unit disk. For $k > 1$, it is conjectured that no such set exists (in contrast to $k = 1$, where they do exist). The case $k = 1$ is the classical Kakeya problem; the higher-k cases are open for all n.
Connections to number theory. The Kakeya conjecture has implications for problems in analytic number theory, including estimates for exponential sums and bounds on the distribution of primes in arithmetic progressions. Montgomery's conjecture on Dirichlet polynomials implies the Kakeya conjecture in all dimensions; the resolution of the three-dimensional case does not immediately yield progress on Montgomery's conjecture, but it strengthens the evidence and provides new techniques that may be applicable. For the number-theoretic connections, see `NateAnalyticNumTheory`.

The resolution of the three-dimensional Kakeya conjecture represents a major advance in geometric measure theory, and the techniques introduced by Wang and Zahl—induction on scales with grain decomposition, the sticky/non-sticky dichotomy, and the interaction with polynomial methods—are expected to have applications beyond the Kakeya problem itself.

The chapter has traced a progression from abstract measure theory (Hausdorff dimension as a refinement of Lebesgue measure) through geometric construction (Besicovitch sets as measure-zero objects with full directional content) to combinatorial analysis (tube intersection estimates and the Kakeya maximal function) and algebra (the polynomial method over finite fields). This progression reflects the interdisciplinary nature of the Kakeya problem: it sits at the intersection of geometric measure theory, combinatorics, algebra, and harmonic analysis. The harmonic-analytic connections—in particular, the relationship between the Kakeya conjecture and the restriction, Bochner–Riesz, and local smoothing conjectures for the Fourier transform—are treated later in this book.

The remaining challenges (higher dimensions, maximal function estimates, Furstenberg sets) provide concrete targets for future work, and the tools developed in this chapter—Hausdorff dimension, Minkowski dimension, tube intersection estimates, and the relationship between directional geometry and volume bounds—are the foundation on which that progress will be built. The reader who wishes to go further is directed to the following references:

1. Wolff, *Lectures on Harmonic Analysis* (AMS, 2003): lecture notes covering the Kakeya problem, restriction estimates, and related topics.
2. Mattila, *Geometry of Sets and Measures in Euclidean Spaces* (Cambridge, 1995): the standard reference for Hausdorff dimension and geometric measure theory.
3. Falconer, *The Geometry of Fractal Sets* (Cambridge, 1985): a more elementary treatment of dimension theory and Besicovitch sets.
4. Wang and Zahl, "Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions" (arXiv:2502.17655, 2025): the proof of the three-dimensional case.

Exercise 5.6.1

1. Verify that Wolff's bound (theorem 5.4.9) gives $\dim_H(K) \geq 5/2$ for Besicovitch sets in $\mathbb{R}^3$, and compare this to the Wang–Zahl result $\dim_H(K) = 3$.
2. Show that if the Kakeya set conjecture holds in $\mathbb{R}^n$, it automatically holds in $\mathbb{R}^m$ for all $m \leq n$. (*Hint*: embed $\mathbb{R}^m$ into $\mathbb{R}^n$ and use proposition 5.1.8(4).)
3. Explain in your own words why the Wang–Zahl argument does not immediately extend to $\mathbb{R}^4$. Identify the specific step that fails and what additional input would be needed.
4. The Besicovitch set K constructed in theorem 5.3.4 has $m(K) = 0$ and $\dim_H(K) = 2$ (by theorem 5.3.6). Show that K must have $\dim_H(K) \geq 2$ directly from the Kakeya maximal function estimate in $\mathbb{R}^2$ (theorem 5.4.8), without appealing to Davies' theorem.
5. Show that a Besicovitch set $K \subseteq \mathbb{R}^3$ with $m(K) = 0$ is nowhere dense (has empty interior). Relate this to definition 1.4.12: K is topologically small (meager) yet has full Hausdorff dimension by the Wang–Zahl theorem. Compare with proposition 1.4.13.

6

Haar Measure

The Riesz-Markov theorem of chapter 4 turned the construction of a measure into the construction of a functional. This chapter cashes that in on the one class of spaces where a canonical functional presents itself: a locally compact group. On such a group the operation of left translation acts on functions, and one asks for a measure that translation cannot see, a *left-invariant* Radon measure. That there is such a measure, unique up to scale, is the theorem of Haar, and it is the single fact on which all of harmonic analysis on groups rests: without it there is no $L^2(G)$ to decompose, no convolution algebra $L^1(G)$, no unitary regular representation. The point-set theory of locally compact groups is developed in [7]; here they enter only as the spaces on which the invariant measure is built.

The construction is Weil's, and it is worth seeing why a construction is needed at all rather than a formula. On $\mathbb{R}^n$ or a Lie group an invariant measure can be written down from a volume form, but a general locally compact group carries no such differentiable structure. Weil's idea replaces the missing formula with a comparison: the *covering number* $(f : g)$ counts how many left-translates of a fixed small bump g are needed to dominate f, a discrete stand-in for the ratio of the integrals of f and g. As the bump g concentrates at the identity this ratio stabilises, and the limit is the invariant functional; Riesz-Markov then converts it into Haar measure. The limit is extracted by a compactness argument, an application of Tychonoff's theorem, and this is one of its most striking uses. The first section carries out the construction and the second proves uniqueness and introduces the modular function that measures the failure of left-invariance to be two-sided.

6.1 Existence of Haar Measure

On the line, Lebesgue measure is singled out by translation invariance: the length of a set does not change when the set is slid along $\mathbb{R}$. A locally compact group carries the same operation of sliding, now written as multiplication by a fixed element, and the question this section answers is whether an invariant measure survives the passage from $\mathbb{R}$ to an arbitrary such group. The affirmative answer is Weil's, and its engine is the Riesz-Markov theorem of chapter 4: rather than build the measure set by set, the argument constructs an invariant way of averaging continuous functions and lets theorem 4.2.1 convert that averaging operation into a measure. The whole labour is therefore in manufacturing a positive, left-invariant linear functional on $C_c(G)$ out of nothing but the group operation and compactness.

Throughout, G is a locally compact Hausdorff group in the sense of [7]: a group carrying a locally compact Hausdorff topology for which multiplication and inversion are continuous. The identity is e. The group acts on functions by *left translation*: for $s \in G$ and $f\colon G \to \mathbb{R}$,

$$L_s f(x) = f(s^{-1}x)$$

the inverse in $s^{-1}x$ being what makes $s \mapsto L_s$ a left action, $L_{st} = L_s L_t$. Left translation carries $C_c(G)$ into itself, since $L_s f$ is continuous and $\operatorname{supp}(L_s f) = s \operatorname{supp} f$ is compact. Write $C_c^+(G)$ for the nonnegative functions in $C_c(G)$. The construction below produces a measure invariant under left translation, a *left* Haar measure; the right-invariant theory is its mirror image, obtained by replacing L_s with right translation throughout, and the comparison of the two is the subject of section 6.2.

The one analytic input, needed only at the final additivity step, is that a compactly supported continuous function cannot oscillate too fast under small translations. On $\mathbb{R}$ this is uniform continuity; on a group the correct formulation is uniform continuity with respect to the group's own notion of smallness, a neighbourhood of the identity.

Lemma 6.1.1: Left-Uniform Continuity

Let G be a locally compact Hausdorff group and $f \in C_c(G)$. For every $\varepsilon > 0$ there is a neighbourhood V of the identity e such that

$$|f(sx) - f(x)| < \varepsilon \quad \text{for all } x \in G \text{ and all } s \in V$$

Equivalently, $\|L_{s^{-1}} f - f\|_\infty < \varepsilon$ for every $s \in V$. The same argument produces a neighbourhood V' of e with the right-translate estimate

$$|f(xt) - f(x)| < \varepsilon \quad \text{for all } x \in G \text{ and all } t \in V'$$

***Proof*:**
The equivalence of the two formulations is a change of variable: $L_{s^{-1}} f(x) = f(sx)$, so $\|L_{s^{-1}} f - f\|_\infty = \sup_x |f(sx) - f(x)|$, and the displayed bound over all x is exactly $\|L_{s^{-1}} f - f\|_\infty \leq \varepsilon$. It suffices to prove the pointwise bound.

Step 1: Localize using continuity at each point. Let $K = \operatorname{supp} f$, a compact set. Fix $\varepsilon > 0$. Because f is continuous, for each point $y \in G$ there is an open neighbourhood of y on which f varies by less than $\varepsilon/2$. Continuity of multiplication lets this neighbourhood be taken in translated form: there is an open neighbourhood W_y of the identity e with

$$|f(wy) - f(y)| < \frac{\varepsilon}{2} \quad \text{for all } w \in W_y$$

since $w \mapsto wy$ is continuous and equals y at $w = e$. By the symmetric-neighbourhood halving lemma of [7], choose for each y a symmetric open neighbourhood V_y of e with $V_y V_y \subseteq W_y$.

Step 2: Cover the support and extract a finite subcover. As y ranges over K, the open sets $V_y y$ cover K, each containing its own y. Compactness of K yields finitely many points $y_1, \dots, y_n$ with

$$K \subseteq \bigcup_{i=1}^{n} V_{y_i} y_i$$

Set $V = \bigcap_{i=1}^{n} V_{y_i}$, an open neighbourhood of e as a finite intersection; shrinking if necessary, take V symmetric. This V is the neighbourhood claimed.

Step 3: The uniform estimate. Fix $s \in V$ and $x \in G$; the goal is $|f(sx) - f(x)| < \varepsilon$. Consider two cases according to whether x or sx meets the support.

Suppose first that $x \in K$. Then $x \in V_{y_i} y_i$ for some i, so $x = v y_i$ with $v \in V_{y_i}$; since V_{y_i} is symmetric this gives $x y_i^{-1} = v \in V_{y_i} \subseteq W_{y_i}$ and hence, applying the Step 1 bound with $w = x y_i^{-1}$,

$$|f(x) - f(y_i)| = |f\big((x y_i^{-1}) y_i\big) - f(y_i)| < \frac{\varepsilon}{2}$$

For the translated point, $sx = (sv) y_i$ with $s \in V \subseteq V_{y_i}$ and $v \in V_{y_i}$, so $sv \in V_{y_i} V_{y_i} \subseteq W_{y_i}$, and the same bound gives

$$|f(sx) - f(y_i)| = |f\big((sv) y_i\big) - f(y_i)| < \frac{\varepsilon}{2}$$

The triangle inequality now yields $|f(sx) - f(x)| < \varepsilon$.

If instead $x \notin K$ but $sx \in K$, apply the previous paragraph with x replaced by sx and s by $s^{-1} \in V$ (using V symmetric): from $sx \in K$ one gets $|f(s^{-1}(sx)) - f(sx)| < \varepsilon$, which is $|f(x) - f(sx)| < \varepsilon$. Finally, if neither x nor sx lies in K, then $f(x) = f(sx) = 0$ and the bound is trivial. In every case $|f(sx) - f(x)| < \varepsilon$, which is the left-translate estimate.

Step 4: The right-translate estimate. The argument for V' is identical with the two sides of every product interchanged. Continuity of f at y and continuity of the map $w \mapsto yw$ furnish an open neighbourhood W'_y of e with $|f(yw) - f(y)| < \varepsilon/2$ for $w \in W'_y$; halving gives a symmetric V'_y with $V'_y V'_y \subseteq W'_y$; the sets $y_i V'_{y_i}$ cover K; and $V' = \bigcap_i V'_{y_i}$, taken symmetric, satisfies $|f(xt) - f(x)| < \varepsilon$ for all x and all $t \in V'$, by the same three-case analysis on whether x or xt meets K. This proves both estimates, completing the proof.

Left-uniform continuity says that translating f by an element close to e perturbs it uniformly little in supremum norm. The role reserved for it is delicate: the covering numbers built next are only *sub*additive in general, and the reason they become exactly additive in the limit is that, over a small enough neighbourhood, a function and its translates are indistinguishable to within ε, so the overcounting in subadditivity is squeezed to zero. Nothing else in the construction uses continuity of f beyond boundedness; this lemma is where the topology of G enters the analysis.

Example 6.1.2: Left-Uniform Continuity on $\mathbb{R}$ and on the Circle

On the additive group $\mathbb{R}$, left translation by s is the shift $L_s f(x) = f(x-s)$, and lemma 6.1.1 reduces to the classical statement that a continuous function of compact support is uniformly continuous: given ε, a symmetric interval $V = (-\delta, \delta)$ works, since $|x - y| < \delta$ forces $|f(x) - f(y)| < \varepsilon$. The neighbourhood V of 0 is exactly the modulus of continuity δ.

On the circle group $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, which is compact, every continuous function has compact support, so $C_c(\mathbb{T}) = C(\mathbb{T})$ and the lemma applies to all continuous functions. A rotation by a small angle moves each point a small arc-length, and a continuous function on the compact circle is uniformly continuous, so again a small symmetric arc V about the identity controls $|f(sx) - f(x)|$ uniformly. The compactness of $\mathbb{T}$ is what removes the compact-support hypothesis; on $\mathbb{R}$ the hypothesis cannot be dropped, as the non-uniformly-continuous $\sin(x^2)$ shows.

With the analytic tool in hand, the combinatorial core of the construction can begin. The idea is to measure the size of one function relative to another by asking how many translated copies of the second are needed to cover the first. If g is thought of as a fixed unit of mass, concentrated near a small region, then covering f by weighted left-translates of g and adding the weights estimates the total mass of f in units of g.

Definition 6.1.3: Haar Covering Number

Let G be a locally compact Hausdorff group and let $f, g \in C_c^+(G)$ with g not identically zero. The *Haar covering number* of f with respect to g is

$$(f : g) = \inf \left\{ \sum_{i=1}^{n} c_i \;\middle|\; n \in \mathbb{N},\ c_i \geq 0,\ s_i \in G,\ f \leq \sum_{i=1}^{n} c_i \, L_{s_i} g \right\}$$

the infimum of the total weight over all finite dominations of f by nonnegative combinations of left-translates of g (including the empty sum, $n = 0$, which dominates $f \equiv 0$ with total weight 0).

The covering number is a discrete stand-in for the ratio of integrals $\left(\int f\right)/\left(\int g\right)$. If a left-invariant integral $\int$ existed, then $f \leq \sum_i c_i L_{s_i} g$ would integrate to $\int f \leq \left(\sum_i c_i\right) \int g$ by invariance, so $\left(\int f\right)/\left(\int g\right) \leq (f : g)$; the covering number is an a priori upper bound for a ratio that does not yet exist, computed purely from the order structure and the group action. The construction extracts the integral by making g shrink toward a point mass at e, at which stage the upper bound becomes exact.

Example 6.1.4: Covering Numbers of Intervals on $\mathbb{R}$

Take $G = \mathbb{R}$. Let g be the tent function of height 1 supported on $[-1, 1]$, equal to $1 - |x|$ there, and let f be the trapezoid of height 2 supported on $[-3, 3]$, equal to 2 on $[-2, 2]$ and tapering linearly to 0 at ± 3. To dominate f by translates $c_i L_{s_i} g(x) = c_i \, g(x - s_i)$, place tents along the interval: translate g to sit centred at each half-integer $-\frac{5}{2}, -2, -\frac{3}{2}, \ldots, \frac{5}{2}$ and give each weight 2. Adjacent unit tents of weight 2 overlap so that their sum stays at least 2 across the plateau $[-2, 2]$ and dominates the tapering flanks, so this domination by eleven translates of weight 2 gives $(f : g) \leq 22$. In the other direction, proposition 6.1.5(f) below gives $(f : g) \geq (\sup f)/(\sup g) = 2/1 = 2$. The true value lies between; its precise size is unimportant, because only the ratio $(f : g)/(f_0 : g)$ against a fixed reference f_0 survives into the limit. What matters is that g concentrated near 0 makes the covering count proportional to the length of the base of f, which is the geometric content of Lebesgue measure.

The covering number obeys a package of order and invariance properties, each read directly off the definition. They are the axioms an integral must satisfy, verified here at the discrete level; the limit will inherit every one of them that survives a passage to a cluster point.

Proposition 6.1.5: Properties of the Covering Number

Let $f, f_1, f_2, g, h \in C_c^+(G)$ with g, h not identically zero. Then:

1. *Finiteness:* $(f : g) < \infty$.
2. *Subadditivity and homogeneity:* $(f_1 + f_2 : g) \leq (f_1 : g) + (f_2 : g)$ and $(cf : g) = c\,(f : g)$ for every $c \geq 0$.
3. *Left invariance:* $(L_s f : g) = (f : g)$ for every $s \in G$.
4. *Submultiplicativity:* $(f : h) \leq (f : g)\,(g : h)$.
5. *Monotonicity:* $f_1 \leq f_2$ implies $(f_1 : g) \leq (f_2 : g)$.
6. *Lower bound:* $\dfrac{\sup f}{\sup g} \leq (f : g)$.

***Proof*:**
(a) Finiteness. The support $K = \mathrm{supp} f$ is compact. Since g is continuous and not identically zero, it is positive somewhere; pick a point x_0 with $g(x_0) = a > 0$ and an open neighbourhood N of x_0 with $g > a/2$ on N. The left-translates sN, as s ranges over G, cover G, so finitely many $s_1 N, \ldots, s_n N$ cover K. For each i the translate $L_{s_i} g$ satisfies $L_{s_i} g(x) = g(s_i^{-1} x) > a/2$ whenever $x \in s_i N$, because then $s_i^{-1} x \in N$, where $g > a/2$. Set $c = 2 \sup f / a$ and consider the domination candidate $\sum_{i=1}^n c\, L_{s_i} g$. At any point $x \in K$, some index i has $x \in s_i N$, so that term alone gives

$$\sum_{i=1}^n c\, L_{s_i} g(x) \geq c\, L_{s_i} g(x) > c \cdot \frac{a}{2} = \sup f \geq f(x)$$

using that all terms are nonnegative; off K the function f vanishes so the inequality $f \leq \sum_i c\, L_{s_i} g$ is automatic there. This exhibits a finite domination of total weight $nc = 2n \sup f / a$, so $(f : g) \leq 2n \sup f / a < \infty$.

(b) Subadditivity and homogeneity. If $f_1 \leq \sum_i c_i L_{s_i} g$ and $f_2 \leq \sum_j d_j L_{t_j} g$ are dominations, then adding them gives $f_1 + f_2 \leq \sum_i c_i L_{s_i} g + \sum_j d_j L_{t_j} g$, a domination of $f_1 + f_2$ with total weight $\sum_i c_i + \sum_j d_j$. Taking infima over the two families separately yields $(f_1 + f_2 : g) \leq (f_1 : g) + (f_2 : g)$. For homogeneity with $c > 0$, scaling a domination of f by c gives a domination of cf and multiplies every admissible total weight by c, so $(cf : g) = c(f : g)$; the case $c = 0$ is $(0 : g) = 0$, taking the empty sum.

(c) Left invariance. Left translation is an order isomorphism commuting with the action: if $f \leq \sum_i c_i L_{s_i} g$ then applying L_s to both sides, and using $L_s L_{s_i} = L_{ss_i}$, gives $L_s f \leq \sum_i c_i L_{ss_i} g$, a domination of $L_s f$ with the same weights. Thus $(L_s f : g) \leq (f : g)$, and applying the same to $L_{s^{-1}}$ gives the reverse inequality, so $(L_s f : g) = (f : g)$.

(d) Submultiplicativity. Suppose $f \le \sum_i c_i L_{s_i} g$ and $g \le \sum_j d_j L_{t_j} h$. Substituting the second domination into the first and using $L_{s_i}\bigl(\sum_j d_j L_{t_j} h\bigr) = \sum_j d_j L_{s_i t_j} h$ (again $L_{s_i} L_{t_j} = L_{s_i t_j}$, and L_{s_i} preserves the pointwise inequality $g \le \sum_j d_j L_{t_j} h$ since it only relabels the argument),

$$f \le \sum_i c_i L_{s_i} g \le \sum_i c_i \sum_j d_j L_{s_i t_j} h = \sum_{i,j} c_i d_j \, L_{s_i t_j} h$$

a domination of f by translates of h with total weight $\bigl(\sum_i c_i\bigr)\bigl(\sum_j d_j\bigr)$. Taking infima over the two families gives $(f : h) \le (f : g)(g : h)$.

(e) Monotonicity. If $f_1 \le f_2$ then any domination $f_2 \le \sum_i c_i L_{s_i} g$ also dominates f_1, so the admissible weights for f_1 include those for f_2, and the infimum can only decrease: $(f_1 : g) \le (f_2 : g)$.

(f) Lower bound. Let $f \le \sum_i c_i L_{s_i} g$ be any domination. Evaluating at a point x^* where $f(x^*) = \sup f$,

$$\sup f = f(x^*) \le \sum_i c_i \, g(s_i^{-1} x^*) \le \Bigl(\sum_i c_i\Bigr) \sup g$$

since each $g(s_i^{-1} x^*) \le \sup g$. Hence $\sum_i c_i \ge (\sup f)/(\sup g)$ for every domination, and taking the infimum gives $(\sup f)/(\sup g) \le (f : g)$, completing the proof.

The six properties are precisely what an invariant integral should do, minus one: the covering number is subadditive, but genuine additivity, $(f_1 + f_2 : g) = (f_1 : g) + (f_2 : g)$, fails, because a single translate of g can straddle the supports of f_1 and f_2 and be counted once toward covering their sum while it would have to be counted twice to cover them separately. Recovering additivity is the whole difficulty, and it is resolved only in the limit as g shrinks. Before taking that limit, the covering number is normalized so that its values live in a fixed compact range independent of the shrinking parameter g, which is what will make a limit available.

Fix once and for all a reference function $f_0 \in C_c^+(G)$ with f_0 not identically zero, and for $f \in C_c^+(G)$ and $g \in C_c^+(G)$ (both nonzero) define the normalized ratio

$$I_g(f) = \frac{(f : g)}{(f_0 : g)}$$

The point of dividing by $(f_0 : g)$ is that the arbitrary unit g cancels to first order, leaving a quantity pinned between two bounds that do not mention g at all. Applying submultiplicativity, proposition 6.1.5(d), twice, once as $(f : g) \le (f : f_0)(f_0 : g)$ and once as $(f_0 : g) \le (f_0 : f)(f : g)$, gives after dividing by $(f_0 : g)$ the two-sided estimate

$$\frac{1}{(f_0 : f)} \le I_g(f) \le (f : f_0) \tag{6.1}$$

valid for every admissible g. The lower bound is nonzero because $(f_0 : f) < \infty$ by proposition 6.1.5(a).

Thus each $I_g(f)$ is confined to the compact interval $[\,1/(f_0 : f),\ (f : f_0)\,]$, whose endpoints depend on f but not on g.

This uniform confinement is exactly what a compactness argument feeds on. As g shrinks, the family $\{I_g\}$ has no reason to converge, but it cannot escape to infinity, and Tychonoff's theorem converts that boundedness into the existence of a limiting functional along a subnet.

The functionals I_g, for g ranging over $C_c^+(G) \setminus \{0\}$, are points of the product space

$$X = \prod_{f \in C_c^+(G) \setminus \{0\}} \left[\tfrac{1}{(f_0 : f)},\ (f : f_0) \right]$$

one coordinate for each nonzero $f \in C_c^+(G)$, the f-coordinate of the point I_g being the number $I_g(f)$; that this is a legitimate point of X is precisely the bound (6.1). Each factor is a compact interval in $\mathbb{R}$, so by *Tychonoff's theorem* [7], which states that an arbitrary product of compact spaces is compact in the product topology, the space X is compact.

To take the limit as g concentrates at e, index the family not by the functions g themselves but by the neighbourhoods of e they can be squeezed into. Let $\mathcal{N}$ be the collection of open neighbourhoods of e, directed by reverse inclusion: declare $V' \succeq V$ when $V' \subseteq V$. This is a genuine directed set, since any two neighbourhoods V_1, V_2 have the common refinement $V_1 \cap V_2 \succeq V_1, V_2$, itself a neighbourhood of e. For each $V \in \mathcal{N}$ choose, once and for all, a nonzero $g_V \in C_c^+(G)$ with $\operatorname{supp} g_V \subseteq V$; such a g_V exists by lemma 4.1.5, applied to a compact neighbourhood of e sitting inside V. The assignment $V \mapsto I_{g_V}$ is then a net in the compact space X. A net in a compact space has a cluster point, so there is a point $I \in X$ that the net $\{I_{g_V}\}$ frequents: for every neighbourhood of I in X and every $V_0 \in \mathcal{N}$, some $V \subseteq V_0$ has I_{g_V} in that neighbourhood. Concretely, since the topology on X is the product topology and the g_V can be squeezed into arbitrarily small neighbourhoods of e, I being a cluster point means that for every finite set of functions $f_1, \ldots, f_k$, every $\varepsilon > 0$, and every neighbourhood U of e, there is a nonzero $g \in C_c^+(G)$ with $\operatorname{supp} g \subseteq U$ and

$$|I_g(f_j) - I(f_j)| < \varepsilon \quad \text{for } j = 1, \ldots, k \tag{6.2}$$

This $I\colon C_c^+(G) \setminus \{0\} \to \mathbb{R}$ is the candidate functional. It remains to show that it has the algebraic properties of an integral, at which point the Riesz-Markov theorem finishes the job. The properties fall into two groups: those that hold for every I_g and are inherited by the cluster point because they are closed conditions, and additivity, which holds for I_g only approximately and becomes exact precisely in the limit.

Closed properties inherited by I. Each of the following holds for every I_g, hence for the cluster point I, because it is a conjunction of equalities and non-strict inequalities among finitely many coordinates, and such conditions are closed in the product topology and preserved under passing to a cluster point. First, normalization: $I_g(f_0) = (f_0 : g)/(f_0 : g) = 1$ for every g, so $I(f_0) = 1$; in particular I is not identically zero. Second, positive homogeneity: $I_g(cf) = cI_g(f)$ for $c \geq 0$ by proposition 6.1.5(b), so $I(cf) = cI(f)$. Third, monotonicity: $f_1 \leq f_2$ gives $I_g(f_1) \leq I_g(f_2)$ by proposition 6.1.5(e), so $I(f_1) \leq I(f_2)$; in particular $I(f) \geq 0$. Fourth, left invariance: $I_g(L_s f) = I_g(f)$ by proposition 6.1.5(c), so $I(L_s f) = I(f)$. Fifth, subadditivity: $I_g(f_1 + f_2) \leq I_g(f_1) + I_g(f_2)$ by proposition 6.1.5(b), so $I(f_1 + f_2) \leq I(f_1) + I(f_2)$.

The one property still missing is the reverse inequality, superadditivity, which together with subadditivity forces additivity. This is the only place the shrinking of g does real work, and the only place left-uniform continuity is used.

Lemma 6.1.6: Approximate Additivity of the Covering Number

Let $f_1, f_2 \in C_c^+(G)$ and $\varepsilon > 0$. There is a neighbourhood U of e such that for every nonzero $g \in C_c^+(G)$ with $\mathrm{supp} g \subseteq U$,

$$(f_1 : g) + (f_2 : g) \le (1+\varepsilon)\,(f_1 + f_2 : g)$$

***Proof*:**
Step 1: A master function separating the supports. Let $K = \mathrm{supp}(f_1 + f_2)$, a compact set, and choose by lemma 4.1.5 a function $\varphi \in C_c^+(G)$ with $\varphi \equiv 1$ on K. Fix a parameter $\delta > 0$ to be chosen at the end, and set

$$F = f_1 + f_2 + \delta\varphi$$

which is strictly positive on K (indeed $F \ge \delta$ there) and lies in $C_c^+(G)$. Define, for $k = 1, 2$, the functions

$$h_k(x) = \begin{cases} \dfrac{f_k(x)}{F(x)} & F(x) > 0 \\ 0 & F(x) = 0 \end{cases}$$

Each h_k is continuous: on the open set $\{F > 0\}$ it is a quotient of continuous functions, so continuous there. At a point x with $F(x) = 0$ the function h_k vanishes together with f_k throughout a whole neighbourhood, so it is continuous at x as well. Indeed $\mathrm{supp} f_k \subseteq \mathrm{supp}(f_1 + f_2) = K$ and $\varphi \equiv 1$ on K, so $F = f_1 + f_2 + \delta\varphi \ge \delta > 0$ on K; hence $F(x) = 0$ forces $x \notin K$, and since K is closed, x has a neighbourhood disjoint from K. On that neighbourhood $f_k \equiv 0$ (as $\mathrm{supp} f_k \subseteq K$) and therefore $h_k \equiv 0$, so h_k is locally constant and in particular continuous at x. Thus $h_k \in C_c^+(G)$, $\mathrm{supp} h_k \subseteq \mathrm{supp} f_k$, and $h_1 + h_2 \le 1$ everywhere, with $h_1 + h_2 = (f_1 + f_2)/F \le 1$ where $F > 0$. Crucially $f_k = h_k F$ pointwise.

Step 2: Uniform continuity of the h_k. By the right-translate estimate of lemma 6.1.1 applied to h_1 and h_2, there is a neighbourhood U of e such that

$$|h_k(xt) - h_k(x)| < \delta \quad \text{for all } x \in G,\ t \in U,\ k = 1, 2 \tag{6.3}$$

Shrinking U, take it symmetric. This U is the neighbourhood claimed; the estimate below holds for any g with $\mathrm{supp} g \subseteq U$.

Step 3: Transport a domination of F into a joint domination of f_1, f_2. Fix a nonzero $g \in C_c^+(G)$ with $\mathrm{supp} g \subseteq U$, and let $F \le \sum_i c_i L_{s_i} g$ be any domination of F by translates of g. The claim is that this single domination controls f_1 and f_2 simultaneously with only an $O(\delta)$ loss. Consider a point x and an index i contributing to the sum, meaning $L_{s_i} g(x) = g(s_i^{-1}x) \ne 0$; then $z := s_i^{-1}x \in \mathrm{supp} g \subseteq U$, so $x = s_i z$ with $z \in U$. Applying (6.3) with argument s_i and $t = z$ gives $|h_k(s_i z) - h_k(s_i)| < \delta$, that is,

$$h_k(x) \le h_k(s_i) + \delta \quad \text{whenever } g(s_i^{-1}x) \ne 0 \tag{6.4}$$

Now estimate $f_k = h_k F$. At any point x, using $f_k = h_k F \le h_k \sum_i c_i L_{s_i} g$ and inserting (6.4) term by term (each term with $L_{s_i} g(x) \ne 0$ satisfies $h_k(x) \le h_k(s_i) + \delta$, and terms with $L_{s_i} g(x) = 0$

contribute nothing),

$$f_k(x) = h_k(x)F(x) \le h_k(x) \sum_i c_i \, L_{s_i} g(x) \le \sum_i c_i \big(h_k(s_i) + \delta\big) L_{s_i} g(x)$$

Hence $f_k \le \sum_i c_i \big(h_k(s_i) + \delta\big) L_{s_i} g$ is a domination of f_k, so by definition of the covering number

$$(f_k : g) \le \sum_i c_i \big(h_k(s_i) + \delta\big)$$

Summing over $k = 1, 2$ and using $h_1(s_i) + h_2(s_i) \le 1$,

$$(f_1 : g) + (f_2 : g) \le \sum_i c_i \big(h_1(s_i) + h_2(s_i) + 2\delta\big) \le (1 + 2\delta) \sum_i c_i$$

This holds for every domination of F, so taking the infimum over such dominations gives

$$(f_1 : g) + (f_2 : g) \le (1 + 2\delta)\,(F : g) \tag{6.5}$$

Step 4: Absorb the master term and choose δ. It remains to compare $(F : g)$ with $(f_1 + f_2 : g)$. By subadditivity and homogeneity, proposition 6.1.5(b),

$$(F : g) = (f_1 + f_2 + \delta\varphi : g) \le (f_1 + f_2 : g) + \delta\,(\varphi : g)$$

The term $(\varphi : g)$ is bounded by $(f_1 + f_2 : g)$ up to a g-independent constant. By submultiplicativity, proposition 6.1.5(d),

$$(\varphi : g) \le (\varphi : f_1 + f_2)\,(f_1 + f_2 : g)$$

and $A = (\varphi : f_1 + f_2)$ is finite by proposition 6.1.5(a) and depends only on f_1, f_2, φ, not on g. Substituting,

$$(F : g) \le \big(1 + \delta A\big)\,(f_1 + f_2 : g)$$

Combining with (6.5),

$$(f_1 : g) + (f_2 : g) \le (1 + 2\delta)(1 + \delta A)\,(f_1 + f_2 : g)$$

Given $\varepsilon > 0$, choose δ small enough that $(1 + 2\delta)(1 + \delta A) \le 1 + \varepsilon$; since A depends only on f_1, f_2, this fixes δ, and with it the master function F and the neighbourhood U from Step 2, before any g is chosen. The resulting inequality is the claim, completing the proof.

Dividing lemma 6.1.6 through by $(f_0 : g)$ turns it into a statement about the normalized functionals: for g supported in the neighbourhood $U = U(\varepsilon)$,

$$I_g(f_1) + I_g(f_2) \le (1 + \varepsilon)\, I_g(f_1 + f_2) \tag{6.6}$$

This is the near-superadditivity that the cluster point converts into exact superadditivity.

Proposition 6.1.7: The Limit Functional Is Additive

The cluster point I of the net $\{I_g\}$ satisfies $I(f_1+f_2) = I(f_1)+I(f_2)$ for all $f_1, f_2 \in C_c^+(G)$.

***Proof*:**
Subadditivity, $I(f_1+f_2) \le I(f_1)+I(f_2)$, was already inherited from proposition 6.1.5(b) as a closed condition. Only superadditivity remains.

Fix $\varepsilon > 0$ and let $U = U(\varepsilon)$ be the neighbourhood provided by lemma 6.1.6, so that (6.6) holds for every g with $\operatorname{supp} g \subseteq U$. Apply the cluster-point property (6.2) to the three functions f_1, f_2, f_1+f_2 with tolerance ε and the neighbourhood U: there is a single g with $\operatorname{supp} g \subseteq U$ such that simultaneously

$$|I_g(f_1) - I(f_1)| < \varepsilon, \qquad |I_g(f_2) - I(f_2)| < \varepsilon, \qquad |I_g(f_1+f_2) - I(f_1+f_2)| < \varepsilon$$

For this g, both (6.6) and these three approximations hold. Combining them,

$$I(f_1)+I(f_2) < I_g(f_1)+I_g(f_2)+2\varepsilon \le (1+\varepsilon)I_g(f_1+f_2)+2\varepsilon < (1+\varepsilon)\big(I(f_1+f_2)+\varepsilon\big)+2\varepsilon$$

The right-hand side is $I(f_1+f_2) + \varepsilon\, I(f_1+f_2) + \varepsilon(1+\varepsilon) + 2\varepsilon$. As $\varepsilon \to 0$ the correction terms vanish, since $I(f_1+f_2)$ is a fixed finite number, and the inequality passes to the limit as $I(f_1)+I(f_2) \le I(f_1+f_2)$. Together with subadditivity this gives $I(f_1)+I(f_2) = I(f_1+f_2)$, as we sought to show.

Additivity is the last missing axiom, and its proof is the reason the whole construction runs through a shrinking net rather than a single covering number: only in the limit do the translated copies of g stop straddling the supports of f_1 and f_2, and lemma 6.1.1, through lemma 6.1.6, is what quantifies the shrinking overcounting. The functional I is now additive and positively homogeneous on $C_c^+(G)$, hence extends by $I(f) = I(f^+) - I(f^-)$ to a positive linear functional on all of $C_c(G)$, real-valued and, by the closed properties above, left-invariant and nonzero.

The extension deserves one line of care, since additivity and homogeneity were established only on the nonnegative cone. Every $f \in C_c(G)$ decomposes as $f = f^+ - f^-$ with $f^\pm \in C_c^+(G)$, and setting $I(f) = I(f^+) - I(f^-)$ is well-defined and linear by the standard argument: if $f = u - v$ with $u, v \in C_c^+(G)$ then $u + f^- = v + f^+$, so additivity on the cone gives $I(u) + I(f^-) = I(v) + I(f^+)$, whence $I(u) - I(v) = I(f^+) - I(f^-)$, and linearity over $\mathbb{R}$ follows from additivity and homogeneity on the cone. Positivity is immediate: $f \ge 0$ means $f = f^+$, so $I(f) = I(f^+) \ge 0$. Left invariance and $I(f_0) = 1 \ne 0$ persist.

Everything is now in place. A positive, left-invariant, nonzero linear functional on $C_c(G)$ has been built from the group structure alone, and theorem 4.2.1 turns any such functional into a measure.

Theorem 6.1.8: Existence of Haar Measure

Every locally compact Hausdorff group G admits a nonzero left-invariant Radon measure μ: a Radon measure, not identically zero, with $\mu(sE) = \mu(E)$ for every $s \in G$ and every Borel set $E \subseteq G$.

***Proof*:**
Let $I: C_c(G) \to \mathbb{R}$ be the positive, left-invariant, nonzero linear functional constructed above, with $I(f_0) = 1$. By the Riesz-Markov representation theorem, theorem 4.2.1, applied to the locally compact Hausdorff space G, there is a unique Radon measure μ on G with

$$I(f) = \int_G f\, d\mu$$

for all $f \in C_c(G)$. The measure is not identically zero, since $\int_G f_0\, d\mu = I(f_0) = 1$ forces μ to give positive mass to the support of f_0.

It remains to transfer left invariance of I to left invariance of μ. Fix $s \in G$ and define the translated measure $\mu_s(E) = \mu(s^{-1}E)$, again a Radon measure, since $E \mapsto s^{-1}E$ is a homeomorphism preserving the Radon axioms. For an indicator, $\int_G \chi_E\, d\mu_s = \mu(s^{-1}E) = \int_G \chi_{s^{-1}E}\, d\mu$, and $\chi_{s^{-1}E}(x) = \chi_E(sx) = L_{s^{-1}}\chi_E(x)$; extending by linearity and monotone limits, $\int_G f\, d\mu_s = \int_G L_{s^{-1}} f\, d\mu$ for every $f \in C_c(G)$. Therefore

$$\int_G f\, d\mu_s = \int_G L_{s^{-1}} f\, d\mu = I(L_{s^{-1}} f) = I(f) = \int_G f\, d\mu$$

where the middle equality is left invariance of I, proposition 6.1.5(c) carried to the limit. Both μ and μ_s are Radon measures representing the same functional I, so by the uniqueness clause of theorem 4.2.1 they coincide: $\mu(s^{-1}E) = \mu(E)$ for every Borel E. Replacing s by s^{-1} gives $\mu(sE) = \mu(E)$, which is left invariance. Thus μ is a nonzero left-invariant Radon measure, completing the proof.

The theorem is the group-theoretic counterpart of the fact that Lebesgue measure exists on $\mathbb{R}$, and the proof exhibits the same mechanism in a form stripped of the line's special features. Nothing was used but the group operation, the topology, and compactness; no coordinates, no metric, and no explicit formula for the measure appear. The measure is defined only implicitly, as the one whose integral averages functions the way the invariant functional I does, which is why the Riesz-Markov theorem is indispensable: it is the passage from the analytic object, an invariant integral, to the geometric object, an invariant measure. The invariant measure so produced is named for its discoverer.

Definition 6.1.9: Haar Measure

Let G be a locally compact Hausdorff group. A *left Haar measure* on G is a nonzero left-invariant Radon measure on G, that is, a Radon measure $\mu \neq 0$ with $\mu(sE) = \mu(E)$ for every $s \in G$ and every Borel set E. A *right Haar measure* is defined identically with right translation, $\mu(Es) = \mu(E)$.

Theorem 6.1.8 is exactly the assertion that a left Haar measure exists on every locally compact Hausdorff group. Existence is only half of what makes Haar measure useful; the measure would be of little value if it depended on the arbitrary choices made in its construction, the reference f_0 and the cluster point I. That it does not, that any two left Haar measures differ only by a positive scalar, is the uniqueness theorem of section 6.2, and the relation between left and right Haar measures is the modular function studied there.

Example 6.1.10: Haar Measure on the Classical Groups
The construction specializes to familiar invariant integrals. On $G = \mathbb{R}^n$ under addition, left translation is $L_s f(x) = f(x - s)$, and Lebesgue measure vol is left-invariant and Radon, so it is a Haar measure; the construction reproduces it up to the scalar fixed by the choice of f_0. On the multiplicative group $G = \mathbb{R}_{>0}$, left translation is $L_s f(x) = f(x/s)$, and the measure $d\mu(x) = dx/x$ is invariant, since substituting $x = su$ gives $d(su)/(su) = du/u$; this is the Haar measure of $\mathbb{R}_{>0}$, and it is the reason the logarithm converts multiplication to translation. On the general linear group $G = \mathrm{GL}_n(\mathbb{R})$, an open subset of $\mathbb{R}^{n^2}$ and hence locally compact, the measure

$$d\mu(A) = \frac{dA}{|\det A|^n}$$

where dA is Lebesgue measure on the n^2 matrix entries, is left-invariant: left multiplication by S is a linear map on the entries with Jacobian $|\det S|^n$, which the denominator exactly cancels. In each case the abstract theorem guarantees existence, while the explicit formula, available here because the group sits concretely inside a Euclidean space, identifies the measure the theorem produces.

6.1.11 Remark: Cartan's Choice-Free Route The construction above extracts I as a cluster point of the net $\{I_g\}$, an appeal to compactness of the product X and hence to Tychonoff's theorem, which rests on the axiom of choice. A second route, due to Cartan, avoids this. Instead of taking a cluster point, one shows directly that the net $\{I_g(f)\}$ is a *Cauchy* net as $\operatorname{supp} g \to \{e\}$: left-uniform continuity, lemma 6.1.1, forces $I_g(f)$ and $I_{g'}(f)$ to agree to within any prescribed tolerance once both g, g' have small enough support, so completeness of $\mathbb{R}$ supplies a genuine limit $I(f) = \lim_g I_g(f)$ with no subnet and no compactness. The Cauchy estimate simultaneously yields additivity and, with a little more work, uniqueness of the limit, so the choice-free argument delivers existence and uniqueness together. The two routes prove the same theorem; the covering-number combinatorics and the role of lemma 6.1.1 are identical, and only the final extraction of the limit differs. See Deitmar and Echterhoff for the Cauchy-net development in full.

6.2 Uniqueness and the Modular Function

The construction of section 6.1 produces a left Haar measure but says nothing about how many there are. A measure is a choice of scale, and an abstract group carries no distinguished unit of volume, so the most one can ask is uniqueness up to a positive multiple. That is what holds: two left Haar measures on the same group differ only by a constant factor. The proof applies the Fubini-Tonelli theorem (theorem 2.5.9) to a product of the two measures, the first place where an invariant measure is integrated against another.

Once uniqueness is available, the failure of a left-invariant measure to be right-invariant can be measured. Right translation carries a left Haar measure to another left Haar measure, so by uniqueness the two differ by a scalar depending on the translating element. The resulting positive function on the group, the modular function, records the discrepancy between left and right; it is identically 1 exactly when the two notions of invariance agree.

Throughout, G is a locally compact Hausdorff group ([7]) with left translation $L_s f(x) = f(s^{-1}x)$, and a left Haar measure is a nonzero left-invariant Radon measure (definition 6.1.9). Two such measures are compared on $C_c^+(G)$, the nonzero nonnegative members of $C_c(G)$, since a Radon measure is determined by its integrals against $C_c(G)$ (the uniqueness clause of Riesz-Markov, theorem 4.2.1) and every $f \in C_c(G)$ is a difference of two members of $C_c^+(G) \cup \{0\}$.

The mechanism of the proof is to show that the ratio $\int f\,d\nu / \int f\,d\mu$ does not depend on f. Once this common value is a single constant c, the two measures satisfy $\int f\,d\nu = c\int f\,d\mu$ on $C_c^+(G)$, hence on $C_c(G)$, and Riesz-Markov upgrades the equality of functionals to $\nu = c\mu$. The independence of f comes from a double integral $\iint f(yx)g(x)\,d\mu(x)\,d\nu(y)$ evaluated two ways: once by left-invariance, which produces $\int f\,d\mu$ times a ν-integral of g, and once by letting g concentrate at the identity, which extracts $\int f\,d\nu$. Reconciling the two limits forces the ratio. Left-uniform continuity (lemma 6.1.1) is what makes the limit converge, and the Fubini-Tonelli theorem is what licenses the two orders.

Theorem 6.2.1: Uniqueness of Haar Measure

Let G be a locally compact Hausdorff group and let μ and ν be left Haar measures on G. Then there is a constant $c > 0$ with $\nu = c\mu$.

***Proof*:**
Integrals over G are written without their domain. Every integrand below is a product of members of $C_c^+(G)$ with translated arguments, hence continuous with compact support; since a Radon measure is finite on compact sets (definition 4.1.9), each integral is a finite real number, and each double integral has a nonnegative continuous integrand of compact support in $G \times G$, so the Tonelli half of theorem 2.5.9 permits evaluating it in either order. Both μ and ν are left-invariant, so $\int \varphi(sx)\,d\mu(x) = \int \varphi\,d\mu$ and likewise for ν, for all $s \in G$ and $\varphi \in C_c(G)$.

Fix $f \in C_c^+(G)$; the claim to be proved is that $\int f\,d\nu / \int f\,d\mu$ is a positive constant independent of f.

Step 1: A double integral evaluated by left-invariance. Let $g \in C_c^+(G)$ be symmetric under inversion, $g(x) = g(x^{-1})$; such g exist, since for any nonzero $\varphi \in C_c^+(G)$ the sum $\varphi(x) + \varphi(x^{-1})$ is symmetric and again in $C_c^+(G)$, inversion being a homeomorphism of G. Consider

$$J = \iint f(yx)\,g(x)\,d\mu(x)\,d\nu(y)$$

Integrating over x first, substitute $x \mapsto y^{-1}x$ in the inner μ-integral, valid by left-invariance of μ. Then $f(yx)g(x)$ becomes $f(x)g(y^{-1}x)$, so

$$\int f(yx)\,g(x)\,d\mu(x) = \int f(x)\,g(y^{-1}x)\,d\mu(x)$$

and hence, by Tonelli,

$$J = \iint f(x)\,g(y^{-1}x)\,d\mu(x)\,d\nu(y) = \int f(x)\left[\int g(y^{-1}x)\,d\nu(y)\right]d\mu(x)$$

The inner integral $S(x) = \int g(y^{-1}x)\,d\nu(y)$ is constant on G: for $u \in G$, substituting $y \mapsto uy$ by left-invariance of ν,

$$S(ux) = \int g(y^{-1}ux)\,d\nu(y) = \int g\big((uy)^{-1}ux\big)\,d\nu(y) = \int g(y^{-1}x)\,d\nu(y) = S(x)$$

As ux ranges over G with u, the value $S(x) = S(e) = \int g(y^{-1})\,d\nu(y) = \int g\,d\nu$ is the same everywhere, the last equality by symmetry of g. Therefore

$$J = \left(\int f\,d\mu\right)\left(\int g\,d\nu\right)$$

Step 2: The same integral evaluated by concentration. Now integrate J in the other order, over y first:

$$J = \int g(x)\left[\int f(yx)\,d\nu(y)\right]d\mu(x) = \int g(x)\,F(x)\,d\mu(x), \qquad F(x) = \int f(yx)\,d\nu(y)$$

The function F is continuous, and $F(e) = \int f\,d\nu$. To bring J close to $F(e)$, specialize g to concentrate near the identity. Fix a compact symmetric neighborhood V_0 of e, available since a locally compact group has a neighborhood base of e consisting of compact symmetric sets [7]. For a symmetric neighborhood $V \subseteq V_0$ of e, choose a symmetric $g_V \in C_c^+(G)$ with $\operatorname{supp} g_V \subseteq V$ and $\int g_V\,d\mu = 1$; such g_V exists by taking a bump function supported in V (lemma 4.1.5), symmetrizing it, and normalizing by its positive μ-integral. With $g = g_V$, Step 1 reads

$$\left(\int f\,d\mu\right)\left(\int g_V\,d\nu\right) = \int g_V(x)\,F(x)\,d\mu(x)$$

Since $g_V \geq 0$, $\int g_V\,d\mu = 1$, and $\operatorname{supp} g_V \subseteq V$,

$$\left|\int g_V(x)\,F(x)\,d\mu(x) - F(e)\right| = \left|\int g_V(x)\big(F(x) - F(e)\big)\,d\mu(x)\right| \leq \sup_{x\in V}|F(x) - F(e)|$$

Step 3: Continuity of F at the identity. By the same argument as lemma 6.1.1 applied through inversion, f is right-uniformly continuous: given $\varepsilon > 0$ there is a neighborhood V of e, which may be taken inside V_0, with $|f(yx) - f(y)| < \varepsilon$ for all $y \in G$ whenever $x \in V$. For such x,

$$|F(x) - F(e)| = \left|\int \big(f(yx) - f(y)\big)\,d\nu(y)\right| \leq \int |f(yx) - f(y)|\,d\nu(y) \leq \varepsilon\,\nu(K)$$

where $K = \operatorname{supp} f \cdot V_0^{-1}$ is a compact set containing the support of every $y \mapsto f(yx) - f(y)$ with $x \in V_0$, and $\nu(K) < \infty$. Hence $\sup_{x\in V}|F(x) - F(e)| \leq \varepsilon\,\nu(K)$, which tends to 0 as V shrinks. Combining with Step 2,

$$\left(\int f\,d\mu\right)\left(\int g_V\,d\nu\right) \longrightarrow F(e) = \int f\,d\nu \qquad \text{as } V \to \{e\}$$

Step 4: The ratio is constant. Dividing the limit of Step 3 by $\int f\,d\mu > 0$ shows that the net $\left(\int g_V\,d\nu\right)$, indexed by the shrinking symmetric neighborhoods V of e, converges to

$$\frac{\int f\,d\nu}{\int f\,d\mu}$$

This net involves only the functions g_V and the measure ν, not f. A second choice $f' \in C_c^+(G)$ runs against the same net, so it converges to $\int f'\,d\nu / \int f'\,d\mu$ as well; since a net has at most one limit, the two ratios agree. Writing c for the common value, $\int f\,d\nu = c\int f\,d\mu$ for every $f \in C_c^+(G)$, and $c > 0$ because $\int f\,d\nu$ and $\int f\,d\mu$ are both positive.

Step 5: Passage to measures. From $\int f\,d\nu = c\int f\,d\mu$ for all $f \in C_c^+(G)$, and by linearity for all $f \in C_c(G)$, the two positive linear functionals $f \mapsto \int f\,d\nu$ and $f \mapsto c\int f\,d\mu$ agree on $C_c(G)$. By the uniqueness clause of Riesz-Markov (theorem 4.2.1), the Radon measures representing them coincide, giving $\nu = c\mu$, as we sought to show.

The uniqueness theorem is what makes "the" Haar measure a meaningful phrase: once a single normalization is fixed, every left-invariant Radon integral on G is determined. The remaining freedom is the choice of scale, and for two classes of groups there is a canonical way to spend it. On a compact group one normalizes so that $\mu(G) = 1$, making Haar measure a probability measure; on a discrete group one normalizes so that each point has mass 1, making Haar measure the counting measure. Between them, on $\mathbb{R}^n$ the normalization $\mu([0,1]^n) = 1$ recovers Lebesgue measure, which is why Lebesgue measure is invisible to translation.

The proof rests on evaluating one double integral two ways. Left-invariance alone gives the clean value $\int f\,d\mu \cdot \int g\,d\nu$, but only when g is symmetric, which is what lets the constant $S(x) = \int g\,d\nu$ emerge from the inner integral. The second evaluation replaces g by an approximate identity concentrated at e, and it is left-uniform continuity (lemma 6.1.1), applied to f through inversion, that forces the smoothed integral $\int g_V(x)F(x)\,d\mu(x)$ to converge to the point value $F(e) = \int f\,d\nu$. The role of the modular function, developed next, is exactly to measure what right-invariance would have supplied for free had it been available; its absence is why the argument routes through a limit rather than a single substitution.

Example 6.2.2: Uniqueness on the Circle

Take $G = \mathbb{T} = \mathbb{R}/\mathbb{Z}$, the circle group, which is compact and abelian. Arc length $d\theta$, normalized to total mass 1, is a left Haar measure, since translation $\theta \mapsto \theta + t$ preserves arc length. Let ν be any other left Haar measure on $\mathbb{T}$. By theorem 6.2.1, $\nu = c\,d\theta$ for some $c > 0$. If ν is also normalized to $\nu(\mathbb{T}) = 1$, then

$$1 = \nu(\mathbb{T}) = c\int_{\mathbb{T}} d\theta = c$$

forces $c = 1$, so $\nu = d\theta$. The normalized Haar measure on $\mathbb{T}$ is thus the unique invariant probability measure, and it assigns to an arc of length ℓ the mass ℓ. This is the fact underlying Fourier series: the characters $\theta \mapsto e^{2\pi i n\theta}$ are orthonormal in $L^2(\mathbb{T}, d\theta)$ precisely because $d\theta$ is the invariant measure.

With uniqueness in hand, right translation can be analyzed. Left translation preserves a left Haar measure by definition, but right translation need not: replacing $f(x)$ by $f(xs)$ transports the measure by a right shift, and there is no reason the outcome should again be μ. What can be said is that the transported measure is still a left Haar measure, so uniqueness pins it down to a scalar multiple of μ. That scalar is the object to be named.

Definition 6.2.3: Right Translation of a Left Haar Measure

Let μ be a left Haar measure on G and $s \in G$. The *right translate* $R_s\mu$ is the Borel measure defined on Borel sets E by

$$(R_s\mu)(E) = \mu(Es)$$

equivalently, by the integration formula $\int f\, d(R_s\mu) = \int f(xs^{-1})\, d\mu(x)$ for $f \in C_c(G)$.

The two descriptions agree: $(R_s\mu)(E) = \mu(Es)$ makes $R_s\mu$ the pushforward of μ under $x \mapsto xs^{-1}$, whose integral of f is $\int f(xs^{-1})\, d\mu(x)$. The right translate $R_s\mu$ is again a Radon measure, and it is left-invariant: for $t \in G$ and a Borel set E, left-invariance of μ gives $(R_s\mu)(tE) = \mu(tEs) = \mu(Es) = (R_s\mu)(E)$, since tEs is the left translate of Es by t. It is nonzero because μ is. Thus $R_s\mu$ is a left Haar measure, and theorem 6.2.1 applies.

Definition 6.2.4: Modular Function

Let μ be a left Haar measure on G. For each $s \in G$, the right translate $R_s\mu$ is a left Haar measure, so by theorem 6.2.1 there is a unique positive constant $\Delta(s)$ with

$$R_s\mu = \Delta(s)\, \mu$$

The function $\Delta\colon G \to \mathbb{R}_{>0}$ so defined is the *modular function* of G.

The constant $\Delta(s)$ does not depend on which left Haar measure defines it: if $\mu' = a\mu$ is another, then $R_s\mu' = aR_s\mu = a\Delta(s)\mu = \Delta(s)\mu'$, so the same $\Delta(s)$ results. The modular function is thus an invariant of G alone, not of the chosen normalization. Its value $\Delta(s)$ measures how a right shift by s scales left-invariant volume, a reading made precise in the proposition below.

The next proposition records the two structural facts about Δ: it is a homomorphism to the multiplicative group of positive reals, and it is continuous. Both follow from uniqueness and the group law, and together they make Δ a continuous character of G valued in $\mathbb{R}_{>0}$.

Proposition 6.2.5: Properties of the Modular Function

Let G be a locally compact Hausdorff group with left Haar measure μ and modular function Δ. Then

1. $\Delta\colon G \to (\mathbb{R}_{>0}, \times)$ is a group homomorphism: $\Delta(st) = \Delta(s)\Delta(t)$ for all $s, t \in G$, and $\Delta(e) = 1$.
2. For every $f \in C_c(G)$ and $s \in G$,
$$\int f(xs)\, d\mu(x) = \Delta(s)^{-1} \int f(x)\, d\mu(x)$$
3. Δ is continuous.

***Proof*:**

1. Right translation is an action of G on measures: for a Borel set E,
$$(R_sR_t\mu)(E) = (R_t\mu)(Es) = \mu(Est) = \mu\big(E(st)\big) = (R_{st}\mu)(E)$$
so $R_sR_t\mu = R_{st}\mu$. Applying definition 6.2.4 twice, $R_sR_t\mu = R_s(\Delta(t)\mu) = \Delta(t)\Delta(s)\mu$, while $R_{st}\mu = \Delta(st)\mu$. Equating the two scalars gives $\Delta(st) = \Delta(s)\Delta(t)$, the homomorphism property. Taking $s = t = e$ gives $\Delta(e) = \Delta(e)^2$, hence $\Delta(e) = 1$.

2. By definition 6.2.3, $\int f\, d(R_s\mu) = \int f(xs^{-1})\, d\mu(x)$, and by definition 6.2.4, $R_s\mu = \Delta(s)\mu$, so $\int f(xs^{-1})\, d\mu(x) = \Delta(s) \int f\, d\mu$. Replacing s by s^{-1} and using $\Delta(s^{-1}) = \Delta(s)^{-1}$ from part (1),
$$\int f(xs)\, d\mu(x) = \Delta(s^{-1}) \int f\, d\mu = \Delta(s)^{-1} \int f\, d\mu$$

3. Normalizing, fix $f \in C_c^+(G)$ with $\int f\, d\mu = 1$; such f exists because μ is nonzero. By part (2), $\Delta(s)^{-1} = \int f(xs)\, d\mu(x)$, so it suffices to show $s \mapsto \int f(xs)\, d\mu(x)$ is continuous, since $\Delta(s)$ is then continuous as the reciprocal of a continuous positive function. Let $s_0 \in G$ and $\varepsilon > 0$. Right-uniform continuity of f, the analogue of lemma 6.1.1 proved by the same argument applied through inversion, gives a neighborhood V of e with $\sup_x |f(xu) - f(x)| < \varepsilon$ for $u \in V$; relabeling the supremum variable $x \mapsto xs_0^{-1}$ gives $\sup_x |f(xs) - f(xs_0)| < \varepsilon$ whenever $s_0^{-1}s \in V$. Moreover, for s ranging over the compact neighborhood $s_0\overline{V_0} = \{s : s_0^{-1}s \in \overline{V_0}\}$ of s_0, with $\overline{V_0} \subseteq V$ compact, all the functions $x \mapsto f(xs)$ are supported in the common compact set $K = \mathrm{supp} f \cdot \overline{V_0}^{-1} s_0^{-1}$. Hence, for s in this neighborhood,
$$\left| \int f(xs)\, d\mu(x) - \int f(xs_0)\, d\mu(x) \right| \le \int |f(xs) - f(xs_0)|\, d\mu(x) \le \varepsilon\, \mu(K)$$
with $\mu(K) < \infty$. As $\varepsilon > 0$ was arbitrary, $s \mapsto \int f(xs)\, d\mu(x)$ is continuous at s_0, and s_0 was arbitrary, so Δ is continuous, completing the proof.

The formula $\int f(xs)\, d\mu(x) = \Delta(s)^{-1} \int f\, d\mu$ is the working form of the modular function: right-translating the argument of f by s scales its integral by $\Delta(s)^{-1}$. In set terms this reads $\mu(Es) = \Delta(s)\mu(E)$, directly from definition 6.2.3 and definition 6.2.4, so right translation of a set by s multiplies its left-invariant volume by $\Delta(s)$. The homomorphism property says these scaling factors compose correctly under the group law, and continuity says they vary continuously with s. A character valued in $\mathbb{R}_{>0}$ is exactly the data needed to correct integrals for the asymmetry between left and right.

The placement of the inverse is fixed once and for all by definition 6.2.4: $R_s\mu = \Delta(s)\mu$ with $(R_s\mu)(E) = \mu(Es)$, so $\mu(Es) = \Delta(s)\mu(E)$ on sets and $\int f(xs)\, d\mu = \Delta(s)^{-1} \int f\, d\mu$ on functions. Some texts adopt the reciprocal, defining the modular function so that $\int f(xs)\, d\mu = \Delta(s) \int f\, d\mu$; the two conventions differ by $s \mapsto s^{-1}$ and both are in use. This book is consistent with definition 6.2.4 throughout.

Example 6.2.6: The Modular Function of an Abelian Group
Let G be abelian with left Haar measure μ. For any $s \in G$ and Borel set E, commutativity gives

$Es = sE$, so

$$(R_s\mu)(E) = \mu(Es) = \mu(sE) = \mu(E)$$

the last step by left-invariance. Thus $R_s\mu = \mu$, and comparison with $R_s\mu = \Delta(s)\mu$ forces $\Delta(s) = 1$ for every s: the modular function of an abelian group is identically 1. Concretely, on $G = \mathbb{R}$ with Lebesgue measure, $\int f(x+s)\,dx = \int f(x)\,dx$, so $\Delta(s)^{-1} = 1$ for all s, recovering translation invariance of the Lebesgue integral in both directions.

The vanishing of the modular discrepancy in the abelian case instances a phenomenon worth naming. Groups on which $\Delta \equiv 1$ are those whose left Haar measure is automatically right-invariant, and they are the setting in which harmonic analysis is cleanest, since a single bi-invariant measure serves.

Definition 6.2.7: Unimodular Group

A locally compact Hausdorff group G is *unimodular* if its modular function is trivial, $\Delta \equiv 1$. Equivalently, G is unimodular if and only if every left Haar measure on G is also right-invariant.

The equivalence in the definition is immediate from proposition 6.2.5(2): right-invariance of μ means $\int f(xs)\,d\mu(x) = \int f\,d\mu$ for all f and s, and this holds if and only if $\Delta(s)^{-1} = 1$ for all s, that is $\Delta \equiv 1$. Unimodularity is therefore the exact condition under which the left and right theories of invariant integration coincide, and a large class of groups satisfies it.

Proposition 6.2.8: Classes of Unimodular Groups

Each of the following locally compact Hausdorff groups is unimodular.

1. Abelian groups.
2. Compact groups.
3. Discrete groups.

***Proof*:**

1. By example 6.2.6, an abelian group has $Es = sE$, so $R_s\mu = \mu$ and $\Delta \equiv 1$.
2. Let G be compact. By proposition 6.2.5(1) and (3), $\Delta\colon G \to \mathbb{R}_{>0}$ is a continuous homomorphism, so its image $\Delta(G)$ is a subgroup of $(\mathbb{R}_{>0}, \times)$ and, as the continuous image of a compact set, is compact. The only compact subgroup of $\mathbb{R}_{>0}$ is $\{1\}$: any subgroup containing a point $a \neq 1$ contains all powers a^n, and either $a > 1$, whence $a^n \to \infty$, or $a < 1$, whence $a^{-n} \to \infty$, so the subgroup is unbounded and not compact. Therefore $\Delta(G) = \{1\}$, that is $\Delta \equiv 1$, and G is unimodular.
3. Let G be discrete. Counting measure μ, which assigns to a finite set its cardinality and to an infinite set ∞, is a left Haar measure: on a discrete space every compact set is finite, so μ is a Radon measure, and $\mu(sE) = |sE| = |E| = \mu(E)$ since left translation is a bijection. Counting measure is equally right-invariant, $\mu(Es) = |Es| = |E| = \mu(E)$, so by definition 6.2.7 G is unimodular. Explicitly, $R_s\mu = \mu$ gives $\Delta(s) = 1$.

Each class thus has trivial modular function, as required.

These three classes cover most groups met in a first course: finite groups, which are compact and discrete at once; $\mathbb{R}^n$, $\mathbb{T}^n$, and $\mathbb{Z}^n$, which are abelian; and the compact matrix groups $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$. Semisimple and reductive Lie groups are also unimodular, though by criteria beyond the elementary ones above. Unimodularity is common enough that its failure can feel exotic, which makes an explicit non-unimodular group essential for calibration.

The standard example is the group of orientation-preserving affine transformations of the line, the $ax + b$ group. It is the smallest natural group on which left and right Haar measures differ, and computing both makes the modular function concrete rather than axiomatic.

Example 6.2.9: The $ax + b$ Group

Let G be the group of affine transformations $x \mapsto ax + b$ of $\mathbb{R}$ with $a > 0$ and $b \in \mathbb{R}$. Writing an element as the pair (a, b), the composition $(a, b)(a', b')$ sends x to $a(a'x+b')+b = (aa')x+(ab'+b)$, so the group law is

$$(a, b)(a', b') = (aa',\ ab' + b)$$

with identity $(1, 0)$ and inverse $(a, b)^{-1} = (a^{-1}, -a^{-1}b)$. As a topological space G is the open half-plane $\{(a, b) : a > 0\} \subseteq \mathbb{R}^2$, a locally compact Hausdorff group.

Left Haar measure. On the half-plane take $d\mu_L = a^{-2}\, da\, db$, meaning $\int f\, d\mu_L = \iint f(a, b)\, a^{-2}\, da\, db$ against ordinary Lebesgue measure $da\, db$. To check left-invariance, fix $s = (a_0, b_0)$; left translation by s replaces (a, b) by $s(a, b) = (a_0a,\ a_0b + b_0)$. Under the substitution $(a', b') = (a_0a, a_0b + b_0)$ the Jacobian is

$$\frac{\partial(a', b')}{\partial(a, b)} = \det\begin{pmatrix} a_0 & 0 \\ 0 & a_0 \end{pmatrix} = a_0^2$$

so $da'\, db' = a_0^2\, da\, db$, while $a' = a_0a$ gives $(a')^{-2} = a_0^{-2}a^{-2}$. Hence

$$(a')^{-2}\, da'\, db' = a_0^{-2}a^{-2} \cdot a_0^2\, da\, db = a^{-2}\, da\, db$$

so $d\mu_L$ is unchanged by left translation, a left Haar measure.

Right Haar measure. By the symmetric computation, take $d\mu_R = a^{-1}\, da\, db$. Right translation by $s = (a_0, b_0)$ replaces (a, b) by $(a, b)s = (aa_0,\ ab_0 + b)$. Under $(a', b') = (aa_0, ab_0 + b)$ the Jacobian is

$$\frac{\partial(a', b')}{\partial(a, b)} = \det\begin{pmatrix} a_0 & 0 \\ b_0 & 1 \end{pmatrix} = a_0$$

so $da'\, db' = a_0\, da\, db$, while $a' = aa_0$ gives $(a')^{-1} = a_0^{-1}a^{-1}$, and

$$(a')^{-1}\, da'\, db' = a_0^{-1}a^{-1} \cdot a_0\, da\, db = a^{-1}\, da\, db$$

so $d\mu_R = a^{-1}\, da\, db$ is right-invariant, a right Haar measure. The two differ by the factor a^{-1}, so no positive constant makes them equal, and G is not unimodular.

The modular function. To read off Δ, apply proposition 6.2.5(2) to μ_L. Right translation by $s = (a_0, b_0)$ acts on $d\mu_L = a^{-2}\, da\, db$ through the Jacobian just computed: $da'\, db' = a_0\, da\, db$ and $(a')^{-2} = a_0^{-2}a^{-2}$, so

$$\int f((a, b)s)\, d\mu_L = \iint f(a', b')\, a^{-2}\, da\, db = \iint f(a', b')\, a_0\, (a')^{-2}\, da'\, db' = a_0 \int f\, d\mu_L$$

Comparing with $\int f(xs)\, d\mu_L(x) = \Delta(s)^{-1} \int f\, d\mu_L$ gives $\Delta(s)^{-1} = a_0$, so

$$\Delta(a_0, b_0) = a_0^{-1}$$

> The modular function depends only on the multiplicative part a_0 and ignores the translation part b_0, consistent with Δ being a homomorphism to $\mathbb{R}_{>0}$: the map $(a_0, b_0) \mapsto a_0^{-1}$ is the projection $(a, b) \mapsto a$ followed by inversion on $\mathbb{R}_{>0}$, both homomorphisms. The relation $d\mu_R = a\, d\mu_L = \Delta^{-1} d\mu_L$ between the right and left measures is the general fact that a right Haar measure is Δ^{-1} times a left Haar measure, here made explicit.

The $ax + b$ group shows that the gap between left and right invariance is real and that the modular function measures it exactly. The reason a group can be non-unimodular is visible in the computation: left translation acts on (a, b) with Jacobian a_0^2 regardless of the base point, while right translation acts with Jacobian a_0, and the discrepancy a_0 between the two is $\Delta(s)^{-1}$. Whenever conjugation on the group distorts volume, the modular function is nontrivial, and the affine group is the first place this occurs.

Part II

Functional Analysis

The transition from Calculus and Linear Algebra to Functional Analysis marks a shift in perspective: in classical analysis individual functions are studied (their derivatives, and their integrals), in linear algebra linear transformation between finite dimensions spaces are studied. *Functional Analysis* treats function f itself as a point in a vector space and looks at transformations between spaces of these functions.

Note the term “functional” historically meant a function F that takes a function and returns a number. We shall show that most linear functionals are given by integrating against a function (section 8.2 and section 8.3), and many important exceptions will be shown to still be given by integrating against distributions.

Historically, the field grew out of a crisis in physics and differential equations. By the early 20th century, physicists dealing with Quantum Mechanics and classical wave equations realized that their problems could not be solved by looking at individual coordinates.

- In Quantum Mechanics, the state of a particle is not a position in $\mathbb{R}^3$, but a wave function ψ. The superposition principle (that if ψ_1 and ψ_2 are states, then $\alpha\psi_1 + \beta\psi_2$ is a state) suggests that these wave functions live in a linear vector space.
- In Partial Differential Equations, viewing the heat or wave equation as an operation on an infinite-dimensional space allows us to decouple the “geometry” of the operator from the specific details of the function.

However, unlike the finite-dimensional linear algebra being essentially governed by $\dim V$ and $\mathrm{GL}_n(k)$, infinite-dimensional spaces have many new properties: bounded sets need not be compact, linear maps need not be continuous, the notion of “convergence” splinters into multiple distinct topologies, analytical indexes becomes non-trivial[1] and many other properties unique to infinite dimensional vector spaces and their (continuous) linear transformations.

Functional Analysis is the rigorous attempt to tame this infinite wilderness. It builds a hierarchy of structures to recover the tools of calculus and algebra in the infinite setting.

The Three Foundations of the Theory

The narrative of this book is driven by the tension between three mathematical forces:

1. The Algebraic Structure (Linearity): At its core, we are doing linear algebra. We want to solve equations like $Lx = y$. If L is an invertible matrix, we invert it. If L is a differential operator (which is a linear operator), we want to do the same. The dream of Functional Analysis at its inception is to treat differential operators as “infinite matrices” and apply the Spectral Theorem to diagonalize them. When this works (as in Hilbert Spaces), we can solve complex PDEs with the same ease as a 2×2 matrix equation.
2. The Topological Structure (Continuity): Algebra alone is insufficient in infinite dimensions. In $\mathbb{R}^n$, all norms are equivalent, and continuity is automatic for linear maps. In infinite dimensions, much information is *lost* if a topology is not imposed, usually a topology induced by a norm or a metric.

[1] An index is $\mathrm{Ind}(\varphi) = (\dim\ker\varphi - \dim \mathrm{coker}\varphi)$ which is always 0 in finite dimensions but can be nonzero for Fredholm operators

3. The Geometric Structure (Distance and Angle): By choosing a norm or metric, we induce geometric structures. The main examples of such structures we'll encounter throughout are:
   (a) *Hilbert Spaces (L^2):* These are "correct" generalization of finite-dimensional euclidean space. They possess an inner product, allowing us to measure angles and define orthogonality.
   (b) *Banach Spaces ($L^p, C(K)$):* They have a notion of size (norm) but lack a notion of angle (i.e. a linear notion of relatedness). The geometry here can be strange: unit balls can be "pointy" or "flat," and the shortest path to a subspace might not be unique.
   (c) *Fréchet and Topological Vector Spaces:* necessary for distributions and generalized functions, where the concept of a "norm" is insufficient and replaced by families of semi-norms (necessary for differential operators to be continuous). They are equivalent to invariant metric spaces

There are more general spaces, but they are beyond an introductory text.

Ultimately, Functional Analysis is the study of *Dualities.* It is the duality between a space and the (linear) functions acting on it. We will see that by lifting concrete problems into these abstract spaces, we gain a new vantage point.

7

Banach and Fréchet Spaces

(these are all spaces key spaces to consider with normed

(alternative title: Fréchet, Banach, and Hilbert Spaces)

7.1 Banach Spaces

Let k be either $\mathbb{R}$ or $\mathbb{C}$, let V or X denote a vector-space over k, with 0 representing the zero vector. Let kx denote the subspace of X spanned by x. If $M, N \subseteq X$, then $M + N = \{m + n \mid m \in M, n \in N\}$.

Definition 7.1.1: Semi-Norm And Norm

Let V be a vector-space. A *semi-norm* on V is a function $x \mapsto \|x$ from V to $[0, \infty)$ satisfying the propeties:

1. $\|x + y\| \leq \|x\| + \|y\|$ (the triangle inequality)
2. $\|\lambda x\| = |\lambda|\|x\|$ (homogeneity of scalars)

If furthermore:

(3) $\|x\| = 0$ if and only if $x = 0$

then the function is called a *norm*

Definition 7.1.2: Normed Vector-Space

Let V be a vector space. Then if there exists a function $\|\cdot\|$ that is defined on V, then $(V, \|\cdot\|)$ is called a *normed vector-space.*

If V is a normed vector-space, we can induce a metric $\rho(x, y) = \|x - y\|$, namely since:

$$\|x - z \leq \|x - y\| + \|y - z\| \qquad \|x - y\| = \|(-1)(x - y)\| = \|y - x\|$$

Since any metric induces a topology, the induced topology by the metric induced by a norm $\|\cdot\|$ is called the *norm topology* on V. A nice consequence of the induced norm topology is that addition and scalar multiplication is continuous. From the axioms of a semi-norm, $\|0\| = 0$. There are norms for which $x \neq 0$ but $\|x\| = 0$, as we'll soon show. Note that the norm $\|\cdot\| : V \to \mathbb{F}$ is evidently continuous in the topology it induces[1]. If $x_n \to x$ in X, then for all $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$, $\|x_n - x\| < \epsilon$. By the reverse triangle inequality:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| < \epsilon$$

showing that $|\|x_n\| - \|x\||$ get's arbitrarily close, and so $\|x_n\| \to \|x\|$, i.e. $\|\cdot\|$ is continuous. This means that the pre-image of $\{0\}$ of $\|\cdot\|$ is closed. Furthermore, it is a vector-space. Denoting

$$K = \{x \in V \mid \|x\| = 0\}$$

Then if $x, y \in K$, $0 \leq \|x+y\| \leq \|x\|+\|y\| = 0$, and $\|cx\| = |c|\|x\| = |c|0 = 0$. Hence, we may consider V/K. What norm to put on V/K that is natural with respect to the projection map $\pi : V \to V/K$ will be addressed soon.

Two norms $\|\cdot\|_1$ and $\|\cdot\|_1$ are said to be *equivalent* if there exits some $C_1, C_2 > 0$ st

$$C_1\|\cdot\|_1 \leq \|\cdot\|_2 \leq C_2\|\cdot\|_1$$

Equivalent norms define equivalent metrics and the same topology, and so the same Cauchy sequences. Recall that in finite dimensions, all norms are equivalent, meaning there is only one notion of distance. As we'll see, this is not the case in infinite dimensions.

Since we have a metric, we have a notion of completeness. As is the case when working with analysis, we will require our spaces to be complete:

Definition 7.1.3: Banach Space

Let V be a normed vector space. Then V is a *Banach space* if it is complete with respect to it's induced metric.

Example 7.1.4: Banach Spaces

1. Every finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ is a Banach space. In this space, all linear map are continuous.

2. Let K be a compact space, and take the set of all continuous function, $C_0(K)$. Then $C(K)$ is a Banach space with the supremum norm $\|f\|_\infty = \sup_{x\in K} |f(x)|$ (also called uniform

[1] Note that in the metric space topology, continuous if and only if sequentially continuous

norm). We will later show (Banach–Mazur theorem) that all real separable Banach spaces are isometric (the isomorphisms in the category of Banach spaces) to a subspace of $C([0,1])$, giving us a theoretical universal way of thinking about Banach spaces. We shall later see that $C([0,1])$ has the Approximation property and has a Schauder basis, but there exists Banach spaces without these properties. This shows that for Banach spaces, it is possible to have nice total spaces with "bad" subspaces. In particular, if X is an isometric embedding into $C([0,1])$ there are not always nl nice projection $P : C([0,1]) \to X$.

If the space is not separable, if the smallest dense set D of a Banach space X has density of cardinality α, then X is isometric to $C_0([0,1]^\alpha, \mathbb{R})$[a]

3. If X is a topological space, then $B(X,k)$ and $BC(X,k)$ (all bounded functionals and bounded continuous functionals) form a Banach space with the uniform norm $\|f\|_u = \sup_{x\in X} |f(x)|$.
4. For p where $1 \le p \le \infty$, the space $L^p(X)$ for some measurable space X is a Banach space with the L^p-norm (recall that $L^p(\mu)$ is the set of equivalence classes functions that agree a.e.; if $L^p(\mu)$ represented only functions, then $\|\cdot\|_p$ would be a semi-norm)
5. Two other common examples of Banach spaces we will encounter later will be special subspaces spaces of differentiable functions: Banach Algebras, and Hilbert Spaces.

There are also examples of spaces that are not Banach Spaces:

1. Let X be a locally compact Hausdorff space and consider $C_c(X)$ with the ∞-norm. Then it is not a Banach space, not being complete. It is easy to see that $C_c(X)$, the set of continuous bounded function, is a Banach space and contains $C_c(X)$, and so it suffices to find the closure of $C_c(X)$ in $C_b(X)$. The closure turns out to be $C_0(X)$.
2. Let $U \subseteq \mathbb{R}^n$ be open and take $C_0(U,\mathbb{C})$. Then This space is *not* a Banach space
3. The space $H(U,\mathbb{C})$ of all holomorphic functions from U to $\mathbb{C}$ is *not* a Banach Space
4. The space $C_c^\infty(\mathbb{R}^n)$ is *not* a Banach space
5. (If you know some PDE's) The space of distributions $\mathcal{D}'$ is *not* a Banach Space

These are all examples of a weaker structure known as a *topological vector space* which we shall cover in section 7.4.

[a] An interesting result using Banach-Mazur's theorem by Luis Rodríguez-Piazza shows that the space of smooth functions with the uniform norm is isometric to the space of nowhere differentiable functions

There is a nice way to check whether a vector space V is a Banach space which requires the following definition (this essentially follows from the fact that continuity if and only if sequential continuity in a metric space and that $\|\cdot\|$ is continuous)

Definition 7.1.5: Absolutely Convergent

Let V be a normed vector space. and $\{x_k\}_k^\infty \subseteq V$ a sequence. Then the sequence is said to *converge absolutely* if

$$\sum_k^\infty \|x_k\| < \infty$$

Lemma 7.1.6: Cauchy and Absolute Convergence

Let V be a normed vector space. Then V is complete if and only if every absolutely convergent series in V converges.

The idea is that $\mathbb{R}$ has an order, making it easier to produce many arguments about sequences.

***Proof*:**
Let's first say that X is complete and that $\sum_{n=1}^\infty \|x\| < \infty$. To show it converges, we take advantage of completeness and show there is a Cauchy sequence. In particular, let $S_n = \sum_{k=1}^n x_k$ be the partial sum. Then since the series is absolutely convergent, there for all $n > m$,

$$\|S_n - S_m\| \leq \sum_{m+1}^n \|x_k\| \to 0 \quad \text{as } m, n \to \infty$$

so the sequence $\{S_n\}$ is Cauchy, and so converges, so $\{x_k\}$ converges.

Conversely, let's say an absolutely convergent series converges, and choose any Cauchy sequence $\{x_k\}$. Since it's a Cauchy sequence, choose a subsequence such that

$$\left\|x_{k_{i+1}} - x_{k_i}\right\| < 2^{-i}$$

Take $y_1 = x_{k_1}$ and $y_i = x_{i+1} - x_i$ for $i > 1$ so that $\sum_{i=1}^n y_i = x_{k_n}$. Then notice that $\{y_i\}$ is in fact absolutely convergent:

$$\sum_{i=1}^\infty \|y_i\| \leq \|y_1\| + \sum_{i=1}^\infty 2^{-i} = \|y_1\| + 1 < \infty$$

So $\sum_{i=1}^\infty y_i = \lim x_{k_n}$ exists. Since the subsequence of every Cauchy sequence approaches the same "point" (just like convergent sequences), we see that $\{x_n\}$ also converges to the same point, showing that Cauchy sequences are in fact convergent sequences, as we sought to show.

Next, we would want to study which linear functions between two vector-spaces are *continuous*. It will *not* be the case that all linear functions will be continuous. The following builds up to the classification of continuous linear functions:

Definition 7.1.7: Bounded Linear Map

Let $T : X \to Y$ be a linear map between two normed vector space. Then if there exists some C such that for all $x \in X$

$$\|T(x)\| \leq C\|x\|$$

Then T is said to be *bounded*, or is a *bounded linear operator*.

Note how this is different then boundedness of a set (ex. the range is bounded if $\|T(x)\| \leq C$). This definition is would be useless to us since no nonzero linear map would satisfy it. All linear map over finite dimensional vector spaces clearly bounded (can you find the C?), so this concept gets interesting in infinite dimesion

Proposition 7.1.8: Bounded $\Leftrightarrow$ Continuous

Let $T : X \to Y$ be a linear map between two normed vector spaces. Then the following are equivalent:

1. T is continuous
2. T is continuous at 0
3. T is bounded

***Proof*:**
If T is the zero map, then we're done, so assume $T \neq 0$

(1) implies (2) by definition. For (2) implies (3), Since T is continuous at 0, for for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x - 0\| < \|x\| < \delta$, then $\|T(x) - T(0)\| = \|T(x) - 0\| = \|T(x)\| < \epsilon$. Choosing $\epsilon = 1$, we get that there exists a δ_1 such that

$$\|x\| < \delta_1 \to \|T(x)\| < 1$$

we know want to show there exists a C such that $\|T(y)\| \leq C\|y\|$ for all y in the domain. We will do this by taking advantage of linearity and shrinking element so that we can take advantage of continuity.
Define $x(y) = \frac{\delta_1}{2\|y\|}y$ so that

$$\|x(y)\| = \left\|\frac{\delta_1}{2\|y\|}y\right\| = \frac{\delta_1}{2}\frac{1}{\|y\|}\|y\| = \frac{\delta_1}{2} < \delta_1$$

Therefore, by continuity and linearity:

$$\|T(x(y))\| = \left\|T\left(\frac{\delta_1}{2\|y\|}y\right)\right\| = \frac{\delta_1}{2\|y\|}\|T(y)\| < 1$$

since this is true for all y, we if we re-arrange we have:

$$\|T(y)\| < \frac{2}{\delta_1}\|y\|$$

showing that T is indeed bounded.

Finally, if $\|T(x)\| \leq C\|x\|$, then for all $\epsilon > 0$, choose $\delta = C^{-1}\epsilon$ so that

$$\|T(y) - T(x)\| = \|T(y-x)\| \leq \epsilon$$

completing the proof.

Note that for $T \in B(X,Y)$ to be an isomorphism, we require that T^{-1} also be bounded[2]. If $\|T(x)\| = \|x\|$ for all x, then T is called an *isometry.*

Since the topology is that of norms, then continuous linear functions between normed vector-spaces are the homomorphisms, meaning they preserve the norm (to the degree of boundedness mentioned in the above proposition) and the vector-space structure (to the degree the map is injective). We will denote the set of bounded linear functions (which we will call linear operators) as $B(X,Y)$, $\mathcal{L}(X,Y)$, or $\mathrm{Hom}_k(X,Y)$, where we understand that since X and Y are normed spaces $\mathrm{Hom}_k(X,Y)$ refers to continuous lienar maps.

Like with most fields of mathematics, we are more often more interested in the functions between spaces rather than the spaces itself, for it is function that ultimatly give us information about the structure we are working with. Thus, if we can use the technics we will be developing on $B(X,Y)$, we may gain many interesting results. For this to be the case, we require to introduce a norm on $B(X,Y)$. Naturally, we want that the norm on $B(X,Y)$ to be compatible with X and Y in some way, especially the codomain Y (as we usually investigate the structure of the image of a map). Sicne we are working with bounded operators, we have that $\|T(x)\| \leq C\|x\|$. Furthermore, since we are working with linear operators, by some scaling we see that we only need to consider when $\|x\| = 1$. This moivates the following norm on $B(X,Y)$:

Definition 7.1.9: Operator Norm

$B(X,Y)$ is a normed vector space with norm[a]

$$\begin{aligned} \|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \inf\{C \mid \|T(x)\| \leq C\|x\| \text{ for all } x\} \\ &= \sup_{\|x\|\neq 0} \frac{\|T(x)\|}{\|x\|} \end{aligned}$$

If $\|T(x)\| = \|x\|$, then T is called an *isometry*

[a] the third equality holds if $V \neq \{0\}$

The operator norm can be thought of as the smallest consant that will satisfy $\|T(x)\| \leq C\|x\|$ when T is bounded, that is $\|T(x)\| \leq \|T\|\|x\|$. In the spacial case where we have $B(X,k)$, we write V^* and call it the dual space with the *norm topology.* It is useful to establish when is $B(X,Y)$ a Banach space

[2] We will show using the open mapping theorem that it suffices to show T is continuous and bijective

Proposition 7.1.10: Completness Of $B(X, Y)$

Let X, Y be normed vector-space and $B(X, Y)$ the normed vector-space of all bounded linear functions from X to Y. If Y is complete, so is $B(X, Y)$.

***Proof*:**
Let (T_n) be a cauchy sequence in $B(X, Y)$. For each $x \in X$, we have a cauchy sequence $(T_n(x)) \subseteq Y$, and thus a limit vector which we will denote $T(x)$. This defines a function $T : X \to Y$ which is clearly linear. The inequality

$$\begin{aligned}\|T(x) - T_n(x)\| &= \lim_m \|T_m(x) - T_n(x)\| \\ &\leq (\limsup_m \|T_m - T_n\|)\|x\|\end{aligned}$$

shows that T is bounded and that $T_n \to T$.

Hence, V^* is always a banach space, since $\mathbb{R}$ and $\mathbb{C}$ are complete. Next, we see what happens with common vector-space constructions like products and quotients.

Subspaces, Product spaces, Quotient Spaces

There are 3 equivalent norms we can put on the product of two normed vector-spaces:

1. $\|(x, y)\| = \max(\|x\|, \|y\|)$
2. $\|(x, y)\| = \|x\| + \|y\|$
3. $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$

The key to these being equivalent is that we are only multiplying by a finite number of spaces, a different strategy would have to be implemented for norms on infinite dimensional vector spaces (recall the different ℓ^p-norms put on the set of all sequences). For subspace, it is clear that if $W \subseteq V$ is a subspace of a normed vector space V, then W has an induced norm with respect to V. What is more interesting is if V is a banach space and we want W to be a Banach space. We then require all sequences to converge. Since V is Hausdorff (even metrizable since there is a norm on V), we see that W is complete if W is a *closed subspace* with respect to the topology of V. For quotients V/W, we would like to define a new norm on V/W so that the natural projeciton map $\pi : V \to V/W$ is continuous. This leads us to the following definition:

Proposition 7.1.11: Banach Quotient Space

Let X be a normed vector space and $Y \subseteq X$ a subspace of X. Then $\pi : X \to X/Y$ is the quotient map. This induces the topology on X/Y by defining:

$$\|\pi(x)\| = \|x + Y\| = \inf_{y \in Y} \|x - y\|$$

which is a *seminorm* on X/Y. If Y is closed, then this is in fact a norm, and if furthermer X is Banach, then X/Y is a banach space.

The intuition in my mind for this definition of quotient norm is to "eliminate" any effect of the quotienting space Y on any element of $x + Y$. Take for example $\mathbb{R}^2/\mathbb{R}$. Then the the norm of $\|(x, y) + \mathbb{R}\|$ should really be $|x|$. Furthermore, π is naturally a continuous map since it is norm-decreasing, meaning $\|\pi(x)\| \leq \|x\|$ (simply choose $C = 1$).

***Proof*:**
For subadditivity, we take advantage of the definition of infimum: for every $x_1, x_2 \in X$, there exists an $\epsilon > 0$ and elements $y_1, y_2 \in Y$ such that

$$\begin{aligned} \|\pi(x_1)\| + \|\pi(x_2)\| + \epsilon &\geq \|x_1 - y_2\| + \|x_2 - y_2\| \\ &\geq \|x_1 + x_2 - (y_1 + y_2)\| \\ &\geq \|\pi(x_1 + x_2)\| \end{aligned}$$

Since ϵ was arbitrary, we have subaditivity. Homogeneity is immediate since π is linear, and hence we have a seminorm.

If $\|\pi(x)\| = \|x + Y\| = 0$, there is a sequence $(y_n) \subseteq Y$ such that $\|x - y_n\| \to 0$. If Y is closed, $\lim_n(y_n) = x \in Y$, meaning $\pi(x) = 0$. It follows that X/Y is a normed space if and only if Y is closed in X.

Now, say $(z_n) \subseteq X/Y$ is a cauchy sequence. Then we can find a subsequence (z_n') were

$$\|z_{n+1}' - z_n'\| < 2^{-n}$$

By induction we can choose $x_n \in X$ such that $\pi(x_n) = z_n'$ and $\|x_{n+1} - x_n\| < 2^{-n}$. If X is banach, (x_n) converges to an element $x \in X$, and since π is "norm decreasing" (the norm the same or smaller), then (z_n') converges to $\pi(x)$. Hence, (z_n) converges to $\pi(x)$, and so X/Y is complete, as we sought to show.

Proposition 7.1.12: Universal Property of Quotient Maps

If $T \in B(X, Y)$ where X and Y are normed spaces, and if $W \subseteq X$ is a closed subspace of X containing $\ker(T)$, then then there exists an operator $\tilde{T} \in B(X/W, Y)$ defined pointwise as $\tilde{T}(\pi(x)) := T(x)$ such that

$$\|\tilde{T}\| = \|T\|$$

in particular, the following diagram commutes in **Ban**:

$$\begin{tikzcd} X \arrow[r, "T"] \arrow[d, "\pi"'] & Y \\ X/W \arrow[ur, "\tilde{T}"'] & \end{tikzcd}$$

***Proof*:**
First, by elementary linear algebra we have that $\tilde{T}$ is an operator on X/W, which is a normed space by the earlier proposition. Since

$$\|\tilde{T}(\pi(x))\| = \|T(x)\| = \|T(x - z)\| \leq \|T\|\|x - z\|$$

for every $z \in W$, it follows form the definition of the quotient norm that

$$\|\tilde{T}(\pi(x)) \leq \|T\|\|\pi(x)\|$$

hence, $\|\tilde{T}\| \leq \|T\|$. The reverse inequality $\|\tilde{T}\| \geq \|T\|$ is clear since Q is norm decreasing.

Proposition 7.1.13: Extending Banach Spaces

Let X be a normed space and Y a closed subspace. Then if Y and X/Y are Banach spaces, then X is a Banach space.

***Proof*:**
Let $(x_n) \subseteq X$ be a cauchy sequence. Since $\pi : X \to X/Y$ is norm-decreasing, $(\pi(x_n)) = (y_n) \subseteq X/Y$ is a cauchy sequence. Since X/Y is complete, it converges to a point $\overline{x} \in X/Y$. Next, by the definition of the quotient norm, for every x_n, there exists an n st

$$\|(x - x_n) - y_n\| \leq \frac{1}{n} + \|\pi(x - x_n)\|$$

Then we get:

$$\begin{aligned} \|y_n - y_m\| &= \|(y_n + x - x_n) - (y_m + x - x_m) + x_n - x_m\| \\ &\geq \frac{1}{n}\|\pi(x - x_n)\| + \frac{1}{m}\|\pi(x - x_m)\| + \|x_n + x_m\| \end{aligned}$$

which is clearly cauchy, and hence since Y is banach it is complete and so this sequence converges to y. Finally, it is easy to see that

$$\|x_n - (x + y)\| \leq \|x_n - (x - y_n)\| + \|y - y_n\|$$

showing that $x - n \to x + y$, as we sought to show.

Finite space naturally have none of the problems introduced in infinite dimensions and preserve all of their nice properties even when subspace of infinite dimensinoal vectorspace:

Proposition 7.1.14: Finite Dimensional Normed Space

Every finite-dimensional subspace Y of a normed space X is a Banach space, and consequently is closed in X. Moreover, if $\dim(Y) = n$, every linear isomrophism of $\mathbb{F}^n$ onto Y is a homeomorphism

***Proof*:**
p. 46 Analysis now.

Working with infinite dimensional space dense subsets become of key interest. As a simple example, the smooth bump functions form a dense subset of L^p for all $1 \leq p < \infty$. The following simplifies for us how we may define functions from spaces given a known dense set:

Proposition 7.1.15: Defining Operator using Dense Sets

Let X and Y be banach spaces and X_0 is a dense subspace of X. Then every operator $T_0 \in B(X_0, Y)$ has a unique extension to an operator $T \in B(X, Y)$ and $\|T\| = \|T_0\|$

***Proof*:**
For each $x \in X$, there is a sequence $(x_0) \subseteq X_0$ that converges to x: $x_n \to x$. Since T_0 is a bounded operator ($\|T_0(x_n)\| \leq C\|x_n\|$), it is clear that $(T_0(x_0))$ is a convergent sequence in Y; let's say it converges to $T(x)$. It should be clear that if we choose another sequence converging to x and do this procedure, then we will get the same value of $T(x)$. Thus, the map $x \to T(x)$ is a operator that is linear, extends T_0, and by construciton

$$\|T\| = \|T_0\|$$

an excellent example would be polynomial over a compact set are dense over continuous functions over compact sets, which themselves are dense over all L^p spaces. Hence in the compact case, for real or complex functions, it suffices to define our function over a the polynomials. For the case of locally compact spaces, Note that smooth bump functions are dense in L^p, and since they have compact support a polynomial sequnce uniformally converges to them. Hence, we may define a lot of operators on different real/complex functions spaces over locally compact spaces using polynomals.

For future reference, it will be important to remember that all normed spaces can be completed and that the norm between a normed space X and its completion $\overline{X}$ is isometric:

Proposition 7.1.16: Unique Extension To Banach Space

For each normed space X, there is a Banach space $\overline{X}$, uniquely determiend up to isometric isomoprhism, such that $\overline{X}$ contains X as a dense subspace

***Proof*:**
p. 47 Analysis now

An important concept in any infinite dimensional space is to figure out which sets are compact. For example, you may recall in real analysis that we proved the Stone-Weistrass theorem, or even simpler that in $\mathbb{R}^n$ a set is compact if it is closed and bounded. The following shows that subsets that are "almost finite-dimensional" are compact.

Proposition 7.1.17: Compact Sets In Banach Spaces

Let $K \subseteq V$ be a subset of a Banach Space V. Then the following are equivalent:

1. K is compact
2. K is sequentially compact
3. K is closed and bounded, and for every $\epsilon > 0$, K lies in the ϵ-neighborhood

$$\{x \in V \mid \|x - y\| < \epsilon \text{ for some } y \in W\}$$

   of a finite dimensional subspace W of V.

Suppose further that there is a nested sequence $V_1 \subseteq V_2 \subseteq \cdots$ of finite dimensional subspaces of V such that $\bigcup_{n=1}^{\infty} V_n$ is dense. Then the following statement is equivalent to the first three:

4. K is closed and bounded, and for every $\epsilon > 0$, there exits an n such that K lies in the ϵ-neighborhood of V_n

***Proof*:**
exercise

Example 7.1.18: Compact Sets In Banach Spaces

Let $1 \leq p < \infty$ and take a set $K \subseteq \ell^p(\mathbb{N})$. In order for it to be compact in the norm topology, it needs to be closed, bounded, and also uniformly p^{th}-power integrable at spatial infinity, that is for every $\epsilon > 0$, there exists an $n > 0$ such that

$$\left(\sum_{m>n} |f(m)|^p \right)^{1/p} \leq \epsilon$$

for all $f \in K$. This shows that the moving bump set is not uniformly p^{th} power integrale and thus not a compact subset, even though it's closed and bounded.

This condition doesn't translate to the continuous equivalent $L^p(\mathbb{R})$. We also need some uniform integrability at a "finte scale", which would be described using harmonic analysis tools like Fourier transforms, and hence we simply mention it here without developing it further.

Given the Banach space $\text{End}_k(V)$, it is natural have a second binary operation of composition and ask if it is also continuous (hence bounded). More generally, if $T \in B(X, Y)$, and $S \in L(Y, Z)$, then

$$\|ST(x)\| \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|$$

and so $ST \in B(X, Z)$. In particular, that makes $B(X, X) = \text{End}_k(X)$ into an algebra, and when X is complete, $B(X, X)$ is a *Banach Algebra*. More generally:

Definition 7.1.19: Banach Algebra

Let X be a normed k-algebra where addition, multiplication, and scalar multiplication are continuous with respect to the induced topology. Then if X is complete, X is a *Banach Algebra*.

For the multiplication to be continuous, it suffices to check that $\|xy\| \leq \|x\|\|y\|$ (can you see why?). If X has a multiplicative unit, it is called *unital*. Not all Banach algebra's have a multiplicative unit, $C_0(X)$ where X is a locally compact (hausdroff) space and $C_0(X)$ is the set of functions from X to $\mathbb{R}$ (or $\mathbb{C}$) that vanish at infinity is *not* unital. On the otherhand, $C_b(X)$, the set of bounded functions with pointwise operations with the supremum norm *is* unital. We will work more with Banach Algebras later.

The Local Geometry of Banach Spaces

While infinite-dimensional Banach spaces can be geometrically wild (as we will see with L^1 and L^∞), their finite-dimensional subspaces exhibit a surprising regularity. We previously noted that all norms on a finite-dimensional space are equivalent. The following theorem, a basic result of the "Local Theory of Banach Spaces," makes a much stronger claim: Euclidean geometry is "locally universal".

Theorem 7.1.20: Dvoretzky's Theorem

Let $(X, \|\cdot\|)$ be an infinite-dimensional Banach space. For every integer $n \geq 1$ and every $\epsilon > 0$, there exists a subspace $E_n \subset X$ with $\dim(E_n) = n$ such that E_n is $(1+\epsilon)$-isomorphic to the Euclidean space ℓ_2^n.
In terms of the Banach-Mazur distance, this means:

$$d(E_n, \ell_2^n) = \inf \|T\|\|T^{-1}\| \leq 1 + \epsilon$$

where the infimum is taken over all isomorphism $T : E_n \to \ell_2^n$.

Geometrically, this is often described as the "slicing problem." If you visualize the unit ball of a Banach space X as a high-dimensional convex body K, Dvoretzky's theorem states that there is always a slice through the center of K (a subspace) that is nearly spherical.

This theorem highlights a tension between local and global structure:

1. *Globally*, X might be very far from a Hilbert space (like ℓ_∞ or the Enflo space).
2. *Locally*, X contains "copies" of Hilbert spaces of all finite dimensions.

Thus, while the "global" geometry of Banach spaces is a vast zoo of structures, the "atomic" local geometry is always Euclidean.

Exercise 7.1.1

1. Let $A, B \subseteq V$ are two closed subspaces. Show that $A + B$ need not be a closed subspace! However, if one of A or B is finite dimensional, then $A + B$ is closed.

2. Show that the open unit ball in X maps onto (by π) the open unit ball of X/Y, but a similar statement about closed unit balls holds only in special cases.
3. For the categorically inclined, it should be verified that the topology on X/Y is the quotient topology corresponding to the equivalent relation $\sim$ where $x_1 \sim x_2$ if and only if $x_1 - x_2 \in Y$.

7.2 Baire Category Theorem

In this section, we introduce some of the basic tools that allow us to work with functions between Banach spaces. Of keen interest to us is the *open mapping theorem*, the *closed graph theorem*, and the *uniform boundedness theorem*. The open mapping theorem makes it sufficient that a surjective continuous linear operator is open, which in particular means that if it is also bijective it is an isomorphism. The closed graph theorem allows us to show that a linear operator $T : X \to Y$ is continuous if the graph $\Gamma(T)$ is closed, i.e. if all sequences in the graph converge. This will lead to great simplifications in showing a function is continuous. Finally, the uniform boundedness theorem will give a powerful criterion for upgrading pointwise boundedness of a family of operators to uniform boundedness.

The underpinning results we need are about dense sets. Let's say we have a countable family of dense sets, which can be thought of as our "generalized basis". Then it is natural that their intersection may be empty, for example taking linearly independent transcendental elements $r_1, r_2, ... \in \mathbb{R}$, then $\bigcap_r \mathbb{Q} + r = \emptyset$. What we need is to ensure that the intersection is non-empty. For that, we require that the limit of an infinite intersection exists (ex. our space is LCH or a complete metric space), and that our sets have a nice property that ensures non-emptiness. The following deals with the case of the space being a complete metric space:

Proposition 7.2.1: Baire Category Theorem I

Let (A_n) be a sequence of open dense subset of a complete metric space (X, d). Then $\bigcap_n A_n$ is dense in X.

***Proof*:**
We want to show that $\bigcap_n A_n$ is dense, so for every ball $B_0 \subseteq X$, $B_0 \cap \bigcap_n A_n \neq \emptyset$.

Let B_0 be any closed ball in X with radius $r > 0$. Then since A_1 is dense in X and is open, the set $A_1 \cap \operatorname{int}(B_0)$ is a nonempty open set, and hence contains a closed ball B_1 with radius at most $r/2$. Since A_2 is dense and open, we can repeat this with $A_2 \cap \operatorname{int}(B_1)$ and find a closed ball B_2 with radius at most $r/2^2$. By induction, we get a sequence of closed balls (B_k) in X such that $B_k \subseteq A_k \cap \operatorname{int}(B_{k-1})$, each with radius at most $r/2^k$. The sequence of centers is Cauchy (since the radii tend to zero and the balls are nested), and since X is complete, there exists a unique point $\{x\}$ such that

$$\{x\} = \bigcap_k B_k \subseteq B_0 \cap \left(\bigcap_k A_k \right)$$

This shows that $\bigcap_k A_k$ intersects every nontrivial ball in X, which implies density, completing the proof.

Note that this is a fully topological proof, and hence is invariant under homeomorphism. In particular, completeness is *not* a topological property (it is a uniform property), however $(0,1) \cong_{\mathbf{Top}} \mathbb{R}$. Furthermore, since each U_n is open, its complement must be nowhere dense. For if some non-empty open set O satisfies $O \subseteq \overline{U_n^c} = U_n^c$ (since U_n^c is closed), then $U_n \subseteq O^c$, and so

$$X = \overline{U_n} \subseteq O^c \subsetneq X$$

a contradiction. This leads to the following very useful corollary:

Corollary 7.2.2: Countable union of Empty Interior Sets

The countable union of closed sets with empty interior has empty interior. If X is a complete metric space, $X \neq \bigcup_i \overline{X_i}$ for nowhere dense sets X_i.

As a consequence, if $X = \bigcup_n X_n$, then there exists some n_0 such that X_{n_0} has nonempty interior.

***Proof*:**
Let (F_n) be a countable collection of closed sets with empty interior. Then each F_n^c is open and dense (open since F_n is closed; dense since F_n has empty interior, meaning F_n^c is dense). By the Baire Category Theorem, $\bigcap_n F_n^c$ is dense in X, and in particular nonempty. Hence $\bigcup_n F_n \neq X$, and since $\bigcup_n F_n$ is contained in a countable union of closed sets with empty interior, it itself has empty interior.

If each X_i is nowhere dense, then each $\overline{X_i}$ is closed with empty interior, so $X \neq \bigcup_i \overline{X_i} \supseteq \bigcup_i X_i$. Contrapositively, if $X = \bigcup_n X_n$, then some X_{n_0} must have nonempty interior, for otherwise each $\overline{X_n}$ would be closed with empty interior, contradicting the above.

This consequence is very interesting, since it shows that through a countable process, we require a set with nonempty interior to form our entire space. This motivates the following definition:

Definition 7.2.3: Meager and Generic Set

Let X be a topological space. Then $A \subseteq X$ is said to be *meager* (or of first-category) if it is a countable union of nowhere dense sets. The complement of a meager set is a *residual* (or co-meager, or generic, or of second-category).

Example 7.2.4: Interesting Meager Sets

1. The Cantor set $\mathcal{C}$ is meager in $\mathbb{R}$. Indeed, $\mathcal{C}$ is closed (as the intersection of closed sets in its construction) and has empty interior (it contains no intervals, since at each stage of the construction every remaining interval is split and its middle third removed). Hence $\mathcal{C}$ is nowhere dense, and in particular meager (as a single-term union of a nowhere dense set).

2. Take $C([0,1])$ with the uniform convergence topology. Let E be the set of all continuous functions that are differentiable somewhere. Then E is in fact a *meager* set in $C([0,1])$. To see this, take:

$$E_n = \{f \mid \exists x_0 \in [0,1],\ |f(x) - f(x_0)| < n|x - x_0|\ \forall x \in [0,1]\}$$

Then all differentiable functions are contained in $\bigcup_n E_n$, and it should be shown that this is a closed set and that there exists functions g such that $\|f-g\|_u < \epsilon$ for any $f \in E_n$ but $g \notin E_n$. Hence, we satisfied the conditions of the Baire Category theorem. As a consequence, the set of continuous nowhere differentiable functions is residual! Thus more generally, smooth and analytic functions are meager.

3. $\mathbb{Q}$ is not a G_δ subset of $\mathbb{R}$. Suppose for contradiction that $\mathbb{Q} = \bigcap_n U_n$ for open sets U_n. Since $\mathbb{Q}$ is dense, each U_n is open and dense. The irrationals $\mathbb{R} \setminus \mathbb{Q}$ can also be written as a countable intersection of open dense sets: $\mathbb{R} \setminus \mathbb{Q} = \bigcap_q (\mathbb{R} \setminus \{q\})$ where q ranges over $\mathbb{Q}$. Then

$$\emptyset = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \left(\bigcap_n U_n \right) \cap \left(\bigcap_q (\mathbb{R} \setminus \{q\}) \right)$$

would be a countable intersection of open dense subsets of the complete metric space $\mathbb{R}$. By the Baire Category Theorem, this intersection must be dense, contradicting the fact that it is empty. Hence $\mathbb{Q}$ is *not* a G_δ. Note however that $\mathbb{R} \setminus \mathbb{Q}$ *is* a G_δ, i.e. the irrationals form a residual set.

The 2nd example above is worth proving in full: the set of continuous functions on $[0,1]$ that are differentiable at *even one point* is meager in $C([0,1])$. Equivalently, the "generic" continuous function is nowhere differentiable. Compare this with the explicit construction of a single such function in [3]; the Baire category argument produces no explicit example, but shows that *almost every* continuous function (in the sense of category) has this pathological behaviour.

Theorem 7.2.5: Genericity of Nowhere Differentiable Functions

Let $X = C([0,1])$ equipped with the uniform norm $\|\cdot\|_u$. Then the set of functions in X that are differentiable at some point of $[0,1]$ is meager. Consequently, the set of continuous nowhere differentiable functions is residual in X.

***Proof*:**
For each $n \in \mathbb{N}$, define:

$$E_n = \{ f \in X \mid \exists x_0 \in [0,1],\ |f(x) - f(x_0)| \leq n|x - x_0| \ \forall x \in [0,1] \}$$

We first check that every function differentiable at some point belongs to $\bigcup_n E_n$. If $f'(x_0)$ exists with $|f'(x_0)| = L$, then there exists $\delta > 0$ such that $|f(x) - f(x_0)| \leq (L+1)|x - x_0|$ whenever $|x - x_0| < \delta$. For $|x - x_0| \geq \delta$, the bound $|f(x) - f(x_0)| \leq 2\|f\|_u \leq (2\|f\|_u/\delta)|x - x_0|$ holds trivially. Taking $n > \max(L+1,\ 2\|f\|_u/\delta)$ gives $f \in E_n$. It now suffices to show each E_n is closed with empty interior.

E_n is closed. Let $(f_k) \subseteq E_n$ converge uniformly to $f \in X$. For each k, choose $x_k \in [0,1]$ witnessing $f_k \in E_n$, i.e. $|f_k(x) - f_k(x_k)| \leq n|x - x_k|$ for all x. By compactness of $[0,1]$, pass to a subsequence (still called f_k) with $x_k \to x_0$. Fix any $x \in [0,1]$. Then:

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_k(x)| + |f_k(x) - f_k(x_k)| + |f_k(x_k) - f(x_0)| \\ &\leq \|f - f_k\|_u + n|x - x_k| + |f_k(x_k) - f(x_0)| \end{aligned}$$

As $k \to \infty$, $\|f - f_k\|_u \to 0$ and $|f_k(x_k) - f(x_0)| \leq \|f_k - f\|_u + |f(x_k) - f(x_0)| \to 0$ by uniform convergence and continuity of f. Hence $|f(x) - f(x_0)| \leq n|x - x_0|$ for all x, showing $f \in E_n$.

E_n has empty interior. Let $f \in X$ and $\epsilon > 0$. We construct g with $\|f - g\|_u < \epsilon$ and $g \notin E_n$. Since f is uniformly continuous on $[0, 1]$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/8$ whenever $|x - y| < \delta$. Choose an integer $M > 4n$ large enough that $w := \epsilon/(2M) < \delta$. Define the *zigzag function* $\varphi : [0, 1] \to \mathbb{R}$ to be the continuous piecewise linear function with slope alternating between $+M$ and $-M$ on consecutive intervals of width w, starting from $\varphi(0) = 0$. Since φ oscillates between 0 and $Mw = \epsilon/2$, we have $\|\varphi\|_u \leq \epsilon/2$.

Set $g = f + \varphi$, so $\|f - g\|_u = \|\varphi\|_u \leq \epsilon/2 < \epsilon$. Suppose for contradiction that $g \in E_n$, witnessed by some $x_0 \in [0, 1]$. Then x_0 lies in some monotone piece of φ of width w. Pick x to be the endpoint of this piece farther from x_0, so that $|x - x_0| \geq w/2$ and φ has constant slope $\pm M$ on the segment from x_0 to x. Since $|x - x_0| \leq w < \delta$:

$$\begin{aligned}|g(x) - g(x_0)| &\geq |\varphi(x) - \varphi(x_0)| - |f(x) - f(x_0)| \\ &= M|x - x_0| - |f(x) - f(x_0)| \\ &> M \cdot \frac{w}{2} - \frac{\epsilon}{8} = \frac{\epsilon}{4} - \frac{\epsilon}{8} = \frac{\epsilon}{8}\end{aligned}$$

On the other hand, $n|x - x_0| \leq nw = n\epsilon/(2M) < \epsilon/8$ since $M > 4n$. Hence $|g(x) - g(x_0)| > n|x - x_0|$, contradicting $g \in E_n$.

Since each E_n is closed with empty interior, $\bigcup_n E_n$ is meager. The set of nowhere differentiable functions contains the complement of $\bigcup_n E_n$ and is therefore residual, completing the proof.

Hence the vast majority (in the sense of Baire category) of continuous functions have this "wild" behaviour: they oscillate so violently at every scale that no tangent line can be drawn at any point. The zigzag perturbation in the proof reflects this oscillation, forced into existence by the Baire Category Theorem without any explicit fractal construction.

The following lemma is the key technical ingredient for the open mapping theorem. It shows that if the image of the unit ball is *dense* in some ball, then it actually *contains* a slightly smaller ball:

Proposition 7.2.6: Open Mapping Lemma

Let $T : V \to W$ be a bounded linear operator between two Banach spaces, $T \in B(V, W)$. Then if the image of the unit ball $B_1(0)$ is dense in some ball $B_r(0)$ with $r > 0$, i.e. $B_r(0) \subseteq \overline{T(B_1(0))}$, then

$$B_{(1-\epsilon)r}(0) \subseteq T(B_1(0))$$

for every $\epsilon > 0$.

***Proof*:**
Let $U = T(B_1(0))$. Since $T(B_1(0))$ is dense in $B_r(0)$, for any $y \in B_r(0)$ and $\epsilon > 0$, there is a $y_1 \in U$ such that $\|y - y_1\| < \epsilon r$. Consider now ϵU. This set is dense in $B_{\epsilon r}(0)$. Hence, there exists a $y_2 \in \epsilon U$ such that $\|(y - y_1) - y_2\| < \epsilon^2 r$. Continuing by induction, we get a sequence (y_n) where $y_n \in \epsilon^{n-1} U$ and

$$\left\| y - \sum_{k=1}^{n} y_k \right\| < \epsilon^n r$$

Choose now $x_n \in B_1(0)$ such that $T(x_n) = y_n$ and $\|x_n\| \leq \epsilon^{n-1}$. Then $\sum x_n = x \in X$ converges, and by definition $T(x) = y$. Since $\|x\| \leq \sum \epsilon^{n-1} = (1-\epsilon)^{-1}$ [a] we have

$$(1-\epsilon)^{-1} U \supseteq B_r(0)$$

completing the proof.

[a] recall that $1 - b^n = (1-b)(1+b+b^2+\cdots+b^{n-1})$

Theorem 7.2.7: Open Mapping Theorem

Let X and Y be Banach spaces and $T \in B(X,Y)$ be a bounded linear operator with $T(X) = Y$. Then T is an open map (i.e. T maps open sets to open sets)

***Proof*:**
Since T is surjective:

$$Y = T(X) = \bigcup_n T(B_n(0)) \subseteq \bigcup_n \overline{T(B_n(0))} = \bigcup_n n\overline{T(B_1(0))}$$

Then by corollary 7.2.2, not all closed sets $n\overline{T(B_1(0))}$ have an empty interior. Since scaling by n is a homeomorphism, it follows that $\overline{T(B_1(0))}$ itself has nonempty interior. Hence, there exist $y_0 \in Y$ and $\delta > 0$ such that

$$B_\delta(y_0) \subseteq \overline{T(B_1(0))}$$

We now show $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$. For any z with $\|z\| < \delta$, both y_0 and $y_0 + z$ lie in $B_\delta(y_0) \subseteq \overline{T(B_1(0))}$. Hence:

$$z = (y_0 + z) - y_0 \in \overline{T(B_1(0))} - \overline{T(B_1(0))} \subseteq \overline{T(B_1(0)) - T(B_1(0))} \subseteq \overline{T(B_2(0))} = 2\overline{T(B_1(0))}$$

where we used linearity of T in the last two steps. It follows that $B_\delta(0) \subseteq 2\overline{T(B_1(0))}$, i.e. $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$, so $T(B_1(0))$ is dense in $B_{\delta/2}(0)$.

By the Open Mapping Lemma (proposition 7.2.6), $B_s(0) \subseteq T(B_1(0))$ for every $s < \delta/2$. Since every open set $U \subseteq X$ is a union of open balls, we conclude from linearity of T that $T(U)$ contains a neighborhood around each of its points, thus $T(U)$ is open, completing the proof.

Corollary 7.2.8: Bijective And Continuous

Let $T : X \to Y$ be a bijective bounded operator. Then it has a bounded inverse, that is T is in fact a Banach-space isomorphism

The following also lets us recover a result similar to the one showing us that there is only one norm on a finite dimensional vector space up to equivalence, namely no (non-equivalent) norm on a Banach space can dominate another:

Corollary 7.2.9: Equivalent Banach Spaces

Let X be a vector-space, and a Banach space under two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then if $\|\cdot\|_a \leq \alpha\|\cdot\|_b$ for some $\alpha > 0$, then there is a $\beta > 0$ such that $\|\cdot\|_b \leq \beta\|\cdot\|_a$

***Proof*:**
Take $\text{id} : (X, \|\cdot\|_b) \to (X, \|\cdot\|_a)$. This is certainly bijective, continuous by the boundedness assumption, and open by the open mapping theorem. Hence it is a homeomorphism, meaning it has a bounded inverse, giving the desired result.

The next powerful result we get is a quick way of proving a function is bounded:

Theorem 7.2.10: Closed Graph Theorem

Let $T : X \to Y$ be an operator between Banach spaces. Then if the graph

$$\Gamma(T) = \{(x, y) \in X \times Y \mid T(x) = y\}$$

is closed in $X \times Y$, then T is bounded

***Proof*:**
Using the ∞-norm $\|(x, y)\| = \max(\|x\|_X, \|y\|_Y)$ on $X \times Y$, assume that $\Gamma(T)$ is closed in $X \times Y$. Since $X \times Y$ is a Banach space (as both X and Y are), and $\Gamma(T)$ is a closed linear subspace, $\Gamma(T)$ is itself a Banach space. Now consider the projection map $\pi_1 : \Gamma(T) \to X$ given by $\pi_1(x, T(x)) = x$. This is bijective (since T is a function) and norm-decreasing: $\|\pi_1(x, T(x))\| = \|x\| \le \max(\|x\|, \|T(x)\|)$. Since π_1 is a bounded bijection between Banach spaces, the Banach isomorphism theorem (corollary 7.2.8) gives that π_1^{-1} is bounded. The other projection $\pi_2 : \Gamma(T) \to Y$ given by $\pi_2(x, T(x)) = T(x)$ is also norm-decreasing by the same reasoning. Hence:

$$T = \pi_2 \circ \pi_1^{-1}$$

is a composition of bounded operators, and therefore bounded, completing the proof.

The key take-away is that if we are verifying the continuity of a linear map $T : X \to Y$ (so $x_n \to x$ implies $T(x_n) \to T(x)$), we just need to verify the closedness of the graph. The closedness of the graph implies that if $x_n \to x$ and $T(x_n) \to y$, then $T(x) = y$. Notice how this simplifies proving a function is continuous, since we did not need to show that $\lim_n T(x_n) = T(x)$, simply that it converges to *something*, namely if $T(x_n) \to y$, then $T(x) = y$. Note how completeness is crucial in both these theorems (exercise 29-31 folland for counter-examples).

The converse of the closed graph theorem, that if $T : X \to Y$ is bounded between Banach spaces, then $\Gamma(T)$ is closed, is elementary: let's say T is bounded and that $T(x_n) \to y$. Since $x_n \to x$, $\lim_n T(x_n) = T(x)$. Then:

$$\begin{aligned}
\|T(x) - y\|_Y &\le \|T(x) - y + T(x_n) - T(x_n)\|_Y \\
&\le \|T(x - x_n)\|_Y + \|T(x_n) - y\|_Y \\
&\le C\|x - x_n\|_X + \|T(x_n) - y\| \xrightarrow{n\to\infty} 0
\end{aligned}$$

hence, $\|T(x) - y\| = 0$, implying $T(x) = y$ and showing $\Gamma(T)$ is closed.

Finally, the following result is a way to pass from pointwise boundedness to uniform boundedness, only requiring the completeness of our domain! It is one of the foundational results of functional analysis:

Theorem 7.2.11: Principle Of Uniform Boundedness

Let X, Y be normed vector spaces and $A \subseteq L(X, Y)$. Then if X is a Banach space and $\sup_{T \in A} \|T(x)\| < \infty$ for all $x \in X$, then $\sup_{T \in A} \|T\| < \infty$

This theorem is also called by some the Banach–Steinhaus Theorem.

***Proof*:**
We want to upgrade from a pointwise bound to a uniform bound, showing that $\sup_{T \in A} \|T\| < \infty$. Let

$$E_n = \left\{ x \in X \;\middle|\; \sup_{T \in A} \|T(x)\| \leq n \right\} = \bigcap_{T \in A} \{x \in X \mid \|T(x)\| \leq n\}$$

Each E_n is closed (as an intersection of closed sets, since each T is continuous) and $X = \bigcup_n E_n$ (since $\sup_{T \in A} \|T(x)\| < \infty$ for each x). Then by the Baire Category Theorem, there must exist some $n \in \mathbb{N}$ such that E_n has nonempty interior. Hence, for some $x_0 \in E_n$ and $r > 0$, $\overline{B_r}(x_0) \subseteq E_n$.

Now, consider $u \in X$ such that $\|u\| \leq 1$ and take any $T \in A$. Since $x_0 + ru \in \overline{B_r}(x_0) \subseteq E_n$, we have $\|T(x_0 + ru)\| \leq n$. Also $x_0 \in E_n$, so $\|T(x_0)\| \leq n$. Then:

$$\begin{aligned}
\|T(u)\|_Y = r^{-1}\|T(ru)\|_Y &= r^{-1}\|T(x_0 + ru) - T(x_0)\|_Y \\
&\leq r^{-1}(\|T(x_0 + ru)\|_Y + \|T(x_0)\|_Y) \\
&\leq r^{-1}(n + n)
\end{aligned}$$

Hence, taking the supremum over u in the unit ball of X and over $T \in A$, we get:

$$\sup_{T \in A} \|T\|_{B(X,Y)} \leq 2r^{-1}n < \infty$$

completing the proof.

If X is simply a normed vector-space, we may keep this theorem by simply taking the set supremum of the set of nonmeager points. The key realization is that $\sup_{T \in A} \|T(x)\|$ is a different supremum for each x, but is finite for all x! That is, we are not working with $\sup_{\substack{T \in A \\ x \in X}} \|T(x)\|$ since that would make the result trivial:

$$\sup_{T \in A} \|T\| = \sup_{\substack{T \in A \\ \|x\| = 1}} \|T(x)\| \leq \sup_{\substack{T \in A \\ x \in X}} \|T(x)\|$$

This is what makes the Uniform boundedness principle so strong, we got a *uniform* bound by having a *local* bound for all x!

Example 7.2.12: Divergence of Fourier Series
One of the most celebrated applications of the uniform boundedness principle is the existence of continuous functions whose Fourier series diverges at a given point. Consider $X = C(\mathbb{T})$, the Banach space of continuous 2π-periodic functions with the supremum norm. For each n, define the functional $\Lambda_n : C(\mathbb{T}) \to \mathbb{C}$ by

$$\Lambda_n(f) = S_n f(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(t) \, dt$$

where $S_n f(0)$ is the n-th partial sum of the Fourier series of f evaluated at 0, and $D_n(t) = \sum_{k=-n}^{n} e^{ikt}$ is the Dirichlet kernel. Each Λ_n is a bounded linear functional with $\|\Lambda_n\| = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_n(t)|\, dt = \|D_n\|_{L^1}$. It is a classical computation that $\|D_n\|_{L^1} \sim \frac{4}{\pi^2}\log n \to \infty$. Hence $\sup_n \|\Lambda_n\| = \infty$. By the *contrapositive* of the uniform boundedness principle, there must exist some $f \in C(\mathbb{T})$ for which $\sup_n |\Lambda_n(f)| = \sup_n |S_n f(0)| = \infty$, i.e. the Fourier series of f diverges at 0. By translation, the same holds at any prescribed point.

Note how the principle is used here: we showed $\sup_n \|\Lambda_n\| = \infty$, and concluded that the pointwise boundedness hypothesis must fail for some f, even though we never constructed such an f explicitly. This is a quintessential non-constructive existence proof via Baire category methods.

Corollary 7.2.13: Pointwise Convergence of Operators

Let X be a Banach space, Y a normed vector space, and $(T_n)_{n\in\mathbb{N}} \subseteq B(X,Y)$ a sequence of bounded operators such that $(T_n(x))_{n\in\mathbb{N}}$ converges in Y for every $x \in X$. Define $T(x) = \lim_n T_n(x)$. Then $T \in B(X,Y)$, i.e. the pointwise limit is itself a bounded linear operator.

***Proof*:**
First, T is linear: for $x, y \in X$ and α, β scalars, $T(\alpha x + \beta y) = \lim_n T_n(\alpha x + \beta y) = \alpha \lim_n T_n(x) + \beta \lim_n T_n(y) = \alpha T(x) + \beta T(y)$.

Since $(T_n(x))$ converges for each x, it is bounded: $\sup_n \|T_n(x)\| < \infty$ for all $x \in X$. Hence, by the principle of uniform boundedness, $\sup_n \|T_n\| = C < \infty$. That is, $\|T_n(x)\| \leq C\|x\|$ for all n and all $x \in X$, where C is independent of n. Taking $n \to \infty$ we get:

$$\|T(x)\| \leq C\|x\|$$

which shows $T \in B(X,Y)$, completing the proof.

More generally, if $(T_\lambda)_{\lambda\in\Lambda} \subseteq B(X,Y)$ is a net such that each $(T_\lambda(x))$ is bounded and converges for every $x \in X$, the same conclusion holds. For sequences, convergence automatically implies boundedness. It also suffices in the principle of uniform boundedness to know that each sequence $(T_n(x))$ is bounded and that the sequence converges for a dense set of elements in X.

We conclude this section with an application of the Baire Category Theorem that does not involve linear operators. It shows that the set of discontinuities of a pointwise limit of continuous functions is, in a precise sense, small:

Theorem 7.2.14: Continuity of Pointwise Limits

Let (f_n) be a sequence of continuous (complex-valued) functions on a complete metric space X such that $f(x) = \lim_{n\to\infty} f_n(x)$ exists for every $x \in X$. Then the set of points at which f is continuous is a dense G_δ (and hence residual) in X.

***Proof*:**
Recall that f is continuous at x if and only if the *oscillation* of f at x is zero, where

$$\omega_f(x) = \inf_{\delta>0} \sup_{y,z\in B_\delta(x)} |f(y) - f(z)|$$

Thus the set of continuity points of f is $\{x : \omega_f(x) = 0\} = \bigcap_{m=1}^{\infty}\{x : \omega_f(x) < 1/m\}$. Each set $O_m = \{x : \omega_f(x) < 1/m\}$ is open (this follows from the definition of oscillation), so the set of continuity points is a G_δ. It remains to show that each O_m is dense.

Fix $m \geq 1$ and let B be any nonempty open ball in X. We must find a point in $B \cap O_m$. For each $n, k \in \mathbb{N}$, define

$$F_{n,k} = \left\{ x \in \overline{B} \;\middle|\; |f_n(x) - f_k(x)| \leq \frac{1}{3m} \right\}$$

Each $F_{n,k}$ is closed (since $f_n - f_k$ is continuous). Since $f_n(x) \to f(x)$ for every x, for each $x \in \overline{B}$ there exists N such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N$. Hence:

$$\overline{B} = \bigcup_{N=1}^{\infty} \bigcap_{n,k \geq N} F_{n,k}$$

The sets $G_N = \bigcap_{n,k\geq N} F_{n,k}$ are closed in $\overline{B}$. Since $\overline{B}$ is a complete metric space (closed subset of X), the Baire Category Theorem implies that some G_{N_0} has nonempty interior in $\overline{B}$. Hence there exists an open ball $B' \subseteq B$ such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N_0$ and all $x \in B'$.

Taking $k \to \infty$, we get $|f_n(x) - f(x)| \leq 1/(3m)$ for all $n \geq N_0$ and $x \in B'$. In particular, f_{N_0} is continuous, so we may shrink B' so that f_{N_0} varies by at most $1/(3m)$ on B'. Then for any $y, z \in B'$:

$$|f(y) - f(z)| \leq |f(y) - f_{N_0}(y)| + |f_{N_0}(y) - f_{N_0}(z)| + |f_{N_0}(z) - f(z)| \leq \frac{1}{3m} + \frac{1}{3m} + \frac{1}{3m} = \frac{1}{m}$$

Hence any point of B' lies in O_m, showing that O_m is dense. Since each O_m is open and dense, $\bigcap_m O_m$ is a dense G_δ by the Baire Category Theorem.

As a consequence, if $f = \lim f_n$ is a pointwise limit of continuous functions (such an f is called a *Baire class 1* function), then f cannot be discontinuous everywhere: its set of continuity points is always residual. This provides another perspective on why "most" functions are badly behaved: a generic continuous function is nowhere differentiable (example 7.2.4), but any pointwise limit of continuous functions must still be continuous on a large set.

7.3 Dual Spaces

The continuous linear functionals on a normed space form its dual, and the first question about them is whether there are enough to be useful. Let X be a Banach space over k, with k either $\mathbb{R}$ or $\mathbb{C}$; its *dual* is $X^* := \operatorname{Hom}(X, k)$, the space of continuous, equivalently bounded, linear functionals on X. In finite dimensions the dual is easily understood: $\dim X^* = \dim X$, so $X^* \cong X$, though not canonically. In infinite dimensions it is far less clear when a functional is continuous at all: d/dx is unbounded, while $\int$ becomes bounded once its domain is restricted. What is wanted is a supply of functionals known in advance to extend to elements of X^*, and the Hahn-Banach extension theorem provides exactly this, showing X^* is always large enough to make its study worthwhile. It is the linear-analytic counterpart of Urysohn's lemma ([7]), which manufactures enough continuous functions to separate points and closed sets in a normal space.

The functionals to be extended are controlled by a one-sided gauge.

Definition 7.3.1: Sublinear Functional

A function $p\colon X \to \mathbb{R}$ is a *sublinear functional* if it is subadditive, $p(x+y) \le p(x)+p(y)$, and positively homogeneous, $p(tx) = t\,p(x)$ for all $t \ge 0$ in $\mathbb{R}$.

A sublinear functional need not be a seminorm: homogeneity is required only for nonnegative scalars, so $p(-x)$ and $p(x)$ may differ. Conversely every seminorm, and hence every norm, is a sublinear functional. For a concrete example that is not a seminorm, $p(x) = \limsup_n x_n$ on the real sequence space ℓ^∞ is subadditive and positively homogeneous, yet $p(-x) = -\liminf_n x_n$ differs from $p(x)$ in general; this p returns below in the construction of Banach limits. The extension theorem is proved first over $\mathbb{R}$; the passage to complex scalars comes afterwards and rests on the following correspondence.

Proposition 7.3.2: Converting Complex and Real Linear Functionals

Let X be a vector space over $\mathbb{C}$. If f is a complex-linear functional on X and $u = \mathrm{Re} f$, then u is a real-linear functional and $f(x) = u(x) - i\,u(ix)$ for all $x \in X$. Conversely, if u is a real-linear functional on X and $f\colon X \to \mathbb{C}$ is defined by $f(x) = u(x) - i\,u(ix)$, then f is complex-linear. When X is normed, $\|u\| = \|f\|$.

***Proof*:**
If f is complex-linear then $u = \mathrm{Re} f$ is real-linear, and $\mathrm{Im} f(x) = -\mathrm{Re}[if(x)] = -u(ix)$, so $f(x) = u(x) - i\,u(ix)$. Conversely, if u is real-linear then $f(x) = u(x) - i\,u(ix)$ is $\mathbb{R}$-linear, and

$$f(ix) = u(ix) - i\,u(-x) = u(ix) + i\,u(x) = if(x)$$

so f is $\mathbb{C}$-linear. Finally, suppose X is normed. Since $|u(x)| = |\mathrm{Re} f(x)| \le |f(x)|$, we have $\|u\| \le \|f\|$. For the reverse inequality, fix x with $f(x) \ne 0$ and set $\alpha = \overline{\mathrm{sgn}(f(x))}$. Then

$$|f(x)| = \alpha f(x) = f(\alpha x) = u(\alpha x)$$

since $f(\alpha x)$ is real, whence $|f(x)| \le \|u\|\,\|\alpha x\| = \|u\|\,\|x\|$ and so $\|f\| \le \|u\|$. Combining the two bounds gives $\|f\| = \|u\|$, as we sought to show.

Theorem 7.3.3: Hahn-Banach Extension Theorem

Let X be a real vector space, p a sublinear functional on X, $M \subseteq X$ a subspace, and f a linear functional on M with $f(x) \le p(x)$ for all $x \in M$. Then there exists a linear functional F on X with $F(x) \le p(x)$ for all $x \in X$ and $F|_M = f$.

***Proof*:**
The argument has two stages: a one-dimensional extension, then a maximality argument by Zorn's lemma (here playing the role of transfinite induction).

Step 1: Extending by one dimension. Let $x \in X \setminus M$; the goal is a linear functional g on $M + \mathbb{R}x$

with $g \le p$ there and $g|_M = f$. For any $y_1, y_2 \in M$,

$$f(y_1) + f(y_2) = f(y_1 + y_2) \le p(y_1 + y_2) \le p(y_1 - x) + p(x + y_2)$$

Rearranging gives $f(y_1) - p(y_1 - x) \le p(x + y_2) - f(y_2)$, and as this holds for all $y_1, y_2 \in M$,

$$\sup_{y \in M} \{f(y) - p(y - x)\} \le \inf_{y \in M} \{p(x + y) - f(y)\}$$

so there is a real α with

$$\sup_{y \in M} \{f(y) - p(y - x)\} \le \alpha \le \inf_{y \in M} \{p(x + y) - f(y)\}$$

Define $g\colon M + \mathbb{R}x \to \mathbb{R}$ by $g(y + \lambda x) = f(y) + \lambda\alpha$. Then g is linear, $g|_M = f$, and $g(y) \le p(y)$ on M (the case $\lambda = 0$). For $\lambda > 0$ and $y \in M$,

$$g(y + \lambda x) = \lambda\big[f(y/\lambda) + \alpha\big] \le \lambda\big[f(y/\lambda) + p(x + y/\lambda) - f(y/\lambda)\big] = p(y + \lambda x)$$

and for $\lambda = -\mu < 0$,

$$g(y + \lambda x) = \mu\big[f(y/\mu) - \alpha\big] \le \mu\big[f(y/\mu) - f(y/\mu) + p(y/\mu - x)\big] = p(y + \lambda x)$$

so $g(z) \le p(z)$ for all $z \in M + \mathbb{R}x$.

Step 2: Maximality. The functional g satisfies the same hypotheses as f, now on the larger domain $M + \mathbb{R}x$. Order the pairs (N, h), with $M \subseteq N \subseteq X$ a subspace and h a linear functional on N extending f and dominated by p, by declaring $(N_1, h_1) \le (N_2, h_2)$ when $N_1 \subseteq N_2$ and $h_2|_{N_1} = h_1$. Every chain has an upper bound, the functional defined on the union of its domains, so Zorn's lemma yields a maximal element (N, F). If $N \ne X$, Step 1 applied to F extends it strictly, contradicting maximality; hence $N = X$, completing the proof.

The extension theorem is most often applied with p a norm, but it does more when p is a one-sided gauge. The construction of *Banach limits* is the standard illustration: a way to assign a limit to every bounded sequence, agreeing with the usual limit when it exists and unchanged by shifting the sequence.

Example 7.3.4: Banach Limits

Let $\ell^\infty = \ell^\infty(\mathbb{N}; \mathbb{R})$ be the bounded real sequences, $c \subseteq \ell^\infty$ the convergent ones, and $S\colon \ell^\infty \to \ell^\infty$ the shift $(Sx)_n = x_{n+1}$. There is a linear functional $\mathrm{LIM}\colon \ell^\infty \to \mathbb{R}$ with:

1. $\mathrm{LIM}\, x = \lim_n x_n$ for every $x \in c$;
2. $\liminf_n x_n \le \mathrm{LIM}\, x \le \limsup_n x_n$ for every $x \in \ell^\infty$; in particular LIM is positive and $\|\mathrm{LIM}\| = 1$;
3. $\mathrm{LIM}(Sx) = \mathrm{LIM}\, x$ for every $x \in \ell^\infty$.

To construct it, let $p(x) = \limsup_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} x_n$, the lim sup of the Cesàro means. Subadditivity of p follows from $\limsup(a_N + b_N) \le \limsup a_N + \limsup b_N$, and positive homogeneity is clear, so p is a sublinear functional. On c the Cesàro means of a convergent sequence converge to its limit, so $\lim_n x_n = p(x)$ there; viewing the ordinary limit as a linear functional on c, it satisfies $\lim \le p$ on c. By theorem 7.3.3 it extends to a linear functional LIM on ℓ^∞ with $\mathrm{LIM} \le p$ everywhere.

Property (1) holds by construction. For (2), the estimate $p(x) \leq \limsup_n x_n$ gives $\mathrm{LIM}\, x \leq \limsup_n x_n$, and applying it to $-x$ gives $\mathrm{LIM}\, x \geq \liminf_n x_n$; taking $x \geq 0$ shows LIM is positive, and since LIM fixes the constant sequence 1 while $|\mathrm{LIM}\, x| \leq \|x\|_\infty$, its norm is exactly 1. For (3), the telescoping average

$$\frac{1}{N}\sum_{n=1}^{N}\big((Sx)_n - x_n\big) = \frac{x_{N+1} - x_1}{N} \xrightarrow[N\to\infty]{} 0$$

forces $p(Sx - x) = 0$ and $p(x - Sx) = 0$, so $\mathrm{LIM}(Sx - x) \leq 0$ and $\mathrm{LIM}(x - Sx) \leq 0$, giving $\mathrm{LIM}(Sx) = \mathrm{LIM}\, x$.

To show this extension, take any bounded sequence periodic with period q, so $x_{n+q} = x_n$. Summing q consecutive shifts, $\sum_{j=0}^{q-1} S^j x$ has the value $x_1 + \cdots + x_q$ at every index, so shift invariance and property (1) give

$$q\ \mathrm{LIM}\, x = \mathrm{LIM}\Big(\sum_{j=0}^{q-1} S^j x\Big) = x_1 + \cdots + x_q$$

that is, $\mathrm{LIM}\, x = \frac{1}{q}(x_1 + \cdots + x_q)$, the average over one period. So $(1, 0, 1, 0, \dots)$ has $\mathrm{LIM}\, x = \frac{1}{2}$ and $(1, 2, 3, 1, 2, 3, \dots)$ has $\mathrm{LIM}\, x = 2$, though neither sequence converges.

Banach limits also settle a duality question. The functional LIM vanishes on the subspace c_0 of sequences tending to 0, yet is not identically zero, so it is not represented by any $a \in \ell^1$ under the pairing $\langle x, a\rangle = \sum_n x_n a_n$ (an $a \in \ell^1$ annihilating all of c_0 must be 0). Thus ℓ^1 is a proper subspace of $(\ell^\infty)^*$, an early sign that ℓ^∞ is not reflexive.

Before drawing the geometric consequences, the extension theorem must be carried to complex scalars. For the domination bound to behave there, p must be a seminorm rather than a general sublinear functional: when p is a seminorm and $f\colon X \to \mathbb{R}$ is linear, the bound $f \leq p$ is equivalent to $|f| \leq p$, since $|f(x)| = \pm f(x) = f(\pm x)$ and $p(-x) = p(x)$.

Proposition 7.3.5: Complex Hahn-Banach Theorem

Let X be a vector space over $\mathbb{C}$, p a seminorm on X, $M \subseteq X$ a subspace, and f a complex-linear functional on M with $|f(x)| \leq p(x)$ for all $x \in M$. Then there exists a complex-linear functional F on X with $|F(x)| \leq p(x)$ for all $x \in X$ and $F|_M = f$.

***Proof*:**
Let $u = \mathrm{Re} f$, a real-linear functional on M with $u \leq |f| \leq p$. Applying theorem 7.3.3 to u produces a real-linear extension U of u to X with $U \leq p$; since $p(-x) = p(x)$, this gives $|U(x)| \leq p(x)$ for all x. Set $F(x) = U(x) - i\,U(ix)$. By proposition 7.3.2, F is a complex-linear extension of f, and for x with $F(x) \neq 0$, taking $\alpha = \overline{\mathrm{sgn}(F(x))}$ gives

$$|F(x)| = \alpha F(x) = F(\alpha x) = U(\alpha x) \leq p(\alpha x) = p(x)$$

completing the proof.

The geometry now follows. The next result collects the consequences that make the dual space a usable tool.

Proposition 7.3.6: Hahn-Banach Consequences

Let X be a normed vector space over k. Then:

1. If $M \subseteq X$ is a closed subspace and $x \in X \setminus M$, there exists $f \in X^*$ with $f|_M = 0$ and $f(x) \neq 0$; it may be chosen so that, with $\delta = \inf_{y \in M} \|x - y\|$,
$$\|f\| = 1 \qquad f(x) = \delta$$
2. If $0 \neq x \in X$, there exists $f \in X^*$ with $\|f\| = 1$ and $f(x) = \|x\|$.
3. The bounded linear functionals on X separate points.
4. For $x \in X$, the evaluation $\hat{x} \colon X^* \to k$, $\hat{x}(f) = f(x)$, defines a linear isometry $x \mapsto \hat{x}$ from X into the double dual X^{**}.

***Proof*:**

1. Let $\delta = \inf_{y \in M} \|x - y\|$, positive because M is closed and $x \notin M$. On $M + kx$ define $f_0(y + \lambda x) = \lambda\delta$, so $f_0(x) = \delta$ and $f_0|_M = 0$. For $\lambda \neq 0$,
$$|f_0(y + \lambda x)| = |\lambda|\,\delta \leq |\lambda|\,\|\lambda^{-1}y + x\| = \|y + \lambda x\|$$
using $\delta \leq \|x + \lambda^{-1}y\|$, so f_0 has norm at most 1 on $M + kx$, with equality since $\delta = \inf$. Taking $p(z) = \|z\|$, theorem 7.3.3 (or proposition 7.3.5 over $\mathbb{C}$) extends f_0 to $F \in X^*$ with $\|F\| = 1$, $F|_M = 0$, and $F(x) = \delta$.
2. Apply part (1) with $M = \{0\}$, for which $\delta = \|x\|$; the resulting f has $\|f\| = 1$ and $f(x) = \|x\|$.
3. If $x \neq y$, part (2) applied to $x - y \neq 0$ gives $f \in X^*$ with $f(x - y) \neq 0$, so $f(x) \neq f(y)$.
4. Both $\hat{x}$ (in f) and $x \mapsto \hat{x}$ are linear. From $|\hat{x}(f)| = |f(x)| \leq \|f\|\,\|x\|$ we get $\|\hat{x}\| \leq \|x\|$, while part (2) supplies f with $\|f\| = 1$ and $\hat{x}(f) = \|x\|$, so $\|\hat{x}\| = \|x\|$; thus $x \mapsto \hat{x}$ is an isometry.

Parts (1) and (2) sharpen Urysohn's separation into a quantitative statement: the separating functional not only exists but has its norm and value pinned to the geometry of the subspace. This motivates the following definition.

Definition 7.3.7: Annihilator

For a subspace $Y \subseteq X$ of a normed space X, the *annihilator* of Y is
$$Y^\perp = \{\varphi \in X^* \mid \varphi(y) = 0 \text{ for all } y \in Y\}$$
Dually, for a subspace $Z \subseteq X^*$, the *pre-annihilator* is
$$^\perp Z = \{x \in X \mid \varphi(x) = 0 \text{ for all } \varphi \in Z\}$$

Part (1) shows that $Y^\perp$ is large enough to detect Y: any point outside the closure of Y is exposed by some functional in $Y^\perp$. Concretely, $Y = {}^\perp(Y^\perp)$ for every norm-closed subspace $Y \subseteq X$. The inclusion $Y \subseteq {}^\perp(Y^\perp)$ is immediate, and conversely, if $x \notin Y$ then part (1) produces $\varphi \in Y^\perp$ with $\varphi(x) \neq 0$, so $x \notin {}^\perp(Y^\perp)$. This identity is the tool behind the closed-range theorem in chapter 11 and its unbounded analogue.

The same functional computes distances, sharpening part (1) into a duality formula.

Corollary 7.3.8: Norm and Distance Duality

Let X be a normed space, $x \in X$, and $M \subseteq X$ a closed subspace. Then

$$\|x\| = \max_{\substack{f\in X^* \\ \|f\|\le 1}} |f(x)| \qquad \text{and} \qquad \inf_{y\in M} \|x-y\| = \max_{\substack{f\in M^\perp \\ \|f\|\le 1}} |f(x)|$$

both maxima attained.

***Proof*:**
Any f with $\|f\| \le 1$ satisfies $|f(x)| \le \|x\|$, and part (2) of proposition 7.3.6 supplies an f of norm 1 with $f(x) = \|x\|$, giving the first identity with the maximum attained. For the second, any $f \in M^\perp$ with $\|f\| \le 1$ has $|f(x)| = |f(x-y)| \le \|x-y\|$ for every $y \in M$, so $|f(x)| \le \inf_y \|x-y\|$, while part (1) supplies an $f \in M^\perp$ of norm 1 attaining $f(x) = \inf_y \|x-y\|$, as we sought to show.

Part (3), that X^* separates points, is the linear face of injectivity: if a construction produces x, y with $f(x) = f(y)$ for every $f \in X^*$, then $x = y$, exactly as coordinates decide equality in finite dimensions. This is the property that makes the weak topology usable. Giving X the coarsest topology in which every $f \in X^*$ is continuous, separation of points is precisely what makes that topology Hausdorff, so weak limits are unique (proposition 7.4.6).

Separation has a second face: bundling all of X^* together represents X as continuous functions on a compact space.

Proposition 7.3.9: Universal Embedding into $C(K)$

Let X be a normed space and $K = (B_{X^*}, w^*)$ the closed unit ball of X^* under the weak* topology. Then K is compact Hausdorff and $x \mapsto \hat{x}|_K$ is a linear isometry $X \to C(K)$. If X is a Banach space, its image is a closed subspace; thus every Banach space is isometrically a closed subspace of some $C(K)$.

***Proof*:**
Compactness of K is Alaoglu's theorem (theorem 7.5.7), proved later in this chapter, and the weak* topology is Hausdorff because X separates the points of X^*. Each $\hat{x}$ is weak* continuous by the definition of that topology, so $\hat{x}|_K \in C(K)$, and

$$\|\hat{x}|_K\|_{C(K)} = \sup_{\|f\|\le 1} |f(x)| = \|x\|$$

by corollary 7.3.8. Hence $x \mapsto \hat{x}|_K$ is a linear isometry. When X is complete, its isometric image is complete and therefore closed in $C(K)$, completing the proof.

7.3.10 Note: Relation to Stone-Weierstrass The embedding places X in $C(K)$ as a *linear* subspace, and $\widehat{X}$ does separate the points of K. It is tempting to invoke the Stone-Weierstrass theorem ([3]) to conclude that X is dense, but that theorem needs a *subalgebra* separating points, and $\widehat{X}$ is not closed under products. What Stone-Weierstrass gives instead is that the closed subalgebra *generated* by $\widehat{X}$ and the constants (with complex conjugates, over $\mathbb{C}$) is dense in $C(K)$; this recovers all of $C(K)$, not X. In general X is a proper closed subspace: linear separation is strictly weaker than the algebraic separation Stone-Weierstrass requires.

The supply of functionals in proposition 7.3.6 depends on the norm being convex in a way the hypotheses quietly use; without local convexity the dual can collapse.

7.3.11 Warning: Enough functionals needs convexity For $0 < p < 1$ the space $L^p([0,1])$, metrized by $d(f,g) = \int_0^1 |f-g|^p$, is a complete topological vector space whose only continuous linear functional is 0. Hahn-Banach produces nothing because its hypotheses fail: d is not a norm and the space is not locally convex. Concretely, let φ be continuous and linear and f arbitrary. Choose c with $\int_0^c |f|^p = \frac{1}{2}\int_0^1 |f|^p$ and split $f = f_1 + f_2$ into the parts on $[0,c]$ and $[c,1]$, so $d(f_i, 0) = \frac{1}{2}d(f,0)$. From $\varphi(f_1) + \varphi(f_2) = \varphi(f)$ one part has $|\varphi(f_j)| \geq \frac{1}{2}|\varphi(f)|$; set $g = 2f_j$, so $|\varphi(g)| \geq |\varphi(f)|$ while $d(g,0) = 2^p \cdot \frac{1}{2}d(f,0) = 2^{p-1}d(f,0)$. As $p < 1$ the factor $2^{p-1} < 1$, so iterating from a fixed f yields g_n with $|\varphi(g_n)| \geq |\varphi(f)|$ but $d(g_n, 0) \to 0$; continuity forces $\varphi(f) = 0$. The abundance of functionals guaranteed by Hahn-Banach is thus a feature of *locally convex* spaces, the class introduced in section 7.4.

7.3.12 Foreshadowing: The geometric form Parts (1) and (3) are the analytic face of Hahn-Banach: extend a functional, separate a point from a subspace. There is a dual geometric face, in which two disjoint convex sets are separated by a closed hyperplane $\{f = c\}$ and a boundary point of a convex body carries a supporting hyperplane. That form belongs to the theory of locally convex spaces begun in section 7.4, where convex neighbourhoods are plentiful enough to run the same gauge argument on an open convex set.

Part (4) invites a look at the double dual. Writing $\widehat{X} = \{\hat{x} \mid x \in X\}$, the space X^{**} is always complete, so the closure $\overline{\widehat{X}}$ inside X^{**} is a Banach space and $x \mapsto \hat{x}$ embeds X into it isometrically with dense image. This exhibits $\overline{\widehat{X}}$ as the *completion* of X: every normed space sits densely inside a Banach space, obtained here with no new construction. If X is already Banach, then $\widehat{X} = \overline{\widehat{X}}$ is closed.

The image $\widehat{X}$ need not exhaust X^{**}. When it does, so that the canonical isometry $X \to X^{**}$ is onto, X is called *reflexive*. Every finite-dimensional space is reflexive, since $\dim X^{**} = \dim X$. In infinite dimensions reflexivity can fail: c_0 has $c_0^{**} = \ell^\infty \neq c_0$, so it is not reflexive, and the same holds for ℓ^1 and ℓ^∞. By contrast the L^p spaces with $1 < p < \infty$ are reflexive, as the duality $(L^p)^* = L^q$ will show.

Exercise 7.3.1

1. Let $T\colon X \to Y$ be a linear map between Banach spaces such that $f \circ T \in X^*$ for every $f \in Y^*$. Show that T is bounded. (Hint: the closed graph theorem and separation of points.)

2. Show that a subspace $M \subseteq X$ of a normed space is dense if and only if $M^\perp = \{0\}$.
3. Deduce from the periodic-sequence value in example 7.3.4 that no Banach limit is multiplicative, i.e. $\mathrm{LIM}(xy) = (\mathrm{LIM}\, x)(\mathrm{LIM}\, y)$ can fail.
4. Let p be a sublinear functional on a real vector space X. Show that $p(x) = \max\{f(x) : f \text{ linear on } X,\ f \leq p\}$ for every x, the maximum attained.
5. For a closed subspace $M \subseteq X$, show that $\psi \mapsto \psi \circ q$, where $q\colon X \to X/M$ is the quotient map, is an isometric isomorphism $(X/M)^* \cong M^\perp$.

7.4 Weak Topologies and TVS's

(This post seems like an excellent post to follow! link here)

There are many topologies on which the operations are continuous (i.e. + and scalar multiplication is continuous) that do not arrise from norms. The topology on a vector-space induced by a norm is in fact quite strong. In general, such a space (in infinite dimensions) doesn't have many compact set, and so we loose many of the nice results of working on comapct or pre-compact sets. This is naturally unfavourable in analysis, since many arguments rely on properties of sequences converging. Looking at functional analysis as the study of sequences, norms induce a particular definition of convergence: $\|x - x_n\|_V \to 0$. However, there are many other notions of convergence that are weaker that are quite useful, for example:

1. pointwise convergence
2. convergence in measure
3. smooth convergence
4. convergence on compact sets

None of these types of convergence are induced by a norm. We will instead be induced new types of topologies on our vector-spaces. The two types of topologies we will usually impose on V will be known as the *weak topology* and the *weak* topology*. We will define analogous topologies on $\mathrm{Hom}_k(V, W)$ which somewhat confusingly will be named *strong operator topology* and *weak operator topology*.

Definition 7.4.1: Topological Vector Space (TVS)

Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$. Then V is a *topological vector space* (TVS) if the operation of addition and scalar multiplication are continuous. A topological vector-space is said to be *locally convex* if there is a base for the topology consisting of convex sets.

It should be verified that the translation map $x \mapsto x_0 + x$ and the dilation map $x \mapsto \lambda x$ are both homeomorphissm for any scalar λ and vector x_0.

Example 7.4.2: Topological Vector Spaces

1. Let V be a normed vector-space with the topology induced by the norm. Then $\|c \cdot x\| \leq |c|\|x\|$ (in fact $=$) and $\| + (x,y)\| \overset{!}{\leq} 2(\|x\| + \|y\|)$ where $\overset{!}{\leq}$ comes from the triangle inequality and the fact that $\|(x,y)\| = \max(\|x\|, \|y\|)$. Hence, V is a topologicla vector space. If $\| - \|$ was a quasi-norm so that we replace the triangle inequality with $\|x+y\| \leq K(\|x\| + \|y\|)$ then a quasi-normed space V is still a topological vector space.

2. Let V be a semi-normed space so that $\|x\| = 0$ does not necessarily imply $x = 0$. Then V is a topological vector space with the topology induced by the semi-norm. V is Hausdroff if and only if the semi-norm is a norm.

3. Any subbspace $W \subseteq V$ of a topological vector space V is a topological vector space with the subspace topology. Similarly for products and quotients.

4. Let V be a vector-space and $(\mathcal{T}_\alpha)_{\alpha \in A}$ be a collection of topologies on V that makes V a topological vector space. Let $\mathcal{T} = \langle \mathcal{T}_\alpha \rangle_{\alpha \in A}$ be the topology generated by $\bigcup_{\alpha \in A} \mathcal{T}_\alpha$ (this is the weakest topology that contains each $\mathcal{T}_\alpha$). Then $(V, \mathcal{T})$ is also a topological vector space. Importantly, a sequence (resp. net) $x_n \to x$ in $\mathcal{T}$ if and only if $x_n \to x$ in $\mathcal{T}_\alpha$ for all $\alpha \in A$ (recall that $x_n \to x$ in a general topology space if for any neighborhood U of x, there exists an N such that for all $n \geq N$, $x_n \in U$). Commonly, we will have $(\mathcal{T}_\alpha)_{i=1}^\infty$ be generated by countable collection of semi-norms. A complete space generated by a countable collection of semi-norms is called a *Fréchet space*.

5. (Pointwise Convergence) Let X be a set and $\mathbb{C}^X$ the collection of all complex valued functions $f : X \to \mathbb{C}$. Then $\mathbb{C}^X$ is a complex-vector-space. For each $x \in X$, define the seminorm:

$$\|f\|_x := |f(x)|$$

Then the topology generated by all of these seminorms is the *topology of pointwise convergence* on $\mathbb{C}^X$ (it is also the product topology on this space). A sequence $(f_n) \subseteq \mathbb{C}^X$ converges to f in this topology if and only if it converges pointwise. Note that if X has more than one point, then none of the semi-norms individually generate a Hausdroff topology, but taken all together the induced topology is Haudsorff.

6. (Uniform Convergence) Let X be a topological space and $C(X)$ the set of complex-valued continuous function $f : X \to \mathbb{C}$. If X is not compact, then in general we would not expect functions in $C(X)$ to be bounded, so the sup-norm will not necessarily make $C(X)$ into a normed vector-space. Nonetheless, we can still define "balls" $B_r(f) \subseteq C(X)$ by:

$$B_r(f) := \left\{ g \in C(X) \;\middle|\; \sup_{x \in X} |f(x) - g(x)| \leq r \right\}$$

these do form a topological structure on $C(X)$, but not a topological vector space structure since scalar multiplication is no longer continuous. Nonetheless, we have that $(f_n) \subseteq C(X)$ converges in this topology to a limit $f \in C(X)$ if and only if f_n converge uniformaly to f. Thus $\sup_{x \in X} |f_n(x) - f(x)|$ is finite for sufficinetly large n and converges to 0 as $n \to \infty$.

7. (Uniform Convergence on Compact sets) Let X and $C(X)$ be as in the above example. For every compact subset $K \subseteq X$, define a seminorm:

$$\|f\|_{C(K)} = \sup_{x \in K} |f(x)|$$

The topology generated by all these seminorms as k ranges over all compact subsets of X is called the *topology of uniform convergence on compact sets.* It is stronger than the topology of pointwise convergence but weaker than the topology of uniform convergence, and is frequently used in complex analysis. A sequence $(f_n) \subseteq C(X)$ converges to $f \in C(X)$ in this toplogy if and only if f_n converges uniformly to f on each compact set. Is the resulting space a topological vector space?

8. (Smooth Convergence) Let $C^\infty([0,1])$ be the space of smooth functions $f : [0,1] \to \mathbb{C}$. Then we can define the C^k-norm on this space for any non-negative integer k by the formula:

$$\|f\|_{C^k} := \sum_{i=1}^{k} \sup_{x\in[0,1]} |f^{(i)}(x)|$$

where $f^{(i)}$ is the ith derivative of f. The topology generated by all C^k norms $k = 0, 1, 2, ...,$ is the *smooth topology.* A sequence f_n converges in this topology to a limit f if $f_n^{(i)}$ converges uniformally to $f^{(i)}$ for each $i \geq 0$. Is this a topological vector space?

9. (Convergence in measure) Let $(X, \mathcal{M}, \mu)$ be a measure space and $L(X)$ the set of measurable functions $f : X \to \mathbb{C}$. Then the sets

$$B_{r,\epsilon}(f) := \{g \in L(X) \mid \mu(\{x \mid |f(x) - g(x)| \geq r\} < \epsilon)\}$$

for $f \in L(X)$, $\epsilon, r > 0$ forms the basis for a topology that makes $L(X)$ into a topological vector space. A sequence $(f_n) \subseteq L(X)$ converges to f in this topology if and only if it converges in measure.

10. Let $[0,1]$ be given the usual Lebesgue measure. Then show that $L^\infty([0,1])$ cannot be given a topological vector space structure in which a sequence $(f_n) \subseteq L^\infty([0,1])$ converges to f in this topology if and only if it converges a.e.

11. Let V be an $\mathbb{R}$-vector space. a subset $U \subseteq V$ is called *algebraically open* if

$$\{t \in \mathbb{R} \mid x + tv \in U\}$$

is open for all $x, v \in V$. Then any open set in a topological vector space is algebraically open. Note that there are algebraically open sets that are not open (take any dense subset of S^1, say D, and $\mathbb{R} \setminus \{D\}$.) This larger set of algebraically open sets forms a topology, but this topology does not give the $\mathbb{R}^2$ structure of a topological vector space.

12. Let X be a LCH space. On $\mathbb{C}^X$, the topology of uniform convergence on compact sets is defined by the seminorms $p_K(f) = \sup_{x\in K} |f(x)|$ as K rangers over compact subsets of X. If X is σ-compact and $\{U_n\}$ are as in proposition ref;HERE (folland p. 168), this topology is defined by the seminorms $p_n(f) = \sup_{x\in\overline{U_n}} |f(x)|$. In this case, $\mathbb{C}^X$ is easly seen to be complete, so it is a Fréchet space. By proposition ref:HERE, so is $C(X)$.

13. The space $L^1_{\text{loc}}(\mathbb{R}^n)$ is a Fréchet space with the topology defined by the seminorms $p_k(f) = \int_{|x|\leq k} |f(x)|dx$ (completeness follows from completness of L^1).

14. Here is another useful place where TVS are defined. Let's say we want to study a space in which d/dx is continuous (and hence bounded) on $C^\infty([0,1])$. However, this is essentially impossible to define the topology through a norm. Indeed, if $f_\lambda(x) = e^{\lambda x}$, then $(d/dx)f_\lambda = \lambda f_\lambda$, and so $\|d/dx\| \geq \lambda$ for all λ, no matter what norm is used on $C^\infty([0,1])$. There are a few ways to rectify this:

(a) One can consider differentiation as an unbounded operator form V to W where W is a suitable banace space and X is a dense subpsac eof W (exercise 30 in Folland)

(b) One can consider differentiation as a bounded linear map from one Banach space V to a different one W such that $V = C^k([0,10])$ and $W = C^{k-1}([0,1])$ (execise 9)

(c) One can consider differentiation as a continuous operator on a locally convex space V whose topology is not given by a norm. It is easy to construct families of seminorms on the space of smooth functions st differentiation becomes continuous almost by definition. For example, the seminorms:

$$p_k(f) = \sup_{0 \leq x \leq 1} |f^{(k)}(x)|$$

where $k \in \{0, 1, 2, ...\}$ makes $C^\infty([0,1])$ into a Fréchet space (the completeness is proved in exercise 9), and d/dx is continuous by proposition 5.15, since $p_k(f') = p_{k+1}(f)$. Other examples are in exer exercsie 45 in chatper 9 of Folland.

15. The space $\mathbb{R}^\infty = \varprojlim \mathbb{R}^n$ is a Fréchet space; can you see the norms?

Proposition 7.4.3: Topological Vector Space And Hausdorff

Let V be a topological vector space. Then V is Hausdroff if and only if $\{0\}$ is closed

***Proof*:**
exercise

We may ask what properties of Banach spaces extend to topological vector space's. Some nice properties of Banach spaces include the uniform boundedness principle and the open mapping theorem. These can be extended to the slighty larger class of *Fréchet spaces* that we mentioned in the examples above. We will single them out and talk about them in the next section.

If V is a topological vector space, we can take V^* to be the set of all continuous functionals from V to the base field $k \in \{\mathbb{R}, \mathbb{C}\}$. Since V has no norm, V^* has no norm topology. We can however still define interesting topologies on it. This topology is dual to a weaker topology we can put on V:

Definition 7.4.4: Weak And Weak* Topologies

Let V be a topological vector space and V^* its dual. Then

1. the *weak* topology* on V^* is the topology generated by the seminorms $\|\lambda\|_x := |\lambda(x)|$ for all $x \in V$

2. the *weak topology* on V is the topology generated by the seminorms $\|x\|_\lambda = |\lambda(x)|$ for all $\lambda \in V^*$

If V is also a normed vector-space, then the weak topology on V is weaker than the norm topology. If V is a normed space, then so is V^* and $(V^*)^*$. the weak* topology on V^* makes V^* into a topological vector space, and is a topology weaker than the weak topology on V^* (which is defined using the double dual $(V^*)^*$). When V is reflexive ($V \cong V^{**}$), then the weak and weak* topologies on V^* are in fact equivalent.

Definition 7.4.5: Weak Convergence

Let V a vector-space endowed with the weak topology. Then a sequence $(x_n) \subseteq V$ that converges in V with the weak topology is called *weakly convergent* and is said to *weakly converge* to a point x. A sequence weakly converges $x_n \to x$ if and only if

$$\lambda(x_n) \to \lambda(x) \qquad \forall \lambda \in V^*$$

and is usually denoted as $x_n \rightharpoonup x$. Similarly, $\lambda_n \rightharpoonup \lambda$ if $\lambda_n(x) \to \lambda(x)$ for all $x \in V$ (λ_n, as a function of V, converges pointwise to λ)

Proposition 7.4.6: Normed Vector Spaces Weak Topology

Let V be a normed vector-space. Then the weak topology on V and the weak* topology on V^* are both Hausdorff. In particular, we conclude that in the weak and weak* limits, when they exist they are unique

***Proof*:**
Use the Hahn-Banach Theorem

Example 7.4.7: Norm, Weak, Weak* Convergence
Let $V = c_0(\mathbb{N})$, $V^* = \ell^1(\mathbb{N})$, and $V^{**} = \ell^\infty(\mathbb{N})$. Let $e_1, e_2, ...$ be the standard basis of any of these vetor-spaces.

1. The sequence (e_i) converges weakly in V to zero, but not not converge in norm in V
2. (easy to think up some examples)

A natural generalization of the weak* topology on X^* can be imposed $B(X, Y)$ for some TVS Y. Just like for X^*, we can impose a weak topology on it, hewer there is another natural topology we may induce on $B(X, Y)$:

Definition 7.4.8: Strong And Weak Operator Topology

Let $B(X, Y)$ be a all continuous operators between Fréchet spaces. Then

1. the *weak operator topology* is the topology induced by the seminorm $T \mapsto |\lambda(T(x))|$ for all $x \in X$ and $\lambda \in Y^*$
2. If furthermore Y is a Banach space, then the *strong operator topology* (or *operator norm topolgoy*) is the topology induced by the seminorms $T \mapsto \|T(x)\|_Y$ for all $x \in X$

Hence, a sequence $(T_n) \subseteq B(X, Y)$ converges in the strong operator topology to a limit T if and only if $T_n(x) \to T(x)$ in the norm topology on Y, and T_n converges to T in the weak operator topology if $T_n(x) \to T(x)$ in e weak topology on x. Hence, we see that the weaker operator topology is weaker than the strong operator topology, which in turn is (somewaht confusingly) weaker than the operator norm topology.

A final result we shall point out is hat we can reprhase the uniform boundedness principle for convergence now as follows:

Proposition 7.4.9: Uniform Boundedness Principle Reformulation

Let $(T_N) \subseteq B(X,Y)$ be a sequence of bounded linear operators from a Banach space X to a normed space Y, and let $T \in B(X,Y)$ be another bounded lienar operator, and let D be a dense subspace of X. Then the following are equivalent:

1. T_n converges in the strong operator topology fo $B(X,Y)$ to T
2. T_n is bounded in the operator norm (i.e. $\|T_n||_{op}$ is bounded) and the restriction of T_n to D converges in the strong operator topology of $B(D,Y)$ to the restriction of T to D

7.5 Fréchet Spaces

When constructing a space, we often cannot define a space simply through a norm; not all elements may be "finite". However, we can often construct a space using countably many normed spaces, or usually semi-normed space $\|x\| = 0$ does not imply $x = 0$, for example if f is an integrable function). Due to the countable nature of this construction, most types of convergences are accomplished by a family of semi-norms. For example, convergence on compact sets, pointwise convergnce, uniform convergence, and smooth convergence are all convergence in Fréchet spaces. In analysis, there will be many spaces that will Fréchet spaces, notably when you encounter distributions and require to define a topology on the dual space. Hence, a moment to get a better undersatnding of them is worth it.

Theorem 7.5.1: Semi-Norms Generate TVS

Let $\{p_\alpha\}_{\alpha\in A}$ be a family of norms on a vector space V. If $x \in X$, $\alpha \in A$, and $\epsilon > 0$, let

$$U_{x,\alpha,\epsilon} = \{y \in V \mid p_\alpha(y - x) < \epsilon\}$$

and let $\mathcal{T}$ be the topology generated by the sets $U_{x,\alpha,\epsilon}$. Then:

1. For each $x \in X$< the inite intersections of the sets $U_{x,\alpha,\epsilon}$ $(\alpha \in A, \epsilon > 0)$ form a neigborhood base at x
2. if $\langle x_i\rangle_{i\in I}$ is a net in V ,then $x_i \to x$ if and only if $p_\alpha(x_i - x) \to 0$ for all $\alpha \in A$.
3. $(V, \mathcal{T})$ is a locally convex topological vector-space

***Proof*:**

1. If $x \in \bigcap_1^k U_{x_j,\alpha_j,\epsilon_j}$, let $\delta_j = \epsilon_j - p_\alpha(x - x_j)$. By the triangle inequality

$$x \in \cap_1^k U_{x,\alpha_j,\delta_j} \subseteq \cap_1^k U_{x-j,\alpha_j,\epsilon_j}$$

Thus, the assertion now follows from the fact that the topology generated consists of $\emptyset$, X, and all unions of finite intersections.

2. By part (1), it suffices to see that $p_\alpha(x_i - x) \to 0$ if and only if $\langle x_i \rangle$ is eventually in $U_{x,\alpha,\epsilon}$ for all $\epsilon > 0$.

3. Continuity is easy from part (b), for if $x_i \to x$ nd $y_i \to y$, then

$$p_\alpha((x_i + y_i) - (x + y)) \le p_\alpha(x_i - x) + p_\alpha(y_i - y) \to 0$$

so $x_i + y_i \to x + y$. If furthermore $\lambda_i \to \lambda$, then eventually $\lambda_i \le C + |\lambda| + 1$, so

$$p + \alpha(\lambda_i x_i - \lambda x) \le p_\alpha(\lambda_i(x_i - x)) + p_\alpha((\lambda_i - \lambda)x) \le CP_\alpha(x_i - x) + |\lambda_i - \lambda| p_\alpha(x)$$

Hence, $\lambda_i x_i \to \lambda x$. Furthermore, the sets $U_{x,\alpha,\epsilon}$ are convex, for if $y, z \in U_{x,\alpha,\epsilon}$, then

$$p_\alpha(x - [t + (1-t)z]) \le p_\alpha(tx - ty) + p_\alpha((1-t)x + (1-t)z) < t\epsilon + (1-t)\epsilon = \epsilon$$

Then local convexity follows from (1)

Definition 7.5.2: Fréchet Space

A complete Hausdorff topological vector-space whose topology is defined by a countable family of seminroms is called a *Fréchet space*.

Proposition 7.5.3: Continuity in Fréchet Space

Let $T : V \to W$ be a linear map between two vector spaces. Let V be a topology induced by a family of semi-norms $(\| - \|_{V_\alpha})_{\alpha \in A}$ and W the topology induced by a family of semi-norm $(\| - \|_{W_\beta})_{\beta \in B}$. Then T is continuous if and only if for each $\beta \in B$, there exists a finite subsets $A_\beta \subseteq A$ and a constant C_β such that

$$\|T(x)\|_{W_\beta} \le C_\beta \sum_{\alpha \in A_\beta} \|x\|_{V_\alpha} \tag{7.1}$$

for all $x \in V$.

***Proof*:**
Let equation 7.1 hold and consider a convergent net $\langle x_i \rangle \subseteq X$. Then by theorem 7.5.1 $p_\alpha(x_i - x) \to 0$ for all α, thus by the inequality $\|T(x_i) - T(x)\|_{S_\beta} \to 0$ for all β, which implies $T(x_i) \to T(x)$. Since T preserves convergent nets, T is continuous.

Conversely, if T is continuous for every $\beta \in B$ choose a neighbor U_β of 0 in V such that $\|T(x)\|_{W_\beta} < 1$ for all $x \in U_\beta$. By theorem 7.5.1, we may assume that $U_\beta = \cap_{1k} U_{x\alpha_{\epsilon_j}}$. Let $\epsilon = \min(\epsilon_1, \epsilon_2, ..., \epsilon_k)$ so that $\|T(x)\|_{W_\beta} < 1$ whenever $p_{\alpha_j}(x) < \epsilon$ for all j. Now, for $x \in X$, either:

1. $p_{\alpha_j})(x) > 0$ for some j. Then let $y = \epsilon x / (\sum_1^k p_{\alpha_j}(x))$ so tht $p_{\alpha_j}(y) < \epsilon$. Then:

$$\|T(x)\|_{W_\beta} = \sum_1^k \epsilon^{-1} p_{\alpha_j}(x) \|T(y)\|_{W_\beta} \le \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

2. If $p_{\alpha_j}(x) = 0$ for all j, then $p_{\alpha_j}(rx) = 0$ for all j and all $r > 0$. Thus $r\|T(x)\|_{W_\beta} = \|T(rx)\|_{W_\beta} < 1$ for all $r > 0$, and os $\|T(x)\|_{W_\beta}$. Then certainly

$$\|T(x)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

covering both cases, we are done.

The following proposition is an interesting characteriztion of Fréchtet spaces:

Proposition 7.5.4: Hausdorff and invariant Metric

Let X be Fŕchet space given by $\{p_\alpha\}_{\alpha \in A}$. Then

1. X is Haudorff if and only if for each $x \neq 0$, there exists $\alpha \in A$ such that $p_\alpha(x) \neq 0$
2. If X is Hausdorff and A is countable, then X is metrizable with a trnslation invariant metric (i.e. $\rho(x, y) = \rho(x + z, y + z)$ for all $x, y, z \in X$)

***Proof*:**
exercise 43 Folland chatper 5

Note that topologically, all these spaces are the same (as the bellow theorem will sketch). So what really matters is what extra information these semi-norms (or this invariant metric) provides:

Theorem 7.5.5: Anderson-Kadec Theorem

Every infinite-dimensional separable Fréchet space is homeomorphic to the product space $\mathbb{R}^{\mathbb{N}}$ (the countable product of real lines equipped with the product topology).

Note that L^∞ is *inseparable* and hence is not included in this theorem.

***Proof*:**
(Sketch). The proof is historically a combination of results by Kadec (1966) and Anderson (1966), establishing a chain of homeomorphisms. We outline the two main steps:

1. The Banach Case: Kadec proved that every infinite-dimensional separable Banach space is homeomorphic to the Hilbert space ℓ_2. The construction typically involves re-norming the space with a strictly convex equivalent norm (often called a Kadec norm) and defining a homeomorphism via radial projection and coordinate mapping.
2. The Product Case: Anderson proved that ℓ_2 is homeomorphic to $\mathbb{R}^{\mathbb{N}}$. This is often shown using "separable absorption" theorems or the "notback-and-forth" method to manage the coordinate systems.

Since every Fréchet space is a projective limit of Banach spaces (and using the assumption of separability), one can extend the homeomorphism to the general case. Consequently, despite their distinct linear and geometric properties, spaces such as the Schwartz space $\mathscr{S}(\mathbb{R})$, the space of holomorphic functions $H(\mathbb{C})$, and $C^\infty[0, 1]$ are all topologically indistinguishable from $\mathbb{R}^{\mathbb{N}}$.

Proposition 7.5.6: dense convergence, then strong convergence

Let $(T_n)_1^\infty \subseteq L(V, W)$ where $\sup_n \|T_n\| < \infty$ and $T \in L(V, W)$. If $\|T_n(x) - T(x)\| \to 0$ for all x in a dense subset D of X, then $T_n \to T$ strongly.

***Proof*:**
We want to show that $\|T_n(x) - T(x)\| \to 0$ for all x. Let $C = \sup\{\|T\|, \|T_1\|, \|T_2\|, ...\}$. Then given $x \in X$ and $\epsilon > 0$, choose $x' \in D$ such that $\|x - x'\| < \epsilon/3C$. Since $T_n(x') \to T(x)$, choose n large enough so that $\|T_n(x') - T(x')\| < \epsilon/3$. Then

$$\begin{aligned}\|T_n(x) - T(x)\| &\leq \|T_n(x) - T_n(x')\| + \|T_n(x') - T(x')\| + \|T(x') - T(x)\| \\ &\leq 2C\|x - x'\| + \frac{1}{3}\epsilon \\ &< \epsilon\end{aligned}$$

but then $T_n(x) \to T(x)$ strongly, as we sought to show.

One important result is that in the weak operator topology, unit balls are always compact:

Theorem 7.5.7: Alaoglu's Theorem

Let X be a normed space, and B^* the closed unit ball of X^*, $B* = \{f \in X^* \mid \|f\| \leq 1\}$. Then B^* is compact in the weak*-topology.

***Proof*:**
We shall show that B^* is a closed subspace of a compact space. For each $x \in X$, define

$$D_x = \{z \in \mathbb{C} \mid \|z\| \leq \|x\|\}$$

and take $D = \prod_{x \in X} D_x$. Then by Tychnoff's Theorem, D is compact. Now, we may exactly identify the elements of D the complex-valued function $\varphi : X \to \mathbb{C}$ such that $|\varphi(x)| \leq \|x\|$ for all $x \in X$. Then B^* would conssit of the elements of D that are linear. Hence, we must just show that B^* is closed. The induced topology B^* from the product topology on D and the weak* topology on X^* are both the topology of pointwise convergence (for D by the product construction), and hence checking for closedness in X is equivalence to checking closedness of D. Take a net $\langle f_\alpha \rangle$ in B^* that converges to $f \in D$ for any $x, y \in X$ and $a, b \in \mathbb{C}$. Then:

$$f(ax + by) = \lim f_\alpha(ax + by) = \lim[af_\alpha(x) + bf_\alpha(y)] = af(x) + bf(y)$$

showing that f is linear, and certainly $\|f\| \leq 1$, hence $f \in B^*$, making it compact and completing the proof.

7.5.8 Note Since X^* is a vector space, one might hope to conclude that it is locally compact (via the "it suffices to show the result at 0" reduction). This is *not* the case, as Exercise 49(b) of Folland's *Real Analysis* shows.

(The following was copied from nLab, and is very informative!)

Every complete locally convex topological vector space T is the cofiltered projective limit of Banach spaces in the category of locally convex spaces. (Note that Fréchet spaces are additionally required to be metrisable, so this is more general.)

(see this here) Now given that a Fréchet space admits a decreasing sequence of convex balanced and absorbing neighborhoods, it follows immediately that: Every Fréchet space is a sequential projective limit of Banach spaces.

(this is also really cool on homomorphisms of Fréchet space here)

8

L^p space

Look at this blog!! https://terrytao.wordpress.com/2009/01/09/245b-notes-3-lp-spaces/
In this chapter, we will be studying Banach spaces which are defined through the integral: L^p spaces. These spaces are a generalization of L^1 spaces, where if $f \in L^p$, then it must decay "faster" at infinity, and f doesn't blow up too fast at any finite point. In other words, L^p spaces capture a type of "speed of growth". As we have seen in the previous chapters, limits of integrals are central to many constructions. For some constructions, being in L^1 was enough, but for others (like Fourier series) we will require that function be more than finite in integral (i.e. more than $f \in L^1$). L^p spaces will give a nice set of examples of Banach spaces that are relatively simple to define.

My current intuition is that they are treated as the intermediate to being able to bound a function: it is sometimes nontrivial to immediately say a function is bounded, but we can build-up to it by saying it's L^3, which will imply L^5, which will imply L^7, and so on allowing us to conclude it's L^∞, which means a functions is bounded up to a zero-measure set. This is a common phenomena in analysis where we may not be able to bet a function to fit exactly in one space or another (in our case L^1 or L^∞ which are bounded a.e. functions), but we can put the function in intermediary buckets.

8.1 Basics

So far, we have limited our attention to measurable functions f such that $\int |f| < \infty$ and have labelled them L^1. Often, what we are looking are function with more precise "convergent properties", in particular:

$$\int |f|^p < \infty$$

for some $p \geq 1$, including ∞ (which we will define shortly). We start by labelling the colletion of all these functions:

Definition 8.1.1: L^p Space

Let f be measurable. for $p \in (1, \infty)$[a], define the *p-norm* to be:

$$\|f\|_p = \left[\int |f|^p \, d\mu\right]^{1/p}$$

From this norm, define:

$$L^p(X, \mathcal{M}, \mu) := \left\{ f : X \to \mathbb{C} \;\middle|\; f \text{ is measurable and } \|f\|_p < \infty \right\}$$

[a]we will define L^∞ later

As with L^1, we often abbreviate to $L^p(\mu)$, $L^p(X)$, or just L^p. As with L^1, we'll let L^p be the equivalence class of the set of functions such that $|f|^p$ is integrable where $f \sim g$ if and only if $\int |f - g|^p = 0$. Just like L^1, two functions will be equal in L^p (i.e. belong in the same equivalence class) if they are equal almost everywhere. Furthermore, we claimed that $\|\cdot\|_p$ is a norm. That $\|f\|_p = 0$ if and only if $f = 0$ a.e. comes almost immediately, since

$$\left(\int |f|^p\right)^{1/p} = 0 \quad \text{iff} \quad \int |f|^p = 0 \quad \text{iff} \quad \int |f| = 0 \quad \text{iff} \quad f = 0 \text{ a.e.}$$

For $\|cf\|_p = |c| \, \|f\|_p$, we take advantage of our normalisation:

$$\|cf\|_p = \left(\int |cf|^p\right)^{1/p} = \left(\int |c|^p |f|^p\right)^{1/p} = (|c|^p)^{1/p} \left(\int |f|^p\right)^{1/p} = |c| \, \|f\|_p$$

However, the triangle inequality is not so trivial. We will in fact take good part of this section to show that the triangle inequality is satisfied. For now, we take a moment to explore an example of L^p spaces and L^p functions to gain an intuition:

Example 8.1.2: L^p spaces

1. Let $f : [0, \infty) \to \mathbb{R}$ $f(x) = \frac{1}{x^{1/p}}$. Then $f \in L^q$ for $q > p$. This comes down to the fact that:

$$x > x^{1/p} \;\Leftrightarrow\; \frac{1}{x} < \frac{1}{x^{1/p}}$$

   and by the fact that $\int \left|\frac{1}{x^n}\right| < \infty$ if $n > 1$. We have therefore found a class of functions that fits into all possible $p \in [1, \infty)$. In fact, we can find functions that fit into any connected subinterval of $[1, \infty)$! In particular, let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \to \mathbb{C}$ a complex measurable function on X. Let

$$E := \left\{ p \in [1, \infty) | \;\middle|\; \int_X |f|^p < \infty \right\}$$

   (a) Then E is connected, that is, if $p, q \in E$ such that $p < r < q$ for some r, then $r \in E$

   Proof. Clearly, if $E = \emptyset$ or $|E| = 1$, then E is connected because the only non-trivial clopen set is $\emptyset$ and E, so let $|E| > 1$.

Pick $p, q \in E$ such that $p < q$, and consider a point r such that $p < r < q$. We'll show that $\int_X |f|^r < \infty$ using the following trick:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^q = \chi_{|f|\geq 1}|f|^q + \chi_{|f|<1}|f|^q$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^q = \int \chi_{|f|\geq 1}|f|^q + \int \chi_{|f|<1}|f|^q$$

It thus suffice show that $\int \chi_{|f|\geq 1}|f|^r < \infty$ and $\int \chi_{|f|<1}|f|^r < \infty$. This is essentially immediate:

$$\chi_{|f|\geq 1}|f|^r \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^p < \infty$$

Therefore

$$\int |f|^r < \infty$$

showing that $r \in E$, as we sought to show. □

(b) Show that we can pick a measure and an f such that $E = (a, b), [a, b), [a, b]$, or $\{a\}$ for $1 \leq a \leq b \leq \infty$. In other words, all connected subsets of $[1, \infty)$ are possible for E (You may freely use the fact that any Riemann integrable function is also integrable with respect to the Lebesgue measure and its Riemann integral is equal to the Lebesgue integral)

Hint: Think about integrability properties of functions like x^{-a} and $\frac{-1}{\log^2(x)x} = \frac{d}{dx}\frac{1}{\log(x)}$

Proof. Throughout this hole proof we'll be using Riemann integration as a substitute for Lebesgue integration (we showed earlier that this is allowed). We'll be using these function's to construct our intervals:

i. $f(x) = x^{-1/a}$ on $(1, \infty)$: Notice that

$$\int_1^\infty x^{-1/a} dx = \lim_{x\to\infty} \frac{x^{-1/a+1}}{-1/a+1} - \frac{1}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in (a, \infty]$.

ii. $f(x) = x^{-1/a}$ on $(0, 1)$: Notice that

$$\int_0^1 x^{-1/a} dx = \frac{1}{-1/a+1} - \lim_{x\to 0} \frac{x^{-1/a+1}}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in [1, a)$.

iii. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(1, \infty)$: This integral doesn't have an elementary solution, however we can still ascertain some basic convergence properties. In particular, if $p = a$, then:

$$\int_1^\infty |f(x)|^a dx = \lim_{x\to\infty}\left[-\frac{1}{\ln(x)}\right] - \lim_{x\to 1}\left[-\frac{1}{\ln(x)}\right]$$

so $\|f\|_p < \infty$ if and only if $p \in [a, \infty]$

iv. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(0, 1)$: This result is similar to the other 3 cases we've proven. The result for this integral shows that $\|f\|_p < \infty$ if and only if $p \in [1, a]$.

Thus, we have proven all possible "sub-rays" of $[1, \infty]$ can exist. To show all possible sub-intervals (closed, open, or half), we simply add the appropriate function's to produce the result. For example, if we want to have the interval $[a, b)$, simply take:

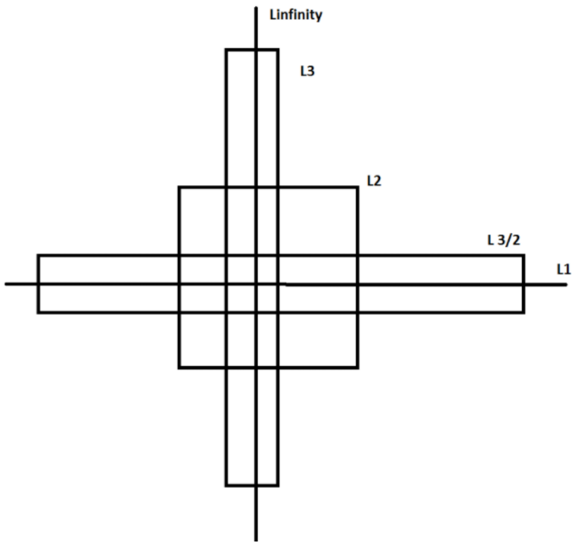
$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(1,\infty)} + x^{-1/b}\chi_{(0,1)}$$

which clearly has interval of convergence of $[a, b)$. Similarly, to get a singleton $\{a\}$, take:

$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(1,\infty)} + \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(0,1)}$$

Thus, we have all possible connected subintervals of $[1, \infty]$. □

So if $f \in L^p$, it does not imply that $f \in L^q$ for $p > q$ or $p < q$! Later on, we will show some of the possible relations between L^p spaces. A more accurate intuition is the following [a]: If $f \in L^p$ for any p, then f must decays fast enough at infinity, and f must not blow up too fast at any finite point. As p increases, the first requirement gets less strict, and the second gets more strict. If X is bounded, the first requirement is vacuous, and if X is discrete, the second is vacuous. Sometimes this visual is used to represent these relations:



2. A particular case that will be common for analyzing sequences is when μ is the counting measure on A. In that case, we usually denote $L^p(\mu)$ by $\ell^p(A)$. If $A = \mathbb{N}$, we often simply

abbreviate notation to ℓ^p. If we write $f : \mathbb{N} \to \mathbb{R},\ f(k) = a_k$, then the sequence f (i.e. $\{a_k\}$ is in ℓ^p if:

$$\sum_{k=1}^{\infty} |a_k|^p < \infty$$

Thus, ℓ^1 is the set of convergence series.

[a] credit to this post on stack-exchange: https://mathoverflow.net/questions/314050/intuition-about-lp-spaces

Notice that we are "normalizing" each integral by a factor of $1/p$. This normalization is done to make sure that the constant can be brought out of the norm. It might seem tempted to say that:

$$\left(\int |f|^p\right)^{1/p} \text{ “=” } \int |f|$$

However, this is most certainly not true (simply take $f(x) = x$). As mentioned before, this makes it non-trivial to show that $\|\cdot\|_p$ is a norm on L^p. However, it is easy to see that L^p is a vector-space. If $f, g \in L^p$, then:

$$|f + g|^p \leq [2\max(|f|, |g|)]^p \leq 2^p(|f|^p + |g|^p)$$

which is finite by assumption, and so $f + g \in L^p$. It is also immediate that $cf \in L^p$. Thus, we may freely add and multiply by L^p functions without worrying about leaving L^p. Because of this, it makes it meaningful to ask whether

$$\|f_1 + f_2\|_p \leq \|f_1\|_p + \|f_2\|_p$$

What we will now do is show that $\|\cdot\|_p$ is does indeed satisfy the triangle inequality for $p \geq 1$. If $p < 1$, then the triangle inequality fails in general:

Example 8.1.3: $p < 1$ then no Triangle Inequality

Take $a, b > 0$ and $0 < p < 1$. Then for any $t \in \mathbb{R}$, we have that:

$$t^{p-1} > (a + t)^{p-1}$$

Integrating both sides from 0 to b, we get:

$$\begin{aligned} \int_0^b t^{p-1} &> \int_0^b (a + t)^{p-1} \\ \frac{b^p}{p} &> \frac{(a + b)^p}{p} \\ b^p &> (a + b)^p \\ a^p + b^p &> (a + b)^p \\ (a^p + b^p)^{1/p} &> a + b \end{aligned}$$

Now, if E and F are disjoint union with positive finite measures in X such that $\mu(E)^{1/p} = a$ and $\mu(E)^{1/p} = b$, then $\chi_E, \chi_F \in L^p$. Taking advantage of the fact that $\chi_E\chi_F = 0$ (i.e. the zero function) since $E \cap F = \emptyset$, then:

$$\|\chi_E + \chi_F\|_p = (a^p + b^p)^{1/p} > a + b = \|\chi_E\|_p + \|\chi_F\|_p$$

Thus, the $\|\cdot\|_p$ for $0 < p < 1$ cannot be a norm.

Thus, we stick to showing the case where $p \geq 1$ First, we will state a simple inequality fact similar Jensen's inequality. This is in fact the key idea behind the entire proof:

Lemma 8.1.4: Young's Inequality

If $a, b \geq 0$ and $0 < \lambda < 1$, then:

$$a^\lambda b^{1-\lambda} \leq \lambda a + (1-\lambda) b$$

with equality if and only if $a = b$. In particular, notice that if $\frac{1}{p} + \frac{1}{q} = 1$, then if $\lambda = p^{-1}$, then $(1-\lambda) = q^{-1}$ and so:

$$a^{1/p} b^{1/q} \leq \frac{1}{p} a + \frac{1}{q} b$$

***Proof*:**
This proof is trying to capture what this image is doing in formal language:

https://www.desmos.com/calculator/jdznjzqcju

If $a = b$, equality clearly holds. If $a = 0$ or $b = 0$, inequality clearly holds. If $a, b \neq 0$, then dividing by b and letting $t = a/b$, we get

$$a^\lambda b^{-\lambda} \leq \lambda(a/b) + (1-\lambda) \quad \Leftrightarrow \quad t^\lambda \leq \lambda t + (1-\lambda) \quad \Leftrightarrow \quad t^\lambda - \lambda t \leq (1-\lambda) \tag{8.1}$$

From here, we use some calculus: we see that $t^\lambda - \lambda t$ is strictly increasing for $0 < t < 1$ (since $a, b \geq 0$, $t \geq 0$) and strictly decreasing for $t > 1$, so the maximal value occurs at $t = 1$, and in fact we get that the maximal value is $1 - \lambda$:

$$1^\lambda - \lambda 1 = 1 - \lambda \leq 1 - \lambda$$

and so the inequalities in equation (8.1) hold! Equality happens when $t = a/b = 1$ which implies $a = b$, completing the proof.

Usually, Young's inequality is stated as

$$a^\alpha b^\beta \leq \alpha a + \beta b \qquad 0 \leq \alpha, \beta \leq 1, \quad \alpha + \beta = 1$$

which is exactly what we have, since if we fix some α, then it must be that $\beta = 1 - \alpha$. Another formulation of Young's inequality is:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with $p^{-1} + q^{-1} = 1$. This proof is a "Jensen equality" like proof that more clearly shows the role of concavity and so I'll dedicate a moment to prove this one too:

***Proof*:**
If $a = 0$ or $b = 0$, the result is immediate, so assume $a, b > 0$. Then notice that since log is monotone:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad \Leftrightarrow \quad \log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$$

which is great, since we know log is concave and so Jensen's inequality applies. Since $p^{-1}+q^{-1}=1$, we get $t=1/p$ and $(1-t)=1/q$, and so by Jensen's inequality:

$$\log(ta^p+(1-t)b^q)\geq t\log(a^p)+(1-t)\log(b^q)=\frac{p}{p}\log(a)+\frac{q}{q}\log(b)=\log(ab)$$

completing the proof.

The second interpretation of $p^{-1}+q^{-1}=1$ in Young's inequality is a nice way to write the straight line (or sub-straight line) equation without the $1-\lambda$ term, but just two numbers. It furthermore let's us relate two numbers that can be much larger. For example, if I choose $p=100$, then it must be that $q=100/99$ so that

$$p^{-1}+q^{-1}=1/100+99/100=1$$

which, if we consider p and q to be the exponents associated to L^p and L^q, we see that this gives us a way to relate different L^p space!

To see this more succinctly, we use this convexity property given by Young's inequality in the "build-up" lemma towards the triangle inequality. This lemma is important later on in functional analysis since it let's us bound the operation of multiplication of tow functions fg, (namely, $\|fg\|_1$ will be less than a value on the right hand side that seperates f and g). For our current purposess, it is important for the triangle inequality:

Theorem 8.1.5: Hölder's Inequality

Let $1\leq p,q\leq\infty$ satisfy:

$$\frac{1}{p}+\frac{1}{q}=1$$

or equivalently $q=p/(p-1)$, where we will interpret $\frac{1}{\infty}$ as 0. Then if f and g are measurable functions on X then:

$$\|fg\|_1\leq\|f\|_p\,\|g\|_q$$

In particular, if $f\in L^p$ and $g\in L^q$ then $fg\in L^1$. Equality holds if and only if $\alpha|f|^p=\beta|g|^q$ a.e. for some constants α,β with $\alpha\beta\neq 0$

Notice that when $p=q=2$ (if $p=2$, it must be that $q=2$ to satisfy the equation), then this inequality is called the *Cauchy-Schwarz inequality*[1]. Notice too that this is the only time when $p=q$; this will become important when we analyze the $p=2$ case later.

I also found this intuition on a stack-exchange: the slower f decays, the faster g needs to decay, and the faster f blows up, the more we need g to not blow up. The previous sentence is equally true if we switch f and g, which convinces us that L^p spaces should be reflexive, and by considering $f=g$, we see that the restrictions on f and g should balance at exactly $p=2$, so L^2 should be self-dual.

Notice too that since we are taking the absolute value of the function on the inside, the integral can be thought of as a sort of "concave function", where the input would be, say in the special case of $\mathbb{R}$ as the domain, some interval. This makes it easier to see why this inequality holds (notice that convex functions must preserve monotonicity, and so does the integral).

[1]Technically we need to define an inner product on L^2 to satisfy this inequality. However, as well see in chapter 11, L^2 is indeed a Hilbert space

***Proof*:**
If $\|f\|_p = 0$ or $\|g\|_p = 0$, then $f = 0$ or $g = 0$ a.e., and so equality holds. Similarly if $\|f\|_p = \infty$ or $\|g\|_p = \infty$, inequality immediately holds. Furthermore, notice that if we find a particular f and g for which the result holds, then the result holds for af and bg since

$$\begin{aligned} &\Leftrightarrow \ \|afbg\|_1 \leq |ab| \, \|af\|_p \, \|bg\|_q \\ &\Leftrightarrow \ |ab| \, \|fg\|_1 \leq |ab| \, \|f\|_p \, \|g\|_q \\ &\Leftrightarrow \ \|fg\|_1 \leq \|f\|_p \, \|g\|_q \end{aligned}$$

So, it suffices to show the result when $\|f\|_p = \|g\|_q = 1$, and equality holds of $|f|^p = |g|^q = 1$. To this end, we use the previous lemma and set $a = |f(x)|^p$ and $b = |g(x)|^q$ and $\lambda = p^{-1}$ (so that $1 - \lambda = 1 - p^{-1} = q^{-1}$). Then by the previous lemma we have:

$$|f(x)|^{p \cdot p^{-1}} |g(x)|^{q \cdot q^{-1}} \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \quad \Leftrightarrow \quad |f(x)g(x)| \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \tag{8.2}$$

Now, integrating both sides, we get:

$$\|fg\|_1 \leq p^{-1} \int |f|^p + q^{-1} \int |f|^q = p^{-1} + q^{-1} = 1 = \|f\|_p \, \|g\|_q$$

where equality holds when equality holds in equation (8.2) a.e., which holds only when $|f|^p = |g|^q$ a.e., as we sought to show.

For a visual conformation, you can check out:

https://www.desmos.com/calculator/gwqvfv4tie

When p and q satisfy the equality in the equation, we call them *conjugate exponents*. As mentioned before, the conjugate exponents is simply another way or writing the equation of a straight line to apply Young's inequality. From this, we can take advantage of this convexity property yet again to prove the triangle inequality of $\|\cdot\|_p$, which has a different name since for the special case of the p-norm due to the fact it is used on other places and needs to be referenced:

Theorem 8.1.6: Minkowski's Inequality

If $1 \leq p < \infty$ and $f, g \in L^p$. Then:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Before proceeding with the proof, a quick word on notation to avoid confusion:

$$\|f\|_p^p = \left(\int |f|^p \right)^{p/p} = \int |f|^p$$

We often work with $\|f\|_p^p = \int |f|^p$ to get rid of the $1/p$ exponent to be able to work with the linearity of the integral, and re-introduce it at the end to get the desried result. We did not have to do this for Hölder's inequality since we reduced to a very nice special case.

***Proof*:**
If $p = 1$, then the result immediately follows by the regular triangle inequality and linearity of the integral. If $f + g = 0$ a.e., then the result also follows, so assume that $f + g \neq 0$ a.e., which using the triangle inequality:

$$|f + g|^p \leq (|f| + |g|)|f + g|^{p-1} = |f||f + g|^{p-1} + |g||f + g|^{p-1}$$

So, we will apply Hölder's inequality using the fact that $q = \frac{p}{p-1}$ and integrating:

$$\begin{aligned}
\|f+g\|_p^p &= \int |f+g|^p d\mu \\
&= \int |f+g| \cdot |f+g|^{p-1} d\mu \\
&\leq \int (|f| + |g|)|f+g|^{p-1} d\mu \\
&= \int |f||f+g|^{p-1} d\mu + \int |g||f+g|^{p-1} d\mu \\
&= \left\| |f|(|f+g|^{p-1}) \right\|_1 + \left\| |g|(|f+g|^{p-1}) \right\|_1 \\
&\leq \||f|\|_p \|f+g\|_q^{p-1} + \|g\|_p \|f+g\|_q^{p-1} && \text{Hölder's Inequality} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^{(p-1)\left(\frac{p}{p-1}\right)} d\mu \right)^{\frac{p-1}{p}} && \text{recall } q = \frac{p}{p-1} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^p d\mu \right)^{\frac{p-1}{p}} \\
&= (\|f\|_p + \|g\|_p) \frac{\|f+g\|_p^p}{\|f+g\|_p}
\end{aligned}$$

Thus, multiplying both sides by $\frac{\|f+g\|_p}{\|f+g\|_p^p}$, we get

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

As we sought to show

Thus, we get that $\|\cdot\|_p$ is indeed a norm! Thus L^p is a normed vector space. In fact, just like L^1, it is a complete normed vector space, i.e., a Banach space. We first recall a lemma to simply our task. To understand the lemma, notice that every norm defines a metric:

$$\rho(x, y) = \|x - y\|$$

and so defines a topology, in particular a normal space, that has the notion of sequences well-defined. Next, recall that if $\{x_n\}$ is a sequence in V, it's series is said to converge if there exists an $x \in V$ such that $\sum_{n=1}^{\infty} x_n = x$, and is said to be absolutely convergence if $\sum_{n=1}^{\infty} \|x_n\| < \infty$ (this is lemma 7.1.6). If you did not cover this result yet, it is important to go see it now to understand the following result:

Theorem 8.1.7: L^p is a Banach Space

For $1 \leq p < \infty$, L^p is a complete normed vector-space, i.e., a Banach space. Moreover, if $f_n \to f$ in L^p, then there exists a subsequence that converges pointwise to f a.e.

***Proof*:**
Let $1 \leq p < \infty$. By the previous lemma, it suffices to show that every absolutely convergent series converges:

$$\sum_{k=1}^{\infty} \|f\|_p = B < \infty \quad \Rightarrow \quad \exists f \text{ s.t. } \sum_{k=1}^{\infty} f_k = f \in L^p$$

Let $G_n = \sum_{k=1}^{n} |f_k|$ be the partial sums of the series and $G = \sum_{k=1}^{\infty} |f_k|$ be the series. Then by Minkowski's inequality, we get:

$$\|G_n\|_p \leq \sum_{k=1}^{n} \|f_k\|_p \leq B \quad \forall n \in \mathbb{N}$$

and so $\|G\|_p = \left(\int |G|^p\right)^{1/p} < \infty$ and so $\left(\int |G|^p\right) < \infty$. Since $G_n \leq G_{n+1}$, and $G_n : X \to [0, \infty)$, by the Monotonce Convergence Theorem:

$$\int \lim_{n\to\infty} G_n^p = \lim_{n\to\infty} \int G_n^p \leq B^p < \infty$$

and so $G \in L^p$. By proposition 2.3.4, $G(x) < \infty$ a.e., so the series $\sum_{k=1}^{\infty} f_k$ converges a.e. If we let $F = \sum_{k=1}^{\infty} f_k$, we have that $|F| \leq G$ a.e., and so $F \in L^p$. From this, we can see that:

$$\left\|F - \sum_{k=1}^{\infty} f_k\right\|_p^p = \int \left|F - \sum_{k=1}^{\infty} f_k\right|^p \to 0$$

Thus, the series $\sum_{k=1}^{\infty} f$ converges (to F) in the p-norm, and so by lemma 7.1.6, L^p is complete, as we sought to show.

This is the proof the prof was doing in class, but I got stuck so found a different proof (the above proof) (Commented it out for now)

This also finally showed that L^1 is a complete space under $\|\cdot\|_1$! Note the importance of taking a subsequence: since $f_n \to f$ in the p-norm, i.e. the "sum-averaging" of the function converges, then there can be lots of wiggle room for hwo the functions f_n approach f:

Example 8.1.8: Taking A Subsequence
Construct a sequence of functions such that all the following hold:

1. $f_n : [0, 1] \to \mathbb{R}$ is continuous
2. $0 \leq f_n(x) \leq 1$ for every x
3. $\lim_{n\to\infty} \int_0^1 f_n dm = 0$
4. For every $x \in [0, 1]$ we have that $\lim_{n\to\infty} f_n(x)$ does not exist

(that is, it is definitely important to take a subsequence in order to deduce pointwise convergence from converge in L^1)

Proof. Essentially, we'll be taking advantage that $\sum_n \frac{1}{n}$ does not converge. Let $f_1(x) = 1$, that is, that constant function with constant 1.

For g_2, we split the interval into 2 equal parts. Define

$$g_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ 0 & x \in (1/2, 1] \end{cases}$$

This function is not continuous. We will make it continuous by simply climbing to it (and for other values, we'll be descending back down too). At the point $1/2$, slope down at a speed of 2^2 until we reach 0:

$$f_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ -2^2 \left(x - \left(1 - \frac{1}{4}\right)\right) & x \in [1/2, 3/4] \\ 0 & x \in (3/4, 1] \end{cases}$$

From here, we will define our function in a sort of "recursive" sense. Starting form the inteval $[0, 1/2)$, we keep the right end point, and add an interval of length $1/3$ to it: $[1/2, 5/6)$. Now, define f_3 to be 1 on $[1/2, 5/6]$, and zero everywhere else. Next, to define f_4, repeat the process by adding $1/4$ to $5/6$, giving us the interveal $[5/6, 13/12]$. Note that $13/12$ goes beyond 1. We thus wrap it around back to 0: $[0, 1/12] \cup [5/6, 1]$. Now, define a function f_3 to be 1 on this set, and zero everywhere else.

In each of these functions, we extend them to be continuous by attaching them a slope which has 2^n as a factor. In this way, the slope shrinks in size quicker than the size of the rectangles, and so we can morally imagine the problem to be one of indicator functions.

We can continue doing this, defining every f_k by appending $1/k$ and wrapping around when we go past 1 and then making the function continuous. Clearly, $\int f_k \to 0$ as $k \to \infty$ since each inteval will be of total length $1/k + 1/2^k$. However, f_k does not converge anywhere: for any $x \in [0, 1]$, we can choose a subsequence from $f_k(x)$ (say $f_{k_l}(x)$) such that $f_{k_l}(x) = 1$. or $f_{k_l}(x) = 0$

To construct such a sequence, it is easier to imagine our construction without the wrapping around, that is, we keep adding intervals of length $1/k$ as though our co-domain was $[0, \infty)$. Then, our first value of l will be the value for which $x + 1$ is in some interval of length $1/k$ for appropriate k. This interval will exist, since $\sum_k 1/k$ diverges. Similarly, our next value of l for which $x + 2$ will be in some interval of length $1/k$ for appropraite k. We can continue doing this indefinitely, showing that there exists a subsequence of f_k that doesn't converge to 0, meaning f_k cannot converge to 0. On the other hand, it is easy to see that we can choose a subsequence that converges to 0 (since f_k is zero after the trapezoid passes by our point).

Since x was arbitrary, this will work any $x \in [0, 1]$. Thus, f does not converge anywhere, as we sought to show. □

Retruning to analyzing L^p spaces, we can now extend theorem 2.3.9 to show that simple function are dense in every L^p!

Theorem 8.1.9: Simple Functions Dense in L^p

For $1 \leq p < \infty$, the set of simple functions $f = \sum_{i=1}^{n} a_i \chi_{E_i}$ where $\mu(E_i) < \infty$ for all i is dense in L^p

***Proof*:**
First of all, the set of simple functions is in L^p for every $1 \leq p < \infty$. Next, choosing any $f \in L^p$, choose some standard representation $\{f_n\}$ where $f_n \to f$ a.e[a]. Then since $|f_n| \leq |f|$, we get that $|f_n| \in L^p$. Furthermore

$$|f - f_n|^p \leq |2f|^p = 2^p |f|^p \in L^1$$

and so by the dominated convergence theorem, we get:

$$\int |f - f_n|^p \to 0 \quad \Rightarrow \quad \left(\int |f - f_n|^p \right)^{1/p} = \|f - f_n\|_p \to 0$$

Moreover, if $f_n = \sum_{k=1}^{\infty} a_k \chi_{E_k}$ was a simple function where all E_i are pairwise disjoint and a_i are nonzero, it must be that $\mu(E_i) < \infty$ since

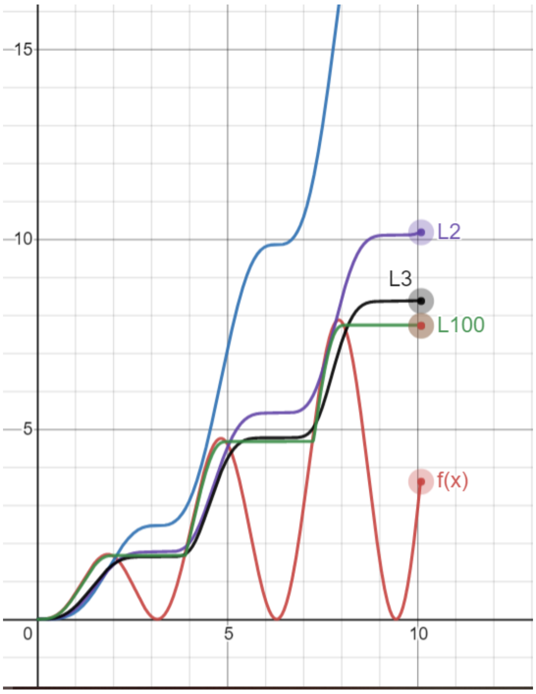
$$\sum_{k=1}^{\infty} |a_k|^p \mu(E_k) = \int |f_n|^p < \infty$$

completing the proof

[a] Recall that the any measurable function has a standard representation, not just integrable functions

Some further remarks about which sets are dense can be enlightening. For $1 \leq p, q < \infty$, L^p is dense in L^q, since each simple set of the form we descriped are subsets of L^p for all $p \geq 1$. If we modify the characteristic function to make them continuous (simply draw a connecting line as we've done in example 8.1.8), then we see that $C_0(\mathbb{R})$ (continuous functions that vanish at infinity) are dense in L^p for every $1 \leq p < \infty$. If we restrict to closed intervals, then $C([a, b])$ is dense in $L^p([a, b])$ since the standard representation becomes uniformaly convergent on bounded domains (see theorem 2.1.18).

So far, we have defined L^p spaces where $0 < p < \infty$ and showed that for $1 \leq p < \infty$, L^p is a Banach space. Interestingly, if we take a look at a graph of $\|f\|_p$ as $p \to \infty$, we'll find that there is a striking pattern that appears:



Notice that as $p \to \infty$, the resulting $\|f\|_p$ value got closer to a max function. In fact, this is almost the case! We might be tempted to then say that $\|f\|_\infty = \sup f$. However, the problem with this is that we might have a measure 0 set that diverges, while the supremum of the rest of the function is attained by a value $k < \infty$. The idea of $\sup f$ on all but a zero measure set will be called the *essential supremum*, and all functions that have a finite essential supremum will be denoted as L^∞. In this way, L^∞ is simply a natural extension of our definition of L^p:

To see the intuition behind this, let us simply consider the p-norm of $\|\vec{v}\|_p = (|3|^p + |10|^p)^{1/p}$

1. Case $p = 1$: $(3^1 + 10^1)^{1/1} = 13$. (The 3 contributes about 23% to the total)
2. Case $p = 2$ (Euclidean): $\sqrt{3^2 + 10^2} = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$. (The 3 matters less now. The 10 is dominating)
3. Case $p = 10$: $(3^{10} + 10^{10})^{1/10} = (59,049 + 10,000,000,000)^{1/10}$. Notice that 3^{10} is negligible dust compared to 10^{10}. $\approx (10^{10})^{1/10} = 10.00000$*something*
4. Case $p = 100$: The 3^{100} is mathematically invisible next to 10^{100}. The sum is basically just 10^{100}. When you take the 1/100 root, you are just returning the base number: 10.

Definition 8.1.10: L^∞ Space

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \to [0, \infty]$ be a measurable function. Let

$$S = \left\{ r \in \mathbb{R} \;\middle|\; \mu(f^{-1}((r, \infty])) = 0 \right\}$$

Then the *essential supremum* of f, denoted $\|f\|_\infty$ or $\operatorname{ess\,sup} f$ is :

$$\|f\|_\infty := \begin{cases} \infty & \text{if } S = \emptyset \\ \inf S & \text{if } S \neq \emptyset \end{cases}$$

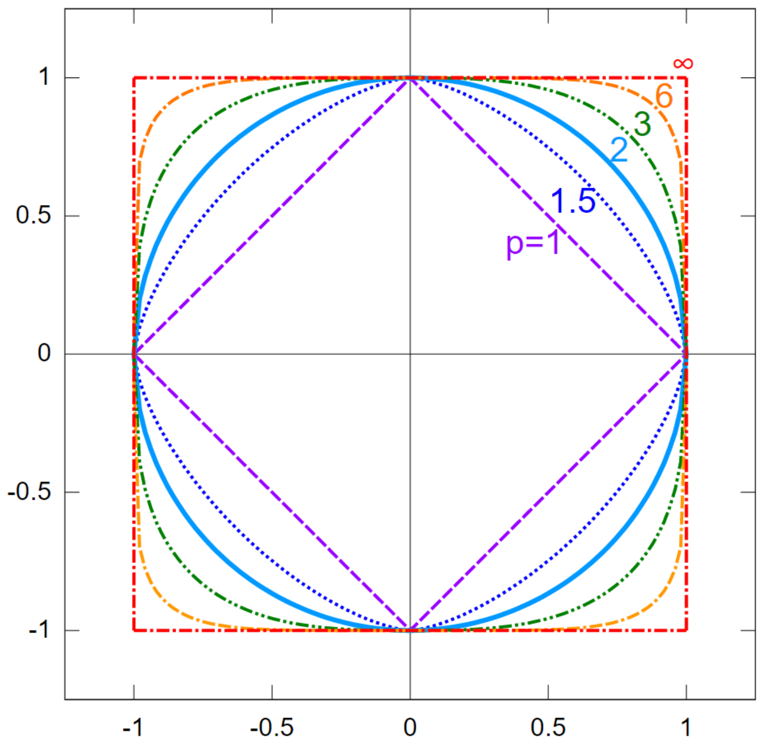
or, equivalently:

$$\|f\|_\infty = \inf_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} \left(\sup_{x \in E} |f(x)| \right)$$

If f is a complex measurable, function, then $\|f\|_\infty = \|\,|f|\,\|_\infty$

Essentially, $\|f\|_\infty$ is the least upper bound of f, ignoring measure zero sets worth of points. Using this definition, we can define L^∞ to be the equivalence classes of functions f such that $\|f\|_\infty < \infty$.

Another quick way of visualizing the relation between all p-norms, $1 \leq p \leq \infty$ is by taking the ℓ^p norm on $\mathbb{R}^2$, taking the function $x \mapsto \|x\|_p = \sqrt[p]{x^p + y^p}$, and considering when $\|x\| = 1$, or in terms of set notation $S_p = \left\{ x \in \mathbb{R}^2 \;\middle|\; \|x\|_p = 1 \right\}$. We'd get:



A couple more quick remarks: $L^\infty(\mu)$ depends on μ only insofar as μ determines which sets have measure zero. If μ and ν are mutually absolutely continuous ($\mu \ll \nu$ and $\nu \ll \mu$[2]) then $L^\infty(\mu) = L^\infty(\nu)$. Secondly, if μ is not semi-finite, it is useful to adpot a slightly useful definition of L^∞ (Folland left this as exercsie 23-25 in chatper 6).

Most properties of L^p for $1 \leq p < \infty$ generalises naturally to the L^∞ case:

Proposition 8.1.11: Properties of L^∞

Let $(X, \mathcal{M}, \mu)$ be a measure and L^∞ the set of measurable function satisfying $\|\cdot\|_\infty < \infty$. Then:

1. If f, g are measurable functions on X, then

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

   If $f \in L^1$ and $g \in L^\infty$, then $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ if and only if $|g(x)| = \|g\|_\infty$ a.e. on the set where $f(x) \neq 0$

2. $\|\cdot\|_\infty$ is a norm on L^∞
3. $\|f_n - f\|_\infty \to 0$ if and only if there exists $E \in \mathcal{M}$ such that $\mu(E^c) = 0$ and $f_n \to f$ uniformally on E
4. L^∞ is a Banach space
5. The simple functions are dense in L^∞

As a consequence of the well-definedness of L^∞, $1^{-1} + \infty^{-1} = 1$ is indeed a well-defined notion (so $\infty^{-1} = 1/\infty = 0$). Thus, 1 and ∞ are conjugate exponents!

***Proof*:**

We will prove (1), from which the rest of the results follow.

1. We'll be using the $\|f\|_\infty = \inf\limits_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} (\sup_{x \in E} |f(x)|)$ definition of the essential supremum to take advantage of the supremum within the definition. Our goal is to show that

$$\int |fg| = \|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

   First, we will limit our attention of $\int |fg|$ to some $E \in \mathcal{M}$ such that $\mu(X \setminus E) = \mu(E^c) = 0$ (i.e. ignoring a zero-measure set). Then:

$$\int |fg| = \int_E |fg| + \int_{E^c} |fg| = \int_E |fg|$$

   Next, we take advantage the fact that we are working with an infimum to see that:

$$\|f\|_\infty \leq \sup_{x \in A} |f(x)| < \|f\|_\infty + \epsilon$$

[2]see definition 3.2.1

Therefore, we can take:

$$\int_E |fg| \leq \int_E \left| f \cdot \sup_{x \in E} |g(x)| \right| = \int_E |f| \cdot \left| \sup_{x \in E} |g(x)| \right| = \left| \sup_{x \in E} |g(x)| \right| \int_E |f|$$

where the last equality comes from the fact that the supremum is a constant. Finally, by the property of the infimum, we have:

$$\left| \sup_{x \in E} |g(x)| \right| \int_E |f| \leq (\|g\|_\infty + \epsilon) \int |f|$$

Notice that this inequality does not depend on ϵ, and so:

$$\|fg\|_1 = \int_E |fg| \leq \|f\|_1 \|g\|_\infty$$

as we sought to show

Notice how similar $\|\cdot\|_\infty$ is to the uniform metric $\|f\|_u = \sup_{x \in X} f(x)$. If we are dealing with Borel measures that assign a positive value to all open sets, then if f is continuous we in fact have $\|f\|_\infty = \|f\|_u$ (since $\{x \mid |f(x)| > a\}$ is open). In this way, we can consider the space of bounded continuous functions as a (closed!) subspace of L^∞.

We conclude with some final remarks about the density of simple functions in L^∞. Notice that for L^p where $p < \infty$, simple functions where $\mu(E_i) < \infty$ where dense. This condition is no longer present for density in L^∞; we in fact require simple functions where $\mu(E) = \infty$ now since the function $1(x) = 1$ is in $L^\infty(\mathbb{R})$. This also means that $C_0(\mathbb{R})$ cannot be dense in L^∞. In the case of $L^\infty([a, b])$ $C([a, b])$, not all funcions can be converged to uniformally dense: consider $f : [0, 1] \to \mathbb{R}$, $f(x) = 0$ for $x \in [0, 1/2]$ and $f(x) = 1$ for $x \in [1/2, 1]$. The no continuous function uniformally converges to f in the ∞-norm.

8.1.1 Relating L^p spaces: Interpolation

In this section, we will show how we can relate L^p spaces with each other. Fundamentally, the way we get any order with L^p spaces is through Hölder's inequality. We take full advantage of it in the following proofs. The end-goal is to show that if $p < q < r$ where we allow $r = \infty$, then:

$$L^p \cap L^r \subseteq L^q \subseteq L^p + L^r$$

This should reflect on the example that we have demonstrated in example 8.1.2. In particular, the larger p is, the less f has decay at infinity, but the more f must not explode at any point. In other words p gets bigger, we can let p decay slower, but it must behave better at points where it can explod3. If our space X is bounded, then we don't have to care about decay (this will give us that $L^p \supseteq L^q$ where $p < q$). If X is discrete, we don't have to care about exploding at any point (this will give us that $\ell^p \subseteq \ell^q$ if $p < q$). We thus explore these relation's in this section. These types of relations bewteen space via addition of functions or intersection of spaces is called *interpolation*.

Proposition 8.1.12: L^p **subset**

Let $0 < p < q < r \leq \infty$. Then

$$L^q \subseteq L^p + L^r$$

In particular, each $f \in L^q$ is the sum of a function in L^p and a function in L^q

***Proof*:**
This proof is really similar to that of example 8.1.2[1]. Let $f \in L^q$. Then consider:

$$|f| = \chi_{|f| \geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f| \geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^r = \chi_{|f| \geq 1}|f|^r + \chi_{|f|<1}|f|^r$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f| \geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^r = \int \chi_{|f| \geq 1}|f|^r + \int \chi_{|f|<1}|f|^r$$

Then we get:

$$\chi_{|f| \geq 1}|f|^p \leq \chi_{|f| \geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^q < \infty$$

Therefore, $\chi_{|f| \geq 1}f \in L^p$ and $\chi_{|f|<1}f \in L^r$. If $r = \infty$, then clearly $\|\chi_{|f| \leq 1}f\|_\infty \leq 1$, showing the result still holds, as we sought to show.

Proposition 8.1.13: L^p **superset**

Let $0 < p < q < r \leq \infty$. Then

$$L^p \cap L^r \subseteq L^q$$

In particular, if if we find that f is bounded in growth from above and bellow by p and r, $\|f\|_p < \infty$ and $\|f\|_r < \infty$, then f is bounded in growth by q.

***Proof*:**
Let $\|f\|_r < \infty$ and $\|f\|_q < \infty$. We need to show that $\|f\|_p < \infty$. Notice that it is equivalent to show that $\|f\|_r^n < \infty$ and $\|f\|_q^m < \infty$ then $\|f\|_p^\ell < \infty$ for $n, m, \ell \in \mathbb{R}_{\geq 0}$ (since if it's finite, then all powers will be finite). We'll prove it in the case where $r = \infty$ and $r < \infty$. In the infinite case, we simply take advantage of bounding properties of the essential supremum, and in the other case we use Hölder's inequality.

Starting with the $r = \infty$ case, notice that

$$|f|^q = |f|^{q-p+p} = |f|^q|f|^{q-p} \leq \|f\|_\infty^{q-p}|f|^p$$

and so, taking the integral and the qth root, we get:

$$\begin{aligned}
\|f\|_q &= \left(\int |f|^q\right)^{1/q} \\
&\leq \left(\int \|f\|_\infty^{q-p}|f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p\right)^{p/pq} \\
&= \|f\|_\infty^{1-\frac{p}{q}} \|f\|_p^{p/q} < \infty
\end{aligned}$$

The fact that the last inequality is less than infinity comes from the that $\|f\|_p < \infty$ and $\|f\|_\infty < \infty$, so any power is less than infinity, and so $\|f\|_q < \infty$. Furthermore, notice that the sum of the power's is 1; this fact will come back for another intuition of $\|f\|_\infty$.

For the $r < \infty$ case the key is to find a way to integrate p and r using conjugate exponents. Since

$$p < q < r$$

then

$$\frac{1}{p} > \frac{1}{q} > \frac{1}{r}$$

therefore, define the continuous function $\lambda \mapsto \frac{1-\lambda}{p} + \frac{\lambda}{r}$. Since $(0,1)$ is open, and $1/p > 1/r$, there exists a λ_0 such that

$$\frac{1-\lambda_0}{p} + \frac{\lambda_0}{r} = \frac{1}{q}$$

manipulating the equation around to get conjugate exponents, notice that:

$$\frac{(1-\lambda_0)q}{p} + \frac{\lambda_0 q}{r} = 1 \quad \Leftrightarrow \quad \frac{1}{\frac{p}{(1-\lambda_0)q}} + \frac{1}{\frac{r}{\lambda_0 q}} = 1$$

Thus, we have that $\frac{p}{(1-\lambda_0)q}$ and $\frac{r}{\lambda_0 p}$. With conjugate exponents established, we proceed with trying to bound $\|f\|_q$. As we commented, it is equivalent to bound $\|f\|_q^m$, so let $m = q$ (this will be a common trick to apply Hölder's inequality). Also, since $1 - \lambda_0 + \lambda_0 = 1$, we have that

$q = \lambda_0 q + (1-\lambda_0)q$, giving us the ability to split up our integral:

$$\begin{aligned}
\|f\|_q^q &= \int |f|^q \\
&= \int |f|^{\lambda_0 q + (1-\lambda_0)q} \\
&= \int |f|^{\lambda_0 q} |f|^{(1-\lambda_0)q} \\
&= \left\| |f|^{\lambda_0 q} |f|^{(1-\lambda_0)q} \right\|_1 \\
&\leq \left\| |f|^{\lambda_0 q} \right\|_{\frac{p}{(1-\lambda_0)q}} \left\| |f|^{(1-\lambda_0)q} \right\|_{\frac{r}{\lambda_0 q}} && \text{Hölder's inequality} \\
&= \left(\int |f|^p \right)^{\frac{(1-\lambda_0)q}{p}} \left(\int |f|^r \right)^{\frac{\lambda_0 q}{r}} \\
&= \|f\|_p^{(1-\lambda_0)q} \|f\|_r^{\lambda_0 q} < \infty
\end{aligned}$$

Since the last two terms are finite, any power of them are also finite, and so we have bounded $\|f\|_q^q < \infty$, and so $\|f\|_q < \infty$, as we sought to show.

Proposition 8.1.14: ℓ^p Covariant Inclusion

If $0 < p < q \leq \infty$, then $\ell^p(A) \subseteq \ell^q(A)$

***Proof*:**
They key part is to take advantage of the properties of the counting measure. Since the only set of measure 0 in the counting measure is the empty-set, we have that the essential supremum is actually the supremum!

Let's say $f \in \ell^p$, so that $\sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$. We'll start by showing that $\|f\|_\infty < \infty$. Since the $\|f\|_\infty$ is in fact the supremum, we have that:

$$\|f\|_\infty^p = \sup_{a\in A} |f(a)|^p \leq \sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$$

Therefore, $\|f\|_\infty < \infty$.

For the $q < \infty$ case, using proposition 8.1.13 with $\lambda = p/q$, we get:

$$\|f\|_q \leq \|f\|_p^\lambda \|f\|_\infty^{1-\lambda} \leq \|f\|_p$$

therefore, $\|f\|_q < \infty$, completing the proof.

Proposition 8.1.15: L^p Contravariant Inclusion in finite measure

Let $(X, \mathcal{M}, \mu)$ be a measure space where $\mu(X) < \infty$. If $0 < p < q \leq \infty$, then $L^p \supseteq L^q$

***Proof*:**
We need to show that if $f \in L^q$ then $f \in L^p$. Since $f \in L^q$, $\|f\|_q < \infty$, so $\|f\|_q^n < \infty$ for all $n \in \mathbb{N}$. If we can show that $\|f\|_p^m \leq \|f\|_q^n$ for some $m, n \in \mathbb{N}$, then the proof is complete.

We'll apply Hölder's inequality in the caes of $q = \infty$ and $q < \infty$. If $q = \infty$, then:

$$\|f\|_p^p = \int |f|^p = \int |f|^p \cdot 1 \overset{\text{H.}}{\leq} \|f\|_\infty^p \int 1 = \|f\|_\infty^p \mu(X) < \infty$$

since $\int 1 = \mu(X)$. If $q < \infty$, then we do a similar argument. First, since $p < q$, notice that:

$$\frac{1}{\frac{q}{p}} + \frac{1}{\frac{q}{q-p}} = \frac{p}{q} + \frac{q-p}{q} = \frac{q}{q} = 1$$

and so $\frac{q}{p}$ and $\frac{q}{p-q}$ are conjugate exponents. Therefore, by Hölder's inequality:

$$\begin{aligned}
\|f\|_p^p &= \int |f|^p \\
&= \int |f|^p \cdot 1 \\
&= \||f|^p \cdot 1\|_1 \\
&\leq \||f|^p\|_{\frac{q}{p}} \|1\|_{\frac{q}{p-q}} && \text{Hölder's Inequality} \\
&= \left(\int |f|^{p\frac{q}{p}}\right)^{\frac{p}{q}} \left(\int 1\right)^{\frac{q-p}{p}} \\
&= \|f\|_q^p \, \mu(X)^{\frac{q-p}{p}} < \infty
\end{aligned}$$

therefore, $\|f\|_p^p < \infty$, so $\|f\|_p < \infty$, showing that $f \in L^p$, as we sought to show.

Proposition 8.1.16: Relating L^p and L^∞

Let $p \in [1, \infty)$ and assume $\|f\|_p < \infty$. Show that:

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty$$

***Proof*:**
For ease of typing, we simplify notation: $\|f\|_p := \|f\|_{L^p(\mu)}$

We'll show that $\lim_{p\to\infty} \|f\|_p \leq \|f\|_\infty$ and $\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$. For the $\leq$ direction, note that $|f| \leq \|f\|_\infty$ a.e., and since x^n is a monotone function on $[0, \infty)$, we have $|f|^n \leq \|f\|_\infty^n$. Then for some fixed λ_0:

$$\begin{aligned}\|f\|_p &= \left(\int |f|^p\right)^{1/p} \\ &= \left(\int |f|^{p+\lambda_0-\lambda_0}\right)^{1/p} \\ &= \left(\int |f|^{p-\lambda_0}|f|^{\lambda_0}\right)^{1/p} \\ &\leq \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}}\end{aligned}$$

where the last inequality is the same trick as part (a) of this question. Since λ_0 is fixed and this inequality holds true for all p, as $p \to \infty$:

$$\lim_{p\to\infty} \|f\|_p \leq \lim_{p\to\infty} \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}} = \|f\|_\infty$$

For the $\geq$ direction, we have to bound $\|f\|_p$ from bellow by something that includes $\|f\|_\infty$, but ultimately the inequality will be independent of the choices we make. In particular, let $\epsilon > 0$. Take all points which are epsilon close to the essential supremum (which exists by the infimum property). Let M_ϵ be our set, in particular $M_\epsilon = \{x \mid |f(x)| \geq \|f\|_\infty - \epsilon\}$. Since $\|f\|_\infty = k$, then $\mu(M_\epsilon) > 0$. Thus:

$$\|f\|_p = \left(\int |f|^p\right)^{1/p} \geq \left(\int_{M_\epsilon} (\|f\|_\infty - \epsilon)^p\right)^{1/p} \stackrel{!}{=} (\|f\|_\infty - \epsilon)\mu(M_\epsilon)^{1/p}$$

where $\stackrel{!}{=}$ comes from the fact that we are integrating over a constant region. Since $\mu(M_\epsilon) > 0$, we have that $\mu(M_\epsilon)^{1/p} > 0$ for all p, and $\mu(M_\epsilon)^{1/p} \xrightarrow{p\to\infty} 1$. Furthermore, notice our inequalities are independent of choice of ϵ, and so since we can let ϵ be as small as we want:

$$\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$$

and since we've shown the $\leq$ direction, we have:

$$\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$$

as we sought to show

Now that we have examined L^p spaces in more detail, we finish off this seciton with some comments on the usefulness these spaces:

1. L^1 is natural for integration; we will essentially work with variation's of L^1 when we integrate. From an algebraic perspective, it can be thought of as a generalizatgion of the notion of "counting" all points in a space, and working in spaces where the count is finite.

2. L^∞ is useful as a broadening of the space of uniform-bound functions. It is a generlization of the idea of working with functions that are bounded.

3. L^p interpolates in between L^1 and L^∞. Many times, L^1 or L^∞ spaces have pathological behaviors (examples of which I yet do not know). L^p spaces behave more nicely, and can sometimes be used to go from L^1 to L^∞ step by step using the inequalities we constructed (ex. we got a function is L^1, we want to show it's L^∞, and we use the proceeding inequalities to build our way up)

4. L^2 is particularly nice since it is a Hilbert space, and so many euclidean intuitions work on L^2 and help us construct many nice properties. A lot of Fourier analysis happens on L^2, though sometimes we do use L^p.

   Another way of thinking of L^2 is that it is a generlization of the notion of a space that *can* have an inner product (this will be made rigorous with a particular universal property, see theorem 11.1.24). In ℓ^2, we would simply get a natural generalization of an inner product on a vector-space, while in L^2, we get the continuous equivalent.

8.2 The dual of Lebesgue Space

By Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. We can thus define a function:

$$\varphi_g \colon L^p \to \mathbb{R} \qquad \varphi_g(f) = \int fg$$

and the operator norm of φ_g is always at most $\|g\|_q$[3]. With the operator norm, the mapping $g \mapsto \varphi_g$ will almost always be an isometry:

Proposition 8.2.1: Operator Norm And Isometry

Let p and q be conjugate exponents, $1 \leq q < \infty$. If $g \in L^q$, then:

$$\|g\|_q = \|\varphi_g\| = \sup\left\{\left|\int fg\right| \;\middle|\; \|f\|_p = 1\right\}$$

If μ is semifinite, this result also holds for $q = \infty$

***Proof*:**
The inequality $\|\varphi_g\| \leq \|g\|_q$ is Hölder's inequality: $\left|\int fg\right| \leq \|f\|_p \|g\|_q$ for every f with $\|f\|_p = 1$. For the reverse inequality we may assume $g \neq 0$ (both sides vanish otherwise).

First suppose $1 < q < \infty$. Set

$$f = \frac{|g|^{q-1}\,\overline{\operatorname{sgn}(g)}}{\|g\|_q^{q-1}}$$

Since $p(q-1) = q$, we have $|f|^p = |g|^q / \|g\|_q^q$, so $\|f\|_p = 1$; and since $\overline{\operatorname{sgn}(g)}\, g = |g|$:

$$\int fg = \frac{1}{\|g\|_q^{q-1}} \int |g|^q = \|g\|_q$$

[3]For $p = 2$ the pairing is usually defined as $\int f\overline{g}$ instead; the same convention can be used for any p, but the conjugation matters most at $p = 2$, where it makes the pairing an inner product (chapter 11).

so the supremum is at least (indeed, attains) $\|g\|_q$. If $q = 1$ (so $p = \infty$), take $f = \overline{\text{sgn}(g)}$ instead: then $\|f\|_\infty = 1$ and $\int fg = \int |g| = \|g\|_1$.

Finally, let $q = \infty$ (so $p = 1$) with μ semifinite, and let $\epsilon > 0$. The set $A = \{x \mid |g(x)| > \|g\|_\infty - \epsilon\}$ has positive measure, so by semifiniteness it contains a subset B with $0 < \mu(B) < \infty$. Taking $f = \overline{\text{sgn}(g)}\,\chi_B/\mu(B)$ gives $\|f\|_1 = 1$ and

$$\int fg = \frac{1}{\mu(B)} \int_B |g| \geq \|g\|_\infty - \epsilon$$

As ϵ was arbitrary, $\|\varphi_g\| \geq \|g\|_\infty$.

Conversely, if $f \mapsto \int fg$ is a bounded linear functional on L^p, then $g \in L^q$ in almost all cases:

Theorem 8.2.2: L^p Linear Functional Implies G In L^q

Let p and q be conjugate exponents. Suppose that g is a measurable function on X such that $fg \in L^1$ for all f in the space Σ of simple functions that vanish outside a set of finite measure, and the quantity

$$M_q(g) = \sup\left\{\left|\int fg\right| \;\middle|\; f \in \Sigma,\ \|f\|_p = 1\right\}$$

is finite. Also, suppose either that $S_g = \{x \mid g(x) \neq 0\}$ is σ-finite or that μ is semifinite. Then $g \in L^q$ and $M_q(g) = \|g\|_q$.

Proof:
We treat real scalars.[a] If $g = 0$ a.e. the claim is trivial, so assume otherwise. Note first that homogeneity gives $\left|\int fg\right| \leq M_q(g)\|f\|_p$ for every $f \in \Sigma$.

Step 1: the semifinite case reduces to σ-finite S_g. Suppose $q < \infty$ and μ is semifinite, and let $A_n = \{x \mid |g(x)| > 1/n\}$. If $B \subseteq A_n$ has finite measure, then $f = \mu(B)^{-1/p}\text{sgn}(g)\chi_B$ lies in Σ with $\|f\|_p = 1$ (when $\mu(B) > 0$), whence

$$M_q(g) \geq \int fg = \mu(B)^{-1/p} \int_B |g| \geq \frac{1}{n}\,\mu(B)^{1/q}$$

so $\mu(B) \leq (nM_q(g))^q$. By semifiniteness $\mu(A_n) = \sup\{\mu(B) \mid B \subseteq A_n,\ \mu(B) < \infty\} \leq (nM_q(g))^q < \infty$, and $S_g = \bigcup_n A_n$ is σ-finite. We may therefore assume S_g is σ-finite in Step 2.

Step 2: $q < \infty$. Let E_n be an increasing sequence of finite-measure sets with $\bigcup_n E_n = S_g$. By theorem 2.1.18 (applied to g^+ and g^- separately, which have disjoint supports) there are simple $\varphi_n \to g$ pointwise with $|\varphi_n| \leq |g|$ and $\text{sgn}(\varphi_n) = \text{sgn}(g)$ wherever $\varphi_n \neq 0$. Set $g_n = \varphi_n\chi_{E_n}$ and, for n large enough that $\|g_n\|_q > 0$,

$$f_n = \frac{|g_n|^{q-1}\text{sgn}(g_n)}{\|g_n\|_q^{q-1}}$$

with the convention $|g_n|^0 = \chi_{\{x \mid g_n(x)\neq 0\}}$ when $q = 1$. Each f_n is simple, vanishes outside E_n, and has $\|f_n\|_p = 1$ exactly as in the proof of proposition 8.2.1. Since $\text{sgn}(g_n) = \text{sgn}(g)$ where $g_n \neq 0$ and $|g_n| \leq |g|$:

$$f_n g = \frac{|g_n|^{q-1}\,|g|\,\chi_{\{x \mid g_n(x)\neq 0\}}}{\|g_n\|_q^{q-1}} \geq \frac{|g_n|^q}{\|g_n\|_q^{q-1}} \geq 0$$

Hence, by Fatou's Lemma (theorem 2.2.11) applied to $|g_n|^q \to |g|^q$:

$$\|g\|_q \leq \liminf_n \|g_n\|_q = \liminf_n \frac{\int |g_n|^q}{\|g_n\|_q^{q-1}} \leq \liminf_n \int f_n g \leq M_q(g)$$

so $g \in L^q$. Hölder's inequality now gives $M_q(g) \leq \|g\|_q$, so $M_q(g) = \|g\|_q$.

Step 3: $q = \infty$. Let $\epsilon > 0$ and $A = \{x \mid |g(x)| \geq M_\infty(g) + \epsilon\}$. If $\mu(A) > 0$, then in either the σ-finite or the semifinite case A contains a subset B with $0 < \mu(B) < \infty$, and $f = \operatorname{sgn}(g)\chi_B/\mu(B) \in \Sigma$ has $\|f\|_1 = 1$ with

$$\int fg = \frac{1}{\mu(B)} \int_B |g| \geq M_\infty(g) + \epsilon$$

contradicting the definition of $M_\infty(g)$. Hence $\mu(A) = 0$ for every $\epsilon > 0$, i.e. $\|g\|_\infty \leq M_\infty(g)$; the reverse inequality is Hölder's, as before.

[a] For complex scalars, replace $\operatorname{sgn}(g)$ by $\overline{\operatorname{sgn}(g)}$ throughout; the only delicate point is Step 2, where f_n must remain simple, which is arranged by using a simple approximation whose phase is a finite discretization of the phase of g.

Finally, the map $g \mapsto \left(f \mapsto \int fg\right)$ is almost always a surjection onto $(L^p)^*$.

Theorem 8.2.3: Description Of Dual Of Lp

Let p and q be conjugate exponents. If $1 < p < \infty$, for each $\varphi \in (L^p)^*$ there exists $g \in L^q$ such that $\varphi(f) = \int fg$ for all $f \in L^p$, and hence L^q is isometrically isomorphic to $(L^p)^*$. The same conclusion holds for $p = 1$ provided μ is σ-finite

***Proof*:**
Folland p. 190

Corollary 8.2.4: Lp Is Reflexive

If $1 < p < \infty$, L^p is reflexive, that is $(L^p)^{**} \cong L^p$

(Some comment in the case of $p = 1$ and $p = \infty$)

8.3 The Dual of $C_0(X)$

Having identified $(L^p)^* \cong L^q$, we turn to a second concrete dual, that of the space of continuous functions vanishing at infinity. The measure-theoretic input is the Riesz-Markov theorem of chapter 4, which represents every *positive* linear functional on $C_c(X)$ by a Radon measure; dropping positivity and completing $C_c(X)$ to a Banach space identifies the full dual with a space of signed measures. Throughout, X is a locally compact Hausdorff (LCH) space.

The completion of $C_c(X)$ under the uniform norm is the space of functions vanishing at infinity.

Definition 8.3.1: Functions Vanishing at Infinity

A continuous function $f\colon X \to \mathbb{R}$ *vanishes at infinity* if for every $\varepsilon > 0$ the set $\{x \in X \mid |f(x)| \geq \varepsilon\}$ is compact. The space of such functions is denoted $C_0(X)$; with the uniform norm $\|f\|_u = \sup_{x\in X} |f(x)|$ it is a Banach space, and $C_c(X)$ is dense in it.

To capture the whole dual, and not just its positive part, measures must be allowed to take both signs.

Definition 8.3.2: The Space $M(X)$ of Finite Radon Measures

Let $M(X)$ be the space of all finite signed regular Borel measures on X: a signed measure μ lies in $M(X)$ if its Jordan variations μ^+, μ^- (theorem 3.1.7) are finite Radon measures (definition 4.1.9). It carries the *total variation norm* $\|\mu\| = |\mu|(X) = \mu^+(X) + \mu^-(X)$.

The identification of the dual is the full Riesz representation theorem, in exact parallel with $(L^p)^* \cong L^q$.

Theorem 8.3.3: Riesz Representation for $C_0(X)^*$

Let X be an LCH space. For every $I \in C_0(X)^*$ there is a unique $\mu \in M(X)$ with

$$I(f) = \int f\, d\mu \qquad \text{for all } f \in C_0(X)$$

and the map $\mu \mapsto I$ is an isometric isomorphism $M(X) \cong C_0(X)^*$, so that $\|I\| = \|\mu\| = |\mu|(X)$.

***Proof*:**
The idea is to split I into positive parts and apply the positive Riesz-Markov theorem to each.

Jordan splitting of the functional. For $f \in C_c(X)$ with $f \geq 0$ set

$$I^+(f) = \sup\{I(g) \mid g \in C_c(X),\ 0 \leq g \leq f\}$$

Boundedness of I gives $I(g) \leq \|I\|\,\|f\|_u$, so the supremum is finite; a routine additivity check extends I^+ to a positive linear functional on $C_c(X)$ (writing $f = f^+ - f^-$). Then $I^- = I^+ - I$ is positive as well, since $I^+(f) \geq I(f)$ for $f \geq 0$.

Representation. By the Riesz-Markov theorem (theorem 4.2.1) there are Radon measures μ^+, μ^- with $I^\pm(f) = \int f\, d\mu^\pm$. Boundedness of I in the uniform norm forces μ^+, μ^- to be finite, so $\mu = \mu^+ - \mu^- \in M(X)$ and $I(f) = \int f\, d\mu$ on $C_c(X)$; density of $C_c(X)$ in $C_0(X)$ and continuity of I extend the representation to all of $C_0(X)$.

Isometry. For $f \in C_0(X)$ with $\|f\|_u \leq 1$,

$$|I(f)| = \left|\int f\, d\mu\right| \leq \int |f|\, d|\mu| \leq |\mu|(X) = \|\mu\|$$

so $\|I\| \leq \|\mu\|$. For the reverse inequality the Hahn decomposition $X = P \cup N$ (theorem 3.1.5) gives, formally, $\int(\chi_P - \chi_N)\, d\mu = |\mu|(X)$; as $\chi_P - \chi_N$ is not continuous, one approximates it in $L^1(|\mu|)$ by $f_n \in C_c(X)$ with $\|f_n\|_u \leq 1$, using the regularity of μ (corollary 4.1.12) and Lusin's theorem, and passes to the limit to obtain $\|I\| \geq \|\mu\|$.

When X is compact, $C_0(X) = C(X)$, so $C(X)^* \cong M(X)$: the finite regular signed measures are the dual of the continuous functions, the measure-theoretic counterpart of the $(L^p)^*$ duality.

8.3.4 Measures as Distributions In chapter 9 it will be natural to replace evaluation of a function with integration against a test function. The present section shows that a continuous functional on $C_0(X)$ is already integration against a measure in $M(X)$, so the measures sit among the distributions as those of order zero.

8.4 Some Useful Inequalities

In this section, we explore more inequalties, since they are at the heart of working with L^p spaces.

We first present a common inequality that shows up often in statistics:

Theorem 8.4.1: Chebychev's Inequality

Let $f \in L^p$ $(0 < p < \infty)$. Then for any $\alpha > 0$

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

***Proof*:**
Let $E_\alpha = m(\{x \mid |f(x)| > \alpha\})$. Then:

$$\|f\|_p^p = \int_\Omega |f|^p \geq \int_{E_\alpha} |f|^p \geq \int_{E_\alpha} \alpha^p = \alpha^p m(E_\alpha)$$

re-arranging gives us:

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

as we sought to show

The next inequality gives us a general statement on being abel to get a function from $L^p(\mu)$ "transformed" into a function of $L^p(\nu)$:

Theorem 8.4.2: Transitioning L^p Spaces

Let$(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measures spaces, K be a $M \otimes N$-measurable function on $X \times Y$, and let's say there exists a $C > 0$ such that

$$\int |K(x,y)| d\mu(x) \leq C \qquad \int |K(x,y)| d\nu(y) \leq C$$

for a.e. x and y. Then if $f \in L^p(\nu)$ $(1 \leq p \leq \infty)$, the integral:

$$Tf(x) = \int K(x,y) f(y) d\nu(y)$$

converges absolutely for a.e. $x \in X$, $Tf \in L^p(\mu)$, and $\|Tf\|_p \leq C\|f\|_p$

***Proof*:**
Let $p \in [1, \infty]$, and let q be it's conjugate exponent. Then:

$$|K(x,y)f(y)| = |K(x,y)||f(y)| = |K(x,y)|^{\frac{1}{p}+\frac{1}{q}}|f(y)| = (|K(x,y)|^{1/p})(|K(x,y)|^{1/q}f(y)|)$$

and applying Hölder to the functions in parenthesis, we get:

$$\begin{aligned}\int |K(x,y)f(y)| d\nu(y) &\leq \left(\int |K(x,y)| d\nu(y)\right)^{1/q} \left(\int |K(x,y)||f(y)|^p d\nu(y)\right)^{1/p} \\ &\leq C^{1/q} \left(\int |K(x,y)|f(y)|^p d\nu(y)\right)^{1/p}\end{aligned}$$

for a.e. $x \in X$. Hence, taking the integral over $X \times Y$, using Tonelli's theorem, and substituting the result we just found, we get:

$$\begin{aligned}\int \left[\int |K(x,y)f(y)d\nu(y)\right]^p d\mu(x) &\leq \int\int C^{p/q}|K(x,y)||f(y)|^p d\nu(y) d\mu(x) \\ &= C^{p/q} \int\int |K(x,y)||f(y)|^p d\mu(x) d\nu(y) \qquad \text{Tonelli} \\ &\leq C^{p/q} \int |f(y)|^p \int K(x,u) d\mu(x) d\nu(y) \\ &= C^{p/q+1} \int |f(y)|^p d\nu(y)\end{aligned}$$

Since $f \in L^p(\nu)$, the last integral is finite. Since this is true for a.e. x, $K(x,\cdot)f \in L^1(\nu)$ for a.e. x. Thus, Tf is well-defined a.e., meaning we can integrate it, and by the inequality we just showed:

$$\int |Tf(x)|^p d\mu(x) = \|Tf\|_p^p \leq C^{p/q+1}\|f\|_p^p$$

taking the pth root of each side get's us the desired inequality

The next result we work towards is a generalization of Minkowski: instead of summing we will be taking integrals. You can also interpret part (a) as you can move the $(|\cdot|^p)^{1/p}$ "down" a layer of integrals, and part (b) as the generalization of monotonicity ($|\int f| \leq \int |f|$).

Theorem 8.4.3: Minkowski's Inequality For Integrals

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite meaure spaces and let f be a [real?] $M \otimes N$-measurable function on $X \times Y$. Then:

1. if $f \geq 0$ and $1 \leq p \leq \infty$, then:

$$\left[\int \left(\int f(x,y) d\nu(y)\right)^p d\nu(x)\right]^{1/p} \leq \int \left[\int f(x,y)^p d\mu(x)\right]^{1/p} d\nu(y)$$

2. If $1 \leq p \leq \infty$, $f(\cdot, y) \in L^p(\mu)$ for a.e. y and the function $y \mapsto \|f(\cdot, y)\|_p$ is in $L^1(\nu)$, then $f(x, \cdot) \in L^1(\nu)$ for a.e. x, the function $x \mapsto \int f(x,y) d\nu(y)$ is in $L^p(\mu)$ and

$$\left\| \int f(\cdot, y) d\nu(y) \right\|_p \leq \int \|f(\cdot, y)\|_p \, d\nu(y)$$

***Proof*:**
If $p = 1$, then this becomes Tonelli's Theorem, so consider $1 < p < \infty$ (the case for $p = \infty$ comes later). Let q be it's conjugate exponent so that $1/p + 1/q = 1$. Suppose $\mu, \nu < \infty$ and $\|f\|_\infty < \infty$ (i.e. it's essential supremum is finite). For notational simplicity, define $H(x) = \int_Y f(x,y) d\nu(y)$. Since the measure is finite and f is esentially bounded, we have that $\|H\|_p < \infty$. Then:

$$\begin{aligned}
\|H\|_p^p &= \int_X \left(\int_Y f(x,y) d\nu(y)\right)^p d\nu(x) \\
&= \int_X \left(\int_Y f(x,y) d\nu(y)\right) (H(x))^{p-1} d\mu(x) && x \text{ indep. of } y \\
&= \int_X \int_Y f(x,y) H(x)^{p-1} d\mu(x) d\nu(y) && \text{Fubini} \\
&\overset{!}{\leq} \int_X \|f(x,\cdot)\|_p \|H\|_p^{p-1} d\nu(y) && \text{Hölder, } q = p/(p-1) \\
&\leq \|H\|_p^{p-1} \int_X \|f(x,\cdot)\|_p d\nu(y)
\end{aligned}$$

where $\overset{!}{\leq}$ I'm not sure how Hölder's inequality was used. Finally, dividng by $\|H(x)\|_p^{p-1}$ get's us:

$$\|H\|_p \leq \int_X \|f(x,\cdot)\|_p d\nu(y)$$

For the general case, Yakov left it as an exercise to do $X = \bigcup_n X_n$, $Y = \bigcup Y_n$, $E_n = \{x \mid f(x) < n\}$, and

$$\tilde{X}_n = X_n \cap E_n \qquad \tilde{Y}_n = Y_n \cap E_n$$

and apply the result to each of these, and the monotone convergence theorem at the end (feels similar to how it was done for Fubini Tonnelli).

For $p = \infty$, This comes down to monotonicity of integrals:

$$\left\| \int f(x,y) d\nu(y) \right\|_\infty \leq \left| \int f(x,y) d\nu(y) \right| \leq \int |f(x,y) d\nu(y) = \int \|f(x,y)\|_\infty d\nu(y)$$

where the last inequality comes from the fact that the integral ignores zero-measure sets.

There are a few more results in the book, but Yakov decided to prove the generalized Hölder's inequality instead for now:

Theorem 8.4.4: Generalized Hölder's Inequality

Let $1 \leq p_i \leq \infty$ for $1 \leq i \leq n$ and $\sum_i^n p_i^{-1} = r^{-1} \leq 1$. If $f_i \in L^{p_i}$, then $\prod_i^n f_i \in L^r$ and

$$\left\| \prod_i^n f_i \right\|_r \leq \prod_i^n \|f_i\|_{p_i}$$

***Proof*:**
If $r = \infty$, then $r^{-1} = 0$, so all the $p_i^{-1} = 0$, so all the $p_i = \infty$, in which case we simply take:

$$\left| \prod_i^n f_i \right| \leq \prod_i^n |f_i|$$

For $r < \infty$, just like for Hölder's inequality, it suffice to assume $r = 1$, since we can do:

$$\sum \frac{1}{\frac{p_i}{r}} = 1$$

In that case, notice that:

$$\begin{aligned} \left\| \prod_i^n f_i \right\|_r^r &= \int |f_1 f_2 \cdots f_n|^r \\ &= \int f_1^r f_2^r \cdots f_n^r \\ &\leq \|f_1\|_{p_1}^r \|f_2\|_{p_2}^r \cdots \|f_n\|_{p_n}^r \end{aligned}$$

Notice that each f_i is raised to the power of p_i/r. (TBD)

Now, we do induction on i. Let the induction hypothesis be:

$$\left(\sum_i^{n-1} p_i^{-1} \right) + p_n^{-1} = r^{-1} + p_n^{-1} = 1$$

Then by Hölder's inequality:

$$\int f_1 \cdots f_n| \leq \|f_1 \cdots f_{n-1}\|_r \|f\|_{p_n} \overset{\text{I.H.}}{\leq} \|f_1\|_{p_1} \cdots \|f_n\|_{p_n}$$

giving our desired result

The final property we'll present a way to represent functions in terms of the integral:

Proposition 8.4.5: Layer Cake Representation

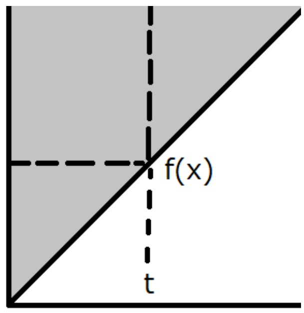
Let $\varphi(t) : [0,\infty] \to [0,\infty]$ be an increasing function which is C^1 on $(0,\infty)$, satisfies $\varphi(0) = 0$ and satisfies $\lim_{t\to\infty} \varphi(t) = \varphi(\infty)$. Let (X, M, μ) be a σ-finite measure space and $f : X \to [0,\infty]$ be a non-negative measurable function. Then:

$$\int_X \varphi(f(x))\, d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) > t\})\varphi'(t)dt$$

One consequence of this equation is that with $\varphi(t) = t^p$ for $1 < p < \infty$ is the following formula for the L^p norm of f:

$$\|f\|_p = \left(p \int_0^\infty \mu(\{x \mid f(x) > t\}\, t^{p-1}\right)^{1/p} dt$$

***Proof*:**
The proof can be captured pictorially as integrating over the following area:



Let $S \subseteq X \times [0,1]$, $S = \{(x,t) \mid f(x) \geq t\}$. With this set, define $F : X \times [0,\infty] \to [0,\infty]$, $F(x,t) = \varphi'(t)\chi_S$. That is, we are going to integrate the "upper triangle". Since φ is an increasing C^1 function and $\varphi'(t) > 0$, we have $F \in L^+(X,[0,\infty])$. Therefore, we can apply Tonneli's Theorem and see that it's equivalent to integrate over the iterated integrals to compute integrals.

To compute it, we need to be clever with the bounds we choose which depend on which iterated integral we choose. The different direction of the iterated integral will ultimately lead to the desired result. If we integrate with respect to $d\mu(x)dt$, then we take slices across t: $(\varphi'(t)\chi_S)_t = \varphi'(t)\chi_{S_t}$, where $(\chi_S)_t = \chi_S(x,t) = \chi_{S_t}$, with $S_t = \{x \mid f(x) \geq t\}$. And so we get:

$$\int_0^\infty \int_X \chi_{S_t}\varphi'(t) = \int_0^\infty \int_{\{x \mid f(x)\geq t\}} \varphi'(t)d\mu(x)dt = \int_0^\infty \mu(\{x \mid f(x) \geq t\})\varphi'(t)dt$$

Now, integrating with respect to $dtd\mu(x)$, we take x slices: $(\varphi'(t)\chi_S)_x = \varphi'(t)\chi_{S_x}$ where

$(\chi_S)_x = \chi_S(x,t) = \chi_{S_x}$, with $S_x = \{x \mid 0 \leq f(x)\}$. And so we get:

$$\int_X \int_0^\infty \chi_{S_x} \varphi'(t) dt d\mu = \int_X \int_0^{f(x)} \varphi'(t) dt d\mu(x)$$

and since $[0, f(x)]$ is bounded and $\varphi(0) = 0$, the Lebesgue integral matches the Riemann integral, and so we get that:

$$\int_X \varphi(f(x)) d\mu(x)$$

and by Tonnelli's theorem:

$$\int_X \varphi(f(x)) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\}) \varphi'(t) dt$$

As we sought to show. Using this, if we let $\varphi(t) = t^p$, then we see that φ is C^1 on $(0, \infty)$ and $\varphi(0) = 0$ for $1 \leq p < \infty$. Thus, the value of $\|f\|_p$ (where we take f^p instead of $|f|^p$ since $f \geq 0$) is:

$$\begin{aligned} \|f\|_{L^p} = \left(\int |f|^p \right)^{1/p} = \left(\int f^p \right)^{1/p} &= \left(\int_X \varphi(f(x)) d\mu(x) \right)^{1/p} \\ &= \left(p \int_0^\infty \mu(\{x \mid f(x) > t)\} \, t^{p-1} \right)^{1/p} dt \end{aligned}$$

Another use of this formula is by taking $\varphi(t) = t$. Then we will get that:

$$\int f(x) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\}) dt$$

This can be useful if we have information of what is happening to the growth of f as f get's larger, in particular if we know that somehow, the amount of points such that $\mu(\{x \mid f(x) \geq t\})$ is shrinking rapidly enough say that $\mu(\{x \mid f(x) \geq t\}) \leq \frac{C}{t^r}$ for some constant C).

8.5 Interpolation: The Riesz-Thorin Theorem

As we saw in the previous section, if a function is in L^p and L^r, it is also in L^q for all $p < q < r$. This is a type of "interpolation" of spaces. A natural question arises: if we have a linear operator T that is bounded on L^p and bounded on L^r, is it also bounded on L^q? The answer is yes, and the Riesz-Thorin theorem provides the precise quantitative relationship between the operator norms.

The proof relies on a powerful complex analysis technique known as the Phragmén-Lindelöf principle, specifically a version called the Three Lines Lemma.

Lemma 8.5.1: Three Lines Lemma

Let φ be a bounded continuous function on the strip $S = \{z \in \mathbb{C} \mid 0 \leq \mathrm{Re}(z) \leq 1\}$ that is holomorphic on the interior of S. If $|\varphi(z)| \leq M_0$ for $\mathrm{Re}(z) = 0$ and $|\varphi(z)| \leq M_1$ for $\mathrm{Re}(z) = 1$, then for all $t \in [0, 1]$ and z with $\mathrm{Re}(z) = t$:

$$|\varphi(z)| \leq M_0^{1-t} M_1^t$$

***Proof*:**

For $\epsilon > 0$, define $\varphi_\epsilon(z) = \varphi(z) M_0^{z-1} M_1^{-z} e^{\epsilon(z^2-1)}$. The term $e^{\epsilon(z^2-1)}$ helps handle the boundedness at infinity within the strip. By the maximum modulus principle applied to rectangles in the strip as the height goes to infinity, one can show $|\varphi_\epsilon(z)| \leq 1$ in the strip. Letting $\epsilon \to 0$ yields the result.

Theorem 8.5.2: Riesz-Thorin Interpolation Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. Let $p_0, p_1, q_0, q_1 \in [1, \infty]$. Define p_t and q_t for $t \in (0, 1)$ by:

$$\frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1}, \qquad \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

Suppose T is a linear operator defined on the set of simple functions Σ such that for all $f \in \Sigma$:

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0} \quad \text{and} \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

Then for all $t \in (0, 1)$, T extends to a bounded operator from $L^{p_t}(\mu)$ to $L^{q_t}(\nu)$ satisfying:

$$\|Tf\|_{q_t} \leq M_0^{1-t} M_1^t \|f\|_{p_t}$$

***Proof*:**

Let $t \in (0, 1)$ be fixed. By duality (Proposition 8.2.1), it suffices to estimate $|\int (Tf) g \, d\nu|$ where f and g are simple functions with $\|f\|_{p_t} = 1$ and $\|g\|_{q_t'} = 1$ (where q_t' is the conjugate exponent of q_t).

We construct a family of functions to apply the Three Lines Lemma. Let $f = |f| e^{i\theta}$ and $g = |g| e^{i\psi}$ be the polar decompositions. For any complex number z in the strip $0 \leq \mathrm{Re}(z) \leq 1$, define:

$$\alpha(z) = (1-z)\frac{1}{p_0} + z\frac{1}{p_1}, \qquad \beta(z) = (1-z)\frac{1}{q_0'} + z\frac{1}{q_1'}$$

Note that for $z = t$, $\alpha(t) = 1/p_t$ and $\beta(t) = 1/q_t'$. Now define deformations of f and g:

$$f_z = |f|^{p_t \alpha(z)} e^{i\theta}, \qquad g_z = |g|^{q_t' \beta(z)} e^{i\psi}$$

(If $f = 0$ or $g = 0$, set the function to 0). Note that $f_t = f$ and $g_t = g$.

Define the function $\Phi(z)$ on the strip $0 \leq \mathrm{Re}(z) \leq 1$:

$$\Phi(z) = \int_Y (Tf_z) g_z \, d\nu$$

Since f and g are simple, f_z is a finite linear combination of terms like $c_k\alpha^z$, making $\Phi(z)$ holomorphic and bounded.

We check the boundary conditions. For $\mathrm{Re}(z) = 0$, $f_z \in L^{p_0}$ and $g_z \in L^{q_0'}$. Specifically:

$$\|f_{iy}\|_{p_0}^{p_0} = \int ||f|^{p_t(\frac{1}{p_0}+i\mathrm{Im}(\alpha))}|^{p_0} = \int |f|^{p_t} = \|f\|_{p_t}^{p_t} = 1$$

Similarly $\|g_{iy}\|_{q_0'} = 1$. Thus by Hölder's inequality and the hypothesis on T:

$$|\Phi(iy)| = \left|\int T(f_{iy})g_{iy}\right| \leq \|Tf_{iy}\|_{q_0}\|g_{iy}\|_{q_0'} \leq M_0\|f_{iy}\|_{p_0} \cdot 1 = M_0$$

Similarly, for $\mathrm{Re}(z) = 1$, we find $|\Phi(1+iy)| \leq M_1$.

By the Three Lines Lemma, $|\Phi(t)| \leq M_0^{1-t}M_1^t$. Since $\Phi(t) = \int(Tf)g$, and this holds for all unit simple functions f, g, we conclude:

$$\|T\|_{L^{p_t}\to L^{q_t}} \leq M_0^{1-t}M_1^t$$

Counter-Example: The Linear Transport Equation

Consider the simple advection equation in one dimension:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

where $c \neq 0$ is a real constant velocity.

1. **The Operator:** The solution is a translation of the initial state $f(x)$:

$$u(x,t) = f(x - ct)$$

   Therefore, the time-shift operator S_t acts as:

$$(S_t f)(x) = f(x - ct)$$

2. **The Adjoint:** To check if S_t is self-adjoint, we compute its adjoint S_t^* using the standard L^2 inner product $\langle f, g\rangle = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\,dx$.

$$\langle S_t f, g\rangle = \int_{-\infty}^{\infty} f(x-ct)\overline{g(x)}\,dx$$

   Perform a change of variables $y = x - ct$, so $x = y + ct$ and $dx = dy$:

$$\langle S_t f, g\rangle = \int_{-\infty}^{\infty} f(y)\overline{g(y+ct)}\,dy$$

$$\langle S_t f, g\rangle = \langle f, S_{-t} g\rangle$$

   This shows that the adjoint is the inverse shift:

$$S_t^* = S_{-t}$$

3. **Conclusion:** For S_t to be self-adjoint, we require $S_t = S_t^*$. This would imply $S_t = S_{-t}$, or $f(x - ct) = f(x + ct)$ for all functions f. This is clearly false for $c \neq 0$ and $t \neq 0$. Thus, S_t **is not self-adjoint.**

General Context

- **Unitary vs. Self-Adjoint:** In conservative systems (like Quantum Mechanics or the Wave Equation), the evolution operator S_t is typically **unitary**, meaning $S_t^* = S_t^{-1} = S_{-t}$. A unitary operator is rarely self-adjoint (unless $S_t^2 = I$, which is not true for general time evolution).
- **When is it Self-Adjoint?** S_t *is* self-adjoint in specific dissipative systems, such as the **Heat Equation** ($u_t = \Delta u$). In this case, $S_t = e^{t\Delta}$. Since the Laplacian Δ is a self-adjoint operator (with appropriate boundary conditions), the operator exponential $e^{t\Delta}$ is also self-adjoint for real t.

Zak Katz: I see. Does that mean that S_t is not always "diagonalizable"?

Raycast AI: No, that is a common misconception! **"Not self-adjoint" does not mean "not diagonalizable."**

The time-shift operator S_t for the Transport Equation (which I used as a counter-example previously) **is** diagonalizable, even though it is not self-adjoint.

Here is the breakdown of why this works and when it fails.

1. The Key Concept: Normal Operators

In linear algebra and functional analysis, a deeper property governs diagonalizability: **Normality**.

An operator A is **Normal** if it commutes with its adjoint:

$$AA^* = A^*A$$

The **Spectral Theorem** states that **any Normal operator is diagonalizable** (specifically, unitarily diagonalizable).

- **Self-Adjoint Operators** are Normal ($A = A^*$, so $AA = AA$).
- **Unitary Operators** are Normal ($A^* = A^{-1}$, so $AA^{-1} = A^{-1}A = I$).

2. Revisiting the Transport Equation (The "Yes" Case)

Let's look at the Transport Equation again ($u_t + u_x = 0$), where S_t shifts the function: $f(x) \to f(x-t)$.

- We found that $S_t^* = S_{-t}$ (the inverse shift).
- Does it commute?
$$S_tS_t^* = S_tS_{-t} = S_0 = I$$
$$S_t^*S_t = S_{-t}S_t = S_0 = I$$

- Because $S_t S_t^* = S_t^* S_t$, the operator is **Normal**.

How do we "diagonalize" it? We use the **Fourier Transform**. In the frequency domain (Fourier space), the shift operator becomes multiplication:

$$\mathcal{F}(S_t f)(k) = e^{-ikt}\hat{f}(k)$$

Multiplication by a scalar (or function) is the infinite-dimensional equivalent of a diagonal matrix. The Fourier basis functions e^{ikx} are the "eigenvectors" of the time-shift operator.

Conclusion: The Transport operator is not self-adjoint, but it is normal, so it **is** diagonalizable.

---

3. When is S_t NOT Diagonalizable? (The "No" Case)

For S_t to fail to be diagonalizable (to be "defective"), the system must be **Non-Normal** in a specific way that creates "Jordan Blocks." This happens when eigenvectors coalesce and you lose a complete basis (this is called an **Exceptional Point** in physics).

Example: A "Defective" System of PDEs Consider a coupled system of two variables $u(x,t)$ and $v(x,t)$:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

Here, the change in u depends on v, but v is constant in time.

1. **Solve it:**
   - $v(x,t) = v(x,0)$ (constant)
   - $u_t = v(x,0) \implies u(x,t) = u(x,0) + t \cdot v(x,0)$
2. **The Time-Evolution Operator S_t:** We can write S_t as a matrix acting on the vector $\begin{pmatrix} f \\ g \end{pmatrix}$:

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Is it Diagonalizable?** Check the matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - The eigenvalues are $\lambda = 1$ (with algebraic multiplicity 2).
   - However, there is only **one** independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we are missing a second eigenvector, this operator is **not diagonalizable** for any $t \neq 0$.

Summary Table

Type of PDE	Operator Property	Self-Adjoint?	Diagonalizable?
Heat Eq. ($u_t = \Delta u$)	Self-Adjoint	**Yes**	**Yes** (Real eigenvalues)
Transport / Wave	Unitary	**No**	**Yes** (Complex eigenvalues)
Defective System	Non-Normal	**No**	**No** (Jordan Block)

Bottom Line: Most standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because they are typically governed by Normal operators. You usually have to construct specific coupled systems or non-Hermitian boundary conditions to break diagonalizability.

Zak Katz: Let me be clear: S_t shifts *in time* not in space. So to be clear: let a PDE $F(x_1, \ldots, x_n, t) = 0$ PDE w.r.t time t, if $u(x,t)$ is a solution and $f_t(x) = u(x,t)$ is a "state" at time t, and Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a. (NOT the space shift operator $f(x) \to f(x+t)$). Is S_t always diagonalizable?

Raycast AI: Thank you for the clarification. I understand the distinction you are making: you are asking about the **time-evolution operator** (often called the Propagator, $U(t)$ or S_t) that maps the initial state $u(\cdot, 0)$ to the state at time t, $u(\cdot, t)$.

The answer remains: **No, S_t is not always diagonalizable.**

While many standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because their generators are Normal (Self-Adjoint or Skew-Adjoint), it is very easy to construct physical systems where S_t is **defective** (non-diagonalizable).

This happens specifically when the system exhibits **Linear Instability** or **Critical Damping** (often associated with "Exceptional Points" in modern physics).

The Mathematical Reason: Jordan Blocks

If a PDE can be written in the form $\frac{\partial}{\partial t}\psi = L\psi$, the time-evolution operator is formally $S_t = e^{Lt}$.

For S_t to be diagonalizable, the generator L must possess a complete basis of eigenfunctions. If L has a **Jordan Block** (i.e., it has generalized eigenvectors but not enough true eigenvectors), then the exponential $S_t = e^{Lt}$ will also possess a Jordan block structure for any $t \neq 0$, making it non-diagonalizable.

Here are two concrete counter-examples.

Counter-Example 1: The "Resonant" System (Algebraic Growth)

Consider a system of two fields, $u(x,t)$ and $v(x,t)$, on a domain Ω. The PDE is:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

where λ is a constant. This is a standard linear PDE system.

1. **Find the Solution:** This is a coupled system.
   - $v_t = \lambda v \implies v(x,t) = e^{\lambda t} v(x,0)$.
   - $u_t = \lambda u + v$. Substituting the solution for v: $u_t = \lambda u + e^{\lambda t} v(x,0)$.
   - The solution for u involves a secular term (a linear t):

   $$u(x,t) = e^{\lambda t} u(x,0) + \mathbf{t}\mathbf{e}^{\lambda \mathbf{t}} v(x,0)$$

2. **The Time-Evolution Operator S_t:** Writing this as an operator matrix acting on the initial vector $\psi_0 = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$:

   $$S_t = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Why it is NOT Diagonalizable:** Look at the matrix part $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - **Eigenvalues:** It has a single eigenvalue $\mu = 1$ (repeated).
   - **Eigenvectors:** It has only **one** linearly independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we need two independent eigenvectors to span the 2D space of fields at every point x, and we only have one, S_t is **not diagonalizable** for any $t \neq 0$.

Physical Interpretation: The term $te^{\lambda t}$ represents "algebraic growth" (or transient growth). Whenever you see a solution of the form $t \times$ (exponential), it is the signature of a non-diagonalizable operator. A diagonalizable operator would only have pure sums of exponentials.

Counter-Example 2: Critical Damping (Classical Mechanics / PDE)

Consider the Damped Harmonic Oscillator equation for a field $u(x,t)$, which is a second-order PDE in time:

$$u_{tt} + 2\gamma u_t + \gamma^2 u = 0$$

(Here $\gamma > 0$ is the damping coefficient).

To analyze S_t, we convert this to a first-order system (state-space form) by defining $v = u_t$:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

The Critical Case: The eigenvalues of the matrix $L = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}$ are found by $\det(L - \omega I) = 0$.

$$\omega^2 + 2\gamma\omega + \gamma^2 = (\omega + \gamma)^2 = 0$$

We have a repeated eigenvalue $\omega = -\gamma$.

Because the matrix L is not the identity matrix but has a repeated eigenvalue, it is a non-trivial Jordan block. The solution is:

$$u(t) = C_1 e^{-\gamma t} + C_2 \mathbf{t}\mathbf{e}^{-\gamma \mathbf{t}}$$

The presence of the $te^{-\gamma t}$ term confirms that the propagator S_t cannot be diagonalized. If it were diagonalizable, we could write the solution purely as $u(t) = Ae^{\omega_1 t} + Be^{\omega_2 t}$, which implies strictly exponential decay without the linear polynomial pre-factor.

Summary

- **Is S_t always diagonalizable? No.**
- **When does it fail?** It fails at **Exceptional Points**, where two or more eigenmodes of the system coalesce (become parallel), making the set of eigenvectors incomplete. This results in solutions that grow or decay with polynomial factors (like t, t^2) multiplying the exponentials.

Zak Katz: Okay. Now, when it *is* diagonalizable, is it *always* the case that we can use Fourier transforms to find an eigenbasis?

Raycast AI: No. Absolutely not.

Just because an operator is diagonalizable does not mean the Fourier Transform provides the basis.

The **Fourier Transform** is a very specific tool that only diagonalizes operators that are **Translation Invariant** (i.e., operators with **Constant Coefficients** defined on the entire space $\mathbb{R}^n$).

If your PDE has **variable coefficients** or is defined on a **bounded domain**, the Fourier basis (e^{ikx}) will generally fail to diagonalize the operator, yet a different eigenbasis might still exist perfectly fine.

Here is the hierarchy of logic:

1. **Diagonalizable:** Means *there exists* a basis $\{\varphi_n(x)\}$ such that the operator acts as multiplication on these functions.
2. **Fourier Diagonalizable:** Means that specific basis is $\{\varphi_k(x) = e^{ikx}\}$.

When does Fourier work?

Fourier analysis works **only** if your operator L commutes with the translation operator T_a (where $T_a f(x) = f(x + a)$).

- **Example:** The Heat Equation $u_t = u_{xx}$.
  - The second derivative behaves the same at $x = 0$ as it does at $x = 100$. Space is homogeneous.
  - Therefore, the eigenfunctions are plane waves e^{ikx}.

When does Fourier fail (but the operator is still diagonalizable)?

Fourier fails whenever **spatial symmetry is broken**. This happens in three common scenarios:

1. Variable Coefficients (Space is not homogeneous) Consider the **Quantum Harmonic Oscillator** (one of the most famous diagonalizable systems in physics).

$$i\frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2\psi$$

- **Is it Diagonalizable?** Yes. The Hamiltonian is Self-Adjoint. It has a discrete, perfect set of eigenfunctions.
- **Are they Fourier Modes? No.** The term x^2 breaks translation invariance (physics looks different at $x = 0$ vs $x = 100$).
- **The Actual Basis:** The eigenfunctions are **Hermite Functions** (a Gaussian multiplied by a polynomial):

  $$\varphi_n(x) = H_n(x)e^{-x^2/2}$$

  If you tried to apply a Fourier Transform to this equation, the x^2 multiplication term turns into a Laplacian in k-space ($-\partial_k^2$), so you effectively just swap the PDE for an identical PDE in frequency space without solving it.

2. Bounded Domains (The "Boundary Condition" Problem) Consider the heat equation $u_t = u_{xx}$ on a finite interval $x \in [0, L]$ with Dirichlet boundary conditions $u(0) = u(L) = 0$.

- **Is it Diagonalizable?** Yes.
- **Is it the Fourier Transform?** No. The Continuous Fourier Transform uses e^{ikx} for any real k.
- **The Actual Basis:** The basis is strictly **Sine functions**: $\sin(\frac{n\pi x}{L})$.
  - While related to Fourier *Series*, the Continuous Fourier *Transform* basis (e^{ikx}) is invalid here because e^{ikx} does not satisfy the boundary conditions (it doesn't vanish at 0 and L). You are forced onto a discrete subset of the frequency spectrum.

3. Different Geometries If your PDE is on a sphere (like weather patterns on Earth):

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

- **Is it Diagonalizable?** Yes.
- **The Actual Basis:** The eigenfunctions are **Spherical Harmonics** $Y_l^m(\theta, \varphi)$.
- Standard planar Fourier modes (e^{ikx}) make no sense here.

Summary Table

PDE Feature	Is S_t Diag.?	Fourier?	The Actual Eigenbasis
Constant Coeffs on $\mathbb{R}$ ($u_t = u_{xx}$)	Yes	**Yes**	Plane Waves (e^{ikx})
Variable Coeffs ($u_t = xu_x$)	Yes	**No**	Power functions (x^λ)
Potential Well ($u_t = i\Delta u + iV(x)u$)	Yes	**No**	Hermite / Legendre / Bessel
Bounded Domain ($x \in [0, 1]$)	Yes	**No**	Fourier *Series* (Sines/Cosines)

Conclusion: Diagonalizability guarantees that *some* coordinate system exists where the operator is simple. It does not guarantee that the coordinate system is the one given by Fourier.

Zak Katz: Okay. Is it a necesssary and sufficient condition that S_t is diagonalizable by the Fourier basis iff the diagonalizable operator is translation invariant? Just to make sure I understand what translation invariant means here, we are talking about *space* translation, and we mean that if S_t is our operator and τ_a is our space-translation, then:

$$S_t(\tau_a(f)) = \tau_a(S_t(f))$$

Raycast AI: Yes, precisely.

For a linear operator S_t acting on functions defined on the entire space $\mathbb{R}^n$ (or a periodic domain), the condition is **necessary and sufficient**:

> **Theorem:** A bounded linear operator S_t is diagonalized by the Fourier Transform if and only if it is **Translation Invariant**.

Your definition of translation invariance is exactly correct:

$$[S_t, \tau_a] = S_t\tau_a - \tau_a S_t = 0 \quad \text{for all } a \in \mathbb{R}^n$$

where τ_a is the spatial shift operator.

Here is the logic for why this is an “if and only if” relationship.

1. Why is it Sufficient? (Translation Invariant $\implies$ Fourier Diagonal)

This relies on a fundamental principle of linear algebra: **Commuting operators share a common eigenbasis.**

1. Consider the family of translation operators $\{\tau_a\}_{a\in\mathbb{R}}$.
2. What are the eigenfunctions of a translation? Let’s verify that the plane waves e^{ikx} are eigenfunctions of τ_a:

   $$\tau_a(e^{ikx}) = e^{ik(x+a)} = e^{ika} \cdot e^{ikx}$$

   The eigenvalue is the constant number e^{ika}, and the eigenfunction is e^{ikx}.

3. Since your time-evolution operator S_t commutes with **every** translation operator τ_a (for all shifts a), it must share the same set of eigenfunctions.
4. Therefore, the eigenfunctions of S_t must also be e^{ikx}.
5. If the eigenfunctions are e^{ikx}, then by definition, the operator is diagonalized by the Fourier Transform.

2. Why is it Necessary? (Fourier Diagonal $\implies$ Translation Invariant)

If an operator is diagonal in Fourier space, it acts as a simple multiplication. Let's call the Fourier transform $\mathcal{F}$.

1. Assume S_t is a Fourier Multiplier: $\widehat{S_t u}(k) = m(k)\widehat{u}(k)$ for some multiplier function $m(k)$.
2. The translation operator τ_a in Fourier space is also just a multiplication (by the phase factor e^{ika}).
3. Since regular multiplication of scalars is commutative ($a \cdot b = b \cdot a$), the multiplication by $m(k)$ commutes with the multiplication by e^{ika}.
4. Therefore, S_t and τ_a commute.

Summary

This concept is the basis of signal processing and linear systems theory (where these are called **Linear Time-Invariant (LTI)** systems, or in your PDE context, "Linear Space-Invariant" systems).

- **Does S_t commute with space shifts?**
  - **Yes:** You can use Fourier Transforms (e.g., Heat Eq, Wave Eq, Schrödinger with constant potential).
  - **No:** You cannot use Fourier Transforms to find the eigenbasis (e.g., Schrödinger with Harmonic Oscillator potential, variable coefficient Wave Eq).

Zak Katz: Right. Is ther a "theory" for diagonalizable operators S_t?

Raycast AI: Yes, there is a vast and rich "theory" for this, but it doesn't have a single name like "The Theory of Diagonalizable Operators." Instead, it is split into different fields of **Functional Analysis** depending on *why* the operator is diagonalizable.

If you are looking for the generalized version of "Fourier Analysis" for operators that are **not** translation invariant (i.e., not solvable by Fourier Transform), here is the map of the theory.

1. The "Gold Standard": The Spectral Theorem

This is the master theory. It applies if your operator S_t (or its generator L) is **Normal** ($LL^* = L^*L$).

- **The Theory: Spectral Theory of Normal Operators**.
- **What it says:** Any normal operator on a Hilbert space is unitarily equivalent to a multiplication operator.
- **Physical Meaning:** There *always* exists an orthonormal basis of eigenfunctions (or a "continuous basis" / spectral measure) that diagonalizes the operator.
- **How to find the basis:** This is the hard part. It usually requires solving the **Sturm-Liouville Problem** (for 1D) or using **Scattering Theory** (for unbounded domains).

2. The "Variable Coefficient" Theory: Sturm-Liouville Theory

If your PDE has variable coefficients but is in 1D (or separable), this is the standard replacement for Fourier analysis.

- **The Setup:** Your generator looks like $L = \frac{d}{dx}[p(x)\frac{d}{dx}] + q(x)$.
- **The Result:** Just like Fourier series uses $\sin(nx)$ and $\cos(nx)$, Sturm-Liouville theory guarantees a set of orthogonal eigenfunctions $\varphi_n(x)$ weighted by a density function.
- **Example:** Legendre polynomials, Bessel functions, Hermite polynomials (Quantum Harmonic Oscillator). These are the "custom Fourier bases" for specific geometries or potentials.

3. The "Abstract" Theory: Semigroup Theory (C_0-Semigroups)

This is the modern mathematical framework for time-evolution operators S_t.

- **The Setup:** You treat S_t as a "one-parameter semigroup" $T(t)$ acting on a Banach space.
- **The Theorem:** The **Hille-Yosida Theorem**.
- **What it does:** It characterizes exactly when a differential operator L generates a stable time-evolution S_t. While Hille-Yosida doesn't guarantee *diagonalizability* (it only guarantees existence and uniqueness of the flow), it is the starting point for analyzing the spectrum of S_t.
- **Spectral Mapping Theorem:** This theorem relates the eigenvalues of the generator L to the eigenvalues of the propagator S_t (basically $\sigma(S_t) = e^{t\sigma(L)}$). If you know the spectrum of L, you know the spectrum of S_t.

4. The "Non-Normal" Theory: Riesz Bases and Spectral Operators

What if S_t is diagonalizable but the eigenvectors are **not orthogonal**? (This happens in non-conservative systems, like convection-diffusion with non-standard boundary conditions).

- **The Theory: Non-Self-Adjoint Spectral Theory** (or the theory of **Riesz Bases**).

- **What it says:** You can still expand solutions in a basis of eigenfunctions, but because the basis is not orthogonal ($\langle \varphi_n, \varphi_m \rangle \neq \delta_{nm}$), you need a "dual basis" $\{\psi_n\}$ to find the coefficients.
  - Expansion: $f = \sum a_n \varphi_n$
  - Coefficients: $a_n = \langle f, \psi_n \rangle$ (instead of $\langle f, \varphi_n \rangle$).
- **Why it matters:** This is currently a very hot topic in fluid dynamics and laser physics, where systems are often "almost" normal but mathematically non-self-adjoint.

Summary: Which "Theory" do you need?

If your operator S_t is...	**The Theory is called...**	**The Basis is...**
Translation Invariant	Harmonic Analysis	Fourier Transform
Self-Adjoint / Normal	Spectral Theory	Orthonormal Eigenbasis
1D Variable Coeffs	Sturm-Liouville Theory	Special Functions
Not Normal (but diag.)	Non-Self-Adjoint Theory	Riesz Basis (Biorthogonal)
Not Diagonalizable	Jordan Form	Generalized Eigenvectors

8.6 Convolution and Approximate Identities

In this section, we introduce a new way of multiplying two functions that will be very useful and very natural. The naturality comes from extending the addition of $\mathbb{R}$ to functions on $\mathbb{R}$, for simplicity we stick to real functions however nothing changes if we use complex functions instead. Consider the function $\delta_a : \mathbb{R} \to \mathbb{R}$

$$\delta_a(x) = \begin{cases} 1 & x = a \\ 0 & \text{otherwise} \end{cases}$$

The naturality comes from the discrete case, and so we will start with it: recall that if we have two polynomials $p(x)$ and $q(x)$ (which can be thought of as formal sums, bringing in the naturality), then the multiplication formula which groups the terms of the same degree is:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{i+j=k} p_i q_j\right) x^k = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{l=0}^{k} p_l q_{k-l}\right) x^k$$

If we set p_k or q_{l-k} to zero if k or $l-k$ is negative, then we can re-write the sum as:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{-\infty}^{\infty} p_l q_{k-l}\right) x^k$$

If we let $p[n]$ and $q[n]$ represent the coefficient of the nth degree term, and $h = pq$, then:

$$h[n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

We can write this as:

$$h[n] = [p * q][n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

where $p*q$ is a compact notation the right hand side. The summation can be thought of as repersenting the discrete measure. Generalized to the continuous case, this is called *convolution*:

Definition 8.6.1: Convolution

Let f, g be measurable on $\mathbb{R}^n$. Then the *convolution* of f and g is a function $f * g : \mathbb{R}^n \to \mathbb{R}$ defined by:

$$f * g(x) := \int f(x-y)g(y)dy$$

for all x such that the integral exists

To see this in action, Let $f = \chi_{[0,1]}$ and $g = \chi_{[0,1]}$, and you'll find that $f * g$ is a a triangle. You can imagine sliding the graph of f and finding the area of the intersection of f and g, and plotting it. There are many ways to make sure that $f * g$ is a.e. defined. For example, if f was bounded and compactly supported, g can be locally integrable. Note that in many cases, we will need that $K(x, y) = f(x - y)$ be measurable on $\mathbb{R}^n \times \mathbb{R}^n$. If we let $s(x, y) = x - y$, then since s is continuous, if f is Borel measurable, so is K. (Folland makes a comment here saying we can "assume" that this is the case, which I'm guessing is f being Borel measurable in the definition of convolution, because of his theorem 2.12, and he says the same can happen for Lebesgue measurability). Usually, we will use convolution to somehow combine the property of two functions f and g. We will soon see that convolution often preserves the "nice" properties the "nicer" function, so that we would convolution, say, a smooth function f with an integrable function g to get a smooth function $f * g$.

I would also be remiss if I don't mention that convolution is like multiplying two functions in a way the respects operation the *domain*. If f, g are real functions, then $f \cdot g$ respects the multiplication in the *codomain* as it multiplies the outputs. As $f * g$ is the continuous case of polynomial multiplication which multiplies, it respects the multiplication happening in the *domain*. To motivate why this is worth pointing out, consider a finite group G, $f \in \mathrm{Hom}_{\mathbf{Set}}(G, \mathbb{R})$, and interpret it as an element of $\mathbb{C}[G]$ so that $f = \sum_i a_i g_i$. Then we can always make $r \in \mathbb{R}$ act on f via:

$$r \cdot f(g_i) = rf(g_i)$$

However, we can also make $g \in G$ act on f:

$$g \cdot f(g_i) = f(g^{-1}g_i)$$

where we inverse for associativity:

$$\begin{aligned}
(g \cdot (h \cdot f))(x) &= (h \cdot f)(g^{-1}x) \\
&= f(h^{-1}(g^{-1}x)) \\
&= f((h^{-1}g^{-1})x) \\
&= f((gh)^{-1}x) \\
&= ((gh) \cdot f)(x)
\end{aligned}$$

This is the same thing as defining $g \cdot f$ given $f \in \mathbb{C}[G]$ since:

$$g \cdot f = g \cdot \sum_{g_i \in G} a_i g_i = \sum_{g_i \in G} a_i (gg_i) = \sum_{g^{-1}g_i \in G} a_i g_i$$

Thus, we may think of the translation operation as an action from the structure of the *domain*, when the domain has a group structure. It shall turn out that the way of characterizing the orthonormal basis for L^2 that gives rise to the Fourier transform is to choose the one that's invariant under τ_y.

Definition 8.6.2: Translation Operator

If $f : \mathbb{R}^n \to \mathbb{R}$, then

$$y * f(x) = \tau_y f(x) = f(x - y)$$

Notice that:

$$\|\tau_y f\|_p = \|f\|_p \qquad \|\tau_y f\|_u = \|f\|_u$$

Using this, we can give another definition of uniformally continuous: f is uniformally continuous if:

$$\|\tau_y f - f\|_u \to 0$$

Lemma 8.6.3: Compact Support Functions And UC

If $f \in C_c(\mathbb{R}^n)$, then f is uniformally continuous

***Proof*:**
should be easy proof (Folland p.238)

Proposition 8.6.4: Translation Is Continuous In Lp

If $1 \leq p < \infty$, then translation is continuous in the L^p norm, in particular if $f \in L^p$ and $z \in \mathbb{R}^n$, then

$$\lim_{y \to 0} \|\tau_{y+z} f - \tau_z f\|_p = 0$$

Note this proposition is false when $p = \infty$

***Proof*:**
Folland p. 238

The following results all work over $\mathbb{R}^n$ or $\mathbb{T}^n$ (interpreted as the unit cube)

Proposition 8.6.5: Algebraic Properties Of Convolution

Assume all the following integrals are defined and converge absolutely. Then:

1. $f * g = g * f$
2. $(f * g) * h = f * (g * h)$ (this result in particular requires absolute convergence)
3. $f * (g + h) = f * g + f * h$
4. $a(f * g) = (af) * g$ for all $a \in \mathbb{C}$
5. for all $z \in \mathbb{R}^n$, $\tau_z(f * g) = (\tau_z f) * g = f * (\tau_z g)$
6. If A is the closure of $\{x + y \mid x \in \text{supp}(f),\ y \in \text{supp}(g)\}$, then $\text{supp}(f * g) \subseteq A$

***Proof*:**

1. $$\begin{aligned} f * g(x) &= \int f(x-y)g(y)dy \\ &= \int f(z)g(x-z)dz && z = x - y,\ y = x - z \\ &= \int g(x-z)f(z)dz \\ &= g * f(x) \end{aligned}$$
2. Follows from (1) from Fubini's theorem
3. Follows from direct computation
4. Follows from direct computation
5. Follows from direct computation and (1)
6. If $x \notin A$, then for any $y \in \text{supp}(g)$, we have $x - y \notin \text{supp}(f)$, thus $f(x-y)g(y) = 0$, i.e. $f * g(x) = 0$

One of the first properties important to establish is the relation between convolution of a function and p-norms:

Theorem 8.6.6: Young's Convolution Inequality

Let $1 \leq p, q, r \leq \infty$ be such that $p^{-1} + q^{-1} = r^{-1} + 1$, and let $f \in L^p$ and $g \in L^q$. Then $f * g(x)$ exists for a.e. x, $f * g \in L^r$ and

$$\|f * g\|_r \leq \|f\|_p \|g\|_q$$

Notice in the statement that we have something a little different than the usual conjugate exponents. To see why, there is another thing worth pointing out, which is that this condition is stable under scaling. In particular, if $f_\lambda(x) = f(\lambda x)$ and $g_\lambda(x) = g(\lambda x)$. Then:

$$\begin{aligned}
\|f_\lambda\|_p &= \left(\int_{\mathbb{R}^n} f(\lambda x)^p\right)^{1/p} \\
&= \left(\int_{\mathbb{R}^n} \lambda^{-n} f(y)^p\right)^{1/p} && y = \lambda x,\ dy = \lambda^n dx \\
&= \lambda^{-n/p}\|f\|_p
\end{aligned}$$

where λ^n is the determinant of the Jacobian matrix. So:

$$\begin{aligned}
(f_\lambda * g_\lambda)(x) &= \int_{\mathbb{R}^n} f(\lambda x - \lambda y) g(\lambda y) dy \\
&= \lambda^{-n}(f * g)(\lambda x)
\end{aligned}$$

Thus we have:

$$\begin{aligned}
\|f_\lambda * g_\lambda\|_r &= \lambda^{-n}\|(f * g)_\lambda\|_r \\
&= \lambda^{-n/r-n}\|f * g\|_r \\
&\leq \lambda^{-n/p-n/q}\|f\|_p\|g\|_q
\end{aligned}$$

Notice that this inequality must hold for *all* $\lambda > 0$. If the exponents of λ on the left and right sides were different, we could force a contradiction by taking $\lambda \to 0$ or $\lambda \to \infty$ (one side would vanish while the other explodes). Thus, the exponents must be equal:

$$-n\left(1 + \frac{1}{r}\right) = -n\left(\frac{1}{p} + \frac{1}{q}\right)$$

Dividing out the $-n$, we recover the necessary condition:

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$$

Intuitively, the "extra" $+1$ comes from the integration dy in the definition of convolution, which adds n dimensions of volume (unlike pointwise multiplication which preserves the exponent balance of Hölder's inequality).

***Proof*:**
First, consider $r = \infty$, meaning $r^{-1} = 0$, $p^{-1} + q^{-1} = 1$. So, we want to show $\|f * g\|_\infty \leq \|f\|_p\|g\|_q$. So:

$$\begin{aligned}
\|f * g\|_\infty &= \sup_x \left|\int f(x - y) g(y) dy\right| \\
&\leq \|f(x - \cdot)\|_p\|g\|_q \\
&\stackrel{!}{=} \|f\|_p\|g\|_q
\end{aligned}$$

where $\stackrel{!}{=}$ comes from $\int |h(x - y)| dy = \int |h(y)| dy$ where $z = x - y$, $dz = (-1)^n dz$ so the $(-1)^n$ dissapears in the absolute value. (Note: I corrected the L^1 typo to L^∞ here to match the case $r = \infty$).

Next, if p or q is 1 (say $p = 1$), then $1 + q^{-1} = r^{-1} + 1$, so $q^{-1} = r^{-1}$. Then we would have:

$$\|f * g\|_q \leq \|f\|_1 \|g\|_q$$

To see that this happens, we use theorem 8.4.2. In particular, let $K(x, y) = f(x - y)$. Then if $\|g\|_q$ exists, $\|K(x, \cdot)\|_1 = \|f\|_1$ and $\|K(\cdot, y)\| = \|f\|_1$, giving us the result.

Now, assume $1 < p < r$, $1 < q < r$. First, we manipulate it to be able to use Hölder's inequality:

$$|f(y)g(x-y)| = (|f(y)|^p |g(x-y)|^q)^{1/r} \, |f(y)|^{1-p/r} |g(x-y)|^{1-q/r}$$

Next, notice that:

$$\frac{1}{r} + \frac{r-q}{rq} + \frac{r-p}{rp} = \frac{1}{r}\left(1 + \frac{r}{q} - 1 + \frac{r}{p} - 1\right) = 1$$

So, we can apply general Hölder's inequality since $r - p, r - q \neq 0$ which gives us:

$$|(f * g)(x)| \leq \left(\int_{\mathbb{R}^n} |f(y)^p||g(x-y)|^q dy\right)^{1/r} \cdot \|f\|_p^{(r-p)/r} \|g\|_q^{(r-q)/r}$$
$$\Rightarrow \; |(f * g)(x)|^r \leq \left(\int_{\mathbb{R}^n} |f(y)^p||g(x-y)|^q dy\right) \cdot \|f\|_p^{r-p} \|g\|_q^{r-q}$$

And now, we can integrate the left hand side with respect to x and use Tonnelli's theorem:

$$\Rightarrow \; \|(f * g)\|_r^r \leq \|f\|_p^p \|f\|_p^{r-p} \|g\|_q^q \|g\|_q^{r-q} = \|f\|_p^r \|g\|_q^r$$

completing the proof.

If we focus on the special case of $p = q = r = 1$, then $f * g \in L^1(\mathbb{R}^n)$. Then proposition 8.6.5 and Young's inequality (theorem 8.6.6) turns $L^1(\mathbb{R}^n)$ into a "commutative associative algebra". Note that this algebra has no multiplicative identity (i.e. We are working over a *rng*). It does have a notion of an identity in a limit, which we will shortly explore. In fact, if we slightly correctly loosen our notion of function to *distribution*, we shall naturally recover an identity (explored in chapter 9). Note that if $f, g \in L^2(\mathbb{R}^n)$ then $f * g \in L^\infty(\mathbb{R}^n)$, namely convoluting forces the functions the result to be bounded.

Before exploring a notion of identity, it must be shown that convoluting two functions where one nicer than the other usually keeps the properties of the nicer function:

Lemma 8.6.7: Convolution Preserves C_c^k

Let $f \in L^1$ and $g \in C_c^k$ (more generally, $\partial^\alpha g$ is bounded for $|\alpha| \leq k$). Then

$$f * g \in C^k$$

and

$$\partial^\alpha (f * g) = (f * \partial^\alpha g)$$

***Proof*:**
In Folland's textbook this follows from differentiation under the integral sign.

This is a matter of computation. We'll do the case of $k = 1$, since the rest is a matter of induction. First, write what you know about $f * g$:

$$\begin{aligned} f * g(x) &= \int_{\mathbb{R}^n} f(x-y)g(y)dy \\ &= \int_{\mathbb{R}^n} f(y)g(x-y)dy \end{aligned}$$

So consider

$$\lim_{h\to 0} \frac{f * g(x+h) - f * g(x)}{h} = \lim_{h\to 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy$$

Notice that:

$$\left| \frac{g(x+h-y) - g(x-y)}{h} \right| \leq \frac{|g'(c)| \cdot h}{h}$$

meaning we have a dominating function, and so we can apply dominated convergence theorem to move the limit in!

$$\begin{aligned} \lim_{h\to 0} \frac{f * g(x+h) - f * g(x)}{h} &= \lim_{h\to 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \lim_{h\to 0} \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \end{aligned}$$

and since $g \in C_c^1$ (in particular, g is differnetiable), this limit exists, and since g has comapct support, the integral is still well-defined and is C^1 (see the remarks we made at the beginning).

From here we can apply the induction, and the proof is done.

Proposition 8.6.8: Smooth Function Dense in L^p

Compact smooth functions are dense in L^p.

***Proof*:**
hint: simple functions are dense in L^p. Convolute them with a smooth bump function.

With this information, we are going to define a notion of "multiplicative idenity"

Definition 8.6.9: Good Kernels

Let $\{\varphi_t\}_{t>0}$ be a set of functions of the form $\varphi_t : \mathbb{R}^n \to \mathbb{C}$. Then they are called a *good kernel* if $t \in (0, \epsilon_0)$ and:

1. $\sup_{t>0} \|\varphi_t\|_1 < \infty$
2. $\int_{\mathbb{R}^n} \varphi_t(x)dx = 1$
3. $\lim_{t\to 0} \int_{|x|>\delta} |\varphi_t(x)|dx = 0$ for all $\delta > 0$

Theorem 8.6.10: Good Kernels Like Mult. Identity

Let $f \in L^p$ for $1 \leq p < \infty$ (note it is $<$, not $\leq$), then:

$$f * \varphi_t \to f$$

in L^p as $t \to \infty$. Furthermore, if f is continuous at some x_0, then:

$$f * \varphi_t(x_0) \to f(x_0)$$

in L^p

***Proof*:**
It is equivalent to show $f * \varphi_t - f \to 0$, so for all x:

$$\begin{aligned}
f * \varphi_t(x) - f(x) &= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - f(x)\cdot 1 \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - f(x)\cdot \int_{\mathbb{R}^n} \varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - \int_{\mathbb{R}^n} f(x)\varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y) - f(x)\varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} [f(x-y) - f(x)]\varphi_t(y)dy \\
&= \int_{|y|\leq\delta} [f(x-y) - f(x)]\varphi_t(y)dy + \int_{|y|>\delta} [f(x-y) - f(x)]\varphi_t(y)dy
\end{aligned}$$

and so, taking the p-norm we get:

$$\|f * \varphi_t - f\|_p \leq \int_{|y|\leq\delta} \|f(\cdot - y) - f(\cdot)\|_p |\varphi_t(y)|dy + \int_{|y|>\delta} 2\|f\|_p |\varphi_t(y)|dy$$

at this point, we are almost done. If we let $c = \sup_{t>0} \int_{\mathbb{R}^n} |\varphi_t(y)|dy$, then we get:

$$\limsup_{t\to 0} \|f * \varphi_t - f\|_p \leq c \limsup_{\delta\to 0,\ |y|\leq\delta} \|f(\cdot - y) - f(\cdot)\|_p = 0$$

where the last inequality comes from the shrinking delta.

It was left as an exercise to do the point-convergence in the continuous case.

Note that if f is uniformally continuous and bouned, then $f * \varphi_t \to f$ uniformally, and if $\int \varphi = a$, then $f * \varphi_t \to af$.

Example 8.6.11: Good Kernels

Let $X(x) \in C_c^\infty(\mathbb{R}^n)$ where $\int X = 1$. Then:

$$\varphi_t(x) = t^{-n} X(x/t)$$

is a good kernel (ex. let $X(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}$). This is just a matter of computation. Note that $\varphi_t = 0$ if $\frac{x}{t} \geq c_0$ for some c_0, and so $t \leq \frac{x}{c_0}$. It [somehow] follows that C_c^∞ is *dense* in L^p $(1 \leq p < \infty)$ (note the $<$ and not $\leq$)

For another Another example of a family of good kernel, take:

$$K_\delta(x) = \delta^{-1/2} e^{\frac{-\pi x^2}{\delta}}$$

These can be visualized here:

https://www.desmos.com/calculator/e8xc37r3ck

Notice that since the function $X \in C_c^\infty(\mathbb{R})$ is smooth on compact supporet then by lemma 8.6.7, $f * X$ is also smooth, and will remain smooth for every φ_n. Since $f * \varphi_n \to f$, we see that a sequence of smooth functions converges to an L^p function, meaning smooth functions are *dense* in L^p!

I found this interesting stack-exchange post on the importance of good Kernels:

https://math.stackexchange.com/questions/3294744/good-kernels-that-exist-in-the-real-world

Using the fact that convolution preserves the "nicer" structure, we strengthen's Urysohn's lemma (or, if you know some differential geometry, generlizes partitions of unity)

Theorem 8.6.12: Smooth Urysohn's Lemma

Let $K \subseteq \mathbb{R}^n$ be compact and $U \subseteq \mathbb{R}^n$ be open and contain K $(U \supseteq K)$. Then there exists a $f \in C_c^\infty$ such that $0 \leq f \leq 1$ and $f \equiv 1$, K and $\operatorname{supp}(K) \subseteq f$.

***Proof*:**

Since K is compact, then $\delta = \rho(K, U^c)$ (the shortest distance from K and U^c) is non-zero. We are going to use a mollification trick: let $V = \left\{x \;\middle|\; \rho(x, K) < \frac{\delta}{3}\right\}$. Choose some φC_c^∞ such that $\int \varphi = 1$ and $\varphi(x) = 0$ for $|x| \geq \delta/3$ (as an example, $\{\psi_t\}$ to be a good kernel and take $\varphi = \left(\int \psi\right)^{-1} \psi_{\delta/3}$).

Now, let $f = \chi_V * \varphi$. Then as we've shown before, $f \in C_c^\infty$, and it is easy to see that $0 \leq f \leq 1$, $f \equiv 1$ on K, and $\operatorname{supp}(f) \subseteq \{x \mid \rho(x, K) \leq 2\delta/3\} \subseteq U$

9

Distribution Theory

Distributions are a generalization of functions that allow us to give more general solutions than classical solutions, similar to how we found a weak solution last week. Locally integrable functions will be distributions (in the appropriate interpretation), but we'll see that there are generalized solutions that don't even fall under the notion of "function".

More broadly, distribution theory is a further exploration of the "dual theory" philosophy that has been a recurring theme in our study of analysis. In functional analysis, we repeatedly found that passing to the dual space reveals structure invisible in the original space: the Riesz representation theorem identified $(L^p)^*$ with L^q, and the Hahn-Banach theorem showed that the dual space is always rich enough to separate points. Distribution theory pushes this idea one step further. Rather than studying functions directly, we study what they do to test functions—that is, we study the linear functionals they induce. By broadening the class of allowable functionals beyond those induced by honest functions, we arrive at a theory in which every distribution is infinitely differentiable, every linear constant-coefficient PDE has a solution, and the Fourier transform becomes an isomorphism. In short, duality gives us completeness of operations that the classical function spaces lack.

Another use we will soon see is the intimate connection between distributions and the Fourier transform. The theory of tempered distributions will allow us to extend the Fourier transform to objects far beyond L^1 or L^2, giving us a powerful and unified framework for studying partial differential equations, signal processing, and harmonic analysis.

9.1 Intuition

For now, let $f \in L^p$. Then as we know, $(L^p)^* \cong L^q$ is an isometric isomorphism where $1 \leq p < \infty$ and q is the conjugate exponent. In particular, f is mapped to the bounded linear functional

$$g \mapsto \int fg.$$

Using the Lebesgue differentiation theorem, by choosing $\varphi_r = m(B_r)^{-1}\chi_{B_r}$, where B_r is a ball of radius r centered at x, we can recover the value of $f(x)$ for a.e. x, as $\lim_{r\to 0} \int f\varphi_r$. In this way, we can think of L^p as being embedded into $(L^q)^*$: the function f is completely determined (up to a.e. equivalence) by the values of the functional it defines.

Now, let us modify the above by letting f be locally integrable (in L^1_{loc}), but restricting our test functions to $\varphi \in C_c^\infty$. Then it is clear that $\varphi \mapsto \int f\varphi$ is still a well-defined linear functional on C_c^∞: the integral converges since φ has compact support and f is integrable on compact sets. Moreover, the pointwise values of f can be recovered a.e. from this functional by the same approximation argument (see Theorem 8.15 in Folland, the Borel Representation Theorem). So locally integrable functions embed faithfully into the space of linear functionals on C_c^∞.

Here is the key observation: unlike in the case for L^p spaces, there are in fact *many* linear functionals on C_c^∞ that are *not* of the form $\varphi \mapsto \int f\varphi$ for any locally integrable f. The most familiar example is the Dirac delta, $\varphi \mapsto \varphi(0)$, which "evaluates at a point"—something no L^1_{loc} function can do. These functionals, subject to some mild continuity conditions we will soon specify, will be our "generalized functions", or *distributions*.

9.2 Definitions

We will first rapid-fire some important definitions to be up to speed with terminology. Recall that $C_c^\infty(E)$ where $E \subseteq \mathbb{R}^n$ is the set of all smooth functions whose support is compact and contained in E. If $U \subseteq \mathbb{R}^n$ is open, $C_c^\infty(U)$ is the union of spaces $C_c^\infty(K)$ where K ranges over all compact subsets of U. Each $C_c^\infty(K)$ is a Fréchet space with the topology defined by the family of seminorms

$$\varphi \mapsto \|\partial^\alpha \varphi\|_u \qquad (\alpha \in \{0, 1, 2, \ldots\}^n).$$

Thus, a sequence $\{\varphi_i\}$ converges in $C_c^\infty(K)$ if and only if $\partial^\alpha\varphi_i \to \partial^\alpha\varphi$ uniformly for all multi-indices α. This is a strong form of convergence: we demand that not only the functions themselves, but all of their derivatives of every order, converge uniformly. The completeness of $C_c^\infty(K)$ under this topology is established through Exercise 9 in Folland, Section 5.1 (the key idea is that a uniform limit of smooth functions, where all derivatives also converge uniformly, is again smooth). With these in hand, we take the following definitions:

1. A sequence $\{\varphi_i\}$ in $C_c^\infty(U)$ *converges in* $C_c^\infty(U)$ to φ if there exists a compact set $K \subseteq U$ such that $\{\varphi_i\} \subseteq C_c^\infty(K)$ and $\varphi_i \to \varphi$ in the topology of $C_c^\infty(K)$, that is, $\partial^\alpha\varphi_i \to \partial^\alpha\varphi$ uniformly for all α. Note the requirement that all the φ_i are supported in a *single* compact set K; the supports are not allowed to "escape to infinity".
2. If X is a locally convex topological vector space and $T : C_c^\infty(U) \to X$ is a linear map, T is *continuous* if $T|_{C_c^\infty(K)}$ is continuous for each compact $K \subseteq U$, that is, if $T(\varphi_i) \to T(\varphi)$

whenever $\varphi_i \to \varphi$ in $C_c^\infty(K)$ for some compact $K \subseteq U$. In other words, continuity is tested "one compact set at a time".

3. A linear map $T : C_c^\infty(U) \to C_c^\infty(U')$ is *continuous* if for each compact $K \subseteq U$, there is a compact $K' \subseteq U'$ such that $T(C_c^\infty(K)) \subseteq C_c^\infty(K')$ and T is continuous from $C_c^\infty(K)$ to $C_c^\infty(K')$. The first condition says that T does not spread out the support in an uncontrolled way; the second says that the map is continuous in the Fréchet topology.

With these, we will define a distribution:

Definition 9.2.1: Distribution

A *distribution* on U is a continuous linear functional on $C_c^\infty(U)$. The space of all distributions on U is denoted by $\mathcal{D}'(U)$, and we set $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$. We impose the weak* topology on $\mathcal{D}'(U)$, that is, the topology of pointwise convergence on $C_c^\infty(U)$.

The *evaluation* of a distribution at a test function φ is often denoted either as:

$$F(\varphi) \qquad \text{or} \qquad \langle F, \varphi \rangle$$

A couple of remarks are in order. First, the $\mathcal{D}'$ notation comes from the fact that Schwartz used the notation $\mathcal{D}$ for C_c^∞, and so $\mathcal{D}'$ denotes the continuous dual of C_c^∞—the space of all continuous linear functionals on it. Next, there is in fact a locally convex topology on $C_c^\infty(U)$ (the so-called *inductive limit topology*) with respect to which sequential convergence is given by the convergence defined above, and the continuity of linear maps $T : C_c^\infty \to X$ and $T : C_c^\infty \to C_c^\infty$ agrees with the definitions in the list above. However, defining this topology rigorously requires the machinery of inductive limits of locally convex spaces, and the construction is somewhat involved without contributing much additional insight to the current exposition of distributions, so it will be omitted. The interested reader may consult Rudin's *Functional Analysis*, Chapter 6, for the full treatment. Furthermore, when we talk about convergence of distributions $F_i \to F$, we will always mean pointwise convergence: $\langle F_i, \varphi \rangle \to \langle F, \varphi \rangle$ for every $\varphi \in C_c^\infty(U)$.

Before giving some examples of distributions, we will remind the reader how to check for continuity of a linear functional in the setting of Fréchet spaces:

Proposition 9.2.2: Distribution Continuity Check

Let $F : C_c^\infty(U) \to \mathbb{R}$ (or $\mathbb{C}$) be linear. Then $u \in \mathcal{D}'(U)$ if and only if for every compact set $K \subseteq U$, there exist constants $C_K > 0$ and $N_K \in \mathbb{N}$ such that

$$|\langle F, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

for all $\varphi \in C_c^\infty(K)$.

***Proof*:**
Suppose first that the estimate holds. We wish to show that u is continuous on each $C_c^\infty(K)$. Let $\varphi_i \to \varphi$ in $C_c^\infty(K)$. Then for every multi-index α, $\partial^\alpha \varphi_i \to \partial^\alpha \varphi$ uniformly, which means

$\|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \to 0$ for each α. Applying the estimate to $\varphi_i - \varphi$:

$$|\langle u, \varphi_i\rangle - \langle u, \varphi\rangle| = |\langle u, \varphi_i - \varphi\rangle| \leq C_K \sum_{|\alpha|\leq N_K} \|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \to 0,$$

so $\langle u, \varphi_i\rangle \to \langle u, \varphi\rangle$, and u is continuous on $C_c^\infty(K)$. Since K was arbitrary, u is a distribution. Conversely, suppose u is continuous on $C_c^\infty(K)$ but no such estimate holds. Then for each $j \in \mathbb{N}$, there exists $\varphi_j \in C_c^\infty(K)$ with

$$|\langle u, \varphi_j\rangle| > j \sum_{|\alpha|\leq j} \|\partial^\alpha \varphi_j\|_{L^\infty}.$$

Set $\psi_j = \varphi_j / |\langle u, \varphi_j\rangle|$. Then $\langle u, \psi_j\rangle = 1$ for all j (taking absolute values and adjusting the sign if needed, we can arrange $|\langle u, \psi_j\rangle| = 1$). But for $|\alpha| \leq j$:

$$\|\partial^\alpha \psi_j\|_{L^\infty} < \frac{1}{j},$$

so $\psi_j \to 0$ in $C_c^\infty(K)$. By continuity of u, we should have $\langle u, \psi_j\rangle \to \langle u, 0\rangle = 0$, contradicting $|\langle u, \psi_j\rangle| = 1$.

Definition 9.2.3: Order of Distribution

Let $F \in \mathcal{D}'(U)$ be a distribution and compact $K \subseteq U$ so that by proposition 9.2.2 there exists a C_K and N_K such that

$$|\langle F, \varphi\rangle| \leq C_K \sum_{|\alpha|\leq N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

then N_K is called the *order* of F on K. If there exists an $N \in \mathbb{N}$ such that for all compact $K \subseteq U$:

$$|\langle F, \varphi\rangle| \leq C_K \sum_{|\alpha|\leq N} \|\partial^\alpha \varphi\|_{L^\infty}$$

then N is called the *order* of F. If no such N exists, then F is said to have infinite order.

The way I like to think about the continuity check is as follows: we will soon see that there are many distribution spaces, for example the Schwartz space $\mathcal{S}$ will have a parallel $\mathcal{S}'$ (the *tempered distributions*). What the continuity condition tells us is that a distribution respects the convergence structure of its test function space. If $\varphi_j \to \varphi$ in the test function topology, then $\langle u, \varphi_j\rangle \to \langle u, \varphi\rangle$. Hence, we can translate convergence information from the test function space to information about the distribution.

Example 9.2.4: Distributions

1. *Locally integrable functions.* For every $f \in L^1_{\text{loc}}(U)$, we can define a distribution on U, namely the functional $\varphi \mapsto \int f\varphi$. This is indeed continuous: if $\varphi \in C_c^\infty(K)$, then

$$\left|\int f\varphi\right| \leq \|f\|_{L^1(K)} \|\varphi\|_{L^\infty},$$

so the estimate in Proposition 9.2.2 holds with $N_K = 0$ and $C_K = \|f\|_{L^1(K)}$. Moreover, two functions f and g define the same distribution precisely when they are equal a.e. (this follows from the fact that C_c^∞ is dense in L^q for the appropriate q, or more directly from the Lebesgue differentiation theorem). In this way, $L^1_{\text{loc}}(U)$ embeds into $\mathcal{D}'(U)$, and we often identify a locally integrable function with the distribution it defines.

2. *Radon measures.* Every Radon measure μ on U defines a distribution $\varphi \mapsto \int \varphi \, d\mu$. The continuity check is similar: for $\varphi \in C_c^\infty(K)$, we have $|\int \varphi \, d\mu| \leq |\mu|(K)\|\varphi\|_{L^\infty}$, where $|\mu|(K)$ is the total variation of μ on K. Again, this satisfies the estimate with $N_K = 0$. Note that this generalizes the previous example, since a locally integrable function f defines a Radon measure $d\mu = f \, dm$ via the Radon-Nikodym theorem. In fact, by the Radon-Nikodym theorem, it is the singular part of the measure that create distributions that are not representable by functions.

3. *Derivatives of point masses.* If $x_0 \in U$ and α is a multi-index, the map $\varphi \mapsto \partial^\alpha \varphi(x_0)$ is a distribution. The continuity estimate is immediate: $|\partial^\alpha \varphi(x_0)| \leq \|\partial^\alpha \varphi\|_{L^\infty}$. This distribution does not arise from a locally integrable function: no L^1_{loc} function can "see" the value of a derivative at a single point. It arises from a Radon measure μ precisely when $\alpha = 0$, in which case $\mu = \delta_{x_0}$ is the point mass at x_0. For $|\alpha| \geq 1$, these distributions are genuinely "beyond" measures—they are the first examples of distributions that cannot be represented by any kind of integration.

4. *measure on cantor set* Another example of a singular distribution that is not the point-mass above is the measure on the cantor set. Once differentiation is defined, we shall see that this distribution is the derivative of the devil's staircase function (or cantor functions), see ref:HERE.

As mentioned earlier, we often associate a function f with the distribution it defines, and even label the distribution by f. For this reason, it might be confusing to write $f(\varphi)$ to mean the distribution defined by f evaluated on φ, and so for historical reasons we use the bracket notation $\langle f, \varphi \rangle$ for the distributional pairing. This notation is suggestive of an inner product (and indeed, when $f \in L^2$ and $\varphi \in C_c^\infty$, it literally is the L^2 inner product), though the reader should keep in mind that the pairing is between two different spaces. The use of the inner product notation should bring to mind Riesz-Representation theorem (theorem 11.1.13) where the inner product is the "perfect pairing" between a space and its dual.

Let's next establish some common notation:

1. First, it is often enough that we will reflect a function. Thus, if we put a tilde over a function, we mean:
$$\tilde{\varphi}(x) = \varphi(-x).$$
This notation will appear frequently in the context of convolutions, where the reflection arises naturally from the change of variables $y \mapsto -y$.

2. Second, the point-mass (or Dirac delta) is a very common distribution, which we will denote by δ:
$$\langle \delta, \varphi \rangle = \varphi(0).$$
More generally, δ_{x_0} denotes the point mass at x_0, defined by $\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0)$. When $x_0 = 0$, we simply write δ.

As an example of the use of the delta function, and to illustrate how convergence of distributions works in practice:

Proposition 9.2.5: Convergence To Delta Function

Let $f \in L^1(\mathbb{R}^n)$ with $\int f = a$, and for $t > 0$, let $f_t(x) = t^{-n} f(t^{-1}x)$. Then $f_t \to a\delta$ in $\mathcal{D}'$ as $t \to 0$.

Before the proof, let us unpack what this says. The rescaling $f_t(x) = t^{-n} f(t^{-1}x)$ concentrates the mass of f near the origin as $t \to 0$: the factor t^{-n} ensures that $\int f_t = \int f = a$ for all t, while the argument $t^{-1}x$ compresses the support. So f_t becomes a taller and narrower spike near 0, with constant total mass a. The proposition says that in the distributional sense, this spike converges to a times the delta function—exactly what our intuition would predict.

***Proof*:**
Let $\varphi \in C_c^\infty$. By Theorem 8.15 (Folland), we have:

$$\langle f_t, \varphi \rangle = \int f_t(x)\varphi(x)\, dx = \int t^{-n} f(t^{-1}x)\varphi(x)\, dx.$$

Performing the substitution $y = t^{-1}x$ (so $x = ty$ and $dx = t^n\, dy$):

$$\int t^{-n} f(t^{-1}x)\varphi(x)\, dx = \int f(y)\varphi(ty)\, dy.$$

Now we recognize the right-hand side as $f_t * \tilde{\varphi}(0)$. As $t \to 0$, $\varphi(ty) \to \varphi(0)$ pointwise, and since $|\varphi(ty)| \le \|\varphi\|_{L^\infty}$ and $f \in L^1$, the dominated convergence theorem gives:

$$\langle f_t, \varphi \rangle = \int f(y)\varphi(ty)\, dy \to \varphi(0) \int f(y)\, dy = a\varphi(0) = a\langle \delta, \varphi \rangle,$$

completing the proof.

For equality of distributions, note that it makes sense to say that $F = G$ on an open set $V \subseteq U$ if and only if $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all $\varphi \in C_c^\infty(V)$. If F and G are continuous functions, this implies they are pointwise equal on V, and if F and G are only locally integrable, this means they are equal a.e. on V.

Since a function in $C_c^\infty(V_1 \cup V_2)$ need not be supported in either V_1 or V_2 individually, it is not immediately obvious that if $F = G$ on V_1 and on V_2, then $F = G$ on $V_1 \cup V_2$. However, this is indeed the case, thanks to partitions of unity:

Proposition 9.2.6: Equality On Support

Let $\{V_\alpha\}$ be a collection of open subsets of U and let $V = \bigcup_\alpha V_\alpha$. If $F, G \in \mathcal{D}'(U)$ and $F = G$ on each V_α, then $F = G$ on V.

***Proof*:**
It suffices to show that $F - G = 0$ on V, so we may assume $G = 0$ and show that $F = 0$ on V. Let $\varphi \in C_c^\infty(V)$. Then $\operatorname{supp}(\varphi)$ is a compact subset of $V = \bigcup_\alpha V_\alpha$. By compactness, finitely many of the V_α cover $\operatorname{supp}(\varphi)$, say $V_{\alpha_1}, \dots, V_{\alpha_m}$. Let $\{\chi_1, \dots, \chi_m\}$ be a smooth partition of unity subordinate to this finite cover, so that each $\chi_j \in C_c^\infty(V_{\alpha_j})$, $0 \le \chi_j \le 1$, and $\sum_{j=1}^m \chi_j = 1$ on a neighborhood of $\operatorname{supp}(\varphi)$. Then $\varphi = \sum_{j=1}^m \chi_j \varphi$, and each $\chi_j \varphi \in C_c^\infty(V_{\alpha_j})$. Since $F = 0$ on each V_{α_j}, we have $\langle F, \chi_j \varphi \rangle = 0$ for each j, and by linearity:

$$\langle F, \varphi \rangle = \sum_{j=1}^m \langle F, \chi_j \varphi \rangle = 0.$$

Since $\varphi \in C_c^\infty(V)$ was arbitrary, $F = 0$ on V.

As a consequence of Proposition 9.2.6, if $F \in \mathcal{D}'(U)$, there is a largest open set on which $F = 0$, namely the union of all open subsets of U on which $F = 0$. The complement of this set in U is called the *support* of F, denoted $\operatorname{supp}(F)$. When F is a locally integrable function, this agrees with the usual notion of [closed] support (up to a set of measure zero).

Definition 9.2.7: Distribution With Compact Support

The set of all distributions on U with compact support is denoted $\mathcal{E}'(U)$

It turns out (though we will not prove it here) that $\mathcal{E}'(U)$ is precisely the dual of $C^\infty(U)$—we no longer need to require that the test functions have compact support, because the distribution itself is compactly supported.

9.3 Operations on Distributions

One of the advantages of generalizing functions to distributions is that even though they are not "functions" in the classical sense, we can perform many of the same operations on them: differentiation, multiplication by a smooth function, convolution, and translation! To achieve this, we use a single unifying principle that we now describe.

Suppose U and V are open subsets of $\mathbb{R}^n$ and T is a linear map from some subspace $\mathcal{V}$ of $L^1_{\mathrm{loc}}(U)$ into $L^1_{\mathrm{loc}}(V)$. Suppose that there is another linear map $T' : C_c^\infty(V) \to C_c^\infty(U)$ (called the *transpose* of T) such that

$$\int (Tf)\varphi = \int f(T'\varphi) \qquad (f \in \mathcal{V},\ \varphi \in C_c^\infty(V)).$$

Suppose further that T' is continuous in the sense defined earlier (i.e., for each compact $K' \subseteq V$, there exists a compact $K \subseteq U$ with $T'(C_c^\infty(K')) \subseteq C_c^\infty(K)$ and T' continuous from $C_c^\infty(K')$ to $C_c^\infty(K)$). Then T can be *extended* to a map from $\mathcal{D}'(U)$ to $\mathcal{D}'(V)$, still denoted by T, by defining:

$$\langle TF, \varphi \rangle = \langle F, T'\varphi \rangle \qquad (F \in \mathcal{D}'(U),\ \varphi \in C_c^\infty(V)).$$

Why does this work? First, TF is a well-defined linear functional on $C_c^\infty(V)$ since $T'\varphi \in C_c^\infty(U)$ and F is a linear functional on $C_c^\infty(U)$. Second, TF is continuous: if $\varphi_i \to \varphi$ in $C_c^\infty(V)$, then $T'\varphi_i \to T'\varphi$ in $C_c^\infty(U)$ by continuity of T', and so $\langle TF, \varphi_i \rangle = \langle F, T'\varphi_i \rangle \to \langle F, T'\varphi \rangle = \langle TF, \varphi \rangle$ since

F is a distribution. Third, this definition is *consistent* with the original map on functions: if F is a locally integrable function in $\mathcal{V}$, then $\langle TF, \varphi\rangle = \langle F, T'\varphi\rangle = \int f(T'\varphi) = \int (Tf)\varphi$, which is exactly the distribution defined by the function Tf.

The continuity of T' also guarantees that the extended map $T : \mathcal{D}'(U) \to \mathcal{D}'(V)$ is continuous with respect to the weak* topology on distributions: if $F_\alpha \to F$ in $\mathcal{D}'(U)$, then for any $\varphi \in C_c^\infty(V)$, $\langle TF_\alpha, \varphi\rangle = \langle F_\alpha, T'\varphi\rangle \to \langle F, T'\varphi\rangle = \langle TF, \varphi\rangle$, so $TF_\alpha \to TF$ in $\mathcal{D}'(V)$.

Using this, we can construct many operations on distributions by transposing the property onto the test functions

1. *(Differentiation)* Let $Tf = \partial^\alpha f$, defined initially on $C^{|\alpha|}(U)$. If $\varphi \in C_c^\infty(U)$, integration by parts gives
$$\int (\partial^\alpha f)\varphi = (-1)^{|\alpha|} \int f(\partial^\alpha \varphi);$$
there are no boundary terms since φ has compact support. Hence $T' = (-1)^{|\alpha|}\partial^\alpha|_{C_c^\infty(U)}$, and we can define the *distributional derivative* $\partial^\alpha F \in \mathcal{D}'(U)$ of any $F \in \mathcal{D}'(U)$ by:
$$\langle \partial^\alpha F, \varphi\rangle := (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle.$$
By this procedure, every distribution is infinitely differentiable! We can define the derivatives of arbitrary locally integrable functions, even when they are not differentiable in the classical sense. When F happens to be a $C^{|\alpha|}$ function, the distributional derivative agrees with the classical one (by integration by parts), so this truly is a generalization. We will compute the distributional derivatives of some important locally integrable functions shortly.
A particularly important consequence: distributional derivatives commute. That is, $\partial^\alpha\partial^\beta F = \partial^{\alpha+\beta} F = \partial^\beta \partial^\alpha F$ for all distributions F and all multi-indices α, β. This is immediate from the definition and the fact that classical derivatives of smooth functions commute.
2. *(Multiplication by a smooth function)* Let $\psi \in C^\infty(U)$ and define $Tf = \psi f$. Then $T' = T|_{C_c^\infty(U)}$ (that is, $T'\varphi = \psi\varphi$), since $\int(\psi f)\varphi = \int f(\psi\varphi)$. Evidently T' maps $C_c^\infty(U)$ to $C_c^\infty(U)$ and is continuous (since multiplication by a smooth function preserves both support and uniform convergence of derivatives, by the Leibniz rule). Thus we define $\psi F \in \mathcal{D}'(U)$ for any $F \in \mathcal{D}'(U)$ by:
$$\langle \psi F, \varphi\rangle := \langle F, \psi\varphi\rangle.$$
Note that we require ψ to be C^∞; multiplying a distribution by a continuous or L^∞ function is in general not well-defined. The Leibniz rule extends to distributions: $\partial^\alpha(\psi F) = \sum_{\beta\le\alpha}\binom{\alpha}{\beta}(\partial^{\alpha-\beta}\psi)(\partial^\beta F)$, which is proved by testing against $\varphi \in C_c^\infty$ and using the classical Leibniz rule on the test function side.
3. *(Translation)* Let $y \in \mathbb{R}^n$ and let $V = U + y = \{x + y : x \in U\}$. Take $T = \tau_y$ (recall that $\tau_y f(x) = f(x - y)$). Via the substitution $x \mapsto x + y$, we get
$$\int f(x-y)\varphi(x)\,dx = \int f(x)\varphi(x+y)\,dx,$$
and so $T' = \tau_{-y}|_{C_c^\infty(V)}$. This is evidently continuous, and so for a distribution $F \in \mathcal{D}'(U)$, we define:
$$\langle \tau_y F, \varphi\rangle := \langle F, \tau_{-y}\varphi\rangle.$$
As a quick sanity check: $\langle \tau_y\delta, \varphi\rangle = \langle \delta, \tau_{-y}\varphi\rangle = (\tau_{-y}\varphi)(0) = \varphi(y) = \langle \delta_y, \varphi\rangle$. So translating the delta function by y gives the point mass at y, exactly as expected.

4. *(Composition with linear maps)* Let S be an invertible linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. Let $V = S^{-1}(U)$ and $Tf = f \circ S$. By the change of variables formula (see Chapter 7), for $\varphi \in C_c^\infty(V)$:
$$\int (f \circ S)(x)\varphi(x)\,dx = \int f(y)\varphi(S^{-1}y)|\det S|^{-1}\,dy.$$
So $T'\varphi = |\det(S)|^{-1}\varphi \circ S^{-1}$. For $F \in \mathcal{D}'(U)$, we define:
$$\langle F \circ S, \varphi\rangle := |\det S|^{-1}\langle F, \varphi \circ S^{-1}\rangle.$$
As a special case, taking $S = -I$ (the map $x \mapsto -x$) gives $\langle \tilde{F}, \varphi\rangle = \langle F, \tilde{\varphi}\rangle$, which defines the *reflection* of a distribution.

Convolutions are interesting and deserve more attention as they smooth distributions and recover *functions*. Consider first convoluting with a function. Let $\psi \in C_c^\infty$, and let
$$V = \{x : x - y \in U \text{ for all } y \in \operatorname{supp}(\psi)\}$$
Note that V is open (but may be empty). If $f \in L^1_{\text{loc}}(U)$, then the integral
$$f * \psi(x) = \int f(x-y)\psi(y)\,dy = \int f(y)\psi(x-y)\,dy = \langle f, \tau_x\tilde{\psi}\rangle$$
is well-defined for all $x \in V$. Notice that we are inputting a *point*. Since the last expression involves only the distributional pairing, we can define for $F \in \mathcal{D}'(U)$:
$$F * \psi(x) = \langle F, \tau_x\tilde{\psi}\rangle$$
This is well-defined because $\tau_x\tilde{\psi} \in C_c^\infty(U)$ for $x \in V$.

Proposition 9.3.1: Convoluting With Function Gives Function

Let F be a distribution and ψ a smooth function with compact support on V. Then $F * \psi$ is a smooth functions

***Proof*:**
Since $\tau_x\tilde{\psi} \to \tau_{x_0}\tilde{\psi}$ in C_c^∞ as $x \to x_0$ (and similarly for all derivatives), continuity of F gives $F * \psi(x) \to F * \psi(x_0)$. More precisely, one can show by differentiating under the pairing that $\partial^\alpha(F * \psi) = F * (\partial^\alpha\psi)$ for all multi-indices α, which establishes that $F * \psi \in C^\infty(V)$.

A pertinent example is the following:
$$\delta * \psi(x) = \langle \delta, \tau_x\tilde{\psi}\rangle = \tau_x\tilde{\psi}(0) = \tilde{\psi}(-x) = \psi(x)$$
so δ is the identity element for convolution! This is one of the most important properties of the delta function and lies at the heart of the theory of Green's functions: if L is a differential operator and E is a distribution satisfying $LE = \delta$ (a *fundamental solution*), then $u = E * f$ formally solves $Lu = f$.

We can also define convolution using the same transpose technique from earlier. Let ψ, $\tilde{\psi}$, and V be as above. If $f \in L^1_{\text{loc}}(U)$ and $\varphi \in C_c^\infty(V)$, then by Fubini's theorem:
$$\begin{aligned}\int (f * \psi)(x)\varphi(x)\,dx &= \iint f(y)\psi(x-y)\varphi(x)\,dy\,dx = \int f(y)\left(\int \psi(x-y)\varphi(x)\,dx\right)dy \\ &= \int f(y)(\varphi * \tilde{\psi})(y)\,dy\end{aligned}$$

Thus $Tf = f * \psi$ maps $L^1_{\text{loc}}(U)$ to $L^1_{\text{loc}}(V)$ with transpose $T'\varphi = \varphi * \tilde{\psi}$. For a distribution $F \in \mathcal{D}'(U)$, we therefore get:

$$\langle F * \psi, \varphi\rangle = \langle F, \varphi * \tilde{\psi}\rangle$$

We should verify that the two definitions of convolution are consistent:

Proposition 9.3.2: Consistency of Convolution

Let $F \in \mathcal{D}'(U)$ and $\psi \in C_c^\infty$, and let V be as above. Then the smooth function $x \mapsto \langle F, \tau_x\tilde{\psi}\rangle$ and the distribution defined by $\langle F, \varphi * \tilde{\psi}\rangle$ agree, in the sense that the 2nd convolution definition is the distribution defined by the locally integrable function from the 1st convolution definition.

***Proof*:**
We need to show that for all $\varphi \in C_c^\infty(V)$:

$$\int \langle F, \tau_x\tilde{\psi}\rangle\varphi(x)\,dx = \langle F, \varphi * \tilde{\psi}\rangle.$$

The left-hand side can be written as the limit of Riemann sums. More precisely, for φ supported in a compact set $K' \subseteq V$, the function $x \mapsto \tau_x\tilde{\psi}$ is a continuous map from K' into $C_c^\infty(K)$ for some compact $K \subseteq U$. One checks (using the Riemann sum approximation and continuity of F) that the integral of a continuous $C_c^\infty(K)$-valued function against a scalar function can be "passed inside" the distributional pairing. But the integral $\int \tau_x\tilde{\psi}\,\varphi(x)\,dx$, computed pointwise, is exactly the function $y \mapsto \int \psi(x-y)\varphi(x)\,dx = (\varphi * \tilde{\psi})(y)$. Hence:

$$\int \langle F, \tau_x\tilde{\psi}\rangle\varphi(x)\,dx = \left\langle F, \int \tau_x\tilde{\psi}\,\varphi(x)\,dx\right\rangle = \langle F, \varphi * \tilde{\psi}\rangle.$$

9.4 Density Results

Next, we will show that smooth compactly supported functions $C_c^\infty(U)$ are dense in $\mathcal{D}'(U)$. This is a remarkable fact: it says that every distribution, no matter how singular, can be approximated (in the weak* topology) by smooth functions. We begin with a preparatory lemma.

Lemma 9.4.1: Build-Up For Density

Suppose that $\varphi \in C_c^\infty$, $\psi \in C_c^\infty$ with $\int \psi = 1$, and let $\psi_t(x) = t^{-n}\psi(t^{-1}x)$ for $t > 0$. Then:

1. Given any open neighborhood U of $\text{supp}(\varphi)$, we have $\text{supp}(\varphi * \psi_t) \subseteq U$ for sufficiently small $t > 0$.
2. $\varphi * \psi_t \to \varphi$ in C_c^∞ as $t \to 0$.

***Proof*:**

1. Note that $\text{supp}(\varphi * \psi_t) \subseteq \text{supp}(\varphi) + \text{supp}(\psi_t) = \text{supp}(\varphi) + t \cdot \text{supp}(\psi)$. Since $\text{supp}(\varphi)$ is compact, the set $\text{supp}(\varphi) + t \cdot \text{supp}(\psi)$ is the Minkowski sum of $\text{supp}(\varphi)$ with a set of diameter shrinking to 0 as $t \to 0$. Since U is an open neighborhood of the compact set $\text{supp}(\varphi)$, there exists $\varepsilon > 0$ such that the ε-neighborhood of $\text{supp}(\varphi)$ is contained in U. For t small enough that $t \cdot \text{supp}(\psi)$ lies within a ball of radius ε, we get $\text{supp}(\varphi * \psi_t) \subseteq U$.

2. We need to show that $\partial^\alpha(\varphi * \psi_t) \to \partial^\alpha \varphi$ uniformly for every multi-index α. Since differentiation commutes with convolution, $\partial^\alpha(\varphi * \psi_t) = (\partial^\alpha \varphi) * \psi_t$. So it suffices to show that $g * \psi_t \to g$ uniformly whenever $g \in C_c^\infty$. We have:

$$(g * \psi_t)(x) - g(x) = \int [g(x-y) - g(x)]\psi_t(y)\, dy = \int [g(x-tz) - g(x)]\psi(z)\, dz,$$

   using the substitution $y = tz$. Since g is smooth and compactly supported, it is uniformly continuous, so for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|g(x-tz) - g(x)| < \varepsilon$ whenever $t|z| < \delta$. For $z \in \text{supp}(\psi)$ and t small enough, this is satisfied, giving $|g * \psi_t(x) - g(x)| \leq \varepsilon \int |\psi|$ for all x. By part (a), all the $\varphi * \psi_t$ are eventually supported in a single compact set, so the convergence is in C_c^∞.

Proposition 9.4.2: Density Of Smooth Compactly Supported Functions In Distributions

Let $U \subseteq \mathbb{R}^n$ be open. Then $C_c^\infty(U)$ is dense in $\mathcal{D}'(U)$ in the weak* topology.

***Proof*:**
Let $F \in \mathcal{D}'(U)$; we must find a net (or sequence) $\{\varphi_i\}$ in $C_c^\infty(U)$ such that $\langle \varphi_i, \eta\rangle \to \langle F, \eta\rangle$ for all $\eta \in C_c^\infty(U)$, where $\langle \varphi_i, \eta \rangle = \int \varphi_i \eta$.

Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\int \psi = 1$ and let $\psi_t(x) = t^{-n}\psi(t^{-1}x)$. We also need a family of cutoff functions: let $\{\chi_j\}_{j=1}^\infty$ be a sequence in $C_c^\infty(U)$ with $0 \leq \chi_j \leq 1$ and $\chi_j \to 1$ pointwise on U (e.g., take $\chi_j = 1$ on $\{x \in U : |x| \leq j \text{ and } \text{dist}(x, U^c) \geq 1/j\}$ and smooth it out). Consider the distributions $F_j = \chi_j F$ (defined by $\langle F_j, \eta\rangle = \langle F, \chi_j \eta\rangle$). Each F_j has compact support in U, and $F_j \to F$ in $\mathcal{D}'(U)$ since for fixed $\eta \in C_c^\infty(U)$, $\chi_j \eta \to \eta$ in $C_c^\infty(U)$ for large j (eventually $\chi_j = 1$ on $\text{supp}(\eta)$).

Now consider $F_j * \psi_t$. By our earlier discussion, this is a C^∞ function. For t sufficiently small, it has compact support in U (since F_j has compact support in U and the convolution only slightly enlarges the support). By Proposition 9.2.5 and the properties of convolution, $F_j * \psi_t \to F_j$ in $\mathcal{D}'$ as $t \to 0$: for any $\eta \in C_c^\infty(U)$,

$$\langle F_j * \psi_t, \eta \rangle = \langle F_j, \eta * \tilde{\psi}_t \rangle \to \langle F_j, \eta \rangle$$

by lemma 9.4.1, since $\eta * \tilde{\psi}_t \to \eta$ in C_c^∞. By a diagonal argument (choosing $t_j \to 0$ slowly enough), we obtain a sequence $g_j = F_j * \psi_{t_j} \in C_c^\infty(U)$ with $g_j \to F$ in $\mathcal{D}'(U)$.

9.5 Distributional Derivatives of Classical Functions

We promised earlier that the distributional derivative lets us differentiate functions that are not classically differentiable, and now we deliver on that promise. The computations in this section are among the most important examples in all of distribution theory, and the reader should internalize them thoroughly.

Recall that for $F \in \mathcal{D}'(\mathbb{R}^n)$ and a multi-index α, the distributional derivative is defined by

$$\langle \partial^\alpha F, \varphi \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \varphi \rangle.$$

Example 9.5.1: Derivative Of The Heaviside Function

Let $H : \mathbb{R} \to \mathbb{R}$ be the Heaviside step function, $H(x) = \mathbf{1}_{[0,\infty)}(x)$. Then $H \in L^1_{\text{loc}}(\mathbb{R})$ and hence defines a distribution. Classically, H is differentiable everywhere except at $x = 0$, where it has a jump discontinuity of height 1. Let us compute H' in the distributional sense. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle H', \varphi \rangle = -\langle H, \varphi' \rangle = -\int_0^\infty \varphi'(x)\, dx = -[\varphi(x)]_{x=0}^{x=\infty} = -\big(\underbrace{\varphi(\infty)}_{=0} - \varphi(0)\big) = \varphi(0) = \langle \delta, \varphi \rangle.$$

Therefore $H' = \delta$ in $\mathcal{D}'(\mathbb{R})$. This is the prototypical example: the distributional derivative detects the jump discontinuity and places a point mass there. More generally, if f is piecewise C^1 with jump discontinuities of height c_j at points x_j, then $f' = f'_{\text{classical}} + \sum_j c_j \delta_{x_j}$ in $\mathcal{D}'$.

Example 9.5.2: Derivative Of $|x|$ And The Sign Function

Consider $f(x) = |x|$ on $\mathbb{R}$. This function is continuous but not differentiable at $x = 0$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned}
\langle f', \varphi \rangle = -\langle f, \varphi' \rangle &= -\int_{-\infty}^\infty |x| \varphi'(x)\, dx \\
&= -\int_{-\infty}^0 (-x)\varphi'(x)\, dx - \int_0^\infty x\varphi'(x)\, dx \\
&= \int_{-\infty}^0 x\varphi'(x)\, dx - \int_0^\infty x\varphi'(x)\, dx.
\end{aligned}$$

Integrating by parts on each piece (the boundary terms at $\pm\infty$ vanish since φ has compact support, and the boundary terms at 0 cancel):

$$= \Big[x\varphi(x)\Big]_{-\infty}^0 - \int_{-\infty}^0 \varphi(x)\, dx - \Big[x\varphi(x)\Big]_0^\infty + \int_0^\infty \varphi(x)\, dx = -\left(\int_{-\infty}^0 \varphi\right) + \left(\int_0^\infty \varphi\right) = \int_{-\infty}^\infty \operatorname{sgn}(x)\varphi(x)\, dx,$$

where $\operatorname{sgn}(x) = \mathbf{1}_{(0,\infty)}(x) - \mathbf{1}_{(-\infty,0)}(x)$ is the sign function. Hence $|x|' = \operatorname{sgn}(x)$ in $\mathcal{D}'$. Note that $\operatorname{sgn}(x) = 2H(x) - 1$, which is consistent: differentiating again gives $|x|'' = \operatorname{sgn}'(x) = 2H'(x) = 2\delta$.

The next computation requires introducing an important distribution that is not given by a locally integrable function in the standard sense.

Definition 9.5.3: Principal Value Distribution

The *principal value of* $1/x$, denoted $\text{p.v.}(1/x)$, is the distribution on $\mathbb{R}$ defined by

$$\left\langle \text{p.v.}\frac{1}{x}, \varphi \right\rangle = \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.$$

We should verify that the limit exists and that this defines a distribution. Since φ is smooth, we can write $\varphi(x) = \varphi(0) + x\psi(x)$ where $\psi(x) = \int_0^1 \varphi'(tx)\, dt$ is also smooth. Then

$$\int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx = \underbrace{\varphi(0)\int_{|x|>\varepsilon} \frac{dx}{x}}_{=0 \text{ by symmetry}} + \int_{|x|>\varepsilon} \psi(x)\, dx \to \int_{-\infty}^{\infty} \psi(x)\, dx$$

as $\varepsilon \to 0$. Since ψ is smooth and φ has compact support, the integral converges. The continuity estimate $|\langle \text{p.v.}(1/x), \varphi\rangle| \le C(\|\varphi\|_\infty + \|\varphi'\|_\infty)$ on any fixed compact set follows from this decomposition, confirming that $\text{p.v.}(1/x) \in \mathcal{D}'(\mathbb{R})$. Notice that the order of this distribution is 1 (it requires control of one derivative), unlike the distributions arising from locally integrable functions, which have order 0.

Example 9.5.4: Derivative Of $\log|x|$

The function $\log|x|$ is locally integrable on $\mathbb{R}$ (since $\int_{-1}^1 |\log|x||\, dx < \infty$) and hence defines a distribution. We claim that its distributional derivative is $\text{p.v.}(1/x)$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned}\langle (\log|x|)', \varphi\rangle = -\langle \log|x|, \varphi'\rangle &= -\int_{-\infty}^{\infty} \log|x|\, \varphi'(x)\, dx \\ &= -\lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \log|x|\, \varphi'(x)\, dx.\end{aligned}$$

Integrating by parts (with the boundary terms at $\pm\infty$ vanishing):

$$\begin{aligned}&= -\lim_{\varepsilon\to 0^+} \left(\Big[\log|x|\, \varphi(x)\Big]_{|x|=\varepsilon}^{|x|=\infty} - \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx \right) \\ &= -\lim_{\varepsilon\to 0^+} \big(-\log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)]\big) + \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.\end{aligned}$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) \to 2\varphi(0)$ and $\varepsilon\log\varepsilon \to 0$, the first term requires more care. We have $\log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)] = 2\varphi(0)\log\varepsilon + O(\varepsilon\log\varepsilon)$. But this term appears with a minus sign outside, so:

$$= \lim_{\varepsilon\to 0^+} \log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)] + \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.$$

For the boundary term, note that $[\log|x|\, \varphi(x)]_{-\infty}^{-\varepsilon} + [\log|x|\, \varphi(x)]_{\varepsilon}^{\infty} = -\log\varepsilon\, \varphi(-\varepsilon) - \log\varepsilon\, \varphi(\varepsilon)$. Thus:

$$= \lim_{\varepsilon\to 0^+} \left[\log\varepsilon \big(\varphi(\varepsilon) + \varphi(-\varepsilon)\big) + \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx \right].$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) = 2\varphi(0) + O(\varepsilon^2)$ (the odd part cancels) and $\int_\varepsilon^1 dx/x = -\log\varepsilon$ (with the symmetric contribution from $[-1, -\varepsilon]$ cancelling), the $\log\varepsilon$ terms cancel exactly, and we are left with

$$\langle (\log|x|)', \varphi\rangle = \lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon}\frac{\varphi(x)}{x}\,dx = \left\langle \text{p.v.}\frac{1}{x}, \varphi\right\rangle.$$

Therefore $(\log|x|)' = \text{p.v.}(1/x)$ in $\mathcal{D}'(\mathbb{R})$.

It is also instructive to see how multiplication interacts with the principal value:

Proposition 9.5.5: Multiplication By x

As distributions on $\mathbb{R}$, we have $x\cdot\text{p.v.}(1/x) = 1$, that is, the distribution $x\cdot\text{p.v.}(1/x)$ agrees with the constant function 1.

***Proof*:**
For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle x\cdot\text{p.v.}(1/x),\, \varphi\rangle = \left\langle \text{p.v.}\frac{1}{x},\, x\varphi\right\rangle = \lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon}\frac{x\varphi(x)}{x}\,dx = \int_{-\infty}^{\infty}\varphi(x)\,dx = \langle 1, \varphi\rangle,$$

since φ is integrable and the integrand $\varphi(x)$ has no singularity.

This is a distributional identity with no classical analogue: the "function" $1/x$ is not locally integrable, so the product $x\cdot(1/x)$ is not defined pointwise in L^1_{loc}. But the distributional version makes perfect sense and gives the expected answer. On the other hand, note that $x\cdot\delta = 0$ (since $\langle x\delta,\varphi\rangle = \langle\delta, x\varphi\rangle = 0$), so x is a zero-divisor in $\mathcal{D}'$—a phenomenon that does not occur for ordinary functions.

10

Sobolev Spaces

The spaces of the previous chapters measure the size of a function; the Sobolev spaces measure a function *together with its derivatives*, and they are where the differential equations of analysis and geometry live. The construction completes the smooth functions in a norm that controls not only $\|f\|_{L^p}$ but $\|\partial^\alpha f\|_{L^p}$ up to a fixed order, so that a limit of smooth functions keeps a definite derivative "in the L^p sense" even where it is nowhere classically differentiable. Two ingredients already in hand make this work: the Lebesgue spaces of chapter 7, which supply the norm, and the distributional derivative of chapter 9, which supplies a notion of derivative robust enough to survive the completion.

This chapter develops what a reader needs before the elliptic and parabolic equations of the partial-differential-equations volumes: the definition and completeness of $W^{k,p}$; the density of smooth functions, proved with the mollifiers of section 8.6; the embedding theorems that trade Sobolev regularity for classical regularity or higher integrability; the trace that assigns boundary values; and the Rellich-Kondrachov compactness behind existence proofs. Everything here is Fourier-free, resting only on L^p and the distributional derivative; the complementary scale H^s, defined through the Fourier transform, is built in the harmonic-analysis part.

10.1 Weak Derivatives and the Spaces $W^{k,p}$

Recall from chapter 9 that every distribution $F \in \mathcal{D}'(U)$ has distributional derivatives $\partial^\alpha F$, defined by $\langle \partial^\alpha F, \varphi\rangle = (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle$ for $\varphi \in C_c^\infty(U)$. A locally integrable function is a distribution, so its derivatives of every order exist as distributions; Sobolev theory is organized by one question: for which α is $\partial^\alpha f$ *again a function*, and how integrable is it?

Definition 10.1.1: Weak Derivative

Let $U \subseteq \mathbb{R}^n$ be open, let $f \in L^1_{\mathrm{loc}}(U)$, and let α be a multi-index. A function $g \in L^1_{\mathrm{loc}}(U)$ is the *weak α-derivative* of f if

$$\int_U f\,\partial^\alpha \varphi\,dx = (-1)^{|\alpha|}\int_U g\,\varphi\,dx \qquad \text{for every } \varphi \in C_c^\infty(U).$$

Equivalently, the distributional derivative $\partial^\alpha f$ is represented by the function g. When such a g exists we write $\partial^\alpha f := g$, and for first-order derivatives $\partial_i f$ and $\nabla f = (\partial_1 f, \dots, \partial_n f)$.

A classical C^k function is weakly differentiable, its weak and classical derivatives agreeing: integration by parts against $\varphi \in C_c^\infty(U)$ produces no boundary terms, which is precisely the identity above. The weak derivative is thus an extension of the classical one, and like the classical derivative it is determined almost everywhere.

Lemma 10.1.2: Uniqueness of the Weak Derivative

If $g_1, g_2 \in L^1_{\mathrm{loc}}(U)$ are both weak α-derivatives of f, then $g_1 = g_2$ almost everywhere.

***Proof*:**
Put $h = g_1 - g_2 \in L^1_{\mathrm{loc}}(U)$. Subtracting the two defining identities gives $\int_U h\,\varphi\,dx = 0$ for every $\varphi \in C_c^\infty(U)$. Fix a compact $K \subset U$ and a mollifier $\{\psi_\varepsilon\}$, a good kernel (definition 8.6.9) supported in $B(0,\varepsilon)$. For $x \in K$ and $\varepsilon < \operatorname{dist}(K, \partial U)$ the function $y \mapsto \psi_\varepsilon(x-y)$ belongs to $C_c^\infty(U)$, so

$$(h * \psi_\varepsilon)(x) = \int_U h(y)\,\psi_\varepsilon(x-y)\,dy = 0 \qquad (x \in K).$$

As $\varepsilon \to 0$, $h * \psi_\varepsilon \to h$ in $L^1(K)$ by theorem 8.6.10, so $h = 0$ almost everywhere on K. Exhausting U by compact sets gives $h = 0$ almost everywhere on U.

Example 10.1.3: Weak Differentiability of a Corner and a Jump

On $U = (-1,1)$ the corner $f(x) = |x|$ has weak derivative $\operatorname{sgn}(x)$: for $\varphi \in C_c^\infty(-1,1)$, integrating by parts over $(-1,0)$ and $(0,1)$ separately (the boundary terms at ± 1 vanish by compact support, and those at 0 cancel),

$$\int_{-1}^{1} |x|\,\varphi'(x)\,dx = \int_{-1}^{0} \varphi\,dx - \int_0^1 \varphi\,dx = -\int_{-1}^{1} \operatorname{sgn}(x)\,\varphi(x)\,dx,$$

so $|x| \in W^{1,p}(-1,1)$ for every p. The jump sgn, by contrast, has *no* weak derivative: $\int_{-1}^1 \operatorname{sgn}(x)\,\varphi'(x)\,dx = -2\varphi(0)$, and no $g \in L^1_{\mathrm{loc}}$ satisfies $-\int g\varphi = -2\varphi(0)$ for all φ (the distributional derivative of sgn is the Dirac mass $2\delta_0$, which is not a function). Hence $|x| \in W^{1,p} \setminus W^{2,p}$: weak differentiability is a real but limited amount of regularity, exactly the feature that makes these spaces useful.

With derivatives in hand we form the spaces. Since the weak derivative is linear in f (immediate from definition 10.1.1), what follows is a linear subspace of L^p.

Definition 10.1.4: Sobolev Space $W^{k,p}$

Let $U \subseteq \mathbb{R}^n$ be open, let k be a nonnegative integer, and let $1 \le p \le \infty$. The *Sobolev space* is

$$W^{k,p}(U) = \{f \in L^p(U) \mid \partial^\alpha f \in L^p(U) \text{ for all } |\alpha| \le k\},$$

the derivatives being weak (definition 10.1.1), normed by

$$\|f\|_{W^{k,p}(U)} = \Big(\sum_{|\alpha|\le k} \|\partial^\alpha f\|^p_{L^p(U)}\Big)^{1/p} \ (p<\infty), \qquad \|f\|_{W^{k,\infty}(U)} = \max_{|\alpha|\le k} \|\partial^\alpha f\|_{L^\infty(U)}.$$

When $p = 2$ we write $H^k(U) := W^{k,2}(U)$, an inner-product space under $\langle f, g\rangle_{H^k} = \sum_{|\alpha|\le k} \int_U \partial^\alpha f\, \overline{\partial^\alpha g}\, dx$.

The norm records a function and all of its weak derivatives up to order k as a single L^p quantity; $W^{0,p} = L^p$, and raising k demands more regularity. That the smooth members of $W^{k,p}$ are dense in it, so that $W^{k,p}$ is the completion of smooth functions promised in the introduction, is the subject of the next section; first we record that the space is already complete as defined.

Theorem 10.1.5: $W^{k,p}$ is a Banach Space

For every open $U \subseteq \mathbb{R}^n$, every nonnegative integer k, and every $1 \le p \le \infty$, $W^{k,p}(U)$ is a Banach space, and $H^k(U)$ is a Hilbert space.

***Proof*:**
The map $f \mapsto \|f\|_{W^{k,p}}$ is a norm: absolute homogeneity is clear, the triangle inequality follows from that of the L^p norms (theorem 8.1.6) combined with the triangle inequality of the ℓ^p sum, and $\|f\|_{W^{k,p}} = 0$ forces $\|f\|_{L^p} = 0$, hence $f = 0$. Only completeness remains.

Let (f_m) be Cauchy in $W^{k,p}(U)$. For each $|\alpha| \le k$,

$$\|\partial^\alpha f_m - \partial^\alpha f_l\|_{L^p} \le \|f_m - f_l\|_{W^{k,p}},$$

so $(\partial^\alpha f_m)$ is Cauchy in $L^p(U)$. As $L^p(U)$ is complete (theorem 8.1.7), there are $g_\alpha \in L^p(U)$ with $\partial^\alpha f_m \to g_\alpha$ in L^p; write $f := g_0$.

It remains to identify g_α as $\partial^\alpha f$. Fix $\varphi \in C_c^\infty(U)$. For each m, the weak-derivative identity reads

$$\int_U f_m\, \partial^\alpha \varphi\, dx = (-1)^{|\alpha|} \int_U (\partial^\alpha f_m)\, \varphi\, dx.$$

Both φ and $\partial^\alpha\varphi$ are bounded with compact support, so they lie in $L^{p'}(U)$; Hölder's inequality (theorem 8.1.5) then passes the limit under each integral, giving $\int_U f_m\, \partial^\alpha\varphi \to \int_U f\, \partial^\alpha \varphi$ and $\int_U (\partial^\alpha f_m)\varphi \to \int_U g_\alpha\, \varphi$. Hence

$$\int_U f\, \partial^\alpha \varphi\, dx = (-1)^{|\alpha|} \int_U g_\alpha\, \varphi\, dx \qquad \text{for all } \varphi \in C_c^\infty(U),$$

which says exactly that $\partial^\alpha f = g_\alpha$ weakly. Thus $f \in W^{k,p}(U)$, and

$$\|f_m - f\|_{W^{k,p}}^p = \sum_{|\alpha|\le k} \|\partial^\alpha f_m - g_\alpha\|_{L^p}^p \to 0,$$

so $f_m \to f$ in $W^{k,p}$ and the space is complete. For $p = 2$ the norm is induced by the inner product of definition 10.1.4, so $H^k(U)$ is a Hilbert space.

The completeness proof used nothing but the completeness of L^p and the stability of the weak-derivative identity under L^p limits, the reward for defining derivatives by duality against test functions rather than pointwise. The next section shows that the members of $W^{k,p}$ are, on reasonable domains, limits of smooth functions, so that theorems can be proved by checking them on C^∞ and passing to the limit.

10.2 Approximation by Smooth Functions

This section proves that smooth functions are dense in $W^{k,p}(U)$, so that a theorem about Sobolev functions can be reduced to the smooth case and recovered by passing to the limit. The mechanism throughout is that mollification commutes with the weak derivative.

Theorem 10.2.1: Local Approximation by Mollification

Let $U \subseteq \mathbb{R}^n$ be open, let $f \in W^{k,p}(U)$ with $1 \le p < \infty$, and fix a mollifier $\{\psi_\varepsilon\}$, a good kernel (definition 8.6.9) with $\psi_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ supported in $B(0,\varepsilon)$. On any open $V \Subset U$, for $\varepsilon < \operatorname{dist}(V, \partial U)$ the mollification $f * \psi_\varepsilon$ is defined and smooth on V, and for every multi-index $|\alpha| \le k$,

$$\partial^\alpha(f * \psi_\varepsilon) = (\partial^\alpha f) * \psi_\varepsilon \qquad \text{on } V$$

Consequently $f * \psi_\varepsilon \to f$ in $W^{k,p}(V)$ as $\varepsilon \to 0$.

***Proof*:**
Fix $V \Subset U$ and write $d = \operatorname{dist}(V, \partial U) > 0$; throughout, $\varepsilon < d$. For $x \in V$ the ball $B(x,\varepsilon)$ lies in U, so

$$(f * \psi_\varepsilon)(x) = \int_U f(y)\,\psi_\varepsilon(x-y)\,dy = \int_{B(x,\varepsilon)} f(y)\,\psi_\varepsilon(x-y)\,dy$$

is a genuine integral against a function supported in U, and it is finite because $f \in L^1_{\mathrm{loc}}(U)$ and ψ_ε is bounded.

Step 1: The commutation identity. Fix $x \in V$ and $|\alpha| \le k$. Since $y \mapsto \psi_\varepsilon(x-y)$ is smooth and, for x in a small neighbourhood of any point of V, supported in a fixed compact subset of U, differentiation under the integral sign applies to the parameter x. Each x-derivative ∂_{x_i} falls on $\psi_\varepsilon(x-y)$, so

$$\partial_x^\alpha(f * \psi_\varepsilon)(x) = \int_U f(y)\,\partial_x^\alpha \psi_\varepsilon(x-y)\,dy$$

Now $\partial_x^\alpha \psi_\varepsilon(x-y) = (-1)^{|\alpha|}\,\partial_y^\alpha \psi_\varepsilon(x-y)$, since each factor of $x - y$ contributes a sign under the switch from an x-derivative to a y-derivative. For $x \in V$ the function $\varphi(y) := \psi_\varepsilon(x-y)$

belongs to $C_c^\infty(U)$: it is smooth and supported in $B(x,\varepsilon) \subseteq U$. The weak derivative identity of definition 10.1.1 applied to this φ therefore reads

$$\int_U f(y)\, \partial_y^\alpha \psi_\varepsilon(x-y)\, dy = (-1)^{|\alpha|} \int_U (\partial^\alpha f)(y)\, \psi_\varepsilon(x-y)\, dy$$

Combining the last three displays, the two factors of $(-1)^{|\alpha|}$ cancel and

$$\partial_x^\alpha (f * \psi_\varepsilon)(x) = \int_U (\partial^\alpha f)(y)\, \psi_\varepsilon(x-y)\, dy = \big((\partial^\alpha f) * \psi_\varepsilon\big)(x)$$

which is the claimed identity. Smoothness of $f * \psi_\varepsilon$ on V is now immediate: since every x-derivative falls on ψ_ε, the identity $\partial_x^\alpha (f * \psi_\varepsilon) = f * \partial_x^\alpha \psi_\varepsilon$ holds for *every* multi-index α (not only $|\alpha| \le k$), so all partial derivatives of $f * \psi_\varepsilon$ exist and, being mollifications of the locally integrable f, are continuous.

Step 2: Convergence in $W^{k,p}(V)$. Since $f \in W^{k,p}(U)$, each $\partial^\alpha f$ with $|\alpha| \le k$ lies in $L^p(U)$, hence in $L^p(W)$ for any W with $V \Subset W \Subset U$; fix such a W. Restricting to $\varepsilon < \operatorname{dist}(V, \partial W)$, the mollification $(\partial^\alpha f) * \psi_\varepsilon$ is computed on V from the values of $\partial^\alpha f$ on W only. By the approximate-identity convergence of theorem 8.6.10, $(\partial^\alpha f) * \psi_\varepsilon \to \partial^\alpha f$ in $L^p(V)$ as $\varepsilon \to 0$. Using the Step 1 identity on the left,

$$\|f * \psi_\varepsilon - f\|_{W^{k,p}(V)}^p = \sum_{|\alpha| \le k} \|\partial^\alpha (f * \psi_\varepsilon) - \partial^\alpha f\|_{L^p(V)}^p = \sum_{|\alpha| \le k} \|(\partial^\alpha f) * \psi_\varepsilon - \partial^\alpha f\|_{L^p(V)}^p$$

and each of the finitely many summands tends to 0. Hence $f * \psi_\varepsilon \to f$ in $W^{k,p}(V)$, completing the proof.

The identity $\partial^\alpha (f * \psi_\varepsilon) = (\partial^\alpha f) * \psi_\varepsilon$ is the whole engine of Sobolev approximation. Mollification is a smoothing operation, so its output is a bona fide C^∞ function to which the classical calculus applies; the identity says that this smooth function carries, as its classical derivatives, exactly the mollified weak derivatives of f. Convergence of each weak derivative in L^p then upgrades automatically to convergence in the Sobolev norm, because that norm is nothing but the L^p norms of the derivatives bundled together. The one limitation is visible in the hypothesis $V \Subset U$: the mollification $(f * \psi_\varepsilon)(x)$ reaches out to the ball $B(x, \varepsilon)$, so near ∂U it would sample points outside U where f is not defined. Local approximation is therefore free; the work in the rest of the section is to recover global statements from it. The restriction $p < \infty$ is inherited from theorem 8.6.10, where the translation $y \mapsto f(\cdot - y)$ fails to be L^∞-continuous.

Example 10.2.2: Mollifying a Corner

Return to $f(x) = |x|$ on $U = (-1, 1)$, which lies in $W^{1,p}(U)$ with weak derivative $\partial f = \operatorname{sgn}(x)$ (example 10.1.3). Fix the standard mollifier $\psi_\varepsilon(x) = \varepsilon^{-1}\psi(x/\varepsilon)$ with $\psi \in C_c^\infty(-1,1)$ even, $\psi \ge 0$, $\int \psi = 1$. On any $V = (-a, a)$ with $a < 1$ and $\varepsilon < 1 - a$, the mollification is

$$(f * \psi_\varepsilon)(x) = \int_{-\varepsilon}^{\varepsilon} |x - y|\, \psi_\varepsilon(y)\, dy$$

a smooth function that agrees with $|x|$ for $|x| > \varepsilon$ (there $|x - y|$ is affine on the support of ψ_ε and the first moment of the even kernel vanishes) and rounds off the corner on $(-\varepsilon, \varepsilon)$. Its derivative is $(f * \psi_\varepsilon)'(x) = (\operatorname{sgn} * \psi_\varepsilon)(x)$, exactly as theorem 10.2.1 predicts: a smooth function increasing from -1 to 1 across the corner, equal to $\operatorname{sgn}(x)$ outside $(-\varepsilon, \varepsilon)$. As $\varepsilon \to 0$ the smoothed corner and its derivative converge to $|x|$ and $\operatorname{sgn}(x)$ in $L^p(V)$, hence $f * \psi_\varepsilon \to f$ in $W^{1,p}(V)$.

Local approximation leaves two gaps. Mollification only produces smooth functions on the interior, so it does not by itself yield a smooth approximant on all of U; and even where it does, the approximant need not extend smoothly to the boundary. The first gap is closed by patching local mollifications together with a partition of unity, which is the content of the Meyers-Serrin theorem. Historically, two definitions of a Sobolev space competed: the space $W^{k,p}(U)$ of L^p functions with weak derivatives in L^p, and the space $H^{k,p}(U)$ obtained as the abstract completion of $\{f \in C^\infty(U) \mid \|f\|_{W^{k,p}} < \infty\}$ in the Sobolev norm. That these two coincide for every open U, with no regularity assumption on the boundary, is the theorem's content, and it is why one no longer distinguishes "H" from "W".

Theorem 10.2.3: Meyers-Serrin ($H = W$)

Let $U \subseteq \mathbb{R}^n$ be open, let $k \geq 0$, and let $1 \leq p < \infty$. Then $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$: for every $f \in W^{k,p}(U)$ and every $\varepsilon > 0$ there is a $g \in C^\infty(U) \cap W^{k,p}(U)$ with $\|f - g\|_{W^{k,p}(U)} < \varepsilon$.

***Proof*:**
The idea is to mollify f separately on each shell of an exhaustion of U, where the previous theorem applies, and to glue the local smooth pieces with a partition of unity, arranging the individual errors to sum to less than ε.

Step 1: An open exhaustion and a partition of unity. Set, for integers $m \geq 1$,

$$U_m = \{x \in U \mid \operatorname{dist}(x, \partial U) > 1/m \text{ and } |x| < m\}$$

with the conventions $U_0 = U_{-1} = \varnothing$; then $U_m \Subset U_{m+1}$ and $\bigcup_m U_m = U$. The open sets

$$V_m = U_{m+1} \setminus \overline{U_{m-1}} \qquad (m \geq 1)$$

form a locally finite open cover of U: each $V_m \Subset U$, and any point of U lies in V_m for at most three consecutive indices. Let $\{\zeta_m\}_{m \geq 1}$ be a smooth partition of unity subordinate to $\{V_m\}$, so $\zeta_m \in C_c^\infty(V_m)$, $0 \leq \zeta_m \leq 1$, and $\sum_m \zeta_m \equiv 1$ on U (the sum being locally finite). Each $\zeta_m f$ then lies in $W^{k,p}(U)$ with support in V_m, because multiplication by the smooth compactly supported ζ_m preserves weak differentiability: for $|\alpha| \leq k$ the Leibniz rule

$$\partial^\alpha(\zeta_m f) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \zeta_m \, \partial^{\alpha - \beta} f$$

holds weakly (the distributional Leibniz rule for multiplication by a smooth function, chapter 9), its right-hand side a finite sum of L^p functions.

Step 2: Mollify each piece with a controlled error. Fix $\varepsilon > 0$. The function $\zeta_m f$ is supported in the compact-in-U set $\overline{V_m}$, so theorem 10.2.1 gives $\zeta_m f * \psi_\delta \to \zeta_m f$ in $W^{k,p}(U)$ as $\delta \to 0$ (the mollification is computed on a neighbourhood of $\overline{V_m}$ still compact in U). Choose $\delta_m > 0$ small enough that

$$g_m := (\zeta_m f) * \psi_{\delta_m} \in C_c^\infty(U), \qquad \|g_m - \zeta_m f\|_{W^{k,p}(U)} < \frac{\varepsilon}{2^m}$$

and, in addition, δ_m small enough that $\operatorname{supp} g_m \subseteq W_m := U_{m+2} \setminus \overline{U_{m-2}}$ (possible because $\operatorname{supp}(\zeta_m f) \subseteq V_m$ sits well inside W_m). The sets W_m are again locally finite: only finitely many W_m meet any given U_l.

Step 3: Sum and estimate. Define

$$g = \sum_{m\geq 1} g_m$$

On any U_l the sum is finite (only the g_m with $W_m \cap U_l \neq \varnothing$ contribute), so g is well-defined and $g \in C^\infty(U)$. Since $\sum_m \zeta_m f = f$ with the same local finiteness, for every $x \in U$

$$g(x) - f(x) = \sum_{m\geq 1} \big(g_m(x) - (\zeta_m f)(x)\big)$$

is a finite sum. Fix l. On U_l only the finitely many indices m with $W_m \cap U_l \neq \varnothing$ or $V_m \cap U_l \neq \varnothing$ enter, and the triangle inequality in $W^{k,p}(U_l)$ gives

$$\|g - f\|_{W^{k,p}(U_l)} \leq \sum_{m\geq 1} \|g_m - \zeta_m f\|_{W^{k,p}(U)} < \sum_{m\geq 1} \frac{\varepsilon}{2^m} = \varepsilon$$

the bound being uniform in l. Letting $l \to \infty$ and applying the monotone convergence theorem to $\sum_{|\alpha|\leq k} |\partial^\alpha (g-f)|^p$ over the increasing exhaustion $U_l \uparrow U$,

$$\|g - f\|_{W^{k,p}(U)} = \lim_{l\to\infty} \|g - f\|_{W^{k,p}(U_l)} \leq \varepsilon$$

In particular $g - f \in W^{k,p}(U)$, so $g \in W^{k,p}(U)$; and $\|g - f\|_{W^{k,p}(U)} \leq \varepsilon$. As $\varepsilon > 0$ was arbitrary, $C^\infty(U) \cap W^{k,p}(U)$ is dense, completing the proof.

The strength of Meyers-Serrin is that it asks nothing of the domain: for any open set whatsoever, however irregular its boundary, the smooth Sobolev functions are dense. This is what dissolves the historical distinction between the completion-of-smooth-functions space $H^{k,p}$ and the weak-derivative space $W^{k,p}$. The price is that the approximating g is smooth only in the *interior* of U: nothing controls its behaviour as x approaches ∂U, and indeed g and its derivatives may blow up there. For many purposes this is enough, since the norm only sees the interior. But to integrate by parts up to the boundary, or to make sense of boundary values, one needs approximants smooth on the closure $\overline{U}$, and that does require the boundary to be regular.

Example 10.2.4: Interior Smoothness Is Not Boundary Smoothness

Take $n = 1$, $U = (0, 1)$, and $f(x) = x^{1/2}$. Then $f \in W^{1,p}(0, 1)$ for $1 \leq p < 2$, since $f' = \frac{1}{2}x^{-1/2} \in L^p(0, 1)$ exactly when $p/2 < 1$. Mollifying f over the exhaustion of $(0, 1)$ as in theorem 10.2.3 produces smooth approximants on the open interval, but they inherit the growth of f' near 0: the mollified derivative $(f' * \psi_\delta)(x)$ is of order $\delta^{-1/2}$ for x within δ of the endpoint, so no interior mollification is bounded up to $x = 0$, and none extends to a function in $C^\infty([0, 1])$. The Meyers-Serrin conclusion is exactly as advertised and no better: density by functions smooth on the *interior*.

The remedy is the inward shift. For small $t > 0$ the translate $f_t(x) = f(x + t) = (x + t)^{1/2}$ is smooth on all of $[0, 1]$, because its only singularity, at $x = -t$, now sits outside the closed interval; and $\|f_t - f\|_{W^{1,p}(0,1)} \to 0$ as $t \to 0$ by L^p-continuity of translation. Thus f_t (already smooth here, no mollification needed) is a $C^\infty([0, 1])$ function within any prescribed $W^{1,p}$ tolerance of f, which is precisely what theorem 10.2.5 delivers in general. The shift trades a function singular *on* the boundary for one singular just outside it, at the cost of a small L^p error, converting interior smoothness into smoothness up to $\overline{U}$.

For a bounded domain with regular boundary, one can do better than interior smoothness. The strategy near a boundary point is to flatten the boundary with a C^1 change of coordinates, translate the function a small distance *into* the domain so that its support pulls away from the boundary, and then mollify at a scale finer than the translation. The translated-and-mollified function is smooth on a neighbourhood of the piece of $\overline{U}$ under consideration, and the translation costs little in $W^{k,p}$ because translation is continuous on L^p.

Theorem 10.2.5: Global Approximation up to the Boundary

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary ∂U, let $k \geq 0$, and let $1 \leq p < \infty$. Then $C^\infty(\overline{U})$ is dense in $W^{k,p}(U)$: for every $f \in W^{k,p}(U)$ and every $\varepsilon > 0$ there is a $g \in C^\infty(\overline{U})$ with $\|f - g\|_{W^{k,p}(U)} < \varepsilon$.

***Proof*:**
Fix $f \in W^{k,p}(U)$ and $\varepsilon > 0$.

Step 1: Localize to a boundary chart and translate inward. Fix a point $x_0 \in \partial U$. Because ∂U is C^1, after relabelling coordinates there is $r > 0$ and a C^1 function $\gamma\colon \mathbb{R}^{n-1} \to \mathbb{R}$ with

$$U \cap B(x_0, r) = \{x \in B(x_0, r) \mid x_n > \gamma(x_1, \dots, x_{n-1})\}$$

so near x_0 the domain lies above the graph of γ. Set $V = U \cap B(x_0, r/2)$. For a point $x \in V$ and $t > 0$, define the inward-shifted point

$$x^t = x + t\, e_n$$

and the shifted function $f_t(x) = f(x^t)$ on V. Choose r small enough (using continuity of $\nabla\gamma$, i.e. the C^1 property: in coordinates where $\nabla\gamma(x_0') = 0$ the graph has small slope over a small ball) that there is a cone opening in the $+e_n$ direction: there is $\lambda > 0$ so that for all $x \in V$ and all $0 < t \leq \lambda$, the ball $B(x^t, t/2)$ lies inside $U \cap B(x_0, r)$. Then $f_t \in W^{k,p}(V)$ with $\partial^\alpha f_t = (\partial^\alpha f)_t$ (translation commutes with weak differentiation, by the change of variables $y \mapsto y - te_n$ in definition 10.1.1), and by the L^p-continuity of translation (a standard consequence of the density of C_c^∞ in L^p, proposition 8.6.8: approximate each $\partial^\alpha f$ by a uniformly continuous compactly supported function, for which the shift converges uniformly),

$$\|f_t - f\|_{W^{k,p}(V)} = \Big(\sum_{|\alpha| \leq k} \|(\partial^\alpha f)_t - \partial^\alpha f\|_{L^p(V)}^p\Big)^{1/p} \to 0 \qquad (t \to 0)$$

Fix $t \leq \lambda$ with $\|f_t - f\|_{W^{k,p}(V)} < \varepsilon/2$.

Step 2: Mollify the shifted function. By construction $B(x^t, t/2) \subseteq U$ for every $x \in V$, so for $\delta < t/2$ the mollification $g^\delta := f_t * \psi_\delta$ samples f only at points of U: for $x \in \overline{V}$,

$$g^\delta(x) = \int_{B(0,\delta)} f(x + te_n - y)\, \psi_\delta(y)\, dy$$

is well-defined because $x + te_n - y \in B(x^t, t/2) \subseteq U$. Thus g^δ is defined and smooth on a neighbourhood of $\overline{V}$ (the mollification of a locally integrable function is smooth, by lemma 8.6.7), including the boundary piece $\partial U \cap B(x_0, r/2)$. By theorem 10.2.1 applied to f_t on V (whose mollification is interior to the slightly larger set where f_t is defined), $g^\delta \to f_t$ in $W^{k,p}(V)$ as

$\delta \to 0$; fix $\delta < t/2$ with $\|g^\delta - f_t\|_{W^{k,p}(V)} < \varepsilon/2$. Then $g^\delta \in C^\infty(\overline{V})$ and, by the triangle inequality,

$$\|g^\delta - f\|_{W^{k,p}(V)} \le \|g^\delta - f_t\|_{W^{k,p}(V)} + \|f_t - f\|_{W^{k,p}(V)} < \varepsilon$$

Step 3: Patch the boundary charts with the interior. The compact boundary ∂U is covered by finitely many such half-ball neighbourhoods $V_1, \dots, V_N$, on each of which Steps 1 and 2 produce a $g_j \in C^\infty(\overline{V_j})$ with $\|g_j - f\|_{W^{k,p}(U\cap V_j)} < \varepsilon$. Adjoin one more open set $V_0 \Subset U$ with $\overline{U} \subseteq V_0 \cup V_1 \cup \cdots \cup V_N$ and, on it, the interior mollification $g_0 \in C^\infty(V_0)$ of theorem 10.2.1 with $\|g_0 - f\|_{W^{k,p}(V_0)} < \varepsilon$. Let $\{\eta_j\}_{j=0}^N$ be a smooth partition of unity subordinate to $\{V_j\}_{j=0}^N$ over a neighbourhood of $\overline{U}$, and set

$$g = \sum_{j=0}^{N} \eta_j \, g_j$$

Each $\eta_j g_j$ is smooth up to $\overline{U}$ (either interior, or smooth on $\overline{V_j}$), so $g \in C^\infty(\overline{U})$. Using $\sum_j \eta_j = 1$ and the Leibniz rule as in the proof of theorem 10.2.3, on U

$$\partial^\alpha (g - f) = \sum_{j=0}^{N} \sum_{\beta \le \alpha} \binom{\alpha}{\beta} \partial^\beta \eta_j \, \partial^{\alpha-\beta}(g_j - f)$$

Since the derivatives of the η_j are bounded (there are finitely many, each in C_c^∞) by a constant M depending only on k and the cover, and each g_j satisfies $\|g_j - f\|_{W^{k,p}(U\cap V_j)} < \varepsilon$,

$$\|g - f\|_{W^{k,p}(U)} \le C(k, N, M)\, \varepsilon$$

for a constant independent of ε. As $\varepsilon > 0$ was arbitrary, $C^\infty(\overline{U})$ is dense in $W^{k,p}(U)$, completing the proof.

The mechanism to remember is the inward shift. Mollification alone fails at the boundary only because it reaches outside U; translating f a distance t into the domain first pulls the relevant values away from ∂U, giving room to mollify at the finer scale $\delta < t/2$ without ever leaving U. The C^1 hypothesis enters exactly to guarantee a consistent inward direction near each boundary point, so that a single shift $+te_n$ (after flattening) works throughout a boundary chart. Domains with cusps or inward spikes, where no such uniform inward cone exists, can fail this theorem; the C^1 regularity is what rules them out. This global approximation is what is used in the extension and trace theorems of the next section, and behind every argument in this chapter that proves an estimate first for smooth functions and then extends it by density.

The density theorems distinguish two natural classes of Sobolev functions: those approximable by smooth functions with no constraint at the boundary, which is all of $W^{k,p}(U)$, and those approximable by smooth functions that *vanish near the boundary.* The latter class is where boundary conditions live, and it deserves a name.

Definition 10.2.6: The Space $W_0^{k,p}$

Let $U \subseteq \mathbb{R}^n$ be open, let $k \geq 0$, and let $1 \leq p < \infty$. The space $W_0^{k,p}(U)$ is the closure of $C_c^\infty(U)$ in the norm of $W^{k,p}(U)$:

$$W_0^{k,p}(U) := \overline{C_c^\infty(U)}^{\|\cdot\|_{W^{k,p}(U)}}$$

When $p = 2$ one writes $H_0^k(U) := W_0^{k,2}(U)$.

By construction $W_0^{k,p}(U)$ is a closed subspace of the Banach space $W^{k,p}(U)$, hence a Banach space in its own right, and $H_0^k(U)$ a Hilbert space. Its elements are the Sobolev functions that are limits, in the Sobolev norm, of functions compactly supported inside U, and one reads this as the functions "with zero boundary values" together with, for $k \geq 2$, vanishing normal derivatives up to order $k-1$. The trace theorem of the next section will make this interpretation precise for $k = 1$ by identifying $W_0^{1,p}(U)$ with the kernel of the boundary-value map (theorem 10.3.9); for now it is a definition, and the one case where it collapses onto the full space is worth recording at once.

Proposition 10.2.7: $W_0^{k,p}(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$

For every $k \geq 0$ and $1 \leq p < \infty$, $W_0^{k,p}(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$.

***Proof*:**

The inclusion $W_0^{k,p}(\mathbb{R}^n) \subseteq W^{k,p}(\mathbb{R}^n)$ is immediate, as $W_0^{k,p}$ is a subspace. For the reverse, it suffices to approximate an arbitrary $f \in W^{k,p}(\mathbb{R}^n)$ by elements of $C_c^\infty(\mathbb{R}^n)$ in the $W^{k,p}$ norm; the smooth approximation runs directly on the whole space, and the only extra work is to make the approximants compactly supported by a cutoff.

Fix $\varepsilon > 0$. First mollify. On $\mathbb{R}^n$ there is no boundary to obstruct the convolution, so the commutation identity holds globally: for $U = \mathbb{R}^n$ every ball $B(x,\delta)$ lies in $\mathbb{R}^n$, so the differentiation-under-the-integral and weak-derivative steps of theorem 10.2.1 Step 1 go through for all x, giving $\partial^\alpha(f * \psi_\delta) = (\partial^\alpha f) * \psi_\delta$ on all of $\mathbb{R}^n$ for $|\alpha| \leq k$. Each $\partial^\alpha f \in L^p(\mathbb{R}^n)$, so the approximate-identity convergence of theorem 8.6.10 on $L^p(\mathbb{R}^n)$ gives $(\partial^\alpha f) * \psi_\delta \to \partial^\alpha f$ in $L^p(\mathbb{R}^n)$ as $\delta \to 0$; summing over $|\alpha| \leq k$, $f * \psi_\delta \to f$ in $W^{k,p}(\mathbb{R}^n)$. Choose $h := f * \psi_\delta \in C^\infty(\mathbb{R}^n) \cap W^{k,p}(\mathbb{R}^n)$ with $\|h - f\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon/2$. Now cut off. Fix a function $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi \equiv 1$ on $B(0,1)$, $0 \leq \chi \leq 1$, supported in $B(0,2)$, and set $\chi_R(x) = \chi(x/R)$, so $\partial^\alpha \chi_R$ is bounded by $C_\alpha R^{-|\alpha|} \leq C_\alpha$ for $R \geq 1$ and $\chi_R \equiv 1$ on $B(0,R)$. Then $\chi_R h \in C_c^\infty(\mathbb{R}^n)$, and by the Leibniz rule

$$\partial^\alpha(\chi_R h - h) = \partial^\alpha\big((\chi_R - 1)h\big) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta(\chi_R - 1)\, \partial^{\alpha-\beta} h$$

where the $\beta = 0$ term is $(\chi_R - 1)\,\partial^\alpha h$ and the $\beta \neq 0$ terms carry a factor $\partial^\beta \chi_R$ supported in the annulus $R \leq |x| \leq 2R$. Since $\chi_R - 1$ vanishes on $B(0,R)$ and is bounded by 1, and each $\partial^\alpha h \in L^p(\mathbb{R}^n)$, dominated convergence gives $\|(\chi_R - 1)\,\partial^\alpha h\|_{L^p(\mathbb{R}^n)} \to 0$ as $R \to \infty$; the annular terms are bounded by $C \sum_{|\gamma|<|\alpha|} \|\partial^\gamma h\|_{L^p(R \leq |x| \leq 2R)} \to 0$ likewise. Summing over $|\alpha| \leq k$, choose R with $\|\chi_R h - h\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon/2$. Then $\chi_R h \in C_c^\infty(\mathbb{R}^n)$ and

$$\|\chi_R h - f\|_{W^{k,p}(\mathbb{R}^n)} \leq \|\chi_R h - h\|_{W^{k,p}(\mathbb{R}^n)} + \|h - f\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon$$

so f lies in the closure of $C_c^\infty(\mathbb{R}^n)$, as we sought to show.

The proposition says that on the whole space there is no boundary to see, so the constraint defining $W_0^{k,p}$ is empty and the two spaces agree. On a bounded domain the situation is the opposite extreme: $W_0^{k,p}(U)$ is a genuinely proper closed subspace of $W^{k,p}(U)$, precisely because a nonzero constant function belongs to $W^{k,p}(U)$ but cannot be approximated by compactly supported functions without paying for the boundary layer where it must drop to zero. The gap between the two spaces is what the trace measures, and the Poincaré inequality of the last section of this chapter will quantify how the vanishing boundary condition of $W_0^{1,p}$ controls a function by its gradient alone.

10.3 Extensions and Traces

A Sobolev function on a domain U is defined only inside U, and two operations that the elliptic theory needs are not visible from the definition alone: extending such a function to all of $\mathbb{R}^n$ without losing a derivative, and restricting it to the boundary ∂U where a function without pointwise values must still be assigned boundary data. This section builds both, resting on the same device, flattening the C^1 boundary in local charts, and on the density of functions smooth up to the boundary proved in the previous section.

Throughout, $U \subseteq \mathbb{R}^n$ is bounded with C^1 boundary ∂U: near each boundary point there is a ball in which, after relabelling coordinates, ∂U is the graph $x_n = \gamma(x')$ of a C^1 function γ of the first $n-1$ coordinates $x' = (x_1, \ldots, x_{n-1})$, with U lying on the side $x_n > \gamma(x')$. The boundary carries its surface measure, and ∂U is a compact $(n-1)$-dimensional C^1 manifold.

The extension is built one boundary chart at a time. In a chart where the boundary has been straightened to a piece of the hyperplane $x_n = 0$, a $W^{1,p}$ function on the upper half is extended to the lower half by reflection. A naive even reflection $f(x', -x_n)$ keeps f continuous across the interface but jumps in $\partial_n f$; the higher-order reflection below instead matches the first weak derivatives across $\{x_n = 0\}$, the condition for the reflected function to lie in $W^{1,p}$ of the full ball. The two lemmas that follow isolate this half-space reflection and the passage across a flattened C^1 graph; the theorem then patches the charts together with a partition of unity.

Write $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n \mid x_n > 0\}$ and $\mathbb{R}^n_- = \{x \in \mathbb{R}^n \mid x_n < 0\}$, and for a ball $B = B(0, r)$ put $B_+ = B \cap \mathbb{R}^n_+$ and $B_- = B \cap \mathbb{R}^n_-$.

Lemma 10.3.1: Higher-Order Reflection Across a Hyperplane

Let $B = B(0, r)$ and let $f \in W^{1,p}(B_+)$ with $1 \le p < \infty$, and suppose f is supported away from the upper cap: $f = 0$ almost everywhere on $B_+ \setminus B(0, \rho)$ for some $\rho < r/2$. Define Ef on B by

$$(Ef)(x) = \begin{cases} f(x', x_n) & x_n > 0 \\ 3f(x', -x_n) - 2f(x', -2x_n) & x_n < 0 \end{cases}$$

where the reflected values are read off as 0 wherever their argument leaves B_+. Then $Ef \in W^{1,p}(B)$, it agrees with f on B_+, is supported in $B(0, \rho)$, and

$$\|Ef\|_{W^{1,p}(B)} \le C\|f\|_{W^{1,p}(B_+)}$$

with C depending only on n.

***Proof*:**

The plan is to prove the estimate and the extension property for $f \in C^1(\overline{B_+})$, where Ef can be differentiated classically, and then remove the smoothness by the density of such functions.

Step 1: The reflection matches value and normal derivative for smooth f. Suppose first $f \in C^1(\overline{B_+})$ vanishing on $B_+ \setminus B(0, \rho)$ with $\rho < r/2$. On B_- set $g(x) = 3f(x', -x_n) - 2f(x', -2x_n)$, so $Ef = f$ on B_+ and $Ef = g$ on B_-. The support hypothesis makes both reflected pieces well-defined on B_-: $f(x', -x_n)$ is nonzero only when $|(x', -x_n)| = |x| < \rho$, and $f(x', -2x_n)$ only when $|(x', -2x_n)|^2 = |x'|^2 + 4x_n^2 < \rho^2$, which forces $|x| < \rho$ as well, so in both cases the argument lies in $B(0, \rho)_+ \subseteq B_+$ (elsewhere the reflected value is 0). In particular g is supported in $B(0, \rho) \cap B_-$, and Ef in $B(0, \rho)$. As $x_n \to 0^-$ both reflected arguments tend to $(x', 0)$, and

$$\lim_{x_n \to 0^-} g(x', x_n) = 3f(x', 0) - 2f(x', 0) = f(x', 0)$$

so Ef is continuous across $\{x_n = 0\}$. Differentiating g classically on B_-,

$$\partial_n g(x) = -3\,\partial_n f(x', -x_n) + 4\,\partial_n f(x', -2x_n)$$

whose boundary limit is $(-3 + 4)\,\partial_n f(x', 0) = \partial_n f(x', 0)$, matching the upper value $\partial_n f(x', 0)$; the tangential derivatives $\partial_i g$ for $i < n$ carry the factors 3 and -2 through the chain rule with total $3 - 2 = 1$, so they too match $\partial_i f(x', 0)$. Thus $Ef \in C^1(\overline{B})$ and its classical gradient is bounded on B by a multiple of $\sup_{B_+}(|f| + |\nabla f|)$.

Step 2: The classical gradient is the weak gradient. Because Ef is continuous across $\{x_n = 0\}$ and C^1 on each of B_+, B_- separately, its weak gradient on all of B is the function that equals the classical gradient off the interface. To see this, fix $\varphi \in C_c^\infty(B)$ and integrate by parts on B_+ and B_- separately:

$$\int_B (Ef)\,\partial_i\varphi\,dx = \int_{B_+} f\,\partial_i\varphi\,dx + \int_{B_-} g\,\partial_i\varphi\,dx$$

Integration by parts on B_+ produces $-\int_{B_+}(\partial_i f)\varphi$ plus a boundary integral over $\{x_n = 0\}$ with integrand $f(x', 0)\varphi\,\nu_i$, where ν is the outward normal of B_+; the same on B_- produces $-\int_{B_-}(\partial_i g)\varphi$ plus a boundary integral with integrand $g(x', 0)\varphi\,(-\nu_i)$. The two boundary integrals cancel because $f(x', 0) = g(x', 0)$ by Step 1. Hence $\int_B (Ef)\,\partial_i\varphi = -\int_B (\partial_i^{\mathrm{cl}} Ef)\,\varphi$, so the weak derivative $\partial_i(Ef)$ is the classical one, as claimed.

Step 3: The norm estimate for smooth f. With the weak and classical derivatives identified, change variables in the reflected pieces. Under $x_n \mapsto -x_n$ the map $(x', x_n) \mapsto (x', -x_n)$ has unit Jacobian, and under $x_n \mapsto -2x_n$ the map $(x', x_n) \mapsto (x', -2x_n)$ has Jacobian $\frac{1}{2}$ in the x_n variable. By Step 1 both substitutions carry the region where the corresponding reflected value is nonzero into $B(0,\rho)_+ \subseteq B_+$, so for the value term

$$\int_{B_-} |g|^p\,dx \le 2^{p-1}\int_{B_-}\big(3^p|f(x',-x_n)|^p + 2^p|f(x',-2x_n)|^p\big)\,dx \le C\int_{B_+}|f|^p\,dx$$

using $(a+b)^p \le 2^{p-1}(a^p+b^p)$ and the substitutions (the second contributing a factor $\frac{1}{2}$ from its Jacobian, and integrating over a region contained in B_+ because f vanishes outside $B(0,\rho)_+$). The identical estimate applied to the derivative formulas of Step 1, whose coefficients are again bounded by constants depending only on n, gives $\int_{B_-}|\nabla g|^p \le C\int_{B_+}|\nabla f|^p$. Summing, $\|Ef\|^p_{W^{1,p}(B)} \le C\|f\|^p_{W^{1,p}(B_+)}$ with $C = C(n)$.

Step 4: Removal of smoothness by density. For general $f \in W^{1,p}(B_+)$ vanishing on $B_+ \setminus B(0,\rho)$, the domain B_+ has C^1 (indeed Lipschitz) boundary, so by theorem 10.2.5 there are $\tilde f_m \in C^\infty(\overline{B_+})$ with $\tilde f_m \to f$ in $W^{1,p}(B_+)$. Fix a cutoff $\chi \in C_c^\infty(B(0,\rho'))$ with $\rho < \rho' < r/2$, $0 \le \chi \le 1$, and $\chi \equiv 1$ on $B(0,\rho)$, and set $f_m := \chi \tilde f_m$. Since $f = \chi f$, the product rule gives $f_m \to f$ in $W^{1,p}(B_+)$, and each $f_m \in C^\infty(\overline{B_+})$ vanishes on $B_+ \setminus B(0,\rho')$ with $\rho' < r/2$, so the reflection of Steps 1 to 3 applies to each f_m. The map E is linear, and Step 3 gives $\|Ef_m - Ef_l\|_{W^{1,p}(B)} \le C\|f_m - f_l\|_{W^{1,p}(B_+)}$, so (Ef_m) is Cauchy in $W^{1,p}(B)$; let F be its limit. Restricting to B_+, where $Ef_m = f_m$, gives $F = f$ on B_+. The estimate passes to the limit as $\|F\|_{W^{1,p}(B)} \le C\|f\|_{W^{1,p}(B_+)}$, and the explicit reflection formula for F on B_- follows because $f_m(x',-x_n) \to f(x',-x_n)$ and $f_m(x',-2x_n) \to f(x',-2x_n)$ in $L^p(B_-)$ under the same changes of variable. Thus $F = Ef$ as defined, and $Ef \in W^{1,p}(B)$ with the stated bound, completing the proof.

The reflection is engineered so that both the value and the full gradient are continuous across the interface, which Step 2 shows is exactly the condition making the two half-space derivatives assemble into a single weak derivative on the whole ball; the even reflection fails precisely at the normal derivative, and the coefficients $3, -2$ are the unique pair (a,b) correcting it while preserving continuity of the value: matching the value forces $a+b=1$ and matching the normal derivative forces $-a-2b=1$ (differentiating the two reflected pieces at $x_n = 0$), two linear equations solved only by $a = 3$, $b = -2$. The support hypothesis, confining the reflected arguments to the half-ball where f lives, costs nothing in the application: the localized pieces that the extension operator reflects are cut off by a partition of unity and already vanish near the top of their chart. This is the entire content of the extension near a straight boundary. The passage to a curved C^1 boundary is a change of variables that straightens it, recorded in the next lemma.

Lemma 10.3.2: Straightening a C^1 Boundary

Let $w \in \partial U$. There is an open ball V about w and a C^1 diffeomorphism $\Phi\colon V \to B(0,r)$ with C^1 inverse $\Psi = \Phi^{-1}$, such that $\Phi(V \cap U) = B(0,r)_+$ and $\Phi(V \cap \partial U) = B(0,r) \cap \{x_n = 0\}$. Composition with Φ and Ψ carries $W^{1,p}(V \cap U)$ boundedly to $W^{1,p}(B(0,r)_+)$ and back, with constants depending on Φ.

***Proof*:**
By the C^1 boundary hypothesis, after relabelling coordinates there is a ball V about w and a C^1 function γ of $x' = (x_1, \ldots, x_{n-1})$ with $V \cap \partial U = \{x \in V \mid x_n = \gamma(x')\}$ and $V \cap U = \{x \in V \mid x_n > \gamma(x')\}$. Define

$$\Phi(x) = (x', x_n - \gamma(x')), \qquad \Psi(y) = (y', y_n + \gamma(y'))$$

Both are C^1 (as γ is C^1) and mutually inverse, with Jacobian determinant identically 1. The shear image $\Phi(V)$ is an open set containing $\Phi(w) = 0$ but need not be a ball, so fix a radius r small enough that $B(0,r) \subseteq \Phi(V)$ and *re-restrict*, replacing V by $\Phi^{-1}(B(0,r))$; on this smaller neighborhood Φ is a C^1 diffeomorphism onto exactly $B(0,r)$, and Φ sends $x_n > \gamma(x')$ to $y_n > 0$ and $x_n = \gamma(x')$ to $y_n = 0$, giving the stated images. For a function u on $V \cap U$, the composition $u \circ \Psi$ is defined on $B(0,r)_+$; by the chain rule for weak derivatives under C^1 changes of variable, $u \in W^{1,p}(V \cap U)$ if and only if $u \circ \Psi \in W^{1,p}(B(0,r)_+)$, with

$$\nabla(u \circ \Psi)(y) = (D\Psi(y))^\top \, (\nabla u)(\Psi(y))$$

and since $D\Phi, D\Psi$ are continuous hence bounded on the relevant compact sets and the Jacobians are 1, the L^p norms of u and its gradient are comparable to those of $u \circ \Psi$ up to a constant depending on $\sup|D\Phi|, \sup|D\Psi|$. The inverse map $v \mapsto v \circ \Phi$ is bounded likewise, as we sought to show.

With the local reflection and the local straightening in hand, the global extension is assembled by a finite partition of unity subordinate to boundary charts, extending each localized piece by reflection in its straightened chart and extending the interior piece by zero.

Theorem 10.3.3: Extension Operator

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 \le p < \infty$. Fix a bounded open set W with $U \Subset W$. There is a bounded linear operator

$$E \colon W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$$

such that for every $f \in W^{1,p}(U)$:

1. $Ef = f$ almost everywhere on U;
2. Ef is supported in W;
3. $\|Ef\|_{W^{1,p}(\mathbb{R}^n)} \le C\|f\|_{W^{1,p}(U)}$, with $C = C(n,p,U,W)$.

***Proof*:**
Step 1: A finite cover by boundary charts. The boundary ∂U is compact, and by lemma 10.3.2 each $w \in \partial U$ has a ball V_w carrying a straightening diffeomorphism $\Phi_w \colon V_w \to B(0, r_w)$. Shrinking each V_w so that $V_w \Subset W$, the balls $\{V_w\}_{w \in \partial U}$ cover ∂U; extract a finite subcover $V_1, \ldots, V_N$ with straightenings $\Phi_1, \ldots, \Phi_N$. The compact set $K = \overline{U} \setminus \bigcup_{j=1}^N V_j$ lies in the open set U, so choose an open V_0 with $K \Subset V_0 \Subset U$. Then $V_0, V_1, \ldots, V_N$ is a finite open cover of $\overline{U}$, all contained in W. Choose a C^∞ partition of unity $\zeta_0, \zeta_1, \ldots, \zeta_N$ subordinate to it: each $\zeta_j \in C_c^\infty(V_j)$, $0 \le \zeta_j \le 1$, and $\sum_{j=0}^N \zeta_j = 1$ on a neighborhood of $\overline{U}$. For each boundary chart

$j \geq 1$, the support of ζ_j is a compact subset of V_j, so its image $\Phi_j(\mathrm{supp}\zeta_j \cap \overline{U})$ is a compact subset of $B(0, r_j)$, contained in some $\overline{B(0, s_j)}$ with $s_j < r_j$. Post-composing Φ_j with the dilation $x \mapsto (r_j/3s_j)\, x$ (a straightening onto $B(0, r_j/3)$, then relabelled to radius r_j) moves this image inside $B(0, r_j/3)$, so that any function supported in it vanishes on $B(0, r_j)_+ \setminus B(0, \rho_j)$ for some $\rho_j < r_j/2$.

Step 2: Extend each localized piece. Given $f \in W^{1,p}(U)$, decompose $f = \sum_{j=0}^N \zeta_j f$ on U. The interior piece $\zeta_0 f$ has compact support in $V_0 \Subset U$; extending it by zero outside U gives a function $E_0 f \in W^{1,p}(\mathbb{R}^n)$ with $\|E_0 f\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|\zeta_0 f\|_{W^{1,p}(U)} \leq C\|f\|_{W^{1,p}(U)}$, the product rule $\nabla(\zeta_0 f) = \zeta_0 \nabla f + f \nabla \zeta_0$ bounding the gradient since $\nabla \zeta_0$ is bounded. For a boundary piece $\zeta_j f$ with $j \geq 1$, transport to the straightened chart: $u_j := (\zeta_j f) \circ \Psi_j$ lies in $W^{1,p}(B(0, r_j)_+)$ by lemma 10.3.2 and, by the radius choice of Step 1, is supported in $B(0, \rho_j)$ with $\rho_j < r_j/2$, so it satisfies the support hypothesis of lemma 10.3.1. That lemma gives a reflection $Eu_j \in W^{1,p}(B(0, r_j))$ satisfying $\|Eu_j\|_{W^{1,p}(B(0,r_j))} \leq C\|u_j\|_{W^{1,p}(B(0,r_j)_+)}$. Transport back: $E_j f := (Eu_j) \circ \Phi_j$, extended by zero outside V_j, is a function on $\mathbb{R}^n$ that agrees with $\zeta_j f$ on $V_j \cap U$, is supported in $V_j \Subset W$, and by the two comparisons of lemma 10.3.2 satisfies

$$\|E_j f\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|Eu_j\|_{W^{1,p}(B(0,r_j))} \leq C\|u_j\|_{W^{1,p}(B(0,r_j)_+)} \leq C\|\zeta_j f\|_{W^{1,p}(V_j \cap U)} \leq C\|f\|_{W^{1,p}(U)}$$

Extension by zero across ∂V_j preserves the $W^{1,p}$ class because Eu_j, hence $E_j f$, already vanishes in a neighborhood of ∂V_j.

Step 3: Assemble and estimate. Set $Ef := \sum_{j=0}^N E_j f$. Each E_j is linear in f (reflection, chart transport, and multiplication by ζ_j are all linear), so E is linear. On U every $E_j f$ agrees with $\zeta_j f$, so $Ef = \sum_j \zeta_j f = f$ almost everywhere on U, giving (1). Each $E_j f$ is supported in $V_j \subseteq W$, so Ef is supported in W, giving (2). Finally, by the triangle inequality and the $(N+1)$ bounds of Step 2,

$$\|Ef\|_{W^{1,p}(\mathbb{R}^n)} \leq \sum_{j=0}^N \|E_j f\|_{W^{1,p}(\mathbb{R}^n)} \leq C\,(N+1)\,\|f\|_{W^{1,p}(U)}$$

with C absorbing the chart constants, giving (3) with $C = C(n, p, U, W)$. Thus E is a bounded linear extension operator, completing the proof.

The extension operator makes the domain U transparent to any theorem already known on $\mathbb{R}^n$: prove an inequality for $W^{1,p}(\mathbb{R}^n)$, apply it to Ef, and restrict back to U using $Ef = f$ there, at the cost of the operator norm C. This is exactly how the Rellich-Kondrachov compactness of section 10.5 will descend from a statement on all of $\mathbb{R}^n$ to bounded C^1 domains. The dependence of C on U is real and not an artifact: the constant blows up as the boundary degenerates (a domain with an outward cusp has no bounded extension operator), which is why the C^1 hypothesis, controlling $\sup |D\Phi_j|$, cannot be dropped.

Example 10.3.4: Extension of a Radial Bump on the Ball

Let $U = B(0, 1) \subseteq \mathbb{R}^n$ and $f(x) = 1 - |x|$, which lies in $W^{1,p}(U)$ for every p: it is Lipschitz, with $\nabla f(x) = -x/|x|$ of modulus 1 almost everywhere, so $\|f\|^p_{W^{1,p}(U)} = \int_U (1 - |x|)^p + \int_U 1 < \infty$. The construction straightens the sphere ∂U in each boundary chart; because f is already Lipschitz on $\overline{U}$ with $f = 0$ on ∂U, the simplest admissible extension is $Ef(x) = \max(1 - |x|, 0)$, which is Lipschitz on all of $\mathbb{R}^n$, equals f on U, is supported in $\overline{B(0, 1)} \Subset W = B(0, 2)$, and satisfies $\|Ef\|_{W^{1,p}(\mathbb{R}^n)} = \|f\|_{W^{1,p}(U)}$ since Ef adds no mass outside U. The reflection machinery is needed only when f does not vanish on the boundary, where an even continuation would break the normal derivative; here $f|_{\partial U} = 0$ lets extension by zero do the job, and the general operator agrees with it up to the boundary chart constants.

The trace is the second boundary operation. A function in $W^{1,p}(U)$ is an equivalence class modulo sets of Lebesgue measure zero, and ∂U has n-dimensional Lebesgue measure zero, so "restrict f to ∂U" has no meaning on the nose. The trace theorem gives it meaning: on functions smooth up to the boundary, restriction is a well-defined map into $L^p(\partial U)$; the estimate below shows this map is bounded for the $W^{1,p}$ norm, so it extends by density to all of $W^{1,p}(U)$. Before the theorem, one preliminary space records where the trace actually lands, which is finer than $L^p(\partial U)$.

The trace of a $W^{1,p}$ function lands in more than $L^p(\partial U)$: it carries a fractional order of differentiability, one derivative shy of the full jump from U (dimension n) to ∂U (dimension $n-1$) discounted by the integrability exponent. That fractional regularity is measured by a seminorm with no reference to integer derivatives, integrating a difference quotient of order s against the surface measure of ∂U. Here $d = n-1$ is the dimension of ∂U.

Definition 10.3.5: Fractional Sobolev Space $W^{s,p}(\partial U)$

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary, write $d = n-1$ for the dimension of ∂U, and let $0 < s < 1$ and $1 \le p < \infty$. The *Gagliardo (Slobodeckij) seminorm* of a measurable $f \colon \partial U \to \mathbb{R}$ is

$$[f]_{W^{s,p}(\partial U)} = \left(\int_{\partial U} \int_{\partial U} \frac{|f(x) - f(y)|^p}{|x-y|^{d+sp}} \, dS(x) \, dS(y) \right)^{1/p}$$

where dS is the surface measure and $|x-y|$ the Euclidean distance in $\mathbb{R}^n$. The *fractional Sobolev space* is

$$W^{s,p}(\partial U) = \left\{ f \in L^p(\partial U) \;\middle|\; [f]_{W^{s,p}(\partial U)} < \infty \right\}$$

normed by $\|f\|_{W^{s,p}(\partial U)} = \left(\|f\|^p_{L^p(\partial U)} + [f]^p_{W^{s,p}(\partial U)} \right)^{1/p}$.

The exponent $d + sp$ in the denominator is calibrated so that the seminorm sees exactly s derivatives: replacing f by $f(\lambda\,\cdot)$ scales $[f]_{W^{s,p}}$ by $\lambda^{s-d/p}$ (the change of variables $u = \lambda x$ turns the p-th power of the seminorm into λ^{sp-d} times itself, matching the scaling of $\|\nabla^s f\|_{L^p}$ for the would-be integer analog), which is what makes s the order of differentiability. Finiteness of the double integral demands that $f(x) - f(y)$ decay faster than $|x-y|^{s+d/p}$ near the diagonal on average, a genuine but fractional smoothness: $W^{s,p}$ sits strictly between L^p and the integer space $W^{1,p}$ as s runs through $(0,1)$.

Example 10.3.6: The Threshold Exponent of a Power Singularity

Take the unit circle $\partial U = \partial B(0,1) \subseteq \mathbb{R}^2$, so $d = 1$, parametrized by arc length $\theta \in [0, 2\pi)$, and the function with a mild corner at $\theta = 0$, $f(\theta) = |\theta|^a$ near 0 for a fixed $0 < a < 1$ (extended smoothly elsewhere). Near the diagonal the seminorm integrand behaves like $|f(\theta) - f(\eta)|^p / |\theta - \eta|^{1+sp}$, and a direct estimate of $\iint$ over a neighborhood of $\theta = \eta = 0$ shows the double integral converges precisely when $a > s$: the difference $|\theta^a - \eta^a|$ is comparable to $|\theta - \eta|^a$ for a corner of this shape, because $t \mapsto |t|^a$ with $0 < a < 1$ is a-Hölder continuous ($||\theta|^a - |\eta|^a| \le |\theta - \eta|^a$ by concavity of t^a) and the bound is saturated when one of the two points sits at the corner (e.g. $\eta \approx 0$, where $||\theta|^a - |\eta|^a| = |\theta|^a \sim |\theta - \eta|^a$), so the integrand is comparable to $|\theta - \eta|^{ap-1-sp}$, integrable across the diagonal exactly when $ap - 1 - sp > -1$, i.e. $a > s$. Thus $f \in W^{s,p}(\partial U)$ if and only if $s < a$: the corner of exponent a has fractional smoothness of every order below a and none above, pinning the seminorm down as a sharp gauge of the singularity.

The trace theorem now states what the boundary restriction is and where it lands. Its heart is a single integration by parts on a boundary chart, bounding the L^p norm of the boundary values by the $W^{1,p}$ norm on the interior; density (theorem 10.2.5) then produces the operator on all of $W^{1,p}(U)$, and a refinement of the same estimate reads off the sharper fractional target.

Theorem 10.3.7: Trace Theorem

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 \le p < \infty$. There is a bounded linear operator

$$T\colon W^{1,p}(U) \to L^p(\partial U)$$

the *trace*, such that $Tf = f|_{\partial U}$ for every $f \in C^1(\overline{U}) \cap W^{1,p}(U)$, and

$$\|Tf\|_{L^p(\partial U)} \le C\|f\|_{W^{1,p}(U)}$$

with $C = C(n,p,U)$. Moreover T maps $W^{1,p}(U)$ into $W^{1-1/p,p}(\partial U)$ for $1 < p < \infty$, boundedly for the fractional norm of definition 10.3.5.

***Proof*:**

Step 1: The boundary estimate on a flat chart. Suppose first $f \in C^1(\overline{U})$, and work in a boundary chart straightened by lemma 10.3.2, so it suffices to bound $\int_{\{x_n=0\}} |f|^p\,dx'$ over a piece of the flattened boundary in terms of the $W^{1,p}$ norm on the half-ball B_+ above it. Fix a cutoff $\xi \in C_c^\infty(B)$ with $\xi \equiv 1$ on the smaller half-ball where the boundary piece sits and $0 \le \xi \le 1$. For x' in that piece, the fundamental theorem of calculus in the x_n-direction gives, since ξf vanishes for large x_n,

$$|f(x',0)|^p = -\int_0^\infty \partial_n\big(\xi(x',x_n)\,|f(x',x_n)|^p\big)\,dx_n$$

Expanding the derivative, $\partial_n(\xi|f|^p) = (\partial_n\xi)|f|^p + \xi\,p|f|^{p-1}\operatorname{sgn}(f)\,\partial_n f$, and integrating in x' over the boundary piece,

$$\int |f(x',0)|^p\,dx' \le \int_{B_+} \Big(|\partial_n\xi|\,|f|^p + p\,\xi\,|f|^{p-1}|\partial_n f|\Big)\,dx$$

The first term is bounded by $C\int_{B_+}|f|^p$. For the second, when $p > 1$ Young's inequality $ab \le \frac{a^{p'}}{p'} + \frac{b^p}{p}$ with $a = |f|^{p-1}$, $b = |\partial_n f|$ (so $a^{p'} = |f|^p$) gives $|f|^{p-1}|\partial_n f| \le \frac{p-1}{p}|f|^p + \frac{1}{p}|\partial_n f|^p$; the degenerate case $p = 1$ is immediate, the integrand $p\,\xi\,|f|^{p-1}|\partial_n f|$ reducing to $\xi\,|\partial_n f| \le |\partial_n f|$ with no exponent to split. Either way,

$$\int |f(x',0)|^p\,dx' \le C\int_{B_+}\big(|f|^p + |\nabla f|^p\big)\,dx = C\|f\|^p_{W^{1,p}(B_+)}$$

Summing over a finite cover of ∂U by such charts, and using that the surface measure dS is comparable on each chart to dx' up to the bounded Jacobian factor of the graph parametrization (lemma 10.3.2), yields

$$\int_{\partial U} |f|^p\,dS \le C\|f\|^p_{W^{1,p}(U)} \qquad (f \in C^1(\overline{U}))$$

Step 2: Extension to all of $W^{1,p}(U)$. Define $Tf = f|_{\partial U}$ for $f \in C^1(\overline{U})$, a linear map into $L^p(\partial U)$ bounded by Step 1. By theorem 10.2.5, $C^\infty(\overline{U})$, hence $C^1(\overline{U})$, is dense in $W^{1,p}(U)$: given $f \in W^{1,p}(U)$ take $f_m \in C^\infty(\overline{U})$ with $f_m \to f$ in $W^{1,p}(U)$. Then (f_m) is Cauchy in $W^{1,p}(U)$, and Step 1 applied to $f_m - f_l$ gives $\|Tf_m - Tf_l\|_{L^p(\partial U)} \le C\|f_m - f_l\|_{W^{1,p}(U)}$, so (Tf_m) is Cauchy in the complete space $L^p(\partial U)$; define $Tf = \lim_m Tf_m$. The limit is independent of the approximating sequence: if $g_m \to f$ as well, interleaving the two sequences is again $W^{1,p}$ convergent to f, so the traces converge to a common limit. The bound $\|Tf\|_{L^p(\partial U)} \le C\|f\|_{W^{1,p}(U)}$ passes to the limit, and T is linear as a limit of linear maps. For $f \in C^1(\overline{U}) \cap W^{1,p}(U)$ the constant sequence $f_m = f$ shows $Tf = f|_{\partial U}$, so T extends pointwise restriction.

Step 3: The sharp fractional target. Fix $1 < p < \infty$; the claim is $[Tf]_{W^{1-1/p,p}(\partial U)} \le C\|f\|_{W^{1,p}(U)}$, so that $s = 1 - 1/p$ is the order of the trace. By Step 2 and density it suffices to prove this for $f \in C^1(\overline{U})$, the seminorm passing to the limit as in Step 2. Work again in a straightened chart, where the boundary is $\{x_n = 0\}$ and $g := f|_{x_n=0}$; the seminorm is comparable to the flat-chart integral $\iint |g(x') - g(y')|^p/|x' - y'|^{d+sp}\, dx'\, dy'$ with $d = n - 1$; the comparability uses that the C^1 graph is a bi-Lipschitz parametrization, so $|x - y|$ on ∂U and $|x' - y'|$ in the chart agree up to a constant (lemma 10.3.2). For $x' \ne y'$ set $h = |x' - y'|$ and pass from $g(x')$ to $g(y')$ through the value of f at height h above each point, splitting the difference into a vertical defect and a horizontal difference that must be handled separately:

$$g(x') - g(y') = \big(g(x') - f(x', h)\big) + \big(f(x', h) - f(y', h)\big) - \big(g(y') - f(y', h)\big)$$

Estimate the two types of piece after raising to the p-th power and dividing by $|x' - y'|^{d+sp} = h^{d+p-1}$ (using $sp = p-1$). *Vertical defect.* Since $g(x') - f(x', h) = -\int_0^h \partial_n f(x', t)\, dt$, integrating y' over the sphere $|y' - x'| = h$ in polar coordinates $dy' = h^{d-1}\, dh\, d\omega$ cancels the h^{d-1} against $h^{-(d+p-1)}$ to leave the weight $h^{-p}\, dh$, so the vertical contribution at fixed x' is

$$\int \frac{|g(x') - f(x', h)|^p}{h^{d+p-1}}\, dy' \le C \int_0^\infty \frac{1}{h^p} \Big| \int_0^h \partial_n f(x', t)\, dt \Big|^p dh \le C \Big(\frac{p}{p-1}\Big)^p \int_0^\infty |\partial_n f(x', t)|^p\, dt$$

the last inequality being the one-dimensional Hardy inequality $\int_0^\infty h^{-p}\big(\int_0^h \psi\big)^p dh \le (p/(p-1))^p \int_0^\infty \psi^p$ applied to $\psi = |\partial_n f(x', \cdot)|$, whose constant is finite precisely because $p > 1$; this is where the logarithm that a direct column bound $\int_0^h |\partial_n f|^p\, dt$ against dh/h would produce is absorbed. Integrating in x' over the boundary piece bounds the vertical contribution by $C\int_{B_+} |\partial_n f|^p$. *Horizontal difference.* The middle term $f(x', h) - f(y', h) = \int_{[x',y']} \nabla' f(w', h) \cdot dw'$ runs along the length-h tangential segment at fixed height h, so Hölder gives $|f(x', h) - f(y', h)|^p \le h^{p-1} \int_{[x',y']} |\nabla' f(w', h)|^p\, dw'$. Dividing by h^{d+p-1} and integrating y' over $|y' - x'| = h$ leaves the weight $h^{-1}\, dh$ against the segment integral; because the integrand is evaluated at the same height $h = |x' - y'|$ that sets the scale, each height is visited once and the h-integral reconstructs the slab integral of $|\nabla' f|^p$ by Fubini, giving

$$\int\!\!\int \frac{|f(x', h) - f(y', h)|^p}{h^{d+p-1}}\, dx'\, dy' \le C \int_{B_+} |\nabla' f|^p$$

with no logarithm, since the horizontal integrand carries no whole-column tail. Combining the vertical and horizontal contributions bounds the flat-chart seminorm by $C\int_{B_+} |\nabla f|^p$. Summing the finitely many charts and using the comparability of dS with dx',

$$[Tf]^p_{W^{1-1/p,p}(\partial U)} \le C \int_U |\nabla f|^p\, dx \le C\|f\|^p_{W^{1,p}(U)}$$

Combined with the L^p bound of Step 1, $\|Tf\|_{W^{1-1/p,p}(\partial U)} \leq C\|f\|_{W^{1,p}(U)}$, so T maps $W^{1,p}(U)$ boundedly into $W^{1-1/p,p}(\partial U)$, completing the proof.

The trace theorem is what makes boundary value problems well posed for Sobolev functions: prescribing $Tf = g$ on ∂U is a closed linear constraint (the trace is bounded), so the Dirichlet condition "$f = g$ on ∂U" has meaning for $f \in W^{1,p}(U)$ even though f has no pointwise values on a measure-zero set. The two targets record two levels of precision: $L^p(\partial U)$ is enough to state boundary conditions and prove the kernel characterization below, while the sharp target $W^{1-1/p,p}(\partial U)$ is the natural home for the boundary data of the inverse problem of extending g inward. Theorem 10.3.7 proves only one half of what makes it sharp, that T maps $W^{1,p}(U)$ boundedly *into* $W^{1-1/p,p}(\partial U)$; the converse, that T is also onto this space, so that every fractional datum $g \in W^{1-1/p,p}(\partial U)$ is the trace of some $W^{1,p}(U)$ function, is a separate and deeper theorem, established by constructing a bounded right inverse (a trace-extension operator) and not proved here.[1] Granting it, $W^{1-1/p,p}(\partial U)$ is exactly the image of T. The loss of $1/p$ of a derivative is the price of the codimension-one restriction, and it is sharp.

Example 10.3.8: Trace of a Radial Bump on the Ball

Return to $U = B(0,1) \subseteq \mathbb{R}^n$ and $f(x) = 1 - |x|$ of example 10.3.4, which lies in $W^{1,p}(U)$ for every p and is continuous up to the boundary, so $f \in C(\overline{U}) \cap W^{1,p}(U)$. The trace is then the pointwise restriction: on $\partial U = \{|x| = 1\}$ every point has $|x| = 1$, so

$$Tf(x) = f|_{\partial U}(x) = 1 - |x| = 0 \qquad (|x| = 1)$$

Hence $Tf = 0$, and f lies in the kernel of the trace. This is consistent with the extension example 10.3.4 continued f by zero across ∂U without breaking the $W^{1,p}$ class: a boundary value of 0 is exactly what lets extension by zero preserve one derivative. By the characterization below, $f \in W_0^{1,p}(U)$, so f is a $W^{1,p}$-limit of functions compactly supported inside the open ball, even though f itself does not vanish in the interior. A function whose boundary trace were a nonzero constant c, by contrast, would have $\|Tf\|^p_{L^p(\partial U)} = |c|^p\,|\partial U| > 0$, placing it outside the kernel.

The kernel of the trace has a clean description: it is exactly the closure of $C_c^\infty(U)$, the space $W_0^{1,p}(U)$ of definition 10.2.6. This says the two competing notions of "vanishing on the boundary" coincide, the analytic one (zero trace) and the approximation-theoretic one (approximable by functions compactly supported inside U).

[1] The extension operator of theorem 10.3.3 does not supply this right inverse: it extends a function already given on U, whereas a right inverse must build a $W^{1,p}(U)$ function from boundary data alone. The standard construction lifts g off ∂U by a convolution extension calibrated to the fractional norm, the boundary trace theorem of Gagliardo; a full proof is beyond the scope of this section.

Theorem 10.3.9: Zero-Trace Characterization

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 < p < \infty$. Then

$$W_0^{1,p}(U) = \ker T = \{ f \in W^{1,p}(U) \mid Tf = 0 \}$$

where T is the trace of theorem 10.3.7 and $W_0^{1,p}(U)$ is the closure of $C_c^\infty(U)$ in $W^{1,p}(U)$ from definition 10.2.6.[a]

[a] The restriction to $p > 1$ enters at the boundary Hardy inequality of Step 3, whose constant $(p/(p-1))^p$ blows up as $p \to 1$; Hardy's inequality genuinely fails at $p = 1$, so the cutoff argument below does not cover that case. The characterization does still hold for $p = 1$, but by a different route (a truncation-and-mollification argument avoiding Hardy), which is not pursued here.

***Proof*:**

Step 1: $W_0^{1,p}(U) \subseteq \ker T$. For $\varphi \in C_c^\infty(U)$ the trace is the pointwise restriction (Step 2 of theorem 10.3.7), and $\varphi \equiv 0$ near ∂U, so $T\varphi = \varphi|_{\partial U} = 0$. Since T is bounded (theorem 10.3.7) and $C_c^\infty(U)$ is dense in $W_0^{1,p}(U)$ by definition (definition 10.2.6), continuity gives $Tf = 0$ for every $f \in W_0^{1,p}(U)$: if $\varphi_m \to f$ in $W^{1,p}(U)$ with $\varphi_m \in C_c^\infty(U)$, then $Tf = \lim_m T\varphi_m = 0$. Hence $W_0^{1,p}(U) \subseteq \ker T$.

Step 2: $\ker T \subseteq W_0^{1,p}(U)$*, reduction to compactly supported functions.* Let $f \in W^{1,p}(U)$ with $Tf = 0$; the goal is to approximate f in $W^{1,p}(U)$ by functions in $C_c^\infty(U)$. By theorem 10.2.5 choose $f_m \in C^\infty(\overline{U})$ with $f_m \to f$ in $W^{1,p}(U)$; then $Tf_m = f_m|_{\partial U} \to Tf = 0$ in $L^p(\partial U)$ by continuity of T. It suffices to produce, for each ε, a function $g \in W^{1,p}(U)$ with compact support in U and $\|f - g\|_{W^{1,p}(U)} < \varepsilon$, since mollifying g (theorem 10.2.1 applied on a neighborhood of supp$g$) then yields a $C_c^\infty(U)$ approximant, placing f in the closure $W_0^{1,p}(U)$.

Step 3: Cutting off near the boundary using the vanishing trace. Work in a boundary chart straightened by lemma 10.3.2, so near a boundary point the domain is the half-ball B_+, the boundary is $\{x_n = 0\}$, and $f \in W^{1,p}(B_+)$ has trace $g = Tf = 0$ on $\{x_n = 0\}$. The vanishing trace gives the boundary Hardy estimate: since the smooth approximants satisfy $f_m(x', 0) = Tf_m \to 0$, the fundamental theorem of calculus $f_m(x', t) = f_m(x', 0) + \int_0^t \partial_n f_m(x', \tau)\, d\tau$ passes to the limit to give, for f with $Tf = 0$,

$$\int_{B_+ \cap \{0 < x_n < h\}} \frac{|f(x)|^p}{x_n^p}\, dx \le C \int_{B_+ \cap \{0 < x_n < 2h\}} |\partial_n f|^p\, dx$$

the one-dimensional Hardy inequality applied in the x_n-variable (its constant $(p/(p-1))^p$ depends only on p and is finite precisely because $p > 1$), using $f(x', 0) = 0$. In particular $|f(x)|/x_n$ is p-integrable near the boundary. Now let $\eta_h(x_n)$ be a smooth cutoff, $\eta_h = 0$ for $x_n \le h$, $\eta_h = 1$ for $x_n \ge 2h$, with $|\eta_h'| \le C/h$, and set $g_h = \eta_h f$. Then g_h vanishes in the strip $0 < x_n < h$, so it has positive distance from the flattened boundary, and

$$\|f - g_h\|_{W^{1,p}(B_+)}^p \le \int_{\{x_n < 2h\}} |f|^p + \int_{\{x_n < 2h\}} |1 - \eta_h|^p |\nabla f|^p + \int_{\{h < x_n < 2h\}} |\eta_h'|^p |f|^p$$

The first two integrals tend to 0 as $h \to 0$ by dominated convergence (the domains shrink to a null set). The third is controlled by the Hardy estimate: $|\eta_h'|^p \le C/h^p \le C/x_n^p$ on $\{h < x_n < 2h\}$, so $\int_{\{h < x_n < 2h\}} |\eta_h'|^p |f|^p \le C \int_{\{h < x_n < 2h\}} |f|^p / x_n^p \to 0$ as $h \to 0$, again because the domain shrinks to a null set while the integrand is dominated by the integrable $|f|^p/x_n^p$. Hence $g_h \to f$ in $W^{1,p}(B_+)$, and each g_h vanishes near the flattened boundary.

Step 4: Globalization. Patch the chart cutoffs with the finite boundary partition of unity $\zeta_1, \ldots, \zeta_N$ of theorem 10.3.3 together with the interior cutoff $\zeta_0 \in C_c^\infty(U)$: transporting each g_h back through the charts and combining as $\sum_{j\geq 1} \zeta_j g_h^{(j)} + \zeta_0 f$, the resulting function g lies in $W^{1,p}(U)$, has support at positive distance from ∂U hence compact in U, and satisfies $\|f - g\|_{W^{1,p}(U)} < \varepsilon$ for h small, by Step 3 in each boundary chart and the boundedness of the transport maps. This furnishes the compactly supported approximant required in Step 2; mollifying it produces a $C_c^\infty(U)$ function within 2ε of f. Since ε was arbitrary, $f \in \overline{C_c^\infty(U)} = W_0^{1,p}(U)$, and $\ker T \subseteq W_0^{1,p}(U)$.

The two inclusions give $W_0^{1,p}(U) = \ker T$, completing the proof.

The characterization anchors $W_0^{1,p}(U)$ as the natural space for the Dirichlet problem: a solution with homogeneous boundary data $f|_{\partial U} = 0$ is sought exactly in $W_0^{1,p}(U)$, and the theorem certifies that this closure of $C_c^\infty(U)$ is the same as the analytically defined set of zero-trace functions. It also explains the caveat in definition 10.2.6 that $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$: on all of $\mathbb{R}^n$ there is no boundary, the trace is the zero map, and its kernel is everything, so every $W^{1,p}(\mathbb{R}^n)$ function is a limit of compactly supported smooth ones. On a bounded domain the boundary is genuine, the trace is nonzero, and $W_0^{1,p}(U)$ is a proper closed subspace on which the Poincaré inequality of section 10.5 will hold.

10.4 Sobolev Embeddings

The completion of section 10.1 records that a Sobolev function has L^p derivatives; it says nothing about the function itself beyond membership in L^p. This section extracts the extra regularity that possession of a weak gradient forces: control of ∇f in L^p gives either higher integrability of f or, past a threshold, outright continuity. Which of the two occurs is decided by comparing the order of differentiation against the dimension n, and the two regimes are governed by the Gagliardo-Nirenberg-Sobolev inequality when the derivative is too weak to reach continuity and by Morrey's inequality when it is strong enough.

The mechanism behind both is a single scaling count. If f is rescaled to $f_\lambda(x) = f(\lambda x)$, then $\|f_\lambda\|_{L^q(\mathbb{R}^n)} = \lambda^{-n/q}\|f\|_{L^q(\mathbb{R}^n)}$ while $\|\nabla f_\lambda\|_{L^p(\mathbb{R}^n)} = \lambda^{1-n/p}\|\nabla f\|_{L^p(\mathbb{R}^n)}$. An inequality of the form $\|f\|_{L^q} \leq C\|\nabla f\|_{L^p}$ can hold for all f and all $\lambda > 0$ only if the two powers of λ agree, forcing $-n/q = 1 - n/p$, that is $q = \frac{np}{n-p}$. This exponent is the only candidate; the content of the theorem is that the inequality holds for it.

Definition 10.4.1: Sobolev Conjugate

Let $1 \leq p < n$. The *Sobolev conjugate* of p is

$$p^* = \frac{np}{n-p}$$

It satisfies $p^* > p$ and $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$.

The relation $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$ is the form to remember: one weak derivative improves the integrability exponent by the fixed amount $\frac{1}{n}$, independent of p. As $p \to n^-$ the conjugate $p^* \to \infty$, the signal that at $p = n$ integrability alone breaks down and a different phenomenon (borderline embedding into every finite exponent, then continuity for $p > n$) takes over.

Example 10.4.2: The Sobolev Conjugate in Low Dimensions

In dimension $n = 3$ the Hilbert exponent $p = 2$ has $p^* = \frac{3 \cdot 2}{3-2} = 6$, so a function on $\mathbb{R}^3$ with gradient in L^2 automatically lies in L^6. In dimension $n = 2$ the same $p = 2$ gives $n - p = 0$, so p^* is undefined and $p = n$ is the borderline case excluded from definition 10.4.1. For $p = 1$ in dimension n one finds $p^* = \frac{n}{n-1}$, the exponent that will appear at the base of the induction below.

The strategy is to prove the gradient estimate first on smooth compactly supported functions, where the fundamental theorem of calculus applies directly, and then to remove the smoothness by the density theory of section 10.2. The smooth estimate rests on a one-dimensional bound integrated against itself in every coordinate direction, tied together by an iterated Hölder inequality of Loomis-Whitney type.

Theorem 10.4.3: Gagliardo-Nirenberg-Sobolev Inequality

Let $1 \le p < n$. There is a constant $C = C(n, p)$ such that

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \le C \, \|\nabla f\|_{L^p(\mathbb{R}^n)} \qquad \text{for every } f \in W^{1,p}(\mathbb{R}^n)$$

where $p^* = \frac{np}{n-p}$. In particular the inclusion $W^{1,p}(\mathbb{R}^n) \hookrightarrow L^{p^*}(\mathbb{R}^n)$ is a bounded linear map.

***Proof*:**

Step 1: The case $p = 1$ for $f \in C_c^\infty(\mathbb{R}^n)$. Here $p^* = \frac{n}{n-1}$. Fix $f \in C_c^\infty(\mathbb{R}^n)$ and $x = (x_1, \dots, x_n)$. For each index i, integrating the derivative along the i-th coordinate line from $-\infty$ and using compact support,

$$f(x) = \int_{-\infty}^{x_i} \partial_i f(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_n) \, dt$$

so, bounding the integrand by its absolute value and extending the range to all of $\mathbb{R}$,

$$|f(x)| \le \int_{-\infty}^{\infty} |\partial_i f(x_1, \dots, t, \dots, x_n)| \, dt =: g_i(\hat{x}_i)$$

where g_i depends on all coordinates except x_i (denoted $\hat{x}_i$). Taking the product over the n directions and raising to the power $\frac{1}{n-1}$,

$$|f(x)|^{\frac{n}{n-1}} \le \prod_{i=1}^{n} g_i(\hat{x}_i)^{\frac{1}{n-1}}$$

The task is to integrate this inequality over $\mathbb{R}^n$. Notice that each factor g_i is independent of the variable x_i, so integrating in x_1 first leaves the factor $g_1^{1/(n-1)}$ constant and applies the generalized Hölder inequality (with $n - 1$ exponents all equal to $n - 1$) to the remaining $n - 1$ factors, each of which depends on x_1:

$$\int_{\mathbb{R}} |f|^{\frac{n}{n-1}} \, dx_1 \le g_1(\hat{x}_1)^{\frac{1}{n-1}} \prod_{i=2}^{n} \Big(\int_{\mathbb{R}} g_i \, dx_1 \Big)^{\frac{1}{n-1}}$$

Integrating successively in $x_2, \dots, x_n$ and applying the same generalized Hölder inequality at each stage, every factor is eventually integrated in the one variable on which it does not already depend, and one arrives at

$$\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}}\, dx \le \prod_{i=1}^{n} \Big(\int_{\mathbb{R}^n} |\partial_i f|\, dx \Big)^{\frac{1}{n-1}}$$

This is the Loomis-Whitney arrangement: an $L^{n/(n-1)}$ bound on f by the geometric mean of the n one-directional gradient integrals. Bounding each $\int_{\mathbb{R}^n} |\partial_i f| \le \int_{\mathbb{R}^n} |\nabla f|$ and taking the resulting product of n equal factors,

$$\Big(\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}}\, dx \Big)^{\frac{n-1}{n}} \le \int_{\mathbb{R}^n} |\nabla f|\, dx$$

that is $\|f\|_{L^{1^*}(\mathbb{R}^n)} \le \|\nabla f\|_{L^1(\mathbb{R}^n)}$, with constant $C(n,1) = 1$.

Step 2: General $1 < p < n$ *for* $f \in C_c^\infty(\mathbb{R}^n)$ *by applying Step 1 to a power of* $|f|$. Fix $t > 1$ to be chosen and apply the $p = 1$ estimate to the function $|f|^t$, which is C^1 with compact support and gradient $\nabla(|f|^t) = t\,|f|^{t-1}\operatorname{sgn}(f)\,\nabla f$. Step 1 gives

$$\Big(\int_{\mathbb{R}^n} |f|^{\frac{tn}{n-1}}\, dx \Big)^{\frac{n-1}{n}} \le t \int_{\mathbb{R}^n} |f|^{t-1}\, |\nabla f|\, dx$$

Apply Hölder's inequality (theorem 8.1.5) to the right side with exponents p and $p' = \frac{p}{p-1}$:

$$\int_{\mathbb{R}^n} |f|^{t-1} |\nabla f|\, dx \le \Big(\int_{\mathbb{R}^n} |f|^{(t-1)p'}\, dx \Big)^{1/p'} \Big(\int_{\mathbb{R}^n} |\nabla f|^p\, dx \Big)^{1/p}$$

Choose t so that the two powers of $|f|$ match, namely $\frac{tn}{n-1} = (t-1)p'$. Solving gives $t = \frac{p(n-1)}{n-p} > 1$, and with this choice

$$\frac{tn}{n-1} = (t-1)p' = \frac{np}{n-p} = p^*$$

Writing $I = \big(\int_{\mathbb{R}^n} |f|^{p^*} \big)^{1/p^*} = \|f\|_{L^{p^*}}$, the left side is $I^{p^* \cdot \frac{n-1}{n}}$ and the first factor on the right is $I^{p^*/p'}$, so the inequality reads

$$I^{\,p^* \frac{n-1}{n}} \le t\, I^{\,p^*/p'}\, \|\nabla f\|_{L^p}$$

A direct computation of exponents gives $p^* \frac{n-1}{n} - \frac{p^*}{p'} = 1$, so dividing by $I^{p^*/p'}$ (which is finite, since $f \in C_c^\infty$ makes I finite) leaves

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \le t\, \|\nabla f\|_{L^p(\mathbb{R}^n)} = \frac{p(n-1)}{n-p}\, \|\nabla f\|_{L^p(\mathbb{R}^n)}$$

giving the inequality on $C_c^\infty(\mathbb{R}^n)$ with constant $C(n,p) = \frac{p(n-1)}{n-p}$.

Step 3: Extension to $W^{1,p}(\mathbb{R}^n)$ *by density.* Let $f \in W^{1,p}(\mathbb{R}^n)$. Since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 10.2.6), there are $f_m \in C_c^\infty(\mathbb{R}^n)$ with $f_m \to f$ in $W^{1,p}(\mathbb{R}^n)$; concretely one mollifies and truncates via theorem 10.2.1. Applying Step 2 to the difference $f_m - f_l$,

$$\|f_m - f_l\|_{L^{p^*}(\mathbb{R}^n)} \le C(n,p)\, \|\nabla f_m - \nabla f_l\|_{L^p(\mathbb{R}^n)} \le C(n,p)\, \|f_m - f_l\|_{W^{1,p}(\mathbb{R}^n)}$$

so (f_m) is Cauchy in $L^{p^*}(\mathbb{R}^n)$ and converges there to some h; passing to a subsequence, $f_m \to h$ almost everywhere, while $f_m \to f$ in L^p forces (along a further subsequence) $f_m \to f$ almost everywhere, so $h = f$ almost everywhere and $f \in L^{p^*}(\mathbb{R}^n)$. Applying Step 2 to f_m and letting $m \to \infty$, with $\|f_m\|_{L^{p^*}} \to \|f\|_{L^{p^*}}$ and $\|\nabla f_m\|_{L^p} \to \|\nabla f\|_{L^p}$,

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq C(n,p)\,\|\nabla f\|_{L^p(\mathbb{R}^n)}$$

which is the claimed inequality on all of $W^{1,p}(\mathbb{R}^n)$. Boundedness of the inclusion is then $\|f\|_{L^{p^*}} \leq C\|\nabla f\|_{L^p} \leq C\|f\|_{W^{1,p}}$, completing the proof.

The inequality controls the full L^{p^*} norm of f by the L^p norm of its gradient alone, with no term for f itself on the right. This is possible only on $\mathbb{R}^n$, where there is no room for constants: a nonzero constant has vanishing gradient but infinite L^{p^*} norm, so the estimate would fail were $\mathbb{R}^n$ replaced by a set of finite measure without also restricting to functions that decay or vanish at the boundary. On a bounded domain the corresponding statement carries a lower-order term, or restricts to $W_0^{1,p}$; the Poincaré inequality of section 10.5 supplies exactly the missing control. The gain in integrability, from p up to p^*, is the quantitative content: one weak derivative is worth a definite jump in the integrability exponent, and iterating this jump is what produces the full hierarchy at the end of the section.

Example 10.4.4: Sharpness of the Exponent

No exponent larger than p^* can appear in theorem 10.4.3. The scaling argument of the section opening shows why: if $\|f\|_{L^q(\mathbb{R}^n)} \leq C\|\nabla f\|_{L^p(\mathbb{R}^n)}$ held for a fixed C and some $q \neq p^*$, testing on $f_\lambda(x) = f(\lambda x)$ gives $\lambda^{-n/q}\|f\|_{L^q} \leq C\lambda^{1-n/p}\|\nabla f\|_{L^p}$, that is

$$\|f\|_{L^q} \leq C\lambda^{n/q+1-n/p}\|\nabla f\|_{L^p}$$

If $q > p^*$ then $n/q + 1 - n/p < 0$ and letting $\lambda \to \infty$ forces $\|f\|_{L^q} = 0$; if $q < p^*$ then the exponent is positive and $\lambda \to 0$ does the same. Only $q = p^*$, where the exponent of λ vanishes, survives.

At $p = n$ the conjugate p^* diverges and the argument above degenerates. Past this threshold, when $p > n$, a weak gradient in L^p is strong enough to force not just integrability but pointwise continuity, indeed Hölder continuity with a definite exponent. The precise target is a Hölder space, which the following definition records.

Definition 10.4.5: Holder Space $C^{0,\gamma}$

Let $0 < \gamma \leq 1$. A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is *Hölder continuous* with exponent γ if the seminorm

$$[f]_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x-y|^\gamma}$$

is finite. The *Hölder space* $C^{0,\gamma}(\mathbb{R}^n)$ is the space of bounded continuous f with $[f]_{C^{0,\gamma}(\mathbb{R}^n)} < \infty$, normed by $\|f\|_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_x |f(x)| + [f]_{C^{0,\gamma}(\mathbb{R}^n)}$. More generally, for a nonnegative integer m, $C^{m,\gamma}(\mathbb{R}^n)$ consists of the $f \in C^m$ whose derivatives up to order m are bounded with $\partial^\alpha f \in C^{0,\gamma}$ for every $|\alpha| = m$, normed by $\|f\|_{C^{m,\gamma}} = \sum_{|\alpha|\leq m} \sup_x |\partial^\alpha f| + \sum_{|\alpha|=m} [\partial^\alpha f]_{C^{0,\gamma}}$.

The exponent γ measures how rapidly f may oscillate: $\gamma = 1$ is Lipschitz continuity, and smaller γ permits sharper local variation. A member of $W^{1,p}(\mathbb{R}^n)$ is only an equivalence class of functions agreeing almost everywhere, so the assertion that it embeds into $C^{0,\gamma}$ means each class has a (necessarily unique) continuous representative lying in that space, and it is this representative the oscillation estimate controls.

Example 10.4.6: The Model Holder Function

On $\mathbb{R}$ the function $f(x) = |x|^\gamma$ (say cut off smoothly away from the origin) has $[f]_{C^{0,\gamma}} < \infty$ but is not Lipschitz for $\gamma < 1$: near 0 the difference quotient $\frac{||x|^\gamma - 0|}{|x|^1} = |x|^{\gamma-1}$ blows up, whereas $\frac{||x|^\gamma - 0|}{|x|^\gamma} = 1$ stays bounded. This is the extremal behavior Morrey's inequality permits, and the exponent $\gamma = 1 - \frac{n}{p}$ produced below is exactly the regularity of such a cusp at the borderline.

The route to continuity runs through a comparison between the value of f at a point and its average over a small ball. Write

$$f_{B(x,r)} = \frac{1}{|B(x,r)|} \int_{B(x,r)} f \, dy$$

for the average of f over the ball $B(x,r)$ of radius r centered at x, where $|B(x,r)|$ is its Lebesgue measure. The key estimate bounds the deviation of f from this average by an integral of the gradient, a Sobolev analogue of the mean value theorem.

Lemma 10.4.7: Averaged Gradient Estimate

Let $p > n$ and $f \in C^1(\mathbb{R}^n)$. For every $x \in \mathbb{R}^n$ and $r > 0$,

$$\frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy \le C(n) \int_{B(x,r)} \frac{|\nabla f(y)|}{|x - y|^{n-1}} \, dy$$

***Proof*:**
Fix x and a point $y = x + s\omega$ in $B(x,r)$, where $|\omega| = 1$ and $0 < s \le r$. The fundamental theorem of calculus along the segment from x to y gives

$$f(x + s\omega) - f(x) = \int_0^s \frac{d}{d\tau} f(x + \tau\omega) \, d\tau = \int_0^s \nabla f(x + \tau\omega) \cdot \omega \, d\tau$$

so $|f(x+s\omega) - f(x)| \le \int_0^s |\nabla f(x + \tau\omega)| \, d\tau$. Integrate this over the unit sphere $\omega \in S^{n-1}$, whose surface measure is written $d\omega$:

$$\int_{S^{n-1}} |f(x + s\omega) - f(x)| \, d\omega \le \int_{S^{n-1}} \int_0^s |\nabla f(x + \tau\omega)| \, d\tau \, d\omega \le \int_{S^{n-1}} \int_0^r |\nabla f(x + \tau\omega)| \, d\tau \, d\omega$$

Insert the factor $\tau^{n-1}/\tau^{n-1} = 1$ into the last integrand and pass to the polar-coordinate variable $z = x + \tau\omega$, for which $dz = \tau^{n-1} \, d\tau \, d\omega$ and $|z - x| = \tau$:

$$\int_{S^{n-1}} \int_0^r |\nabla f(x + \tau\omega)| \, d\tau \, d\omega = \int_{S^{n-1}} \int_0^r \frac{|\nabla f(x + \tau\omega)|}{\tau^{n-1}} \tau^{n-1} \, d\tau \, d\omega = \int_{B(x,r)} \frac{|\nabla f(z)|}{|z - x|^{n-1}} \, dz$$

Now multiply the resulting inequality

$$\int_{S^{n-1}} |f(x + s\omega) - f(x)| \, d\omega \le \int_{B(x,r)} \frac{|\nabla f(z)|}{|z - x|^{n-1}} \, dz$$

by s^{n-1} and integrate in s from 0 to r. The right side is independent of s, so it picks up the factor $\int_0^r s^{n-1}\,ds = r^n/n$, while the left side becomes $\int_{B(x,r)} |f(y)-f(x)|\,dy$ in polar coordinates. Thus

$$\int_{B(x,r)} |f(y)-f(x)|\,dy \le \frac{r^n}{n}\int_{B(x,r)} \frac{|\nabla f(z)|}{|z-x|^{n-1}}\,dz$$

Dividing by $|B(x,r)| = \omega_n r^n$, where ω_n is the volume of the unit ball, gives the claim with $C(n) = \frac{1}{n\,\omega_n}$, as we sought to show.

With the deviation from an average controlled, the oscillation of f between two points follows by placing both points inside a common ball and comparing each to the average over it.

Theorem 10.4.8: Morrey's Inequality

Let $n < p < \infty$ and set $\gamma = 1 - \frac{n}{p} \in (0,1)$. There is a constant $C = C(n,p)$ such that every $f \in W^{1,p}(\mathbb{R}^n)$ has a representative in $C^{0,\gamma}(\mathbb{R}^n)$, and this representative satisfies

$$|f(x)-f(y)| \le C\,|x-y|^\gamma\,\|\nabla f\|_{L^p(\mathbb{R}^n)} \qquad \text{for all } x,y \in \mathbb{R}^n$$

together with $\sup_x |f(x)| \le C\|f\|_{W^{1,p}(\mathbb{R}^n)}$. Consequently the inclusion $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n)$ is a bounded linear map.

***Proof*:**

Step 1: The oscillation estimate for $f \in C^1(\mathbb{R}^n)$. Fix $x, y \in \mathbb{R}^n$, set $r = |x-y|$, and let $W = B(x,r) \cap B(y,r)$, a set of measure $|W| \ge c_n r^n$ for a dimensional constant $c_n > 0$ (both balls have radius r and centers at distance r, so their intersection contains a fixed fraction of a ball). The triangle inequality through the average $f_W = \frac{1}{|W|}\int_W f$ gives

$$|f(x)-f(y)| \le |f(x)-f_W| + |f_W - f(y)| = \frac{1}{|W|}\left|\int_W (f(x)-f(z))\,dz\right| + \frac{1}{|W|}\left|\int_W (f(y)-f(z))\,dz\right|$$

Estimate the first term. Since $W \subseteq B(x,r)$ and $|W| \ge c_n r^n \ge \frac{c_n}{\omega_n}|B(x,r)|$,

$$\frac{1}{|W|}\int_W |f(x)-f(z)|\,dz \le \frac{\omega_n}{c_n}\cdot\frac{1}{|B(x,r)|}\int_{B(x,r)} |f(x)-f(z)|\,dz \le C(n)\int_{B(x,r)} \frac{|\nabla f(z)|}{|x-z|^{n-1}}\,dz$$

the last step by lemma 10.4.7. Apply Hölder's inequality (theorem 8.1.5) with exponents p and p' to this integral over $B(x,r)$:

$$\int_{B(x,r)} \frac{|\nabla f(z)|}{|x-z|^{n-1}}\,dz \le \|\nabla f\|_{L^p(\mathbb{R}^n)}\Big(\int_{B(x,r)} |x-z|^{-(n-1)p'}\,dz\Big)^{1/p'}$$

The remaining integral is finite precisely because $p > n$: in polar coordinates centered at x,

$$\int_{B(x,r)} |x-z|^{-(n-1)p'}\,dz = \left|S^{n-1}\right|\int_0^r \tau^{-(n-1)p'}\tau^{n-1}\,d\tau = \left|S^{n-1}\right|\int_0^r \tau^{(n-1)(1-p')}\,d\tau$$

and the exponent satisfies $(n-1)(1-p') > -1$ exactly when $p > n$, so the integral converges to a constant times $r^{(n-1)(1-p')+1}$. Raising to the power $1/p'$ and simplifying the exponent,

$\frac{(n-1)(1-p')+1}{p'} = 1 - \frac{n}{p} = \gamma$, so

$$\Big(\int_{B(x,r)} |x-z|^{-(n-1)p'}\,dz\Big)^{1/p'} = C(n,p)\,r^{\gamma}$$

Combining, the first term is bounded by $C(n,p)\,r^{\gamma}\|\nabla f\|_{L^p}$. The second term is identical with y in place of x and $B(y,r)$ in place of $B(x,r)$, giving the same bound. Adding the two and recalling $r = |x-y|$,

$$|f(x) - f(y)| \le C(n,p)\,|x-y|^{\gamma}\,\|\nabla f\|_{L^p(\mathbb{R}^n)}$$

Step 2: The sup bound for $f \in C^1(\mathbb{R}^n) \cap W^{1,p}$. Fix x and take $r = 1$. Then

$$|f(x)| \le |f(x) - f_{B(x,1)}| + |f_{B(x,1)}|$$

The first term is bounded via lemma 10.4.7 and the Hölder step of Step 1 (with $r = 1$) by $C(n,p)\|\nabla f\|_{L^p}$; the second is $\big|\frac{1}{|B(x,1)|}\int_{B(x,1)} f\big| \le \frac{1}{|B(x,1)|}\|f\|_{L^1(B(x,1))} \le C(n)\|f\|_{L^p(\mathbb{R}^n)}$ by Hölder on the unit ball. Taking the supremum over x,

$$\sup_x |f(x)| \le C(n,p)\big(\|\nabla f\|_{L^p} + \|f\|_{L^p}\big) \le C(n,p)\|f\|_{W^{1,p}(\mathbb{R}^n)}$$

Step 3: Passage to $W^{1,p}(\mathbb{R}^n)$ by density. Let $f \in W^{1,p}(\mathbb{R}^n)$ and choose $f_m \in C_c^{\infty}(\mathbb{R}^n)$ with $f_m \to f$ in $W^{1,p}(\mathbb{R}^n)$, which is possible since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 10.2.6) and by theorem 10.2.1. Steps 1 and 2 apply to each f_m; applied to $f_m - f_l$ they give

$$\sup_x |f_m(x) - f_l(x)| \le C\|f_m - f_l\|_{W^{1,p}}, \qquad [f_m - f_l]_{C^{0,\gamma}} \le C\|\nabla f_m - \nabla f_l\|_{L^p}$$

so (f_m) is Cauchy in the norm of $C^{0,\gamma}(\mathbb{R}^n)$. This space is complete, so $f_m \to \tilde{f}$ in $C^{0,\gamma}(\mathbb{R}^n)$, hence uniformly, and $\tilde{f}$ is continuous with $[\tilde{f}]_{C^{0,\gamma}} < \infty$. Uniform convergence gives $f_m \to \tilde{f}$ pointwise, while $f_m \to f$ in L^p gives $f_m \to f$ almost everywhere along a subsequence; thus $\tilde{f} = f$ almost everywhere and $\tilde{f}$ is the sought continuous representative. Letting $m \to \infty$ in the estimates of Steps 1 and 2, now for $\tilde{f}$, yields the oscillation bound and the sup bound, and boundedness of the inclusion follows, completing the proof.

Morrey's inequality is the mechanism by which analysis recovers classical smoothness from weak hypotheses: a function known only to have a p-th power integrable gradient, $p > n$, is after modification on a null set genuinely continuous, with a quantitative modulus of continuity. The exponent $\gamma = 1 - \frac{n}{p}$ is sharp and degenerates to 0 as $p \to n^+$, matching the divergence of p^* from the other side. The two theorems together partition the range of p at the dimension: below n one gains integrability, above n one gains continuity, and the borderline $p = n$ sits between, embedding into every finite exponent but into no Hölder space (the model cusp $|x|^{\gamma}$ with $\gamma \to 0$ shows why continuity fails there, cf. example 10.4.6).

The two first-order theorems combine to describe the entire scale $W^{k,p}$. The organizing quantity is kp, the total order of differentiation weighted against the dimension, and the outcome is decided by comparing kp to n. When $kp < n$ each derivative can be traded, via theorem 10.4.3, for a fixed gain in the integrability exponent, and iterating k times lands in an L^q space with a computable q; when $kp > n$ enough derivatives can be spent to cross the Morrey threshold and land in a classical Hölder space; and when $kp = n$ the borderline holds, with membership in every finite L^q but not, in general, continuity.

Theorem 10.4.9: General Sobolev Embedding

Let $k \geq 1$ be an integer and $1 \leq p < \infty$, and let $f \in W^{k,p}(\mathbb{R}^n)$.

1. If $kp < n$, then $f \in L^q(\mathbb{R}^n)$ for $q = \frac{np}{n-kp}$, and there is a constant $C = C(n,p,k)$ with $\|f\|_{L^q(\mathbb{R}^n)} \leq C\|f\|_{W^{k,p}(\mathbb{R}^n)}$. Consequently $W^{k,p}(\mathbb{R}^n) \hookrightarrow L^q(\mathbb{R}^n)$ for every $p \leq q \leq \frac{np}{n-kp}$.
2. If $kp = n$, then $f \in L^q(\mathbb{R}^n)$ for every $p \leq q < \infty$, with $\|f\|_{L^q(\mathbb{R}^n)} \leq C(n,p,k,q)\|f\|_{W^{k,p}(\mathbb{R}^n)}$.
3. If $kp > n$, write $m = k - \lfloor n/p \rfloor - 1$ and let

$$\gamma = \begin{cases} \lfloor n/p \rfloor + 1 - \dfrac{n}{p} & \text{if } n/p \notin \mathbb{Z} \\ \text{any } \gamma \in (0,1) & \text{if } n/p \in \mathbb{Z} \end{cases}$$

   Then f has a representative in $C^{m,\gamma}(\mathbb{R}^n)$, with $\|f\|_{C^{m,\gamma}(\mathbb{R}^n)} \leq C(n,p,k,\gamma)\|f\|_{W^{k,p}(\mathbb{R}^n)}$. Hence $W^{k,p}(\mathbb{R}^n) \hookrightarrow C^{m,\gamma}(\mathbb{R}^n)$.

***Proof*:**
The argument iterates the first-order theorems, tracking the exponent one derivative at a time. Throughout, if $f \in W^{j,p}$ then each $\partial^\alpha f$ with $|\alpha| \leq j-1$ lies in $W^{1,p}$, so the first-order estimates apply to f and to all its lower-order derivatives at once.

Step 1: The subcritical case $kp < n$. Apply theorem 10.4.3 to f and to each first derivative. Since $f \in W^{k,p}$ has $f, \partial^\alpha f \in W^{1,p}$ for $|\alpha| \leq k-1$, the theorem gives $f \in W^{k-1,p_1}$ with $\frac{1}{p_1} = \frac{1}{p} - \frac{1}{n}$: each derivative of order at most $k-1$ gains one unit of integrability, so $\|f\|_{W^{k-1,p_1}} \leq C\|f\|_{W^{k,p}}$. Repeating, after j steps $f \in W^{k-j,p_j}$ with

$$\frac{1}{p_j} = \frac{1}{p} - \frac{j}{n}$$

valid as long as $p_j < n$ at each stage, that is $\frac{1}{p} - \frac{j}{n} > \frac{1}{n}$, which holds for $j \leq k-1$ because $kp < n$ means $\frac{1}{p} - \frac{k-1}{n} > \frac{1}{n}$. After k steps $f \in W^{0,p_k} = L^{p_k}$ with $\frac{1}{p_k} = \frac{1}{p} - \frac{k}{n} = \frac{n-kp}{np}$, so $p_k = \frac{np}{n-kp} = q$ and $\|f\|_{L^q} \leq C\|f\|_{W^{k,p}}$. The embedding into intermediate L^q for $p \leq q \leq \frac{np}{n-kp}$ follows by interpolation: $f \in L^p \cap L^q$ gives $f \in L^{q'}$ for every q' between with $\|f\|_{L^{q'}} \leq \|f\|_{L^p}^{1-\theta}\|f\|_{L^q}^{\theta} \leq C\|f\|_{W^{k,p}}$.

Step 2: The critical case $kp = n$. Iterate as in Step 1 while $p_j < n$. The obstruction is the final step: after $k-1$ iterations $f \in W^{1,p_{k-1}}$ with $\frac{1}{p_{k-1}} = \frac{1}{p} - \frac{k-1}{n} = \frac{1}{n}$, so $p_{k-1} = n$ and theorem 10.4.3 no longer applies with a finite conjugate. The following borderline fact finishes the argument: if $g \in W^{1,n}(\mathbb{R}^n)$ then $g \in L^q(\mathbb{R}^n)$ for every $n \leq q < \infty$, with $\|g\|_{L^q} \leq C(n,q)\|g\|_{W^{1,n}}$. To prove it, run the power substitution behind theorem 10.4.3 at $p = n$. For $g \in C_c^\infty(\mathbb{R}^n)$ and $t > 1$, the $p = 1$ Gagliardo estimate applied to $|g|^t$ gives

$$\left(\int_{\mathbb{R}^n} |g|^{\frac{tn}{n-1}}\right)^{\frac{n-1}{n}} \leq t\int_{\mathbb{R}^n} |g|^{t-1}|\nabla g| \leq t\left(\int_{\mathbb{R}^n} |g|^{(t-1)n'}\right)^{1/n'}\|\nabla g\|_{L^n}$$

by Hölder (theorem 8.1.5) with exponents n and $n' = \frac{n}{n-1}$. Since $(t-1)n' = (t-1)\frac{n}{n-1}$ and $\frac{tn}{n-1}$ differ, this bounds $\|g\|_{L^{tn/(n-1)}}$ by a power of $\|g\|_{L^{(t-1)n/(n-1)}}$ times $\|\nabla g\|_{L^n}$. Starting from $t = n$, where the input exponent $(t-1)n/(n-1) = n$ is controlled by $\|g\|_{W^{1,n}}$, the output exponent is $\frac{n^2}{n-1} > n$; feeding the output exponent back in as the new input and raising t at each stage, the output exponent $\frac{tn}{n-1}$ increases without bound, so after finitely many steps it exceeds any prescribed $q < \infty$, and interpolation between that exponent and n (as in Step 1) yields $\|g\|_{L^q} \le C(n,q)\|g\|_{W^{1,n}}$. Density of C_c^∞ in $W^{1,n}$ (definition 10.2.6, theorem 10.2.1) extends the bound to all $g \in W^{1,n}(\mathbb{R}^n)$, exactly as in Step 3 of theorem 10.4.3. Applying this to $g = \partial^\alpha f$ for $|\alpha| = k-1$, which lie in $W^{1,n}$, gives $\partial^\alpha f \in L^q$ for every finite q and every $|\alpha| \le k-1$; combined with $f \in L^p$ this gives $f \in L^q$ for every finite q, with the stated bound.

Step 3: The supercritical case $kp > n$, with $n/p \notin \mathbb{Z}$. Let $\ell = \lfloor n/p \rfloor$, so $\ell < n/p < \ell + 1$ and $\ell < k$ (because $kp > n$ gives $k > n/p > \ell$). Iterate theorem 10.4.3 exactly ℓ times: at each stage the current exponent r_j satisfies $\frac{1}{r_j} = \frac{1}{p} - \frac{j}{n}$, which stays positive and below $\frac{1}{n}$ only while $j < n/p$, so all ℓ steps are legitimate and land $f \in W^{k-\ell,r}$ with $\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n}$. Here $r > n$ precisely because $\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n} < \frac{1}{n}$, using $\ell < n/p$. Now $k - \ell = k - \lfloor n/p \rfloor = m+1$, so $f \in W^{m+1,r}$ with $r > n$. Each $\partial^\alpha f$ with $|\alpha| \le m$ then lies in $W^{1,r}$, $r > n$, so theorem 10.4.8 gives $\partial^\alpha f \in C^{0,\gamma}(\mathbb{R}^n)$ with $\gamma = 1 - \frac{n}{r} = \ell + 1 - \frac{n}{p}$ and the estimate $[\partial^\alpha f]_{C^{0,\gamma}} \le C\|\nabla \partial^\alpha f\|_{L^r} \le C\|f\|_{W^{k,p}}$, together with $\sup|\partial^\alpha f| \le C\|f\|_{W^{k,p}}$. Assembling over all $|\alpha| \le m$ gives $f \in C^{m,\gamma}(\mathbb{R}^n)$ with $\|f\|_{C^{m,\gamma}} \le C\|f\|_{W^{k,p}}$.

Step 4: The supercritical case $kp > n$, with $n/p \in \mathbb{Z}$. Now $\lfloor n/p \rfloor = n/p$ and $m = k - \frac{n}{p} - 1$. Iterate theorem 10.4.3 exactly $\frac{n}{p} - 1$ times, which is legitimate since each intermediate exponent stays below n; this lands $f \in W^{k-(n/p-1),n}$, that is, $f \in W^{m+2,n}$. Each $\partial^\alpha f$ with $|\alpha| \le m+1$ then lies in $W^{1,n}$, so by the borderline fact of Step 2 it lies in $L^s(\mathbb{R}^n)$ for every finite s, with $\|\partial^\alpha f\|_{L^s} \le C(n,s)\|f\|_{W^{k,p}}$. Fix any $s > n$; then each $\partial^\beta f$ with $|\beta| \le m$ lies in $W^{1,s}$ (its derivative $\partial^\alpha f$, $|\alpha| = |\beta| + 1 \le m+1$, is in L^s, and $\partial^\beta f$ itself is in L^s), so theorem 10.4.8 gives $\partial^\beta f \in C^{0,\gamma}(\mathbb{R}^n)$ with $\gamma = 1 - \frac{n}{s}$, any value in $(0,1)$ being attainable by the choice of $s > n$, and $\|\partial^\beta f\|_{C^{0,\gamma}} \le C\|\partial^\beta f\|_{W^{1,s}} \le C\|f\|_{W^{k,p}}$. Assembling over $|\beta| \le m$ gives $f \in C^{m,\gamma}(\mathbb{R}^n)$ for the chosen $\gamma \in (0,1)$, with the stated estimate, completing the proof.

The theorem converts the abstract membership $f \in W^{k,p}$ into concrete information a differential equation can use, either an integrability exponent or a number of classical derivatives with a Hölder modulus. The count kp against n is the single dial. In the elliptic regularity theory of the partial-differential-equations volumes, a solution is first shown to lie in some $W^{k,p}$ by energy estimates; the embedding then reads off how many classical derivatives it actually has, and the passage from $kp < n$ to $kp > n$ as k increases is precisely the bootstrap that turns a weak solution into a smooth one. The Rellich-Kondrachov theorem of section 10.5 sharpens the subcritical case further, upgrading the bounded inclusion into L^q to a compact one for $q < p^*$, which is what existence proofs by compactness require.

Example 10.4.10: Reading Off Regularity in Dimension Three

Take $n = 3$ and $p = 2$, the setting of most of mathematical physics. For $k = 1$ one has $kp = 2 < 3$, so theorem 10.4.9(1) gives $H^1(\mathbb{R}^3) = W^{1,2} \hookrightarrow L^q$ for $2 \le q \le \frac{3\cdot 2}{3-2} = 6$, recovering the L^6 bound of example 10.4.2. For $k = 2$ one has $kp = 4 > 3$, so case (3) applies with $\lfloor n/p \rfloor = \lfloor 3/2 \rfloor = 1$, hence $m = 2 - 1 - 1 = 0$ and $\gamma = 1 + 1 - \frac{3}{2} = \frac{1}{2}$; thus $H^2(\mathbb{R}^3) \hookrightarrow C^{0,1/2}(\mathbb{R}^3)$, and a function with two weak derivatives in L^2 is Hölder-$\frac{1}{2}$ continuous. This is why H^2 is the natural space in which to seek continuous solutions of second-order equations in three dimensions.

10.5 Compactness and the Poincaré Inequality

The embeddings of section 10.4 are bounded but not compact on $\mathbb{R}^n$: a fixed bump translated to infinity keeps its Sobolev norm while wandering off, so no subsequence converges. On a bounded domain there is no room to escape, and the same one-derivative gain that produced the embedding now produces *compactness*. This section proves that on a bounded domain the embedding of $W^{1,p}$ into L^q is compact below the critical exponent, and extracts the two inequalities that turn this compactness into existence theorems: the Poincaré inequality, which controls a function by its gradient once boundary values are killed, and its mean-zero variant, the Poincaré-Wirtinger inequality.

The compactness statement underlies existence proofs in the calculus of variations and the elliptic theory: a bounded sequence of approximate solutions, bounded in $W^{1,p}$ by an energy estimate, has a subsequence converging *strongly* in L^q, and strong convergence is what lets one pass to the limit in a nonlinear term. What makes it work is that a $W^{1,p}$ bound controls not just the size of a function but the size of its translates: a gradient bound forces $\|\tau_h f - f\|_{L^p}$ to be small uniformly, and uniform smallness of translates is exactly the hypothesis of the Fréchet-Kolmogorov compactness criterion in L^p.

Theorem 10.5.1: Rellich-Kondrachov Compactness

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary, and let $1 \leq p < n$. Then for every q with $1 \leq q < p^*$ the embedding

$$W^{1,p}(U) \hookrightarrow L^q(U)$$

is compact: every bounded sequence in $W^{1,p}(U)$ has a subsequence converging in $L^q(U)$.

***Proof*:**
Let (f_m) be a sequence with $\|f_m\|_{W^{1,p}(U)} \leq M$ for all m; the goal is a subsequence Cauchy in $L^q(U)$. Throughout, C denotes a constant that may change from line to line but never depends on m.

Step 1: Extend to compactly supported functions on $\mathbb{R}^n$. By the extension theorem (theorem 10.3.3) there is a bounded linear $E\colon W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$ with $Ef = f$ almost everywhere on U. A cutoff makes E land in functions supported in a fixed bounded open set V with $\overline{U} \subseteq V$: multiply by a fixed cutoff $\zeta \in C_c^\infty(V)$ equal to 1 on a neighborhood of $\overline{U}$, which changes E only by a bounded operator and does not disturb $Ef = f$ on U. Write $g_m := Ef_m$. Then each g_m is supported in V and

$$\|g_m\|_{W^{1,p}(\mathbb{R}^n)} \leq \|E\| \, \|f_m\|_{W^{1,p}(U)} \leq CM$$

so (g_m) is bounded in $W^{1,p}(\mathbb{R}^n)$, with all g_m supported in the one bounded set V. A subsequence of (f_m) converges in $L^q(U)$ as soon as the corresponding subsequence of (g_m) converges in $L^q(V)$, since $f_m = g_m$ on U; it therefore suffices to extract a convergent subsequence of (g_m) in $L^q(V)$.

Step 2: A uniform bound at the critical exponent. Each $g_m \in W^{1,p}(\mathbb{R}^n)$, so by the Gagliardo-Nirenberg-Sobolev inequality (theorem 10.4.3),

$$\|g_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C \, \|\nabla g_m\|_{L^p(\mathbb{R}^n)} \leq C \, \|g_m\|_{W^{1,p}(\mathbb{R}^n)} \leq CM$$

Thus (g_m) is bounded in $L^{p^*}(\mathbb{R}^n)$, uniformly in m. This is the reservoir of integrability that the interpolation in Step 5 will spend.

Step 3: Control of translates. Fix $h \in \mathbb{R}^n$ and write $\tau_h g(x) = g(x-h)$. For a smooth compactly supported φ the fundamental theorem of calculus along the segment from x to $x-h$ gives

$$\varphi(x) - \varphi(x-h) = \int_0^1 h \cdot \nabla\varphi(x - th)\, dt$$

so, taking absolute values and applying Minkowski's integral inequality (theorem 8.1.6),

$$\|\tau_h\varphi - \varphi\|_{L^p(\mathbb{R}^n)} \le |h| \int_0^1 \|\nabla\varphi(\cdot - th)\|_{L^p(\mathbb{R}^n)}\, dt = |h|\, \|\nabla\varphi\|_{L^p(\mathbb{R}^n)}$$

the last equality because translation preserves the L^p norm. Since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 10.2.6), every element of $W^{1,p}(\mathbb{R}^n)$ is a limit in the $W^{1,p}$ norm of such $\varphi \in C_c^\infty(\mathbb{R}^n)$; both sides of the estimate are continuous in that norm, so it passes to the limit and

$$\|\tau_h g_m - g_m\|_{L^p(\mathbb{R}^n)} \le |h|\, \|\nabla g_m\|_{L^p(\mathbb{R}^n)} \le |h|\, CM$$

holds for every m and every h. The translates of the g_m are thus small uniformly in m: given $\varepsilon > 0$, taking $|h| \le \varepsilon/(CM)$ makes $\|\tau_h g_m - g_m\|_{L^p(\mathbb{R}^n)} \le \varepsilon$ for all m at once.

Step 4: Precompactness in $L^1(W_0)$. The estimate of Step 3, together with the common support in V and the uniform L^p bound (which gives a uniform $L^1(V)$ bound since V is bounded), is precisely the hypothesis of the Fréchet-Kolmogorov criterion for precompactness in L^1. Rather than invoke it, mollify directly. Let W_0 be the closed 1-neighborhood of V, a fixed compact set, and let $\{\psi_\delta\}_{0<\delta\le 1}$ be the good kernel (definition 8.6.9) with $\psi_\delta(x) = \delta^{-n}\psi(x/\delta)$ for a fixed $\psi \in C_c^\infty(\mathbb{R}^n)$, $\psi \ge 0$, $\int \psi = 1$, supported in the unit ball; then g_m and $g_m * \psi_\delta$ are all supported in W_0. For each m,

$$(g_m * \psi_\delta)(x) - g_m(x) = \int_{\mathbb{R}^n} \big(g_m(x-y) - g_m(x)\big)\, \psi_\delta(y)\, dy$$

so integrating and using $\int \psi_\delta = 1$, Minkowski's integral inequality (theorem 8.1.6), and Step 3,

$$\begin{aligned}
\|g_m * \psi_\delta - g_m\|_{L^1(\mathbb{R}^n)} &\le \int_{|y|\le\delta} \|\tau_y g_m - g_m\|_{L^1(\mathbb{R}^n)}\, \psi_\delta(y)\, dy \\
&\le \int_{|y|\le\delta} |W_0|^{1-1/p}\, \|\tau_y g_m - g_m\|_{L^p(\mathbb{R}^n)}\, \psi_\delta(y)\, dy \le C\,\delta
\end{aligned}$$

where Hölder on the common bounded support W_0 turned the L^1 norm into an L^p norm at the cost of $|W_0|^{1-1/p}$, and the bound is uniform in m. Fix $\varepsilon > 0$ and choose δ so that $C\delta \le \varepsilon/3$. For this frozen δ the mollified family $(g_m * \psi_\delta)_m$ is uniformly bounded and equicontinuous on W_0: uniformly bounded because

$$|(g_m * \psi_\delta)(x)| \le \|g_m\|_{L^1(\mathbb{R}^n)}\, \|\psi_\delta\|_{L^\infty(\mathbb{R}^n)} \le C\delta^{-n}$$

and equicontinuous because

$$|(g_m * \psi_\delta)(x) - (g_m * \psi_\delta)(x')| \le \|g_m\|_{L^1(\mathbb{R}^n)}\, \|\tau_{x'-x}\psi_\delta - \psi_\delta\|_{L^\infty(\mathbb{R}^n)}$$

and ψ_δ is uniformly continuous. By the Arzelà-Ascoli theorem a subsequence of $(g_m * \psi_\delta)_m$ converges uniformly on W_0, hence in $L^1(W_0)$. Combining with the uniform bound $\|g_m * \psi_\delta - g_m\|_{L^1} \le \varepsilon/3$, along this subsequence

$$\limsup_{k,l} \|g_{m_k} - g_{m_l}\|_{L^1(W_0)} \le \frac{2\varepsilon}{3} + \limsup_{k,l} \|g_{m_k} * \psi_\delta - g_{m_l} * \psi_\delta\|_{L^1(W_0)} = \frac{2\varepsilon}{3} < \varepsilon$$

A diagonal argument over $\varepsilon = 1, \frac{1}{2}, \frac{1}{3}, \dots$ then produces a single subsequence, still written (g_m), that is Cauchy in $L^1(W_0)$, hence in $L^1(V)$ since $V \subseteq W_0$.

Step 5: Interpolate up to $L^q(V)$. Fix q with $1 \le q < p^*$. Interpolation between L^1 and L^{p^*} writes $\frac{1}{q} = \frac{\theta}{1} + \frac{1-\theta}{p^*}$ for a unique $\theta \in (0,1]$, and Hölder's inequality (theorem 8.1.5) gives, for any g,

$$\|g\|_{L^q(V)} \le \|g\|_{L^1(V)}^{\theta} \, \|g\|_{L^{p^*}(V)}^{1-\theta}$$

Applying this to differences of the extracted subsequence, and using the uniform L^{p^*} bound of Step 2 (which controls $\|g_{m_k} - g_{m_l}\|_{L^{p^*}(V)} \le 2CM$),

$$\|g_{m_k} - g_{m_l}\|_{L^q(V)} \le \|g_{m_k} - g_{m_l}\|_{L^1(V)}^{\theta} \, (2CM)^{1-\theta}$$

The right side tends to 0 as $k, l \to \infty$ because the sequence is Cauchy in $L^1(V)$ and $\theta > 0$. Hence (g_{m_k}) is Cauchy, so convergent, in $L^q(V)$, and its restrictions to U give a subsequence of (f_m) convergent in $L^q(U)$, as we sought to show.

The mechanism is worth isolating. A $W^{1,p}$ bound gives two things: an integrability gain past p^*, and control of translates through the gradient. The first is a fixed reservoir of smallness at high exponents; the second, via mollification, upgrades the reservoir to genuine L^1 compactness on a bounded set. Interpolation then converts L^1 compactness plus an L^{p^*} bound into L^q compactness for every q strictly below p^*. The exclusion of the endpoint $q = p^*$ is not an artifact of the proof: at the critical exponent, concentration can defeat compactness. Fixing a bump $u \in C_c^\infty(\mathbb{R}^n)$ and rescaling by $u_\lambda(x) = \lambda^{(n-p)/p} u(\lambda x)$ keeps both $\|\nabla u_\lambda\|_{L^p}$ and $\|u_\lambda\|_{L^{p^*}}$ constant in λ while concentrating the mass to a point as $\lambda \to \infty$, so no subsequence converges in L^{p^*}; the compact embedding is a subcritical phenomenon.

Example 10.5.2: Compactness on the Interval

Take $U = (0,1) \subset \mathbb{R}$, so $n = 1$. Here the subcritical range $1 \le p < n$ is empty, but the compactness is still visible directly through Morrey's inequality (theorem 10.4.8), which for $n = 1$ and $p > 1 = n$ turns a $W^{1,p}(0,1)$ bound into a uniform Hölder bound. Concretely, a sequence (f_m) bounded in $W^{1,2}(0,1)$ satisfies, for $x < y$,

$$|f_m(y) - f_m(x)| = \left| \int_x^y f_m'(t)\, dt \right| \le \|f_m'\|_{L^2(0,1)} \, |y-x|^{1/2} \le M \, |y-x|^{1/2}$$

by Cauchy-Schwarz, so (f_m) is equicontinuous with a uniform Hölder-1/2 modulus; it is uniformly bounded because the Morrey embedding turns the $W^{1,2}(0,1)$ bound into a uniform sup-bound $\|f_m\|_{L^\infty(0,1)} \le CM$. Arzelà-Ascoli then extracts a uniformly convergent subsequence, in particular convergent in $L^q(0,1)$ for every $q < \infty$: in one dimension a gradient bound is a Hölder bound, and Hölder bounds are precompact. The proof of theorem 10.5.1 replaces this pointwise Hölder estimate, unavailable when $p \le n$, by the L^p control of translates in Step 3.

The first payoff of compactness is a coercivity estimate. For a function that vanishes at the boundary, the gradient alone controls the full L^p norm: there is nothing left to compensate a large function with a small gradient once constants are forbidden. This is the inequality that makes the Dirichlet energy $\int_U |\nabla f|^2$ an equivalent norm on $W_0^{1,2}(U)$, and with it the existence theory for the Dirichlet problem. Its proof needs no compactness at all, only the one-dimensional fundamental theorem of calculus, so it precedes the Wirtinger variant.

Theorem 10.5.3: Poincaré Inequality

Let $U \subseteq \mathbb{R}^n$ be bounded and let $1 \leq p < \infty$. There is a constant $C = C(n, p, U)$ such that

$$\|f\|_{L^p(U)} \leq C \,\|\nabla f\|_{L^p(U)}$$

for every $f \in W_0^{1,p}(U)$.

***Proof*:**
Since U is bounded, choose $a > 0$ with $U \subseteq \{x \in \mathbb{R}^n \mid |x_1| < a\}$, the slab of width $2a$ in the first coordinate. By the definition of the zero-trace space (definition 10.2.6), $C_c^\infty(U)$ is dense in $W_0^{1,p}(U)$, so it suffices to prove the inequality for $f \in C_c^\infty(U)$ and then pass to the limit; extend such an f by zero to all of $\mathbb{R}^n$, keeping it smooth and compactly supported inside the slab.

Fix $x = (x_1, x') \in U$ with $x' = (x_2, \ldots, x_n)$. Because $f(-a, x') = 0$, the fundamental theorem of calculus in the first variable gives

$$f(x_1, x') = \int_{-a}^{x_1} \partial_1 f(t, x')\, dt$$

Take absolute values and apply Hölder's inequality (theorem 8.1.5) in t over the interval $(-a, a)$, of length $2a$:

$$|f(x_1, x')| \leq \int_{-a}^{a} |\partial_1 f(t, x')|\, dt \leq (2a)^{1/p'} \left(\int_{-a}^{a} |\partial_1 f(t, x')|^p\, dt\right)^{1/p}$$

Raise to the pth power and integrate in x_1 over $(-a, a)$. The right side does not depend on x_1, so integrating contributes a factor $2a$:

$$\int_{-a}^{a} |f(x_1, x')|^p\, dx_1 \leq (2a)^{p/p'}\,(2a) \int_{-a}^{a} |\partial_1 f(t, x')|^p\, dt = (2a)^p \int_{-a}^{a} |\partial_1 f(t, x')|^p\, dt$$

using $p/p' + 1 = p$. Now integrate in the remaining variables x' and recall f vanishes outside the slab:

$$\|f\|_{L^p(U)}^p = \int_{\mathbb{R}^{n-1}} \int_{-a}^{a} |f(x_1, x')|^p\, dx_1\, dx' \leq (2a)^p \int_{\mathbb{R}^{n-1}} \int_{-a}^{a} |\partial_1 f|^p\, dt\, dx' = (2a)^p \,\|\partial_1 f\|_{L^p(U)}^p$$

Since $\|\partial_1 f\|_{L^p(U)} \leq \|\nabla f\|_{L^p(U)}$, taking pth roots gives $\|f\|_{L^p(U)} \leq 2a\,\|\nabla f\|_{L^p(U)}$ with $C = 2a$. For general $f \in W_0^{1,p}(U)$, choose $f_m \in C_c^\infty(U)$ with $f_m \to f$ in $W^{1,p}(U)$; the inequality holds for each f_m, and both $\|f_m\|_{L^p(U)} \to \|f\|_{L^p(U)}$ and $\|\nabla f_m\|_{L^p(U)} \to \|\nabla f\|_{L^p(U)}$, so it passes to the limit, completing the proof.

The constant depends only on the width of the smallest slab containing U, not on its shape: Poincaré's inequality holds on any bounded domain, and its constant degrades only as the domain grows in one direction. On $W_0^{1,p}(U)$ the estimate says $\|\nabla f\|_{L^p(U)}$ is a norm equivalent to the full $\|f\|_{W^{1,p}(U)}$ norm, since $\|f\|_{W^{1,p}}^p = \|f\|_{L^p}^p + \|\nabla f\|_{L^p}^p \leq (1 + C^p)\|\nabla f\|_{L^p}^p$. The gradient alone therefore measures everything, which is why the Dirichlet energy is the natural object on the zero-trace space. The hypothesis that boundary values vanish is essential: on all of $W^{1,p}(U)$ the inequality is false, since a nonzero constant function has zero gradient but positive L^p norm. The constant functions are the only obstruction, and quotienting them out restores the estimate, which is the content of the next theorem.

Example 10.5.4: The Optimal Constant on an Interval

On $U = (0, L) \subset \mathbb{R}$ with $p = 2$, the Poincaré inequality reads $\|f\|_{L^2(0,L)} \le C\,\|f'\|_{L^2(0,L)}$ for $f \in H_0^1(0, L)$. The proof above gives $C = 2a = L$ after centering the interval, but the sharp constant is smaller. Expand $f \in H_0^1(0, L)$ in the sine basis $e_k(x) = \sqrt{2/L}\,\sin(k\pi x/L)$, which vanishes at both endpoints and diagonalizes $-d^2/dx^2$ with eigenvalues $(k\pi/L)^2$. Writing $f = \sum_k c_k e_k$, Parseval gives $\|f\|_{L^2}^2 = \sum_k c_k^2$ and $\|f'\|_{L^2}^2 = \sum_k (k\pi/L)^2 c_k^2 \ge (\pi/L)^2 \sum_k c_k^2$, so

$$\|f\|_{L^2(0,L)} \le \frac{L}{\pi}\,\|f'\|_{L^2(0,L)}$$

with equality for $f = e_1$. The sharp constant L/π is the reciprocal square root of the first Dirichlet eigenvalue $(\pi/L)^2$; the crude slab estimate $C = L$ overshoots it by the factor π. The general principle, that the best Poincaré constant is the reciprocal square root of the first eigenvalue of the Dirichlet Laplacian, connects these inequalities to spectral theory: the compactness of theorem 10.5.1 is exactly what makes $-\Delta$ with zero boundary values have compact resolvent, so its spectrum is a discrete sequence of Poincaré-type constants.

When boundary values are not prescribed the constant functions must be removed by hand, subtracting the average. The resulting inequality controls the deviation of a function from its mean by the gradient, and it holds on any bounded connected domain with reasonable boundary. Unlike the Poincaré inequality, no explicit constant is available from a one-dimensional estimate: the argument is by compactness and contradiction, and it is here that Rellich-Kondrachov is used.

Theorem 10.5.5: Poincaré-Wirtinger Inequality

Let $U \subseteq \mathbb{R}^n$ be bounded, connected, and with C^1 boundary, and let $1 \le p < \infty$. Write $f_U = \frac{1}{|U|}\int_U f\,dx$ for the average of f over U. There is a constant $C = C(n, p, U)$ such that

$$\|f - f_U\|_{L^p(U)} \le C\,\|\nabla f\|_{L^p(U)}$$

for every $f \in W^{1,p}(U)$.

***Proof*:**

Suppose the inequality fails for every constant. Then for each m there is $f_m \in W^{1,p}(U)$ with

$$\|f_m - (f_m)_U\|_{L^p(U)} > m\,\|\nabla f_m\|_{L^p(U)}$$

Normalize by setting

$$g_m := \frac{f_m - (f_m)_U}{\|f_m - (f_m)_U\|_{L^p(U)}}$$

which is legitimate because the left side of the failed inequality is positive. Then $g_m \in W^{1,p}(U)$ has three properties: its average $(g_m)_U = 0$ (subtracting a constant sets the mean to zero, and normalizing preserves it), its L^p norm $\|g_m\|_{L^p(U)} = 1$, and dividing the failed inequality by its own left side,

$$\|\nabla g_m\|_{L^p(U)} = \frac{\|\nabla f_m\|_{L^p(U)}}{\|f_m - (f_m)_U\|_{L^p(U)}} < \frac{1}{m}$$

so $\nabla g_m \to 0$ in $L^p(U)$. In particular

$$\|g_m\|^p_{W^{1,p}(U)} = \|g_m\|^p_{L^p(U)} + \|\nabla g_m\|^p_{L^p(U)} \leq 1 + \frac{1}{m^p} \leq 2$$

so (g_m) is bounded in $W^{1,p}(U)$.

The compactness of the embedding $W^{1,p}(U) \hookrightarrow L^p(U)$ furnishes a convergent subsequence in every case. If $p < n$, then $q = p < p^*$ lies in the compact range of the Rellich-Kondrachov theorem (theorem 10.5.1), so a subsequence of (g_m) converges in $L^p(U)$. If $p > n$, the $kp > n$ case of theorem 10.4.9 embeds $W^{1,p}(U)$ into a Hölder space, and a bounded sequence there is precompact in $C(\overline{U})$ by Arzelà-Ascoli, hence in $L^p(U)$. The two boundary cases $p = n$ remain. For $p = n \geq 2$, pick $p' < n$ with $(p')^* > p$ (possible since $(p')^* \to \infty$ as $p' \uparrow n$); then $W^{1,p}(U) \hookrightarrow W^{1,p'}(U)$ on the bounded set U, and $p < (p')^*$ puts $q = p$ in the compact range for exponent p', so theorem 10.5.1 again applies. The last case is $n = 1$, $p = 1$: here neither route is available, but the Rellich mechanism applies directly. Extend the g_m to a common bounded support in $\mathbb{R}$ as in Step 1 of the proof of theorem 10.5.1; the translate estimate of its Step 3 gives $\|\tau_h g_m - g_m\|_{L^1(\mathbb{R})} \leq |h|\,\|\nabla g_m\|_{L^1}$, uniformly bounded in m, which together with the common support is the hypothesis of the Fréchet-Kolmogorov criterion for precompactness in L^1, yielding a subsequence convergent in $L^1(U) = L^p(U)$. In every case, pass to a subsequence, still called (g_m), with $g_m \to g$ in $L^p(U)$ for some $g \in L^p(U)$.

The limit g inherits both normalizations by L^p convergence: $\|g\|_{L^p(U)} = \lim \|g_m\|_{L^p(U)} = 1$, and averaging is L^p-continuous on the bounded set U (by Hölder against the constant $1 \in L^{p'}(U)$), so $g_U = \lim (g_m)_U = 0$. It remains to identify the weak gradient of g. For any $\varphi \in C_c^\infty(U)$ and any index i, the weak-derivative identity (definition 10.1.1) for g_m reads $\int_U g_m\, \partial_i \varphi = -\int_U \partial_i g_m\, \varphi$. Let $m \to \infty$: the left side tends to $\int_U g\, \partial_i \varphi$ since $g_m \to g$ in L^p and $\partial_i \varphi \in L^{p'}$, while the right side tends to 0 since $\partial_i g_m \to 0$ in L^p and $\varphi \in L^{p'}$. Hence $\int_U g\, \partial_i \varphi = 0$ for all φ, which says $\partial_i g = 0$ weakly for every i, so $\nabla g = 0$. Thus $g \in W^{1,p}(U)$ with vanishing weak gradient on the connected set U.

A $W^{1,p}$ function with zero weak gradient on a connected open set is constant almost everywhere. Indeed, mollifying, $g * \psi_\delta$ is smooth with classical gradient $\nabla(g * \psi_\delta) = (\nabla g) * \psi_\delta = 0$ on any $V \Subset U$ (by theorem 10.2.1, which commutes the derivative past the mollifier), hence constant on each component of V; letting $\delta \to 0$ and using connectedness of U shows g equals a single constant almost everywhere. But then $g = g_U = 0$ almost everywhere, contradicting $\|g\|_{L^p(U)} = 1$. The assumed failure is therefore impossible, and the inequality holds with some finite constant C, completing the proof.

The proof is nonconstructive: it produces a finite constant but no formula for it, in contrast to the explicit $2a$ of the Poincaré inequality. This is the price of the compactness argument, and it is characteristic of the method: an inequality is proved by ruling out its failure, and the witness to failure is destroyed by extracting a convergent subsequence whose limit cannot exist. Connectedness is used exactly once, in the final step, to force a gradient-free function to be a single constant rather than a locally constant one; on a disconnected domain the average over the whole set is the wrong reference value, and the inequality fails with a function equal to different constants on different components. The mean f_U is the natural replacement for the boundary value of the Poincaré inequality: both the zero-trace condition and the zero-mean condition eliminate the constants, which are the sole functions the gradient cannot see, and the estimate holds on the quotient by constants in either case.

Example 10.5.6: Mean-Zero Functions on a Ball

On the unit ball $U = B(0,1) \subset \mathbb{R}^n$ with $p = 2$, the Poincaré-Wirtinger inequality $\|f - f_U\|_{L^2(U)} \leq C\,\|\nabla f\|_{L^2(U)}$ again has a spectral reading. Its sharp constant is the reciprocal square root of the first *nonzero* Neumann eigenvalue μ_1 of $-\Delta$ on the ball: the mean-zero constraint $f_U = 0$ is exactly orthogonality to the constant eigenfunction, whose Neumann eigenvalue is $\mu_0 = 0$, so on the orthogonal complement the Rayleigh quotient $\|\nabla f\|_{L^2}^2 / \|f\|_{L^2}^2$ is bounded below by μ_1. A mean-zero f therefore satisfies $\|f\|_{L^2}^2 \leq \mu_1^{-1}\|\nabla f\|_{L^2}^2$, giving $C = \mu_1^{-1/2}$. The contradiction argument of theorem 10.5.5 recovers the finiteness of this constant without computing μ_1: it shows only that the infimum of the Rayleigh quotient over mean-zero functions is positive, which is exactly $\mu_1 > 0$, the statement that 0 is a simple Neumann eigenvalue on a connected domain.

11

Hilbert Spaces

There are many reasons why Hilbert spaces are interesting. We shall start with a motivation stemming from looking for the ideal space where the product of two functions is still integrable, and then take a moment to elaborate on some more advanced intuitions of Hilbert spaces for readers with more background.

We are naturally interested in integrating products of functions:

$$\int_X fg$$

The problem we may encounter when working over L^1 is that $\int_X fg$ need not be finite. We may be tempted to say "let us work with the functions for which this integral *is* finite". That is, let $L'(X)$ be the space of all measurable functions where for any choice $f, g \in L'(X)$:

$$\int_X |f\overline{g}| < \infty \qquad \text{i.e.} \qquad f\overline{g} \in L^1$$

Then we have just made a circular-reasoning mistake, for we are quantifying existence of elements in $L'(X)$ using elements of $L'(X)$. Instead, we may ask: given a set $S \subseteq \mathcal{M}(X)$, what is the subset of measurable functions $T \subseteq \mathcal{M}(X)$ such that:

$$T = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X |f\overline{g}| < \infty,\ \forall g \in S \right\}$$

We can label this space $L^1_S(X)$. The two extremes of this are if $S = \{0\}$, then $L^1_{\{0\}}(X) = \mathcal{M}(X)$ is all measurable functions, and if S is all measurable functions, then $L^1_{\mathcal{M}(X)}(X) = \{0\}$ only contains the zero function. The ideal is a middle ground choice of S where $L^1_S(X) = S$, so that we don't need to keep track of the two spaces we are choosing our functions from. Fortunately, this is *exactly* the space L^2:[1] taking $S = L^2(X)$,

$$L^1_{L^2(X)}(X) = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X |f\overline{g}| < \infty \;\forall g \in L^2 \right\} = L^2$$

Half of this is elementary: by the Cauchy-Schwarz inequality[2]

$$\int_X |fg| \leq \left(\int_X |f|^2 \right)^{1/2} \left(\int_X |g|^2 \right)^{1/2} = \|f\|_2 \|g\|_2$$

every $f \in L^2$ pairs integrably with all of L^2, showing $L^2(X) \subseteq L^1_{L^2(X)}(X)$. The substantive half is the reverse inclusion: if $\int_X |f\overline{g}| < \infty$ for *every* $g \in L^2$, then f itself must lie in L^2. This is the *converse* of Hölder's inequality, theorem 8.2.2.[3] Hence $L^1_{L^2(X)}(X) = L^2$: the space L^2 is a fixed point of $S \mapsto L^1_S(X)$.

It is, moreover, the *only* fixed point. Suppose $L^1_S(X) = S$. Every $f \in S$ then pairs with itself, so $\int_X |f|^2 = \int_X |f\overline{f}| < \infty$; that is, $S \subseteq L^2$. But $S \mapsto L^1_S(X)$ is inclusion-reversing, so $S = L^1_S(X) \supseteq L^1_{L^2(X)}(X) = L^2$. Hence $S = L^2$.

From this perspective, L^2 is the ideal space in which to multiply functions and they remain integrable (i.e. $f, g \in L^2$, $fg \in L^1$)! We shall call such multiplication the *inner product*:

$$\langle f, g \rangle = \int_X f\overline{g}$$

The conjugation in the definition is used so that $f\overline{f} = |f|^2$. Note that we are not looking for the product of two integrable functions ($f, g \in L^1$) to remain within L^1. As we saw in section 8.6, *convolution* has that property: if $f, g \in L^1(G)$ then $f * g \in L^1(G)$ for G a locally compact group, making $L^1(G)$ a *group algebra*.

11.0.1 Post-Hoc Motivation A lot of the following motivation came *a posteriori*, that is, I felt much more comfortable with Hilbert spaces after learning about L^p spaces, and generalizing some results from Riemannian Geometry to Banach spaces to notice that L^2 is the only space with flat geometry, hence really is the "correct" generalization of Euclidean space/geometry in infinite dimensions.

Banach spaces provide a robust framework for measuring the magnitude of elements, allowing us to quantify convergence and bound operations via the triangle and Minkowski inequalities. However, a fundamental deficiency in general Banach spaces is the lack of a linear measurement theory to quantify *similarity*. While we can measure the total size of a sum $\|f + g\|$, we often lack the granular information required to isolate the contribution of f from g. For instance, determining how much a vector projects onto a specific basis element is straightforward in Euclidean space but algebraically obstructed in general norms.

[1] We assume in this motivation that the measure space is σ-finite. On a measure space consisting of a single atom of infinite measure, say, $L^2 = \{0\}$ while $L^1_{\{0\}}(X) = \mathcal{M}(X)$, and no fixed point exists at all.

[2] Proven as theorem 11.1.2 in section 11.1; the reader has most likely already encountered it, so we use it here for intuition.

[3] Strictly, theorem 8.2.2 also assumes the finiteness of the supremum M_q; that the integrability hypothesis alone forces this bound is a closed-graph argument, which we omit here.

To make this deficiency precise, consider how the inner product arises in Euclidean geometry. Define the *energy* of a vector as $E(x) = \frac{1}{2}\|x\|_2^2$. Then the directional derivative of this energy at x in the direction of a perturbation y is:

$$\frac{d}{dt}E(x+ty)\Big|_{t=0} = \frac{d}{dt}\frac{1}{2}\langle x+ty, x+ty\rangle\Big|_{t=0} = \langle x, y\rangle$$

Thus, the inner product $\langle x, y\rangle$ physically represents the "sensitivity" of the energy of x to a perturbation by y. In particular, the energy satisfies the clean linear decomposition:

$$E(x+y) = E(x) + E(y) + \langle x, y\rangle$$

Note the interaction term $\langle x, y\rangle$ is *bilinear*. In general Banach spaces, such a linear decomposition of the interaction is impossible due to the non-flat geometry of the unit ball.

The natural question, then, is: *what replaces the inner product in L^p?* The correct generalization is to replace the quadratic energy with the *p-energy*:

$$F(f) = \frac{1}{p}\|f\|_p^p = \frac{1}{p}\int |f|^p$$

and ask: how does $F(f)$ change if we (linearly) perturb f by a function g? To answer this, we must compute the gradient ∇F, which requires first understanding the natural duality of L^p spaces.

What does it mean to go in the "same direction" in a function space? (For simplicity we work throughout with real L^p spaces for $1 < p < \infty$.[4]) Rigorously, given $f \neq 0$, we are asking for the continuous linear functional $\varphi_f \in (L^p)^*$ of unit norm that *maximizes* the evaluation at f. We can interpret this φ_f as the *optimal filter* or sensor for f, i.e. it captures the full "energy" of f without loss ($\varphi_f(f) = \|f\|_p$ and $\|\varphi_f\|_{(L^p)^*} = 1$). Such a functional exists in any Banach space by the Hahn-Banach Theorem (theorem 7.3.3); that it is *unique* is special to $1 < p < \infty$, as the next computation shows (at $p = 1$ or $p = \infty$ the maximizer need not be unique).

By theorem 8.2.3 (and the equality condition of Hölder's inequality), this functional is represented by a positive multiple of:

$$g(x) = |f(x)|^{p-1}\operatorname{sgn}(f(x))$$

Note that $g \in L^q$ since:

$$|g|^q = |f|^{q(p-1)} = |f|^{qp-q} \overset{!}{=} |f|^p$$

where $\overset{!}{=}$ holds since $1/p + 1/q = 1$ iff $p + q = pq$ iff $p = q(p-1)$. Hence as $f \in L^p$:

$$\int |g|^q = \int |f|^p < \infty$$

The same computation fixes the multiple: $\int fg = \int |f|^p = \|f\|_p^p$ while $\|g\|_q = \|f\|_p^{p/q} = \|f\|_p^{p-1}$, so the unit-norm sensor is represented by $\|f\|_p^{1-p}\, g$, and indeed $\varphi_f(f) = \|f\|_p^{1-p}\int fg = \|f\|_p$.

For the geometry to come, however, it is the unnormalized representer that matters. We define the *duality map*:

$$J_p \colon L^p \to L^q, \qquad J_p(f) = |f|^{p-1}\operatorname{sgn}(f)$$

[4]The reasoning can extend to nuclear Fréchet spaces, which admit descriptions as projective limits of Hilbert spaces. For general metric spaces, a different approach is needed via CAT(k) geometry; see [15] for details. For complex scalars, replace $\operatorname{sgn}(f)$ below by $\overline{\operatorname{sgn}(f)}$; at $p = 2$ the duality map then becomes conjugation $f \mapsto \overline{f}$, which is why the Hermitian inner product conjugates one of its slots.

so that $J_p(f)$ represents the functional $\|f\|_p^{p-1}\varphi_f$ for $f \neq 0$, and $J_p(0) = 0$. As $(L^p)^* \cong L^q$, we can equivalently view this as a map $J_p \colon L^p \to (L^p)^*$, $f \mapsto \big(h \mapsto \int J_p(f)\, h\big)$.

The key is to realize that the duality map J_p is the *geometric normal* to the level sets of the p-energy. To see this, consider the function:

$$F(x) = \frac{1}{p}\|x\|_p^p = \frac{1}{p}\sum |x_i|^p$$

(where the ℓ^p norm is used for visualization, but the logic holds for integrals). From multivariable calculus, the unit sphere is the level set $F(x) = \frac{1}{p}$, and the gradient $\nabla F(x)$ is the vector strictly perpendicular (normal) to the level set at point x. Computing the gradient (the Fréchet derivative):

$$\frac{\partial}{\partial x_i}\left(\frac{1}{p}|x_i|^p\right) = |x_i|^{p-1}\mathrm{sgn}(x_i)$$

So, the gradient vector is:

$$\nabla F(x) = \big(|x_1|^{p-1}\mathrm{sgn}(x_1), \ldots, |x_n|^{p-1}\mathrm{sgn}(x_n)\big)$$

This is exactly our duality map:

$$\boxed{\nabla F(x) = J_p(x)}$$

This identification is the key to the entire theory. It tells us that the optimal functional for f (the element of $(L^p)^*$ that best "measures" f) points precisely along the gradient of the energy at f: the direction of steepest increase in p-energy. The duality map is a *geometric* object, not only an algebraic one.

The p-Pairing and Its Interpretation. With the gradient in hand, we can now answer our original question: how does the p-energy $F(f)$ change under a perturbation by g? The first-order variation is[5]:

$$dF(f; g) = \langle \nabla F(f), g\rangle_{\text{dual}} = \int \nabla F(f) \cdot g$$

Substituting $\nabla F = J_p(f)$, this defines a natural pairing:

$$\langle f, g\rangle_p := \int J_p(f) \cdot g$$

Note that it is important to pass through $J_p(f)$ since $\int fg$ need not be finite in general. However, by Hölder's inequality:

$$\int |J_p(f) \cdot g| < \infty$$

This pairing has a precise geometric interpretation. The gradient $J_p(f)$ is a cotangent vector (a 1-form $dF|_f \in T_f^* L^p$) and the perturbation g is a tangent vector $g \in T_f L^p$. Thus the pairing is the evaluation of a 1-form on a tangent vector, under the canonical evaluation $T_f^* L^p \times T_f L^p \to \mathbb{R}$:

$$\langle f, g\rangle_p = dF|_f(g)$$

[5]The dF here can be thought of as the Gâteaux derivative, commonly defined for functions between locally convex topological vector spaces.

This canonical evaluation pairing between the tangent and cotangent spaces exists in *every* geometric setting without any choice of metric. What J_p adds is the ability to *produce* a specific cotangent vector from a tangent vector, turning the pairing from a bilinear form on $T_f^* L^p \times T_f L^p$ into a (generally nonlinear) form on $T_f L^p \times T_f L^p$: that is, internalizing the measurement within the space itself.

The Infinitesimal Meaning of V^*. We are now in a position to make a precise statement about the geometric content of the dual space. In a bare vector space without additional structure, the dual V^* is only an algebraic object: a collection of linear maps with no geometric significance beyond probing linearly. However, in L^p equipped with its norm, the identity $J_p = \nabla F$ reveals that *every element of $(L^p)^*$ arises as the gradient of the energy functional*, a differential $dF|_f$. This endows $(L^p)^*$ with genuine infinitesimal meaning: its elements are not just linear maps, but *cotangent vectors* that encode the first-order sensitivity of the geometry.

Crucially, cotangent vectors are a different species of infinitesimal object than tangent vectors. A tangent vector $g \in T_f L^p \cong L^p$ represents an *infinitesimal direction of motion*: it answers "which way?". A cotangent vector $\varphi \in T_f^* L^p \cong L^q$ represents an *infinitesimal measurement*: it answers "how fast is this quantity changing?". These are related by the evaluation pairing $\varphi(g) = dF|_f(g)$, but they transform in opposite ways: tangent vectors push forward under maps, while cotangent vectors pull back. The duality map $J_p : T_f L^p \to T_f^* L^p$ provides a (nonlinear) identification between these two types of infinitesimal data, allowing us to transfer the directional interpretation onto the dual space. However, this transfer *depends essentially on the norm geometry*: it is canonical only for $p = 2$.

Geometrically, if we view the identity map $f \mapsto f$ as the radial vector field (a *section* of the tangent bundle TL^p), then the duality map J_p transforms this into a covector field (a *section* of the cotangent bundle T^*L^p). In the language of Riemannian geometry, this transformation is the *musical isomorphism* $\flat$ (flat), which maps a vector field X to a 1-form $X^\flat$ by lowering indices via the metric tensor. The four corollaries that follow are all consequences of this single geometric fact.

Corollary 1: Nonlinearity of the Musical Isomorphism. The duality map is *not* linear:

$$J_p(2f) = |2f|^{p-1}\mathrm{sgn}(2f) = 2^{p-1}|f|^{p-1}\mathrm{sgn}(f) = 2^{p-1}J_p(f)$$

This is only linear in the special case of $p = 2$, since:

$$J_2(nf) = |nf|^{2-1}\mathrm{sgn}(nf) = nJ_2(f)$$

In a general Banach space ($p \neq 2$), the musical isomorphism is *nonlinear*: the operation of lowering an index depends on the magnitude of the vector itself, meaning the "metric" changes as we move through the space. However, when $p = 2$, this map is linear. This implies a linear bundle isomorphism $TL^2 \cong T^*L^2$, effectively trivializing the distinction between vectors and covectors. Only in L^2 is the musical isomorphism a linear operator, meaning the "force" required to correct an error is linearly proportional to the error itself.

Corollary 2: Non-Symmetry of the Pairing. The pairing $\langle f, g\rangle_p$ is *not* symmetric for $p \neq 2$. This is an immediate consequence of the nonlinearity of J_p: the 1-form $dF|_f$ evaluated on g has no reason to equal the 1-form $dF|_g$ evaluated on f, since the gradient depends nonlinearly on its base point. If however $p = 2$, then $J_p(x)$ maps identically to itself and we recover the symmetric form $\int fg$. This asymmetry is the first signal that the geometry of L^p is *not flat* for $p \neq 2$.

Corollary 3: Failure of Rotational Invariance. This lack of symmetry in the pairing $\langle f, g\rangle_p$ is directly linked to the fact that L^2 is the unique L^p space that is *rotationally invariant*. Geometrically, rotational invariance means that the properties of a vector depend only on its length, not on its alignment with the coordinate axes. Mathematically, this requires that the duality map commutes with rotations. If R is a Euclidean rotation, we expect the "normal vector" of the rotated point to be the rotated "normal vector" of the original point:

$$J_p(Rf) \stackrel{?}{=} R(J_p(f))$$

In L^2, since J_2 is linear (essentially the identity), this holds trivially: $J_2(Rf) = Rf = R(J_2(f))$. However, for $p \neq 2$, the map $f \mapsto |f|^{p-1}\text{sgn}(f)$ applies a non-linear distortion component-wise. Rotating the vector mixes the components, and applying the power law *after* mixing is not the same as applying it *before*.

To see this concretely, consider ℓ^3 $(p = 3)$ in $\mathbb{R}^2$ and let $f = (1, 0)$. If we first compute the functional and then rotate by $\pi/4$ (45°), we get:

$$R(J_3(f)) = R(1^2, 0^2) = R(1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

However, if we rotate the vector first, the components become $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Applying the duality map J_3 (squaring components) yields:

$$J_3(Rf) = J_3\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Since $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \neq (\frac{1}{2}, \frac{1}{2})$, we have $J_p(Rf) \neq R(J_p(f))$. This confirms that for $p \neq 2$, the geometry is *anisotropic*: the shape of the unit ball looks different depending on the angle from which you view it. Equivalently, for $p \neq 2$ a Euclidean rotation is not an isometry of ℓ^p; the genuine isometries of ℓ^p (signed permutations of the coordinates) do commute with J_p.

Corollary 4: Birkhoff-James Orthogonality. With this understanding of the position-dependent metric, we can clarify the notion of orthogonality. In Euclidean geometry, orthogonality is symmetric ($x \perp y \iff y \perp x$) and additive. In general Banach spaces, without a global bilinear form, we must rely on *Birkhoff-James orthogonality*: we say $x \perp y$ if moving in the direction of y does not decrease the length of x ($\|x + \lambda y\| \geq \|x\|$ for all scalars λ). Geometrically, this means the line passing through x in the direction of y is tangent to the unit sphere at x.

Analytically, utilizing the duality map J_p, this condition is equivalent to requiring that the tangent vector y be annihilated by the cotangent vector generated by x:

$$\langle J_p(x), y\rangle = 0$$

This is precisely the statement that y lies in the kernel of the 1-form $dF|_x$, a natural geometric condition. However, it reveals the fundamental asymmetry in non-Hilbert geometry. In L^2, because J_2 is linear, the condition is symmetric. However, in L^p $(p \neq 2)$, the non-linearity of J_p breaks reciprocity. The set of vectors orthogonal to x (kernel of $J_p(x)$) does not align symmetrically with the set of vectors orthogonal to y. Consequently, it is common to have $x \perp y$ but $y \not\perp x$: in ℓ^3 on $\mathbb{R}^2$, take $x = (2, 1)$ and $y = (1, -4)$; then $\langle J_3(x), y\rangle = 4 \cdot 1 + 1 \cdot (-4) = 0$ while $\langle J_3(y), x\rangle = 1 \cdot 2 + (-16) \cdot 1 = -14 \neq 0$. This lack of symmetry illustrates why Hilbert spaces are the unique setting where Euclidean geometric intuition remains valid.

The Riemannian Perspective. The corollaries above all concern the first derivative of the energy (the gradient J_p). The full geometry of the space, however, is determined by how this gradient *changes*: the second derivative, which acts as the metric tensor. In this framework, the metric tensor $g_{ij}(x)$ at a point x dictates how to measure the length of tangent vectors. In our context, this metric is generated by the Hessian of the energy function $F(x) = \frac{1}{p}\|x\|_p^p$. For L^2, this Hessian is proportional to the identity operator everywhere; the metric is *constant* (flat). However, for L^p ($p \neq 2$), the second derivative depends on position x, specifically scaling as $|x|^{p-2}$. This implies that the "metric" changes as we move around the space.

It is worth noting why the theory breaks down at the boundary cases. For L^1, the unit ball is a diamond with corners; at these corners, the gradient is not unique (it is a set-valued subgradient). For L^∞, the unit ball has flat faces; on these faces, the gradient is non-unique or vanishes in certain directions. This lack of smoothness (L^1) or rotundity (L^∞) prevents a well-defined duality map, leaving L^p $(1 < p < \infty)$ as the range where smooth geometry.

Topological vs. Geometric Structure. To visualize the practical consequences of this non-flat geometry, it is helpful to consider the "shape" of the unit ball in infinite dimensions. In finite dimensions, all norms are equivalent, meaning the unit ball of one norm can always be fit inside a scaled unit ball of another. In infinite dimensions, this equivalence breaks. The curvature of the norm dictates how "energy" is allowed to concentrate without bound. The L^∞ unit ball acts like a strict box, while the L^1 unit ball allows for infinitely long thin spikes.

Ideally, we would like to know if these distortions are topological or only geometric. This is answered by the *Mazur map*:

$$M_{p,q} \colon L^p \to L^q, \qquad M_{p,q}(f) = \operatorname{sgn}(f) \cdot |f|^{p/q}$$

which restricts to a *uniform* homeomorphism between the unit spheres of L^p and L^q: we can continuously deform the "spikes" of the L^1 sphere into the round sphere of L^2, with uniform control in both directions. The distinction between spheres and whole spaces here is essential: as topological spaces all separable infinite-dimensional L^p are homeomorphic (theorem 7.5.5), but the whole spaces are *not* uniformly homeomorphic to one another for $p \neq q$, since uniform homeomorphisms preserve enough local linear structure to distinguish the L^p scale (a rigidity phenomenon initiated by Ribe; see Benyamini and Lindenstrauss [1] for both the Mazur map and this rigidity theory). However, while the Mazur map preserves the uniform structure of the spheres, it destroys the linear structure (additivity): it is homogeneous of degree p/q rather than degree one. The special role of L^2 in this family is thus not topological but geometric: it is the unique L^p whose duality map is linear, as the preceding analysis showed.

This geometric disparity explains why convergence depends on the norm. Convergence $f_n \to 0$ is not about movement, but about entering arbitrarily small ϵ-balls. Consider the sequence $f_n(x) = x^n$ on $[0, 1]$. In the L^1 geometry, this sequence travels down one of the "spikes" of the unit ball: the functions get thinner and thinner, their area shrinking to zero despite their height remaining constant. In the L^∞ geometry, however, these functions are trapped on the surface of the unit sphere; they cannot enter the ϵ-ball because the L^∞ "box" does not extend down the "spikes" of concentration.

Consequences for Hilbert Spaces. The unique geometry of Hilbert spaces (self-duality, isotropy, and the inner product) unlocks powerful results unavailable in general Banach spaces. The preceding analysis explains *why*: in L^2, the duality map J_2 is the identity, making the musical isomorphism linear, the pairing symmetric, and the geometry flat. The following are key consequences we will explore:

1. *The Projection Theorem (Best Approximation):* In general Banach spaces, finding the element in a subspace M closest to a point x is difficult (the closest point might not be unique, e.g.,

the flat edge of L^1). In Hilbert spaces, the strict convexity and "roundness" of the unit ball guarantee that for any closed subspace M, there exists a *unique* element $P_M x$ minimizing the distance $\|x - m\|$. Furthermore, the error vector $x - P_M x$ is orthogonal to the subspace.

2. *The Riesz Representation Theorem (Self-Duality):* The duality map $J\colon H \to H^*$, $y \mapsto \langle -, y\rangle$, is bijective and conjugate-linear (linear, over the reals). This means we can completely identify linear functionals with vectors. Every continuous linear functional φ acts exactly like an inner product: $\varphi(x) = \langle x, y\rangle$ for a unique y. This collapses the study of H^* to the study of H.

3. *Orthonormal Bases and Fourier Series:* In L^p, we may have a Schauder basis[6], but determining the coefficients is non-trivial. In Hilbert spaces, orthogonality allows us to decouple coefficients. This is key for Fourier series, where any element is reconstructed by projection: $x = \sum \langle x, e_n\rangle e_n$. Additionally, the *Parseval Identity* allows us to measure size by summing squared coefficients: $\|x\|^2 = \sum |\langle x, e_n\rangle|^2$.

4. *Existence of Adjoints:* In Banach spaces, the adjoint of $T : X \to Y$ maps $Y^* \to X^*$. Because Hilbert spaces are self-dual, we can pull this operator back to the original space. Thus, for every bounded operator T on a Hilbert space, there is a unique adjoint T^* on the *same space* satisfying $\langle Tx, y\rangle = \langle x, T^* y\rangle$. This is foundational for spectral theory.

In summary, while Banach spaces provide the necessary topological framework (completeness), Hilbert spaces add the rigid geometric structure required to generalize linear algebra to infinite dimensions. The duality map J_p, understood as the gradient of the p-energy, reveals the precise mechanism: it is the *cotangent structure* of the space (the infinitesimal measurement theory) that determines the geometry. The canonical evaluation pairing between tangent and cotangent vectors is the universal source of "derivative-like" phenomena; J_p is the geometric data that internalizes this pairing within the space itself. In L^2 alone, this internalization is linear, making vectors and covectors (directions and measurements) two aspects of a single object.

11.1 Basic definition and properties

Definition 11.1.1: Inner Product Space

Let H be a complex vector space with an inner product $\langle \cdot, \cdot\rangle : H \times H \to \mathbb{C}$ where

1. $\langle x, y\rangle = \overline{\langle y, x\rangle}$
2. $\langle x + y, z\rangle = \langle x, z\rangle + \langle x, z\rangle$
3. $\langle ax, y\rangle = a\langle x, y\rangle$, $a \in \mathbb{C}$
4. $\langle x, x\rangle \in (0, \infty)$ for all nonzero $x \in H$
5. $\langle x, x\rangle = 0$ if and only if $x = 0$

We'll define $\|x\|_2 = \sqrt{\langle, x, x\rangle}$ where the 2 subscript hints that this will become the L^2-norm and is added to not hold ambiguity with the L_1 norm. This is naturally a norm since:

[6] A sequence $\{b_n\}$ where every $v \in V$ has a unique expansion $v = \sum \alpha_n b_n$.

1. $\|x\|_2 = \sqrt{\langle x, x\rangle} = 0$ if and only if $x = 0$
2. $\|ax\|_2 = \sqrt{\langle ax, ax\rangle} = \sqrt{a\bar{a}\langle x, x\rangle} = \sqrt{a^2\langle x, x\rangle} = |a|\sqrt{\langle x, x\rangle} = |a|\|x\|_2$

Proving the triangle inequality is a bit more difficult: IF we square both sides, we it is equivalent to show:

$$\|x+y\|_2^2 \leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2$$

expanding out the left hand side, we get:

$$\|x+y\|_2^2 = \|x\|_2^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|_2^2$$

so we have to show that $\mathrm{Re}\langle x, y\rangle \leq \|x\|_2\|y\|_2$. In fact, a bit more is true: $|\langle x, y\rangle| \leq \|x\|_2\|y\|_2$. Though we are using this inequality to prove the triangle inequality, notice that this in fact gives us a general bound on the value of the inner product! This inequality is one of the central tools used in the study of Hilbert spaces:

Theorem 11.1.2: Cauchy-Schwarz Inequality

$$|\langle x, y\rangle| \leq \|x\|_2\|y\|_2$$

***Proof*:**
If $\|y\|_2 = 0$ then $y = 0$, and then $|\langle x, 0\rangle| = 0$, so assume $\|y\|_2 \neq 0$. Then for all $t \in \mathbb{R}$

$$\begin{aligned} 0 &\leq \langle x - ty, x - ty\rangle \\ &= \|x\|_2^2 + 2t\mathrm{Re}\langle x, y\rangle + t^2\|y\|_2^2 \\ &= \|x\|_2^2 + \frac{2\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} + \frac{\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} \qquad & t = \frac{\mathrm{Re}\langle x, y\rangle}{\|y\|_2^2} \\ \Rightarrow |\mathrm{Re}\langle x, y\rangle| &\leq \|x\|_2\|y\|_2 \end{aligned}$$

To finish the proof, pick $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that

$$|\mathrm{Re}(\alpha\langle x, y\rangle)| = |\langle x, y\rangle|$$

Notice this does not break the inequality, and so we can repeat the entire process but with α taken into account with t, and so:

$$|\langle x, y\rangle \leq \|x\|_2\|y\|_2$$

as we sought to show

With this, the rest of the proof that $\|-\|_2$ induced by $\langle -, -\rangle$ is a norm follows

Theorem 11.1.3: Triangle Inequality For Induced Norm

$$\|x+y\|_2 \leq \|x\|_2 + \|y\|_2$$

***Proof*:**

$$\begin{aligned}\|x+y\|_2^2 &= \langle x+y, x+y\rangle \\ &\leq \|x\|_2^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|_2^2 \\ &\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2\end{aligned}$$

The Cauchy-Shwartz inequality is the tool we use to bring in geometry as we know it into Hilbert space. For example, in real inner product spaces, we will define the angle between two vectors to be θ where θ satisfies:

$$\langle u, v\rangle = \cos(\theta)\|u\|_2\|v\|_2$$

Many other cool properties of the Cauchy-Shwartz inequality can be found here: (commented)

Now, notice we can define a metric by doing $\rho(x, y) = \|x - y\|_2$:

1. $\rho(x, x) = \|x - x\|_2 = 0$ if and only if $x - x = 0$ so $x = x$
2. $\rho(x, y) = \|x - y\|_2 \geq 0$
3. $\rho(x, y) = \|x - y\|_2 = \|x - y + y - z\|_2 \leq \|x - y\|_2 + \|y - z\|_2 = \rho(x, y) = \rho(y, z)$

Thus, we can define a natural topology on any pre-Hilbert space. If that topology is complete, we give it a name:

Definition 11.1.4: Hilbert Space

Let H be an inner product space. Then H is a Hilbert space if it is *complete* with respect to the induced metric.

Example 11.1.5: Hilbert Spaces

1. $\mathbb{R}^n$ is a (real) Hilbert space with the inner product being:

$$\langle x, y\rangle = \sum_i x_i y_i$$

verifying this is a Hilbert space is left as an exercise.

2. $L^2_\mu(X)$ for a complex measure μ is a Hilbert space with the inner product being:

$$\langle f, g\rangle_{L^2} = \int_X f(x)\overline{g(x)}d\mu(x)$$

To show it's well-defined, recall that if $a, b \geq 0$:

$$2|ab| = 2|a||b| \leq |a|^2 + |b|^2$$

Thus:

$$|f(x)\overline{g(x)}| = |f(x)||g(x)| \leq |f(x)|^2 + |g(x)|^2$$

As integrating keeps inequality when integrating positive functions:

$$\|f\overline{g}\|_1 \leq \int_X |f(x)|^2 + \int_X |g(x)|^2 < \infty$$

showing it is well-defined. Alternatively, using Cauchy-Schwarz:

$$\|f\overline{g}\|_1 \leq \|f\|_2\|g\|_2 < \infty$$

3. As an exercise, show that $\ell^2(\mathbb{Z})$ is a Hilbert Space.

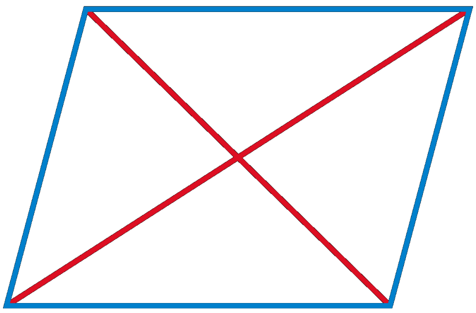
Hilbert spaces are very euclidean in nature, for example:

Theorem 11.1.6: Parallelogram Law

Let $\| - \|$ be a norm. Then $\| - \|$ is induced by an inner product if and only if:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Visually, this means that the "area" of a parallelogram is can be measured in two ways:



***Proof*:**

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= 2\langle x,x\rangle + \langle x,y\rangle - \langle x,y\rangle + \langle y,x\rangle - \langle y,x\rangle + 2\langle y,y\rangle \\
&= 2\langle x,x\rangle + 2\langle y,y\rangle \\
&= 2\|x\|^2 + 2\|y\|^2
\end{aligned}$$

It turns out that a norm satisfying the parallelogram law will in fact define an inner product. This shows how norms define geometries that are more general than euclidean:

Example 11.1.7: Exploring Parallelogram Law

1. (The polarization Identity)$\mathcal{H}$ be a Hilbert space. For any $x, y \in \mathcal{H}$,

$$\langle x, y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2)$$

Proof. This is simply a matter of computation:

$$\begin{aligned}
&\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2) \\
&= \frac{1}{4}(\langle x+y, x+y\rangle - \langle x-y, x-y\rangle + i\langle x+iy\rangle - i\langle x-iy, x-iy\rangle) \\
&= \frac{1}{4}(\langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle - (\langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle) \\
&\quad + i(\langle x,x\rangle + \langle x,iy\rangle + \langle iy,x\rangle + \langle iy,iy\rangle - (\langle x,x\rangle - \langle x,iy\rangle - \langle iy,x\rangle + \langle iy,iy\rangle)) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + i(2\langle x,iy\rangle + 2\langle iy,x\rangle)) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + i(-i)2\langle x,y\rangle + 2ii\langle y,x\rangle) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + 2\langle x,y\rangle - 2\langle y,x\rangle) \\
&= \frac{1}{4}4\langle x,y\rangle \\
&= \langle x,y\rangle
\end{aligned}$$

□

2. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary if and only if it is isometric and surjective

Proof. Let U be unitary so that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$ for any $x, y \in \mathcal{H}$. Letting $x = y$ gives us $\|Ux\|_2 = \|x\|_1$, giving us that it's an isometry. Since U is invertible, it is surjective, and so U is a surjective isometry.

Conversely, let U be a surjective isometry. We need to show its invertible (it suffices to show it's injective), an isometry, and that U and U^{-1} are bounded. Clearly, U is bounded since $\|U(x)\| \leq 1\|x\|$. For injectivity, let $Ux = Uy$, so $Ux - Uy = 0$. Taking the norm on both sides, we get:

$$0 = \|0\|_2 = \|Ux - Uy\|_2 = \|U(x-y)\|_2 = \|x-y\|_1$$

where the last equality is due to U being an isometry. Since $\|z\| = 0$ if and only if $z = 0$, we have $x - y = 0$, i.e. $x = y$, proving that U is injective. Along with surjectivity, that makes U invertible. With this, we can also show that U^{-1} is bounded: for any $y \in \mathcal{H}'$, there exists an $x \in \mathcal{H}$ such that $U(x) = y$. Thus:

$$\|U^{-1}(y)\| = \|x\| \stackrel{!}{=} \|U(x)\| = \|y\|$$

where $\stackrel{!}{=}$ comes from U being an isometry, showing that $\|U^{-1}(y)\| \leq 1 \cdot \|y\|$, making it also bounded.

Finally, to show that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$, we will use part (a):

$$\begin{aligned}
\langle x, y\rangle &= \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2) \\
&\overset{!}{=} \frac{1}{4}(\|U(x+y)\|^2 - \|U(x-y)\|^2 + i\|U(x+iy)\|^2 - i\|U(x-iy)\|^2) \\
&= \frac{1}{4}(\|U(x)+U(y)\|^2 - \|U(x)-U(y)\|^2 + i\|U(x)+iU(y))\|^2 - i\|U(x)-iU(y)\|^2) \\
&= \langle U(x), U(y)\rangle
\end{aligned}$$

where $\overset{!}{=}$ comes from U being an isometry. Thus:

$$\langle Ux, Uy\rangle = \langle x, y\rangle$$

As we sought to show. □

Proposition 11.1.8: Inner Product is Continuous

Let V be a Hilbert space. Then $\langle -, -\rangle : V \times V \to \mathbb{R}$ (or $\mathbb{C}$) is continuous

***Proof*:**
As $V \times V$ has the product topology, it is enough to show that if $x_n \to x$ and $y_n \to y$ in V then

$$\langle x_n, y_n\rangle \to \langle x, y\rangle$$

Take:

$$\begin{aligned}
|\langle x, y\rangle - \langle x_n, y_n\rangle| &= |\langle x, y\rangle - \langle x_n, y\rangle + \langle x_n, y\rangle - \langle x_n, y_n\rangle| \\
&= |\langle x - x_n, y\rangle + \langle x_n, y - y_n\rangle| \\
&\leq |\langle x - x_n, y\rangle| + |\langle x_n, y - y_n\rangle| \qquad \leq \|x - x_n\|\|y\| + \|x_n\|\|y - y_n\| \\
&\to 0
\end{aligned}$$

where the last line comes since $x_n \to x$ and $y_n \to y$, completing the proof.

The next concept we consider is a way of defining two vectors being somehow "unrelated" or going in some notion of "direction" that doesn't give information for the other vector:

Definition 11.1.9: Orthogonal

We say that x is *orthogonal* to y if $\langle x, y\rangle = 0$. We will write $x \perp y$

If two vectors are orthogonal, the triangle inequality becomes instant:

Theorem 11.1.10: Pythagoras

$$x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$$

***Proof*:**

$$\|x+y\|^2 = \langle x+y, x+y\rangle = \|x\|^2 + \|y\|^2$$

We might wonder if we take $E \subseteq H$, can we find all vectors in H that are somehow "dual" to E? Define it's *orthogonal compliment* to be:

$$E^\perp = \{x \in H \mid \langle x, y\rangle = 0 \ \forall y \in E\}$$

Notice that E is *always* a closed subspace:

Proof. The fact it's a subspace follows from linearity of the second component.t

for closed, suppose there is a sequecne $\{x_n\} \subseteq E^\perp$ where $x_n \to x \in H$. Then:

$$|\langle x, y\rangle| \leq |\langle x - x_n, y\rangle| \leq \|x - x_n\|\|y\| \to 0 \quad \text{as } n \to \infty$$

□

We can also find closed (and not closed) subspace of hilbert spaces:

Example 11.1.11: Subspaces Examples

1. If $\mathcal{H}$ is a finite-dimensional Hilbert space, then all subspace are closed
2. If $\mathcal{H}$ is infinite-dimensional, consider $\mathcal{H} = L^1([0,1])$ with the L^2 inner product. and take $P([0,1]) \subseteq L^1([0,1])$ to be the subspace of all polynomial functions. This is is certainly not equal to $L^1([0,1])$, however polynomial functions are dense in $L^1([0,1])$ (they can arbitraily approximate continuous bump characteristic functions, and hence simple functions). However, their closure is $L^1([0,1])$, showing that $P([0,1])$ cannot be closed.
3. Assume $E \subseteq \mathcal{H}$ is an open subspace. Then since $0 \in E$, there exists a $r > 0$ such that

   $$B_r(0) \subseteq E$$

   However, for any nonzero element $x \in \mathcal{H}$,

   $$\frac{r}{2\|x\|}x \in B_r(0) \subseteq E$$

   Since E is closed under scalar multiplication, this shows that $x \in E$, so $\mathcal{H} \subseteq E$, so $\mathcal{H} = E$, showing that the only open subspace of $\mathcal{H}$ is $\mathcal{H}$ itself.

Thus, we are interested in closed subspace since are closed under limits, meaning we can use our tools from analysis to get the following decomposition result:

Theorem 11.1.12: Hilbert Space Decomposition

Let $E \subseteq \mathcal{H}$ be a closed subspace. Then:

$$\mathcal{H} = E \oplus E^\perp$$

In particular, for any $x \in H$, there is a unique $y \in E$ and $z \in E^\perp$ such that $x = y + z$.

***Proof*:**
Let $x \in \mathcal{H}$, and let $\delta = \inf\{\|x - y\| = \rho(x, y) \mid y \in E\}$ (which exists by completeness of $\mathbb{R}^n$ and $\mathcal{H}$). let $\{y_n\} \subseteq E$ be a sequence such that $\|x - y_n\| \to \delta$. We will show that $\{y_n\}$ is a cauchy sequence, and then find a limit point for it. By the parralelogram law:

$$2(\|x - y_n\| + \|x - y_m\|) = \|y_n - y_m\|^2 + \|(y_n + y_n) - 2x\|^2$$

Since $y_n + y_m \in E$, for sufficinetly large N so that $n, m > N$:

$$\begin{aligned}\|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\|\frac{1}{2}(y_n + y_n) - x\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2\end{aligned}$$

and as $n, m \to \infty$, this sequence goes to $4\delta^2 - 4\delta^2 = 0$, and so $\|y_n - y_m\|\rho(y_n, y_m) \to 0$, meaning $\{y_i\}_i$ is a cauchy sequence. Take $y \lim y_n$. Then since E is closed, $y \in E$. By construction $\|x - y\| = \delta$. Now let $z = x - y$.

I claim that $z \in E^\perp$, that is $\langle z, u\rangle = 0$ for all $u \in E$. So, consider $\langle u, z\rangle$ for any $u \in E$. If $\langle u, z\rangle$ is complex, multiply it by a [complex] unit-length scalar to make it real ($\langle u, z\rangle = 0$ if and only if $a\langle u, z\rangle = 0$). With the fact that it's real, we will do a sneaky trick where we take advantage of optimization of differentiable functions. In particular, let

$$f : \mathbb{R} \to \mathbb{R} \quad f(t) = \|z + tu\|^2 = \|z\|^2 + 2t\langle z, u\rangle + \|u\|^2$$

f is certainly a differentiable function, being a polynomial with a square-root on $\mathbb{R}$. Furthermore, $z = x - y$ is the minimal value of $\|x - y\|$, and so the minimal value of $\|x - y\|^2$. It follows that $z + ut$ is not minimal, and so this equation is has a minimum when $t = 0$. Thus, we get $0 = f'(0) = 2\langle u, z\rangle$, implying that $\langle u, z\rangle = 0$. Since u was arbitrary, we get that $z \in E^\perp$.

To show that z is the minimal distance from x, let's say there was another $z' \in E^\perp$ that satisfied the equation. Then by the Pythagorean theorem:

$$\begin{aligned}\|x - z'\| &= \|(x - z) + (z - z')\|^2 \\ &= \|x - z\|^2 + \|z - z'\|^2 && x - y \in E, z - z' \in E^\perp \\ &\geq \|x - z\|^2\end{aligned}$$

So equality holds if and only if $z = z'$, meaning z is of minimal distance to x. A similar reasoning shows the same for y.

To show uniqueness, let's say $y + z = x = y' + z'$. Then $y - y' = z - z' \in E \cap E^\perp = \{0\}$, so $y - y' = z - z' = 0$, which can be used to show that $y = y'$ and $z = z'$, completing the proof.

We said that orthogonal vector are somehow dual to each other. However, there is the more classical notion of "dual" vector spaces. Let $\mathcal{H}^*$ be set of all linear functions from $\mathcal{H}$ to the underlying field (usually $\mathbb{C}$, but sometimes $\mathbb{R}$); such linear functions are called functionals. It turns out that all linear functionals can be written in the form $f_y(x) = \langle x, y\rangle$, and we can define a norm on $\mathbb{H}^*$ such that $\|f_y\| = \|y\|$. Thus, $y \mapsto f_y$ will define a conjugate-linear isometry from $\mathcal{H}$ to $\mathcal{H}^*$. The following theorem shows tha this map is surjective:

Theorem 11.1.13: Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)

Let $f \in \mathcal{H}^*$. Then there exists a $y \in \mathcal{H}$ such that $f(x) = \langle x, y\rangle$ for all $x \in X$.

***Proof*:**
It easy easy to check uniqueness: if $\langle x, y\rangle = \langle x, y'\rangle$ for all x, then let $x = y - y'$. Manipulating the equation, we get that $\|y - y'\| = 0$, so $y = y'$.

If f is the zero functional, then take $y = 0$. If otherwise, then let $\mathcal{M} = \{x \mid f(x) = 0\}$ be the kernel of f. It is clearly a subspace, and any squence will also necesarily converge within $\mathcal{M}$, and so it is also closed. Furthermore, $\mathcal{M}$ is a proper subspace, since $f \neq 0$. Thus, $\mathcal{M}^\perp \neq \{0\}$ by the previous proof.

Now, pick any $z \in \mathcal{M}^\perp$ with $\|z\| = 1$. Then if we set $u = f(x)z - f(z)x$, we have that $u \in \mathcal{M}$. Hence:

$$0 = \langle u, z\rangle = f(x)\|z\|^2 - f(z)\langle x, z\rangle = f(x) - \langle x, \overline{f(z)}z\rangle$$

re-arrangine and setting $y = \overline{f(z)}z$, we get

$$f(x) = \langle y, z\rangle$$

as we sought to show.

The uniqueness makes this map to be injective, and by linearity of f and the first component of the inner product makes it is an isomorphism. Furthermore, since $\|f_y\| = \|y\|$, limits are also preserved, and so this in fact defines an isomorphism between $\mathcal{H} \cong \mathcal{H}^*$. Notice how this differs from vector-spaces where there is no natural isomorphism between V and V^* (there is one between V and V^{**}).

Since we are working over a vector-space, we should be able to define a basis. Since we are studying infinite dimensional vector-spaces, we often care more about "countable linear combination" rather than "finite linear combination" as we have for vector-spaces. To distinguish the two, we will call *Hamel basis* a set of linearly independent elements for which all *finite linear combinations* can represent every element of the vector space. We will call a *Schauder basis* a set of linearly independent elements for which all *countable inear combinations* can represent every element of the vector space. If every element in a Schauder basis has norm 1 and is orthogonal, we call it an *orthonormal basis.* Since we have an inner-product that is complete, it would be nicest if this vector-space has an orthonormal basis. We strive to this end in the following build-up:

Theorem 11.1.14: Bessel's Inequality

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$:

$$\sum_{\alpha \in A} |\langle x, u_\alpha\rangle|^2 \leq \|x\|^2$$

As a consequence, $\{\alpha \mid \langle x, u_\alpha\rangle \neq 0\}$ is countable.

***Proof*:**
It suffices to show that $\sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 \le \|x\|^2$ for any finite subset $F\subseteq A$, since if it fails for some countable amount, it will fail for some finite amount. Then we see that:

$$\begin{aligned}
0 &\le \left\| x - \sum_{\alpha\in F}\langle x,u_\alpha\rangle u_\alpha \right\| \\
&= \|x\|^2 - 2\mathrm{Re}\left\langle x, \sum_{\alpha\in F}\langle x,u_\alpha\rangle u_\alpha \right\rangle + \left\| \sum_{\alpha\in F}\langle x,u_\alpha\rangle u_\alpha \right\| \\
&= \|x\|^2 - 2\sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 + \sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 && \text{Pythagorean Theorem} \\
&= \|x\|^2 - \sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2
\end{aligned}$$

and manipulating the final equation we get:

$$\sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 \le \|x\|^2$$

as we sought to show

This in particular is important for uncountable sets. In the following we establish when equality holds, and thus when an orthonormal set is an orthonormal basis

Theorem 11.1.15: Orthonormal Basis Condition

Let $\{u_\alpha\}_{\alpha\in A}$ be an orthonormal set in $\mathcal{H}$. Then the following are equivalent:

1. Finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$.
2. (Completeness) If $\langle x,u_\alpha\rangle = 0$ for all α, then $x=0$
3. For each $x\in\mathcal{H}$, $x=\sum_{\alpha\in A}\langle x,u_\alpha\rangle u_\alpha$ where the right hand side sum has only countably many nonzero terms and converges in the norm topology no matter the order of the terms
4. (Parseval's Identity) $\|x\|^2 = \sum_{\alpha\in A}|\langle x,u_\alpha\rangle|^2$ for all $x\in\mathcal{H}$

***Proof*:**

(1) ⇒ (2) Assume $\langle f,u_\alpha\rangle = 0$. As finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$, for any $f\in\mathcal{H}$ there is a sequence g_n such that $\|f-g_n\| \xrightarrow{n\to\infty} 0$. As $\langle f,u_\alpha\rangle = 0$, then $\langle f,-g_n\rangle = 0$. By bi-linearity and Cauchy-Schwarz:

$$\|f\|^2 = \langle f,f\rangle = \langle f,f-g_n\rangle \le \|f\|\|f-g_n\| \xrightarrow{x\to\infty} 0 \tag{11.1}$$

hence $f=0$

(2) ⇒ (3) As the nonzero portion of the sum is countable, we can reduce to the countable

case. For any $f \in \mathcal{H}$, let:

$$S_n(f) = \sum_{k=1}^{n} a_k u_k \qquad a_k = \langle f, u_k \rangle$$

Let's first show $S_n(f) \to g$ for some $g \in \mathcal{H}$, then show $g = f$. First, we show an alternative proof of Bessel's identity with a satisfying trick. Notice that certainly $S_n(f) \perp (f - S_n(f))$. Thus:

$$\|f\|^2 = \|S_n(f)\|^2 + \|f - S_n(f)\|^2 = \sum_{k=1}^{n} a_k u_k + \|f\|^2$$

Taking $n \to \infty$ we get:

$$\|f\| \geq \sum_{k=1}^{\infty} a_k u_k$$

Importantly, $\sum_{k=1}^{\infty}$ *converges*, and hence $S_n(f)$ forms a Cauchy sequence in $\mathcal{H}$, so by completeness $S_n(f) \to g$. To show $f = g$, notice that for any k, for sufficiently large n:

$$\langle f - S_n(f), u_k \rangle = a_k - a_k = 0$$

Thus:

$$\langle f - g, u_k \rangle = 0 \qquad \forall k$$

so by (2), $f - g = 0$, so $f = g$.

(3) ⇒ (4) Assuming (3) so that $= \sum_k^{\infty} \langle f, u_k \rangle$, then $\|f - S_n(f)\| \to 0$ so plugging this into equation (11.1):

$$\|f\|^2 = \sum_{k}^{\infty} |a_k|^2$$

(4) ⇒ (1) Notice that $S_n(f)$ is a finite linear combinations elements u_k, hence using equation (11.1) again we get the desired implication.

Any orthonormal set satisfying these conditions is given a name:

Definition 11.1.16: Orthonormal Basis

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set. Then if the set satisfies any of the 3 conditions in theorem 11.1.15, it is called an *orthonormal basis.*

An argument similar to the one for vector-spaces gives that every Hilbert spaces has an orthonormal basis, essentially an orthonormal basis is an orthonormal set that is maximal:

Theorem 11.1.17: Hilbert Space Basis

Every Hilbert space has an orthonormal basis.

***Proof*:**
Application of Zorn's lemma: The collection of ordered orthonormal sets ordered by inclusion has a maximal element, and maximality is equivalent to the first property of theorem 11.1.15.

Example 11.1.18: Orthonormal Basis
Given a Hilbert space $L^2(G)$ where $G = \mathbb{R}^n$ or $\mathbb{T}^n$, we would want to find an orthonormal basis to get decomposition of functions. go over all of these!

1. Legendre basis (can be thought of as "global" Taylor approximation) (must add integrate image. Also a good example of orthonormal basis that is not the Fourier one)

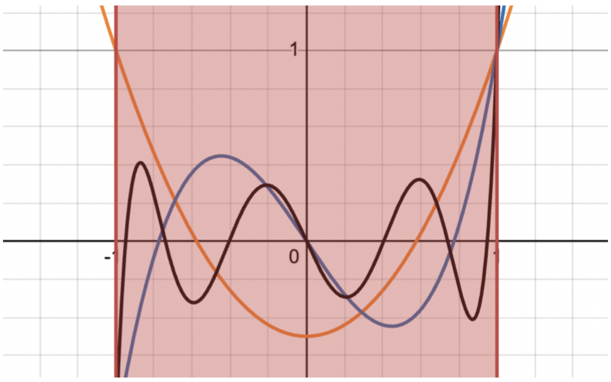


Figure 11.1: Legendre polynomials integrate to 0

2. Hermite basis
3. Jacobi Basis
4. Chebyshev

11.1.19 Basis On Banach Space Without an inner product, we cannot define an orthonormal basis on L^p. We can ask if we can at least define a *Schauder basis*, i..e any $f \in L^p$ is a (potentially) linear of a *single* sequence $\{b_n\}$ with different weights:

$$f = \sum a_n b_n$$

This was actually a very difficult problem and is related to the following question: is every compact operator on a Banach space the limit of finite operators. It isn't too hard to check this is the case in Hilbert spaces (exercise ref:HERE). However, this was proven in the negative by Per Enflo [9] (for a refinement of the proof see [11]). A consequence of this is that not every Banach space has a Schauder basis [2]. Another consequence of this paper for Banach spaces is that not all Banach space embed into $\ell^p(\mathbb{Z})$, $L^p(\mathbb{R}^d)$, or $L^p(K)$ for a compact set K; the family of Banach spaces is in fact larger (ex. $C(K)$ Is a Banach space not belonging to any of these spaces) .

Now that we have established a basis for Hilbert spaces, we will separate them into 3 categories: Those with finite, countable, and uncountable basis. In most cases, we will be working with countable basis due to the following property:

Proposition 11.1.20: Separable Hilbert Spaces

A Hilbert space $\mathcal{H}$ is separable if and only if it has a countable orthonormal basis, in which case every orthonormal basis for $\mathcal{H}$ is countable.

***Proof*:**
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Though all separable Hilbert spaces are "equivalent" (isometric), it is often the specific manifestation of the Hilbert space that matters as it frames how operators on the space transform functions (ex. the choice of basis, the choice of inner product).

Theorem 11.1.21: Enflo's Rigidity Theorem

A Banach space is uniformly homeomorphic to a Hilbert space if and only if it is linearly isomorphic to a Hilbert space.

***Proof*:**
This was a seminal result by Per Enflo (1969) [10]. It settled the question of whether the "uniform" structure of a space determines its linear structure in the context of Hilbert spaces. Since L^p (for $p \neq 2$) is not isomorphic to L^2, this theorem implies they are not uniformly homeomorphic either.

11.1.22 Hierarchy of Sameness It is useful to summarize how "same" two infinite-dimensional separable spaces can be. Let X and Y be infinite-dimensional separable Banach spaces (e.g., L^p and L^2).

1. Topological Homeomorphism : They are homeomorphic by the *Anderson-Kadec Theorem*, i.e. all such spaces are homeomorphic to $\mathbb{R}^{\mathbb{N}}$. They are topologically indistinguishable.
2. Uniform Homeomorphism: Generally no. By *Enflo's Rigidity Theorem*, if X is a Hilbert space and Y is not linearly isomorphic to it (like $L^p, p \neq 2$), they are *not* uniformly homeomorphic. *Note:* Their unit balls *are* uniformly homeomorphic via the **Mazur Map**, but this does not extend to the whole space.
3. Linear Isomorphism (The "Algebraic" level): Is there a linear bijection? Depends. L^p is not isomorphic to L^q for $p \neq q$.
4. Isometric Isomorphism (The "Geometry" level): They don't they preserve distance. The geometry of the unit ball (round vs. diamond vs. box) is strictly preserved here.

We next talk about the isomorphisms in the category of Hilbert spaces:

Definition 11.1.23: Unitary Map

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces with inner products $\langle\cdot,\cdot\rangle_1$ and $\langle\cdot,\cdot\rangle_2$, and let $U : \mathcal{H}_1 \to \mathcal{H}_2$. Then U is called an *isometry* if if it is a linear map that preserves inner products:

$$\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$$

If it is also bijective, it is a *unitary map*.

Note that by taking $x = y$, we get that $\|x\|_1 = \|Ux\|_2$, i.e. it is an isometry. In fact, the converse is true too: every surjective isometry is unitary. With the definition of an isomorphism, we can see that all Hilbert spaces are isomorphic to $\ell^2(A)$ given basis A:

Theorem 11.1.24: Classifying Hilbert Spaces

Let $\{e_\alpha\}_{\alpha\in A}$ be an orthonormal basis for $\mathcal{H}$. Then the correspondence

$$x \mapsto \hat{x} \qquad \hat{x}(\alpha) = \langle x, u_\alpha\rangle$$

is a unitary map from $\mathcal{H}$ to $\ell^2(A)$.

***Proof*:**
We'll show it's a surjective isometry, which is equivalent to it being unitary.

First off, the map $x \to \hat{x}$ is linear since the inner product is linear in each component, and it is an isometry by Parseval identity: $\|x\|^2 = \sum |\hat{x}(x)|^2$.

For surjectivity, let $f \in \ell^2(A)$, so $\sum_{\alpha\in A} |f(\alpha)^2| < \infty$, so the Pythagorean shows that all partial sums of the series $\sum f(\alpha)u_\alpha$ are Cauchy [a], so $x = \sum_\alpha f(\alpha)u_\alpha$ exists in $\mathcal{H}$ and $\hat{x} = f$.

[a] Note that only countably many terms are nonzero

As an application that ties together several threads from previous chapters, we close off with the proof a theorem of Grothendieck showing that the Hilbert space structure of L^2 imposes rigid constraints on subspaces of L^p. The result says, roughly, that a closed subspace of L^p cannot consist entirely of bounded functions unless it is finite-dimensional. The proof combines the closed graph theorem from the Baire category section (theorem 7.2.10), the L^p inclusion theory (proposition 8.1.13), and the Bessel inequality developed above.

Theorem 11.1.25: Grothendieck's Theorem on Closed Subspaces of L^p

Let $(X, \mathcal{F}, \mu)$ be a finite measure space, i.e. $\mu(X) < \infty$. Suppose that E is a closed subspace of $L^p(X, \mu)$ for some $1 \le p < \infty$, and that $E \subseteq L^\infty(X, \mu)$. Then E is finite-dimensional.

***Proof*:**
Step 1: The L^∞ norm is controlled by the L^p norm on E. Consider the inclusion $\iota : (E, \|\cdot\|_p) \to L^\infty(X, \mu)$ defined by $\iota(f) = f$. We verify that its graph is closed. Suppose $f_n \to f$ in L^p with $f_n \in E$, and simultaneously $f_n \to g$ in L^∞. Since E is closed in L^p, $f \in E$. Convergence in L^p gives $f_{n_k} \to f$ a.e. along a subsequence, while convergence in L^∞ gives $f_n \to g$ a.e. Hence $f = g$

a.e., showing the graph is closed. By the closed graph theorem (theorem 7.2.10), ι is bounded: there exists $C > 0$ such that:

$$\|f\|_\infty \leq C\|f\|_p \qquad \text{for all } f \in E$$

Step 2: E carries a Hilbert space structure. Since $\mu(X) < \infty$ and $E \subseteq L^\infty$, the L^2 norm is finite on E. We show the L^p and L^2 norms are equivalent on E.

For one direction: $\|f\|_2^2 = \int |f|^2 \leq \|f\|_\infty^2 \mu(X) \leq C^2\|f\|_p^2\mu(X)$, giving $\|f\|_2 \leq C\mu(X)^{1/2}\|f\|_p$.

For the reverse: if $p \leq 2$, then Hölder directly gives

$$\|f\|_p \leq \mu(X)^{1/p-1/2}\|f\|_2$$

If $p > 2$, then interpolate:

$$\|f\|_p^p \leq \|f\|_\infty^{p-2}\|f\|_2^2 \leq (C\|f\|_p)^{p-2}\|f\|_2^2$$

so that $\|f\|_p^2 \leq C^{p-2}\|f\|_2^2$, i.e. $\|f\|_p \leq C^{(p-2)/2}\|f\|_2$.

In either case, the L^p and L^2 norms are equivalent on E. Since E is complete in $\|\cdot\|_p$, it is complete in $\|\cdot\|_2$, so $(E, \langle\cdot,\cdot\rangle_{L^2})$ is a Hilbert space. Write C' for a constant satisfying $\|f\|_\infty \leq C'\|f\|_2$ for all $f \in E$ (obtained by combining Step 1 with the norm equivalence).

Step 3: Bounding the dimension via Bessel's inequality. Let $\{e_1, \ldots, e_N\}$ be any orthonormal set in $(E, \|\cdot\|_2)$. Note that N is independent of the choice of C'; this will be the key fact to bound $\dim(E)$. For almost every $x \in X$, the evaluation functional $\varphi_x : E \to \mathbb{C}$ defined by $\varphi_x(f) = f(x)$ satisfies:

$$|\varphi_x(f)| \leq \|f\|_\infty \leq C'\|f\|_2$$

so φ_x is a bounded linear functional on the Hilbert space E with $\|\varphi_x\| \leq C'$. By the Riesz representation theorem (theorem 11.1.13), there exists $K_x \in E$ with $\varphi_x(f) = \langle f, K_x\rangle$ and $\|K_x\|_2 = \|\varphi_x\| \leq C'$. In particular,

$$\varphi_x(e_n) = e_n(x) = \langle e_n, K_x\rangle$$

for each n. Applying Bessel's inequality (theorem 11.1.14) to K_x:

$$\sum_{n=1}^{N} |e_n(x)|^2 = \sum_{n=1}^{N} |\langle K_x, e_n\rangle|^2 \leq \|K_x\|_2^2 \leq C'^2$$

Integrating over X and using $\|e_n\|_2 = 1$:

$$N = \sum_{n=1}^{N} \|e_n\|_2^2 = \int_X \sum_{n=1}^{N} |e_n(x)|^2 \, d\mu \leq C'^2\mu(X)$$

Since N was the size of an arbitrary orthonormal set, $\dim(E) \leq C'^2\mu(X) < \infty$.

The element K_x appearing in the proof is a *reproducing kernel*: it allows one to recover pointwise values of functions in E via the inner product. The argument shows that the existence of such a kernel, together with the finiteness of the measure, forces the space to be finite-dimensional. The bound $\dim(E) \leq C'^2\mu(X)$ is quantitative: the larger the ratio $\|f\|_\infty/\|f\|_2$ can be on E, the fewer orthonormal directions the space can accommodate.

It is worth pausing to appreciate how three seemingly unrelated tools conspire here. The closed graph theorem (theorem 7.2.10) upgrades the set-theoretic inclusion $E \subseteq L^\infty$ into a norm inequality. The L^p interpolation inequalities (proposition 8.1.13) transfer this to equivalence of all L^q norms on E, revealing a hidden Hilbert space. And Bessel's inequality (theorem 11.1.14) converts the uniform pointwise bound into a dimension bound via integration.

11.2 Operators on Hilbert Spaces

In this section, we try to generalize many notion's we've seen before, including:

1. adjoints, self-adjoint, and normal operators
2. sign and absolute value and when can we decompose a function into this form
3. diagonalizable
4. polar decomposition (think of how for every $z \in \mathbb{C}$, $z = \arg(z)|z|$)

Adjoints

Hilbert spaces have very simple dual structure, since $\mathbb{H} \cong \mathbb{H}^*$ in a very natural way with the inner product of $\mathbb{H}$. Thus, the problem of finding the structure of dual spaces is fully finished for hilbert spaces. Due to this very simple dual structure, we will be able to properly define an "opposite" operator given any operator $T : \mathbb{H}_1 \to \mathbb{H}_2$. In essense, there is an interesting symmetry in the structure of any map: Given a map $T : \mathbb{H}_1 \to \mathbb{H}_2$ between two hilbert spaces, there is a natural dual map $T^* : \mathbb{H}_2^* \to \mathbb{H}_1^*$. Hence, we have

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2^*, \mathbb{H}_1^*)$$

and since in Hilbert spaces $\mathbb{H}_i \cong \mathbb{H}_i^*$, we may envision this map as "flipping" the direction of arrows in a natural way:

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2, \mathbb{H}_1)$$

Not only does this map exist, but it is an isometric isomrophism (in **Ban**)! We will start with the special case of $\mathrm{Hom}(\mathbb{H}, k) = \mathbb{H}^*$. We will find that $(-)^*$ is a map

$$(-)^* : \mathrm{Hom}(\mathbb{H}, \mathbb{H}) \to \mathrm{Hom}(\mathbb{H}, \mathbb{H})$$

The following shows how we can define such a dual and some consequences:

Lemma 11.2.1: Endomorphism and And Inner Products

Let $\mathbb{H}$ be a complex hilbert space. Then there is a bijective isometric correspondence between $\mathrm{End}_k(\mathbb{H})$ and bounded sesquilinear forms on $\mathbb{H}$ given by tne map $T \to B_T$ where

$$B_T(x, y) = \langle x, T(y) \rangle$$

***Proof*:**
Let $T \in \mathrm{End}_k(\mathbb{H})$. Then certainly B_T is a sequilinear form on $\mathbb{H}$ bounded by $\|T\|$ since

$$\begin{aligned}\|B_T\| &= \sup\{|B_T(x,y)| \mid \|x\| \leq 1,\ \|y\| \leq 1\}\\ &= \sup\{|\langle x, T(y)\rangle| \mid \|x\| \leq 1, \|y\| \leq 1\}\\ &\leq \|T\|\end{aligned}$$

On the other hand

$$\begin{aligned}\|T(x)\|^2 &= \langle T(x), T(x)\rangle\\ &= B_T(T(x), x)\\ &\leq \|B_T\|\|T(x)\|\|x\|\\ &\leq \|B_T\|\|T\|\|x\|^2\end{aligned}$$

showing that $\|T\| \leq \|B_T\|$, and so $\|T\| = \|B_T\|$.

Theorem 11.2.2: Existence Of Adjoint

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then there exists a unique T^* in $\mathrm{End}_k(\mathbb{H})$ such that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

for all $x, y \in \mathbb{H}$. The map $T \mapsto T^*$ is conjugate linear, contravariant with respect to multiplication, an isometry of $\mathrm{End}_k(\mathbb{H})$ onto itself, an involution, and satisfies

$$\|T^*T\| = \|T\|^2$$

***Proof*:**
Fix $T \in \mathrm{End}_k(\mathbb{H})$, and let $B_T(x,y)$ be a continuous sesquilinear form defined as $(x,y) \mapsto \langle T(x), y\rangle$. Then by lemma 11.2.1, there exists a $T^* \in \mathrm{End}_k(\mathbb{H})$ such that $T^* \to B_T$. In particular

$$B_T(x,y) = \langle x, T^*(y)\rangle$$

Thus,

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

giving us existence of such an operator. We now establish all the properties named in the theorem. By conjugate symmetry

$$\begin{aligned}\langle x, T^{**}(y)\rangle &= \langle T^*(x), y\rangle\\ &= \overline{\langle y, T^*(x)\rangle}\\ &= \overline{\langle T(y), x\rangle}\\ &= \langle x, T(y)\rangle\end{aligned}$$

Thus, T^{**} and T correspond to the same sesquilienar form, thus $T^{**} = T$. This also shows that $T \mapsto T^*$ is an involution. Furthermore:

$$\langle x(ST)^*y\rangle = \langle S(T(x)), y\rangle = \langle T(x), S^*(y)\rangle = \langle x, T^*(S^*(y))\rangle$$

thus $(S \circ T)^* = T^* \circ S^*$ showing $(-)^*$ is a contravariant endofunctor. Since the operator norm is submiltiplicative, we have $\|T^*T\| \leq \|T\|\|T^*\|$. On the other hand, applying $\angle T(x), y\rangle = \langle x, T^*(y)\rangle$ to T^* we get

$$\|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle T^*(T(x)), x\rangle = \|T^*T\|\|x\|^2$$

showing that $\|T\|^2 \leq \|T^*T$
. Combining these we get $\|T\| \leq \|T^*\|$, hence $\|T\| = \|T^*\|$ (since $T^{**} = T$). This finishes the proof.

Note that if $\mathbb{H}$ is a finite dimensional complex Hilbert space, then each operator T has a matrix representation, say (α_{ij})). Then $T^* = (\overline{\alpha_{ji}})$, i.e. $T^* = \overline{T^t}$ where T^t is the transpose given the matrix representation.

Definition 11.2.3: Adjoint And Self-Adjoint

Let $T \in B(\mathbb{H})$ be a continuous linear operator between Hilbert spaces. Then T^* is called the *adjoint* of T. If $T = T^*$, then T is said to be *self adjoint* (or *Hermitian*).

Proposition 11.2.4: Adjoint Properties I

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then

$$\ker(T^*) = (T(\mathbb{H}))^\perp$$

***Proof*:**
Since $\langle T(x), y\rangle = \langle x, T^*(y)\rangle$ we have $y \in \ker(T^*)$ when $y \in (T(\mathbb{H}))^\perp$. Conversely, if $y \in (T(\mathbb{H}))^\perp$, then $T^*(y) \in \mathbb{H}^\perp = \{0\}$.

For ease of reference, we shall list some of the results we have proven in the theorem:

Proposition 11.2.5: Adjoint Properties II

Let $T, S \in B(\mathbb{H})$. Then:

1. $(T + S)^* = T^* + S^*$
2. $(cT)^* = \overline{c}T^*$
3. $(TS)^* = S^*T^*$
4. $(T^*)^* = T$
5. $I^* = I$

***Proof*:**
If you're not convinced of any of these, double check the theorem or try proving it again.

Proposition 11.2.6: Adjoint Properties III

Let $T \in \text{End}_k(\mathbb{H})$. Then the following are equivalent:

1. T is invertible (so $T^{-1} \in \text{End}_k(\mathbb{H})$)
2. T^* is invertible
3. T and T^* are bounded away from zero.
4. T and T^* are injective, and $T(\mathbb{H})$ is closed
5. T is injective and $T(\mathbb{H}) = \mathbb{H}$

***Proof*:**
Since T is invertible, $T^{-1}T = TT^{-1} = I$, and so $(T^{-1})^*$ is the inverse of T^*, giving (1) $\Leftrightarrow$ (2).

For (1) $\Rightarrow$ (3), we say that an operator F is bounded away from zero if

$$\|x\| \leq C\|F(x)\|$$

for some constant C. In our case:

$$\|x\| = \|T^{-1} \circ T(x)\| \leq \|T^{-1}\|\|T(x)\|$$

Since (1) $\Leftrightarrow$ (2), the same holds for T^*.

For (3) $\Rightarrow$ (4), We have $\|T(x)\| \geq \epsilon\|x\|$ and $\|T^*(x)\| \geq \epsilon\|x\|$ for some $\epsilon > 0$ and all $x \in \mathbb{H}$. Thus certainly both T and T^* are injective: if $T(x) = T(y)$ then

$$0 = \|T(x) - T(y)\| = \|T(x - y)\| \geq \epsilon\|x - y\|$$

hence $\|x - y\| = 0$, so $x = y$. Furthermore, $\|T(x) - T(y)\| \geq \|x - y\|$ imples that $T(\mathbb{H})$ is complete, hence a closed subspace of $\mathbb{H}$.

For (4) $\Rightarrow$ (5), by corollary **??** ($\overline{Y} = (Y^\perp)^\perp$) and proposition 11.2.4 we get

$$\overline{T(\mathbb{H})} = (T(\mathbb{H})^\perp)^\perp = (\ker(T^*))^\perp = \{0\}^\perp = \mathbb{H}$$

where the last equality come from T^* bein injective.

Finally, for (5) $\rightarrow$ (1), we use the open mapping theorem.

This allows us to generlize this result to the followin

Proposition 11.2.7: Inversing Maps

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be hilbert spaces and consider an operator $T \in \mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$. Then there exists a unique operator $T^* \in B(\mathbb{H}_2, \mathbb{H}_1)$ such that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

show that the map $T \to T^*$ is conjugate linear isometry of $\mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$ to $\mathrm{Hom}_k(\mathbb{H}_2, \mathbb{H}_1)$ and satisfies

$$\|T^*T\| = \|T^2 = \|TT^*\|$$

***Proof*:**
Let

$$\hat{T} = \begin{pmatrix} 0 & 0 \\ T & 0 \end{pmatrix} \in \mathrm{End}_k(\mathbb{H}_1 \oplus \mathbb{H}_2)$$

By theorem 11.2.2, there is a unique function $\hat{T}^*$ such that

$$\langle \widehat{T}(x_1, x_2), (y_1, y_2)\rangle = \langle (x_1, x_2), \widehat{T}^*(y_1, y_2)\rangle$$

By definition of $\widehat{T}$, we see

$$\widehat{T}^* = \begin{pmatrix} 0 & T^* \\ 0 & 0 \end{pmatrix}$$

Hence we have that T^* is a function with domain and codomain $T^* : \mathbb{H}_2 \to \mathbb{H}_1$. It follows that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

and by theorem 11.2.2, T and T^* respect all the desired properties, completing the proof.

For the rest of this seciton, we will focus on $T \in B(\mathbb{H})$ insead of $T \in B(\mathbb{H}_1, \mathbb{H}_2)$. The reason for this is to find suitable decompositino theorems of T generlizing the notion of eigenvalues.

Recall that over finite dimensional vector spaces, $TT^* = T^*T$ implied T was diagonalizable, with the usual special case of $T = T^*$ (self-adjoint) being looked into. We would like to know if this notion of diagonalizability for normal operators remains true for infinite dimensional Hilbert spaces. We shall see that this is not the case, some more conditions is required (for ex. a special case of digaonalizable operators will be comapct normal operators). Before we get into the technicalities, I want to put a word on how I think about $TT^* = T^*T$. In my mind, when two functions commute, it means that one function is somehow "invariant" with respect to the other. Think of PDE's with equations that commute with the differential operator that makes it invariant to the Lorenzian metric. In this case, I like to think of T being somehow "symmetric". Since $TT^*(x) = T^*T(x)$ for all $x \in \mathbb{H}$, and T^* is in some ways the dual (think of ref:HERE, the generalization), we see that T is "invariant" to the effect of the dual T^*. We shall see in a moment that the invariant to the dual is translated to T and T^* being metrically identical. In the simpleset case of $T = T^*$, it is evident why this is the case. If $T = T^*$, we can deepen our intuition with complex numbers: if we consider $T : \mathbb{C} \to \mathbb{C}$ where $T(z) = z$, then $T^*(z) = \overline{z}$. Hence, $T = T^*$ if and only if $T(\mathbb{C}) \subseteq \mathbb{R}$, since only the real number are invariant under conjugation.

Definition 11.2.8: Normal Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *normal* if it commutes with its adjoint, i.e. if $T^*T = TT^*$

As an immedaite consequence, we have

$$\|T(x)\| = \langle T^* \circ T(x), x\rangle^{1/2} = \langle T \circ T^*(x), x\rangle^{1/2} = \|T^* x\|$$

showing that T and T^* are metrically identical. In fact, by the palalalogram law (also known as polarization identity) if $T \in \text{End}_k(\mathbb{H})$ such that T and T^* ar emetrically identical, we can show $\langle T^* \circ T(x), y\rangle = \langle T \circ T^*(x), y\rangle$ for all x, y and so T is normal by lemma 11.2.1. By proposition 11.2.6, an operator is normal if and only if it is bounded away from zero.

Positive Operators

Definition 11.2.9: Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *positive* if $T = T^*$ and $\langle T(x), x\rangle \geq 0$ for all $x \in \mathbb{H}$. If $k = \mathbb{C}$, the positivey condition implies self-adjointness by the polarization identity.

Certainly, the sum of two positive operators is positive (geometrically, positive operatros form a cone in $\text{End}_k(\mathbb{H})$). However the product is not positive, and need not be self-adjoint.

Example 11.2.10: Positive And Non-Positive Operators

here

However, if the opeartors commute (if the product is self-adjoint), we see that there is a product that is positive since ($ST = S^{1/2}TS^{1/2} \geq 0$ by what we'll show in the next proposition)

Let $B(\mathbb{H})_{sa} \subseteq \text{End}_{\mathbb{R}}(\mathbb{H})$ be the collection of self-adjoint operators. Then this is a closed subspace, this subspace intersected with the conte of positive operators defines an ordder $S \leq T$ if $T - S \geq 0$. By taking the case of 2 dimensional real subspaces and $\text{End}_{\mathbb{R}}(\mathbb{R}^2)$, this order is not total, not even a lattice order.

We require the following lemma to gain some important results (which will be an immedate consequence of the spectral theorem in ref:HERE)

Lemma 11.2.11: Square Root Lemma

If $S \leq T$ in $B(\mathbb{H})_{sa}$, then $A^*SA \leq A^*TA$ for every $A \in \text{End}_k(\mathbb{H})$. If morever $) \leq S$, then $\|S\| \leq \|T\|$

***Proof*:**
Since $S \leq T$, we know that $\langle (T-S)(y), y\rangle \geq 0$ for evey $y \in \mathbb{H}$. Replacing y with $A(x)$ for appropraite $x \in \mathbb{H}$, it follows that $A^*(T-S)A \geq 0$, so $A^*SA \leq A^*TA$.

If $0 \leq S \leq T$, by the Cauchy-Schwarz inequality on the positive sesquilinear form $\langle S(\cdot), \cdot\rangle$, we get that for every pair of unit vectors $x, y \in \mathbb{H}$

$$|\langle S(x), y\rangle|^2 \leq \langle S(x), x\rangle\langle S(y), y\rangle \leq \langle T(x), x\rangle\langle T(y), y\rangle \leq \|T\|^2$$

Hence, $\|S\|^2 \leq \|T\|^2$ by lemma 11.2.1.

Lemma 11.2.12: Approximation Lemma

There is a sequence $(p_n) \subseteq \mathbb{R}[x]$ of polynomials with positive coefficients such that $\sum p_n$ converges uniformally on $[0, 1]$ to

$$t \mapsto 1 - (1-t)^{1/2}$$

***Proof*:**
This is immediate from real analysis, but we can also construct it explicitly.If $q_0 = 0$ and $q_n(t) = \frac{1}{2}(t + q_{n-1}(t)^2)$, then we can see that each coefficent of q_n is positive and that $q_n(t) \leq 1$ for every $t \in [0, 1]$. Morever, since

$$2(q_{n+1} - q_n) = q_n^2 - q_{n-1}^2 = (q_n - q_{n-1})(q_n + q_{n-1})$$

it follows by induction that each polynomial $p_n = q_n - q_{n-1}]$ has positive coefficients. Taking the q_n as elements of $C([0, 1])$, we see they converge pointwise to $[0, 1]$ to a function q such that $2q(t) = t + q(t)^2$. Thus, $q(t) = 1 - (1-t)^{1/2}$. Sicne $q \in C([0, 1])$ and $[0, 1]$ is compact, by the monotone ocnvergence $q_n \nearrow q$ si in fact uniform by Dini's lemma, hence

$$\sum p_n = \lim q_n = q$$

completing the proof.

Proposition 11.2.13: Square-Rooting Positive Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$ be a positive operator. Then there is a unique positive operator $T^{1/2}$ such that $(T^{1/2})^2 = T$. Moreover, $T^{1/2}$ commutes with every operator commuting with T.

***Proof*:**
Analysis Now p.92

Proposition 11.2.14: Invertible Positive Operator

A positive operator T is invertible in $\mathrm{End}_k(\mathbb{H})$ if and only if $T \geq \epsilon I$ for some $\epsilon > 0$. I that case, $T^-1 \geq$ and $T^{1/2}$ is invertib,e and $(T^{-1})^{1/2} = (T^{1/2})^{-1}$. If $T \leq S$, then $S^{-1} \leq T^{-1}$

***Proof*:**
Analayis now p.94

Projection

Let $W \subseteq \mathbb{H}$ be a closed subspace of a hilbert space H. Then we know that $\mathbb{H} = W + W^{\perp}$. In particular, for eah $y \in \mathbb{H}$, we have $y = w + w^{\perp}$, $w \in W$ and $w^{\perp} \in W^{\perp}$. Since

$$\alpha y_1 + \beta y_2 = (\alpha x_1 + \beta x_2) + (\alpha x_1^{\perp} + \beta x_2^{\perp})$$

is the decomposition of a linear combination, it follows that $P : \mathbb{H} \to \mathbb{H}$ given by $P(y) = x$ is an operator in $\mathrm{End}_k(\mathbb{H})$ with $\|P\| \leq 1$. Note P is idempotent ($P^2 = P$), self-adjoint, and positive since

$$\langle P(y_1), y_2\rangle = \langle x_1 < x_2 + x_2^{\perp}\rangle = \langle x_1, x_2\rangle = \langle x_1 + x_1^{\perp}, x_2\rangle = \langle y_1, P(y_2)\rangle$$

and

$$\langle P(y), y\rangle = \langle x, x + x^{\perp}\rangle = \|x\|^2 \geq 0$$

Definition 11.2.15: (Orthogonal) Projection

The map P in the above is called the *(orthogonal) projection.*

If P is a self-adjoint and idempotent in $\mathrm{End}_k(\mathbb{H})$, then

$$W = \{x \in \mathbb{H} \mid P(x) = x\} = P(\mathbb{H})$$

is a closed subsapce of $\mathbb{H}$. If $w^{\perp} \in W^{\perp}$, we have (since $P = P^*$) that

$$\|P(w^{\perp})\|^2 = \langle w^{\perp}, p^2(w^{\perp})\rangle = 0$$

Thus, P is an orthogonal projection of $\mathbb{H}$ onto W. Note that $I - P$ is the projection on $W^{\perp}$.

Projections are some of the east operators we may construct, and can be thought as the "characteristic function" from real analysis. The analogue of simple function is then obtained by decomposing $\mathbb{H}$ into a finite orthogonal sum

$$\mathbb{H} = W_1 \oplus W_2 \oplus \cdots \oplus W_n$$

corresponding to the projections $P_1, ..., P_n$ which are pairwise orthogonal ($P_iP_j = 0$ for $i \neq j$ satisfying $\sum P_i = I$). We may then consider

$$T = \sum \lambda_n P_n$$

for scalars $\lambda_1, \lambda_2, ..., \lambda_n \in k$. This may look familiar to the notion of *diagonalization.* Indeed, we may generalize the concept to infintie dimensions like so:

Definition 11.2.16: Diagonalizable

Let $T \in \mathrm{End}_k(\mathbb{H})$ be an operator. Then T is said to be *diagonalizable* if there is an orthonormal basis $\{e_i \mid i \in I\}$ for $\mathbb{H}$ and a bounded set $\{\lambda_i \mid i \in I\} \subseteq k$ such that

$$T(x) = \sum \lambda_i \langle x, e_i\rangle e_i$$

The number $\langle x, e_i\rangle$ are the coordiantes for x in the basis $\{e_i\}$ and each λ_i is an *eigenvalue* for T corresponding to the eignevector e_i> Denotign P_i to be the projecti nofo $\mathbb{H}$ onto ke_i, we have $P_i(x) = \langle x, e_i\rangle e_i$, so we may write the operator as:

$$T = \sum \lambda_i P_i$$

Note that the above sum is not necessarily convergent in the norm topology on $\mathrm{End}_k(\mathbb{H})$, but only pointwise convergent (i.e. in convergent in the strong operator topology). If T is diagonalizable, then T^* is diagonalizable with
$$T^*(x) = \sum \overline{\lambda_i}\langle x, e_i\rangle e_i$$
with eigenvalues $\{\overline{\lambda_i}\}$. Furthermore, if T is diagonalizable, $T^*T = TT^*$ (with eigenvalues $|\lambda_i|^2$ with basis $\{e_i\}$), and hence T is normal. T is self-adjoint if and only if all λ_i are real and T is positive if and only if $\lambda_i \geq 0$ for all i. If $\mathbb{H}$ is finite-dimensional, every normal operator is diagonalizable. This is not generally true for infinite dimensions. In general, most "interesting" (normal) operators on infinite-dimensional hilbert sapces are nondiagonalizable, and must be handled with a continuous analogue of the concept of a basis (this will be the spectrum, see section 11.3 and chapter 12).

Unitary Operators

Recall $T : \mathbb{H} \to \mathbb{H}$ is unitary if T is an isometric isomorphism from $\mathbb{H}$ onto itself (if $k = \mathbb{R}$ we say it is an *orthogonal operator*). By the polarization identity, we see that $U^*U = UU^* = I$, so that U is normal and invertible with $U^{-1} = U^*$. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^{-1} = u^*$ then U is unitary since
$$\|U(x)\|^2 = \langle U^*U(x), x\rangle = \|x\|^2$$
So U is an isometry and is surjective since U is invertible. If $\{e_i \mid i \in I\}$ is an orthonormal basis for $\mathbb{H}$ and U is unitary, then $\{Ue_i \mid i \in I\}$ is a basis for $\mathbb{H}$. Conversely, we saw that every transition between orthonormal bases is given by a unitary operator. For future reference the denote the group of unitary operators $U(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ (it should be evident it forms a group).

Definition 11.2.17: Unitary Equivalent

Let $S, T \in \mathrm{End}_k(\mathbb{H})$. Then we say that S and T are *unitarily equivalnet* of $S = UTU^*$ for some unitary operator U.

It should be clear that unitary equivalent preserves norm, self-adjointness, normality, diagonalizability, and unitarity.

Definition 11.2.18: Partial Isometry

Let $U \in \mathrm{End}_k(\mathbb{H})$. Then U is said to be a *partial isometry* if there is a closed subpspace $X \subseteq \mathbb{H}$ such that $U|_X$ is isometric and $U|_{X^\perp} = 0$

This immediately implies that $X^\perp = \ker(U)$, hence $X = \bar{(}U^*(\mathbb{H}))$. Setting $P = U^*U$, then
$$\langle P(x), x\rangle = \|U(x)^2 = \|x\|^2$$
for every $x \in X$, hence $P(x) = x$ by the Cauchy-Schwarz inequality. Furthermore, $P(x^\perp) = 0$ for every $x^\perp \in X^\perp$, so P is a projetion of $\mathbb{H}$ onto X. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^*U = P$ for some projeciotn P, taking $X = P(\mathbb{H})$ and see from the equation $\|U(x)$
$^2 = \langle p(x), x\rangle$ that U is isometric on X and 0 on $X^\perp$, and so U is a partial isometry. Using this, we can also show that U^* is a partial isometry. Since $U(I - P) = 0$,
$$(UU^*)^2 = UU^*UU^* = UPU^* = UU^*$$

so UU^* is self-adjoint idempodent, i.e. a projection. Note that U^* is the projectio nof $U(\mathbb{H})$ back onto $U^*(\mathbb{H})$ so that U^* is a *partial inverse* of U.

Be weary that there exist isometries that are not unitary (since they need not be surjective). For example, if $\mathbb{H}$ is a separable and has countable basis $\{e_n \mid n \in \mathbb{N}\}$, we have

$$S\left(\sum \alpha_n e_n\right) = \sum \alpha_n e_{n+1}$$

Then S (the unilateral shift operator) is an isometry of $\mathbb{H}$ ont othe subspace $\{e_1\}^|perp$ and consequently S^* is a patial isometry of $\{e_1\}^\perp$ onto $\mathbb{H}$. In particular, $S^*S = I$, wheas SS^* is the projection onto $\{e_1\}^\perp$.

Theorem 11.2.19: Polar Decomposition of T

Let $T \in \text{End}_k(\mathbb{H})$. Then there is a unique positive operator $|T| \in \text{End}_k(\mathbb{H})$ such that

$$\|T(x)\| = \||T|(x)\|$$

and $|T| = (T^*T)^{1/2}$. Moreover, there is a unique partial isometry U with kernel $\ker(U) = \ker(T)$ and $U \circ |T| = T$. In particular

$$U^*U|T| = |T| \qquad U^*T = |T| \qquad UU^*T = T$$

***Proof*:**
Analysis Now p.96

We may think of U as the generalized "sign" of T and $|T|$ as the generalized "absolute value" of T. By the formula

$$T^* = |T|U^* = U^*(U|T|U^*)$$

and the unique of polar decomposition, we see that U^* is the sign of T^*, whereas $U|T|U^*$ is the absolute value fo T^*. In general, the absolute value of a sum or product of noncommuting operators doens't have much relation to the sum or product of the absolute values. Similarly, the sign (in the finite dimensional case), can't always be chosen to be unitary (the unilateral shift is an immedaite counter example). For T to have the form $U|T|$ for some unitary operator U, it is necessary and sufficient that the cloesd subspaces $\ker(T)$ and $\ker(T^*)$ have the same dimension (finite or infinite). Easy cases of this general theorem are estalblisehd in the next result

Proposition 11.2.20: Polar Decomposition For Isomorphisms

Let $T \in \text{End}_k(\mathbb{H})$ be invertible. Then the partial isometry in its polar decomposition is unitary

***Proof*:**
Analysis Now p.97

Proposition 11.2.21: Polar Decomposition For Normal Operators

Let $T \in \text{End}_k(\mathbb{H})$ be normal. Then there is a unitary operator W commuting with T, T^*, and $|T|$ such that $T = W|T|$

***Proof*:**
Analysis Now p.97

I want to add one more thing, I just don't konw where: for any $T \in B(\mathbb{H})$, $T = T_1 + iT_2$ for self-adjoint operators T_1, T_2. This lets us generalize the spectral theorem for noraml operators, since we will konw it for self-adjoint operator.

Lemma 11.2.22: lemma I

If $T = T^*$ and $\|T\| \leq 1$, the operator $U = T + i(I - T^2)^{1/2}$ is unitary and $T = \frac{1}{2}(U + U^*)$

***Proof*:**
Analysis Now p.97

Lemma 11.2.23: Lemma II

Let $S \in \text{End}_k(\mathbb{H})$ with $\|S\| < 1$. Then for every unitary U, there are unitaries U_1, V_1 such that

$$S + U = U_1 + V_1$$

***Proof*:**
Analysis Now p.98

Theorem 11.2.24: Russo-Dye-Gardner Theorem

Let $T \in \text{End}_k(\mathbb{H})$ with

$$\|T\| < 1 - \frac{2}{n} \qquad n > 2$$

Then there are unitary operators $U_1, U_2, ..., U_n$ such that

$$T = \frac{1}{n}(U_1 + U_2 + \cdots + U_n)$$

***Proof*:**
Analysis Now p.98

Since the self-adjoint elements span $\text{End}_k(\mathbb{H})$, we see from lemma 11.2.22 that every $T \in \text{End}_k(\mathbb{H})$ is the linear combination of (at least 4) unitary operators The estimate in the Russo-Dye-Gardner Theorem is much more precise with respect to the norm, and shows that every element in the open unit ball of $\text{End}_k(\mathbb{H})$ is a convex combination (indeed, a mean) of unitary operators. On the surface of the ball this need not be the case, in fact the unilateral shift cannot be expressed as a convex combination of unitaries (exercise 3.2.8 in Gert K. Pederson Analysis Now).

Proposition 11.2.25: Numerical Radius Of Operator

Let $\mathbb{H}$ be a complex hilbert space and $T \in \operatorname{End}_{\mathbb{C}}(\mathbb{H})$. Then we deifn eth *numerical radius* of T as

$$|||T||| = \sup \{ \langle T(x), x \rangle \mid x \in \mathbb{H}, \|x\| \leq 1 \}$$

Then $\|\cdot\|$ is a norm on $\operatorname{End}_{\mathbb{C}}(\mathbb{H})$ satisfying $\frac{1}{2}\|\cdot\| \leq |||\cdot||| \leq \|\cdot\|$. More over, $|||T^2||| \leq |||T|||^2$ for every T and $|||T||| = \|T$
if T is normal

***Proof*:**
Analysis Now p.99

11.3 Compact Operators

Throughout this section, let $\mathbb{H}$ be a hilbert space over $k \in \{\mathbb{R}, \mathbb{C}\}$ unless otherwise specified.

Definition 11.3.1: Finite Rank Operator

Let $T \in \operatorname{End}_k(\mathbb{H})$. Then T is said to have *finite rank* if $T(\mathbb{H})$ is finite-dimensional subspace of $\mathbb{H}$. Denote by $B_f(\mathbb{H}) \subseteq \operatorname{End}_k(\mathbb{H})$ the set of all finite rank opeartors.

Clearly, $B_f(\mathbb{H})$ is a subspace, even a subalgebra, even an ideal of $\operatorname{End}_k(\mathbb{H})$. If $T \in B_f(\mathbb{H})$, by proposition 11.2.4, we get an orthogonal decomposition $\mathbb{H} = T(\mathbb{H}) \oplus \ker(T^*)$, which shows that $T^*(\mathbb{H}) - T^*T(\mathbb{H})$, so T^* has finite rank. Thus, $B_f(\mathbb{H})$ is *self-adjoint* ideal in $\operatorname{End}_k(\mathbb{H})$ ($(B_f(\mathbb{H}))^* = B_f(\mathbb{H})$ as a set). The subset $B_f(\mathbb{H}) \subseteq \operatorname{End}_k(\mathbb{H})$ is similar in relation to $C_c(X) \subseteq C_b(X)$ when X is a locally compact Hausdorff space. In particular, these classes describe the local phenomena on $\mathbb{H}$ and on X. Passing to a limit in norm may destroy the "locality". but enogh structure is preservd to make these "quasilocal" operators and functions very useful and interesting. We shall stud the closure of $B_f(\mathbb{H})$ in this section as a noncommutative analogue of $C_0(X)$ in function theory.

Lemma 11.3.2: Net And Compact Operators

There is a net $(P_\lambda)_{\lambda \in \Lambda}$ of projection in $B_f(\mathbb{H})$ such that $\|P_\lambda(x) - x\| \to 0$ for each $x \in \mathbb{H}$.

that is, the projection are "dense" in $B_f(\mathbb{H})$.

***Proof*:**
We want to find a next P_n such that $\|P_n(x) - x\| \to 0$. Let $\{e_i \mid i \in I\}$ be an orthonormal basis for $\mathbb{H}$ and let N be a net of finite subsets of I ordered by inclusion. For each $n \in N$, let P_n denote the projection of $\mathbb{H}$ onto $\operatorname{span}\{e_i \mid i \in n\}$ so that $(P_n)_{n \in N}$ is a net in $B_f(\mathbb{H})$.

Now, if $x \in \mathbb{H}$, then $x = \sum a_i e_i$, thus

$$\|P_n(x) - x\|^2 = \sum_{i \notin n} |a_i|^2$$

By Parseval's identity, this tends to zero, completing the proof.

Theorem 11.3.3: Closed Unit Balls In Hilbert Spaces

Let $B \subseteq \mathbb{H}$ be a closed unit ball in a hilbert space $\mathbb{H}$. Then the following conditions on an operator $T \in \text{End}_k(\mathbb{H})$ are equivalent

1. $T \in \overline{B_f(\mathbb{H})}$
2. $T|_B$ is a weak-norm continuous function from B into $\mathbb{H}$
3. $T(B)$ is compact in $\mathbb{H}$
4. $\overline{T(B)}$ is compact in $\mathbb{H}$
5. each net in B has a subnet whose iamge under T converges in $\mathbb{H}$

***Proof*:**
Analysis Now p.105

Definition 11.3.4: Compact Operators

Let $T \in B(\mathbb{H})$. Then T is compact if $T(B)$ is compact. The set of compact operators is denoted by $B_0(\mathbb{H})$ to show that these operators "vanish at infinity".

Lemma 11.3.5: Diagonalizable Operator Is Compact

Let $T \in \text{End}_k(\mathbb{H})$. Then T is comapct if and only if its eigenvalues $\{\lambda_i \mid i \in I\}$ belongs to $c_0(I)$

***Proof*:**
Let $T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$. If $T \in B_0(\mathbb{H})$, then for any $\epsilon > 0$, define

$$I_\epsilon = \{i \in I \mid |\lambda_i| \geq \epsilon\}$$

If I_ϵ is infinite, the net $(e_i)_{i \in I_\epsilon}$ converges weakly to zero for any well-ordering of I_ϵ since $\langle e_i, x \rangle \to 0$ by Parsaval's identity. Sicne $\|T(e_i)\| = |\lambda_i| \geq \epsilon$ for $i \in I_\epsilon <$ this contradicts theorem 11.3.3(3). Thus, I_ϵ must be finite for each $\epsilon > 0$, hence λ_i must vanish at infinity.

Conversely, if I_ϵ is finite for all $\epsilon > 0$. et

$$T_\epsilon = \sum_{i \in I_\epsilon} \lambda_i \langle \cdot, e_i \rangle e_i$$

Then T_ϵ has finite rank, and

$$\begin{aligned}\|(T - T_\epsilon)(x)\|^2 &= \left\| \sum_{i \notin I_\epsilon} \lambda_i \langle x, e_i \rangle e_i \right\|^2 \\ &= \sum_{i \notin I_\epsilon} |\lambda_i|^2 |\langle x, e_i \rangle|^2 \\ &\leq \epsilon^2 \|x\|^2\end{aligned}$$

hence, $\|T - T_\epsilon\| \leq \epsilon$, thus $T \in B_0(\mathbb{H})$ by theorem 11.3.3(1), completing the proof.

Lemma 11.3.6: Eigen Vectors Between Adjoint

Let x be an eigenvector for a normal operator $T \in \mathrm{End}_k(\mathbb{H})$ corresponding to an eigenvalue λ. Then x is an eigenvector for T^* corresponding to the eigenvalue $\overline{\lambda}$. Eigenvectors for T correspondign to different eigenvalues are orthogonal.

***Proof*:**
Note that the operator $T - \lambda I$ is normal and it's adjoitn is $T^* - \overline{\lambda} I$. COnversely,

$$\|(T^* - \overline{\lambda} I)(x)\| = \|(T - \lambda I)(x)\| = 0$$

If $T(y) = \gamma y$ where $\gamma \neq \lambda$, then we may certainly assume $\lambda \neq 0$ so that

$$\begin{aligned}\langle x, y \rangle &= \lambda^{-1} \langle T(x), y \rangle \\ &= \lambda^{-1} \langle x, T^*(y) \rangle \\ &= \lambda^{-1} \langle x, \overline{\gamma} y \rangle \\ &= \lambda^{-1} \gamma \langle x, y \rangle\end{aligned}$$

Hence, $\langle x, y \rangle = 0$, showing they are different eigenvectors, completing the proof.

Lemma 11.3.7: Normal Compact Operator, then Eivenvalue Exists

Every normal, compact operator T on a complex Hilbert space $\mathbb{H}$ has an eigenvalue λ with $|\lambda| = \|T\|$

***Proof*:**
Analysis now p. 107

Theorem 11.3.8: Normal Compact, Then Diagonalizable

Every normal compact operator $T \in \mathrm{End}_k(\mathbb{H})$ is diagonalizable and its eigenvalues (counted-with multiplicity) vanish at infinite. Conversely, each such operator is normal and compact.

***Proof*:**
Analysis now p.108

It will be convenient for future sections to introduce a new notation: Let $x \odot y$ represent the rank one operator in $\mathrm{End}_k(\mathbb{H})$ determined by the vectors $x, y \in \mathbb{H}$ and the formula

$$(x \odot y)(z) = \langle z, y \rangle x$$

The map $(x, y) \mapsto x \odot y$ is sesquilinear map of $\mathbb{H} \times \mathbb{H}$ into $B_f(\mathbb{H})$. If $\|e\| = 1$ then $e \odot e$ is the one-dimensional projection of $\mathbb{H}$ on $\mathbb{C}e$. Every normal compact operator $\mathbb{H}$ can now be written by the above theorem in the form

$$T = \sum \lambda_i e_i \odot e_j$$

For suitable orthonormal basis $\{e_i \mid i \in I\}$. The sum will converge in norm, since either the set $I_0 = \{i \in I \mid \lambda_i \neq 0\}$ is finite (so that $T \in B_f(\mathbb{H})$) or else countably infinite in which case the sequence $\{\lambda_i \mid i \in I_0\}$ converges to zero.

Definition 11.3.9: Spectrum

We say that the compact set

$${}^{(}T) = \{\lambda_i \mid i \in I_0\} \cup \{0\}$$

is the *spectrum* of T.

For every continuous function f on ${}^{(}T)$, define

$$f(T) = \sum f(\lambda_i) e_i \odot e_i$$

Then $f(T)$ is compact if and only if $f(0) = 0$ and the map $f \mapsto f(T)$ Is an isometric $*$-preserving homomorhpism of $C({}^{(}T))$ into $\mathrm{End}_k(\mathbb{H})$. If $f(z) = \sum \alpha_{nm} z^n \overline{z}^m$ is a polynomial of two commuting variables z and $\overline{z}$, then $f(T) = \sum \alpha_{nm} T^n (T^*)^m$. This result is the *spectrial (mapping) theorem* for normal compact operators. In the next chapter, we shall generalize this to any normal operator.

Calkin Algebra and Index

Since $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$ is a closed ideal, we may define $B(\mathbb{H})/B_0(\mathbb{H})$ with the quotient norm[7]. This quotient is given a name

Definition 11.3.10: Calkin Algebra

Let $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$. Then the algebra

$$B(\mathbb{H})/B_0(\mathbb{H})$$

is called the *Calkin Algebra* of $\mathbb{H}$

We shall see that many properties of $B(\mathbb{H})$ are conveniently expressed in terms of the Calkin algebra.

[7]This is even a C^*-algebra

Definition 11.3.11: Compact Pertubation

If S and T are elements in $B(\mathbb{H})$ and $S - T \in B_0(\mathbb{H})$, we say that S is a *compact pertubation* of T.

This is antoher way of saying that S and T hav the same image in the Calkin algebra. Such "local" pertubation happen frequently in application and properties of an operator that are invariatn under compact pertubations are highly used. The best known of these we shall study know is known as the *index*

Theorem 11.3.12: Atkinson's Theorem

Let $T \in B(\mathbb{H})$. Then the following are equivalent:

1. There is a unique operator $S \in B(\mathbb{H})$ such that ST and TS are the projections on $(\ker(T))^\perp$ and $\ker(T^*)^\perp$, respectively, and both projections have finite co-rank
2. For some operator $S \in B(\mathbb{H})$ both operators $ST - I$ and $TS - I$ are compact
3. the image of T is invertible in the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$
4. Both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional subsapces and $T(\mathbb{H})$ is closed

***Proof*:**
Analysis now p.109

In essence, Fredholm operators are operators that are are almost invertible up to some finite-dimension beign ignored, in particular they are invertible modulo compact operators.

Definition 11.3.13: Fredholm Operator

The operator satisfying the condition's in Atkinson's theorem are called *Fredholm operators*, and the class of them is dneoted $F(\mathbb{H})$.

Definition 11.3.14: Index

For each $T \in F(\mathbb{H})$, define the *index* of T to be

$$\text{index}(T) = \dim(\ker(T)) - \dim(\ker(T^*))$$

If we shooe S for T as in (1) of Atkinson's theorem and write $suchthat = I - P$ and $TS = I - Q$ with P and Q projection of rank, then another way of writing this would be

$$\text{index}(T) = \text{rank}(P) - \text{rank}(Q)$$

By (3) of Atkinson's Theorem, the product of Fredholm operators is again a Fredholm operator. In particular, every product $RT \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ and R is invertible in $B(\mathbb{H})$, and in this case

$$\text{index}(RT) = \text{index}(TR) = \text{index}(T)$$

since R is bijective. It should also be clear that $T^* \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ with index $\operatorname{index}(T^*) = -\operatorname{index}(T)$. Finally, the partial inverse S to T in (1) of Atkinson's theorem is a Fredholm operator with index $S) = -T)$. F

Definition 11.3.15: Indexed Fredholm Operators

For each $n \in \mathbb{Z}$ define

$$F_n(\mathbb{H}) = \{T \in F(\mathbb{H}) \mid T) = n\}$$

All of these subsets are nonemtpy, since if S is the unilateral shift, then $S^n \in F_{-n}(\mathbb{H})$ and $(S^*)^n \in F_n(\mathbb{H})$ for every $n > 0$.

Lemma 11.3.16: Finite Rank And 0 Index

If $A \in B_f(\mathbb{H})$, then $I + A \in F_0(\mathbb{H})$

***Proof*:**
Analysis now p.110

Corollary 11.3.17: Fredholm Alternative

If $A \in B_0(\mathbb{H})$ and $\lambda \in \mathbb{C} \setminus \{9\}$, then either $\lambda I - A$ is invertible in $B(\mathbb{H})$ or λ is an eigenvalue of A with finite multiplicity, in which case $\overline{\lambda}$ is an eigenvalue for A^* with the same multiplicity

***Proof*:**
Let $T + I - \lambda^{-1}A$. Since $T \in F_0(\mathbb{H})$, by lemma 11.3.16 we know that $T(\mathbb{H})$ is closed and that

$$\dim(\ker(T)) = \dim(\ker(T^*)) < \infty$$

If these dimension are 0, then $T(\mathbb{H}) = (\ker(T^*))^\perp = \mathbb{H}$ and $\ker(T) = \{0\}$, hence T is invertible by proposition 11.2.6, completing the proof.

This lemma implies that the spectrum of a compact operator (i.e. those λ such that $\lambda I - T$ is not invertible, as we'll see in the next chapter) consists of $\{0\}$ and the eigenvalues for T. IN particualr, it is a countable subset of $\mathbb{C}$ with 0 as the only possible accumulatpion point.

Theorem 11.3.18: Index Invariant Under Compact Operator

Let $T \in F(\mathbb{H})$ and A be a compact operator. Then

$$T + A) = T)$$

***Proof*:**
Analysis now p.111

Proposition 11.3.19: Fredholm Class Is Open

Each Fredholm class $F_n(\mathbb{H})$ for each $n \in \mathbb{H}$ is open in $B(\mathbb{H})$

***Proof*:**
analysis now p.112

Proposition 11.3.20: Grading Structure Of Freidholm Operators

If $T_1 \in F_n(\mathbb{H})$ and $T_2 \in F_m(\mathbb{H})$ then $T_1T_2 \in F_{n+m}(\mathbb{H})$

***Proof*:**
Analysis now p. 112

We close with an interesting topological fact that brings some light to the index .Let G denote the group of invertible elements of the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$. and G_0 is the connected component of the identity (which is the group generated by the elements of the form $\exp(A)$ for some $A \in B(\mathbb{H})/B_0(\mathbb{H})$). Then G_0 is an open and closed subgroup of G and G/G_0 is a discrete group that labels the connected components in G. In this case, G/G_0 is isomorphic to $\mathbb{Z}$, and the index of a Fredholm operator T is the image of T under the compsoed quotinet maps from $F(\mathbb{H})$ to G and from G to G/G_0 $(= \mathbb{Z})$

11.4 Trace

11.5 *Intuition to add

This is a conversation I had with Gemini that I really want to incorporate somewhere as it gave some good insights. I will save it here for now:

Zak Katz: Let $F(x_1, \ldots, x_n, t)$ represent a PDE w.r.t time. Let's say $u(x,t)$ represent a solution and $f_t(x) = u(x,t)$ is a "state" at time 0. Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a.

My question is: Is S_t always self-adjoint? If yes, can you prove it, and if not, can you give a counter-example?

Raycast AI:

`Tool: web_search`

No, the time-shift operator S_t is not always self-adjoint.

In fact, for many common PDEs (like the wave equation, transport equation, or Schrödinger equation), S_t is almost never self-adjoint. It is usually unitary (energy-conserving) or a contraction semi-group (dissipative).

12

Spectral Theory

In short, Spectral theory generalizes the notion of eigen vectors and values to infinite dimensional operators. In the previous chapter, we saw how there is a notion of diagonalizability for compact normal operators. In this chapter, we extend this result to normal operators. We also along the way learn some important dualities such as the Gelfand duality which has some useful applications, and is an intersting motivator for classical algebraic geometry.

12.1 Banach Algebras

Recall the definition of a banach algebra

Definition 12.1.1: Banach Algebra Revisited

A *Banach Alagebra* is an algebra A with a submultiplicative norm (so that it is continuous) st the unerlying vector space is a banach space.

All the same applies for Banach Algebras as we expect. Note that for A/I to be a Banach algebra for an ideal I, I has to be closed, just like for subspaces. If A is unital and $A \neq \{0\}$, then it is certainly unique and $\|I\| \geq 1$ (why not =?? Analysis now p.128) If A has no units, we may try to isometrically embed it into a larger Banach Algebra $\widehat{A}$ such that $A \subseteq \widehat{A}$ becomes an ideal such that every nonzero ideal of $\widehat{A}$ has a nonzero intersection with A (such an ideal is called an *essential ideal*). This can be thought as the algebraic counterpart of the notion of compactification of a topological space. For classical Banach algebra, there is usually a natural way of *adjoining a unit*, but there is always a general method of doing so that is the counter-part of one-point compactification. Take the direct sum $A \oplus k = \widehat{A}$ along with the product

$$(x, \alpha)(y, \beta) = (xy + \alpha y + \beta x, \alpha\beta)$$

and the submultiplicative 1-norm. Then $A \oplus k$ is a unital Banach algebra with $I(0,1)$ and contains A as an ideal of co-dimension 1.

Definition 12.1.2: Regular Representation

Let A be a Banch Algebra. Then the (left) *regular representation* $\rho : A \to B(A)$ by $\rho(x)(y) = xy$ for $x, y \in A$.

It should be verified that ρ is a norm decreasing algebra homomorphism, and if A is unital, the inequalities

$$\|A\| \leq \|\rho(A)\| \|I\| \leq \|A\| \|I\|$$

show that ρ is a homeomrophism. Renorming A by using the equivalent norm from $B(A)$, we may assume that $\|I\| = 1$ (if $A \neq \{0\}$), which we do.

Definition 12.1.3: Approximate Unit

A net $(E_\lambda)_{\lambda \in \Lambda}$ in the unit ball of a Banah algebra A is an *approximate unit* if

$$\lim E_\lambda x = \lim x E_\lambda = x$$

for all $x \in A$.

Example 12.1.4: Approximate Units

Can you think of approximate units for the non-unital Banach algebra $C_0(X)$, $B_0(\mathbb{H})$ and $L^1(\mathbb{R}^n)$? Show that the existence of an approximate unit guarantees that the regular representation ρ is an isometry

We shall denote $A^\times = \mathrm{GL}(A)$ the set of all invertible elements of A (in algebra, we'd usually call these elements unital, but the term unit in functional analysis is more commonly reserved for the identity)

Lemma 12.1.5: Almost Invertible Elements

Let $x \in A$ where A is a unital banach algebra with $\|A\| < 1$. Then $I - A \in \mathrm{GL}(A)$ and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

***Proof*:**
Since the norm is submultiplicative $\|x^n\| \leq \|x\|^n$, thus the series $\sum x^n$ converges in x to an element y. Sicne $xy = yx = y - I$ by manipulating the above power series, it follwos that $y = (I - x)^{-1}$, completing the proof.

Proposition 12.1.6: Properties Of $\mathrm{GL}(A)$

Let A be a unital algebra and $\mathrm{GL}(A)$ be it's multiplicative group. Then $\mathrm{GL}(A)$ is an open subset of A, an the map $x \mapsto x^{-1}$ is a homeomorphism of $\mathrm{GL}(A)$

***Proof*:**
Let $x \in \mathrm{GL}(A)$ and $y \in A$, Then by lemma 12.1.5

$$y = x - (x - y) = x(I - x^{-1}(x - y)) \in \mathrm{GL}(A)$$

provided that $\|A^{-1}(A - B)\| < 1$. In particular, the ball $B_\epsilon(x)$ is contained in $\mathrm{GL}(A)$ for $\epsilon < \|x^{-1}\|^{-1}$, which proves that $\mathrm{GL}(A)$ is open. If $y \to a$, we see from the above equation and lemma 12.1.5 that $y^{-1} \mapsto a^{-1}$, thus inverting is a continuous process and therefore a homeomorphism since $(A^{-1})^{-1} = A$, completing the proof.

From ehre on out, all Banach algebra will be over $\mathbb{C}$, i.e. all Banach algebras will be *complex*. See exercise ref:HERE on the complexicication of general real Banach algebras (or E.2.1.13 in Analysis Now).

Definition 12.1.7: Spectrum

Let A be a complex unital Banach algebra. Then for every $x \in A$, define the *spectrum* of x t obe the set

$${}^{(}x) = \{\lambda \in \mathbb{C} \mid \lambda I - x \notin \mathrm{GL}(A)\}$$

The comliment of ${}^{(}x)$ is the *resolvent set*.

On the resolvent set, we can define the resolvent function

$$R(x, \lambda) = (\lambda I - A)^{-1}$$

Definition 12.1.8: Spectral Radius

Let A be a complex unital Banach algera, $x \in A$, and ${}^{(}x)$ be its spectrum. Then the smalleest $r \geq 0$ such that ${}^{(}x) \subseteq B_r(0)$ is called the *spectral radius* of x, and is denoted $r(x)$. THus

$$r(x) = \sup \left\{ |\lambda| \;\middle|\; \lambda \in^{(} x) \right\}$$

Lemma 12.1.9: Build-Up Lemma I

Let $x \in A$ and $f(z) = \sum \alpha_n z^n$ be a holomorphic function in a region contined in the closed disk $B_r(0)$ with $\|x\| \leq r$. then we can define $f(x) = \sum \alpha_n x^n$ in A. Morever, if $\lambda \in^{(} x)$ and $|\lambda| \leq r$, then $f(\lambda) \in^{(} f(x))$

***Proof*:**
The series $f(x)$ is absolutely convergence ($\sum |\alpha_n| \|x\|^n < \infty$), so certainly $f(x) \in A$. Moerover

$$\begin{aligned} f(\lambda)I - f(x) &= \sum \alpha_n(\lambda^n I - x^n) \\ &= (\lambda I - x)\sum \alpha P_{n-1}(\lambda, x) \\ &= (\lambda I - x)y \end{aligned}$$

where $P_{n-1}(\lambda, x) = \sum_{k=0}^{n-1} \lambda^k A^{n-k-1}$ so that $\|P_{n-1}(\lambda, x)\| \leq nr^{n-1}$, which implies the series of polynomials converges in A to an element y commuting with x. We see form this computation that if $f(\lambda)I - f(x) \in \mathrm{GL}(A)$ with inverse v, then yv will be the inverse of $\lambda I - x$, so tha $\lambda I - x \in \mathrm{GL}(A)$

Lemma 12.1.10: Build-Up Lemma II

For every $x\ inA$, we have

$$r(x) \leq \inf \|A^n\|^{1/n}$$

***Proof*:**
If $|\lambda| > \|A\|$, thn by lemma 12.1.5

$$(\lambda I - x)^{-1} = \lambda^{-1}(I - \lambda^{-1}x)^{-1} = \sum \lambda^{-n-1}x^n$$

which shows that $r(x) \leq \|x\|$. Now if $\lambda \in^(x)$, then $\lambda^n \in^(x^n)$ by lemma 12.1.9. Thus $|\lambda^n| \leq \|x^n\|$ from what we've just shown. It follows that $r(x) \leq \|x^n\|^{1/n}$ for every n, completing the proof.

Theorem 12.1.11: Spectrum Is Compact

Let $x \in A$ be an element of a complex unital Banach algebra A. Then the spectrum of x, $^(x)$ is a compact nonempty subset of $\mathbb{C}$, and the spectral radius of x is the limit of the convergent sequence $\|x^n\|^{1/n}$

***Proof*:**
Compactness. If $|\lambda| > \|x\|$ then $\lambda I - x = \lambda(I - \lambda^{-1}x)$ is invertible, its inverse being the norm-convergent Neumann series $\sum_{n \geq 0} \lambda^{-n-1}x^n$; hence $^(x) \subseteq \{\lambda : |\lambda| \leq \|x\|\}$ is bounded. The map $\lambda \mapsto \lambda I - x$ is continuous and the group $\mathrm{GL}(A)$ of invertibles is open, so the resolvent set $\rho(x) = \mathbb{C} \setminus^(x)$ is open and $^(x)$ is closed. A closed bounded subset of $\mathbb{C}$ is compact.

Nonemptiness. Write $R(\lambda) = (\lambda I - x)^{-1}$ for the resolvent on $\rho(x)$. Near any $\lambda_0 \in \rho(x)$ the factorization $\lambda I - x = (\lambda_0 I - x)\big(I + (\lambda - \lambda_0)R(\lambda_0)\big)$ gives the norm-convergent expansion $R(\lambda) = \sum_{n \geq 0}(\lambda_0 - \lambda)^n R(\lambda_0)^{n+1}$, so R is an A-valued analytic function on $\rho(x)$. Suppose $^(x) = \varnothing$, so $\rho(x) = \mathbb{C}$ and R is entire. From $R(\lambda) = \lambda^{-1}(I - \lambda^{-1}x)^{-1}$ we get $\|R(\lambda)\| \to 0$ as $|\lambda| \to \infty$. For any φ in the dual A', the scalar function $\varphi \circ R$ is then entire and vanishes at infinity, hence is bounded, hence constant by Liouville's theorem ([12]); vanishing at infinity forces that constant to be 0. As $\varphi \in A'$ was arbitrary, proposition 7.3.6 gives $R(\lambda) = 0$ for every

λ, contradicting the invertibility of $\lambda I - x$. Hence $^{(}x) \neq \varnothing$.

The spectral radius formula. The estimate $r(x) \leq \|x^n\|^{1/n}$ for every n was shown above (from $\lambda^n \in^{(} x^n)$, lemma 12.1.9), so $r(x) \leq \liminf_n \|x^n\|^{1/n}$. For the reverse inequality fix $\varphi \in A'$. On $\{|\lambda| > \|x\|\}$ the Neumann series gives the Laurent expansion $\varphi(R(\lambda)) = \sum_{n\geq 0} \varphi(x^n)\lambda^{-n-1}$; since $\varphi \circ R$ is analytic on the larger region $\{|\lambda| > r(x)\} \subseteq \rho(x)$, uniqueness of Laurent coefficients extends this expansion to all $|\lambda| > r(x)$. For each such λ the terms $\varphi(x^n)\lambda^{-n-1}$ are then bounded in n, so $\sup_n |\varphi(\lambda^{-n}x^n)| < \infty$. This holds for every $\varphi \in A'$, so by the principle of uniform boundedness (theorem 7.2.11) the sequence $(\lambda^{-n}x^n)_n$ is norm-bounded, say $\|x^n\| \leq C_\lambda |\lambda|^n$; hence $\limsup_n \|x^n\|^{1/n} \leq |\lambda|$. Letting $|\lambda| \downarrow r(x)$ yields $\limsup_n \|x^n\|^{1/n} \leq r(x)$. Combining, $\lim_n \|x^n\|^{1/n} = r(x)$.

An immediate consequence is the classical Gelfand-Mazur theorem, which pins down the only complex Banach division algebra:

Corollary 12.1.12: Gelfand-Mazur

Let A be a unital Banach algebra over $\mathbb{C}$. Then if A is a division ring, $A = \mathbb{C}$

***Proof*:**
For each $x \in A$, there is some $\lambda \in^{(} x)$ by the theorem. Thus, $\lambda I - x \notin \mathrm{GL}(A)$, hence $x = \lambda I$.

12.1.1 The Gelfand Transform

Gelfand-Mazur pins down $\mathbb{C}$ as the only complex Banach division algebra. For a *commutative* Banach algebra this one fact reorganizes the whole algebra into a space of functions: every maximal ideal has quotient $\mathbb{C}$, so it is the kernel of a homomorphism onto $\mathbb{C}$, and evaluating an element against all such homomorphisms turns it into a function on the set of homomorphisms. This assignment is the Gelfand transform, and it is the mechanism behind every functional calculus.

Definition 12.1.13: Character and Character Space

Let A be a commutative Banach algebra over $\mathbb{C}$. A *character* (or *multiplicative functional*) of A is a nonzero algebra homomorphism $\varphi\colon A \to \mathbb{C}$: a linear map with $\varphi(xy) = \varphi(x)\varphi(y)$ and $\varphi \neq 0$. The set of all characters is the *character space* (equivalently the *Gelfand spectrum* or *maximal ideal space*) of A, written $\Delta(A)$ and topologized as a subspace of $(A', \text{weak-}*)$; this subspace topology is the *Gelfand topology.*

If A is unital then $\varphi(1) = 1$ for every character (apply φ to $1 \cdot 1 = 1$ and cancel, using $\varphi \neq 0$), and φ cannot vanish on an invertible element (from $\varphi(x)\varphi(x^{-1}) = 1$). The characters turn out to be exactly the quotient maps of A onto $\mathbb{C}$.

Proposition 12.1.14: Characters, Maximal Ideals, and Compactness

Let A be a commutative Banach algebra over $\mathbb{C}$.

1. Every character is contractive: $\|\varphi\| \leq 1$. Thus $\Delta(A)$ lies in the closed unit ball of A'.
2. If A is unital, then $\varphi \mapsto \ker\varphi$ is a bijection of $\Delta(A)$ onto the set of maximal ideals of A, and $\Delta(A)$ is weak-$*$ compact and Hausdorff.
3. If A is non-unital, then $\Delta(A) \cup \{0\}$ is weak-$*$ compact, so $\Delta(A)$ is locally compact and Hausdorff.

***Proof*:**

(1) Adjoin a unit: on $\widetilde{A} = A \oplus \mathbb{C}$ (definition 12.1.1) the map $\widetilde{\varphi}(x, \lambda) = \varphi(x) + \lambda$ is a unital character extending φ. If $|\varphi(x)| > \|x\|$ for some $x \in A$, put $\lambda = \varphi(x) \neq 0$; then $\|x/\lambda\| < 1$, so $1 - x/\lambda$ is invertible in $\widetilde{A}$ by lemma 12.1.5, yet $\widetilde{\varphi}(1 - x/\lambda) = 1 - \varphi(x)/\lambda = 0$ - a character vanishing on an invertible, which is impossible. Hence $|\varphi(x)| \leq \|x\|$.

(2) The kernel of a character has codimension 1 (since $A/\ker\varphi \cong \mathbb{C}$), so it is a maximal ideal. Conversely let M be a maximal ideal. It is closed: $\mathrm{GL}(A)$ is open (proposition 12.1.6) and disjoint from M, so the unit has a neighbourhood missing M, whence the closure $\overline{M}$ is a proper ideal containing M, forcing $\overline{M} = M$. Then A/M is a unital Banach algebra and a field, hence a Banach division algebra, so $A/M \cong \mathbb{C}$ by Gelfand-Mazur (corollary 12.1.12); the quotient map is a character with kernel M. The two assignments are mutually inverse. For compactness, $\Delta(A)$ is the set of φ in the closed unit ball of A' meeting the conditions $\varphi(xy) = \varphi(x)\varphi(y)$ for all x, y and $\varphi(1) = 1$; each is weak-$*$ closed (the maps $\varphi \mapsto \varphi(xy)$ and $\varphi \mapsto \varphi(x)\varphi(y)$ are weak-$*$ continuous), so $\Delta(A)$ is weak-$*$ closed in the ball, which is weak-$*$ compact by Alaoglu's theorem (theorem 7.5.7).

(3) Characters of A correspond to the unital characters of $\widetilde{A}$ other than the augmentation $(x, \lambda) \mapsto \lambda$; that is, $\Delta(\widetilde{A}) = \Delta(A) \cup \{0\}$, writing 0 for the character that vanishes on A. By (2) applied to $\widetilde{A}$ this set is weak-$*$ compact, and $\Delta(A)$ is its complement of the single point 0, hence open in a compact Hausdorff space and so locally compact Hausdorff.

Evaluation now turns each element of A into a function on $\Delta(A)$.

Theorem 12.1.15: The Gelfand Transform

Let A be a commutative Banach algebra over $\mathbb{C}$. For $x \in A$ define $\widehat{x}\colon \Delta(A) \to \mathbb{C}$ by $\widehat{x}(\varphi) = \varphi(x)$. Then:

1. $\widehat{x} \in C_0(\Delta(A))$, and the *Gelfand transform* $\Gamma\colon A \to C_0(\Delta(A))$, $x \mapsto \widehat{x}$, is a norm-decreasing algebra homomorphism.
2. If A is unital, then for every $x \in A$ the range of $\widehat{x}$ is exactly the spectrum,

$$^{(}x) = \widehat{x}(\Delta(A)) = \{\varphi(x) \mid \varphi \in \Delta(A)\}$$

and in particular $\|\widehat{x}\|_\infty = r(x) \leq \|x\|$.

***Proof*:**
Continuity of $\widehat{x}$ is the definition of the weak-$*$ topology, and Γ is linear and multiplicative because each φ is. For vanishing at infinity, pass to $\widetilde{A}$: there $\Delta(\widetilde{A}) = \Delta(A) \cup \{0\}$ is compact Hausdorff (proposition 12.1.14), the extended function $\widehat{x}$ is continuous on it, and it takes the value 0 at the added character; continuity at that point is precisely $\widehat{x} \in C_0(\Delta(A))$.

For (2), with A unital: $\lambda \in^{(} x)$ iff $x - \lambda$ is not invertible, iff $x - \lambda$ generates a proper ideal and so lies in some maximal ideal M (Zorn), iff by proposition 12.1.14(2) there is a character φ with $\ker \varphi = M$, that is $\varphi(x - \lambda) = 0$, i.e. $\varphi(x) = \lambda$. Hence $^{(}x) = \{\varphi(x) \mid \varphi \in \Delta(A)\}$, and $\|\widehat{x}\|_\infty = \sup \{|\lambda| \bigm| \lambda \in^{(} x)\} = r(x) \leq \|x\|$ by theorem 12.1.11.

For a general commutative Banach algebra Γ is neither injective nor isometric: its kernel is the *radical* $\bigcap_{\varphi \in \Delta(A)} \ker \varphi$, and the norm it induces is the spectral radius $r(x)$, not $\|x\|$. Both defects vanish for a commutative C^*-algebra, where the C^*-identity $\|A^*A\| = \|A\|^2$ forces $r = \|\cdot\|$ and Γ becomes an isometric $*$-isomorphism onto $C_0(\Delta(A))$ - the commutative Gelfand-Naimark theorem, taken up in the C^*-algebras companion. The next section is the first payoff: the singly generated algebra $C^*(T)$ is commutative, and identifying it with $C(^{T})$ via Γ is exactly what upgrades the polynomial calculus to a continuous one.

12.2 The Continuous Functional Calculus

For a self-adjoint operator $T \in B(\mathbb{H})$, the polynomial $p(T) = \alpha_0 I + \alpha_1 T + \cdots + \alpha_n T^n$ is unambiguous: it is built from T by the algebra operations alone. The question this section answers is how far the assignment $p \mapsto p(T)$ can be pushed. If T were a diagonalizable matrix with eigenvalues $\lambda_1, \ldots, \lambda_k$, then $p(T)$ would act as multiplication by $p(\lambda_i)$ on the ith eigenspace, and this description would make sense for *any* function f on the eigenvalues, not just polynomials: $f(T)$ multiplies the ith eigenspace by $f(\lambda_i)$. The eigenvalues are the spectrum, so the natural domain of f is $\sigma(T)$. The continuous functional calculus makes this precise in infinite dimensions, where T need not have eigenvectors: it produces an operator $f(T)$ for every continuous f on $\sigma(T)$, compatible with the algebra operations and with complex conjugation, and it does so isometrically. The mechanism is a single inequality, $\|p(T)\| = \sup_{\sigma(T)} |p|$, which lets the discrete algebra of polynomials be completed to the continuous algebra $C(\sigma(T))$ by density.

Two facts about the spectrum of a self-adjoint operator drive everything. First, its spectrum is real, so $\sigma(T)$ is a compact subset of $\mathbb{R}$ and $C(\sigma(T))$ is an algebra of functions of a real variable. Second, its spectral radius equals its norm, so the spectrum "sees" the full size of T. Both follow from the single algebraic identity $\|T^2\| = \|T\|^2$, which is the operator-norm signature of self-adjointness.

Theorem 12.2.1: Spectrum of a Self-Adjoint Operator

Let $T \in B(\mathbb{H})$ be self-adjoint. Then $\sigma(T) \subseteq \mathbb{R}$, and the spectral radius equals the norm:

$$r(T) = \|T\|$$

***Proof*:**
Step 1: The spectral radius equals the norm. For self-adjoint T the adjoint identity $\|S^*S\| = \|S\|^2$ (proposition 11.2.7) applied to $S = T$ gives $\|T^2\| = \|T^*T\| = \|T\|^2$. Since T^2 is again self-adjoint ($(T^2)^* = (T^*)^2 = T^2$), the identity iterates: replacing T by T^2 gives $\|T^4\| = \|T^2\|^2 = \|T\|^4$, and by induction

$$\|T^{2^k}\| = \|T\|^{2^k}$$

for every $k \geq 0$. Taking 2^kth roots, $\|T^{2^k}\|^{1/2^k} = \|T\|$ for all k. The spectral radius is the limit of the convergent sequence $\|T^n\|^{1/n}$ (theorem 12.1.11), so its value is read off from any subsequence:

$$r(T) = \lim_{k\to\infty} \|T^{2^k}\|^{1/2^k} = \|T\|$$

Step 2: The spectrum is real. Fix $\lambda = a + ib$ with $a, b \in \mathbb{R}$ and $b \neq 0$; the claim is that $\lambda I - T$ is invertible, so $\lambda \notin \sigma(T)$. For any $x \in \mathbb{H}$,

$$\|(\lambda I - T)x\|^2 = \langle(\lambda I - T)x, (\lambda I - T)x\rangle$$

Expand using $T = T^*$ and $\langle Tx, x\rangle = \langle x, Tx\rangle = \overline{\langle Tx, x\rangle}$, so that $\langle Tx, x\rangle$ is real. Writing $\lambda I - T = (aI - T) + ibI$ and using that $aI - T$ is self-adjoint,

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 + 2\operatorname{Re}\langle(aI - T)x, ibx\rangle$$

The cross term is $2\operatorname{Re}\big(\overline{(ib)}\,\langle(aI - T)x, x\rangle\big) = 2\operatorname{Re}\big(-ib\,\langle(aI - T)x, x\rangle\big)$, and $\langle(aI - T)x, x\rangle$ is real, so the whole term is $2\operatorname{Re}(-ib \cdot (\text{real})) = 0$. Hence

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 \geq b^2\|x\|^2 \tag{12.1}$$

so $\|(\lambda I - T)x\| \geq |b|\,\|x\|$. Thus $\lambda I - T$ is bounded below, hence injective with closed range (proposition 11.2.6). The same bound applies to $(\lambda I - T)^* = \overline{\lambda}I - T$, since $\overline{\lambda} = a - ib$ has the same nonzero imaginary part in modulus; so $\overline{\lambda}I - T$ is injective as well. By proposition 11.2.4 the range of $\lambda I - T$ is dense: $\overline{\operatorname{ran}(\lambda I - T)} = \ker((\lambda I - T)^*)^\perp = \ker(\overline{\lambda}I - T)^\perp = \{0\}^\perp = \mathbb{H}$. A dense closed subspace is everything, so $\lambda I - T$ is a bijection, invertible by the open mapping theorem. Therefore no λ with nonzero imaginary part lies in $\sigma(T)$, giving $\sigma(T) \subseteq \mathbb{R}$, as we sought to show.

The two conclusions play different roles. Reality of the spectrum is what makes $C(\sigma(T))$ an algebra of functions on a compact subset of the real line, so that a function like $\sqrt{\cdot}$ or $|\cdot|$ can be applied to T whenever the spectrum lies where the function is continuous. The equality $r(T) = \|T\|$ is the quantitative heart: it says the operator norm is recovered from the spectrum alone, $\|T\| = \sup\{|\lambda| : \lambda \in \sigma(T)\}$, so nothing about the size of T is invisible to its spectrum. For a general (non-self-adjoint) operator this fails, as the lower bound (13.6) exhibits directly: a nilpotent matrix has spectrum $\{0\}$ yet nonzero norm, so $r < \|\cdot\|$ strictly.

Example 12.2.2: A Multiplication Operator
Let $\mathbb{H} = L^2([0,1])$ and let M_g be multiplication by a real-valued $g \in C([0,1])$: $(M_g f)(t) = g(t)f(t)$. This is self-adjoint, since $\langle M_g f, h\rangle = \int_0^1 g(t)f(t)\overline{h(t)}\,dt = \langle f, M_g h\rangle$ using that g is real. Its spectrum is the range of g: for $\lambda \in \mathbb{C}$, the operator $\lambda I - M_g$ is multiplication by $\lambda - g$, invertible exactly

when $\lambda - g$ is bounded away from zero, that is when $\lambda \notin g([0,1])$. Hence

$$\sigma(M_g) = g([0,1]) \subseteq \mathbb{R}$$

which is a compact real interval (or union of intervals), consistent with theorem 12.2.1. The norm is $\|M_g\| = \sup_t |g(t)| = \max_{\lambda \in \sigma(M_g)} |\lambda| = r(M_g)$, confirming $r = \|\cdot\|$ concretely. When $g(t) = t$, the spectrum is the full interval $[0,1]$ and M_g has no eigenvalues at all: $\lambda f = tf$ forces $f = 0$ almost everywhere. This is the model to keep in mind. The spectrum can be a solid interval with no eigenvectors, and the functional calculus below is exactly what replaces the missing eigenspace decomposition.

With reality and $r(T) = \|T\|$ in hand, the polynomial calculus can be normed. A polynomial p with complex coefficients defines $p(T) \in B(\mathbb{H})$ by substitution; the point is that its operator norm is computed by evaluating $|p|$ on the spectrum. This is where the real spectrum matters: to make $p(T)$ self-adjoint (so that Step 1 of the previous proof applies to it) one restricts to polynomials with *real* coefficients, or more usefully passes to $\overline{p}p$, whose value on the real spectrum is $|p|^2$.

Proposition 12.2.3: The Polynomial Functional Calculus

Let $T \in B(\mathbb{H})$ be self-adjoint. For a polynomial $p(z) = \sum_{k=0}^n \alpha_k z^k$ with $\alpha_k \in \mathbb{C}$, set $p(T) = \sum_{k=0}^n \alpha_k T^k$. Then $p \mapsto p(T)$ is an algebra homomorphism from $\mathbb{C}[z]$ into $B(\mathbb{H})$ with $(\overline{p})(T) = p(T)^*$, where $\overline{p}$ has the conjugate coefficients, and

$$\|p(T)\| = \sup_{\lambda \in \sigma(T)} |p(\lambda)| \tag{12.2}$$

***Proof*:**
That $p \mapsto p(T)$ respects sums and products is immediate from the fact that powers of T commute: $(pq)(T) = p(T)q(T)$ because both sides are the operator obtained by multiplying out the polynomial pq and substituting T. Taking adjoints term by term, $p(T)^* = \sum_k \overline{\alpha_k}(T^*)^k = \sum_k \overline{\alpha_k} T^k = (\overline{p})(T)$, using $T = T^*$.

For (12.2), consider the operator $\overline{p}(T)\,p(T) = (\overline{p}p)(T)$. It is self-adjoint, since $((\overline{p}p)(T))^* = (\overline{\overline{p}p})(T) = (\overline{p}p)(T)$ as $\overline{p}p$ has real values on $\mathbb{R}$ and its coefficient conjugate is itself. By the C^*-identity proposition 11.2.7,

$$\|p(T)\|^2 = \|p(T)^* p(T)\| = \|(\overline{p}p)(T)\|$$

Since $\overline{p}p$ is a self-adjoint operator's defining polynomial, theorem 12.2.1 gives $\|(\overline{p}p)(T)\| = r\big((\overline{p}p)(T)\big)$. Now apply the polynomial spectral mapping theorem: for the polynomial $q = \overline{p}p$,

$$\sigma(q(T)) = q(\sigma(T)) = \{q(\lambda) : \lambda \in \sigma(T)\} \tag{12.3}$$

This is the two-sided refinement of lemma 12.1.9, which supplies the inclusion $q(\sigma(T)) \subseteq \sigma(q(T))$; the reverse inclusion holds because for $\mu \notin q(\sigma(T))$ one factors $q(z) - \mu = c\prod_j (z - \mu_j)$ over $\mathbb{C}$, and each root μ_j satisfies $q(\mu_j) = \mu$, so no μ_j lies in $\sigma(T)$, whence each $\mu_j I - T$ is invertible and $q(T) - \mu I = c\prod_j(\mu_j I - T)$ is a product of invertibles, invertible. Thus $\mu \notin \sigma(q(T))$, proving

(12.3). Taking suprema of $|\cdot|$ over both sides of (12.3) for $q = \overline{p}p$, and using that $q(\lambda) = |p(\lambda)|^2$ for real λ (since $\lambda \in \sigma(T) \subseteq \mathbb{R}$ makes $\overline{p}(\lambda) = \overline{p(\lambda)}$),

$$r\big((\overline{p}p)(T)\big) = \sup_{\mu \in \sigma(q(T))} |\mu| = \sup_{\lambda \in \sigma(T)} |q(\lambda)| = \sup_{\lambda \in \sigma(T)} |p(\lambda)|^2$$

Chaining the equalities, $\|p(T)\|^2 = \sup_{\lambda \in \sigma(T)} |p(\lambda)|^2$, and taking square roots gives (12.2), completing the proof.

Equation (12.2) is the linchpin. Written in the sup norm on $C(\sigma(T))$, it reads $\|p(T)\|_{B(\mathbb{H})} = \|p\|_\infty$, where $\|p\|_\infty = \sup_{\sigma(T)} |p|$: the assignment $p \mapsto p(T)$ is an *isometry* from the polynomials, viewed as a subspace of $C(\sigma(T))$, into $B(\mathbb{H})$. An isometry defined on a dense subspace of a complete space extends uniquely to the whole space, and the polynomials are dense in $C(\sigma(T))$ by Stone-Weierstrass. This is the entire content of the next theorem: the extension exists because the norm is already controlled on the dense subspace.

One subtlety deserves attention before the extension. In (12.2) the norm of $p(T)$ depends on p only through its restriction to $\sigma(T)$: if two polynomials agree on the spectrum, they produce the same operator. This is forced by isometry, $\|(p-q)(T)\| = \sup_{\sigma(T)} |p-q| = 0$ when $p = q$ on $\sigma(T)$, and it is exactly what makes $C(\sigma(T))$, rather than $\mathbb{C}[z]$, the correct domain: the calculus factors through evaluation on the spectrum.

Theorem 12.2.4: The Continuous Functional Calculus

Let $T \in B(\mathbb{H})$ be self-adjoint. There is a unique map

$$\Phi\colon C(\sigma(T)) \to B(\mathbb{H})$$

that is an isometric $*$-homomorphism with $\Phi(1) = I$ and $\Phi(\mathrm{id}) = T$, where 1 is the constant function and $\mathrm{id}(\lambda) = \lambda$. Explicitly, Φ is linear and multiplicative, satisfies $\Phi(\overline{f}) = \Phi(f)^*$, and

$$\|\Phi(f)\| = \|f\|_\infty = \sup_{\lambda \in \sigma(T)} |f(\lambda)|$$

for every $f \in C(\sigma(T))$. One writes $f(T) := \Phi(f)$.

***Proof*:**
Step 1: Density of polynomials. The spectrum $\sigma(T)$ is a compact subset of $\mathbb{R}$ (theorem 12.2.1). The polynomials in the real variable λ form a subalgebra of $C(\sigma(T))$ that contains the constants, separates points (the function id already does), and is closed under complex conjugation (the conjugate of a polynomial is the polynomial with conjugated coefficients). By the Stone-Weierstrass theorem this subalgebra is dense in $C(\sigma(T))$ for the sup norm. Every $f \in C(\sigma(T))$ is thus a uniform limit $f = \lim_m p_m$ of polynomials.

Step 2: Definition of Φ by extension. On the dense subalgebra of polynomials, define $\Phi(p) = p(T)$. By (12.2) this is an isometry into $B(\mathbb{H})$: $\|\Phi(p) - \Phi(q)\| = \|(p-q)(T)\| = \|p - q\|_\infty$. Given $f \in C(\sigma(T))$ and polynomials $p_m \to f$ uniformly, the sequence (p_m) is Cauchy in $\|\cdot\|_\infty$, so $(\Phi(p_m)) = (p_m(T))$ is Cauchy in $B(\mathbb{H})$; since $B(\mathbb{H})$ is complete (it is a Banach space, $\mathbb{H}$ being complete), the limit

$$\Phi(f) := \lim_{m\to\infty} p_m(T)$$

exists in $B(\mathbb{H})$. The limit is independent of the approximating sequence: if $p_m \to f$ and $q_m \to f$, then $\|p_m(T) - q_m(T)\| = \|p_m - q_m\|_\infty \to 0$, so the two limits coincide. Thus Φ is well defined on all of $C(\sigma(T))$.

Step 3: Φ is an isometric $*$-homomorphism. Each property holds for polynomials and passes to the limit because all the operations involved are continuous. For $f = \lim p_m$ and $g = \lim q_m$:

- *Linearity.* $p_m + \alpha q_m \to f + \alpha g$ uniformly, and $\Phi(p_m + \alpha q_m) = p_m(T) + \alpha\, q_m(T) \to \Phi(f) + \alpha\Phi(g)$, so $\Phi(f + \alpha g) = \Phi(f) + \alpha\Phi(g)$.
- *Isometry.* The norm is continuous, so $\|\Phi(f)\| = \lim \|p_m(T)\| = \lim \|p_m\|_\infty = \|f\|_\infty$, using $\|p_m\|_\infty \to \|f\|_\infty$.
- *Multiplicativity.* $p_m q_m \to fg$ uniformly (uniform convergence is stable under products of uniformly bounded sequences, and $\|p_m\|_\infty, \|q_m\|_\infty$ are bounded as convergent sequences). Multiplication in $B(\mathbb{H})$ is jointly continuous on bounded sets: $\|p_m(T)q_m(T) - \Phi(f)\Phi(g)\| \leq \|p_m(T)\|\,\|q_m(T) - \Phi(g)\| + \|p_m(T) - \Phi(f)\|\,\|\Phi(g)\| \to 0$. Hence $\Phi(fg) = \lim(p_m q_m)(T) = \lim p_m(T)q_m(T) = \Phi(f)\Phi(g)$.
- *$*$-preservation.* $\overline{p_m} \to \overline{f}$ uniformly and adjunction is a (conjugate-linear) isometry, hence continuous, so $\Phi(\overline{f}) = \lim \overline{p_m}(T) = \lim p_m(T)^* = (\lim p_m(T))^* = \Phi(f)^*$, using $(\overline{p})(T) = p(T)^*$ from proposition 12.2.3.

The normalizations $\Phi(1) = I$ and $\Phi(\mathrm{id}) = T$ hold already at the polynomial level, since the constant 1 and id are polynomials with $1(T) = I$ and $\mathrm{id}(T) = T$.

Step 4: Uniqueness. Suppose $\Psi\colon C(\sigma(T)) \to B(\mathbb{H})$ is any isometric $*$-homomorphism with $\Psi(1) = I$ and $\Psi(\mathrm{id}) = T$. Multiplicativity and linearity force $\Psi(\lambda^k) = T^k$ for every k, so $\Psi(p) = p(T) = \Phi(p)$ on every polynomial p. Two continuous maps that agree on a dense set agree everywhere; both Ψ and Φ are continuous, being isometries, and the polynomials are dense by Step 1. Hence $\Psi = \Phi$, completing the proof.

The theorem converts the algebra $C(\sigma(T))$ into an algebra of operators, faithfully and without distortion of norms. Three features of the map $f \mapsto f(T)$ are worth isolating. It is an *algebra* map, so composition of the calculus with algebra operations is transparent: $(f+g)(T) = f(T) + g(T)$, $(fg)(T) = f(T)g(T)$, and in particular any two operators $f(T), g(T)$ produced by the calculus commute, because $C(\sigma(T))$ is commutative. It is a *$*$-map*, so real-valued f give self-adjoint $f(T)$ and $|f| = 1$ on the spectrum gives unitary $f(T)$; the calculus respects the involution as well as the multiplication. And it is *isometric*, so estimates on operators reduce to estimates on scalar functions, $\|f(T)\| = \|f\|_\infty$, which is the single most useful computational consequence. Together these say that studying the self-adjoint T through functions of it is the same as studying scalar functions on $\sigma(T)$, with no loss.

The construction is a completion argument, so it is worth naming what it gives over the polynomial calculus: functions that are continuous but not polynomial. The most consequential of these is the modulus.

Example 12.2.5: The Absolute Value and the Positive Part

Let $T \in B(\mathbb{H})$ be self-adjoint, so $\sigma(T) \subseteq \mathbb{R}$. The function $f(\lambda) = |\lambda|$ is continuous on $\sigma(T)$ but is not a polynomial; the calculus produces an operator

$$|T| := f(T)$$

the *absolute value* of T. By the $*$-property, $|T|$ is self-adjoint (since f is real-valued), and by multiplicativity $|T|^2 = (f^2)(T)$ where $f(\lambda)^2 = \lambda^2$, so $|T|^2 = T^2$: the operator $|T|$ is a square root of T^2. It is a positive operator, since $|T| = g(T)^* g(T)$ for $g(\lambda) = \sqrt{|\lambda|}$ (continuous on $\sigma(T)$), giving $\langle |T|x, x\rangle = \|g(T)x\|^2 \geq 0$. Likewise the positive and negative parts $\lambda^+ = \max(\lambda, 0)$ and $\lambda^- = \max(-\lambda, 0)$ are continuous on $\sigma(T)$ and yield positive operators $T^+ = f^+(T)$, $T^- = f^-(T)$ with

$$T = T^+ - T^-, \qquad |T| = T^+ + T^-, \qquad T^+T^- = 0$$

the last because $\lambda^+\lambda^- = 0$ identically, so multiplicativity forces $T^+T^- = (f^+f^-)(T) = 0$. These identities decompose a self-adjoint operator into its positive and negative parts exactly as one decomposes a real function, and none of $|T|$, T^+, T^- is reachable by polynomials in T: the continuous calculus is what makes them available. For the multiplication operator M_g of example 12.2.2 with g real-valued, $|M_g| = M_{|g|}$ and $M_g^{\pm} = M_{g^{\pm}}$, the calculus acting by composition on the multiplier.

The isometry (12.2) was the input to the construction; the same identity, now for arbitrary continuous f, is the output, and it upgrades to a full description of the spectrum of $f(T)$. The polynomial spectral mapping theorem used inside the proof of proposition 12.2.3 extends verbatim to the continuous calculus: the spectrum of $f(T)$ is the image of $\sigma(T)$ under f.

Proposition 12.2.6: Continuous Spectral Mapping

Let $T \in B(\mathbb{H})$ be self-adjoint and $f \in C(\sigma(T))$. Then

$$\sigma(f(T)) = f(\sigma(T)) = \{f(\lambda) : \lambda \in \sigma(T)\}$$

***Proof*:**

Both sides are compact subsets of $\mathbb{C}$: $f(\sigma(T))$ is the continuous image of the compact set $\sigma(T)$, and $\sigma(f(T))$ is compact by theorem 12.1.11. It suffices to prove the two inclusions.

Step 1: $f(\sigma(T)) \subseteq \sigma(f(T))$. Fix $\lambda_0 \in \sigma(T)$ and set $\mu = f(\lambda_0)$; the claim is $\mu \in \sigma(f(T))$, that is $f(T) - \mu I$ is not invertible. Suppose it were, with inverse S. Choose polynomials $p_m \to f$ uniformly on $\sigma(T)$ (theorem 12.2.4, Step 1). Then $p_m(T) \to f(T)$ in norm, so for large m the operator $p_m(T) - \mu I$ is invertible (the invertibles are open, proposition 12.1.6), with inverses bounded uniformly near S. But $p_m(\lambda_0) \to f(\lambda_0) = \mu$, and by the polynomial spectral mapping (12.3), $p_m(\lambda_0) \in p_m(\sigma(T)) = \sigma(p_m(T))$, so $p_m(T) - p_m(\lambda_0)I$ is *not* invertible. Since $p_m(\lambda_0) \to \mu$ and inversion is continuous where defined, the non-invertible elements $p_m(T) - p_m(\lambda_0)I$ converge to $f(T) - \mu I$; the non-invertibles form a closed set (the complement of the open group of invertibles), so the limit $f(T) - \mu I$ is not invertible. This contradicts the supposition, so $\mu \in \sigma(f(T))$.

Step 2: $\sigma(f(T)) \subseteq f(\sigma(T))$. Let $\mu \notin f(\sigma(T))$; the claim is $\mu \notin \sigma(f(T))$, that is $f(T) - \mu I$ is invertible. Because μ is not in the compact set $f(\sigma(T))$, the function

$$g(\lambda) = \frac{1}{f(\lambda) - \mu}$$

is continuous on $\sigma(T)$: its denominator $f(\lambda) - \mu$ never vanishes on $\sigma(T)$, so $g \in C(\sigma(T))$. Applying the calculus and its multiplicativity (theorem 12.2.4) to the identity $g \cdot (f - \mu) = 1$ in $C(\sigma(T))$,

$$g(T)\,(f(T) - \mu I) = (f(T) - \mu I)\,g(T) = I$$

so $g(T)$ is a two-sided inverse of $f(T) - \mu I$. Hence $f(T) - \mu I$ is invertible and $\mu \notin \sigma(f(T))$, as we sought to show.

The spectral mapping theorem is the statement that the calculus is compatible with the spectrum in the strongest possible sense: applying f to T applies f to its spectrum. It repackages the isometry $\|f(T)\| = \|f\|_\infty$ (take suprema of $|\cdot|$ over $\sigma(f(T)) = f(\sigma(T))$ and use $r(f(T)) = \|f(T)\|$ when f is real-valued) and it makes the calculus *internally consistent*: applying a second function $h \in C(\sigma(f(T)))$ to $f(T)$ agrees with applying the composite $h \circ f \in C(\sigma(T))$ to T, since both are computed by uniform approximation and the spectra match. For the multiplication operator M_g of example 12.2.2, $\sigma(f(M_g)) = f(\sigma(M_g)) = f(g([0,1])) = (f \circ g)([0,1])$, which is $\sigma(M_{f\circ g})$, recovering $f(M_g) = M_{f\circ g}$ and matching the concrete calculation there. The one gap the continuous calculus leaves open is discontinuous f such as indicators of subsets of the spectrum, which are exactly the functions that would carve $\mathbb{H}$ into spectral subspaces; supplying those is the task of the Borel calculus in the next section.

12.3 The Borel Functional Calculus and Spectral Projections

The continuous functional calculus (theorem 12.2.4) lets a continuous function f act on a self-adjoint operator T, but continuity is a severe restriction on the functions available. To slice the operator apart the way a diagonal matrix is sliced by its eigenprojections, the indicator χ_Ω of a Borel set $\Omega \subseteq \sigma(T)$ must be admissible, and χ_Ω is almost never continuous. The task of this section is to enlarge the calculus $f \mapsto f(T)$ from $C(\sigma(T))$ to the bounded Borel functions $B_b(\sigma(T))$. The enlargement is forced: each pair of vectors $x, y \in \mathbb{H}$ turns the continuous calculus into a bounded functional on $C(\sigma(T))$, and the Riesz representation theorem (theorem 8.3.3) converts that functional into a complex measure against which any bounded Borel function may be integrated. The indicators χ_Ω then produce the *spectral projections* $E(\Omega)$, the objects that will assemble into the projection-valued measure of the spectral theorem.

Throughout, $T \in B(\mathbb{H})$ is self-adjoint, so $\sigma(T) \subseteq \mathbb{R}$ is a compact subset of the line (theorem 12.2.1), and $f \mapsto f(T)$ denotes the isometric $*$-homomorphism $C(\sigma(T)) \to B(\mathbb{H})$ of theorem 12.2.4. Write $B_b(\sigma(T))$ for the bounded Borel functions on $\sigma(T)$, a commutative C^*-algebra under pointwise operations, complex conjugation, and the supremum norm $\|f\|_\infty = \sup_{\lambda \in \sigma(T)} |f(\lambda)|$; it contains

$C(\sigma(T))$ as a subalgebra.

The first step is to attach a measure to each pair of vectors. Fix $x, y \in \mathbb{H}$. The continuous calculus already gives a map $C(\sigma(T)) \to \mathbb{C}$ by

$$f \longmapsto \langle f(T)x, y \rangle$$

This is linear in f (the calculus is an algebra homomorphism, hence linear), and it is bounded, because the calculus is isometric:

$$|\langle f(T)x, y \rangle| \leq \|f(T)x\| \, \|y\| \leq \|f(T)\| \, \|x\| \, \|y\| = \|f\|_\infty \|x\| \, \|y\|$$

So $f \mapsto \langle f(T)x, y \rangle$ is a bounded linear functional on $C(\sigma(T))$ of norm at most $\|x\| \, \|y\|$. Since $\sigma(T)$ is compact, $C(\sigma(T)) = C_0(\sigma(T))$, and theorem 8.3.3 identifies its dual with the complex regular Borel measures on $\sigma(T)$. There is therefore a unique such measure, which we call $\mu_{x,y}$, representing the functional.

Definition 12.3.1: Spectral Measures $\mu_{x,y}$

Let $T \in B(\mathbb{H})$ be self-adjoint and $x, y \in \mathbb{H}$. The *spectral measure* associated to the pair (x, y) is the unique complex regular Borel measure $\mu_{x,y}$ on $\sigma(T)$ furnished by theorem 8.3.3 for the bounded functional $f \mapsto \langle f(T)x, y \rangle$ on $C(\sigma(T))$; that is,

$$\langle f(T)x, y \rangle = \int_{\sigma(T)} f \, d\mu_{x,y} \qquad (f \in C(\sigma(T)))$$

Its total variation satisfies $\|\mu_{x,y}\| \leq \|x\| \, \|y\|$.

The bound $\|\mu_{x,y}\| \leq \|x\| \, \|y\|$ is exactly the operator norm of the functional, which theorem 8.3.3 equates with the total variation $|\mu_{x,y}|(\sigma(T))$. The family $\{\mu_{x,y}\}$ inherits the sesquilinearity of the inner product: for fixed y the map $x \mapsto \mu_{x,y}$ is linear, and for fixed x the map $y \mapsto \mu_{x,y}$ is conjugate-linear, since $\langle f(T)x, y \rangle$ has these properties in x and y and the representing measure is unique. This is the raw material for the extension: any bounded Borel f can be integrated against $\mu_{x,y}$, and the resulting number will play the role of $\langle f(T)x, y \rangle$.

Example 12.3.2: Spectral Measures for a Diagonal Operator

Let $\mathbb{H} = \ell^2(\mathbb{N})$ and let T be the diagonal operator $Te_n = \lambda_n e_n$ on the standard orthonormal basis (e_n), where (λ_n) is a bounded real sequence; T is self-adjoint and $\sigma(T) = \overline{\{\lambda_n \mid n \in \mathbb{N}\}}$. For $f \in C(\sigma(T))$ the continuous calculus acts diagonally, $f(T)e_n = f(\lambda_n)e_n$, so for $x = \sum_n a_n e_n$ and $y = \sum_n b_n e_n$,

$$\langle f(T)x, y \rangle = \sum_n f(\lambda_n) \, a_n \overline{b_n} = \int_{\sigma(T)} f \, d\left(\sum_n a_n \overline{b_n} \, \delta_{\lambda_n} \right)$$

where δ_{λ_n} is the point mass at λ_n. Thus $\mu_{x,y} = \sum_n a_n \overline{b_n} \, \delta_{\lambda_n}$, a purely atomic measure supported on the eigenvalues, with total variation $\sum_n |a_n| \, |b_n| \leq \|x\| \, \|y\|$ by Cauchy-Schwarz. Taking $x = y = e_k$ gives $\mu_{e_k, e_k} = \delta_{\lambda_k}$: the diagonal spectral measure at a basis vector is the point mass at its eigenvalue. Integrating $\chi_{\{\lambda_k\}}$ against it will recover the eigenprojection onto $\mathbb{C}e_k$.

The measures $\mu_{x,y}$ now drive the extension of the calculus to bounded Borel functions. The definition $\langle f(T)x, y \rangle = \int f \, d\mu_{x,y}$ for Borel f must be shown to determine a genuine bounded operator $f(T)$, and then to respect the algebra operations. Boundedness comes from the estimate on the total variation; the operator is produced by the correspondence between bounded sesquilinear forms and bounded operators.

Theorem 12.3.3: The Borel Functional Calculus

Let $T \in B(\mathbb{H})$ be self-adjoint. There is a unique linear map

$$\Phi_b \colon B_b(\sigma(T)) \longrightarrow B(\mathbb{H}), \qquad \Phi_b(f) =: f(T)$$

extending the continuous calculus of theorem 12.2.4 and characterized by

$$\langle f(T)x, y\rangle = \int_{\sigma(T)} f \, d\mu_{x,y} \qquad (x, y \in \mathbb{H},\ f \in B_b(\sigma(T))) \tag{12.4}$$

It has the following properties.

1. Φ_b is a $*$-homomorphism: $(fg)(T) = f(T)g(T)$ and $\overline{f}(T) = f(T)^*$, with $1(T) = I$.
2. Φ_b is a contraction: $\|f(T)\| \leq \|f\|_\infty$.
3. Φ_b is boundedly convergent: if $(f_n) \subseteq B_b(\sigma(T))$ satisfies $\sup_n \|f_n\|_\infty < \infty$ and $f_n \to f$ pointwise, then $f_n(T)x \to f(T)x$ for every $x \in \mathbb{H}$ (convergence in the strong operator topology).
4. If $f \geq 0$ then $f(T) \geq 0$; consequently $f(T)$ is self-adjoint when f is real-valued.

***Proof*:**
The proof runs in five steps: build the operator from the sesquilinear form and prove the norm bound (Step 1), establish the adjoint identity (Step 2), promote bounded convergence from the continuous calculus to Borel functions weakly (Step 3), deduce multiplicativity from that weak convergence by a monotone-class argument (Step 4), and finally upgrade the convergence to the strong topology (Step 5). Positivity and uniqueness follow.

Step 1: $f(T)$ is a well-defined bounded operator. Fix $f \in B_b(\sigma(T))$ and define

$$B(x, y) = \int_{\sigma(T)} f \, d\mu_{x,y}$$

The assignment $(x, y) \mapsto \mu_{x,y}$ is linear in x and conjugate-linear in y, and f is fixed, so B is sesquilinear (linear in x, conjugate-linear in y). It is bounded:

$$|B(x, y)| = \left| \int f \, d\mu_{x,y} \right| \leq \|f\|_\infty \, |\mu_{x,y}|(\sigma(T)) = \|f\|_\infty \|\mu_{x,y}\| \leq \|f\|_\infty \|x\| \, \|y\|$$

By the correspondence between bounded sesquilinear forms and bounded operators on a Hilbert space, there is a unique $f(T) \in B(\mathbb{H})$ with $\langle f(T)x, y\rangle = B(x, y)$ for all x, y, and $\|f(T)\| = \sup_{\|x\|, \|y\| \leq 1} |B(x, y)| \leq \|f\|_\infty$. This gives (12.4), property (2), and, by uniqueness of the representing operator, linearity of Φ_b in f. When $f \in C(\sigma(T))$ the defining property of $\mu_{x,y}$ makes $B(x, y) = \langle f(T)x, y\rangle$ for the *continuous*-calculus $f(T)$, so Φ_b agrees with theorem 12.2.4 there and is a genuine extension.

Step 2: the adjoint identity. For $f \in B_b(\sigma(T))$ and any x, y,

$$\langle \overline{f}(T)x, y\rangle = \int \overline{f}\, d\mu_{x,y} = \overline{\int f\, d\overline{\mu_{x,y}}}$$

The measure $\overline{\mu_{x,y}}$ (conjugate of each complex value) represents $g \mapsto \int g\, d\mu_{y,x}$ over continuous g: for such g, $\overline{\langle \overline{g}(T)x, y\rangle} = \langle y, \overline{g}(T)x\rangle = \langle g(T)y, x\rangle = \int g\, d\mu_{y,x}$, where the middle step uses $\overline{g}(T)^* = g(T)$ from the continuous calculus. By uniqueness of the representing measure, $\overline{\mu_{x,y}} = \mu_{y,x}$. Hence $\langle \overline{f}(T)x, y\rangle = \overline{\int f\, d\mu_{y,x}} = \overline{\langle f(T)y, x\rangle} = \langle x, f(T)y\rangle = \langle f(T)^*x, y\rangle$ for all x, y, so $\overline{f}(T) = f(T)^*$. Applying this to a real f gives $f(T) = f(T)^*$, the self-adjointness in (4).

Step 3: bounded convergence, weakly. Let (f_n) be uniformly bounded, $\|f_n\|_\infty \le M$, with $f_n \to f$ pointwise on $\sigma(T)$. For fixed x, y the sequence integrates against the finite measure $\mu_{x,y}$: each $|f_n| \le M$ is dominated by the constant $M \in L^1(|\mu_{x,y}|)$ (finite since $|\mu_{x,y}|(\sigma(T)) \le \|x\|\,\|y\|$), so dominated convergence gives

$$\langle f_n(T)x, y\rangle = \int f_n\, d\mu_{x,y} \longrightarrow \int f\, d\mu_{x,y} = \langle f(T)x, y\rangle$$

Thus $f_n(T) \to f(T)$ in the weak operator topology whenever $\|f_n\|_\infty$ is bounded and $f_n \to f$ pointwise. (Strong convergence is deferred to Step 5, once multiplicativity is available.)

Step 4: multiplicativity, by a bounded-convergence / monotone-class argument. Multiplicativity $(fg)(T) = f(T)g(T)$ holds on $C(\sigma(T))$ by theorem 12.2.4. Extend it in two stages, each using only the weak convergence of Step 3.

First fix $g \in C(\sigma(T))$ and let

$$\mathcal{M}_g = \{f \in B_b(\sigma(T)) \mid (fg)(T) = f(T)g(T)\}$$

This is a linear subspace containing $C(\sigma(T))$, and it is closed under uniformly bounded pointwise limits: if $f_n \in \mathcal{M}_g$ with $\sup_n \|f_n\|_\infty < \infty$ and $f_n \to f$ pointwise, then $f_n g \to fg$ pointwise and boundedly, so by Step 3 both $(f_n g)(T) \to (fg)(T)$ and $f_n(T) \to f(T)$ weakly. Testing against x, y and using that $g(T)$ is bounded,

$$\langle (fg)(T)x, y\rangle = \lim_n \langle (f_n g)(T)x, y\rangle = \lim_n \langle f_n(T)\, g(T)x, y\rangle = \langle f(T)g(T)x, y\rangle$$

the last step applying weak convergence of $f_n(T)$ to the fixed vector $g(T)x$; so $f \in \mathcal{M}_g$. A linear class of bounded functions that contains $C(\sigma(T))$ and is closed under uniformly bounded pointwise limits contains every bounded Borel function. Indeed, for an open $U \subseteq \sigma(T)$ the Urysohn functions $g_n \in C(\sigma(T))$ with $0 \le g_n \uparrow \chi_U$ are uniformly bounded and converge pointwise to χ_U, so $\chi_U \in \mathcal{M}_g$; the collection $\mathcal{D} = \{\Omega \subseteq \sigma(T) \mid \chi_\Omega \in \mathcal{M}_g\}$ then contains the open sets, contains $\sigma(T)$ (as $1 \in C(\sigma(T)) \subseteq \mathcal{M}_g$), is closed under complements ($\chi_{\sigma(T)\setminus\Omega} = 1 - \chi_\Omega \in \mathcal{M}_g$ by linearity), and is closed under countable disjoint unions (for pairwise disjoint Ω_n the indicators $\chi_{\bigcup_{n\le N}\Omega_n} = \sum_{n\le N} \chi_{\Omega_n}$ are uniformly bounded and increase pointwise to $\chi_{\bigcup_n \Omega_n}$, a bounded pointwise limit). Thus $\mathcal{D}$ is a Dynkin system. The open sets are closed under finite intersection, so they form a π-system generating the Borel σ-algebra; by Dynkin's π-λ theorem $\mathcal{D}$ contains the σ-algebra they generate, that is, every Borel set. (Closure under complements and disjoint unions alone gives only a Dynkin system, not a σ-algebra; the π-λ theorem supplies the passage to all Borel sets, since products $\chi_A\chi_B$ are not yet known to lie in $\mathcal{M}_g$.) Every bounded Borel function is a uniformly bounded pointwise limit of simple functions built from these indicators, so $\mathcal{M}_g = B_b(\sigma(T))$. Thus $(fg)(T) = f(T)g(T)$ for all $f \in B_b(\sigma(T))$ and $g \in C(\sigma(T))$.

Now fix $f \in B_b(\sigma(T))$ and repeat with $\mathcal{N}_f = \{g \in B_b(\sigma(T)) \mid (fg)(T) = f(T)g(T)\}$. The stage just completed shows $C(\sigma(T)) \subseteq \mathcal{N}_f$, and the identical monotone-class argument shows $\mathcal{N}_f$ closed under uniformly bounded pointwise limits, so $\mathcal{N}_f = B_b(\sigma(T))$. Hence $(fg)(T) = f(T)g(T)$ for all bounded Borel f, g, which with Step 2 and $1(T) = I$ (inherited from the continuous calculus) makes Φ_b a $*$-homomorphism, property (1).

Positivity. If $f \geq 0$ is bounded Borel, write $f = h^2$ with $h = \sqrt{f} \in B_b(\sigma(T))$ real; then $f(T) = h(T)^2 = h(T)^* h(T)$ by property (1) and Step 2, so $\langle f(T)x, x\rangle = \|h(T)x\|^2 \geq 0$, giving property (4).

Step 5: upgrading to strong convergence. With multiplicativity in hand, the weak convergence of Step 3 improves to strong. Keep $\|f_n\|_\infty \leq M$ and $f_n \to f$ pointwise, and fix x. The scalar measure $\mu_{x,x}$ is finite and positive with total mass $\|x\|^2$: for real Borel $u \geq 0$, writing $u = w^2$ with $w = \sqrt{u}$ real Borel gives $\int u \, d\mu_{x,x} = \langle u(T)x, x\rangle = \langle w(T)^* w(T)x, x\rangle = \|w(T)x\|^2 \geq 0$ by property (1) and Step 2, and $\mu_{x,x}(\sigma(T)) = \langle 1(T)x, x\rangle = \|x\|^2$. Applying the $*$-homomorphism to $|f_n - f|^2 = \overline{(f_n - f)}(f_n - f)$,

$$\|f_n(T)x - f(T)x\|^2 = \langle (f_n - f)(T)^*(f_n - f)(T)x, x\rangle = \int |f_n - f|^2 \, d\mu_{x,x}$$

The integrand is dominated by the constant $(2M)^2 \in L^1(\mu_{x,x})$ and tends to 0 pointwise, so dominated convergence sends the integral to 0. Hence $f_n(T)x \to f(T)x$ in norm for every x, which is property (3).

Uniqueness. Any linear extension satisfying (12.4) is determined on every pair (x, y) by the measure $\mu_{x,y}$, hence is unique, completing the proof.

The extension trades one property for another. The continuous calculus is an *isometry*; the Borel calculus is only a *contraction* (property (2)), because a discontinuous f may have $\|f(T)\| < \|f\|_\infty$ once f is altered off the "mass" of the measures $\mu_{x,x}$ (a function supported where T has no spectrum in the measure-theoretic sense contributes nothing). What is gained is closure under bounded pointwise limits (property (3)): the calculus now commutes with the limit operations of measure theory, not with uniform limits alone. That single gain is what admits the indicators χ_Ω, and with them the entire machinery of projections and integration developed next. The $*$-homomorphism property (1) says the algebraic structure survives intact: products, adjoints, and the unit all transport from functions to operators exactly as for continuous f.

Example 12.3.4: Borel Calculus on a Diagonal Operator

Continue with the diagonal $Te_n = \lambda_n e_n$ on $\ell^2(\mathbb{N})$ from example 12.3.2. For a bounded Borel f on $\sigma(T)$, formula (12.4) with $\mu_{e_n,e_m} = \delta_{\lambda_n}$ if $n = m$ and $\mu_{e_n,e_m} = 0$ for $n \neq m$ gives

$$\langle f(T)e_n, e_m\rangle = \int_{\sigma(T)} f \, d\mu_{e_n,e_m} = f(\lambda_n)\, \delta_{nm}$$

so $f(T)e_n = f(\lambda_n)e_n$: the Borel calculus still acts diagonally, now for *any* bounded Borel f, not just continuous f. Taking $f = \chi_{\{\lambda_k\}}$ (the indicator of a single eigenvalue, a genuinely discontinuous function when λ_k is a limit of other eigenvalues) yields the operator $e_n \mapsto \chi_{\{\lambda_k\}}(\lambda_n)e_n$, which annihilates every e_n with $\lambda_n \neq \lambda_k$ and fixes those with $\lambda_n = \lambda_k$. This is precisely the orthogonal projection onto the eigenspace $\overline{\operatorname{span}}\,\{e_n \mid \lambda_n = \lambda_k\}$. The isometry defect is visible here: if some λ_k is isolated with a one-dimensional eigenspace, $\|\chi_{\{\lambda_k\}}\|_\infty = 1 = \|\chi_{\{\lambda_k\}}(T)\|$, but χ_Ω for a Borel Ω disjoint from $\{\lambda_n \mid n\} \cup \{0\}$ has $\|\chi_\Omega\|_\infty = 1$ while $\chi_\Omega(T) = 0$.

With the calculus in hand, the indicators are now legitimate arguments, and their operators are the spectral projections. These are the building blocks that a general $f(T)$ integrates against, the operator analogue of the sets a scalar integral integrates over.

Definition 12.3.5: Spectral Projection

Let $T \in B(\mathbb{H})$ be self-adjoint. For a Borel set $\Omega \subseteq \sigma(T)$, the *spectral projection* of T over Ω is the operator

$$E(\Omega) := \chi_\Omega(T)$$

the Borel calculus (theorem 12.3.3) applied to the indicator function χ_Ω.

Each $E(\Omega)$ deserves its name: it is an orthogonal projection in the sense of definition 11.2.15.

Proposition 12.3.6: Spectral Projections Are Orthogonal Projections

For every Borel $\Omega \subseteq \sigma(T)$, the operator $E(\Omega) = \chi_\Omega(T)$ is an orthogonal projection: $E(\Omega) = E(\Omega)^* = E(\Omega)^2$.

***Proof*:**
The indicator satisfies $\chi_\Omega^2 = \chi_\Omega$ and $\overline{\chi_\Omega} = \chi_\Omega$ (it is real-valued and idempotent). Applying the $*$-homomorphism Φ_b of theorem 12.3.3, part (1),

$$E(\Omega)^2 = \chi_\Omega(T)^2 = (\chi_\Omega^2)(T) = \chi_\Omega(T) = E(\Omega), \qquad E(\Omega)^* = \overline{\chi_\Omega}(T) = \chi_\Omega(T) = E(\Omega)$$

so $E(\Omega)$ is self-adjoint and idempotent, hence an orthogonal projection by definition 11.2.15, as we sought to show.

An orthogonal projection is completely described by its range, a closed subspace of $\mathbb{H}$, and $I - E(\Omega) = \chi_{\sigma(T)\setminus\Omega}(T) = E(\sigma(T) \setminus \Omega)$ is the projection onto the orthogonal complement. The range of $E(\Omega)$ is the "part of $\mathbb{H}$ that T places over Ω", made precise in example 12.3.7 for the diagonal case and, in general, by the spectral theorem of the next section. When $\Omega = \{\lambda\}$ is a single point and λ is an eigenvalue, $E(\{\lambda\})$ is the orthogonal projection onto the eigenspace $\ker(T - \lambda I)$; when λ is in the spectrum but not an eigenvalue (continuous spectrum), $E(\{\lambda\}) = 0$ even though small neighborhoods of λ carry nonzero projections. This is the mechanism by which the calculus handles operators with no eigenvectors at all.

Example 12.3.7: Spectral Projections of a Diagonal Operator

For the diagonal $Te_n = \lambda_n e_n$ on $\ell^2(\mathbb{N})$, example 12.3.4 gives $E(\Omega)e_n = \chi_\Omega(\lambda_n)e_n$, so $E(\Omega)$ is the orthogonal projection onto the closed span of the basis vectors whose eigenvalue lies in Ω:

$$\operatorname{ran} E(\Omega) = \overline{\operatorname{span}}\,\{e_n \mid \lambda_n \in \Omega\}$$

Two special cases fix the intuition. With $\Omega = \sigma(T)$ every $\lambda_n \in \Omega$, so $E(\sigma(T)) = I$; with $\Omega = \varnothing$ no eigenvalue qualifies, so $E(\varnothing) = 0$. For $\Omega = [0, \infty)$ the projection $E([0, \infty))$ cuts $\mathbb{H}$ into the closed span of the eigenvectors with nonnegative eigenvalue and its complement, the "positive part" of the operator. These are the countable analogue of grouping the diagonal entries of a matrix by which subset of the spectrum they fall in and reading off the corresponding coordinate projection.

Assembled over all Borel sets, these projections form a set function. The map $\Omega \mapsto E(\Omega)$ carries the Boolean algebra of Borel sets, with intersection and complement, to a commuting family of projections, with operator product and orthogonal complement. The correspondence is exact, and its countable additivity in the strong topology is what makes E a measure valued in projections rather than a mere finitely additive assignment.

Proposition 12.3.8: Properties of the Spectral Projections

Let $T \in B(\mathbb{H})$ be self-adjoint and $\Omega, \Omega', (\Omega_n)_{n\in\mathbb{N}}$ Borel subsets of $\sigma(T)$. The spectral projections satisfy:

1. $E(\varnothing) = 0$ and $E(\sigma(T)) = I$;
2. $E(\Omega \cap \Omega') = E(\Omega)E(\Omega')$; in particular the projections commute, and if $\Omega \cap \Omega' = \varnothing$ then $E(\Omega)E(\Omega') = 0$ (their ranges are orthogonal);
3. (countable strong additivity) if the Ω_n are pairwise disjoint with union $\Omega = \bigcup_n \Omega_n$, then

$$E(\Omega)x = \sum_{n=1}^{\infty} E(\Omega_n)x \qquad (x \in \mathbb{H})$$

the series converging in the norm of $\mathbb{H}$.

***Proof*:**
Part (1). The empty set has $\chi_\varnothing = 0$, so $E(\varnothing) = 0(T) = 0$; the full set has $\chi_{\sigma(T)} = 1$, so $E(\sigma(T)) = 1(T) = I$ by the unit property in theorem 12.3.3(1).

Part (2). Indicators multiply as $\chi_\Omega \chi_{\Omega'} = \chi_{\Omega\cap\Omega'}$ pointwise. Applying the multiplicativity of theorem 12.3.3(1),

$$E(\Omega)E(\Omega') = \chi_\Omega(T)\chi_{\Omega'}(T) = (\chi_\Omega\chi_{\Omega'})(T) = \chi_{\Omega\cap\Omega'}(T) = E(\Omega \cap \Omega')$$

Since $\Omega\cap\Omega' = \Omega'\cap\Omega$, this operator equals both $E(\Omega)E(\Omega')$ and $E(\Omega')E(\Omega)$, so the two projections commute. When $\Omega \cap \Omega' = \varnothing$ the product is $E(\varnothing) = 0$; then for any x, y, $\langle E(\Omega)x, E(\Omega')y\rangle = \langle E(\Omega')^*E(\Omega)x, y\rangle = \langle E(\Omega')E(\Omega)x, y\rangle = 0$, so the ranges are orthogonal.

Part (3). Set $S_N = \bigcup_{n=1}^N \Omega_n$, a Borel set, so $\chi_{S_N} = \sum_{n=1}^N \chi_{\Omega_n}$ (the Ω_n are disjoint) and by linearity of the calculus $E(S_N) = \sum_{n=1}^N E(\Omega_n)$. As $N \to \infty$ the indicators converge pointwise and monotonically to χ_Ω,

$$\chi_{S_N} \uparrow \chi_\Omega, \qquad 0 \le \chi_{S_N} \le 1$$

so the family (χ_{S_N}) is uniformly bounded (by 1) and converges pointwise to χ_Ω. The bounded convergence property theorem 12.3.3(3) gives $E(S_N)x \to E(\Omega)x$ in norm for every $x \in \mathbb{H}$, that is,

$$\sum_{n=1}^N E(\Omega_n)x = E(S_N)x \longrightarrow E(\Omega)x$$

which is the asserted norm-convergent series. (The convergence is at the level of vectors, the strong operator topology, not in operator norm: for disjoint Ω_n with nonzero projections the partial sums are projections onto growing subspaces and cannot be operator-norm Cauchy.) This completes the proof.

Part (3) is the reason E is called a projection-valued *measure*: it is countably additive on disjoint unions, in the strong operator topology. The failure of norm convergence noted at the end is not a defect but the correct behavior; it mirrors the fact that in scalar measure theory a countable partition of the space gives a series of measures $\mu(\Omega_n)$ summing to $\mu(\Omega)$, and the operator statement upgrades numbers to orthogonal projections onto orthogonal subspaces. The Pythagorean identity is exact here: for disjoint Ω_n, part (2) makes the ranges of $E(\Omega_n)$ mutually orthogonal, so

$$\|E(\Omega)x\|^2 = \left\|\sum_n E(\Omega_n)x\right\|^2 = \sum_n \|E(\Omega_n)x\|^2$$

and applying this to $\Omega = \sigma(T)$ with any partition decomposes $\|x\|^2 = \|E(\sigma(T))x\|^2$ into the masses the vector places over the pieces of the spectrum. The scalar measure $\Omega \mapsto \langle E(\Omega)x, x\rangle = \|E(\Omega)x\|^2$ is then a genuine finite positive Borel measure of total mass $\|x\|^2$, the diagonal spectral measure $\mu_{x,x}$ of definition 12.3.1 recovered from the projections. This positive measure is the object the spectral theorem integrates λ against to write $T = \int \lambda\, dE(\lambda)$, the construction taken up in section 12.4.

12.4 The Spectral Theorem for Bounded Self-Adjoint Operators

The Borel calculus of the previous section produced, for each self-adjoint $T \in B(\mathbb{H})$ and each Borel set $\Omega \subseteq \sigma(T)$, an orthogonal projection $E(\Omega) = \chi_\Omega(T)$, and recorded its arithmetic (proposition 12.3.8): the projections multiply the way the sets intersect, and they add up strongly over disjoint unions. This section reads that family the other way around. Packaged as a single object, the assignment $\Omega \mapsto E(\Omega)$ is a measure whose values are projections; against it one integrates a bounded Borel function to recover $f(T)$, and integrating the coordinate function λ itself returns T. The identity $T = \int_{\sigma(T)} \lambda\, dE(\lambda)$ is the resolution of the identity, and it is the exact infinite-dimensional replacement for writing a Hermitian matrix as $\sum_i \lambda_i P_i$ over its eigenvalues. A second reading of the same data trades the abstract projections for concrete multiplication operators: every bounded self-adjoint operator is, up to a unitary change of coordinates, multiplication by the independent variable on a direct sum of L^2 spaces.

The projection family from proposition 12.3.8 has exactly the structure that makes it integrable. Abstracting that structure gives the central object of the section.

Definition 12.4.1: Projection-Valued Measure

Let $K \subseteq \mathbb{R}$ (or $K \subseteq \mathbb{C}$) be compact and let $\mathcal{B}(K)$ denote its Borel σ-algebra. A *projection-valued measure* on K is a map $E\colon \mathcal{B}(K) \to B(\mathbb{H})$ assigning to each Borel set Ω an orthogonal projection $E(\Omega)$ (definition 11.2.15), such that

1. $E(\varnothing) = 0$ and $E(K) = I$;
2. $E(\Omega \cap \Omega') = E(\Omega)E(\Omega')$ for all Borel Ω, Ω';
3. for every countable family $\{\Omega_n\}$ of pairwise disjoint Borel sets and every $x \in \mathbb{H}$,

$$E\Big(\bigcup_n \Omega_n\Big)x = \sum_n E(\Omega_n)x$$

the series converging in the norm of $\mathbb{H}$.

Condition (2) with $\Omega = \Omega'$ recovers idempotence $E(\Omega)^2 = E(\Omega)$, which is built into "projection" already; its content is that for disjoint Ω, Ω' the ranges are orthogonal, since $E(\Omega)E(\Omega') = E(\varnothing) = 0$. The convergence in (3) is strong, not in operator norm: for a diagonal operator the partial sums $\sum_{n \le N} E(\Omega_n)$ are projections onto growing subspaces and never approach $E(\bigcup_n \Omega_n)$ uniformly unless the tail is eventually trivial. So a projection-valued measure behaves like an ordinary measure evaluated one vector at a time, and it is exactly this vectorwise reading that the integral below exploits. The spectral projections of definition 12.3.5 are the motivating instance.

Example 12.4.2: The Spectral Projections of a Self-Adjoint Operator

Let $T \in B(\mathbb{H})$ be self-adjoint and set $K = \sigma(T) \subseteq \mathbb{R}$ (theorem 12.2.1). Define $E(\Omega) = \chi_\Omega(T)$ for Borel $\Omega \subseteq K$, the spectral projection of definition 12.3.5. Then E is a projection-valued measure on K: each $E(\Omega)$ is an orthogonal projection, and conditions (1)-(3) are precisely the properties collected in proposition 12.3.8 ($E(\varnothing) = \chi_\varnothing(T) = 0$, $E(\sigma(T)) = \chi_{\sigma(T)}(T) = I$, multiplicativity from $\chi_{\Omega\cap\Omega'} = \chi_\Omega\chi_{\Omega'}$, and strong countable additivity). For a self-adjoint operator on $\mathbb{C}^n$ with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ and eigenprojections $P_1, \ldots, P_m$, the spectrum is the finite set $\{\lambda_1, \ldots, \lambda_m\}$ and $E(\Omega) = \sum_{\lambda_i \in \Omega} P_i$: the projection-valued measure simply sums the eigenprojections whose eigenvalues land in Ω. Every projection-valued measure below is modelled on this example, and theorem 12.4.6 shows that this example is the only one that arises.

To integrate against E one first turns it back into ordinary scalar measures by pairing with vectors, exactly as in the construction of the Borel calculus. Fix $x, y \in \mathbb{H}$ and set

$$E_{x,y}(\Omega) := \langle E(\Omega)x, y\rangle \tag{12.5}$$

Additivity of E in the strong sense makes $\Omega \mapsto E_{x,y}(\Omega)$ countably additive, so $E_{x,y}$ is a complex Borel measure on K; its total variation is at most $\|x\|\,\|y\|$. Bounding each term $|E_{x,y}(\Omega_i)| \le \|x\|\,\|y\|$ would not control the sum, since a countable partition carries infinitely many terms; the bound instead uses the orthogonality of the ranges. For any Borel partition $\{\Omega_i\}$ of K, choose unimodular scalars α_i with $\alpha_i E_{x,y}(\Omega_i) = |E_{x,y}(\Omega_i)|$; then, because the projections $E(\Omega_i)$ have pairwise orthogonal ranges,

$$\sum_i |E_{x,y}(\Omega_i)| = \Big\langle \sum_i \alpha_i E(\Omega_i)x, y\Big\rangle \le \Big\|\sum_i \alpha_i E(\Omega_i)x\Big\|\,\|y\| = \Big(\sum_i \|E(\Omega_i)x\|^2\Big)^{1/2}\|y\| \le \|x\|\,\|y\|$$

using $|\alpha_i| = 1$ and $\sum_i \|E(\Omega_i)x\|^2 = \|x\|^2$. Taking the supremum over partitions gives $|E_{x,y}|(K) \leq \|x\|\,\|y\|$. When $y = x$ the measure

$$E_{x,x}(\Omega) = \langle E(\Omega)x, x\rangle = \langle E(\Omega)^2 x, x\rangle = \|E(\Omega)x\|^2 \geq 0$$

is a positive measure of total mass $E_{x,x}(K) = \langle E(K)x, x\rangle = \|x\|^2$, using $E(\Omega) = E(\Omega)^* = E(\Omega)^2$. These scalar measures are the raw material for the integral.

Definition 12.4.3: Integral against a Projection-Valued Measure

Let E be a projection-valued measure on a compact $K \subseteq \mathbb{C}$ and let $f \in B_b(K)$ be a bounded Borel function. The *integral of f against E* is the operator $\int_K f\,dE \in B(\mathbb{H})$ determined by

$$\Big\langle \Big(\int_K f\,dE \Big) x, y \Big\rangle = \int_K f\,dE_{x,y} \qquad (x, y \in \mathbb{H}) \tag{12.6}$$

where $E_{x,y}$ is the complex measure of (12.5).

The definition asserts that a single operator is pinned down by (12.6); the following proposition supplies the operator and its basic estimate.

Proposition 12.4.4: The Integral against a Projection-Valued Measure

Let E be a projection-valued measure on a compact $K \subseteq \mathbb{C}$. For each $f \in B_b(K)$ there is a unique operator $\int_K f\,dE \in B(\mathbb{H})$ satisfying (12.6), and

$$\Big\| \int_K f\,dE \Big\| \leq \|f\|_\infty$$

Moreover $f \mapsto \int_K f\,dE$ is linear, and $\int_K \chi_\Omega\,dE = E(\Omega)$ for every Borel set $\Omega \subseteq K$.

***Proof*:**
Fix $f \in B_b(K)$ and consider the map $B(x, y) := \int_K f\,dE_{x,y}$ on $\mathbb{H} \times \mathbb{H}$. In the first argument $x \mapsto E(\Omega)x$ is linear for each Ω, so $x \mapsto E_{x,y}$ is linear as a map into complex measures and hence $x \mapsto B(x, y)$ is linear; in the second argument the conjugate-linearity of $\langle\cdot,\cdot\rangle$ makes $y \mapsto B(x, y)$ conjugate-linear. Thus B is a sesquilinear form. It is bounded, because

$$|B(x, y)| = \Big| \int_K f\,dE_{x,y} \Big| \leq \|f\|_\infty\, |E_{x,y}|(K) \leq \|f\|_\infty\, \|x\|\, \|y\|$$

using the total-variation bound $|E_{x,y}|(K) \leq \|x\|\,\|y\|$ noted above. A bounded sesquilinear form is represented by a unique bounded operator: by the Riesz representation of the Hilbert space (theorem 11.1.13), for each fixed x the conjugate-linear functional $y \mapsto \overline{B(x, y)}$ is $y \mapsto \langle y, Sx\rangle$ for a unique vector Sx, and $x \mapsto Sx$ is linear with $\|Sx\| \leq \|f\|_\infty \|x\|$. Writing $S = \int_K f\,dE$ gives $\langle Sx, y\rangle = B(x, y)$, which is (12.6), and the operator norm bound $\|S\| \leq \|f\|_\infty$. Uniqueness is immediate: an operator is determined by all inner products $\langle Sx, y\rangle$. Linearity in f follows from linearity of $f \mapsto \int_K f\,dE_{x,y}$ for the fixed measures $E_{x,y}$. Finally, for $f = \chi_\Omega$ the defining identity reads $\langle (\int_K \chi_\Omega\,dE)x, y\rangle = E_{x,y}(\Omega) = \langle E(\Omega)x, y\rangle$ for all x, y, so $\int_K \chi_\Omega\,dE = E(\Omega)$, completing the proof.

The integral turns the algebra of bounded Borel functions into an algebra of operators, and it does so multiplicatively. This is the property that makes it a functional calculus rather than a mere assignment of operators to functions.

Proposition 12.4.5: Multiplicativity of the Projection-Valued Integral

Let E be a projection-valued measure on a compact $K \subseteq \mathbb{C}$. For $f, g \in B_b(K)$,

$$\int_K fg\,dE = \Big(\int_K f\,dE\Big)\Big(\int_K g\,dE\Big)$$

and $\int_K \overline{f}\,dE = \big(\int_K f\,dE\big)^*$. In particular $f \mapsto \int_K f\,dE$ is a $*$-homomorphism from $B_b(K)$ into $B(\mathbb{H})$.

***Proof*:**
Step 1: Indicators. For Borel sets Ω, Ω' the product $\chi_\Omega \chi_{\Omega'} = \chi_{\Omega\cap\Omega'}$, so by proposition 12.4.4 and the multiplicativity of E (definition 12.4.1(2)),

$$\int_K \chi_\Omega \chi_{\Omega'}\,dE = E(\Omega\cap\Omega') = E(\Omega)E(\Omega') = \Big(\int_K \chi_\Omega\,dE\Big)\Big(\int_K \chi_{\Omega'}\,dE\Big)$$

By linearity in each factor, the product identity extends to all pairs of simple Borel functions (finite linear combinations of indicators).

Step 2: Passage to bounded Borel functions. Every $f \in B_b(K)$ is a uniform limit of simple functions s_n with $\|s_n\|_\infty \le \|f\|_\infty$: partition the bounded range of f into finitely many small cells and let s_n take a sample value on each preimage. By the norm bound of proposition 12.4.4, $\int_K s_n\,dE \to \int_K f\,dE$ in operator norm, and likewise $\int_K t_n\,dE \to \int_K g\,dE$ for simple $t_n \to g$ uniformly. Since multiplication is jointly norm-continuous on bounded sets and $s_n t_n \to fg$ uniformly,

$$\int_K fg\,dE = \lim_n \int_K s_n t_n\,dE = \lim_n \Big(\int_K s_n\,dE\Big)\Big(\int_K t_n\,dE\Big) = \Big(\int_K f\,dE\Big)\Big(\int_K g\,dE\Big)$$

where the middle equality is Step 1 applied to the simple functions s_n, t_n.

Step 3: The involution. Each $E(\Omega)$ is self-adjoint, so for a simple $s = \sum_j c_j \chi_{\Omega_j}$ one has $(\int_K s\,dE)^* = \sum_j \overline{c_j} E(\Omega_j) = \int_K \overline{s}\,dE$. Passing to the uniform limit and using continuity of the adjoint gives $\int_K \overline{f}\,dE = (\int_K f\,dE)^*$ for all $f \in B_b(K)$. Together with linearity and $\int_K 1\,dE = E(K) = I$, the map $f \mapsto \int_K f\,dE$ preserves products, adjoints, and the unit, hence is a $*$-homomorphism, as we sought to show.

The projection-valued integral is now available for any projection-valued measure. Applied to the spectral projections of a self-adjoint operator, it reconstructs both the operator and its entire functional calculus, and this is the spectral theorem. The measure E carries all the information in T, distributed across the Borel sets of the spectrum, and integration reassembles it.

Theorem 12.4.6: The Spectral Theorem (Resolution of the Identity)

Let $T \in B(\mathbb{H})$ be self-adjoint and let $E(\Omega) = \chi_\Omega(T)$ be its spectral projection-valued measure on $\sigma(T) \subseteq \mathbb{R}$. Then, with all integrals taken against E over $\sigma(T)$:

1. For every bounded Borel function $f \in B_b(\sigma(T))$, the Borel calculus of theorem 12.3.3 agrees with integration against E,
$$f(T) = \int_{\sigma(T)} f\, dE$$
2. Taking $f(\lambda) = \lambda$ gives the resolution of the identity
$$T = \int_{\sigma(T)} \lambda\, dE(\lambda)$$
3. For every $x \in \mathbb{H}$,
$$\|x\|^2 = \int_{\sigma(T)} d\langle E(\lambda)x, x\rangle$$

The projection-valued measure E with these properties is unique.

***Proof*:**
Step 1: The calculus is integration against E. Both $f \mapsto f(T)$ (the Borel calculus of theorem 12.3.3) and $f \mapsto \int_{\sigma(T)} f\, dE$ are linear maps $B_b(\sigma(T)) \to B(\mathbb{H})$. They agree on indicators: $\chi_\Omega(T) = E(\Omega)$ by the definition of the spectral projection (definition 12.3.5), while $\int_{\sigma(T)} \chi_\Omega\, dE = E(\Omega)$ by proposition 12.4.4. By linearity they agree on simple Borel functions. For general $f \in B_b(\sigma(T))$, choose simple $s_n \to f$ uniformly with $\|s_n\|_\infty \le \|f\|_\infty$. The Borel calculus satisfies $\|s_n(T) - f(T)\| \le \|s_n - f\|_\infty \to 0$ (theorem 12.3.3), and the projection-valued integral satisfies $\|\int s_n\, dE - \int f\, dE\| \le \|s_n - f\|_\infty \to 0$ (proposition 12.4.4); as the two limits agree on the s_n, they agree on f. This proves (1).

Step 2: Recovering T. Apply (1) to the coordinate function $\iota(\lambda) = \lambda$, which is continuous, hence in $B_b(\sigma(T))$. The continuous calculus of theorem 12.2.4 sends ι to T (it sends id to T by construction), and the Borel calculus extends the continuous one, so $\iota(T) = T$. Thus
$$T = \iota(T) = \int_{\sigma(T)} \iota\, dE = \int_{\sigma(T)} \lambda\, dE(\lambda)$$
which is (2).

Step 3: The norm identity. Apply (1) to $f = 1$, the constant function. Then $1(T) = I$ by the continuous calculus, so $\int_{\sigma(T)} 1\, dE = I$. Pairing with x and unwinding the defining identity (12.6),
$$\|x\|^2 = \langle Ix, x\rangle = \left\langle \left(\int_{\sigma(T)} 1\, dE \right) x, x \right\rangle = \int_{\sigma(T)} 1\, dE_{x,x} = \int_{\sigma(T)} d\langle E(\lambda)x, x\rangle$$
which is (3); the integrand 1 integrates the total mass of the positive measure $E_{x,x}$, already seen to equal $\|x\|^2$.

Step 4: Uniqueness. Suppose E' is a projection-valued measure on $\sigma(T)$ with $\int \lambda\, dE' = T$. Then $\int p\, dE' = p(T)$ for every polynomial p: by proposition 12.4.5 the map $f \mapsto \int f\, dE'$ is a $*$-homomorphism sending $\lambda \mapsto T$ and $1 \mapsto I$, so it agrees with the polynomial calculus (proposition 12.2.3) on polynomials. By Stone-Weierstrass, polynomials in λ (real spectrum, so $\overline{\lambda} = \lambda$ and complex conjugates add nothing) are uniformly dense in $C(\sigma(T))$, and both $\int (\cdot)\, dE'$ and the continuous calculus are norm-continuous, so $\int f\, dE' = f(T)$ for all $f \in C(\sigma(T))$. Fixing x, y, the two complex measures $E'_{x,y}$ and $E_{x,y}$ therefore satisfy $\int f\, dE'_{x,y} = \langle f(T)x, y\rangle = \int f\, dE_{x,y}$ for every $f \in C(\sigma(T))$. A complex Borel measure on a compact set is determined by its integrals against continuous functions (theorem 8.3.3), so $E'_{x,y} = E_{x,y}$ for all x, y, whence $E'(\Omega) = E(\Omega)$ for every Borel Ω. Thus E is unique, completing the proof.

The three statements are three readings of one fact. Statement (1) says the projection-valued measure is a repackaging of the Borel calculus with no loss: every $f(T)$, and in particular every continuous function of T, is an integral against E. Statement (2) is the exact analogue of the finite-dimensional $T = \sum_i \lambda_i P_i$: the eigenprojections have become the projection-valued measure, and the finite sum over eigenvalues has become an integral over the spectrum. The passage from sum to integral is what accommodates operators with no eigenvalues at all, such as multiplication by x on $L^2[0,1]$, whose spectrum $[0,1]$ is entirely continuous and carries no point masses. Statement (3) records that the positive measure $\lambda \mapsto \langle E(\lambda)x, x\rangle$ has total mass $\|x\|^2$; it is the operator-theoretic Parseval identity, the continuous replacement for $\|x\|^2 = \sum_i \|P_i x\|^2$ (theorem 11.1.15), distributing the squared length of x over the spectrum. Uniqueness is what earns the phrase "the" resolution of the identity: the data of T and the data of E are equivalent.

Example 12.4.7: The Multiplication Operator on $L^2[0,1]$

Let $\mathbb{H} = L^2[0,1]$ and let T be multiplication by the independent variable, $(Tf)(t) = tf(t)$. Then T is self-adjoint (multiplication by a real bounded function equals its own adjoint) with $\|T\| = 1$, and $\sigma(T) = [0,1]$: for $\lambda \notin [0,1]$ the function $1/(t-\lambda)$ is bounded on $[0,1]$ and multiplication by it inverts $T - \lambda I$, while for $\lambda \in [0,1]$ the operator $T - \lambda I$ has dense but non-closed range and no bounded inverse. The spectral projection $E(\Omega)$ is multiplication by χ_Ω, the indicator of $\Omega \subseteq [0,1]$: one checks $\chi_\Omega(T)f = \chi_\Omega \cdot f$ directly, since for continuous g the operator $g(T)$ is multiplication by g, and the Borel calculus extends this. The scalar measures are then $\langle E(\Omega)f, f\rangle = \int_\Omega |f(t)|^2\, dt$, so $\langle E(\cdot)f, f\rangle$ is the measure $|f(t)|^2\, dt$ on $[0,1]$. The resolution of the identity reads

$$\langle Tf, f\rangle = \int_0^1 t\, |f(t)|^2\, dt = \int_{[0,1]} \lambda\, d\langle E(\lambda)f, f\rangle$$

which is just the statement that T acts as multiplication by t. This operator has no eigenvalues: $Tf = \lambda f$ forces $(t-\lambda)f(t) = 0$ almost everywhere, hence $f = 0$ in L^2. The spectral theorem thus captures an operator that the eigenvalue picture cannot reach, and the projection-valued measure E has no atoms.

The last example is not a special case but the general shape. Multiplication by the coordinate on an L^2 space is what every bounded self-adjoint operator looks like once the right coordinates are chosen. Making this precise requires decomposing $\mathbb{H}$ into pieces on which T has a single generating vector, and on each such piece the Borel calculus produces an isometry onto an L^2 space that conjugates T into multiplication.

The building block is the closed subspace generated by one vector under T.

Definition 12.4.8: Cyclic Vector and Cyclic Subspace

Let $T \in B(\mathbb{H})$ be self-adjoint. For $x \in \mathbb{H}$, the *cyclic subspace* generated by x is the closed linear span

$$\mathbb{H}_x := \overline{\operatorname{span}}\,\{p(T)x \mid p \text{ a polynomial}\}$$

A vector x is a *cyclic vector* for T if $\mathbb{H}_x = \mathbb{H}$, and T is a *cyclic operator* if it has a cyclic vector.

Because T is self-adjoint, $p(T)^* = \overline{p}(T)$ is again a polynomial in T, so $\mathbb{H}_x$ is invariant under both T and T^*; its orthogonal complement is then invariant as well, which is what lets the space be split off cleanly. On a cyclic subspace the operator is as simple as possible, and the following lemma identifies that simple model.

Lemma 12.4.9: Cyclic Operators are Multiplication Operators

Let $T \in B(\mathbb{H})$ be self-adjoint with a cyclic vector x. Then there is a finite positive Borel measure μ on $\sigma(T)$ and a unitary operator $U\colon \mathbb{H} \to L^2(\sigma(T), \mu)$ with $Ux = 1$ (the constant function) such that

$$(UTU^{-1}g)(\lambda) = \lambda\, g(\lambda) \qquad (g \in L^2(\sigma(T), \mu))$$

so T is unitarily equivalent to multiplication by λ on $L^2(\sigma(T), \mu)$.

***Proof*:**
Step 1: The measure. Let $\mu := E_{x,x}$ be the positive Borel measure $\mu(\Omega) = \langle E(\Omega)x, x\rangle = \|E(\Omega)x\|^2$ on $\sigma(T)$, of total mass $\|x\|^2$ (theorem 12.4.6(3)). For a continuous function $f \in C(\sigma(T))$, the Borel calculus gives

$$\|f(T)x\|^2 = \langle f(T)^* f(T)x, x\rangle = \langle |f|^2(T)x, x\rangle = \int_{\sigma(T)} |f|^2\, dE_{x,x} = \int_{\sigma(T)} |f|^2\, d\mu = \|f\|^2_{L^2(\mu)} \tag{12.7}$$

using $f(T)^* f(T) = \overline{f}(T)f(T) = |f|^2(T)$ (theorem 12.2.4) and the defining identity (12.6) for the measure $E_{x,x}$.

Step 2: The isometry on a dense set. Define $V\colon C(\sigma(T)) \to \mathbb{H}$ by $Vf = f(T)x$. Its range $\{f(T)x \mid f \in C(\sigma(T))\}$ contains all $p(T)x$ for polynomials p, hence is dense in $\mathbb{H}_x = \mathbb{H}$. By (12.7), V is an isometry from $(C(\sigma(T)), \|\cdot\|_{L^2(\mu)})$ into $\mathbb{H}$. Since $C(\sigma(T))$ is dense in $L^2(\sigma(T), \mu)$ (continuous functions are dense in L^2 of a finite Borel measure on a compact metric space) and V is isometric, V extends uniquely to an isometry $W\colon L^2(\sigma(T), \mu) \to \mathbb{H}$. Its range is closed (an isometry has closed range) and dense (it contains the dense set $\{f(T)x \mid f \in C(\sigma(T))\}$), hence all of $\mathbb{H}$; so W is unitary. Set $U := W^{-1}\colon \mathbb{H} \to L^2(\sigma(T), \mu)$. The constant function $1 \in C(\sigma(T))$ maps to $W1 = 1(T)x = x$, so $Ux = 1$.

Step 3: Conjugating T into multiplication. It suffices to check UTU^{-1} on the dense subspace $C(\sigma(T)) \subseteq L^2(\sigma(T), \mu)$. For $f \in C(\sigma(T))$,

$$UTU^{-1}f = UT(f(T)x) = U\big((\iota f)(T)x\big) = \iota f$$

where $\iota(\lambda) = \lambda$ is the coordinate function, using $Tf(T) = (\iota f)(T)$ by multiplicativity of the calculus (theorem 12.2.4) and $U(g(T)x) = g$ for $g \in C(\sigma(T))$. Thus UTU^{-1} acts as $f \mapsto \iota f$, multiplication by λ, on a dense subspace, and by continuity on all of $L^2(\sigma(T), \mu)$, as we sought to show.

A general self-adjoint operator need not be cyclic, but the whole space breaks into an orthogonal sum of cyclic pieces. Assembling the models from each piece gives the multiplication form of the spectral theorem. The direct sum is indexed by however many cyclic subspaces are needed to exhaust $\mathbb{H}$, which for a separable space is at most countable.

Theorem 12.4.10: The Spectral Theorem (Multiplication-Operator Form)

Let $T \in B(\mathbb{H})$ be self-adjoint. Then there is a family of finite positive Borel measures $\{\mu_n\}_{n\in N}$ on $\sigma(T)$ and a unitary operator

$$U : \mathbb{H} \to \bigoplus_{n\in N} L^2(\sigma(T), \mu_n)$$

such that UTU^{-1} is multiplication by λ,

$$\big(UTU^{-1}(g_n)_n\big)(\lambda) = (\lambda\, g_n(\lambda))_n$$

If $\mathbb{H}$ is separable, the index set N may be taken finite or countably infinite. In particular every bounded self-adjoint operator is unitarily equivalent to multiplication by the independent variable on a direct sum of L^2 spaces.

***Proof*:**
Step 1: An orthogonal cyclic decomposition. Consider the collection of sets $\{\mathbb{H}_{x_i}\}_{i\in I}$ of mutually orthogonal nonzero cyclic subspaces (definition 12.4.8), ordered by inclusion of the index families. If $\mathbb{H} = \{0\}$ the statement is vacuous, so assume $\mathbb{H} \neq \{0\}$; then a single nonzero cyclic subspace exists, so the collection is nonempty. Any chain has an upper bound (its union), so by Zorn's lemma there is a maximal family $\{\mathbb{H}_{x_n}\}_{n\in N}$ of pairwise orthogonal cyclic subspaces. Let $K = \overline{\bigoplus_n \mathbb{H}_{x_n}}$ be the closure of their orthogonal sum. If $K \neq \mathbb{H}$, pick a nonzero $y \in K^\perp$. Each $\mathbb{H}_{x_n}$ is invariant under T and $T^* = T$, so K is invariant under T, hence $K^\perp$ is invariant under $T^* = T$; then $\mathbb{H}_y \subseteq K^\perp$ is a nonzero cyclic subspace orthogonal to every $\mathbb{H}_{x_n}$, contradicting maximality. Therefore $K = \mathbb{H}$ and

$$\mathbb{H} = \bigoplus_{n\in N} \mathbb{H}_{x_n}$$

an orthogonal direct sum of cyclic subspaces.

Step 2: Model each summand. For each n, the restriction $T|_{\mathbb{H}_{x_n}}$ is self-adjoint on $\mathbb{H}_{x_n}$ with cyclic vector x_n, so by lemma 12.4.9 there is a finite positive Borel measure μ_n on $\sigma(T|_{\mathbb{H}_{x_n}}) \subseteq \sigma(T)$ (the inclusion holds because $\mathbb{H}_{x_n}$ reduces T, so $T = T|_{\mathbb{H}_{x_n}} \oplus T|_{\mathbb{H}_{x_n}^\perp}$ splits the spectrum as a union; extend μ_n by zero to a measure on $\sigma(T)$) and a unitary $U_n : \mathbb{H}_{x_n} \to L^2(\sigma(T), \mu_n)$ with $U_n(T|_{\mathbb{H}_{x_n}})U_n^{-1}$ equal to multiplication by λ.

Step 3: Assemble. Define $U := \bigoplus_n U_n : \mathbb{H} \to \bigoplus_n L^2(\sigma(T), \mu_n)$ by acting as U_n on the summand $\mathbb{H}_{x_n}$. As an orthogonal sum of unitaries between the summands it is unitary, and since T preserves each $\mathbb{H}_{x_n}$,

$$UTU^{-1}(g_n)_n = \big(U_n T|_{\mathbb{H}_{x_n}} U_n^{-1} g_n\big)_n = (\lambda\, g_n)_n$$

which is multiplication by λ on the direct sum. When $\mathbb{H}$ is separable, each $\mathbb{H}_{x_n}$ is a nonzero closed subspace, and mutually orthogonal nonzero subspaces of a separable space are at most countable in number (each contributes an independent direction to a countable orthonormal basis), so N is finite or countably infinite. This completes the proof.

The two forms of the spectral theorem are the same statement dressed differently. The resolution of the identity is coordinate-free and intrinsic, built from projections attached to the operator; the multiplication form is a concrete normal form, exhibiting T as an operator one can compute with directly. The measures μ_n record how the spectrum is "filled in" along each cyclic direction: a point mass at λ signals a genuine eigenvector, while a diffuse part signals continuous spectrum with no eigenvector, as in example 12.4.7. On a finite-dimensional space each μ_n is a finite sum of point masses and the direct sum recovers the diagonalization $T = \sum_i \lambda_i P_i$; the theorem is the statement that the same picture survives verbatim into infinite dimensions once "sum over eigenvalues" is read as "integral over the spectrum."

The self-adjointness of T entered only through two facts: its spectrum is real, and the functional calculus is a $*$-homomorphism on which conjugation of functions matches the adjoint of operators. Both survive for a normal operator, at the cost of moving from the real line to the complex plane. A normal T (definition 11.2.8) has $\sigma(T) \subseteq \mathbb{C}$ compact, and the unital C^*-subalgebra $C^*(T, T^*) \subseteq B(\mathbb{H})$ generated by T and T^* is commutative. The Gelfand transform (theorem 12.1.15) identifies its character space with $\sigma(T)$ and maps it onto $C(\sigma(T))$; this transform is isometric because every self-adjoint element $S = S^*$ of the commutative algebra $C^*(T, T^*)$ has $r(S) = \|S\|$ (the argument of theorem 12.2.1 applies to S inside the algebra), and the C^*-identity $\|A^*A\| = \|A\|^2$ propagates the equality to every element A. The transform therefore gives $C^*(T, T^*) \cong C(\sigma(T))$ carrying T to the coordinate function $z \mapsto z$ on $\sigma(T) \subseteq \mathbb{C}$. This continuous calculus over $\mathbb{C}$ replaces theorem 12.2.4, and the Riesz extension of theorem 12.3.3 produces a projection-valued measure E on the compact set $\sigma(T) \subseteq \mathbb{C}$ verbatim.

Theorem 12.4.11: The Spectral Theorem for Normal Operators

Let $T \in B(\mathbb{H})$ be normal. Then there is a unique projection-valued measure E on the compact set $\sigma(T) \subseteq \mathbb{C}$ such that

$$T = \int_{\sigma(T)} \lambda \, dE(\lambda)$$

and $f(T) = \int_{\sigma(T)} f \, dE$ for every $f \in B_b(\sigma(T))$, where $f \mapsto f(T)$ is the Borel calculus over $C^*(T, T^*) \cong C(\sigma(T))$. Equivalently, T is unitarily equivalent to multiplication by λ on a direct sum $\bigoplus_n L^2(\sigma(T), \mu_n)$ of L^2 spaces over $\sigma(T) \subseteq \mathbb{C}$.

***Proof Key ideas*:**
The self-adjoint argument transposes without change once the continuous calculus is supplied over $\mathbb{C}$. The commutative C^*-algebra $C^*(T, T^*)$ has, by the Gelfand transform (theorem 12.1.15), character space homeomorphic to $\sigma(T) \subseteq \mathbb{C}$, with Gelfand transform a unital $*$-homomorphism onto $C(\sigma(T))$ sending T to z and T^* to $\overline{z}$. The transform is isometric: every self-adjoint $S = S^*$ in the algebra satisfies $r(S) = \|S\|$ (the $\|S^2\| = \|S\|^2$ argument of theorem 12.2.1, applied to S inside $C^*(T, T^*)$), and the C^*-identity $\|A^*A\| = \|A\|^2$ upgrades this to $\|A\|^2 = \|A^*A\| = r(A^*A)$ for every element A, matching the algebra norm to the sup norm on $\sigma(T)$, so the transform is an isometric $*$-isomorphism onto $C(\sigma(T))$. This is the input played by theorem 12.2.4 in the self-adjoint case; the polynomial calculus is now the calculus of polynomials in the two

> commuting variables $z, \overline{z}$. From here the Riesz-measure extension to $B_b(\sigma(T))$ (theorem 12.3.3), the projection-valued integral (proposition 12.4.4, proposition 12.4.5), the resolution of the identity and its uniqueness (theorem 12.4.6), and the cyclic decomposition (theorem 12.4.10) all repeat verbatim, reading $\sigma(T) \subseteq \mathbb{C}$ in place of $\sigma(T) \subseteq \mathbb{R}$ throughout, completing the proof.

The normal case is where the theory reaches its natural boundary. A self-adjoint operator is normal with real spectrum; a unitary operator is normal with spectrum on the unit circle (definition 11.2.8); both are now handled by one theorem, integration of λ over a compact subset of $\mathbb{C}$ against a projection-valued measure. The engine driving the generalization is the identification of the commutative C^*-algebra $C^*(T, T^*)$ with $C(\sigma(T))$, carried out here by the Banach-algebra Gelfand theory of the first section together with the C^*-identity. This identification is the single-operator instance of the commutative Gelfand-Naimark duality, which the abstract theory of C^*-algebras develops in full generality. Pushing past bounded operators, to the unbounded self-adjoint operators that carry the differential operators of quantum mechanics and of partial differential equations, requires domains, closedness, and a projection-valued measure supported on an unbounded subset of $\mathbb{R}$; that extension is the subject of the next chapter, and the resolution of the identity proved here is the template it generalizes.

13

Unbounded Operators and Semigroups

The operators of analysis are rarely bounded. Differentiation, the Laplacian, and the position and momentum operators of quantum mechanics are each defined only on a dense subspace of a Hilbert space and are discontinuous there, so the bounded theory of the preceding chapters does not apply to them directly. Its central result survives the extension, however: the spectral theorem holds for unbounded self-adjoint operators and remains the organizing tool of the subject. This chapter builds the framework around it and then reads off the two ways a self-adjoint or dissipative operator generates a dynamics, one unitary and one contractive.

The chapter opens with the vector-valued integration it will need: a solution of an evolution equation is a curve $t \mapsto u(t)$ in a Banach space, and the formulas that produce and analyze it integrate such curves, which the Bochner integral makes precise. On that base we develop unbounded operators and their adjoints, carry the spectral theorem of chapter 12 to the unbounded self-adjoint case through the Cayley transform, and prove the two generation theorems: Stone's theorem, that the strongly continuous one-parameter unitary groups are exactly the e^{itA} for self-adjoint A, and the Hille-Yosida theorem, which characterizes the generators of C_0-semigroups and underwrites the parabolic evolution equations of the later volumes.

13.1 Bochner Integration

This section builds the integral of a Banach-space-valued function of a real variable, records the criterion that decides when it exists, and proves the two properties the later sections lean on: that a closed operator passes through the integral, and that integrating a continuous curve and then differentiating returns the curve.

Throughout, (Ω, Σ, μ) is a measure space and X a Banach space; for the applications Ω is an interval of $\mathbb{R}$ with Lebesgue measure and X is the Hilbert space $\mathbb{H}$ or the Banach space carrying a semigroup.

The Lebesgue integral is built from simple functions, and the vector-valued theory begins the same way. A *simple function* $s\colon \Omega \to X$ takes finitely many values $x_1, \dots, x_n \in X$, each on a measurable set $A_k = s^{-1}(x_k)$; it is written $s = \sum_{k=1}^n x_k \mathbf{1}_{A_k}$. The subtlety absent in the scalar case is that a general X-valued function need not be an everywhere limit of simple functions unless X is separable along the range: a function into a nonseparable space can fail to be approximable even when each scalar coordinate is measurable. The right notion of measurability therefore builds separability into its definition.

Definition 13.1.1: Strongly Measurable Function

Let (Ω, Σ, μ) be a measure space and X a Banach space. A function $f\colon \Omega \to X$ is *strongly measurable* if there is a sequence of simple functions $s_n\colon \Omega \to X$ with

$$\|s_n(\omega) - f(\omega)\| \to 0 \quad \text{for } \mu\text{-almost every } \omega \in \Omega$$

The function f is *weakly measurable* if $\omega \mapsto \langle f(\omega), \varphi \rangle$ is measurable for every φ in the dual X^*, and *essentially separably valued* if there is a μ-null set N with $f(\Omega \setminus N)$ separable in X.

Strong measurability is the exact hypothesis under which the norm-limit construction of the integral will go through: it supplies the approximating simple functions directly. The two weaker notions in the definition matter because strong measurability is often awkward to check by hand, whereas weak measurability reduces to the scalar theory one functional at a time, and essential separable-valuedness is frequently visible from the range. Pettis's theorem, below, says these two together are equivalent to the strong notion, so one may verify strong measurability by the two convenient conditions.

A first consequence records how the norm interacts with strong measurability: if f is strongly measurable then $\omega \mapsto \|f(\omega)\|$ is a measurable scalar function, being the almost-everywhere limit of the measurable functions $\omega \mapsto \|s_n(\omega)\|$. This is what makes the phrase "$\|f(\cdot)\| \in L^1$" meaningful in the integrability criterion.

Example 13.1.2: A Step-Modulated Operator Orbit

Fix a Banach space X, an operator $B \in B(X)$ with $\|B\| \le 1$, and a vector $x \in X$. On $\Omega = [0, \infty)$ with Lebesgue measure, define $f\colon [0, \infty) \to X$ by $f(t) = e^{-t} B^{\lfloor t \rfloor} x$, where $\lfloor t \rfloor$ is the integer part. To exhibit approximating simple functions, partition $[0, \infty)$ at the dyadic points $t_{n,k} = n + k2^{-N}$ for integers $n \ge 0$ and $0 \le k < 2^N$, and let s_N take on each half-open interval $[t_{n,k}, t_{n,k+1})$ the constant value $e^{-t_{n,k}} B^n x$; on $[N, \infty)$ set $s_N = 0$. Each s_N is a simple function. For $t < N$, writing $t \in [t_{n,k}, t_{n,k+1})$ with $n = \lfloor t \rfloor$,

$$\|f(t) - s_N(t)\| = |e^{-t} - e^{-t_{n,k}}| \, \|B^n x\| \le (e^{-t_{n,k}} - e^{-t_{n,k+1}}) \|x\| \le 2^{-N} \|x\|$$

using $\|B^n\| \le 1$ and the mean value theorem on e^{-t}. Hence $s_N(t) \to f(t)$ at every t, so f is strongly measurable. Its range lies in the separable subspace spanned by $\{B^n x \mid n \ge 0\}$, which is the separability that Pettis's theorem will read off directly.

The equivalence promised above is Pettis's measurability theorem. It is the vector-valued analogue of the fact that a scalar function is measurable iff its preimages of intervals are measurable: it trades the hard-to-check pointwise-limit condition for two conditions that decompose along the two structures of X, its linear functionals and its separable subspaces.

Theorem 13.1.3: Pettis's Measurability Theorem

Let X be a Banach space and $f\colon \Omega \to X$. Then f is strongly measurable if and only if f is weakly measurable and essentially separably valued.

***Proof*:**
Necessity. Suppose f is strongly measurable, with simple functions $s_n \to f$ almost everywhere. For each $\varphi \in X^*$ the map $\omega \mapsto \langle s_n(\omega), \varphi\rangle$ is a scalar simple function, and $\langle s_n(\omega), \varphi\rangle \to \langle f(\omega), \varphi\rangle$ almost everywhere by continuity of φ, so the limit $\omega \mapsto \langle f(\omega), \varphi\rangle$ is measurable; thus f is weakly measurable. For essential separable-valuedness, let N be the null set off which $s_n \to f$. Each s_n takes finitely many values, so the countable set D of all values taken by all s_n is a countable subset of X, and for $\omega \notin N$ the vector $f(\omega) = \lim_n s_n(\omega)$ lies in the closure $\overline{D}$. Hence $f(\Omega \setminus N)$ is contained in the separable set $\overline{D}$, so f is essentially separably valued.

Sufficiency. Suppose f is weakly measurable and essentially separably valued. Discarding a null set, assume $f(\Omega)$ lies in a separable closed subspace $X_0 \subseteq X$; it suffices to produce the approximating simple functions with values in X_0. Let $\{d_j \mid j \geq 1\}$ be a countable dense subset of X_0.

Step 1: a countable norming set. The separable space X_0 carries a countable set $\{\varphi_i \mid i \geq 1\} \subseteq X^*$ with $\|\varphi_i\| \leq 1$ that is *norming*, meaning $\|z\| = \sup_i |\langle z, \varphi_i\rangle|$ for every $z \in X_0$. To construct it, enumerate a countable dense subset $\{d_j \mid j \geq 1\}$ of X_0 and, for each j with $d_j \neq 0$, use the Hahn-Banach theorem to pick $\varphi_j \in X^*$ with $\|\varphi_j\| = 1$ and $\langle d_j, \varphi_j\rangle = \|d_j\|$. For $z \in X_0$ and $\varepsilon > 0$, choosing d_j with $\|z - d_j\| < \varepsilon$ gives $|\langle z, \varphi_j\rangle| \geq \|d_j\| - \varepsilon \geq \|z\| - 2\varepsilon$, while $|\langle z, \varphi_i\rangle| \leq \|z\|$ for every i; letting $\varepsilon \to 0$ yields $\sup_i |\langle z, \varphi_i\rangle| = \|z\|$.

The norm to a fixed vector is measurable. Fix $x \in X_0$. Applying the norming identity to $z = f(\omega) - x \in X_0$ gives

$$\|f(\omega) - x\| = \sup_i |\langle f(\omega), \varphi_i\rangle - \langle x, \varphi_i\rangle|$$

Each $\omega \mapsto \langle f(\omega), \varphi_i\rangle$ is measurable by weak measurability, so $\omega \mapsto \|f(\omega) - x\|$ is a countable supremum of measurable functions, hence measurable.

Step 2: nearest-point simple approximants. Fix $n \geq 1$. For each ω, at least one of the first n dense points $d_1, \ldots, d_n$ lies within $\inf_{1\leq j\leq n} \|f(\omega) - d_j\| + 1/n$ of $f(\omega)$; let $j_n(\omega)$ be the least index $j \leq n$ achieving this, and set $s_n(\omega) = d_{j_n(\omega)}$. The sets $\{\omega \mid j_n(\omega) = j\}$ are measurable by Step 1 (they are defined by finitely many inequalities among the measurable functions $\omega \mapsto \|f(\omega) - d_j\|$), so s_n is a simple function with values in $\{d_1, \ldots, d_n\}$.

Step 3: convergence. Fix ω and $\varepsilon > 0$. By density of $\{d_j \mid j\}$ in X_0 there is a j_0 with $\|f(\omega) - d_{j_0}\| < \varepsilon/2$. For $n \geq \max(j_0, 2/\varepsilon)$ the choice in Step 2 gives

$$\|f(\omega) - s_n(\omega)\| \leq \inf_{1\leq j\leq n} \|f(\omega) - d_j\| + \frac{1}{n} \leq \|f(\omega) - d_{j_0}\| + \frac{1}{n} < \varepsilon$$

Hence $s_n(\omega) \to f(\omega)$ for every ω, and f is strongly measurable, as we sought to show.

Pettis's theorem is the working criterion for strong measurability. In every application below the range of the function is visibly separable, either because X itself is separable or because the function is a norm-continuous curve $t \mapsto u(t)$ whose image is the continuous picture of an interval; weak measurability is then automatic, and the theorem delivers strong measurability with no further work.

With the class of integrands fixed, the integral is defined by the same two-stage passage as the Lebesgue integral: first on simple functions, where it is a finite sum, then extended by a norm-limit. The one point requiring care is that the limit of the integrals of approximating simple functions exists and does not depend on the approximating sequence.

For a simple function $s = \sum_{k=1}^{n} x_k \mathbf{1}_{A_k}$ with the A_k of finite measure and disjoint, the integral is the finite sum

$$\int_\Omega s\, d\mu = \sum_{k=1}^{n} \mu(A_k)\, x_k \in X$$

which is independent of the representation of s, by the same additivity computation as in the scalar case. Call a simple function *integrable* if each A_k with $x_k \neq 0$ has finite measure; then $\int_\Omega \|s\|\, d\mu = \sum_k \mu(A_k)\|x_k\|$ is finite, and the triangle inequality gives the basic estimate $\|\int_\Omega s\, d\mu\| \le \int_\Omega \|s\|\, d\mu$.

Definition 13.1.4: Bochner Integral

A strongly measurable function $f \colon \Omega \to X$ is *Bochner integrable* if there is a sequence of integrable simple functions $s_n \colon \Omega \to X$ with

$$\int_\Omega \|s_n - f\|\, d\mu \to 0$$

The *Bochner integral* of such an f is

$$\int_\Omega f\, d\mu = \lim_{n\to\infty} \int_\Omega s_n\, d\mu$$

the limit taken in the norm of X. For a measurable $E \subseteq \Omega$ one writes $\int_E f\, d\mu = \int_\Omega \mathbf{1}_E f\, d\mu$.

The defining condition asks the approximants to converge to f in the L^1-sense of the integrated norm, which is stronger than the almost-everywhere convergence of strong measurability. The estimate for simple functions shows the sequence of integrals is Cauchy, so the limit exists; that it is independent of the approximating sequence is the content of the following well-definedness check, after which the integral is an unambiguous element of X.

Proposition 13.1.5: The Bochner Integral is Well-Defined

Let $f \colon \Omega \to X$ be strongly measurable and Bochner integrable, with integrable simple functions (s_n) as in definition 13.1.4. Then $(\int_\Omega s_n\, d\mu)_n$ is a Cauchy sequence in X, its limit is independent of the choice of (s_n), and it agrees with $\int_\Omega s\, d\mu$ when $f = s$ is itself an integrable simple function.

***Proof*:**
Cauchy. For indices m, n, the function $s_m - s_n$ is an integrable simple function, so by the basic

estimate and the triangle inequality inside the integral,

$$\left\|\int_\Omega s_m\,d\mu - \int_\Omega s_n\,d\mu\right\| = \left\|\int_\Omega (s_m - s_n)\,d\mu\right\| \le \int_\Omega \|s_m - s_n\|\,d\mu \le \int_\Omega \|s_m - f\|\,d\mu + \int_\Omega \|f - s_n\|\,d\mu$$

Both terms on the right tend to 0 by the defining condition, so the sequence of integrals is Cauchy; since X is complete, it converges.

Independence of the sequence. Let (s_n) and (t_n) be two approximating sequences for the same f. Interleave them into the single sequence $s_1, t_1, s_2, t_2, \ldots$, which also satisfies $\int_\Omega \|\cdot - f\|\,d\mu \to 0$ and hence, by the first part, has a convergent sequence of integrals. A convergent sequence has a unique limit, and both $(\int s_n)$ and $(\int t_n)$ are subsequences of the interleaved integrals, so they share that limit. The Bochner integral is therefore independent of the approximating sequence.

Consistency. If $f = s$ is an integrable simple function, the constant sequence $s_n = s$ approximates it, and $\int_\Omega f\,d\mu = \lim_n \int_\Omega s\,d\mu = \int_\Omega s\,d\mu$, so the two meanings of the symbol agree, completing the proof.

The construction is exactly the scalar one with $|\cdot|$ replaced by $\|\cdot\|$ and completeness of $\mathbb{R}$ replaced by completeness of X; every place the scalar argument used the absolute value, the norm serves in its stead. Completeness of X is what makes the Cauchy sequence converge, and it is the only property of X the definition uses beyond its linear and normed structure.

The definition presupposes that a suitable approximating sequence exists, and one wants a criterion deciding this from f alone. In the scalar theory a measurable function is integrable iff its absolute value has finite integral; the vector-valued statement is identical with the norm in place of the absolute value, and it is the criterion invoked whenever an integral is formed in the sequel.

Theorem 13.1.6: Bochner's Integrability Criterion

A strongly measurable function $f\colon \Omega \to X$ is Bochner integrable if and only if

$$\int_\Omega \|f(\omega)\|\,d\mu(\omega) < \infty$$

that is, $\|f(\cdot)\| \in L^1(\mu)$. In that case

$$\left\|\int_\Omega f\,d\mu\right\| \le \int_\Omega \|f\|\,d\mu$$

***Proof*:**
Necessity. Suppose f is Bochner integrable, with integrable simple functions s_n satisfying $\int_\Omega \|s_n - f\|\,d\mu \to 0$. Each $\|s_n\|$ is a nonnegative integrable simple function, and the reverse triangle inequality gives, pointwise, $\big|\|f\| - \|s_n\|\big| \le \|f - s_n\|$, so

$$\int_\Omega \big|\|f\| - \|s_n\|\big|\,d\mu \le \int_\Omega \|f - s_n\|\,d\mu \to 0$$

Thus $\|f\|$ is the L^1-limit of the integrable functions $\|s_n\|$, hence $\|f\| \in L^1(\mu)$, and $\int_\Omega \|f\|\,d\mu = \lim_n \int_\Omega \|s_n\|\,d\mu < \infty$.

Sufficiency. Suppose f is strongly measurable with $\int_\Omega \|f\|\,d\mu < \infty$. Let (σ_n) be simple functions with $\sigma_n \to f$ almost everywhere, as furnished by strong measurability. These need not be integrable, so they are truncated. Set

$$s_n(\omega) = \begin{cases} \sigma_n(\omega) & \text{if } \|\sigma_n(\omega)\| \le 2\|f(\omega)\| \\ 0 & \text{otherwise} \end{cases}$$

Each s_n is simple, since it takes each nonzero value of σ_n on the intersection of a measurable set with the measurable set $\{\omega \mid \|\sigma_n(\omega)\| \le 2\|f(\omega)\|\}$; and $\|s_n(\omega)\| \le 2\|f(\omega)\|$ everywhere, so $\int_\Omega \|s_n\|\, d\mu \le 2\int_\Omega \|f\|\, d\mu < \infty$, whence each s_n is integrable.

Pointwise convergence of s_n to f. Fix ω with $\sigma_n(\omega) \to f(\omega)$ (all but a null set of ω). If $f(\omega) \neq 0$, then $\|\sigma_n(\omega)\| \to \|f(\omega)\| < 2\|f(\omega)\|$, so $\|\sigma_n(\omega)\| \le 2\|f(\omega)\|$ for all large n, and thus $s_n(\omega) = \sigma_n(\omega) \to f(\omega)$. If $f(\omega) = 0$, then the truncation forces $\|s_n(\omega)\| \le 2\|f(\omega)\| = 0$, so $s_n(\omega) = 0 = f(\omega)$. In both cases $s_n(\omega) \to f(\omega)$, and moreover

$$\|s_n(\omega) - f(\omega)\| \le \|s_n(\omega)\| + \|f(\omega)\| \le 3\|f(\omega)\|$$

L^1 convergence via dominated convergence. The scalar functions $\omega \mapsto \|s_n(\omega) - f(\omega)\|$ converge to 0 pointwise almost everywhere and are dominated by the integrable function $3\|f\|$. By the dominated convergence theorem of the scalar theory,

$$\int_\Omega \|s_n - f\|\, d\mu \to 0$$

So the (s_n) witness Bochner integrability of f.

The norm estimate. With such (s_n), the basic estimate for simple functions and the continuity of the norm give

$$\left\|\int_\Omega f\, d\mu\right\| = \lim_n \left\|\int_\Omega s_n\, d\mu\right\| \le \lim_n \int_\Omega \|s_n\|\, d\mu = \int_\Omega \|f\|\, d\mu$$

the last equality by the necessity computation applied to the sequence (s_n), completing the proof.

The criterion reduces the existence of a vector-valued integral to a single scalar integrability question, which the reader already knows how to answer. In practice one checks that $t \mapsto \|f(t)\|$ is integrable, often by a crude bound such as continuity on a compact interval or an exponential decay estimate, and the Bochner integral then exists automatically. The accompanying norm inequality is the vector triangle inequality passed through the integral, controlling the size of an integrated curve by the integral of its length.

Example 13.1.7: The Laplace Transform of a Bounded Curve

Let X be a Banach space and $u\colon [0,\infty) \to X$ a norm-continuous curve with $\|u(t)\| \le Me^{\omega t}$ for constants $M \ge 0$ and $\omega \in \mathbb{R}$. For a scalar λ with $\operatorname{Re}\lambda > \omega$, consider the integrand $f(t) = e^{-\lambda t}u(t)$ on $[0,\infty)$. Continuity of u makes its range separable, so f is strongly measurable by Pettis's theorem (theorem 13.1.3); and

$$\int_0^\infty \|f(t)\|\, dt = \int_0^\infty e^{-(\operatorname{Re}\lambda)t}\|u(t)\|\, dt \le M\int_0^\infty e^{-(\operatorname{Re}\lambda-\omega)t}\, dt = \frac{M}{\operatorname{Re}\lambda - \omega}$$

is finite. By theorem 13.1.6, the Bochner integral $\int_0^\infty e^{-\lambda t}u(t)\,dt$ exists, and its norm is at most $M/(\operatorname{Re}\lambda-\omega)$. This is the Laplace transform of the curve u; section 13.5 identifies exactly this integral, taken against a semigroup orbit $u(t)=T(t)x$, with the resolvent $R(\lambda,A)x$ of the generator.

The integral would be of little use in operator theory if the unbounded operators of the chapter could not be moved across it. They can, provided the operator is closed: a closed operator applied to a Bochner integrable curve, whose image is again Bochner integrable, commutes with the integral. This is the vector-valued substitute for differentiation under the integral sign, and the semigroup theory derives its resolvent identities through it.

Proposition 13.1.8: Closed Operators Commute with the Bochner Integral

Let X and Y be Banach spaces and $T\colon \operatorname{dom}T\to Y$ a closed linear operator with domain $\operatorname{dom}T\subseteq X$. Let $f\colon\Omega\to X$ be Bochner integrable with $f(\omega)\in\operatorname{dom}T$ for almost every ω, and suppose the function $\omega\mapsto Tf(\omega)$ is Bochner integrable in Y. Then

$$\int_\Omega f\,d\mu\in\operatorname{dom}T \qquad\text{and}\qquad T\int_\Omega f\,d\mu=\int_\Omega Tf\,d\mu$$

***Proof*:**
Recall that T is closed means its graph $\Gamma(T)=\{(x,Tx)\mid x\in\operatorname{dom}T\}$ is a closed subspace of $X\oplus Y$, the direct sum carrying the norm $\|(x,y)\|=\|x\|+\|y\|$. The strategy is to integrate the graph curve $\omega\mapsto(f(\omega),Tf(\omega))$ into $X\oplus Y$ and read off that its integral lands in the closed subspace $\Gamma(T)$.

Step 1: the graph curve is Bochner integrable in $X\oplus Y$. Define $g\colon\Omega\to X\oplus Y$ by $g(\omega)=(f(\omega),Tf(\omega))$ where $f(\omega)\in\operatorname{dom}T$, and $g(\omega)=0$ on the null set where $f(\omega)\notin\operatorname{dom}T$. Both coordinate functions are strongly measurable, so g is strongly measurable, and

$$\int_\Omega\|g\|\,d\mu=\int_\Omega\big(\|f\|+\|Tf\|\big)\,d\mu=\int_\Omega\|f\|\,d\mu+\int_\Omega\|Tf\|\,d\mu<\infty$$

both terms finite by the integrability of f and of Tf and theorem 13.1.6. Hence g is Bochner integrable in $X\oplus Y$.

Step 2: the integral respects the two coordinate projections. The coordinate projections $\pi_X: X\oplus Y\to X$ and $\pi_Y: X\oplus Y\to Y$ are bounded linear maps. A bounded linear map S commutes with the Bochner integral: for integrable simple functions $s_n\to g$ in the L^1-norm one has $\int_\Omega Ss_n\,d\mu=S\int_\Omega s_n\,d\mu$ term by term, and passing to the limit through the bounded S gives $\int_\Omega Sg\,d\mu=S\int_\Omega g\,d\mu$. Applying this to π_X and π_Y,

$$\int_\Omega g\,d\mu=\left(\int_\Omega f\,d\mu,\ \int_\Omega Tf\,d\mu\right)$$

Step 3: the integral lies in the graph. Each value $g(\omega)$ lies in $\Gamma(T)$, which is a closed subspace of $X\oplus Y$. The integral of a $\Gamma(T)$-valued Bochner integrable function lies in $\Gamma(T)$, by the Hahn-Banach route through the annihilator. Let $\psi\in(X\oplus Y)^*$ annihilate $\Gamma(T)$, so $\psi(g(\omega))=0$ for every ω. Since ψ is a bounded linear functional, it passes through the integral by the bounded-map interchange of Step 2 (a functional is a bounded linear map into the scalars), giving

$$\psi\left(\int_\Omega g\,d\mu\right) = \int_\Omega \psi(g)\,d\mu = \int_\Omega 0\,d\mu = 0$$

Thus $\int_\Omega g\,d\mu$ is annihilated by every $\psi \in (X \oplus Y)^*$ that annihilates $\Gamma(T)$. A closed subspace equals the annihilator of its annihilator, by the Hahn-Banach separation theorem: any vector outside the closed subspace $\Gamma(T)$ is separated from it by some such ψ with $\psi \neq 0$ on that vector. Hence $\int_\Omega g\,d\mu \in \Gamma(T)$. Therefore

$$\left(\int_\Omega f\,d\mu,\ \int_\Omega Tf\,d\mu\right) = \int_\Omega g\,d\mu \in \Gamma(T)$$

By the definition of the graph, this says exactly that $\int_\Omega f\,d\mu \in \operatorname{dom} T$ and $T\int_\Omega f\,d\mu = \int_\Omega Tf\,d\mu$, as we sought to show.

The proposition is what lets an unbounded operator be handled by integrating its values rather than its action: to know T on the integral it suffices to integrate Tf. Closedness cannot be dropped, for an operator that is only densely defined can send a convergent sequence of integrals to a divergent one; the closed graph is precisely the hypothesis that survives the limit. Every resolvent formula of section 13.5, where T is the generator of a semigroup and f is a weighted orbit, rests on this interchange.

The last property is the one that makes the Bochner integral a genuine antiderivative. For a continuous curve on an interval, the integral from a fixed base point is differentiable in the upper limit, with the integrand as derivative. This is the fundamental theorem of calculus, and its vector-valued form is what turns the integral formulas of the semigroup theory into differential equations. The derivative here is the norm derivative: $t \mapsto v(t)$ has derivative w at t_0 if $\|(v(t) - v(t_0))/(t - t_0) - w\| \to 0$ as $t \to t_0$.

Theorem 13.1.9: Fundamental Theorem for the Bochner Integral

Let X be a Banach space and $g\colon [a,b] \to X$ continuous. Then the function

$$G\colon [a,b] \to X, \qquad G(t) = \int_a^t g(\tau)\,d\tau$$

is well-defined, continuously differentiable, and satisfies $G'(t) = g(t)$ for every $t \in [a,b]$.

***Proof*:**
Well-definedness. A continuous g on the compact interval $[a,b]$ has separable range, so it is strongly measurable by theorem 13.1.3, and it is bounded, say $\|g(\tau)\| \le M$, so $\int_a^t \|g\|\,d\tau \le M(b-a) < \infty$. By theorem 13.1.6 the Bochner integral defining $G(t)$ exists for each t.

The difference quotient. Fix $t_0 \in [a,b]$ and $t \neq t_0$ in $[a,b]$. By additivity of the integral over adjacent intervals,

$$G(t) - G(t_0) = \int_{t_0}^t g(\tau)\,d\tau$$

and since the constant function $g(t_0)$ integrates to $(t-t_0)g(t_0)$ over $[t_0, t]$,

$$\frac{G(t)-G(t_0)}{t-t_0} - g(t_0) = \frac{1}{t-t_0}\int_{t_0}^{t} \big(g(\tau)-g(t_0)\big)\,d\tau$$

Estimating by continuity. Let $\varepsilon > 0$. By continuity of g at t_0 there is a $\delta > 0$ with $\|g(\tau)-g(t_0)\| < \varepsilon$ whenever $|\tau - t_0| < \delta$. For $0 < |t-t_0| < \delta$ the integrand above satisfies $\|g(\tau)-g(t_0)\| < \varepsilon$ on the whole interval of integration, so by the norm estimate of theorem 13.1.6,

$$\left\|\frac{G(t)-G(t_0)}{t-t_0} - g(t_0)\right\| \le \frac{1}{|t-t_0|}\left|\int_{t_0}^{t} \|g(\tau)-g(t_0)\|\,d\tau\right| \le \frac{1}{|t-t_0|}\cdot \varepsilon\,|t-t_0| = \varepsilon$$

Hence the difference quotient converges to $g(t_0)$ in the norm of X as $t \to t_0$, so G is differentiable at t_0 with $G'(t_0) = g(t_0)$.

Continuity of the derivative. The derivative $G' = g$ is continuous by hypothesis, so G is continuously differentiable, completing the proof.

The theorem says the Bochner integral inverts differentiation exactly as in the scalar case: to solve $u'(t) = h(t)$ with a continuous right side, integrate h. The generation theorems of this chapter are read through this lens. A semigroup orbit $t \mapsto T(t)x$ has, for x in the domain of the generator, derivative $AT(t)x$, and the equation $\frac{d}{dt}T(t)x = AT(t)x$ that identifies A is exactly a statement that a Bochner integral of the right side recovers the orbit. The multiplication-operator and heat-semigroup models of the later sections will each be verified by differentiating an explicit Bochner integral of this kind.

13.2 Unbounded Operators and Their Adjoints

This section sets up the operators the spectral theorem will act on and locates the self-adjoint ones among them. An unbounded operator carries a domain as part of its data, and the two structural notions that make it manageable, closedness and the adjoint, both read cleanly off its graph. Symmetry is the naive analogue of self-adjointness and is strictly weaker; the section closes by identifying, through the ranges of $T \pm i$, exactly which symmetric operators are self-adjoint.

An operator on a Hilbert space $\mathbb{H}$ need not be defined on all of $\mathbb{H}$. The three operators named in the chapter opening make the point: momentum $-i\,d/dx$ is defined only on differentiable functions, and the value $-i\,f'$ of such a function need not be square-integrable even when f is. So an unbounded operator is a linear map defined on a subspace, and that subspace is not an afterthought but part of the operator's identity.

Definition 13.2.1: Densely-Defined Operator and Its Graph

An *operator* T on a Hilbert space $\mathbb{H}$ is a linear map $T\colon \operatorname{dom} T \to \mathbb{H}$ defined on a linear subspace $\operatorname{dom} T \subseteq \mathbb{H}$, its *domain*. The operator is *densely defined* if $\operatorname{dom} T$ is dense in $\mathbb{H}$. Its *range* is $\operatorname{ran} T = \{Tx \mid x \in \operatorname{dom} T\}$, and its *graph* is the subspace

$$\Gamma(T) = \{(x, Tx) \mid x \in \operatorname{dom} T\} \subseteq \mathbb{H} \oplus \mathbb{H}$$

Two operators are equal when they have the same domain and agree on it; equivalently, when their graphs coincide. Write $S \subseteq T$, and call T an *extension* of S, when $\Gamma(S) \subseteq \Gamma(T)$, that is, $\operatorname{dom} S \subseteq \operatorname{dom} T$ and T restricts to S on $\operatorname{dom} S$.

The graph encodes the operator completely, and the inclusion order on operators is the inclusion order on graphs. This is the reorganization that makes the theory work: statements about domains and values become statements about a single subspace of $\mathbb{H} \oplus \mathbb{H}$, on which the Hilbert-space geometry of the ambient product is available. The word "unbounded" is used loosely for an operator defined on a proper dense domain; a genuinely unbounded operator is discontinuous, but the domain, not the discontinuity, is what the definitions track.

Example 13.2.2: The Momentum Operator

Let $\mathbb{H} = L^2(\mathbb{R})$ and let $T = -i\,d/dx$ act on the domain $\operatorname{dom} T = C_c^\infty(\mathbb{R})$ of smooth compactly supported functions, so $Tf = -if'$. The domain is dense in $L^2(\mathbb{R})$, so T is densely defined. It is unbounded: with $f(x) = e^{-x^2/2}$ scaled to $f_\lambda(x) = f(\lambda x)$, so that $f_\lambda'(x) = \lambda f'(\lambda x)$, direct computation gives $\|f_\lambda'\|_2 = \lambda^{1/2}\|f'\|_2$ while $\|f_\lambda\|_2 = \lambda^{-1/2}\|f\|_2$, so the ratio $\|Tf_\lambda\|_2/\|f_\lambda\|_2$ grows like λ and is not bounded over the domain. (A smooth cutoff makes each f_λ compactly supported without changing the growth.) The factor $-i$ is chosen so that T is symmetric, as the next definitions will show: integration by parts moves the derivative across the inner product and the i absorbs the sign.

Example 13.2.3: The Position (Multiplication) Operator

Let $\mathbb{H} = L^2(\mathbb{R})$ and let M be multiplication by the coordinate, $(Mf)(x) = xf(x)$, on the natural domain $\operatorname{dom} M = \{f \in L^2(\mathbb{R}) \mid xf(x) \in L^2(\mathbb{R})\}$, the largest domain on which the formula lands back in $\mathbb{H}$. It contains $C_c^\infty(\mathbb{R})$ and so is dense. The operator is unbounded, since for f supported near a large R the factor $x \approx R$ multiplies the norm by roughly R. This is the model unbounded operator: the multiplication operator by an unbounded measurable function is the concrete form to which every self-adjoint operator will reduce (theorem 13.3.6).

For the multiplication operator the maximal domain above is forced: it is the set of f for which Mf makes sense as an element of $\mathbb{H}$, and there is no room to enlarge it. For a differential operator the domain is a genuine choice, and different choices give different operators with different adjoints and different spectral behaviour. Managing that choice is what the rest of the section is about; the first requirement to impose is that the graph be closed.

Definition 13.2.4: Closed and Closable Operators

An operator T is *closed* if its graph $\Gamma(T)$ is a closed subspace of $\mathbb{H}\oplus\mathbb{H}$. Concretely, T is closed when for every sequence (x_n) in $\operatorname{dom} T$ with $x_n \to x$ and $Tx_n \to y$ in $\mathbb{H}$, one has $x \in \operatorname{dom} T$ and $Tx = y$. The operator T is *closable* if the closure $\overline{\Gamma(T)}$ of its graph in $\mathbb{H}\oplus\mathbb{H}$ is again the graph of an operator, that is, if $\overline{\Gamma(T)}$ contains no point $(0, y)$ with $y \neq 0$. In that case the operator whose graph is $\overline{\Gamma(T)}$ is the *closure* $\overline{T}$ of T, the smallest closed extension of T.

Closedness is the substitute for continuity available in the unbounded setting. A bounded operator extends by continuity to the closure of its domain, so a densely defined bounded operator is closed after that extension; the closed graph theorem says the two conditions coincide for operators defined on all of $\mathbb{H}$. For an unbounded operator continuity fails, but closedness can survive, and it is exactly enough for the arguments that matter: it lets one pass to limits inside the operator. The obstruction to closability is a subspace $\overline{\Gamma(T)}$ containing $(0, y)$ with $y \neq 0$, which would force $T\,0 = y \neq 0$ on any operator with that graph; closability rules this out, and then $\overline{T}$ is the operator read off the closed subspace $\overline{\Gamma(T)}$.

Example 13.2.5: The Momentum Operator Is Closable

Return to $T = -i\,d/dx$ on $C_c^\infty(\mathbb{R})$ from example 13.2.2. It is not closed: the limit of a sequence $f_n \in C_c^\infty(\mathbb{R})$ with $f_n \to f$ and $-if_n' \to g$ in L^2 need only have $g = -if'$ in the weak (distributional) sense, and such an f need not lie in $C_c^\infty(\mathbb{R})$. But T is closable. Suppose $f_n \to 0$ and $-if_n' \to g$ in L^2. For any $\varphi \in C_c^\infty(\mathbb{R})$, integration by parts on the compactly supported f_n gives $\langle -if_n', \varphi\rangle = \langle f_n, -i\varphi'\rangle$, and letting $n \to \infty$ the right side tends to $\langle 0, -i\varphi'\rangle = 0$ while the left tends to $\langle g, \varphi\rangle$. Thus g is orthogonal to the dense set $C_c^\infty(\mathbb{R})$, so $g = 0$. Hence $\overline{\Gamma(T)}$ contains no $(0, g)$ with $g \neq 0$, and T is closable. Its closure is $-i\,d/dx$ on the Sobolev space $H^1(\mathbb{R})$ (chapter 10), the functions with a square-integrable weak derivative.

Closability was checked by pairing against a dense set of test functions and pushing the derivative across the inner product. That maneuver is the adjoint, made into an operator. For a bounded T the adjoint is defined on all of $\mathbb{H}$ by the Riesz representation theorem (proposition 11.2.7); for an unbounded T the same defining relation still determines the values of the adjoint, but now it also determines the domain, since the relation can be satisfied only for those y that make $x \mapsto \langle Tx, y\rangle$ continuous.

Definition 13.2.6: Adjoint of a Densely-Defined Operator

Let T be densely defined on $\mathbb{H}$. The *adjoint* T^* is the operator with domain

$$\operatorname{dom} T^* = \{y \in \mathbb{H} \mid x \mapsto \langle Tx, y\rangle \text{ is continuous on } \operatorname{dom} T\}$$

defined as follows. For $y \in \operatorname{dom} T^*$ the functional $x \mapsto \langle Tx, y\rangle$, continuous on the dense subspace $\operatorname{dom} T$, extends uniquely to a bounded linear functional on $\mathbb{H}$, so by the Riesz representation theorem (theorem 11.1.13) there is a unique $z \in \mathbb{H}$ with $\langle Tx, y\rangle = \langle x, z\rangle$ for all $x \in \operatorname{dom} T$; set $T^*y = z$. Thus T^*y is characterized by

$$\langle Tx, y\rangle = \langle x, T^*y\rangle \qquad \text{for all } x \in \operatorname{dom} T \tag{13.1}$$

Density of $\operatorname{dom} T$ is what makes T^*y well defined: two vectors representing the same functional on a dense subspace agree on all of $\mathbb{H}$, so z is unique. Without density the value T^*y would be ambiguous, which is why the adjoint is defined only for densely defined operators. The domain of T^* is part of the construction: a vector y belongs to it precisely when the pairing $\langle Tx, y\rangle$ is controlled by $\|x\|$ alone, uniformly over the domain, and how large this set is depends delicately on how large $\operatorname{dom} T$ was. Shrinking the domain of T enlarges the domain of T^*, since fewer constraints on x make continuity easier to achieve; this inverse relationship is the source of the gap between symmetric and self-adjoint operators below.

Example 13.2.7: The Adjoint of the Momentum Operator

The adjoint T^* of $T = -i\,d/dx$ on $C_c^\infty(\mathbb{R})$ is identified as follows. Let $y \in L^2(\mathbb{R})$ be such that $f \mapsto \langle -if', y\rangle$ is continuous on $C_c^\infty(\mathbb{R})$. Integration by parts gives, for $f \in C_c^\infty(\mathbb{R})$,

$$\langle Tf, y\rangle = \int_{\mathbb{R}} (-if'(x))\,\overline{y(x)}\,dx$$

and the requirement that this be a bounded functional of f in the L^2-norm says exactly that y has a weak derivative in L^2, that is, $y \in H^1(\mathbb{R})$, and then $\langle Tf, y\rangle = \langle f, -iy'\rangle$. Hence $\operatorname{dom} T^* = H^1(\mathbb{R})$ and $T^*y = -iy'$. The adjoint is the same differential expression on the strictly larger domain $H^1(\mathbb{R}) \supsetneq C_c^\infty(\mathbb{R})$: the operator is a proper restriction of its own adjoint, the exact situation the next definition names.

The momentum example shows the adjoint agreeing with the operator as a formula while acting on a larger domain. When the operator sits inside its adjoint this way the pairing $\langle Tx, y\rangle$ is symmetric on the original domain, and when the two operators coincide outright the spectral theorem will apply. These are the two conditions to name.

Definition 13.2.8: Symmetric, Self-Adjoint, Essentially Self-Adjoint

Let T be densely defined on $\mathbb{H}$.

1. T is *symmetric* if $T \subseteq T^*$, equivalently if $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in \operatorname{dom} T$.
2. T is *self-adjoint* if $T = T^*$, that is, symmetric and with $\operatorname{dom} T^* = \operatorname{dom} T$.
3. T is *essentially self-adjoint* if it is symmetric and its closure $\overline{T}$ is self-adjoint.

The equivalence in (1) is immediate from (13.1): symmetry says $\langle Tx, y\rangle = \langle x, Ty\rangle$ on the domain, which is exactly the statement that every $y \in \operatorname{dom} T$ lies in $\operatorname{dom} T^*$ with $T^*y = Ty$, i.e. $T \subseteq T^*$. Self-adjointness demands the reverse inclusion as well, and by the inverse relationship noted above this is a real constraint on the domain, not on the formula. Essential self-adjointness is the practical middle ground: a symmetric operator defined on a convenient small domain (test functions, say) is rarely self-adjoint there, but if its closure is self-adjoint then the small domain already determines a unique self-adjoint operator, and the small domain is called a *core*. The distinction is not pedantry, as the following example shows: symmetry is genuinely weaker than self-adjointness, and the difference is measured by domains.

Example 13.2.9: Symmetric but Not Self-Adjoint

Let $\mathbb{H} = L^2([0,1])$ and let $T = -i\,d/dx$ act on

$$\operatorname{dom} T = \big\{ f \in H^1([0,1]) \;\big|\; f(0) = f(1) = 0 \big\}$$

the smooth-enough functions vanishing at both endpoints. For $f, g \in \operatorname{dom} T$, integration by parts gives

$$\langle Tf, g\rangle = \int_0^1 (-if')\,\overline{g}\,dx = \big[-if\,\overline{g}\big]_0^1 + \int_0^1 f\,\overline{(-ig')}\,dx = \langle f, Tg\rangle$$

the boundary term vanishing because $f(0) = f(1) = 0$. So T is symmetric. But it is not self-adjoint. Computing T^*, the boundary term $[-if\,\overline{g}]_0^1$ vanishes for all f with $f(0) = f(1) = 0$ regardless of the values $g(0), g(1)$, so no boundary condition on g is forced: $\operatorname{dom} T^* = H^1([0,1])$ with $T^*g = -ig'$, the full first Sobolev space. Thus $\operatorname{dom} T^* \supsetneq \operatorname{dom} T$ and $T \subsetneq T^*$; the operator is symmetric but not self-adjoint. Imposing instead the periodic condition $f(0) = f(1)$ gives an operator T_{per} that is self-adjoint, since then the boundary term forces the matching condition on g too. The moral is that boundary conditions, encoded in the domain, decide self-adjointness; symmetry alone does not see them.

Two operators that agree as differential expressions can therefore differ as self-adjoint operators, and the difference is invisible to symmetry. To detect self-adjointness one needs an intrinsic criterion, and to state it one first needs the two structural facts about the adjoint: it is always closed, and its density governs whether T itself is closable. Both come from a single geometric observation about the graph.

The adjoint has a graph, and that graph is a rotation of the graph of T inside $\mathbb{H} \oplus \mathbb{H}$. Introduce the unitary

$$V : \mathbb{H} \oplus \mathbb{H} \to \mathbb{H} \oplus \mathbb{H}, \qquad V(x, y) = (-y, x) \tag{13.2}$$

a quarter-turn of the product space; it is unitary, with $V^2 = -I$ and $V^{-1}(x, y) = (y, -x)$. The defining relation (13.1) is an orthogonality statement between the two graphs under V, which is the content of the next proposition and the basis of everything after it.

Proposition 13.2.10: The Adjoint Is Closed; Closability via the Adjoint

Let T be densely defined on $\mathbb{H}$, and let V be the unitary of (13.2). Then:

1. $\Gamma(T^*) = \big(V\,\Gamma(T)\big)^\perp$ in $\mathbb{H} \oplus \mathbb{H}$. In particular T^* is closed.
2. T is closable if and only if T^* is densely defined, and then $T^{**} = \overline{T}$.

***Proof*:**

Step 1: The graph of T^* as an orthogonal complement. A pair $(y, z) \in \mathbb{H} \oplus \mathbb{H}$ lies in $\big(V\,\Gamma(T)\big)^\perp$ precisely when it is orthogonal to $V(x, Tx) = (-Tx, x)$ for every $x \in \operatorname{dom} T$, that is,

$$0 = \langle (y, z), (-Tx, x)\rangle = -\langle y, Tx\rangle + \langle z, x\rangle$$

for all $x \in \operatorname{dom} T$. Conjugating, this reads $\langle Tx, y\rangle = \langle x, z\rangle$ for all $x \in \operatorname{dom} T$. By (13.1) this holds exactly when $y \in \operatorname{dom} T^*$ and $z = T^*y$, i.e. when $(y, z) \in \Gamma(T^*)$. Hence $\Gamma(T^*) = (V\,\Gamma(T))^\perp$. An orthogonal complement is a closed subspace, so $\Gamma(T^*)$ is closed and T^* is a closed operator. This proves (1).

Step 2: V preserves orthogonal complements. Since V is unitary it carries closed subspaces to closed subspaces and commutes with taking orthogonal complements: $V(M^\perp) = (VM)^\perp$ for any subspace M. Moreover $V^2 = -I$ acts as multiplication by a scalar of modulus one on each factor, so $V^2M = M$ for every subspace M. Both facts are used below.

Step 3: Closability equals density of $\operatorname{dom} T^*$. The closure of $\Gamma(T)$ is recovered from (1) by taking complements twice. Since $M^{\perp\perp} = \overline{M}$ for any subspace M, and using Step 2,

$$\overline{\Gamma(T)} = \Gamma(T)^{\perp\perp} = \big(V^2\Gamma(T)\big)^{\perp\perp} = \big(V(V\Gamma(T))\big)^{\perp\perp}$$

Applying $V(M^\perp) = (VM)^\perp$ and part (1),

$$\overline{\Gamma(T)} = \big(V\,(V\Gamma(T))^\perp\big)^\perp = \big(V\,\Gamma(T^*)\big)^\perp$$

Now T is closable if and only if $\overline{\Gamma(T)}$ contains no $(0,w)$ with $w \neq 0$ (definition 13.2.4). A pair $(0,w)$ lies in $(V\,\Gamma(T^*))^\perp$ exactly when it is orthogonal to $V(y, T^*y) = (-T^*y, y)$ for every $y \in \operatorname{dom} T^*$, i.e. when $\langle w, y\rangle = 0$ for all $y \in \operatorname{dom} T^*$; equivalently $w \in (\operatorname{dom} T^*)^\perp$. Thus $\overline{\Gamma(T)}$ contains a nonzero $(0,w)$ if and only if $(\operatorname{dom} T^*)^\perp \neq \{0\}$, that is, if and only if $\operatorname{dom} T^*$ is *not* dense. Contrapositively, T is closable if and only if T^* is densely defined.

Step 4: $T^{**} = \overline{T}$. Suppose T is closable, so by Step 3 T^* is densely defined and $T^{**} = (T^*)^*$ makes sense. Applying part (1) to T^* in place of T,

$$\Gamma(T^{**}) = \big(V\,\Gamma(T^*)\big)^\perp$$

which is exactly $\overline{\Gamma(T)}$ by the displayed identity of Step 3. Two operators with the same graph are equal, so $T^{**} = \overline{T}$, completing the proof.

The proposition is the payoff of moving to graphs. The adjoint is the orthogonal complement of a rotated graph, so it is automatically closed, and taking the adjoint twice returns the closure of the original operator, provided the first adjoint is densely defined. For a symmetric operator $T \subseteq T^*$ this last proviso is free: T is densely defined and contained in T^*, so T^* is densely defined and T is closable, with $\overline{T} = T^{**} \subseteq T^*$. Every symmetric operator is thus automatically closable, and its closure is still symmetric. This is why one may always replace a symmetric operator by its closure without loss, and why essential self-adjointness is the right target for operators given on a small core.

For a self-adjoint operator the chain collapses: $T = T^*$ forces $T = T^* = T^{**} = \overline{T}$, so a self-adjoint operator is closed. A symmetric operator has the descending chain $T \subseteq \overline{T} = T^{**} \subseteq T^*$, and it is self-adjoint exactly when the whole chain collapses to a single operator. The criterion below measures the gap in the chain by the size of two subspaces and converts self-adjointness into a surjectivity statement one can check.

The idea is to test $T \pm i$ rather than T itself. Adding $\pm i$ to a symmetric operator makes it bounded below in a way that rules out kernel and forces closed range, so the only thing that can fail is surjectivity; and surjectivity of both $T + i$ and $T - i$ is precisely enough to close the gap between T and T^*. The two kernels $\ker(T^* \mp i)$, which measure the failure of surjectivity, are the deficiency spaces.

Theorem 13.2.11: Basic Criterion for Self-Adjointness

Let T be a symmetric operator on $\mathbb{H}$. The following are equivalent.

1. T is self-adjoint.
2. T is closed and $\ker(T^* - i) = \ker(T^* + i) = \{0\}$.
3. $\operatorname{ran}(T + i) = \mathbb{H}$ and $\operatorname{ran}(T - i) = \mathbb{H}$.

The dimensions $n_\pm = \dim\ker(T^* \mp i)$, the *deficiency indices* of T, thus measure the failure of self-adjointness: a closed symmetric operator is self-adjoint if and only if $n_+ = n_- = 0$.

***Proof*:**
Step 1: The fundamental estimate. For $x \in \operatorname{dom} T$ and the symmetry $\langle Tx, x\rangle = \langle x, Tx\rangle = \overline{\langle Tx, x\rangle}$, the number $\langle Tx, x\rangle$ is real. Hence

$$\|(T \pm i)x\|^2 = \|Tx\|^2 \pm \langle Tx, ix\rangle \pm \langle ix, Tx\rangle + \|x\|^2 = \|Tx\|^2 + \|x\|^2$$

the two cross terms $\mp i\langle Tx, x\rangle \pm i\overline{\langle Tx, x\rangle}$ cancelling because $\langle Tx, x\rangle$ is real. In particular

$$\|(T \pm i)x\| \geq \|x\| \qquad \text{for all } x \in \operatorname{dom} T \tag{13.3}$$

so $T \pm i$ is injective and bounded below. If moreover T is closed, then $T \pm i$ is closed (its graph is that of T sheared by the bounded map $x \mapsto \pm ix$), and a closed operator bounded below has closed range: if $(T \pm i)x_n \to w$, then (13.3) makes (x_n) Cauchy, so $x_n \to x$, and closedness gives $x \in \operatorname{dom} T$ with $(T \pm i)x = w$. Thus for closed symmetric T the ranges $\operatorname{ran}(T \pm i)$ are closed subspaces.

Step 2: The range-kernel duality. For any densely defined operator, a vector w is orthogonal to $\operatorname{ran}(T + i)$ if and only if $\langle (T + i)x, w\rangle = 0$ for all $x \in \operatorname{dom} T$, i.e. $\langle Tx, w\rangle = -\langle ix, w\rangle = \langle x, iw\rangle$ for all such x. By (13.1) this says exactly $w \in \operatorname{dom} T^*$ with $T^*w = iw$, that is, $w \in \ker(T^* - i)$. Hence

$$\operatorname{ran}(T + i)^\perp = \ker(T^* - i), \qquad \operatorname{ran}(T - i)^\perp = \ker(T^* + i) \tag{13.4}$$

the second identity by the same computation with $-i$ in place of i.

Step 3: (1) ⇒ (2). If T is self-adjoint then $T = T^* = \overline{T}$ is closed (proposition 13.2.10). Suppose $T^*w = iw$ for some w. Then, using $T = T^*$ and the reality of $\langle Tw, w\rangle$ from Step 1,

$$i\|w\|^2 = \langle iw, w\rangle = \langle T^*w, w\rangle = \langle Tw, w\rangle \in \mathbb{R}$$

forcing $\|w\|^2 = 0$, so $w = 0$; likewise $\ker(T^* + i) = \{0\}$. This gives (2).

Step 4: (2) ⇒ (3). Assume T closed with both deficiency kernels trivial. By Step 1 the ranges $\operatorname{ran}(T \pm i)$ are closed, so $\operatorname{ran}(T + i) = \overline{\operatorname{ran}(T + i)} = \big(\operatorname{ran}(T + i)^\perp\big)^\perp = \ker(T^* - i)^\perp = \{0\}^\perp = \mathbb{H}$, using (13.4); the same argument gives $\operatorname{ran}(T - i) = \mathbb{H}$. This is (3).

Step 5: (3) ⇒ (1). Assume $\operatorname{ran}(T \pm i) = \mathbb{H}$. Since T is symmetric, $T \subseteq T^*$, and it remains only to show $\operatorname{dom} T^* \subseteq \operatorname{dom} T$. Let $y \in \operatorname{dom} T^*$. Because $\operatorname{ran}(T + i) = \mathbb{H}$, there is $x \in \operatorname{dom} T$ with $(T + i)x = (T^* + i)y$; here $(T^* + i)y$ makes sense as $y \in \operatorname{dom} T^*$. As $T \subseteq T^*$, also $x \in \operatorname{dom} T^*$ and $(T^* + i)x = (T + i)x = (T^* + i)y$, so

$$(T^* + i)(y - x) = 0$$

that is, $y - x \in \ker(T^* + i) = \operatorname{ran}(T - i)^\perp$ by (13.4). But $\operatorname{ran}(T - i) = \mathbb{H}$ by hypothesis, so its orthogonal complement is $\{0\}$, giving $y = x \in \operatorname{dom} T$. Hence $\operatorname{dom} T^* = \operatorname{dom} T$ and $T = T^*$, as we sought to show.

Deficiency indices. By (13.4) the deficiency space $\ker(T^* \mp i)$ is the orthogonal complement of $\operatorname{ran}(T \pm i)$, whose vanishing (for closed symmetric T) is equivalent to surjectivity of $T \pm i$ by Steps 1 and 4. Thus $n_\pm = 0$ simultaneously characterizes self-adjointness among closed symmetric operators, which is the final assertion.

The criterion converts a domain question, whether T^* overreaches T, into a range question, whether $T \pm i$ hits all of $\mathbb{H}$. Its usefulness is twofold. First, ranges are often computable: verifying $\operatorname{ran}(T \pm i) = \mathbb{H}$ means solving the equations $(T \pm i)x = w$ for every w, which for a differential operator is a boundary-value problem. Second, the deficiency indices (n_+, n_-) organize the failure. When they are equal and finite the symmetric operator has a finite-dimensional family of self-adjoint extensions parametrized by the unitaries between the two deficiency spaces (the von Neumann extension theory), and when one of them is nonzero while the other vanishes no self-adjoint extension exists at all. The endpoint conditions of example 13.2.9 are the visible face of this: the two boundary values give $n_+ = n_- = 1$ there, and the circle of self-adjoint extensions is the circle of admissible matching conditions $f(1) = e^{i\theta} f(0)$.

The equivalence with essential self-adjointness follows by applying the criterion to the closure. Since $\overline{T} = T^{**}$ has the same adjoint as T (both equal T^*, as $(T^{**})^* = \overline{T^*} = T^*$ using proposition 13.2.10 and the closedness of T^*), the deficiency spaces of T and of $\overline{T}$ coincide, and $\overline{T}$ is self-adjoint exactly when $\ker(T^* \mp i) = \{0\}$. This gives the working test used throughout the sequel: a symmetric operator is essentially self-adjoint if and only if $\ker(T^* - i) = \ker(T^* + i) = \{0\}$, equivalently $\operatorname{ran}(T + i)$ and $\operatorname{ran}(T - i)$ are both dense in $\mathbb{H}$. For the momentum operator on the whole line this is the statement that $-i\, d/dx$ on $C_c^\infty(\mathbb{R})$ is essentially self-adjoint, its closure being the self-adjoint operator on $H^1(\mathbb{R})$ of example 13.2.5.

13.3 The Spectral Theorem for Unbounded Self-Adjoint Operators

The spectral theorem for bounded self-adjoint operators (theorem 12.4.6) is not re-derived here. Instead a self-adjoint T is converted into a bounded unitary operator, its Cayley transform, the bounded theorem is applied to that, and the resulting resolution of the identity is transported back to T, yielding the functional calculus.

The first task is to say what the spectrum of an unbounded operator is at all. For a bounded operator the resolvent $(\lambda I - T)^{-1}$ was a bounded inverse in the algebra $B(\mathbb{H})$, and the spectrum was the compact set where no such inverse exists. An unbounded T is defined only on $\operatorname{dom} T$, so $\lambda I - T$ is a map from $\operatorname{dom} T$ into $\mathbb{H}$, and inverting it means finding a bounded operator on all of $\mathbb{H}$ that undoes it. The closedness of T (definition 13.2.4) is exactly the hypothesis that makes such a bounded inverse automatic once $\lambda I - T$ is a bijection, by the closed graph theorem.

Definition 13.3.1: Resolvent Set and Spectrum of a Closed Operator

Let T be a closed, densely-defined operator on $\mathbb{H}$. The *resolvent set* $\rho(T)$ is the set of $\lambda \in \mathbb{C}$ for which $\lambda I - T\colon \operatorname{dom} T \to \mathbb{H}$ is a bijection with bounded inverse; that inverse is the *resolvent*

$$R(\lambda, T) = (\lambda I - T)^{-1} \in B(\mathbb{H})$$

The *spectrum* is the complement $\sigma(T) = \mathbb{C} \setminus \rho(T)$.

For a closed operator the boundedness of the inverse is not an extra demand. If $\lambda I - T$ is a bijection of $\operatorname{dom} T$ onto $\mathbb{H}$, its inverse is a closed operator defined on all of $\mathbb{H}$: the graph of $R(\lambda, T)$ is the graph of $\lambda I - T$ with its two coordinates swapped, and swapping coordinates preserves closedness. A closed operator defined on the whole space is bounded by the closed graph theorem, so $R(\lambda, T)$ is automatically bounded. Thus for closed T one has $\lambda \in \rho(T)$ precisely when $\lambda I - T$ is a bijection $\operatorname{dom} T \to \mathbb{H}$; the bounded-inverse clause in the definition is a consequence, recorded for emphasis. This is the first place the standing assumption of closedness is used.

A concrete unbounded operator makes the spectrum visible.

Example 13.3.2: The Momentum Operator on the Line

Let $\mathbb{H} = L^2(\mathbb{R})$ and let $T = -i\, d/dx$ on the domain $\operatorname{dom} T = \{f \in L^2(\mathbb{R}) \mid f \text{ absolutely continuous}, f' \in L^2(\mathbb{R})\}$, the maximal domain on which the distributional derivative lands back in L^2. This T is self-adjoint, and its spectrum is the whole real line, $\sigma(T) = \mathbb{R}$, with no eigenvalues. For $\lambda \in \mathbb{C}$ with $\operatorname{Im} \lambda \neq 0$ the equation $-if' - \lambda f = g$ is solved by the bounded convolution

$$f(x) = R(\lambda, T)g(x) = \int_{\mathbb{R}} k_\lambda(x - y)\, g(y)\, dy$$

whose kernel k_λ decays exponentially because $\operatorname{Im} \lambda \neq 0$, so $R(\lambda, T)$ is a bounded operator and $\lambda \in \rho(T)$. For real λ the candidate eigenfunction $e^{i\lambda x}$ solves $-if' = \lambda f$ but is not in $L^2(\mathbb{R})$, so λ is not an eigenvalue; nevertheless $\lambda I - T$ fails to be onto (its range is a dense proper subspace, as the plane waves $e^{i\lambda x}$ occupy the "continuous spectrum"), so $\lambda \in \sigma(T)$. The spectrum is real and unbounded, and it consists entirely of continuous spectrum.

The example already exhibits the two structural facts that make the self-adjoint case tractable: the spectrum lies on the real axis, and the resolvent is controlled by the distance from λ to that axis. Both follow from a single inequality, valid for symmetric operators, that bounds $\|(\lambda I - T)x\|$ below by $|\operatorname{Im} \lambda|\, \|x\|$.

Proposition 13.3.3: The Spectrum of a Self-Adjoint Operator is Real

Let T be self-adjoint (definition 13.2.8). Then $\sigma(T) \subseteq \mathbb{R}$, and for every λ with $\operatorname{Im} \lambda \neq 0$,

$$\|R(\lambda, T)\| \leq \frac{1}{|\operatorname{Im} \lambda|} \tag{13.5}$$

***Proof*:**
Write $\lambda = a + ib$ with $a, b \in \mathbb{R}$ and $b \neq 0$. For $x \in \operatorname{dom} T$ the vector $(\lambda I - T)x = (aI - T)x + ibx$ has squared norm

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 + 2\operatorname{Re}\langle (aI - T)x,\, ibx\rangle$$

The cross term vanishes: since T is symmetric and a is real, $(aI - T)$ is symmetric, so $\langle (aI-T)x, x\rangle$ is real, and $\operatorname{Re}\langle (aI - T)x,\, ibx\rangle = \operatorname{Re}\big(-ib\,\overline{\langle (aI - T)x, x\rangle}\big) = 0$. Hence

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 \geq b^2\|x\|^2 \tag{13.6}$$

so $\|(\lambda I - T)x\| \geq |b|\,\|x\| = |\operatorname{Im}\lambda|\,\|x\|$.

This lower bound has three consequences. First, $\lambda I - T$ is injective, since $(\lambda I - T)x = 0$ forces $x = 0$. Second, its range is closed: if $(\lambda I - T)x_n \to y$, the bound makes (x_n) Cauchy, so $x_n \to x$ for some x, and closedness of T (a self-adjoint operator is closed, being an adjoint, definition 13.2.4) gives $x \in \operatorname{dom} T$ with $(\lambda I - T)x = y$, so $y \in \operatorname{ran}(\lambda I - T)$. Third, the range is dense: its orthogonal complement is $\ker(\overline{\lambda} I - T^*) = \ker(\overline{\lambda} I - T)$ (using $T^* = T$), and the lower bound applied to $\overline{\lambda}$ (which also has nonzero imaginary part) shows $\overline{\lambda} I - T$ is injective, so this kernel is $\{0\}$. A closed dense subspace is the whole of $\mathbb{H}$, so $\lambda I - T$ is a bijection onto $\mathbb{H}$, i.e. $\lambda \in \rho(T)$. Every λ with $\operatorname{Im}\lambda \neq 0$ therefore lies in the resolvent set, which is to say $\sigma(T) \subseteq \mathbb{R}$. Finally, for such λ the inverse inherits the reciprocal bound: putting $x = R(\lambda, T)y$ into $\|(\lambda I - T)x\| \geq |\operatorname{Im}\lambda|\,\|x\|$ gives $\|y\| \geq |\operatorname{Im}\lambda|\,\|R(\lambda, T)y\|$ for all y, which is (13.5), completing the proof.

Reality of the spectrum is the unbounded echo of the bounded fact that a self-adjoint operator has real spectrum, proved in the bounded chapter (theorem 12.4.6). The resolvent bound (13.5) is the quantitative refinement: the resolvent blows up no faster than the reciprocal of the distance to the real axis, so $R(\lambda, T)$ is a bounded holomorphic function of λ off the line. It is this bound, evaluated at $\lambda = \pm i$, that will make the two operators $T \pm iI$ boundedly invertible and open the door to the Cayley transform.

The resolvent bound tells us where T is invertible but not what T looks like. To import the bounded spectral theorem we must manufacture a bounded operator out of T that retains all of its spectral information. The device is due to Cayley: the map $z \mapsto (z - i)/(z + i)$ carries the real axis bijectively onto the unit circle minus the point 1, sending ∞ to 1; applying the same fractional linear transformation to the operator T turns a self-adjoint operator (spectrum on $\mathbb{R}$) into a unitary operator (spectrum on the circle). The exceptional point 1 on the circle corresponds to the point at infinity, which is exactly the room an unbounded operator needs.

Theorem 13.3.4: The Cayley Transform

Let T be self-adjoint. The operators $T \pm iI\colon \operatorname{dom} T \to \mathbb{H}$ are bijections onto $\mathbb{H}$, and the *Cayley transform*

$$U = (T - iI)(T + iI)^{-1}$$

is a unitary operator on $\mathbb{H}$ for which 1 is not an eigenvalue (equivalently, $\operatorname{ran}(I - U)$ is dense). Conversely, if U is unitary and 1 is not an eigenvalue of U, then

$$T = i(I + U)(I - U)^{-1}$$

defined on $\operatorname{dom} T = \operatorname{ran}(I - U)$, is self-adjoint and has U as its Cayley transform. The assignment $T \mapsto U$ is a bijection between self-adjoint operators on $\mathbb{H}$ and unitary operators on $\mathbb{H}$ with no eigenvalue 1.

***Proof*:**
Step 1: $T + iI$ and $T - iI$ are bijections onto $\mathbb{H}$. By theorem 13.2.11, a symmetric operator T is self-adjoint if and only if $\operatorname{ran}(T+iI) = \operatorname{ran}(T-iI) = \mathbb{H}$; the self-adjointness of T is thus precisely the surjectivity of $T \pm iI$. Injectivity is the lower bound (13.6) at $\lambda = \mp i$: $\|(T \pm iI)x\| \geq \|x\|$, so $T \pm iI$ is injective with $(T \pm iI)^{-1}$ bounded of norm at most 1. Hence both $T + iI$ and $T - iI$ are bijections of $\operatorname{dom} T$ onto $\mathbb{H}$, and $\pm i \in \rho(T)$ as proposition 13.3.3 already recorded.

Step 2: U is an isometry. For $x \in \operatorname{dom} T$ set $y = (T + iI)x$, so that $x = (T + iI)^{-1}y$ ranges over all of $\mathbb{H}$ as y does, and $Uy = (T - iI)x$. The symmetry of T makes $\langle Tx, x\rangle$ real, so expanding as in (13.6),

$$\|(T + iI)x\|^2 = \|Tx\|^2 + \|x\|^2 = \|(T - iI)x\|^2$$

which says $\|y\| = \|Uy\|$ for every $y \in \mathbb{H}$: U is a linear isometry of $\mathbb{H}$ into itself.

Step 3: U is onto, hence unitary. The range of U is $\operatorname{ran}(T - iI) = \mathbb{H}$ by Step 1, so U is a surjective isometry, i.e. unitary (definition 11.1.23); its inverse is $U^{-1} = (T + iI)(T - iI)^{-1}$.

Step 4: 1 is not an eigenvalue. Compute $I - U$ on a vector $y = (T + iI)x$:

$$(I - U)y = (T + iI)x - (T - iI)x = 2ix \tag{13.7}$$

so $\operatorname{ran}(I - U) = \operatorname{dom} T$, which is dense. If $Uy = y$ then $(I - U)y = 0$ forces $x = 0$ by (13.7), whence $y = (T + iI) \cdot 0 = 0$; so 1 is not an eigenvalue of U. Likewise

$$(I + U)y = (T + iI)x + (T - iI)x = 2Tx \tag{13.8}$$

Comparing (13.8) with (13.7) recovers T: for $y = (T + iI)x$ they read $i(I + U)y = i \cdot 2Tx = T \cdot (2ix) = T(I - U)y$, so $i(I + U) = T(I - U)$ as maps on $\mathbb{H}$, and inverting $I - U$ on its dense range $\operatorname{ran}(I - U) = \operatorname{dom} T$ yields $T = i(I + U)(I - U)^{-1}$.

Step 5: The converse. Let U be unitary with 1 not an eigenvalue, so $I - U$ is injective and $\operatorname{ran}(I - U)$ is dense (its complement is $\ker(I - U^*) = \ker(I - U^{-1}) = \ker(U - I) = \{0\}$, using $U^* = U^{-1}$). Define $T = i(I + U)(I - U)^{-1}$ on $\operatorname{dom} T = \operatorname{ran}(I - U)$. For $x = (I - U)u$ and $x' = (I - U)u'$ in $\operatorname{dom} T$,

$$\langle Tx, x'\rangle - \langle x, Tx'\rangle = i\langle (I + U)u, (I - U)u'\rangle + i\langle (I - U)u, (I + U)u'\rangle$$

Expanding both inner products and cancelling using $\langle Uu, Uu'\rangle = \langle u, u'\rangle$ (unitarity) leaves 0, so T is symmetric. Moreover $T + iI = i(I + U)(I - U)^{-1} + i = i\big[(I + U) + (I - U)\big](I - U)^{-1} =$

$2i(I-U)^{-1}$ and similarly $T - iI = 2iU(I-U)^{-1}$; both are onto $\mathbb{H}$ (since $I-U$ and U are bijections of $\mathbb{H}$ and $\operatorname{dom} T$), so by theorem 13.2.11 T is self-adjoint. Its Cayley transform is $(T-iI)(T+iI)^{-1} = 2iU(I-U)^{-1} \cdot \left(2i(I-U)^{-1}\right)^{-1} = U$, recovering U. Thus $T \mapsto U$ and $U \mapsto T$ are mutually inverse, establishing the claimed bijection, as we sought to show.

The Cayley transform is a discrete analogue of the exponential map: where $e^{i\theta}$ wraps the real line onto the circle in the theory of one-parameter groups, the fractional linear map $\lambda \mapsto (\lambda - i)/(\lambda + i)$ wraps the spectral line onto the circle in one algebraic step, trading the unbounded self-adjoint T for the bounded unitary U with no loss. The single excluded point $z = 1$ is the image of $\lambda = \infty$; it is an eigenvalue of U exactly when T would have to be defined "at infinity," which a genuine operator cannot be, and this is why the correspondence lands on unitaries missing the eigenvalue 1. The formulas (13.7) and (13.8) are worth remembering: $I - U$ recovers the domain and $I + U$ recovers the action of T, so the whole of T is encoded in the two boundary operators $I \pm U$.

Example 13.3.5: The Cayley Transform of Multiplication by x

Let $\mathbb{H} = L^2(\mathbb{R})$ and let T be multiplication by the coordinate, $(Tf)(x) = xf(x)$, on the domain $\operatorname{dom} T = \left\{f \mid xf \in L^2(\mathbb{R})\right\}$. This T is self-adjoint with $\sigma(T) = \mathbb{R}$. Its Cayley transform is again a multiplication operator, now by the scalar function $x \mapsto (x-i)/(x+i)$:

$$(Uf)(x) = \frac{x-i}{x+i}\, f(x)$$

The multiplier $(x-i)/(x+i)$ has modulus 1 for every real x, so U is multiplication by a unimodular function and hence unitary, as the theorem demands. As x runs over $\mathbb{R}$ the value $(x-i)/(x+i)$ traces the unit circle: at $x = 0$ it is -1, and as $x \to \pm\infty$ it approaches 1 without attaining it, so the value 1 is never taken. Consequently $U - I$ is multiplication by a function that vanishes nowhere on $\mathbb{R}$, hence injective, and 1 is not an eigenvalue of U, matching the theorem exactly. The excluded point 1 on the circle is the image of the missing spectral point at infinity.

With the Cayley transform in hand the spectral theorem for unbounded self-adjoint operators is now within reach: apply the bounded spectral theorem to the unitary U, obtaining a projection-valued measure on the unit circle, and push it forward to the real line through the inverse Cayley map $z \mapsto i(1+z)/(1-z)$. The pushed-forward measure integrates the coordinate λ to reconstruct T, exactly as in the bounded case, with the sole new feature that λ is now unbounded, so the integral $\int \lambda\, dE$ produces an unbounded operator with the natural domain that keeps $\int \lambda^2\, d\langle E(\lambda)x, x\rangle$ finite.

Theorem 13.3.6: The Spectral Theorem for Unbounded Self-Adjoint Operators

Let T be self-adjoint on $\mathbb{H}$. There is a unique projection-valued measure E on the Borel σ-algebra of $\mathbb{R}$ (a map assigning orthogonal projections to Borel sets, satisfying the axioms of definition 12.4.1 with $E(\mathbb{R}) = I$) such that

$$T = \int_{\mathbb{R}} \lambda \, dE(\lambda) \tag{13.9}$$

in the sense that

$$\operatorname{dom} T = \left\{ x \in \mathbb{H} \;\middle|\; \int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)x, x\rangle < \infty \right\}$$

and $\langle Tx, y\rangle = \int_{\mathbb{R}} \lambda \, d\langle E(\lambda)x, y\rangle$ for all $x \in \operatorname{dom} T$ and $y \in \mathbb{H}$. The support of E is $\sigma(T)$: a real point λ_0 lies in $\sigma(T)$ if and only if $E(\Omega) \neq 0$ for every neighbourhood Ω of λ_0, and $\sigma(T)$ is the smallest closed set of full measure.

***Proof*:**
Step 1: The spectral measure of the Cayley transform. Let $U = (T - iI)(T + iI)^{-1}$ be the Cayley transform (theorem 13.3.4), a unitary operator, hence normal (definition 11.2.8), with spectrum contained in the unit circle $\mathbb{T} = \{z \in \mathbb{C} \mid |z| = 1\}$. The bounded spectral theorem, applied to the normal operator U (theorem 12.4.6), supplies a unique projection-valued measure F on $\mathbb{T}$ with

$$U = \int_{\mathbb{T}} z \, dF(z)$$

and, for every bounded Borel g on $\mathbb{T}$, $g(U) = \int_{\mathbb{T}} g \, dF$ by the Borel functional calculus (theorem 12.3.3). Because 1 is not an eigenvalue of U, the single point $\{1\}$ carries no mass: $F(\{1\})$ is the orthogonal projection onto $\ker(U - I) = \{0\}$, so $F(\{1\}) = 0$. Thus F lives on $\mathbb{T} \setminus \{1\}$.

Step 2: Transport to the line. The Cayley map and its inverse

$$z = \frac{\lambda - i}{\lambda + i}, \qquad \lambda = i\,\frac{1+z}{1-z}$$

are mutually inverse homeomorphisms between $\mathbb{R}$ and $\mathbb{T} \setminus \{1\}$. Define a projection-valued measure E on $\mathbb{R}$ by pushing F forward along $\lambda(z) = i(1+z)/(1-z)$: for a Borel set $\Omega \subseteq \mathbb{R}$,

$$E(\Omega) = F\big(\{z \in \mathbb{T} \setminus \{1\} \mid \lambda(z) \in \Omega\}\big)$$

Since $\lambda(\cdot)$ is a Borel isomorphism of $\mathbb{T} \setminus \{1\}$ onto $\mathbb{R}$ and $F(\{1\}) = 0$, the assignment E satisfies all the projection-valued-measure axioms ($E(\varnothing) = 0$, $E(\mathbb{R}) = F(\mathbb{T} \setminus \{1\}) = I$, multiplicativity and strong countable additivity transported from F). For any bounded Borel h on $\mathbb{R}$, the change of variables $z \mapsto \lambda(z)$ gives

$$\int_{\mathbb{R}} h(\lambda) \, dE(\lambda) = \int_{\mathbb{T}} h\big(\lambda(z)\big) \, dF(z) = (h \circ \lambda)(U) \tag{13.10}$$

so integration of h against E is the Borel functional calculus of U applied to the bounded Borel function $h \circ \lambda$ on $\mathbb{T}$.

Step 3: The integral recovers T. The coordinate λ is unbounded on $\mathbb{R}$, so its integral must be built as a limit of bounded truncations. For $n \in \mathbb{N}$ let $\lambda_n(\lambda) = \lambda\, \chi_{[-n,n]}(\lambda)$, a bounded Borel function, and set $S_n = \int_{\mathbb{R}} \lambda_n \, dE = \int_{[-n,n]} \lambda \, dE(\lambda) \in B(\mathbb{H})$. Define

$$D = \left\{ x \in \mathbb{H} \;\middle|\; \int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)x, x\rangle < \infty \right\}$$

and, for $x \in D$, note $\|S_n x - S_m x\|^2 = \int_{n<|\lambda|\le m} \lambda^2 \, d\langle E(\lambda)x, x\rangle \to 0$ as $n, m \to \infty$ (the tail of a convergent integral), so $S_n x$ converges; call the limit Sx. This S is the operator $\int_{\mathbb{R}} \lambda \, dE$ on $\operatorname{dom} S = D$. It remains to prove $S = T$.

Fix $x \in \operatorname{dom} T$ and put $y = (T + iI)x$, so $x = (T + iI)^{-1} y$. The bounded operator $(T + iI)^{-1}$ is $r(U)$ for the function r continuous on $\mathbb{T} \setminus \{1\}$,

$$r(z) = \frac{1 - z}{2i}, \qquad \text{since } (T + iI)^{-1} = \tfrac{1}{2i}(I - U)$$

by (13.7). Since $\lambda(z) + i = 2i/(1 - z)$, the function $r(z) = (1 - z)/(2i)$ equals $1/(\lambda(z) + i)$, so by the transport (13.10) applied to $h(\lambda) = 1/(\lambda + i)$, $(T + iI)^{-1} = r(U) = \int_{\mathbb{R}} \frac{1}{\lambda+i} \, dE(\lambda)$. Likewise (13.8) gives, since $\lambda(z)/(\lambda(z) + i) = (1 + z)/2$,

$$T(T + iI)^{-1} = \tfrac{1}{2}(I + U) = \int_{\mathbb{R}} \frac{\lambda}{\lambda + i} \, dE(\lambda) \tag{13.11}$$

a bounded operator. Now fix $x = (T + iI)^{-1} y \in \operatorname{dom} T$. First, $x \in D$: the scalar measure satisfies $d\langle E(\lambda)x, x\rangle = |1/(\lambda + i)|^2 \, d\langle E(\lambda)y, y\rangle$ (from $x = \int \frac{1}{\lambda+i} \, dE \, y$ and multiplicativity of the bounded calculus), so

$$\int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)x, x\rangle = \int_{\mathbb{R}} \frac{\lambda^2}{\lambda^2 + 1} \, d\langle E(\lambda)y, y\rangle \le \|y\|^2 < \infty$$

Thus $Sx = \int_{\mathbb{R}} \lambda \, dE(\lambda) \, x$ is defined. To identify it with Tx, apply the bounded, multiplicative calculus to the two bounded factors $\lambda/(\lambda+i)$ and $(\lambda+i)$ truncated: for each n, $S_n x = \int_{[-n,n]} \lambda \, dE \cdot \int_{\mathbb{R}} \frac{1}{\lambda+i} \, dE \, y = \int_{[-n,n]} \frac{\lambda}{\lambda+i} \, dE \, y$, which converges as $n \to \infty$ to $\int_{\mathbb{R}} \frac{\lambda}{\lambda+i} \, dE \, y = T(T + iI)^{-1} y = Tx$ by (13.11). Hence $Sx = Tx$, and since x ranges over all of $\operatorname{dom} T$ as y ranges over $\mathbb{H}$, this gives $T \subseteq S$. Conversely S is symmetric (each S_n is self-adjoint and $S_n \to S$ strongly on D) and T is self-adjoint, so a self-adjoint operator has no proper symmetric extension: $T \subseteq S$ with both symmetric and T maximal-symmetric forces $S = T$. Thus $T = \int_{\mathbb{R}} \lambda \, dE(\lambda)$ with the stated domain.

Step 4: Uniqueness. Suppose E' is another projection-valued measure on $\mathbb{R}$ with $\int_{\mathbb{R}} \lambda \, dE' = T$ and the corresponding domain. Reversing the transport, define F' on $\mathbb{T} \setminus \{1\}$ by $F'(\Gamma) = E'\big(\{\lambda \in \mathbb{R} \mid z(\lambda) \in \Gamma\}\big)$ where $z(\lambda) = (\lambda - i)/(\lambda + i)$, and set $F'(\{1\}) = 0$. Then $\int_{\mathbb{T}} z \, dF' = \int_{\mathbb{R}} \frac{\lambda-i}{\lambda+i} \, dE' = (T - iI)(T + iI)^{-1} = U$, using the two calculus identities above applied to E'. By the uniqueness in the bounded spectral theorem for the normal operator U (theorem 12.4.6), $F' = F$, and transporting back gives $E' = E$. This proves uniqueness.

Step 5: The support is $\sigma(T)$. By (13.10) the spectrum of U is the support of F on $\mathbb{T}$, and $\lambda(\cdot)$ carries the support of F (minus the point 1) onto the support of E; since $\lambda(z(\lambda)) = \lambda$ conjugates U's spectral data to T's, a real point λ_0 lies in $\sigma(T)$ if and only if $E(\Omega) \neq 0$ for every neighbourhood Ω of λ_0, which is the statement that $\operatorname{supp} E = \sigma(T)$. This completes the proof.

The theorem is the exact analogue of theorem 12.4.6, with two differences forced by unboundedness.

First, the measure lives on all of $\mathbb{R}$ rather than on a compact set, because $\sigma(T)$ need no longer be bounded; multiplication by x (example 13.3.5) and the momentum operator (example 13.3.2) both have $\sigma(T) = \mathbb{R}$. Second, the integral $\int \lambda \, dE$ is genuinely improper: it converges only on the domain D where the "energy" $\int \lambda^2 \, d\langle E(\lambda)x, x\rangle$ is finite, and this convergence condition *is* the definition of $\operatorname{dom} T$. The domain is thus not an arbitrary technical choice but is dictated by the spectral measure: x lies in $\operatorname{dom} T$ precisely when its spectral distribution has a finite second moment. The Cayley transform did the entire job of converting the theorem into a bounded statement, and the reality bound (13.5) was what guaranteed $\pm i \in \rho(T)$ so that the transform existed at all.

Example 13.3.7: The Spectral Measure of the Multiplication Operator

Return to $\mathbb{H} = L^2(\mathbb{R})$ with T multiplication by the coordinate, $(Tf)(x) = xf(x)$ (example 13.3.5). Its projection-valued measure is the most transparent possible: for a Borel set $\Omega \subseteq \mathbb{R}$,

$$(E(\Omega)f)(x) = \chi_\Omega(x) \, f(x)$$

multiplication by the indicator of Ω. Each $E(\Omega)$ is the orthogonal projection onto $\{f \mid f = 0 \text{ a.e. off } \Omega\}$, and $E(\mathbb{R}) = I$. The resolution (13.9) reads

$$\langle Tf, f\rangle = \int_{\mathbb{R}} x \, |f(x)|^2 \, dx = \int_{\mathbb{R}} \lambda \, d\langle E(\lambda)f, f\rangle$$

since $d\langle E(\lambda)f, f\rangle = |f(\lambda)|^2 \, d\lambda$ is the scalar measure with density $|f|^2$. The domain condition $\int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)f, f\rangle = \int_{\mathbb{R}} x^2 |f(x)|^2 \, dx < \infty$ says exactly that $xf \in L^2(\mathbb{R})$, which is the domain of T named at the outset. The spectral theorem here recovers, with no new information, the fact that multiplication operators are already "diagonal"; its force is that *every* self-adjoint operator is unitarily one of these, as the multiplication form below makes explicit.

Once E is available, any Borel function of T, bounded or not, is defined by integrating that function against E, exactly as continuous and bounded Borel functions of a bounded operator were. This is the Borel functional calculus for unbounded operators: it lets one form e^{itT}, $(T - \lambda)^{-1}$, $\chi_\Omega(T)$, and every other spectral gadget of the later sections by a single integral, and it is the calculus that Stone's theorem and the semigroup theory will run on.

Theorem 13.3.8: The Borel Functional Calculus for Unbounded Operators

Let T be self-adjoint with projection-valued measure E on $\mathbb{R}$ (theorem 13.3.6). For each Borel function $f\colon \mathbb{R} \to \mathbb{C}$ the operator

$$f(T) = \int_{\mathbb{R}} f(\lambda)\, dE(\lambda), \qquad \operatorname{dom} f(T) = \left\{ x \in \mathbb{H} \;\middle|\; \int_{\mathbb{R}} |f(\lambda)|^2\, d\langle E(\lambda)x, x\rangle < \infty \right\}$$

is well-defined and satisfies:

1. the map $f \mapsto f(T)$ is a $*$-homomorphism in the sense that $(f+g)(T) \supseteq f(T) + g(T)$, $(fg)(T) \supseteq f(T)g(T)$, and $\overline{f}(T) = f(T)^*$, with equality of domains when f, g are bounded;
2. $f(T) \in B(\mathbb{H})$ if and only if f is (essentially) bounded on $\sigma(T)$, and then $\|f(T)\| = \sup_{\lambda \in \sigma(T)} |f(\lambda)|$ (the essential supremum against E);
3. $f(T)$ is self-adjoint when f is real-valued, and unitary when $|f| = 1$ on $\sigma(T)$; in particular $\chi_\Omega(T) = E(\Omega)$;
4. *(multiplication model)* there is a measure space (M, μ), a unitary $W\colon \mathbb{H} \to L^2(M, \mu)$, and a real-valued measurable $\varphi\colon M \to \mathbb{R}$ with $WTW^{-1} = M_\varphi$ multiplication by φ, and then $Wf(T)W^{-1} = M_{f\circ\varphi}$ for every Borel f.

***Proof*:**
Step 1: Definition and the domain. For bounded Borel f the integral $\int_{\mathbb{R}} f\, dE$ is the bounded operator produced by transporting the bounded calculus of U through (13.10), namely $f(T) = (f \circ \lambda)(U)$, and $\|f(T)\| \le \sup_{\mathbb{R}} |f|$. For unbounded f set $f_n = f\,\chi_{\{|f|\le n\}}$ and, on the domain $D_f = \{x \mid \int |f|^2\, d\langle Ex, x\rangle < \infty\}$, define $f(T)x = \lim_n f_n(T)x$; the limit exists because $\|f_n(T)x - f_m(T)x\|^2 = \int_{n<|f|\le m} |f|^2\, d\langle E(\lambda)x, x\rangle \to 0$ (the calculation is (13.10) paired with x, using multiplicativity to turn $|f_n - f_m|^2$ into the integrand), and D_f is dense because $\bigcup_n E(\{|f| \le n\})\mathbb{H}$ is dense (its closure is $E(\mathbb{R})\mathbb{H} = \mathbb{H}$). This defines $f(T)$ with the stated domain.

Step 2: The algebraic properties. Each identity in (1) holds for bounded f, g by the corresponding property of the bounded calculus of U (multiplicativity, $*$-preservation, unit) transported through (13.10). For unbounded f, g the operators $f(T) + g(T)$ and $f(T)g(T)$ are defined only where both summands or the composite make sense, so their domains can be strictly smaller than that of $(f+g)(T)$ or $(fg)(T)$; passing to the strong limits $f_n(T), g_n(T)$ and using that a self-adjoint (hence closed) operator contains the closure of these limits yields the inclusions, with equality when the functions are bounded so that all domains are $\mathbb{H}$. The adjoint identity $\overline{f}(T) = f(T)^*$ holds because $\overline{f_n}(T) = f_n(T)^*$ for the bounded truncations and the closed operator $f(T)$ is the closure of $\bigcup_n f_n(T)$, whose adjoint is the intersection, computed to be $\overline{f}(T)$.

Step 3: Boundedness and the norm. If f is bounded on $\sigma(T)$ with $M = \sup_{\sigma(T)} |f|$, then $f = f\,\chi_{\sigma(T)}$ up to an E-null set (since E is supported on $\sigma(T)$, theorem 13.3.6), and $f(T) = (f\chi_{\sigma(T)})(T)$ is bounded of norm $\le M$; conversely if f is unbounded on every set of full measure then $\int |f|^2\, d\langle Ex, x\rangle = \infty$ for suitable x supported spectrally where $|f|$ is large, so $\operatorname{dom} f(T) \ne \mathbb{H}$ and $f(T)$ is unbounded. The norm equals the essential supremum $\operatorname{ess\,sup}_E |f|$ by the corresponding statement for the bounded calculus of U, transported. This is (2), and (3) is the specialization to real-valued f (then $f(T) = f(T)^*$ by the adjoint identity, and

$\operatorname{dom} f(T) = \operatorname{dom} f(T)^*$, so $f(T)$ is self-adjoint), to $|f| = 1$ (then $f(T)^* f(T) = f(T) f(T)^* = I$), and to $f = \chi_\Omega$ (then $f(T) = \int \chi_\Omega \, dE = E(\Omega)$).

Step 4: The multiplication model. Apply the multiplication form of the bounded spectral theorem (theorem 12.4.10) to the normal operator U: there is a unitary $W : \mathbb{H} \to L^2(M, \mu)$ and a measurable $\psi : M \to \mathbb{T}$ with $WUW^{-1} = M_\psi$. Since 1 is not an eigenvalue of U, $\psi \neq 1$ μ-a.e., so $\varphi := i(1+\psi)/(1-\psi)$ is a well-defined real-valued measurable function on M (real because $|\psi| = 1$ makes $i(1+\psi)/(1-\psi)$ real). Then $WTW^{-1} = i(I + M_\psi)(I - M_\psi)^{-1} = M_\varphi$ by the inverse Cayley formula of theorem 13.3.4 read pointwise, and for any Borel f, transporting $f(T) = (f \circ \lambda)(U)$ through W gives $Wf(T)W^{-1} = (f \circ \lambda)(M_\psi) = M_{f \circ \lambda \circ \psi} = M_{f \circ \varphi}$, since $\lambda \circ \psi = \varphi$. This establishes (4), completing the proof.

The multiplication model is the concrete payoff. It says that every self-adjoint operator, however singular its original description as a differential operator, becomes multiplication by a real function φ once the space is re-coordinatized by the unitary W, and that any Borel function of the operator is then just multiplication by $f \circ \varphi$. Boundedness of $f(T)$ becomes the elementary question of whether $f \circ \varphi$ is essentially bounded, and the operator e^{itT} that generates a dynamics is multiplication by the unimodular function $e^{it\varphi}$. This diagonalization is what makes the spectral theorem a working tool rather than an existence statement: the resolvent $(\lambda - T)^{-1} = M_{1/(\lambda - \varphi)}$, the projection $\chi_\Omega(T) = M_{\chi_{\varphi^{-1}(\Omega)}}$, and the unitary group $e^{itT} = M_{e^{it\varphi}}$ are all read off from a single real function on a measure space.

Example 13.3.9: Functions of the Laplacian

Let $\mathbb{H} = L^2(\mathbb{R}^n)$ and let $T = -\Delta$ be the (positive) Laplacian, self-adjoint on the domain $\{ f \in L^2 \mid \Delta f \in L^2 \}$ (the distributional Laplacian landing in L^2). Although the Fourier transform is the fastest diagonalizer, the spectral theorem alone already fixes the shape: $\sigma(-\Delta) = [0, \infty)$, and the projection-valued measure E is supported on $[0, \infty)$, so $-\Delta$ is unitarily multiplication by a nonnegative φ. Two Borel functions of it govern the fundamental evolution equations. Taking $f(\lambda) = e^{-t\lambda}$ for $t > 0$, which is bounded by 1 on $[0, \infty)$, gives the bounded operator

$$e^{-t(-\Delta)} = \int_{[0,\infty)} e^{-t\lambda} \, dE(\lambda)$$

the *heat semigroup*, whose norm is $\sup_{\lambda \geq 0} e^{-t\lambda} = 1$ (attained at $\lambda = 0$), consistent with (2). Taking $f(\lambda) = e^{-it\lambda}$, which has modulus 1 on the spectrum, gives the *unitary* Schrödinger group $e^{it\Delta} = \int e^{-it\lambda} \, dE(\lambda)$ of (3). The heat operator smooths (its multiplier decays for large λ, damping high spectral modes) while the Schrödinger operator rotates (its multiplier has modulus 1), and this single distinction, bounded-decaying versus unimodular multiplier, separates parabolic from dispersive dynamics.

13.4 Stone's Theorem

The functional calculus of section 13.3 produces from a self-adjoint operator A the family $U(t) = e^{itA}$, a group of unitaries read here as a dynamics. Stone's theorem is the converse: every such group arises this way, from a self-adjoint operator uniquely determined as its infinitesimal generator.

Definition 13.4.1: Strongly Continuous One-Parameter Unitary Group

A *strongly continuous one-parameter unitary group* on a complex Hilbert space $\mathbb{H}$ is a map $U: \mathbb{R} \to B(\mathbb{H})$ such that

1. each $U(t)$ is unitary (definition 11.1.23);
2. $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$, and $U(0) = I$;
3. for every $x \in \mathbb{H}$ the orbit $t \mapsto U(t)x$ is continuous from $\mathbb{R}$ to $\mathbb{H}$.

The first two conditions make U a group homomorphism from $(\mathbb{R}, +)$ into the unitary group of $\mathbb{H}$; in particular $U(t)^{-1} = U(-t) = U(t)^*$. The third condition, *strong* continuity, is the correct regularity requirement here. Norm continuity of $t \mapsto U(t)$ would be far too strong: it forces the generator to be bounded, excluding every operator the chapter exists to treat. Strong continuity asks only that the group move each vector continuously, which is what a solution of an evolution equation does, and it is automatically upgraded from a single point: because the $U(t)$ are uniformly bounded (each has norm 1) and satisfy the group law, continuity of every orbit at $t = 0$ already gives continuity everywhere, since $U(t+h)x - U(t)x = U(t)(U(h)x - x)$ and $\|U(t)\| = 1$.

The motivating example is the translation group, which will also fix the reading of the generator as a differential operator.

Example 13.4.2: The Translation Group on $L^2(\mathbb{R})$

On $\mathbb{H} = L^2(\mathbb{R})$ define, for each $t \in \mathbb{R}$, the translation

$$(U(t)f)(x) = f(x-t)$$

Each $U(t)$ is unitary: translation is a bijection of $\mathbb{R}$ preserving Lebesgue measure, so $\|U(t)f\|_2 = \|f\|_2$, and its inverse is $U(-t)$. The group law $U(s+t) = U(s)U(t)$ is the identity $f(x-s-t) = (U(s)f)(x-t)$, and $U(0) = I$. For strong continuity, fix $f \in L^2(\mathbb{R})$. Continuous functions of compact support are dense in $L^2(\mathbb{R})$, and such a g is uniformly continuous, so $\|U(t)g - g\|_2 \to 0$ as $t \to 0$ by dominated convergence (the integrand is bounded by an integrable dominating function once $|t| \le 1$, since the supports stay inside a fixed compact set). For general f, given $\varepsilon > 0$ pick such a g with $\|f - g\|_2 < \varepsilon$; then

$$\|U(t)f - f\|_2 \le \|U(t)(f-g)\|_2 + \|U(t)g - g\|_2 + \|g - f\|_2 < 2\varepsilon + \|U(t)g - g\|_2$$

using $\|U(t)\| = 1$, and the middle term tends to 0. So U is a strongly continuous one-parameter unitary group. Its generator, computed below, is $A = i\,d/dx$ on the natural domain, and the group is the flow of the transport equation $\partial_t u = -\partial_x u$.

The generator is the object that recovers A from the group $U(t) = e^{itA}$ by differentiating at $t = 0$. Since $e^{it\lambda} = 1 + it\lambda + o(t)$, one expects $(U(t)x - x)/(it) \to Ax$, but only for those x on which the limit exists. That set of x, not prescribed in advance, is exactly the domain of the generator.

Definition 13.4.3: Generator of a Unitary Group

Let U be a strongly continuous one-parameter unitary group on $\mathbb{H}$. Its *generator* is the operator A with domain

$$\operatorname{dom} A = \left\{ x \in \mathbb{H} \;\middle|\; \lim_{t \to 0} \frac{U(t)x - x}{it} \text{ exists in } \mathbb{H} \right\}$$

and $Ax = \lim_{t\to 0}(U(t)x - x)/(it)$ for $x \in \operatorname{dom} A$.

The factor i in the denominator is a convention that makes A self-adjoint rather than skew-adjoint: with it, $U(t) = e^{itA}$ pairs a real spectral parameter λ with the phase $e^{it\lambda}$, matching the functional calculus of theorem 13.3.8. The domain is genuinely a proper subspace in the unbounded case, but it is never small: the following lemma produces a dense supply of vectors in $\operatorname{dom} A$ by averaging the orbit against smooth weights, the standard device for regularizing a group that is only continuous.

Lemma 13.4.4: Density of the Generator's Domain

Let U be a strongly continuous one-parameter unitary group with generator A. Then $\operatorname{dom} A$ is dense in $\mathbb{H}$, and A is symmetric (definition 13.2.8).

***Proof*:**
Step 1: A dense set of smoothed vectors lies in the domain. For $x \in \mathbb{H}$ and $\varphi \in C_c^\infty(\mathbb{R})$ (smooth, compactly supported) define the averaged vector

$$x_\varphi = \int_{\mathbb{R}} \varphi(s)\, U(s)x\, ds$$

a Bochner integral of the continuous, compactly supported $\mathbb{H}$-valued function $s \mapsto \varphi(s)U(s)x$. Apply the group law to $U(t)x_\varphi$ and substitute $s \mapsto s - t$ inside the integral:

$$U(t)x_\varphi = \int_{\mathbb{R}} \varphi(s)\, U(s+t)x\, ds = \int_{\mathbb{R}} \varphi(s-t)\, U(s)x\, ds$$

Hence

$$\frac{U(t)x_\varphi - x_\varphi}{it} = \int_{\mathbb{R}} \frac{\varphi(s-t) - \varphi(s)}{it}\, U(s)x\, ds$$

As $t \to 0$ the difference quotient $(\varphi(s-t) - \varphi(s))/t$ converges uniformly to $-\varphi'(s)$, with all supports inside a fixed compact interval, so the integrand converges in the norm of $\mathbb{H}$ uniformly in s and the integral converges. Therefore $x_\varphi \in \operatorname{dom} A$ and

$$Ax_\varphi = i \int_{\mathbb{R}} \varphi'(s)\, U(s)x\, ds$$

Choosing $\varphi = \varphi_n$ an approximate identity (nonnegative, integral 1, supports shrinking to 0) gives $x_{\varphi_n} = \int \varphi_n(s)U(s)x\, ds \to U(0)x = x$ by strong continuity, since $\|x_{\varphi_n} - x\| \le \int \varphi_n(s)\|U(s)x - x\|\, ds \to 0$. So every $x \in \mathbb{H}$ is a limit of vectors $x_{\varphi_n} \in \operatorname{dom} A$, proving $\operatorname{dom} A$ dense.

Step 2: A is symmetric. Fix $x, y \in \operatorname{dom} A$. Because each $U(t)$ is unitary, $\langle U(t)x, U(t)y\rangle = \langle x, y\rangle$ for all t, so

$$0 = \frac{d}{dt}\bigg|_{t=0} \langle U(t)x, U(t)y\rangle$$

Computed as a limit of difference quotients, using $\langle U(t)x, U(t)y\rangle = \langle x, y\rangle$ and the definition of A,

$$0 = \left\langle \lim_{t\to 0} \frac{U(t)x - x}{t}, y\right\rangle + \left\langle x, \lim_{t\to 0} \frac{U(t)y - y}{t}\right\rangle = \langle iAx, y\rangle + \langle x, iAy\rangle$$

where the two limits exist precisely because $x, y \in \operatorname{dom} A$. Thus $i\langle Ax, y\rangle - i\langle x, Ay\rangle = 0$, that is $\langle Ax, y\rangle = \langle x, Ay\rangle$ for all $x, y \in \operatorname{dom} A$. So $A \subseteq A^*$ (definition 13.2.6), which is symmetry, completing the proof.

The lemma has done the analytic work: the generator of any unitary group is a densely-defined symmetric operator. To promote symmetry to self-adjointness one direction of Stone's theorem needs the deficiency criterion of theorem 13.2.11, and to invert the construction the other direction needs the exponential e^{itA} to be defined and to depend continuously on t, which the functional calculus supplies. Both are assembled in the theorem.

Before the theorem, one more preparation. In the converse direction A is given self-adjoint and $U(t) := e^{itA}$ is defined through theorem 13.3.8 as $\int_{\mathbb{R}} e^{it\lambda}\, dE(\lambda)$ against the spectral measure E of A. The functions $e_t(\lambda) = e^{it\lambda}$ are bounded of modulus 1, so each $U(t)$ is unitary, and their pointwise products and limits control the group law and the strong continuity through the calculus.

Theorem 13.4.5: Stone's Theorem

Let $\mathbb{H}$ be a complex Hilbert space. There is a bijection between strongly continuous one-parameter unitary groups on $\mathbb{H}$ and self-adjoint operators on $\mathbb{H}$, given in the two directions as follows.

1. If A is self-adjoint, then $U(t) := e^{itA} = \int_{\mathbb{R}} e^{it\lambda}\, dE(\lambda)$, defined through the functional calculus of theorem 13.3.8 against the spectral measure E of A, is a strongly continuous one-parameter unitary group.

2. Conversely, every strongly continuous one-parameter unitary group U is of this form for a unique self-adjoint operator A, namely its generator (definition 13.4.3),

$$Ax = \lim_{t\to 0} \frac{U(t)x - x}{it}$$

   on its natural domain.

The two constructions are mutually inverse: the generator of e^{itA} is A, and the exponential of the generator of U is U.

***Proof*:**
Step 1: a self-adjoint A yields a group (part 1). Let E be the spectral measure of A (theorem 13.3.6) and $e_t(\lambda) = e^{it\lambda}$. Each e_t is a bounded Borel function, so $U(t) = e_t(A) = \int_{\mathbb{R}} e^{it\lambda}\, dE(\lambda)$ is a bounded operator by theorem 13.3.8. The functional calculus is a $*$-homomorphism on bounded Borel functions, and $\overline{e_t} = e_{-t}$ with $e_t \cdot e_{-t} = 1$, so

$$U(t)^* U(t) = e_{-t}(A) e_t(A) = (e_{-t} e_t)(A) = 1(A) = I$$

and likewise $U(t)U(t)^* = I$, so each $U(t)$ is unitary. The pointwise identity $e_{s+t} = e_s e_t$ gives $U(s+t) = U(s)U(t)$ through the same homomorphism property, and $e_0 = 1$ gives $U(0) = I$. It remains to check strong continuity. Fix $x \in \mathbb{H}$ and let $\mu_x(\Omega) = \langle E(\Omega)x, x\rangle$ be the associated finite Borel measure on $\mathbb{R}$, with $\mu_x(\mathbb{R}) = \|x\|^2$. The functional calculus gives, for the difference $U(t)x - x = (e_t - 1)(A)x$,

$$\|U(t)x - x\|^2 = \int_{\mathbb{R}} |e^{it\lambda} - 1|^2 \, d\mu_x(\lambda)$$

As $t \to 0$ the integrand $|e^{it\lambda} - 1|^2$ tends to 0 pointwise in λ and is bounded by the μ_x-integrable constant 4, so dominated convergence gives $\|U(t)x - x\| \to 0$. Strong continuity at $t = 0$, hence everywhere by the group law, follows. This proves part 1.

Step 2: the generator of e^{itA} is A (half of mutual inversion). Let $U(t) = e^{itA}$ as in Step 1 and let B denote its generator (definition 13.4.3). For $x \in \operatorname{dom} A$, that is x with $\int_{\mathbb{R}} \lambda^2 \, d\mu_x(\lambda) < \infty$, estimate the difference quotient against $Ax = \int \lambda \, dE(\lambda)\, x$:

$$\left\| \frac{U(t)x - x}{it} - Ax \right\|^2 = \int_{\mathbb{R}} \left| \frac{e^{it\lambda} - 1}{it} - \lambda \right|^2 d\mu_x(\lambda)$$

The integrand tends to 0 pointwise as $t \to 0$; moreover $|(e^{it\lambda} - 1)/(it)| \le |\lambda|$ for all t (the chord of $e^{i\theta}$ is no longer than its arc), so the integrand is bounded by $(2|\lambda|)^2 = 4\lambda^2$, which is μ_x-integrable exactly because $x \in \operatorname{dom} A$. Dominated convergence gives the limit 0, so $x \in \operatorname{dom} B$ and $Bx = Ax$; thus $A \subseteq B$. But A is self-adjoint, hence maximally symmetric, and B is symmetric (lemma 13.4.4) with $A \subseteq B \subseteq B^* \subseteq A^* = A$, forcing $B = A$. So the generator of e^{itA} is exactly A.

Step 3: the generator is self-adjoint, and $U = e^{itA}$ (part 2). Now let U be an arbitrary strongly continuous one-parameter unitary group and A its generator. By lemma 13.4.4, A is densely defined and symmetric, so theorem 13.2.11 reduces self-adjointness to showing $\operatorname{ran}(A \pm i) = \mathbb{H}$, equivalently $\ker(A^* \mp i) = 0$. Suppose $y \in \operatorname{dom} A^*$ with $A^* y = iy$; the goal is $y = 0$ (the case $A^* y = -iy$ is identical). For any $x \in \operatorname{dom} A$ the orbit $t \mapsto U(t)x$ is differentiable with derivative

$$\frac{d}{dt} U(t)x = iA\, U(t)x = i\, U(t)Ax \tag{13.12}$$

the first equality because $U(t)x \in \operatorname{dom} A$ with $AU(t)x = U(t)Ax$ (the group commutes with its generator, since $U(t)$ commutes with each $U(h)$ and hence with the difference quotient defining A), and both expressions agreeing by that commutation. Consider the scalar function $g(t) = \langle U(t)x, y \rangle$. Using (13.12) and $A^* y = iy$,

$$g'(t) = \langle iA\, U(t)x, y \rangle = i\langle AU(t)x, y \rangle = i\langle U(t)x, A^* y \rangle = i\langle U(t)x, iy \rangle = \langle U(t)x, y \rangle = g(t)$$

So g solves $g' = g$, giving $g(t) = g(0)e^t$. But $|g(t)| = |\langle U(t)x, y \rangle| \le \|U(t)x\| \, \|y\| = \|x\| \, \|y\|$ is bounded in t, while $g(0)e^t$ is unbounded as $t \to +\infty$ unless $g(0) = 0$. Hence $\langle x, y \rangle = g(0) = 0$ for every x in the dense set $\operatorname{dom} A$, so $y = 0$. This kills both deficiency spaces, and theorem 13.2.11 makes A self-adjoint.

It remains to identify U with e^{itA}. Both are strongly continuous unitary groups; write $V(t) = e^{itA}$ for the group of Step 1, whose generator is A by Step 2, and fix $x \in \operatorname{dom} A$. The vector $w(t) = U(t)x - V(t)x$ is differentiable in t, and by (13.12) applied to each group (both have generator A, and $U(t)x, V(t)x \in \operatorname{dom} A$),

$$\frac{d}{dt} w(t) = iA\, U(t)x - iA\, V(t)x = iA\, w(t)$$

Its squared norm has derivative

$$\frac{d}{dt}\|w(t)\|^2 = \langle iAw(t), w(t)\rangle + \langle w(t), iAw(t)\rangle = i\big(\langle Aw(t), w(t)\rangle - \langle w(t), Aw(t)\rangle\big) = 0$$

using the symmetry of A on $\operatorname{dom} A$. So $\|w(t)\|^2$ is constant; since $w(0) = x - x = 0$, it is identically 0, giving $U(t)x = V(t)x$ for all t and all $x \in \operatorname{dom} A$. As $\operatorname{dom} A$ is dense and $U(t), V(t)$ are bounded (norm 1), $U(t) = V(t) = e^{itA}$ for every t. This proves part 2 and, together with Step 2, the mutual inversion.

Step 4: uniqueness. If A_1, A_2 are self-adjoint with $e^{itA_1} = e^{itA_2}$ for all t, then by Step 2 both equal the generator of the common group, so $A_1 = A_2$. The two constructions are therefore inverse bijections, completing the proof.

Stone's theorem is the exact analogue, in infinite dimensions, of writing a one-parameter group of unitary matrices as e^{itH} for a Hermitian H. The content beyond the matrix case is entirely in the domain bookkeeping: the generator A is unbounded, defined only where the difference quotient converges, and the whole force of the theorem is that this a-priori-partial operator turns out to be self-adjoint, so the spectral calculus applies to it and reconstitutes the group. The self-adjointness in Step 3 is where symmetry is not enough; the boundedness of the orbit $\langle U(t)x, y\rangle$ against a putative eigenvector of A^* at $\pm i$ is exactly what a symmetric but non-self-adjoint operator could fail to provide, and it is the unitarity of U that supplies it. In physical terms, A is the Hamiltonian (up to sign and units), the group $U(t) = e^{itA}$ is the time evolution, and the theorem says that the reversible dynamics on a Hilbert space are in bijection with the self-adjoint observables generating them.

The theorem also settles the translation group, whose generator was announced but not yet computed.

Example 13.4.6: The Momentum Operator as Generator of Translation

Return to the translation group $(U(t)f)(x) = f(x-t)$ on $L^2(\mathbb{R})$ (example 13.4.2). Its generator A is computed directly from definition 13.4.3: for f in the domain,

$$(Af)(x) = \lim_{t\to 0} \frac{f(x-t) - f(x)}{it} = \frac{1}{i} \lim_{t\to 0} \frac{f(x-t) - f(x)}{t} = -\frac{1}{i} f'(x) = if'(x)$$

so $A = i\,d/dx$ acts as i times the derivative. The domain is not all of $L^2(\mathbb{R})$: it is exactly the set of $f \in L^2(\mathbb{R})$ for which the limit converges in L^2, which is the Sobolev space $\operatorname{dom} A = \{f \in L^2(\mathbb{R}) \mid f' \in L^2(\mathbb{R})\}$ where f' is the weak derivative (section 10.1), that is $H^1(\mathbb{R})$. On this domain $A = i\,d/dx$ is self-adjoint, by Stone's theorem applied to the unitary group U, without a separate integration-by-parts verification. Up to sign this A is the momentum operator of quantum mechanics, $P = -i\,d/dx = -A$: Stone's theorem identifies it as the self-adjoint generator of spatial translation, and $U(t) = e^{itA} = e^{-t\,d/dx}$ is the unitary that translates a wavefunction by t.

The identification $A = i\,d/dx$ closes the loop with the chapter's opening: the differential operator that is unbounded and discontinuous, and to which the bounded theory did not apply, is recovered here as the generator of a bounded, well-behaved group of unitaries. Its self-adjointness was earned through the group rather than assumed, and the same route, from a group of symmetries to its self-adjoint generator, is how the position, momentum, and energy observables of a quantum system acquire the spectral calculus that makes their measurement statistics meaningful.

13.5 C_0-Semigroups and the Hille-Yosida Theorem

This section takes up one-directional dynamics on a Banach space X: which operators arise as the infinitesimal generator of a C_0-semigroup. The Hille-Yosida theorem answers this completely for the contractive case, reconstructing the semigroup from the resolvent of the generator by an approximation due to Yosida.

Definition 13.5.1: C_0-Semigroup

A *C_0-semigroup* (or *strongly continuous semigroup*) on a Banach space X is a family $(T(t))_{t\geq 0}$ of bounded operators $T(t) \in B(X)$ such that

1. $T(0) = I$;
2. $T(s+t) = T(s)T(t)$ for all $s, t \geq 0$ (the semigroup law);
3. for each $x \in X$ the orbit $t \mapsto T(t)x$ is continuous from $[0, \infty)$ to X (strong continuity).

The semigroup is a *contraction semigroup* if $\|T(t)\| \leq 1$ for all $t \geq 0$.

The three axioms encode a deterministic, time-homogeneous, one-directional dynamics: the state at time $s+t$ is obtained from the state at time s by evolving for a further time t, and the evolution depends only on the elapsed time, not on the absolute clock. Strong continuity is the minimal regularity that makes the orbit a genuine trajectory in X while still admitting the unbounded generators of analysis; requiring norm continuity of $t \mapsto T(t)$ instead would force the generator to be bounded and exclude every differential operator. The two stronger conditions suppressed here, that T extend to all of $\mathbb{R}$ and preserve norms, are exactly what distinguishes the unitary groups of section 13.4 from the present setting.

The semigroup law forces the norms to grow at most exponentially. This bound is the first structural fact and the reference point against which the resolvent is later defined.

Proposition 13.5.2: Exponential Bound

Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on X. There are constants $M \geq 1$ and $\omega \in \mathbb{R}$ with

$$\|T(t)\| \leq Me^{\omega t} \qquad (t \geq 0) \tag{13.13}$$

***Proof*:**
Step 1: Local boundedness. The claim is that $\sup_{0\leq t\leq 1} \|T(t)\| = M < \infty$. Suppose not; then there are $t_n \in [0, 1]$ with $\|T(t_n)\| \to \infty$. For each fixed $x \in X$ the orbit $t \mapsto T(t)x$ is continuous on the compact interval $[0, 1]$, hence bounded, so $\sup_n \|T(t_n)x\| < \infty$. The family $\{T(t_n)\}$ is therefore pointwise bounded, and by the Banach-Steinhaus uniform boundedness principle it is uniformly bounded, $\sup_n \|T(t_n)\| < \infty$, contradicting the choice of t_n. Thus $M := \sup_{0\leq t\leq 1} \|T(t)\|$ is finite, and $M \geq \|T(0)\| = 1$.

Step 2: Exponential growth from the semigroup law. Fix $t \geq 0$ and write $t = n + r$ with $n \in \mathbb{N}$ (or $n = 0$) and $r \in [0, 1)$. The semigroup law gives $T(t) = T(1)^n T(r)$, so

$$\|T(t)\| \leq \|T(1)\|^n \|T(r)\| \leq M^{n+1} = M \cdot M^n \leq M \cdot M^t$$

using $\|T(1)\| \le M$, $\|T(r)\| \le M$, and $n \le t$ with $M \ge 1$. Setting $\omega = \log M$ gives $M^t = e^{\omega t}$, hence (13.13), as we sought to show.

The pair (M, ω) is not unique; the infimum of the admissible ω is the growth bound of the semigroup. When ω may be taken to be 0 and $M = 1$ the semigroup is a contraction semigroup, the case the Hille-Yosida theorem below characterizes. The general case reduces to it by the rescaling $S(t) = e^{-\omega t}T(t)$, which is again a C_0-semigroup with a smaller growth bound; only a bounded shift of the generator separates the two.

The prototype is the heat semigroup, where the abstract axioms become the concrete statement that heat diffuses and the total is conserved.

Example 13.5.3: The Heat Semigroup

On $X = L^2(\mathbb{R})$ define, for $t > 0$,

$$(T(t)f)(x) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-|x-y|^2/(4t)} f(y)\, dy$$

and $T(0) = I$. The kernel $g_t(x) = (4\pi t)^{-1/2} e^{-|x|^2/(4t)}$ is the Gaussian of variance $2t$, so $T(t)f = g_t * f$ is convolution with g_t. Young's inequality gives $\|T(t)f\|_2 \le \|g_t\|_1 \|f\|_2 = \|f\|_2$, since $\int_{\mathbb{R}} g_t = 1$; thus each $T(t)$ is a contraction. The Gaussians satisfy the convolution identity $g_s * g_t = g_{s+t}$ (the variances add), which is exactly the semigroup law $T(s)T(t) = T(s+t)$. Strong continuity $\|T(t)f - f\|_2 \to 0$ as $t \to 0^+$ is the statement that $(g_t)_{t>0}$ is an approximate identity on L^2 (section 8.6). The orbit $u(t, \cdot) = T(t)f$ solves the heat equation $\partial_t u = \partial_{xx} u$ with initial datum f: this is the reason the generator, computed below on suitable f, is the second-derivative operator ∂_{xx}.

The whole theory of a C_0-semigroup is encoded in the rate of change of its orbits at time zero. That rate is a densely defined operator, generally unbounded, and it plays the role for the semigroup that a differential equation's right-hand side plays for its flow.

Definition 13.5.4: Generator of a C_0-Semigroup

Let $(T(t))_{t\ge 0}$ be a C_0-semigroup on X. Its *generator* is the operator A with domain

$$\operatorname{dom} A = \left\{ x \in X \;\middle|\; \lim_{t\to 0^+} \frac{T(t)x - x}{t} \text{ exists in } X \right\}$$

defined on that domain by

$$Ax = \lim_{t\to 0^+} \frac{T(t)x - x}{t}$$

The generator is the right derivative of the orbit at the origin, and by time homogeneity it controls the derivative at every later time. For a differential-equation reader, A is the "vector field" whose flow is $(T(t))$: the orbit $u(t) = T(t)x$ is the solution of the abstract Cauchy problem $u'(t) = Au(t)$, $u(0) = x$, whenever $x \in \operatorname{dom} A$. The content of the next results is that this domain is large (dense) and that A is closed, so that despite being unbounded it is a well-behaved object to which the resolvent machinery applies.

For the heat semigroup the generator is the operator anticipated by the heat equation.

Example 13.5.5: Generator of the Heat Semigroup

For the heat semigroup of example 13.5.3 on $L^2(\mathbb{R})$, the difference quotient $(T(t)f - f)/t$ converges in L^2 precisely when $f'' \in L^2$ (interpreting the derivative weakly), and its limit is f''. Thus the generator is

$$Af = f'', \qquad \operatorname{dom} A = \left\{ f \in L^2(\mathbb{R}) \mid f'' \in L^2(\mathbb{R}) \right\}$$

the second-derivative operator on the Sobolev space $\left\{ f \in L^2 \mid f, f' \text{ absolutely continuous}, f'' \in L^2 \right\}$. This domain is dense (it contains the Schwartz functions), and A is unbounded. To witness this, fix a nonzero Schwartz bump φ and set $f_k(x) = e^{ikx}\varphi(x)$ for $k \in \mathbb{N}$. Modulation preserves the L^2 norm, so $\|f_k\|_2 = \|\varphi\|_2$ is fixed, while $f_k'' = e^{ikx}(\varphi'' + 2ik\varphi' - k^2\varphi)$ has $\|f_k''\|_2 \geq k^2\|\varphi\|_2 - 2k\|\varphi'\|_2 - \|\varphi''\|_2 \to \infty$. Thus $\|Af_k\|_2/\|f_k\|_2 \to \infty$ on a family of fixed norm, so $\sup_{\|f\|_2=1} \|Af\|_2 = \infty$: the second derivative amplifies high frequencies without bound, which on the Fourier side is the unboundedness of the multiplier ξ^2. The abstract Cauchy problem $u' = Au$ is the heat equation, recovering the interpretation of example 13.5.3.

The differentiation properties of the orbits are collected next; they are the computational engine behind density, closedness, and the resolvent formula. The key identity is that integrating the orbit lands one back in the domain of A, so that A can be studied through the everywhere-defined averaging operators $x \mapsto \int_0^t T(s)x\, ds$.

Lemma 13.5.6: Orbit Calculus

Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on X with generator A.

1. For every $x \in X$ and $t \geq 0$ the vector $\int_0^t T(s)x\, ds$ lies in $\operatorname{dom} A$, and
$$A \int_0^t T(s)x\, ds = T(t)x - x \tag{13.14}$$
2. If $x \in \operatorname{dom} A$ then $T(t)x \in \operatorname{dom} A$ for every $t \geq 0$, the orbit $t \mapsto T(t)x$ is continuously differentiable, and
$$\frac{d}{dt} T(t)x = AT(t)x = T(t)Ax \tag{13.15}$$

***Proof*:**

The integral $\int_0^t T(s)x\, ds$ is the Bochner integral of the continuous X-valued function $s \mapsto T(s)x$ (theorem 13.1.9); continuity makes it Bochner integrable on $[0, t]$ by theorem 13.1.6.

Part (1). Fix x and t, and let $h > 0$. Since each $T(h)$ is bounded, it commutes with the Bochner integral (a bounded operator is closed, so proposition 13.1.8 applies, or directly by approximation by simple functions), giving

$$\frac{T(h) - I}{h} \int_0^t T(s)x\, ds = \frac{1}{h} \int_0^t \big(T(s+h)x - T(s)x\big)\, ds$$

Substituting $\sigma = s + h$ in the first term and splitting the intervals,

$$\frac{1}{h} \int_0^t T(s+h)x\, ds - \frac{1}{h} \int_0^t T(s)x\, ds = \frac{1}{h} \int_t^{t+h} T(\sigma)x\, d\sigma - \frac{1}{h} \int_0^h T(s)x\, ds$$

As $h \to 0^+$ each average converges to the value of the continuous integrand at the endpoint, by the fundamental theorem of calculus for Bochner integrals (theorem 13.1.9): the first tends to $T(t)x$ and the second to $T(0)x = x$. Hence $\int_0^t T(s)x\,ds \in \operatorname{dom} A$ and (13.14) holds.

Part (2). Let $x \in \operatorname{dom} A$. For $h > 0$, the semigroup law and boundedness of $T(t)$ give

$$\frac{T(h) - I}{h} T(t)x = T(t)\frac{T(h) - I}{h} x \to T(t)Ax \quad (h \to 0^+)$$

the limit existing because $x \in \operatorname{dom} A$ and $T(t)$ is continuous. Therefore $T(t)x \in \operatorname{dom} A$ with $AT(t)x = T(t)Ax$, and the right derivative of $t \mapsto T(t)x$ is $T(t)Ax$. To identify this as a two-sided derivative, consider for $t > 0$ and $0 < h < t$ the left difference quotient

$$\frac{T(t)x - T(t-h)x}{h} = T(t-h)\frac{T(h)x - x}{h}$$

The factor $(T(h)x - x)/h \to Ax$, while $T(t-h) \to T(t)$ strongly, and the two convergences combine on the bounded family $\{T(t-h)\}_{0 \le h \le t/2}$ (bounded by proposition 13.5.2): writing $T(t-h)y_h - T(t)Ax = T(t-h)(y_h - Ax) + (T(t-h) - T(t))Ax$ with $y_h = (T(h)x - x)/h$, the first term is bounded in norm by $(\sup \|T(t-h)\|)\|y_h - Ax\| \to 0$ and the second tends to 0 by strong continuity. Thus the left quotient also converges to $T(t)Ax$, so $t \mapsto T(t)x$ is differentiable with derivative $T(t)Ax = AT(t)x$, which is continuous in t; this is (13.15) and shows the orbit is C^1, completing the proof.

Density and closedness now follow by reading off consequences of the orbit calculus.

Proposition 13.5.7: The Generator is Densely Defined and Closed

The generator A of a C_0-semigroup $(T(t))_{t \ge 0}$ on X is densely defined and closed.

***Proof*:**

Density. For $x \in X$ set $x_t = \frac{1}{t}\int_0^t T(s)x\,ds$ for $t > 0$. By lemma 13.5.6(1), $x_t \in \operatorname{dom} A$. As $t \to 0^+$ the average of the continuous function $s \mapsto T(s)x$ over $[0, t]$ converges to its value $T(0)x = x$ at the endpoint (theorem 13.1.9). Hence every $x \in X$ is a limit of elements $x_t \in \operatorname{dom} A$, so $\operatorname{dom} A$ is dense.

Closedness. Let $x_n \in \operatorname{dom} A$ with $x_n \to x$ and $Ax_n \to y$ in X. By (13.15) the orbit of each x_n is C^1 with $\frac{d}{ds}T(s)x_n = T(s)Ax_n$, so the fundamental theorem of calculus for Bochner integrals (theorem 13.1.9) gives

$$T(t)x_n - x_n = \int_0^t T(s)Ax_n\,ds$$

Fix $t > 0$. On the compact interval $[0, t]$ the operators $T(s)$ are uniformly bounded by $C = \sup_{0 \le s \le t} \|T(s)\| < \infty$ (proposition 13.5.2), so $\|T(s)Ax_n - T(s)y\| \le C\|Ax_n - y\| \to 0$ uniformly in s. Passing to the limit under the integral (uniform convergence of the integrand on a finite interval) and using $x_n \to x$, $T(t)x_n \to T(t)x$,

$$T(t)x - x = \int_0^t T(s)y\,ds$$

Dividing by t and letting $t \to 0^+$, the right side converges to $T(0)y = y$ by theorem 13.1.9, while the left side is the difference quotient defining Ax. Hence $x \in \operatorname{dom} A$ and $Ax = y$, so A is closed, as we sought to show.

That A is closed and densely defined is what admits it to the resolvent theory of section 13.2: its spectrum and resolvent are defined exactly as for the closed operators studied there. The next result computes the resolvent explicitly and reveals the analytic identity organizing the whole subject. The resolvent of the generator is the Laplace transform of the semigroup.

The motivation is the scalar identity $\int_0^\infty e^{-\lambda t} e^{at}\,dt = (\lambda - a)^{-1}$, valid for $\operatorname{Re}\lambda > a$. Reading e^{at} as the semigroup $T(t)$ and $(\lambda - a)^{-1}$ as the resolvent $R(\lambda, A)$ suggests that the resolvent is the operator Laplace transform of the semigroup, and this is exactly the theorem.

Theorem 13.5.8: The Resolvent as Laplace Transform

Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on X with generator A and bound $\|T(t)\| \leq Me^{\omega t}$ as in proposition 13.5.2. Then every λ with $\operatorname{Re}\lambda > \omega$ lies in the resolvent set $\rho(A)$, the improper Bochner integral

$$R(\lambda)x = \int_0^\infty e^{-\lambda t} T(t)x\,dt \qquad (x \in X) \tag{13.16}$$

converges, and $R(\lambda) = R(\lambda, A) = (\lambda I - A)^{-1}$. Moreover $\|R(\lambda, A)\| \leq M/(\operatorname{Re}\lambda - \omega)$.

***Proof*:**
Step 1: Convergence and the norm bound. Fix λ with $\operatorname{Re}\lambda > \omega$ and $x \in X$. The integrand $t \mapsto e^{-\lambda t}T(t)x$ is continuous, hence strongly measurable, and

$$\|e^{-\lambda t}T(t)x\| \leq e^{-(\operatorname{Re}\lambda)t} M e^{\omega t}\|x\| = M\|x\|e^{-(\operatorname{Re}\lambda - \omega)t}$$

whose integral over $[0, \infty)$ is $M\|x\|/(\operatorname{Re}\lambda - \omega) < \infty$. By the Bochner integrability criterion (theorem 13.1.6) the integral (13.16) converges and $\|R(\lambda)x\| \leq M\|x\|/(\operatorname{Re}\lambda - \omega)$, giving the norm bound on the bounded operator $R(\lambda)$.

Step 2: $R(\lambda)$ maps into $\operatorname{dom} A$ *and* $(\lambda I - A)R(\lambda) = I$. For $h > 0$, commuting the bounded operator $(T(h) - I)/h$ past the Bochner integral and shifting as in lemma 13.5.6,

$$\frac{T(h) - I}{h} R(\lambda)x = \frac{1}{h}\int_0^\infty e^{-\lambda t}\big(T(t+h)x - T(t)x\big)\,dt = \frac{e^{\lambda h} - 1}{h} R(\lambda)x - \frac{e^{\lambda h}}{h}\int_0^h e^{-\lambda t}T(t)x\,dt$$

where the substitution $\sigma = t+h$ in the first term produces the factor $e^{\lambda h}$ and the boundary integral over $[0, h]$. Letting $h \to 0^+$: the first term tends to $\lambda R(\lambda)x$, and the average $\frac{1}{h}\int_0^h e^{-\lambda t}T(t)x\,dt \to x$ by theorem 13.1.9, so the second tends to x. Hence $R(\lambda)x \in \operatorname{dom} A$ and

$$AR(\lambda)x = \lambda R(\lambda)x - x$$

that is, $(\lambda I - A)R(\lambda)x = x$ for all $x \in X$.

Step 3: $R(\lambda)(\lambda I - A) = I$ on $\operatorname{dom} A$. Let $x \in \operatorname{dom} A$. Since A is closed and commutes with the semigroup on its domain (lemma 13.5.6(2)), A commutes with the integral defining $R(\lambda)$ by proposition 13.1.8: the functions $t \mapsto e^{-\lambda t}T(t)x$ and $t \mapsto e^{-\lambda t}T(t)Ax = A(e^{-\lambda t}T(t)x)$ are both Bochner integrable, so $R(\lambda)x \in \operatorname{dom} A$ and $AR(\lambda)x = R(\lambda)Ax$. Therefore

$$R(\lambda)(\lambda I - A)x = \lambda R(\lambda)x - R(\lambda)Ax = \lambda R(\lambda)x - AR(\lambda)x = x$$

the last equality by Step 2. Steps 2 and 3 exhibit $R(\lambda)$ as a two-sided inverse of $\lambda I - A$, so $\lambda \in \rho(A)$ and $R(\lambda) = R(\lambda, A)$, completing the proof.

Two consequences of (13.16) organize what follows. First, the resolvent bound $\|R(\lambda, A)\| \leq M/(\operatorname{Re}\lambda - \omega)$ is the operator reflection of the exponential bound on the semigroup; for a contraction semigroup ($M = 1$, $\omega = 0$) it reads $\|R(\lambda, A)\| \leq 1/\lambda$ for $\lambda > 0$, precisely the inequality that will characterize generators. Second, the theorem is not yet a construction: it computes the resolvent from a semigroup already in hand. The Hille-Yosida theorem runs the implication the other way, building the semigroup from an operator whose resolvent satisfies that bound, and it is the substantive result of the section.

The reconstruction rests on approximating the unbounded generator A by bounded operators whose exponentials can be formed directly. The device is the Yosida approximation.

Definition 13.5.9: Yosida Approximation

Let A be a densely defined closed operator on X with $(0, \infty) \subseteq \rho(A)$. For $\lambda > 0$ the *Yosida approximation* of A is the bounded operator

$$A_\lambda = \lambda A R(\lambda, A) = \lambda^2 R(\lambda, A) - \lambda I$$

where the second expression uses $AR(\lambda, A) = \lambda R(\lambda, A) - I$.

The Yosida approximation replaces the unbounded A by the bounded operator $\lambda A R(\lambda, A)$, which one should read as A composed with the "mollifier" $\lambda R(\lambda, A)$. The second expression $A_\lambda = \lambda^2 R(\lambda, A) - \lambda I$ shows A_λ is bounded, so its exponential $e^{tA_\lambda} = \sum_{n \geq 0} (tA_\lambda)^n/n!$ converges in operator norm and defines a genuine semigroup for each λ. The plan is to show these approximating semigroups converge strongly as $\lambda \to \infty$ and that the limit is the semigroup generated by A. The next lemma records the two facts that make the plan work: the mollifier tends to the identity, and A_λ tends to A.

Lemma 13.5.10: Convergence of the Yosida Approximation

Let A be densely defined and closed with $(0, \infty) \subseteq \rho(A)$ and $\|R(\lambda, A)\| \leq 1/\lambda$ for all $\lambda > 0$. Then:

1. $\|\lambda R(\lambda, A)\| \leq 1$ for all $\lambda > 0$, and $\lambda R(\lambda, A)x \to x$ as $\lambda \to \infty$, for every $x \in X$;
2. $A_\lambda x \to Ax$ as $\lambda \to \infty$, for every $x \in \operatorname{dom} A$.

***Proof*:**
Part (1). The bound $\|\lambda R(\lambda, A)\| \leq \lambda \cdot (1/\lambda) = 1$ is immediate. For $x \in \operatorname{dom} A$, the resolvent identity $AR(\lambda, A) = \lambda R(\lambda, A) - I$ rearranges to $\lambda R(\lambda, A)x - x = R(\lambda, A)Ax$, whence

$$\|\lambda R(\lambda, A)x - x\| = \|R(\lambda, A)Ax\| \leq \frac{1}{\lambda}\|Ax\| \to 0 \quad (\lambda \to \infty)$$

Thus $\lambda R(\lambda, A)x \to x$ on the dense subspace $\operatorname{dom} A$. Since the operators $\lambda R(\lambda, A)$ are uniformly bounded by 1, the convergence extends to all $x \in X$: given $x \in X$ and $\varepsilon > 0$, choose $y \in \operatorname{dom} A$ with $\|x - y\| < \varepsilon$; then $\|\lambda R(\lambda, A)x - x\| \leq \|\lambda R(\lambda, A)(x - y)\| + \|\lambda R(\lambda, A)y - y\| + \|y - x\| \leq 2\varepsilon + \|\lambda R(\lambda, A)y - y\|$, and the middle term tends to 0.

Part (2). For $x \in \operatorname{dom} A$, the operators A and $R(\lambda, A)$ commute on $\operatorname{dom} A$, so $A_\lambda x =$

$\lambda AR(\lambda, A)x = \lambda R(\lambda, A)Ax$. By Part (1) applied to the vector $Ax \in X$, $\lambda R(\lambda, A)Ax \to Ax$ as $\lambda \to \infty$. Hence $A_\lambda x \to Ax$, completing the proof.

Everything is in place for the theorem. It characterizes exactly which operators generate a C_0-contraction semigroup, and its content is that the resolvent conditions read off from theorem 13.5.8 are not only necessary but sufficient. The proof constructs the semigroup from the resolvent via the Yosida approximation.

Theorem 13.5.11: Hille-Yosida

Let A be an operator on a Banach space X. Then A is the generator of a C_0-contraction semigroup $(T(t))_{t\geq 0}$ on X (that is, $\|T(t)\| \leq 1$ for all $t \geq 0$) if and only if

1. A is densely defined and closed;
2. $(0, \infty) \subseteq \rho(A)$; and
3. $\|R(\lambda, A)\| \leq 1/\lambda$ for every $\lambda > 0$.

When these hold the semigroup is unique, and $T(t)x = \lim_{\lambda\to\infty} e^{tA_\lambda}x$ for every $x \in X$, uniformly for t in compact intervals, where A_λ is the Yosida approximation of definition 13.5.9.

***Proof*:**
Necessity. Suppose A generates a C_0-contraction semigroup. Then A is densely defined and closed by proposition 13.5.7. With $M = 1$ and $\omega = 0$ in proposition 13.5.2, theorem 13.5.8 gives $(0, \infty) \subseteq \rho(A)$ and $\|R(\lambda, A)\| \leq 1/\lambda$ for $\lambda > 0$. This is (1), (2), (3).

Sufficiency. Assume (1), (2), (3). The construction proceeds in four steps: form the bounded-operator semigroups e^{tA_λ}, bound them uniformly, prove they converge strongly, and verify the limit is a C_0-contraction semigroup with generator A.

Step 1: The approximating semigroups are contractions. Fix $\lambda > 0$. Since $A_\lambda = \lambda^2 R(\lambda, A) - \lambda I$ is bounded, $e^{tA_\lambda} = e^{-\lambda t}e^{t\lambda^2 R(\lambda,A)}$ is defined by its norm-convergent power series. Using $\|\lambda R(\lambda, A)\| \leq 1$ from lemma 13.5.10(1),

$$\|e^{tA_\lambda}\| \leq e^{-\lambda t}\sum_{n=0}^{\infty}\frac{(t\lambda^2)^n}{n!}\|R(\lambda, A)\|^n \leq e^{-\lambda t}\sum_{n=0}^{\infty}\frac{(t\lambda^2)^n}{n!}\left(\frac{1}{\lambda}\right)^n = e^{-\lambda t}e^{t\lambda} = 1$$

so each e^{tA_λ} is a contraction, for every $t \geq 0$ and $\lambda > 0$.

Step 2: The approximations commute. For fixed $\lambda, \mu > 0$ the operators A_λ and A_μ are polynomials in the commuting resolvents $R(\lambda, A)$ and $R(\mu, A)$ (which commute by the resolvent identity), hence commute with each other and with each e^{sA_μ}.

Step 3: Strong convergence of e^{tA_λ}. Fix $x \in \operatorname{dom} A$ and $t \geq 0$, and let $\lambda, \mu > 0$. The operators e^{sA_λ} and $e^{(t-s)A_\mu}$ commute (Step 2), so the telescoping identity

$$e^{tA_\lambda}x - e^{tA_\mu}x = \int_0^t \frac{d}{ds}\left(e^{sA_\lambda}e^{(t-s)A_\mu}x\right) ds = \int_0^t e^{sA_\lambda}e^{(t-s)A_\mu}(A_\lambda - A_\mu)x\, ds$$

holds, the integrand being continuous in s. Since e^{sA_λ} and $e^{(t-s)A_\mu}$ are contractions (Step 1),

$$\|e^{tA_\lambda}x - e^{tA_\mu}x\| \leq \int_0^t \|(A_\lambda - A_\mu)x\|\, ds = t\|A_\lambda x - A_\mu x\|$$

By lemma 13.5.10(2) the sequence $(A_\lambda x)$ is Cauchy as $\lambda \to \infty$ (it converges to Ax), so $(e^{tA_\lambda} x)$ is Cauchy in X, uniformly for t in any compact interval $[0, t_0]$. Define $T(t)x = \lim_{\lambda\to\infty} e^{tA_\lambda} x$ for $x \in \operatorname{dom} A$. Each $T(t)$ is a contraction on $\operatorname{dom} A$, and since $\operatorname{dom} A$ is dense (proposition 13.5.7) and the e^{tA_λ} are uniformly bounded by 1, the same density argument as in lemma 13.5.10(1) extends $T(t)$ to a contraction on all of X with $e^{tA_\lambda} x \to T(t)x$ for every $x \in X$, uniformly on compact t-intervals.

Step 4: The limit is a C_0-contraction semigroup generated by A. The semigroup properties pass to the limit: $T(0)x = \lim e^0 x = x$, and for the semigroup law, each $e^{(s+t)A_\lambda} = e^{sA_\lambda} e^{tA_\lambda}$, so letting $\lambda \to \infty$ and using uniform boundedness gives $T(s+t)x = T(s)T(t)x$. Strong continuity holds because $e^{tA_\lambda} x \to T(t)x$ uniformly on compact intervals and each $t \mapsto e^{tA_\lambda} x$ is continuous, so the uniform limit $t \mapsto T(t)x$ is continuous. Thus $(T(t))$ is a C_0-contraction semigroup; let B denote its generator.

It remains to identify B with A. For $x \in \operatorname{dom} A$, the orbit calculus applied to each approximating semigroup gives $e^{tA_\lambda} x - x = \int_0^t e^{sA_\lambda} A_\lambda x \, ds$. The integrand converges to $T(s)Ax$ uniformly in $s \in [0, t]$: indeed $\|e^{sA_\lambda} A_\lambda x - T(s)Ax\| \le \|e^{sA_\lambda}(A_\lambda x - Ax)\| + \|(e^{sA_\lambda} - T(s))Ax\| \le \|A_\lambda x - Ax\| + \|(e^{sA_\lambda} - T(s))Ax\|$, and both terms tend to 0 uniformly on $[0, t]$ by lemma 13.5.10(2) and Step 3. Passing to the limit,

$$T(t)x - x = \int_0^t T(s)Ax \, ds$$

Dividing by t and letting $t \to 0^+$, the right side tends to $T(0)Ax = Ax$ by theorem 13.1.9, so $x \in \operatorname{dom} B$ and $Bx = Ax$. Hence $A \subseteq B$.

For the reverse inclusion, note that $1 \in \rho(A)$ by (2) and $1 \in \rho(B)$ since B generates a contraction semigroup (necessity applied to B). Now $I - A$ maps $\operatorname{dom} A$ bijectively onto X, and $I - B$ maps $\operatorname{dom} B$ bijectively onto X; since $A \subseteq B$, the operator $I - B$ restricted to $\operatorname{dom} A$ already surjects onto X. If $x \in \operatorname{dom} B$, pick $y \in \operatorname{dom} A$ with $(I - A)y = (I - B)x$; then $(I - B)y = (I - A)y = (I - B)x$ because $A \subseteq B$, and injectivity of $I - B$ forces $y = x$, so $x = y \in \operatorname{dom} A$. Thus $\operatorname{dom} B \subseteq \operatorname{dom} A$ and $B = A$.

Uniqueness. Suppose $(S(t))$ is another C_0-contraction semigroup with generator A. Fix $x \in \operatorname{dom} A$ and $t > 0$, and consider the X-valued function $s \mapsto S(t - s)T(s)x$ on $[0, t]$, where $(T(t))$ is the semigroup constructed above. By lemma 13.5.6(2) it is differentiable with

$$\frac{d}{ds} S(t - s)T(s)x = -S(t - s)AT(s)x + S(t - s)AT(s)x = 0$$

the two terms cancelling because $T(s)x \in \operatorname{dom} A$ and both semigroups have generator A. A function with zero derivative on $[0, t]$ is constant, so evaluating at $s = 0$ and $s = t$ gives $S(t)x = T(t)x$ for all $x \in \operatorname{dom} A$, and by density $S(t) = T(t)$. This establishes uniqueness and completes the proof.

The theorem is the exact analogue for irreversible dynamics of Stone's theorem for the reversible case. Stone's theorem realizes every strongly continuous unitary group as e^{itA} for a self-adjoint A (theorem 13.4.5); Hille-Yosida realizes every C_0-contraction semigroup as, informally, e^{tA} for an operator A satisfying the resolvent bound, with the exponential made precise by the Yosida limit $e^{tA} = \lim_\lambda e^{tA_\lambda}$. On a Hilbert space the two theorems meet: a self-adjoint $A \le 0$ satisfies the Hille-Yosida conditions (its spectrum lies in $(-\infty, 0]$, so $(0, \infty) \subseteq \rho(A)$ with the required bound), and the contraction semigroup it generates is the $e^{tA} = \int_{(-\infty,0]} e^{t\lambda} \, dE(\lambda)$ of the functional calculus of theorem 13.3.8. The heat semigroup of example 13.5.3 is this case with $A = \partial_{xx} \le 0$.

The general (non-contractive) case follows by the rescaling noted after proposition 13.5.2: A generates a C_0-semigroup with $\|T(t)\| \le Me^{\omega t}$ if and only if $A - \omega I$ generates a bounded (M-bounded) semigroup, and the sharp characterization of that case requires bounding all powers $\|R(\lambda, A)^n\| \le M/(\lambda - \omega)^n$ rather than the resolvent alone. This is the Feller-Miyadera-Phillips theorem, whose proof is the same Yosida construction with the contraction bound of Step 1 replaced by the power bound; the contraction case above carries all the ideas, and the general statement is developed where it is needed, in the evolution-equation volumes.

One refinement of the generation theory is indispensable for parabolic problems and is recorded here for orientation, its development deferred to the volumes that use it. A C_0-semigroup is *analytic* when the orbit map $t \mapsto T(t)$ extends holomorphically from the positive real axis to a sector of the complex plane. Such semigroups are exactly those whose generators have resolvent bounds on a sector, and they carry the smoothing that makes parabolic equations regularize their data instantly.

Proposition 13.5.12: Analytic Semigroups and Sectorial Generators

Let A be a densely defined closed operator on a Banach space X, and for $\theta \in (0, \pi/2]$ let $\Sigma_\theta = \{\lambda \in \mathbb{C} \setminus \{0\} \mid |\arg \lambda| < \pi/2 + \theta\}$. Suppose $\Sigma_\theta \subseteq \rho(A)$ and there is a constant C with

$$\|R(\lambda, A)\| \le \frac{C}{|\lambda|} \qquad (\lambda \in \Sigma_\theta)$$

(such an A is called *sectorial*). Then A generates a C_0-semigroup $(T(t))_{t \ge 0}$ that extends to a holomorphic map $z \mapsto T(z)$ on the open sector $\{z \in \mathbb{C} \setminus \{0\} \mid |\arg z| < \theta\}$, bounded on each subsector $|\arg z| \le \theta' < \theta$, with $T(z)X \subseteq \operatorname{dom} A$ for every z in the sector. The semigroup is given by the Dunford integral $T(z) = \frac{1}{2\pi i} \int_\Gamma e^{z\lambda} R(\lambda, A)\, d\lambda$ over a suitable contour Γ in Σ_θ.

The sectorial resolvent bound is what turns the Laplace-transform representation of theorem 13.5.8 into a contour integral that may be differentiated in the time variable arbitrarily often, so an analytic semigroup maps X into $\operatorname{dom} A$ (indeed into $\bigcap_n \operatorname{dom} A^n$) for every $t > 0$. This is the abstract form of parabolic smoothing: the solution $T(t)f$ of $u' = Au$ is infinitely differentiable in t and lies in the domain of every power of A for $t > 0$, however rough the datum f. The Laplacian on L^p and the elliptic operators of the theory of linear parabolic equations are the governing examples, and the contour-integral construction, the perturbation theory, and the maximal-regularity estimates that accompany it belong to *Linear Elliptic and Parabolic PDE* [5], where the heat and reaction-diffusion equations are solved by exactly this machinery.

Part III

Fourier Analysis and Hard Analysis on $\mathbb{R}^n$

The first part is Fourier analysis in its classical, Euclidean form together with the hard analysis it powers. It opens with the Fourier transform and Fourier series on $\mathbb{R}^n$ and $\mathbb{T}^n$ and their convergence theory, extends the transform to the tempered distributions on which it acts most naturally, and develops the operators this makes available: singular integrals and Calderón-Zygmund theory, the Sobolev scale, and applications to partial differential equations. Throughout, the group underneath is $\mathbb{R}^n$ under translation; the abstract theory of part IV is what remains when that group is replaced by an arbitrary locally compact abelian one. The foundational theory of distributions as a topological vector space is developed in [13]; this part uses only its Fourier-facing layer.

14

Fourier Analysis

In the 19th century, Joseph Fourier was trying to find out how to model the movement of heat through solid bodies, in particular model how heat seems to dissipate and homogenise within its medium. For simplicity, let's say this body was a rode, and we can represent any point on the rode with the interval $[a, b]$. If $u : [0, t] \times [a, b] \to \mathbb{R}$ is a function that at any given point $c \in [a, b]$ and at any given time $t_0 \in [0, t]$ give you the temperature, then Fourier devised that the function u must respect the following differential equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

which can be interpreted that in any movement in time for some point $c \in [a, b]$, the temperature at c will equal to the *average* of its neighbor (see 3Blue1Brown's video on Fourier analysis for an excellent visualization of this). After choosing the value at $u|_{t_0=0}$ which represent the initial distribution of heat in the rode $[a, b]$, as t_0 changes $u|_{t_0}$ represents the temperature of the rod at every point at time t_0 . Working with this equation, Fourier saw that trigonometric identities are a solution for u given the restraint (with suitable scaling[1]). The next insight Fourier had was that a huge amount of functions can be interpreted as an infinite sum of trigonometric functions, and even better, he found how to find the appropriate coefficients for the trigonometric functions! In particular, this family of equations is *dense* in the space of all L^2 functions on a compact interval (or even better, they form an orthonormal basis), and since the PDE in question is *linear*, we can use the Dominated convergence theorem to show that the resulting sequence can mean that $u(0, \cdot)$ can be any L^2-integrable function (or really, simply integrable since the domain is compact). This is known as the *Fourier Series*, and going form a function to its Fourier series is called the *Fourier Transform*.

This was the beginning of Fourier Analysis, also known as Harmonic Analysis[2]. Since then, it was

[1] In particular, some care must be given to the boundary conditions, but this is something that would be covered in more detail in a proper class in PDE's

[2] Sometimes, Harmonic Analysis is used as a generalization of Fourier Analysis in the case of non-abelian groups, something which will be made more clear in this chapter

asked what spaces of functions admit a Fourier transform, can we go back and forth between the Fourier series and the original function, and can we generalize the domain of the functions (this will be pushed to be compact abelian group, and more generally locally compact groups). The great advantage of being able to find the space of functions that have a Fourier transform is that (under suitable conditions) derivatives work really well with infinite series, trigonometric functions have some of the simple derivatives (they are periodic of order 4), and the Fourier Transform allows us to transfer our problem into a potentially nice space (similarly to how in Representation Theory, it is much easier to find the representation of the monster group to study it rather than directly study it).

These days, Harmonic Analysis has found a large range of applications, from solving my different types of PDE's, to having connections to algebraic number theory (the pursuit of this connection is known as *Langlands Program*). In this chapter, we will cover the elementary theory required to understand the theory of Fourier analysis.

14.1 Basic Definitions for Fourier Analysis

In this chapter, we will always work with $\mathbb{R}^n$ with the Lebesgue measure, denoted m. If $E \subseteq \mathbb{R}^n$ is measurable, $L^p(E)$ (or $L^p(E,m)$) is the set of all L^p functions from E to $\mathbb{R}$, $C^k(E)$ is the set of all k-times continuously differentiable functions from E to $\mathbb{R}$, $C^\infty(E)$ (the set of all smooth functions form E to $\mathbb{R}$) is equal to $\cap_{k=1}^\infty C^k(E)$, and $C_c^\infty(E)$ is the set of all smooth function whose compact support is in E. If $E = \mathbb{R}^n$, we will often drop the domain of the function space, so:

$$L^p = L(\mathbb{R}^n), \qquad C^\infty = C^\infty(\mathbb{R}^n) \qquad C_c^\infty = C_c^\infty(\mathbb{R}^n)$$

The inner product and norm on $\mathbb{R}^n$ is the usual dot product and euclidean norm:

$$x \cdot y = \sum_i^n x_i y_i \qquad \|x\| = |x| = \sqrt{x \cdot x}$$

Derivative play a prominent role in this chapter, and so we establish important notation. We will assume the standard basis and write:

$$\partial_j = \frac{\partial}{\partial x^j}$$

and for higher order mixed derivatives, we will use a multi-index, which is an ordered n-tuple $\alpha = (\alpha_1, ..., \alpha_n)$. As usual:

$$|\alpha| = \sum_1^n \alpha_i \quad \alpha! = \prod_1^n \alpha_i! \quad \partial^\alpha = \left(\frac{\partial}{\partial x_i}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}$$

and for any $x = (x_1, x_2, ..., x_n) \in \mathbb{R}^n$:

$$x^\alpha = \prod_1^n x_i^{\alpha_i}$$

Be aware that $|\alpha|$ and $|x|$ do not means the same thing; the meaning should be clear from context. For reference, the product rule on a multi-index expands to:

$$\partial^\alpha(fg) = \sum_{\beta+\gamma=\alpha} \frac{\alpha!}{\beta!\gamma!} (\partial^\beta f)(\partial^\gamma g)$$

A common function we'll use a lot in C_c^∞ is:

$$\psi(x) = \eta(1 - |x|^2) = \begin{cases} \exp((|x|^2 - 1)^{-1})) & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

where $\eta(t) = e^{1/t}\chi_{(0,\infty)}(t)$. Another common subset of C^∞ that will be of use is the following which ensures that a function and all its derivatives decay faster than polynomial time:

Definition 14.1.1: Schwartz Space

The set $\mathcal{S} \subseteq C^\infty$ is the set of all smooth functions and all their derivatives which vanish faster than any power of $|x|$, that is, for any $n \in \mathbb{N}$ and any multi-index α, if we define:

$$\|f\|_{(N,\alpha)} = \sup_{x \in \mathbb{R}^n} (1 + |x|)^n |\partial^\alpha f(x)|$$

Then:

$$\mathcal{S} = \left\{ f \in C^\infty \;\middle|\; \|f\|_{(N,\alpha)} < \infty \text{ for all } N, \alpha \right\}$$

Example 14.1.2: Schwartz Functions

1. Naturally, $C_c^\infty \subseteq \mathcal{S}$
2. $f_\alpha(x) = x^\alpha e^{-|x|^2}$ for any multi-index α is in $\mathcal{S}$

Note that if $f \in \mathcal{S}$, then $\alpha f \in L^p$ for all α and all $p \in [1, \infty]$ since $|\partial^\alpha f(x)| \leq C_N(1 + |x|)^{-N}$ for all $N \in \mathbb{N}$, and $(1 + |x|)^{-N} \in L^p$, for all $N > n/p$

Proposition 14.1.3: $\mathcal{S}$ is a Fréchet Space

The Schwartz space $\mathcal{S}$ is a Fréchet space with the topology defined by the norms $\|\cdot\|_{(N,\alpha)}$

***Proof*:**
All is easy to prove with the possible exception of completeness. Let $\{f_k\}$ be a cauchy sequence in $\mathcal{S}$ so that $\|f_i - f_j\|_{(N,\alpha)} \to 0$ for all N, α. In particular, for each α the sequence $\{\partial^\alpha f_k\}$ converges uniformly to some g_α. Letting e_i denote the standard ith basis vector, we get:

$$f_k(x + te_i) - f_k(x) = \int_0^t \partial_i f_k(x + se_i) ds$$

Letting $k \to \infty$, we get:

$$g_0(x + te_i) - g_0(x) = \int_0^t g_{e_i}(x + se_i) ds$$

and the fundamental theorem of calculus tells us that $g_{e_i} = \partial_i g_0$, and so by induction we get $g_\alpha = \partial^\alpha g_0$ for all α. From this, it is now easy to verify that $\|f_k - g_0\|_{(N,\alpha)} \to 0$ for all α and $N \in \mathbb{N}$.

A useful characterization of $\mathcal{S}$ is also the following:

Proposition 14.1.4: Characterizing Schwartz Functions

If $f \in C^\infty$, Then $f \in \mathcal{S}$ if and only if $x^\beta \partial^\alpha f$ is bounded for all multi-indices α, β, if and only if $\partial^\alpha (x^\beta f)$ is bounded for all multi-indices α, β.

***Proof*:**
If $f \in \mathcal{S}$, then naturally $|x^\beta| \leq (1+|x|)^N$ for $\beta \leq N$. Conversely, notice that $\sum_1^n |x_i|^N$ is strictly positive on the unit sphere $|x| = 1$ (which is compact), and hence has a positive minimum, say δ. It follows that $\sum_1^n |x_i|^N \geq \delta |x|^N$ for all x since both sides are homogeneous of degree N, and hence:

$$(1+|x|)^N \leq 2^N(1+|x|^N) \leq 2^N \left[1 + \delta^{-1} \sum_1^n \left|x_i^N\right|\right] \leq 2^N \delta^{-1} \sum_{|\beta| \leq N} |x^\beta|$$

Giving the first equivalence. For the second equivalence, notice that for every α, $\partial^\alpha(x^\beta f)$ is a linear combination of terms of the form $x^\gamma \partial^\delta f$ and vice versa, by the product rule, completing the proof.

Next we define a periodic function to be a function for which $f(x+k) = f(x)$ for all $x \in \mathbb{R}^n$ and $k \in \mathbb{Z}^n$. Without loss of generality the period of the functions will be of size 1. Thus, we can consider the space on which we define periodic functions to be $\mathbb{R}^n/\mathbb{Z}^n \cong (\mathbb{R}/\mathbb{Z})^n \equiv \mathbb{T}^n = S^1 \times S^1 \times \cdots \times S^1$, i.e. we can think of defining period functions on an n-torus (note that $\mathbb{T}^1 = S^1$, but $\mathbb{T}^2 \neq S^2$). Naturally, $\mathbb{T}^n$ is a compact Hausdorff space. It can be identified with all points $z = (z_1, z_2, ..., z_n) \in (\mathbb{C}^n)^*$ with $|z_i| = 1$ via the map

$$(x_1, x_2, ..., x_n) \mapsto (e^{2\pi i x_1}, ..., e^{2\pi i x_n})$$

For measure-theoretic purposes, we will sometimes identify $\mathbb{T}^n$ with the unit cube:

$$Q = \left[\frac{-1}{2}, \frac{1}{2}\right)^n$$

and when we talk about the Lebesgue measure on $\mathbb{T}^n$, we will mean the induced measure from Q. This means that $m(\mathbb{T}^n) = 1$. We will think of functions on $\mathbb{T}^n$ as either periodic functions on $\mathbb{R}^n$ or as functions on Q (the context will make it clear which we pick).

Exercise 14.1.1

1. Is the sum of two periodic functions periodic?

14.1.1 Convolution

The convolution $f * g$, Young's convolution inequality, translation-continuity in L^p, the smoothing lemma, approximate identities (good kernels), and the density of smooth functions in L^p are developed as foundational L^p operations in the functional-analysis part, section 8.6. Here we record only the fact special to the Schwartz class, which the Fourier theory below uses:

Proposition 14.1.5: Convolution in Schwartz Space

Let $f, g \in \mathcal{S}$. Then $f * g \in \mathcal{S}$

***Proof*:**
First, $f * g \in C^\infty$ by lemma 8.6.7. Next, we establish the following for the coming chain of inequalities:

$$1 + |x| \overset{\triangle\text{ineq.}}{\leq} 1 + |x-y| + |y| \leq (1 + |x-y|)(1 + |y|)$$

and so:

$$\begin{aligned}(1+|x|)^N |\partial^\alpha (f * g)(x)| &\leq \int (1 + |x-y|)^N |\partial^\alpha f(x-y)| (1+|y|)^N |g(y)| dy \\ &= \|f\|_{(N,\alpha)} \|g\|_{(N+n+1,\alpha)} \int (1+|y|)^{-n-1} dy\end{aligned}$$

which is certainly finite

14.2 Fourier Transform

Consider again the heat equation on a compact interval:

$$\frac{du}{dt} = \frac{\partial^2 u}{\partial x^2} \tag{14.1}$$

where u is a solution. If u represents a rod that is at time $t = 0$ half hot on the left and half cold on the right, with the two halves in contact at the midpoint, then as we let time move forward, the temperature will equalize due to diffusion. The shape goes from a step function, to a smoothed profile resembling the error function $\text{erf}(x)$, and as time $\to \infty$ it converges to a constant function representing the average temperature. These are all functions with distinctly different behaviours.

Now, assume that the heat of the rod at $t = 0$ has the shape of $\sin(x)$ or $\cos(x)$. Then as we let time propagate these functions dampen uniformly, notably *they are always the same function*, simply scaled. This is essentially the behaviour of *eigenvectors*, namely as time moves forward we are scaling by a factor. To have eigenvectors, we need a linear transformation between vector spaces. Let S_t be the *time evolution operator* that takes in a state f_{t_0} (a function on the domain that is a t_0-slice of a solution $u(x, t_0)$) and outputs f_{t_0+t}, in particular $S_t : \mathscr{P}(X) \to \mathscr{P}(X)$ where $\mathscr{P}(X)$ is the state space over X and S_t is the evolve by t operator:

$$S_t(u(-, t_0)) = u(-, t_0 + t)$$

Or, viewed as an (abelian) monoid action[3]:

$$S : [0, \infty) \times \mathscr{P}(X) \to \mathscr{P}(X) \qquad t \cdot f_{t_0} = f_{t_0+t}$$

[3]The heat equation is well-posed forward in time but ill-posed backward: reversing diffusion amplifies noise catastrophically (failing Hadamard's conditions for well-posedness; see [5]). Hence S_t forms a monoid $[0, \infty)$ rather than a group—time evolution is irreversible.

Note that it isn't "obvious" what type of space $\mathscr{P}(X)$ is, we will assume that this is a Hilbert space; see [5] for the justification of this assumption. At minimum it is certainly the case that any (smooth) L^2-integrable function is a valid starting state that can then evolve according to the heat equation. If the reader can accept that there is a reasonable sense in which integrable functions can be differentiated as shall be demonstrated in chapter 9, the reader can let $\mathscr{P}(X) = L^2([0,1])$.

The choice of L^2 is not arbitrary. As we saw in the preceding section, L^2 is the unique L^p space where the duality map J_p is linear, the geometry is flat, and vectors and covectors are canonically identified. The Fourier transform exploits this: it is an *isometry* of L^2 (Plancherel's theorem), which is possible precisely because the "rotation" from position coordinates to frequency coordinates preserves the inner product. On L^p for $p \neq 2$, the Fourier transform does not stay within the same space; instead, the Hausdorff-Young inequality gives

$$\|\hat{f}\|_{L^q} \leq C\|f\|_{L^p}, \qquad \tfrac{1}{p} + \tfrac{1}{q} = 1, \quad 1 \leq p \leq 2$$

mapping L^p into its dual L^q—reflecting the nonlinear duality of those spaces. Only at $p = 2$ does the transform remain an endomorphism, because only there is $V \cong V^*$ linearly.

Now, certainly S_t is linear as the heat equation is linear and homogeneous[4]. We can thus ask if it has eigenvectors, that is functions $f_r \in \mathscr{P}(X)$ where

$$S_t f_r(x) = \lambda_t f_r(x)$$

If we let t vary, we can define a function $\lambda(t)$ dependent on the time. Given a compact domain $X = [a, b]$, by direct computations we can check that $\sin(kx)$ is an eigenvector, where this is indeed a state as it is a t-slice of:

$$u(x,t) = e^{-k^2 t}\sin(kx)$$

For example at $t = 0$:

$$S_t(\sin(kx)) = e^{-k^2 t}\sin(kx)$$

With the assumption that $\mathscr{P}$ is a Hilbert space, if these eigenfunctions form an orthonormal basis, we can decompose a function into a "weighted sum" of these well-behaved functions with respect to time. For the rest of this section we shall take for granted $\{\sin(kx)\}_{k\in\mathbb{N}}$ form an orthonormal basis for S_t (proven in theorem 14.2.4) after appropriate normalization depending on the domain.

It must be pointed out that the same argument works for $\cos(kx)$:

$$S_t(\cos(kx)) = e^{-k^2 t}\cos(kx)$$

and these *also* form an orthonormal basis that *also* diagonalizes S_t. These two sets are individually complete orthogonal bases for $L^2([0,1])$ (with appropriate boundary conditions), so their union is *overcomplete*—it contains "twice as many" basis elements as needed. Moreover, the combined set is not even orthogonal:

$$\int_0^1 \sin(\pi x)\cos(2\pi x)dx = \frac{-2}{3\pi} \neq 0$$

More fundamentally, as we shall see, neither set alone diagonalizes the *translation* operator—they only block-diagonalize it—and the passage to complex exponentials will be forced by the requirement of full diagonalization.

[4] It is also continuous and normal, but for that we would need to choose whether we consider $\mathscr{P}$ as a Banach space or a distribution space; intuitively the heat equation dissipates energy so $\|S_t f\| \leq \|f\|$

Thus, the question shifts into one about finding a *natural* eigenbasis for S_t. A-priori, it is not necessarily the case that these are the only eigenbases. What we need is for the eigenbasis to "respect" some inherent geometry of the space. Phrased differently, the eigenbasis needs to be *invariant* with respect to some operator(s), where the operators represent some symmetry of the space. To get a better grasp of invariance of operators, we may start looking at other PDEs and considering diagonalizing the operator S_t with respect to the solutions of those PDEs.

For example, if we solve the Heat equation on a sphere:

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

the operator can still be diagonalized (see [5]). What we shall want is for an eigenbasis that respects the spherical geometry, namely for it to be invariant to an $SO(3)$ action. Such eigenfunctions do exist, they are the *Spherical Harmonics* $Y_l^m(\theta, \varphi)$, and they are invariant to rotation[5]

$$\Delta(f \circ R) = (\Delta f) \circ R$$

To give another example that is famous in physics, consider the Quantum Harmonic Oscillator (QHO):

$$i\frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2 \psi$$

This PDE is diagonalizable (see [5]), but it is *not* diagonalizable by the Fourier basis e^{ikx}. The reason, in the language of our framework, is that the potential $V(x) = x^2$ *breaks translation invariance*: since $x^2 \neq (x+a)^2$, the QHO Hamiltonian $H = -\partial_x^2 + x^2$ does not commute with the translation operator τ_a. By the general principle we are developing—different symmetries produce different eigenbases—the translation eigenbasis cannot diagonalize the QHO.

Instead it was shown to be diagonalized (uniquely) by the *Hermite Functions* (a product of the Hermite polynomials H_n with a Gaussian):

$$\varphi_n(x) = H_n(x) e^{-x^2/2}$$

The relevant symmetry is the *Heisenberg-Weyl algebra* generated by the ladder operators $a = \frac{1}{\sqrt{2}}(x + \partial_x)$ and $a^\dagger = \frac{1}{\sqrt{2}}(x - \partial_x)$, whose eigenbasis is precisely the Hermite functions.

A revealing computation shows *why* the Fourier basis fails here. If we attempt to apply the Fourier transform, the $-\partial_x^2$ term becomes multiplication by k^2 in frequency space, but the x^2 multiplication term becomes a Laplacian $-\partial_k^2$ in k-space. The result is the *identical* PDE in frequency space:

$$i\frac{\partial \hat{\psi}}{\partial t} = k^2 \hat{\psi} - \frac{\partial^2 \hat{\psi}}{\partial k^2}$$

The Fourier transform swaps the PDE for a copy of itself without solving it. In physics, this reflects the fact that the QHO Hamiltonian is *invariant* under the Fourier transform—it treats position and momentum symmetrically—creating a fascinating link between abstract harmonic analysis and the concrete geometry of quantum mechanics, a connection to be explored in another book.

[5] Important technicality: $SO(3)$ is not abelian, hence it cannot be that *all* rotation operators simultaneously commute. The best solution is to pick a "maximal set" of commuting symmetries: usually the Total Angular Momentum (L^2, representing the amount of rotation invariance) and Azimuthal Symmetry (L_z, rotation around the z-axis). This choice of a maximal commuting family is the analytic form of selecting a *maximal torus* (Cartan subalgebra) in the Lie algebra $\mathfrak{so}(3)$.

So what is the invariance we want to impose for the operator S_t with respect to the heat equation? In the example of the sphere, we had the group $SO(3)$. A compact interval has no inherent invariance simply because it is not a group. However, by gluing the end points of the interval we get a 1-*torus* (a circle)[6]. We now have the rotation group operation on $\mathbb{T}^1$, or equivalently the translation operation on $\mathbb{R}/\mathbb{Z}$[7]. This does shift the geometry and hence the inherent dynamics of our PDEs; the problem of solving a PDE on a compact interval will be a subset of solving PDEs on a torus as shall be addressed shortly. Let us first simply consider S_t on $L^2(\mathbb{T}^1)$ and see how it interacts with the translation operator. Since we are working on the torus, we now have that $\sin(2\pi kx)$ and $\cos(2\pi lx)$ *are* orthogonal (where the 2π-scaling was to ensure a full period of each lies within the torus), and $\{\sin(kx), \cos(lx)\}_{k,l\in\mathbb{N}}$ is an orthonormal basis (after appropriate scaling and stretching). Now, note that sin and cos bases are *not* translation invariant, for example:

$$\tau_a(\sin(kx)) = \sin(k(x+a)) = \sin(kx)\cos(ka) + \cos(kx)\sin(ka)$$

and so they are not eigenvectors of τ_a. However, they still *block-diagonalize* the translation operator τ_a with blocks of size 2:

$$\begin{pmatrix} \cos(ka) & \sin(ka) \\ -\sin(ka) & \cos(ka) \end{pmatrix}$$

If we look at these blocks, we shall see that they are all rotation matrices by an angle ka; in other words the imposition is that we did not have enough eigenvalues. By extending the codomain to $\mathbb{C}$, we can readily verify that:

$$\tau_a(e^{iz}) = e^{ia} \cdot e^{iz}$$

and so $\{e^{ikz}\}_{k\in\mathbb{Z}}$ is our desired eigenbasis for τ_a. Now, the question remains whether translations of these functions commute with S_t, namely:

$$S_t(\tau_a(f)) = \tau_a(S_t(f)) \qquad \forall a \in \mathbb{T}^1$$

and indeed:

$$\begin{aligned}
\tau_a(S_t(e^{ikx})) &= \tau_a(e^{-k^2 t} e^{ikx}) \\
&= e^{-k^2 t} e^{ik(x+a)} \\
&= e^{-k^2 t} e^{ika} e^{ikx} \\
&= e^{ika} S_t(e^{ikx}) \\
&= S_t(e^{ika} e^{ikx}) && S_t \text{ is linear} \\
&= S_t(\tau_a(e^{ikx}))
\end{aligned}$$

What this means physically is that the heat equation dissipates the same way no matter where you are on a ring-shaped rod. If the rod was half wood and half metal, then this translation invariance shall fail.

This principle is the spectral-theoretic form of *Noether's theorem*: continuous symmetries of a physical system correspond to conservation laws. Translation invariance corresponds to conservation of *momentum*, and the Fourier decomposition $f = \sum \hat{f}(k) e^{ikx}$ is precisely the decomposition into

[6] More generally, any rectangular domain is homeomorphic to $[0,1]^n$ which then can be glued to become $\mathbb{T}^n = (S^1)^n$, and so this same translation structure appears in higher dimensions

[7] Really $\mathbb{C}/\mathbb{Z}$ as we shall justify shortly

states of definite momentum k. Each Fourier mode evolves independently—its amplitude changes but its frequency is preserved—because momentum is conserved by any translation-invariant dynamics.

Now, notice that as $\tau_a \circ S_t = S_t \circ \tau_a$ for all $a \in \mathbb{T}^1$ (or in commutator bracket notation $[\tau_a, S_t] = 0$ for all a, t), the eigenvectors of τ_a are the same eigenvectors as S_t. The eigenvectors of S_t are in general hard to compute, but the eigenvectors of τ_a are *precisely* the functions e^{ikx}. In fact, any (linear) operator L where $\tau_a \circ L = L \circ \tau_a$ will have the same eigenvectors. Thus, we are actually finding eigenvectors of the translation operators τ_a and then finding transformations that commute with translation. Note that any Linear Differential Operator with Constant Coefficients

$$L = a_n \frac{d^n}{dx^n} + \cdots + a_1 \frac{d}{dx} + a_0$$

and any convolution operator:

$$L(f) = f * g = \int f(y)g(x-y)dy$$

will commute with translation. This is also why the Heat Equation, Wave Equation, and Schrödinger Equation can all be solved (or transformed) using the Fourier transform.

From Local to Global: The Exponential Map

We have established that the Fourier basis functions e^{ikx} are the eigenfunctions of the "global" spatial translation operator τ_a. However, the reader likely recognizes these functions primarily as the solutions to the differential equation $f' = \lambda f$. This connection between the *global* translation τ_a and the *local* derivative $D = \frac{d}{dx}$ is a central theme of Lie Theory, and it provides the structural explanation for *why* exponentials are the natural Fourier basis.

In chapter 9, it shall be shown that it makes sense to infinitely differentiate integrable functions. For now, we rely on the intuition that any continuous function can be approximated by a smooth function (and locally by a polynomial).

Consider the Taylor expansion of a function f around a point x:

$$f(z) = f(x) + f'(x)(z-x) + \frac{f''(x)}{2!}(z-x)^2 + \ldots$$

If we wish to evaluate f at a shifted point $x + y$, the Taylor series gives:

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{2!}f''(x) + \ldots$$

Notice that we can re-write this using the differentiation operator $D = \frac{d}{dx}$:

$$f(x+y) = \left(I + yD + \frac{y^2 D^2}{2!} + \ldots \right) f(x)$$

Recognizing the series inside the parenthesis as the definition of the matrix exponential, we arrive at the compact operator identity:

$$\tau_y = e^{yD}$$

This equation states that *differentiation generates translation.*

Now, consider the spectral consequences. If an operator A has an eigenvector v with eigenvalue λ ($Av = \lambda v$), then the operator exponential e^A acts on v as:

$$e^A v = \left(I + A + \frac{A^2}{2!} + \dots\right) v = \left(1 + \lambda + \frac{\lambda^2}{2!} + \dots\right) v = e^\lambda v$$

Applying this to our spatial operators: since $f(x) = e^{ikx}$ is an eigenfunction of the derivative D with eigenvalue ik (i.e., $De^{ikx} = ike^{ikx}$), it must automatically be an eigenfunction of the translation operator $\tau_y = e^{yD}$ with eigenvalue e^{iyk}.

This confirms our earlier finding: the Fourier basis e^{ikx} simultaneously diagonalizes the derivative (and by extension the Laplacian $\Delta = D^2$) and the translation group.

Theorem 14.2.1: Eigenvector Mapping Lemma

Any eigenvector with eigenvalue λ of an operator A is an eigenvector of the operator e^A with eigenvalue e^λ.

***Proof*:**
Follows from the power series definition of the matrix exponential shown above.

Remark. The full *Spectral Mapping Theorem* is the stronger statement $\sigma(e^A) = e^{\sigma(A)}$ for the entire spectrum, not just eigenvalues. The eigenvector case stated above is the special case sufficient for our purposes.

To see this explicitly, observe the action of these operators on the basis function $f_k(x) = e^{ikx}$:

- *Derivative:* $Df_k = (ik)f_k$ (Eigenvalue ik)
- *Laplacian:* $\Delta f_k = D^2 f_k = (ik)^2 f_k = -k^2 f_k$ (Eigenvalue $-k^2$)
- *Translation:* $\tau_a f_k = e^{aD} f_k = e^{a(ik)} f_k$ (Eigenvalue e^{ika})

In the language of representation theory, the differentiation operator is the *Lie Algebra* generator $\mathfrak{g}$, and the translation operator is the *Lie Group* element $e^{\mathfrak{g}}$. The Fourier basis provides the unique coordinate system where the group action and its generator are both simple scalings.

The Fourier Variable as a Cotangent Coordinate. The eigenvalue list above reveals a deeper geometric structure that connects directly to the duality theory presented in chapter 11. Observe the pattern: the derivative $D = \frac{d}{dx}$ is a *tangent* operation (it acts on the spatial/position variable x), yet its eigenvalue ik is a scalar that labels the *frequency* k. In physics, x is position and k is momentum—and these are canonically dual variables, related as tangent and cotangent coordinates.

The Fourier transform makes this duality explicit. Under the transform:

- The derivative $\frac{d}{dx}$ (a tangent operation) becomes *multiplication by ik* (a cotangent scalar).
- Multiplication by x (a tangent/position operation) becomes $-i\frac{d}{dk}$ (a cotangent derivative).

The Fourier transform *exchanges tangent and cotangent roles.* In the language of symplectic geometry, it is a canonical transformation swapping position and momentum coordinates on the phase space $T^*\mathbb{R}$. Where the duality map J_p from chapter 11 provides a *pointwise, infinitesimal* passage from tangent to cotangent data (via the gradient of the energy), the Fourier transform achieves this passage *globally* via integration against the characters e^{ikx}.

This perspective also explains the QHO's Fourier invariance noted above. The QHO Hamiltonian $H = -\partial_x^2 + x^2$ maps under the Fourier transform to $k^2 - \partial_k^2 = -\partial_k^2 + k^2$: the identical operator with x replaced by k. This is precisely because H treats position and momentum *symmetrically*—it sits on the diagonal of the tangent-cotangent duality, which is why the Fourier transform (which swaps the two) leaves it invariant.

Boundary Problem

Having understood why the Fourier basis is structurally natural, we now examine two important practical consequences: how boundary conditions on an interval arise from the symmetry of the torus, and how the Fourier transform decouples partial differential equations.

The immediate point that must be clarified when switching to $\mathbb{T}^1$ is that this shifts the dynamics of any PDE we are solving, as now points at the boundary are affected by their neighbourhood. However, for PDEs that will respect the group structure of $\mathbb{T}^1$, the effects at the boundary can be "cancelled out" in two separate ways that shall recover the original two bases $\{\sin(kx)\}_{k\in\mathbb{Z}}$ and $\{\cos(kx)\}_{k\in\mathbb{Z}}$ we saw before (again, appropriately normalized). The key insight is that the interval $[0,1]$ can be viewed as a *sub-domain* of the torus $\mathbb{T}^1$ (identified with $[-1,1]$) subject to a symmetry constraint. While the torus possesses full translation invariance, the interval possesses *reflection invariance.* By imposing a symmetry on the torus, we create "virtual boundaries" at the fixed points of the reflection ($x = 0$ and $x = 1$). This gives rise to two distinct orthonormal bases for $L^2([0,1])$, each corresponding to a different physical boundary condition:

- *Odd Symmetry (Dirichlet Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *odd*, satisfying $u(-x) = -u(x)$. At the fixed point $x = 0$, this symmetry forces $u(0) = -u(0)$, implying $u(0) = 0$. Physically, the heat flowing from the right is perfectly cancelled by the "negative heat" flowing from the left, creating a virtual cold wall. The eigenfunctions respecting this symmetry are precisely the sine functions:

$$\sin(n\pi x) = \frac{e^{in\pi x} - e^{-in\pi x}}{2i}$$

  Restricting these functions to $[0,1]$ yields the *Sine Basis*, which diagonalizes the Laplacian subject to $u(0) = u(1) = 0$.

- *Even Symmetry (Neumann Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *even*, satisfying $u(-x) = u(x)$. At the fixed point, the derivative must be odd, forcing $u'(0) = 0$. Physically, this represents an insulated boundary where heat reflects back without loss. The eigenfunctions respecting this symmetry are the cosine functions:

$$\cos(n\pi x) = \frac{e^{in\pi x} + e^{-in\pi x}}{2}$$

  Restricting these to $[0,1]$ yields the *Cosine Basis*, which diagonalizes the Laplacian subject to $u'(0) = u'(1) = 0$.

This leads to a result in Harmonic Analysis known as *half-range expansions*: while sines and cosines *together* form a basis for the full torus, *either* set alone forms a complete orthonormal basis for the interval $[0, 1]$.

Decoupling PDE

To understand the consequences better, take a solution $u : \mathbb{T}^1 \times [0, \infty) \to \mathbb{C}$. Then, think of each (space) neighborhood in the usual heat equation as influencing its infinitesimal neighbourhood, namely the $\frac{\partial^2 u}{\partial x^2}$ term determines how each neighbouring point affects each other as time propagates. We can think of this as being uncountably many dependent equations.

Let us now change the "basis" of our state space from the position basis $\{\delta_x\}_{x \in \mathbb{T}^1}$ to the eigenbasis of the spatial operator $\{e^{ikx}\}_{k \in \mathbb{Z}}$. By taking the Fourier transform, i.e. transforming to the translation-invariant orthonormal basis, we change a solution to:

$$u(x,t) : \mathbb{T}^1 \times [0, \infty) \to \mathbb{C} \qquad \xmapsto{\mathcal{F}_x} \qquad \hat{u}(k,t) : \mathbb{Z} \times [0, \infty) \to \mathbb{C}$$

where the $\mathbb{Z}$ is the index for the eigenbasis $\{f_k\}_{k \in \mathbb{Z}}$. Then if $\hat{u}_k$ is the function whose input is the index of the eigenvector and output is its coefficient, we see that as $\hat{u}_k$ propagates through time, $\hat{u}_k(t)$ is only dependent on the original function up to scaling. In fact we shall show that the heat equation PDE decomposes into the equations:

$$\frac{d\hat{u}_k}{dt} = -k^2 \hat{u}_k \qquad \forall k \in \mathbb{Z}$$

where now we have *countably many decoupled* equations.

Non-Compact Domain

Notice that $\sin, \cos, e^z$ are not L^2-integrable on $\mathbb{R}^n$ (resp. $\mathbb{C}^n$), hence these cannot form an orthonormal basis on $L^2(\mathbb{R}^n)$. Thus, the initial foray into Fourier will focus on functions with domain $\mathbb{T}^n$. In section 14.2.2 we shall show how to generalize the tools to the non-compact case of Euclidean space $\mathbb{R}^n$; the passage to a general locally compact abelian group is the subject of part IV.

The structural reason for this shift is illuminated by *Pontryagin duality* (section 16.2). The characters of a group G themselves form a group $\hat{G}$ (the *Pontryagin dual*) under pointwise multiplication, and the Fourier transform is a map $L^2(G) \to L^2(\hat{G})$. For the compact group $G = \mathbb{T}^1$, the dual group is discrete: $\hat{G} = \mathbb{Z}$ (indexed by frequency k), and the Fourier transform produces a *discrete* sequence of coefficients $\hat{f}(k)$. Conversely, for the discrete group $G = \mathbb{Z}$, the dual is compact: $\hat{G} = \mathbb{T}^1$. And remarkably, $G = \mathbb{R}$ is *self-dual*: $\hat{\mathbb{R}} \cong \mathbb{R}$, so the Fourier transform maps functions of a continuous variable to functions of a continuous variable, replacing the discrete sum $\sum \hat{f}(k)e^{ikx}$ with the continuous integral $\int \hat{f}(\xi)e^{2\pi i \xi x}\, d\xi$. Pontryagin duality ($\hat{\hat{G}} \cong G$) is then the abstract statement that the Fourier transform is involutive.

Generalizing using Representation Theory

In the preceding discussion, our motivation was to solve a specific time-dependent PDE (the heat equation). We discovered that the key to solving it lay not in the time evolution S_t itself, but in the underlying symmetry of the spatial domain (translation invariance). The basis functions e^{ikx} worked because they diagonalized the translation operator τ_a, which we then understood via the exponential map $\tau_a = e^{aD}$ as the passage from the Lie algebra (differentiation) to the Lie group (translation).

We now abstract this principle. We no longer need a specific "time" equation; instead, we look for the basis that diagonalizes the action of the symmetry group itself. By finding the irreducible representations of the symmetry group, we construct a "universal" eigenbasis that will automatically diagonalize *any* linear operator respecting that symmetry (be it the Laplacian, the Heat Operator, or others).

To make this concrete, let us recall the representation theory of finite groups. If G is a finite group, we can represent G on a vector space V (over a field k) by taking a homomorphism $\rho : G \to \mathrm{GL}(V)$.

$$G \curvearrowright V \qquad \rho(g)v = g \cdot v$$

As we are not interested in any arithmetic limitations, we take $k = \mathbb{C}$. Then by Maschke's theorem (see [14, chapter 24]) the induced $\mathbb{C}[G]$-module structure on V is semi-simple, meaning:

$$V \cong_G \bigoplus_i V_i$$

where V_i are irreducible representations. Given the basis for the above isomorphism, any $g \in G$ will give a matrix representation $\rho(g)$ which respects this block-diagonal decomposition:

$$\rho(g) = \begin{bmatrix} \mathbf{M}_1(g) & 0 & 0 & 0 \\ 0 & \mathbf{M}_2(g) & 0 & 0 \\ 0 & 0 & \mathbf{M}_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

where $\mathbf{M}_i(g)$ represents the matrix block acting on the subspace V_i. Crucially, if G is *abelian*, Schur's Lemma implies that *all* irreducible representations are 1-dimensional. Hence, the matrices become scalars:

$$\rho(g) = \begin{bmatrix} \lambda_1(g) & 0 & 0 & 0 \\ 0 & \lambda_2(g) & 0 & 0 \\ 0 & 0 & \lambda_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

Since G is abelian, the operators $\{\rho(g)\}_{g \in G}$ commute, and the construction above shows they are *simultaneously diagonalizable* by a single basis.

Now, we generalize this to function spaces. While for a finite group the vector space of functions $\mathbb{C}[G]$ is finite-dimensional, equipping it with an inner product structure is essential to define unitarity and orthogonality. A finite group G carries a natural unique measure (the counting measure), which we can normalize to be a probability measure (the Haar measure $\mu(g) = 1/|G|$). This allows us to define the Hilbert space $L^2(G)$ with inner product:

$$\langle f, h \rangle = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{h(x)}$$

We define the *Right Regular Representation*[8] of G on this space:

$$G \curvearrowright L^2(G) \qquad (R_g \cdot f)(x) = f(xg)$$

[8] or, Left Regular Representations $(L_g \cdot f)(x) = f(g^{-1}x)$

We verify that this action is unitary (and hence can be diagonalized with an orthonormal basis). Since right-multiplication by g is a bijection on the group G (it permutes the elements), the sum over x is identical to the sum over $y = xg$:

$$\begin{aligned}\langle R_g f, R_g h\rangle &= \frac{1}{|G|}\sum_{x\in G} f(xg)\overline{h(xg)}\\ &= \frac{1}{|G|}\sum_{y\in G} f(y)\overline{h(y)}\\ &= \langle f, h\rangle\end{aligned}$$

Thus, the operators R_g are unitary. Since G is abelian, the operators $\{R_g\}_{g\in G}$ form a commuting family of unitary operators. By the spectral theorem for unitary operators, there exists an orthonormal basis $\mathcal{B}$ of $L^2(G)$ that simultaneously diagonalizes the action. If $\varphi \in \mathcal{B}$ is such a basis vector, then:

$$R_g(\varphi) = \chi_\varphi(g)\varphi$$

where $\chi_\varphi(g) \in \mathbb{C}$ is the eigenvalue. Since the representation is a homomorphism ($R_{gh} = R_g R_h$), the eigenvalues must satisfy the multiplicative property $\chi_\varphi(gh) = \chi_\varphi(g)\chi_\varphi(h)$. Moreover, since R_g is unitary, $|\chi_\varphi(g)| = 1$ and hence the codomain is S^1. These homomorphisms $\chi : G \to S^1$ are precisely the *characters* of the group. Thus,

> the group-theoretic eigenbasis for $L^2(G)$ is simply the set of characters!

We shall explore this more in section 16.2.

14.2.2 Advanced: Operators Commuting with Translations We focused on finding the basis that diagonalizes translations (R_g). But what are the operators that commute with all translations?

$$T \circ R_g = R_g \circ T \qquad \forall g \in G$$

A fundamental result in Harmonic Analysis states that any such linear operator T is a *Convolution Operator*. That is, there exists a function $k \in L^1(G)$ (the kernel) such that $Tf = f * k$.
The deeper algebraic structure is this: the algebra of all translation-invariant operators on $L^2(G)$ is isomorphic to the *group algebra* $L^1(G)$ acting by convolution. The Fourier transform simultaneously diagonalizes this entire algebra—it is the *Gelfand transform* of the commutative Banach algebra $L^1(G)$, mapping the group algebra to its spectrum (the space of characters). This is the structural reason why the Fourier transform converts convolution to pointwise multiplication: it maps the group algebra to its maximal ideal space, where every element acts as a scalar.
This explains why the Heat Operator, the Laplacian, and other physical operators can all be written as convolutions: they are simply the most general linear maps that respect the symmetry of the space.

The Infinite Compact Case

In the case where the domain is infinite (like the torus $\mathbb{T}^n$), we must be more careful. First, let us stick to compact domains so that we generalize the finite case directly. Functions defined on a rectangular domain with periodic boundary conditions are equivalent to functions on the torus $\mathbb{T}^n = (S^1)^n$. Our state space is the Hilbert space $L^2(\mathbb{T}^n, \mathbb{C})$.

We want to act by translation by $\mathbb{R}^n$, but since the domain is periodic, this descends to a faithful action of the compact group $\mathbb{R}^n/\mathbb{Z}^n \cong \mathbb{T}^n$. It is easy to check that translation is unitary ($\langle \tau_y f, \tau_y g\rangle = \langle f, g\rangle$) using the translation invariance of the Lebesgue measure.

We now invoke the *Peter-Weyl Theorem*, which is the generalization of Maschke's theorem to compact topological groups (see [8]). It states that the unitary representation of a compact group on $L^2(G)$ decomposes into a Hilbert direct sum of finite-dimensional irreducible representations:

$$L^2(G) \cong_G \widehat{\bigoplus_i} V_i$$

Since $\mathbb{T}^n$ is abelian, all irreducible representations V_i are *one-dimensional*. Being one-dimensional and unitary, they are given by characters $\chi : \mathbb{T}^n \to S^1$.

Thus, the Peter-Weyl theorem guarantees that the characters of $\mathbb{T}^n$ form a complete orthonormal basis for $L^2(\mathbb{T}^n)$. We have recovered our Fourier basis purely from the algebraic structure of the domain, without reference to any specific differential equation!

14.2.1 Fourier Transform on n-Torus

Finally, we look at the details and formally define the Fourier transform. Let us first formalize the eigenvectors of the translation operator:

Theorem 14.2.3: Eigenvector of Translation Functions

Let φ be a measurable function on $\mathbb{R}^n$ (resp. $\mathbb{T}^n$) such that $\varphi(x+y) = \varphi(x)\varphi(y)$ and $|\varphi(x)| = 1$. Then there exists a $\xi \in \mathbb{R}^n$ (resp. $\xi \in \mathbb{T}^n$) such that $\varphi(x) = e^{2\pi i \xi \cdot x}$ where $\xi \cdot x$ is the dot product.

***Proof*:**
We prove it for $\mathbb{R}$, the result naturally generalizing for $\mathbb{R}^n$ and automatically working for $\mathbb{T}^n$. Let $a \in \mathbb{R}$ such that $\int_0^a \varphi(t)dt \neq 0$. Such an a exists, or else by the Lebesgue Differentiation Theorem $\varphi = 0$ a.e. Setting $A = \left(\int_0^a \varphi(t)dt\right)^{-1}$, we get:

$$\varphi(x) = \varphi(x)AA^{-1} = A\int_0^a \varphi(x)\varphi(t)dx = A\int_0^a \varphi(x+t)dt = A\int_x^{x+a} \varphi(t)dt$$

Thus, φ must be continuous, being the integral of a locally integrable function. But that makes the function inside the integral continuous, and so φ is in fact C^1 (in fact C^∞, but we will not need this fact). Now, notice that:

$$\varphi'(x) = A[\varphi(x+a) - \varphi(x)] = B\varphi(x)$$

where $B = A(\varphi(a) - 1)$. This is a simple ODE, which can be solved to show we get an exponential (and is unique by the uniqueness of a solution to an ODE). If you don't know about ODEs, then notice that

$$\frac{d}{dx}(e^{-Bx}\varphi(x)) = 0$$

so $e^{-Bx}\varphi(x)$ is constant. Since $\varphi(0) = 1$, we have $\varphi(x) = e^{Bx}$, and since $|\varphi(x)| = 1$, B must be purely imaginary meaning $B = 2\pi i\xi$ for some $\xi \in \mathbb{R}$, completing the proof for $\mathbb{R}$. For $\mathbb{T}$, we would consider φ to be periodic with period 1 and $e^{2\pi i\xi} = 1$ if and only if $\xi \in \mathbb{Z}$

For the n-dimensional case, let $e_1, e_2, ..., e_n$ be the standard basis for $\mathbb{R}^n$ and define the coordiante functions $\varphi_i(t) = \varphi(te_i)$. Naturally, each φ_i satisifes the property $\varphi_i(t+s) = \varphi_i(t)\varphi_i(s)$ on $\mathbb{R}$, and so $\varphi_i(t) = e^{2\pi i\xi_i t}$, and so:

$$\varphi(x) = \varphi\left(\sum_1^n x_i e_i\right) = \prod_1^n \varphi_i(x_i) = e^{2\pi i\xi\cdot x}$$

completing the proof.

Hence, a function satisfying this translation invariance is of the form $f(x) = ce^{2\pi i\xi\cdot x}$ for some constant $c \in \mathbb{R}$. With the classification of these functions, we will want to use these to decompose reasonably arbitrary functions in $\mathbb{R}^n$ and $\mathbb{T}^n$ (i.e. we will show they form a dense set). In the case of L^2 with $\mathbb{T}^n$, we get a very precise result:

Theorem 14.2.4: Orthonormality Of Translation Functions

Let $E_k(x) = e^{2\pi ik\cdot x}$. Then the set $\{E_k \mid k \in \mathbb{Z}^n\}$ is an orthonormal basis for $L^2(\mathbb{T}^n)$

***Proof*:**
For orthonormality, notice that:

$$\int_0^1 e^{2\pi ikt}dt = \begin{cases} 1 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

using this, notice that if $n \neq m$, then:

$$\begin{aligned} \langle E_a, E_b\rangle &= \int_0^1 e^{2\pi iat}\overline{e^{2\pi ibt}}dt \\ &= \vdots \\ &= \frac{1}{2\pi i(a-b)}(e^{2\pi i(a-b)} - e^0) \\ &= \frac{1}{2\pi i(a-b)}(1-1) \\ &= 0 \end{aligned}$$

and if $n = m$, then we get

$$\int_0^1 e^{2\pi ikt}e^{2\pi i-kt}dt = \int_0^1 dt = 1$$

For density, since $E_xE_y = E_{x+y}$, the set of finte linear combinations of the E_k's form an algebra which clearly seperates points on $\mathbb{T}^n$, and $E_0 = 1$, $\overline{E_k} = E_{-k}$. Since $\mathbb{T}^n$ compact, by the Stone-Weiestrass theorem, this algebra is dense in $C(\mathbb{T}^n)$ in the uniform norm and hence in the L^2 norm, and $C(\mathbb{T}^n)$ is dense in $L^2(\mathbb{T}^n)$, and so $\{E_k\}$ is a orthonormal basis for $L^2(\mathbb{T}^n)$, as we sought to show.

Using this, we can now define the Fourier transform:

Definition 14.2.5: Fourier Transform And Fourier Series on Torus

Let $f \in L^2(\mathbb{T}^n)$, $\langle -, - \rangle$ be the induced inner product, and E_k the orthonormal basis from theorem 14.2.4. Then the *Fourier Transform* $\hat{f} : \mathbb{Z}^n \to \mathbb{C}$ is the function defined as:

$$\hat{f}(k) = \langle f, E_k \rangle = \int_{\mathbb{T}^n} f(x) e^{-2\pi i k \cdot x} dx$$

The map $f \mapsto \hat{f}$ is called the *Fourier transform*. The representation

$$f = \sum_{k \in \mathbb{Z}^n} \hat{f}(k) E_k$$

is called the *Fourier series* of f.

Example 14.2.6: Discrete Spectrum Examples (Torus)

1. *The Pure Tone (Finite Spectrum):* Let $f(x) = \sin(2\pi m x)$ for some integer m. Using Euler's identity:
$$f(x) = \frac{e^{2\pi i m x} - e^{-2\pi i m x}}{2i}$$
The Fourier transform is a pair of spikes: $\hat{f}(m) = \frac{1}{2i}$, $\hat{f}(-m) = -\frac{1}{2i}$, and $\hat{f}(k) = 0$ everywhere else. A smooth, oscillating function on the torus has a *finite* number of non-zero coefficients.

2. *The Square Wave (Algebraic Decay):* Consider the periodic function equal to $+1$ on $[0, 1/2)$ and -1 on $[1/2, 1)$.
$$\hat{f}(k) = \frac{1 - (-1)^k}{\pi i k} \quad \text{for } k \neq 0$$
This gives non-zero values only for odd k, decaying as $1/k$. Discontinuities in the function lead to coefficients that decay slowly (algebraically). This slow decay causes the partial sums to oscillate violently near the jump (Gibbs Phenomenon).

3. *The Periodized Gaussian (Jacobi Theta Function):* Instead of a single Gaussian (which doesn't live on a torus), we sum infinitely many copies:
$$f(x) = \sum_{n \in \mathbb{Z}} e^{-\pi (x-n)^2}$$
Remarkably, the Fourier series coefficients are sampled from the Gaussian itself:
$$\hat{f}(k) = e^{-\pi k^2}$$
This connects local geometry to global geometry. Even though the function wraps around the circle, its coefficients still decay super-fast (exponentially), reflecting the smoothness of the function.

4. *The Dirac Comb (Flat Spectrum):* Let $f = \delta_0$ be the Dirac delta distribution at the origin (periodized).
$$\hat{f}(k) = \int_{\mathbb{T}} \delta_0(x) e^{-2\pi i k x} dx = 1$$
Extreme localization in space (a single point) leads to extreme delocalization in frequency (all frequencies contribute equally).

By the definition of an orthonormal basis, the Fourier series of f converges to f in the L^2-norm. In the next sections, we will address point-wise convergence. As this is simply a conversion of basis:

Theorem 14.2.7: Parseval's Identity On $\mathbb{T}^n$

$$\|\hat{f}\|_2 = \|f\|_2$$

***Proof*:**
Theorem 14.2.4 shows that $L^2(\mathbb{T}^n)$ surjects onto $\ell^2(\mathbb{Z}^n)$ and forms an orthonormal basis, hence by theorem 11.1.15
$$\|\hat{f}\|_2 = \|f\|_2$$

Let us take some time concretely interpreting $\hat{f}(k)$. First, $\hat{f}(0)$ represents the "average total energy" in the system:
$$\hat{f}(0) = \int_{\mathbb{T}^n} f(x) dx$$
where as $m(\mathbb{T}^n) = 1$ the normalization is hidden. If we visualized this, we can think of it as the center of mass of the graph of a compact function wrapped around the circle. Then $\hat{f}(n)$ for $n \in \mathbb{Z}^k$ is that same graph wrapped around in each cardinal direction n_i number of times, and then taking its average:
$$\hat{f}(n) = \int_{\mathbb{T}^n} f(x) e^{2\pi i x \cdot n} dx$$

Now, let's say that instead of trying to take the formal approach we've done, we motivate the coefficient formula geometrically. Consider the inner product as a projection that will project onto one of the axis given by one of the terms in the series:
$$f = \cdots + c_{-1} e^{-1 \cdot 2\pi i t} + c_0 e^{0 \cdot 2\pi i t} + c_1 e^{1 \cdot 2\pi i t} + c_2 e^{2 \cdot 2\pi i t} + \cdots$$
How do we find the coefficients? First, by the DCT:
$$\begin{aligned} \int f(t) dt &= \int \left(\cdots + c_{-1} e^{-1 \cdot 2\pi i t} + c_0 e^{0 \cdot 2\pi i t} + c_1 e^{1 \cdot 2\pi i t} + c_2 e^{2 \cdot 2\pi i t} \right) dt \\ &= \cdots + \int c_{-1} e^{-1 \cdot 2\pi i t} dt + \int c_0 e^{0 \cdot 2\pi i t} dt + \int c_1 e^{1 \cdot 2\pi i t} dt + \int c_2 e^{2 \cdot 2\pi i t} dt + \cdots \end{aligned}$$

Now, notice that all terms except the one containing c_0 are disappear, since the rotation of by the exponential will always guarantee to make the integrals average to zero! Thus, c_0 can be thought of as the average mass point of f. To find the other terms, we simply have to multiply f at the beginning by $e^{k2\pi i t}$
$$\int f(t) e^{-k2\pi i t} dt$$

and then we have shifted over all of our terms except the kth term, which we can now find the integral of! In fact, notice we have just recovered:

$$\hat{f}(k) = \int f(t)e^{-k2\pi it}dt$$

giving us a geometric intuition for this formula!

Now, this gives the Fourier transform on $L^2(\mathbb{T}^n, \mathbb{C})$, but we often want to work with different L^p spaces. First, the definition of the Fourier transform $\hat{f}(k)$ makes sense if $f \in L^1(\mathbb{T}^n)$ since:

$$|\hat{f}(\xi)| = \left|\int f(x)e^{2\pi ix\cdot\xi}dx\right| \leq \int \left|f(x)e^{2\pi ix\cdot\xi}\right| dx = \int |f(x)| = \|f\|_1$$

Hence for each $\xi \in \mathbb{Z}^n$, if $f \in L^1(\mathbb{T}^n)$ then $|\hat{f}(\xi)| \leq \|f\|_1$ showing us that $\hat{f}$ must be essentially bounded:

$$\|\hat{f}\|_\infty$$

so the Fourier series would then extend to a norm-decreasing map from $L^1(\mathbb{T}^n)$ to $\ell^\infty(\mathbb{Z}^n)$. In particular, this shows us that there is a sort of duality between the speed of decay of a function f and its Fourier transform $\hat{f}$. The nature of this duality is made explicit in the following theorem:

Theorem 14.2.8: Hausdorff-Young Inequality over $\mathbb{T}^n$

Let $1 \leq p \leq 2$ and q be the conjugate exponent of p so that $\frac{1}{p} + \frac{1}{q} = 1$. If $f \in L^p(\mathbb{T}^n)$, then

$$\hat{f} \in \ell^q(\mathbb{Z}^n) \qquad \text{and} \qquad \|\hat{f}\|_q \leq \|f\|_p$$

***Proof*:**

Since $\|\hat{f}\|_\infty \leq \|f\|_1$ and $\|\hat{f}\|_2 = \|f\|_2$ for $f \in L^1$ and $f \in L^2$, this follows from the Riesz-Thorin interpolation theorem (theorem 8.5.2)

Example 14.2.9: Advanced: The Theta Function and Modularity

1. *The Heat Kernel Trace:* Recall example 14.2.6.3, where we periodized the Gaussian to get $K_t(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}e^{2\pi inx}$. If we evaluate this at the identity ($x = 0$), we obtain the classical *Theta Function* $\theta(it)$:

   $$\theta(it) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$$

   In the language of Number Theory, this is a modular form (specifically of weight 1/2). The fact that the Fourier transform of a Gaussian is a Gaussian translates directly into the *functional equation* of the Theta function via the Poisson Summation Formula:

   $$\theta(t^{-1}) = \sqrt{t}\theta(t)$$

   This identity is the "analytic engine" that proves the analytic continuation of the Riemann Zeta function $\zeta(s)$.

2. *Gauss Sums (Arithmetic Fourier Transform):* If we replace our field $\mathbb{C}$ with a finite field $\mathbb{F}_p$ and our torus with the additive group of the field, the Fourier Transform becomes the *Gauss*

Sum:
$$g(a) = \sum_{x \in \mathbb{F}_p} \chi(x) e^{2\pi i a x/p}$$

Just as $\int |f|^2 = \int |\hat{f}|^2$ (Plancherel), we have $|g(a)| = \sqrt{p}$. The "square root cancellation" of Fourier coefficients in analytic number theory is the discrete analogue of the decay rates we study in analysis.

14.2.2 Fourier Transform on Euclidean Space

We often want to work with $L^2(\mathbb{R}^n)$. The problem is that $e^{2\pi i k}$ is *not* an element of $L^2(\mathbb{R}^n)$ as its integral is infinite. To generalize the domain to $\mathbb{R}^n$, we need to shift perspectives. Recall that all unitary representation of $\mathbb{T}^n$ are given by functions in:

$$\mathrm{Hom}(\mathbb{T}^n, S^1) \cong \mathbb{Z}^n$$

This is the *Pontryagin dual* of $\mathbb{T}^n$ and represents all the characters. We can generalize this and choose a locally compact (topological) groups and consider the Pontryagin dual $\mathrm{Hom}(G, S^1)$. As shall be shown in chapter 16, there shall always be a unique invariant Borel measure defined on any locally compact group up to scaling (ex. on $\mathbb{R}$ it is the Lebesgue measure and on $\mathbb{Z}$ it is the counting measure). It can be shown that $\mathrm{Hom}(G, S^1)$ is also a locally compact group and hence has a unique measure. When G is compact, $\mathrm{Hom}(G, S^1)$ is *discrete* and the natural Haar measure is the counting measure. If G is not compact but locally compact, $\mathrm{Hom}(G, S^1)$ is *continuous*, most importantly:

$$\mathrm{Hom}(\mathbb{R}^d, S^1) \cong \mathbb{R}^d \qquad \text{given by} \qquad (\xi \mapsto e^{2\pi i x \cdot \xi}) \mapsto x$$

Notice that if G is compact and abelian, $\mathrm{Hom}(G, S^1)$ represents all the characters. Hence if G is *locally compact*, then $\hat{G} = \mathrm{Hom}(G, S^1)$, can interpreted as the generalization of characters.

14.2.10 Advanced: The Notion of Points If you have seen Grothendieck's notion of the *Functor of Points*, then the "right" way to see points is not as elements of a set, but as *morphisms* from a test object. In algebraic geometry, a point of a scheme X is determined by the morphisms $T \to X$. In harmonic analysis, this philosophy manifests as *Tannaka-Krein Duality*. We can reconstruct a group G solely from its category of representations $\mathrm{Rep}(G)$. A "geometric point" $g \in G$ corresponds precisely to a tensor automorphism of the fiber functor (the forgetful functor to vector spaces). Thus, the Fourier Transform is simply the mechanism that translates between the geometric perspective (points as elements) and the spectral perspective (points as representations), proving they are dual aspects of the same underlying structure.

We can thus generalize the orthonormal basis interpretation of the Fourier transform to

$$\mathcal{F} : L^2(G, \mathbb{C}) \to L^2(\hat{G}, \mathbb{C})$$

Of course we have not proven it is well-defined, namely since $\hat{G}$ need not be an orthonormal basis anymore. For example if $f \in L^2(\mathbb{R}^d)$, it is not necessarily the case that $f \in L^1(\mathbb{R}^d)$ however as we've defined it so far we integrate against $e^{2\pi i x \cdot \xi}$:

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{2\pi i x \cdot \xi} dx$$

Simply take $(1+|x|)^{-1} \in L^2(\mathbb{R})$ which not in $L^1(\mathbb{R})$. However, $\mathcal{F}$ shall still be a well-defined map between L^2 spaces as shall be shown in upcoming sections. Note that in the case of $G = \mathbb{T}^n$ this is exactly what we had before:

$$\mathcal{F} : L^2(\mathbb{T}^n, \mathbb{C}) \to \ell^2(\mathbb{Z}^n, \mathbb{C})$$

There is a lot of details that are brushed off for now; this is all covered in chapter 16.

Now, if $f \in L^1$, $\hat{f}(\xi)$ will certainly be bounded (by $\|f\|_1$), and hence it is essentially bounded:

$$\|\mathcal{F}(f)\|_\infty < \infty$$

We thus have our first definition of Fourier transform on $\mathbb{R}^d$:

Definition 14.2.11: Fourier Transform on L^1

Let $f \in L^1(\mathbb{R}^n)$. Then the *Fourier Transform* $\hat{f} : \mathbb{R}^n \to \mathbb{C}$ is the function defined as:

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i \xi \cdot x} dx$$

The map $f \mapsto \hat{f}$ is called the *Fourier transform*.

As it turns out, $\mathcal{F}(f)$ is also continuous, hence it is not only essentially bounded (in L^∞) but *actually* bounded (in BC):

Theorem 14.2.12: Riemann-Lebesgue Lemma

$$\mathcal{F}(L^1(\mathbb{R}^n)) \subseteq \mathrm{BC}(\mathbb{R}^n)$$

***Proof*:**

Let $f \in L^1(\mathbb{R}^n)$ and consider $\{\xi_k\}_{k=1}^\infty \subset \mathbb{R}^n$ such that $\xi_k \to \xi$. It suffices to show $\lim_{k\to\infty} \hat{f}(\xi_k) = \hat{f}(\xi)$.

By the definition of the Fourier transform,

$$\hat{f}(\xi_k) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i \xi_k \cdot x}\, dx.$$

Let $F_k(x) = f(x)e^{-2\pi i \xi_k \cdot x}$. Since the complex exponential is continuous, we have pointwise convergence for every $x \in \mathbb{R}^n$:

$$\lim_{k\to\infty} F_k(x) = f(x) \lim_{k\to\infty} e^{-2\pi i \xi_k \cdot x} = f(x)e^{-2\pi i \xi \cdot x}.$$

Next, for all k and x, the integrand is bounded by the magnitude of f:

$$|F_k(x)| = \left|f(x)e^{-2\pi i \xi_k \cdot x}\right| = |f(x)| \cdot 1 = |f(x)|.$$

Since $f \in L^1(\mathbb{R}^n)$, $|f|$ dominates it. Thus, by the DCT:

$$\lim_{k\to\infty} \int_{\mathbb{R}^n} F_k(x)\, dx = \int_{\mathbb{R}^n} \lim_{k\to\infty} F_k(x)\, dx = \int_{\mathbb{R}^n} f(x)e^{-2\pi i \xi \cdot x}\, dx = \hat{f}(\xi).$$

showing $\hat{f}$ is continuous, as we sought to show.

Proposition 14.2.13: Properties Of Fourier Transform

Let $f, g \in L^1(\mathbb{R}^n)$. Then:

1. $\widehat{(\tau_y f)}(\xi) = e^{-2\pi i \xi \cdot y}\hat{f}(\xi)$ and $\tau_\eta(\hat{f}) = \widehat{h}$ where $h(x) = e^{2\pi i \eta \cdot x}f(x)$
2. If T is an invertible linear transformation on $\mathbb{R}^n$ and $S = (T^*)^{-1}$ is its inverse transpose, Then $\widehat{(f \circ T)} = |\det(T)|^{-1}\hat{f} \circ S$. In particular, if T is a rotation, then $\widehat{(f \circ T)} = \hat{f} \circ T$. Furthermore, if $T(x) = t^{-1}x$ (for $t > 0$), then $\widehat{(f \circ T)}(\xi) = t^n \hat{f}(t\xi)$.

   Note: This implies the Heisenberg Uncertainty Principle qualitatively: dilating (widening) f results in contracting (narrowing) $\hat{f}$.
3. Convolution becomes multiplication after the Fourier Transform:
$$\widehat{(f * g)} = \hat{f}\hat{g}$$
4. (*Conjugate Symmetry*) if f is real-valued, then $\hat{f}$ (which is still complex-valued) satisfies the Hermitian symmetry condition:
$$\hat{f}(-\xi) = \overline{\hat{f}(\xi)}$$
   This implies the information in the codomain is 'half' redundant.
5. If $x^\alpha f \in L^1$ for $|\alpha| \leq k$ then $\hat{f} \in C^k$ and
$$\partial^\alpha \hat{f}(\xi) = [\widehat{(-2\pi i x)^\alpha f}](\xi)$$
6. if $f \in C^k$, $\partial^\alpha f \in L^1$ for $|\alpha| \leq k$, and $\partial^\alpha f \in C_0$, for $|\alpha| \leq k-1$, then
$$\widehat{(\partial^\alpha f)}(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$$
7. upgraded Riemann-Lebesgue lemma:
$$\mathcal{F}(L^1(\mathbb{R}^n)) \subseteq C_0(\mathbb{R}^n)$$

***Proof*:**

1. Computing directly:
$$\widehat{(\tau_y f)}(\xi) = \int f(x-y)e^{-2\pi i \xi \cdot x}dx$$
   Let $z = x - y$, then $dx = dz$ and $x = z + y$:
$$= \int f(z)e^{-2\pi i \xi \cdot (z+y)}dz = e^{-2\pi i \xi \cdot y}\int f(z)e^{-2\pi i \xi \cdot z}dz = e^{-2\pi i \xi \cdot y}\hat{f}(\xi)$$
   For the second part, let $h(x) = e^{2\pi i \eta \cdot x}f(x)$.
$$\hat{h}(\xi) = \int e^{2\pi i \eta \cdot x}f(x)e^{-2\pi i \xi \cdot x}dx = \int f(x)e^{-2\pi i(\xi - \eta)\cdot x}dx = \hat{f}(\xi - \eta) = \tau_\eta(\hat{f})(\xi)$$

2. Apply theorem 2.6.4 (Change of Variables). Let $y = Tx$, then $x = T^{-1}y$ and $dx = |\det T|^{-1}dy$:

$$\begin{aligned}\widehat{(f \circ T)}(\xi) &= \int f(Tx)\exp(-2\pi i\xi \cdot x)dx \\ &= |\det T|^{-1}\int f(y)\exp(-2\pi i\xi \cdot T^{-1}y)dy\end{aligned}$$

Using the definition of the adjoint (transpose), $\xi \cdot T^{-1}y = (T^{-1})^*\xi \cdot y = S\xi \cdot y$:

$$\begin{aligned}&= |\det T|^{-1}\int f(y)\exp(-2\pi iS\xi \cdot y)dy \\ &= |\det T|^{-1}\hat{f}(S\xi)\end{aligned}$$

3. By Fubini's Theorem (justified since $f, g \in L^1$):

$$\begin{aligned}\widehat{(f * g)}(\xi) &= \int \left(\int f(x-y)g(y)dy\right) e^{-2\pi i\xi\cdot x}dx \\ &= \int g(y)\left(\int f(x-y)e^{-2\pi i\xi\cdot x}dx\right)dy\end{aligned}$$

Applying the translation property (proven in 1) to the inner integral where f is translated by y:

$$\begin{aligned}&= \int g(y)\left(e^{-2\pi i\xi\cdot y}\hat{f}(\xi)\right)dy \\ &= \hat{f}(\xi)\int g(y)e^{-2\pi i\xi\cdot y}dy \\ &= \hat{f}(\xi)\hat{g}(\xi)\end{aligned}$$

4. exercise.

5. By Differentiating under the integral sign (justified by the Dominated Convergence Theorem and the assumption $x^\alpha f \in L^1$):

$$\partial^\alpha\hat{f}(\xi) = \partial_\xi^\alpha\int f(x)e^{-2\pi i\xi\cdot x}dx = \int f(x)\partial_\xi^\alpha(e^{-2\pi i\xi\cdot x})dx = \int f(x)(-2\pi ix)^\alpha e^{-2\pi i\xi\cdot x}dx$$

6. Assuming first that $n = 1$ and $k = 1$. Since $f \in C_0$, f vanishes at boundaries. We integrate by parts:

$$\widehat{f'}(\xi) = \int_{-\infty}^{\infty} f'(x)e^{-2\pi i\xi x}dx$$

Let $u = e^{-2\pi i\xi x}$ and $dv = f'(x)dx$. Then $du = -2\pi i\xi e^{-2\pi i\xi x}dx$ and $v = f(x)$.

$$= \left[f(x)e^{-2\pi i\xi x}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x)(-2\pi i\xi)e^{-2\pi i\xi x}dx$$

The boundary term is 0 because $f \in C_0$.

$$= 2\pi i\xi \int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi x}dx = 2\pi i\xi \hat{f}(\xi)$$

For $n > 1$, we compute $\widehat{(\partial_j f)}$ by integrating by parts with respect to the j-th variable x_j. The boundary terms vanish because $f \in C_0(\mathbb{R}^n)$. The general case follows by induction on $|\alpha|$.

7. By the previous part, if $f \in C_c^1$ (Compactly supported C^1 functions), then $\widehat{\partial_j f}(\xi) = 2\pi i\xi_j \hat{f}(\xi)$. Since $\widehat{\partial_j f}$ is bounded (as the Fourier transform of an L^1 function), $|\xi_j \hat{f}(\xi)|$ is bounded. This implies $\hat{f}(\xi)$ decays as $|\xi| \to \infty$, so $\hat{f} \in C_0$.

   By a similar argument to proposition 8.6.8, C_c^1 is dense in L^1. Let $f \in L^1$ and $f_n \in C_c^1$ such that $f_n \to f$ in L^1. The Fourier transform is a bounded linear map from L^1 to L^∞ (specifically $\|\hat{g}\|_\infty \leq \|g\|_1$), so:

   $$\|\hat{f}_n - \hat{f}\|_\infty \leq \|f_n - f\|_1 \to 0$$

   Thus $\hat{f}_n \to \hat{f}$ uniformly. Since C_0 is closed under the uniform norm (a uniform limit of functions vanishing at infinity also vanishes at infinity), we conclude $\hat{f} \in C_0$.

Notice that (5) and (6) give an interesting relation between the smoothness of f and the decay of $\hat{f}$: the smoother f is, the faster $\hat{f}$ decays (and vice-versa).

Proposition 14.2.14: Regularity and Decay Duality

The smoothness of a function determines the decay of its Fourier transform, and the decay of a function determines the smoothness of its Fourier transform. In particular:

1. *Decay $\Rightarrow$ Smoothness:* If $(1+|x|)^k f(x) \in L^1(\mathbb{R}^n)$ for some integer $k \geq 1$, then $\hat{f} \in C^k(\mathbb{R}^n)$. Specifically, differentiation converts to multiplication by a polynomial in the spatial domain:

   $$\partial^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^n} (-2\pi i x)^\alpha f(x) e^{-2\pi i \xi \cdot x}\, dx$$

   for all multi-indices $|\alpha| \leq k$.

2. *Smoothness $\Rightarrow$ Decay:* If $f \in C^k(\mathbb{R}^n)$ and $\partial^\alpha f \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$ (and f vanishes at infinity), then $\hat{f}$ decays rapidly:

   $$|\hat{f}(\xi)| \leq \frac{C}{(1+|\xi|)^k}$$

   for some constant C.

***Proof*:**

1. **(Decay $\Rightarrow$ Smoothness):** Formally differentiating:

$$\frac{\partial}{\partial \xi_j}\hat{f}(\xi) = \frac{\partial}{\partial \xi_j}\int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}\,dx = \int_{\mathbb{R}^n} f(x)\frac{\partial}{\partial \xi_j}\left(e^{-2\pi i\xi\cdot x}\right)\,dx.$$

Calculating the derivative of the exponential gives $(-2\pi i x_j)e^{-2\pi i\xi\cdot x}$. Thus the integrand becomes $(-2\pi i x_j)f(x)e^{-2\pi i\xi\cdot x}$. To justify this operation rigorously, we need a dominating function. Observe that:

$$\left|(-2\pi i x_j)f(x)e^{-2\pi i\xi\cdot x}\right| = 2\pi|x_j||f(x)| \le 2\pi|x||f(x)|.$$

By our hypothesis, $(1+|x|)^k f \in L^1$, which implies $|x|f \in L^1$. Thus, the integrand is dominated by an integrable function, and the Dominated Convergence Theorem allows differentiation. Repeating this for higher derivatives up to order k yields the result.

2. **(Smoothness $\Rightarrow$ Decay):** We use the property that $\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$. Since $\partial^\alpha f \in L^1$, by the standard Riemann-Lebesgue Lemma (or simply the boundedness of the Fourier transform), we know that its transform is bounded:

$$\|\widehat{\partial^\alpha f}\|_\infty \le \|\partial^\alpha f\|_1 < \infty.$$

Substituting the identity, we have:

$$|(2\pi i\xi)^\alpha \hat{f}(\xi)| \le \|\partial^\alpha f\|_1.$$

This implies that for any multi-index α with $|\alpha| \le k$, the product $|\xi^\alpha \hat{f}(\xi)|$ is bounded. Since a function dominated by $1/|\xi|^k$ is equivalent to being bounded when multiplied by $|\xi|^k$, we conclude:

$$|\hat{f}(\xi)| \le C(1+|\xi|)^{-k}.$$

Corollary 14.2.15: Paley-Wiener Theorem

If $f \in C_c^\infty(\mathbb{R}^n)$ has compact support (i.e., there exists $R > 0$ such that $f(x) = 0$ for $|x| > R$), then its Fourier transform $\hat{f}(\xi)$ extends to a holomorphic (complex analytic) function on $\mathbb{C}^n$.

***Proof*:**

Let $z \in \mathbb{C}^n$. Define the complex extension of the Fourier transform as:

$$F(z) = \int_{|x|\le R} f(x)e^{-2\pi i x\cdot z}\,dx$$

Since f has compact support, the integration range is effectively bounded by the ball $B_R(0)$. To show $F(z)$ is holomorphic, it suffices to show it is complex differentiable with respect to each component z_j. We compute the partial derivative by differentiating under the integral sign:

$$\frac{\partial}{\partial z_j}F(z) = \int_{|x|\le R} f(x)\frac{\partial}{\partial z_j}\left(e^{-2\pi i x\cdot z}\right)\,dx = \int_{|x|\le R} f(x)(-2\pi i x_j)e^{-2\pi i x\cdot z}\,dx$$

To justify integral converges, let $z = \xi + i\eta$. The magnitude of the exponential term is:

$$\left|e^{-2\pi i x\cdot(\xi+i\eta)}\right| = \left|e^{-2\pi i x\cdot\xi}\right|\left|e^{2\pi x\cdot\eta}\right| = e^{2\pi x\cdot\eta}$$

Since the domain of integration is bounded ($|x| \le R$), this term is bounded by $e^{2\pi R|\eta|}$. Furthermore, f and x_j are bounded on this domain. Thus, the integrand is bounded for any fixed z, and the integral is well-defined.

Since the derivative exists at every point $z \in \mathbb{C}^n$, the function $F(z)$ is entire (holomorphic everywhere). Consequently, the Fourier transform $\hat{f}(\xi)$ is simply the restriction of this entire function to the real subspace $\mathbb{R}^n$, implying that $\hat{f}$ is real-analytic.

Notice that $C_c^\infty(\mathbb{R}^n)$ does not map into compact spaces, since an analytic function cannot be non-zero and bounded. The closest equivalent is Schwartz space:

Corollary 14.2.16: Fourier Transform On Schwartz Space

The map $\mathcal{F}$ maps the Schwartz space $\mathcal{S}$ into itself

***Proof*:**
If $f \in \mathcal{S}$, then $x^\alpha\partial^\beta f \in L^1 \cap C_0$ for all α, β, so by proposition 14.2.13, $\hat{f}$ is C^∞ and

$$(\widehat{x^\alpha\partial^\beta f}) = (-1)^{|\alpha|}(2\pi i)^{|\beta|-|\alpha|}\partial^\alpha(\xi^\beta\hat{f}).$$

Thus $\partial^\alpha(\xi^\beta\hat{f})$ is bounded for all α, β, whence $\hat{f} \in \mathcal{S}$. Moreover, since $\int(1+|x|)^{-n-1}dx < \infty$,

$$\|(\widehat{x^\alpha\partial^\beta f})\|_u \le \|x^\alpha\partial^\beta f\|_1 \le C\|(1+|x|)^{n+1}x^\alpha\partial^\beta f\|_u.$$

It then follows that $\|\hat{f}\|_{(N,\beta)} \le C_{N,\beta}\sum_{|\gamma|\le|\beta|}\|f\|_{(N+n+1,\gamma)}$ by proposition 14.1.4, so the Fourier transform is continuous on $\mathcal{S}$.

14.2.17 Distribution Theory and Compactly Supported In distribution theory (chapter 9), it shall be shown that $C(\mathbb{R}^n)$ can be recovered by integrating against smooth functions with compact support. Thus, C_c^∞ would be ideal space for the Fourier transform *if* they mapped back to themselves. However, by the boundedness argument above it is not the case. *However*, in over totally disconnected topological groups, such as those present in the representation theory of absolute galois groups, the equivalent of Schwartz space *is* C_c^∞; see [6] for more details.

Example 14.2.18: Continuous Spectrum Examples (Euclidean)

1. *The Box Function (The Sinc Function):* Let $f(x) = \mathbb{1}_{[-1/2,1/2]}(x)$ (The indicator function of width 1).
$$\hat{f}(\xi) = \int_{-1/2}^{1/2} 1\cdot e^{-2\pi i\xi x}dx = \frac{e^{-\pi i\xi} - e^{\pi i\xi}}{-2\pi i\xi} = \frac{\sin(\pi\xi)}{\pi\xi} \equiv \operatorname{sinc}(\xi)$$
Unlike the Square Wave on the torus (which had discrete coefficients $1/k$), this has a *continuous* spectrum that oscillates and decays as $1/\xi$. The compact support in space ($f = 0$

outside the box) forces the transform to extend infinitely (it cannot be compactly supported).

2. *The Two-Sided Exponential (Lorentzian):* Let $f(x) = e^{-2\pi|x|}$. This is a continuous L^1 function with a "cusp" at $x = 0$.
$$\hat{f}(\xi) = \frac{1}{\pi}\frac{1}{1+\xi^2}$$
This is the Cauchy (or Lorentzian) distribution. Notice the duality: Exponential decay in space leads to Polynomial ($1/\xi^2$) decay in frequency. The lack of smoothness at $x = 0$ (the cusp) prevents the frequency decay from being exponential.

3. *The Gaussian (Self-Dual):* Let $f(x) = e^{-\pi x^2}$. As proven in theorem 14.3.1:
$$\hat{f}(\xi) = e^{-\pi\xi^2}$$
The Gaussian is the unique eigenfunction of the Fourier Transform. It is the perfect balance point of the Uncertainty Principle—it is equally localized in both space and frequency.

4. *The Shifted Function (Modulation):* Let $f(x)$ be a standard Gaussian centered at $x = 100$ rather than 0: $g(x) = f(x - 100)$.
$$\hat{g}(\xi) = e^{-2\pi i(100)\xi}\hat{f}(\xi) = e^{-2\pi i 100\xi}e^{-\pi\xi^2}$$
In Euclidean space, shifting a function far away does not change the "energy envelope" of its spectrum (it's still a Gaussian), but it causes the complex phase to wind rapidly.

Example 14.2.19: Advanced uses of Fourier Transform

1. *The Poincaré Bundle:* In our definition of the Fourier Transform $\hat{f}(\xi) = \int f(x)e^{-2\pi i\xi\cdot x}dx$, the kernel $K(x,\xi) = e^{-2\pi i\xi\cdot x}$ plays the central role. In Algebraic Geometry, if A is an abelian variety (say a complex torus V/Λ), its dual $\hat{A}$ parametrizes line bundles on A. The universal line bundle $\mathcal{P}$ on $A \times \hat{A}$ (the Poincaré bundle) is the geometric geometrization of the kernel $e^{-2\pi i\xi\cdot x}$. The Fourier transform generalizes to the *Fourier-Mukai Transform*, an equivalence of derived categories of coherent sheaves:
$$\Phi_{\mathcal{P}}(\mathcal{F}^\bullet) = \mathbf{R}\pi_{2,*}(\pi_1^*\mathcal{F}^\bullet \otimes \mathcal{P})$$
Here, the integration $\int dx$ is replaced by the pushforward $\mathbf{R}\pi_{2,*}$, and the product is the tensor product.

2. *The Spectrum of the Primes (Riemann's Explicit Formula):* You may have heard that the zeros of the Riemann Zeta function constitute the "spectrum" of the prime numbers. This is literal, not metaphorical. Consider the Chebyshev function $\psi(x) = \sum_{p^k\leq x}\log p$, which counts primes with a logarithmic weight. Riemann's Explicit Formula links this step function to the zeros $\rho = \frac{1}{2} + i\gamma$ of $\zeta(s)$:
$$\psi(x) = x - \sum_{\rho}\frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2})$$

If we make the logarithmic change of variables $x = e^u$, the sum over zeros becomes:

$$\sum_\rho \frac{e^{u(\frac{1}{2}+i\gamma)}}{\frac{1}{2}+i\gamma} = e^{u/2} \sum_\gamma \left(\frac{1}{\frac{1}{2}+i\gamma} \right) e^{i\gamma u}$$

This is a Fourier series (or integral) in the variable u!

- The "time" Domain signal is the distribution of primes (spikes at $u = \log p$).
- The "Frequency" Domain is the set of imaginary parts of zeta zeros (γ).

Just as a musical chord is "recovered" by summing its constituent frequencies, the location of prime numbers is recovered by summing the waves defined by the Riemann Zeros. This duality is an instance of Fourier Duality on the ideles.

3. *Pontryagin Duality as a Moduli Problem:* We defined $\hat{G} = \mathrm{Hom}(G, S^1)$. For a number theorist, we can view the frequency domain $\hat{G}$ as a *moduli space* of rank-1 unitary local systems on G.

   - If $G = \mathbb{R}$, the moduli space is $\mathbb{R}$ (Euclidean Fourier Transform).
   - If $G = S^1$, the moduli space is $\mathbb{Z}$ (Fourier Series).
   - If $G = \mathbb{Z}$, the moduli space is S^1 (Discrete Time Fourier Transform).

   The "Duality" implies that the moduli space of the moduli space recovers the original space.

4. *The Circle Method (Diophantine Equations):* How many integer solutions are there to $n_1^k + \cdots + n_s^k = N$? (Waring's Problem). This is a combinatorial problem that Fourier Analysis turns into an analytic one. Construct the generating function (a trigonometric polynomial):

$$f(\alpha) = \sum_{m=1}^{N^{1/k}} e^{2\pi i m^k \alpha}$$

   The number of solutions $R(N)$ is exactly the N-th Fourier coefficient of the function $f(\alpha)^s$:

$$R(N) = \int_0^1 (f(\alpha))^s e^{-2\pi i N\alpha} d\alpha$$

   The Key insight of Hardy and Littlewood was to split this integral (the torus) into "Major Arcs" (near rational numbers with small denominators, where the sum behaves predictably) and "Minor Arcs" (where the sum oscillates and cancels out, similar to the Riemann-Lebesgue lemma).

5. *Spectral Geometry (Can You Hear the Shape of a Drum?):* The set of eigenvalues $\{\lambda_n\}$ of the Laplacian Δ on a compact Riemannian manifold (M, g) is called the *spectrum*. This is the set of "pure frequencies" the manifold can support. The Fourier Transform allows us to recover geometric data from this spectral data via the trace of the Heat Kernel:

$$\mathrm{Tr}(e^{t\Delta}) = \sum_{n=0}^{\infty} e^{-\lambda_n t} \sim \frac{1}{(4\pi t)^{d/2}} \left(\mathrm{Vol}(M) + \frac{t}{6} \int_M \mathrm{Scal}_g + O(t^2) \right)$$

   As $t \to 0$ (high frequencies), the asymptotics of the heat kernel allow us to "hear" the dimension, the volume, and the total scalar curvature of the manifold.

Note: Milnor gave examples of isospectral tori (16-dimensional) that are not isometric, proving that you cannot hear the *entire* shape, but Fourier analysis recovers the local invariants.

14.3 Fourier Integral and Series

Let us now show that up to measure zero the Fourier transform is injective. Equivalently, let us now look into finding the inverse Fourier transform. Let $f \in L^1$. Then define:

$$f^\vee(x) = \hat{f}(-x) = \int f(\xi) e^{2\pi i \xi \cdot x} d\xi$$

We'll show that if $f, \hat{f} \in L^1$, then $(\hat{f})^\vee = f$. This cannot be done by appealing to Fubini's theorem as the integral:

$$(\hat{f})^\vee(x) = \int \int f(y) e^{2\pi i \cdot x} e^{2\pi i \cdot y} dy d\xi$$

is not in $L^1(\mathbb{R}^n \times \mathbb{R}^n)$ (it will a.e. diverge). Instead, we do the usual trick of sequences that converge to the desired values. The solution is the usual solution of taking a limit of functions that satisfy the property and insuring that the desired properties are preserved in the limit. First, a technical lemma that also has an amazing tangential property that simply must be mentioned:

Theorem 14.3.1: Fourier Transform Fixed Point

Let $f(x) = e^{-\pi a |x|^2}$ where $a > 0$. Then

$$\hat{f}(\xi) = a^{-n/2} e^{-\pi |\xi|^2 / a}$$

In particular if $a = 1$ so that $f(x) = e^{-\pi |x|^2}$, then

$$\hat{f} = f$$

***Proof*:**

First consider the case $n = 1$. Since the derivative of $e^{-\pi a x^2}$ is $-2\pi a e^{-\pi a x^2}$, by proposition 14.2.13(e) we have

$$(\hat{f})'(\xi) = (-2\pi i x e^{-\pi a x^2})^\wedge(\xi) = \frac{i}{a}(f')^\wedge(\xi) = \frac{i}{a}(2\pi i \xi)\hat{f}(\xi) = -\frac{2\pi}{a}\xi \hat{f}(\xi).$$

Then:

$$\begin{aligned}\frac{d}{d\xi}\left(e^{\pi\xi^2/a}\cdot\hat{f}(\xi)\right) &= \left[\frac{d}{d\xi}\left(e^{\pi\xi^2/a}\right)\right]\cdot\hat{f}(\xi) + e^{\pi\xi^2/a}\cdot\hat{f}'(\xi)\\ &= e^{\frac{\pi}{a}\xi^2}\cdot\left(\frac{2\pi}{a}\xi\right)\cdot\hat{f}(\xi) + e^{\pi\xi^2/a}\cdot\hat{f}'(\xi)\\ &= e^{\pi\xi^2/a}\left[\frac{2\pi}{a}\xi\hat{f}(\xi) + \hat{f}'(\xi)\right]\\ &= e^{\pi\xi^2/a}\left[\frac{2\pi}{a}\xi\hat{f}(\xi) + \left(-\frac{2\pi}{a}\xi\hat{f}(\xi)\right)\right]\\ &= e^{\pi\xi^2/a}[0] = 0\end{aligned}$$

It follows that $(d/d\xi)(e^{\pi\xi^2/a}\hat{f}(\xi)) = 0$, so that $e^{\pi\xi^2/a}\hat{f}(\xi)$ is constant. To evaluate the constant, set $\xi = 0$. Then computing the integral:

$$\hat{f}(0) = \int e^{-\pi a x^2}dx = a^{-1/2}.$$

The n-dimensional case follows by Fubini's theorem, since $|x|^2 = \sum_1^n x_j^2$:

$$\begin{aligned}\hat{f}(\xi) &= \prod_1^n \int \exp(-\pi a x_j^2 - 2\pi i \xi_j x_j)dx_j\\ &= \prod_1^n [a^{-1/2}\exp(-\pi\xi_j^2/a)] = a^{-n/2}\exp(-\pi|\xi|^2/a).\end{aligned}$$

completing the proof.

Theorem 14.3.2: The Fourier Inversion Theorem

Let $f \in L^1$ and $\hat{f} \in L^1$. Then f agrees a.e. with a continuous function f_0, and

$$(\hat{f})^\vee = (f^\vee)^\wedge = f_0$$

***Proof*:**
Given $t > 0$ and fixed $x \in \mathbb{R}^n$, set

$$\varphi(\xi) = \exp(2\pi i\xi\cdot x - \pi t^2|\xi|^2).$$

We use the standard Gaussian $g(x) = e^{-\pi|x|^2}$ and the scaling notation $g_t(y) = t^{-n}g(y/t)$. Using the translation and dilation properties of the Fourier transform on the Gaussian, we have:

$$\hat{\varphi}(y) = t^{-n}\exp(-\pi|x-y|^2/t^2) = g_t(x-y).$$

Since $f, \varphi \in L^1$, Fubini's theorem justifies the multiplication formula $\int\hat{f}\varphi = \int f\hat{\varphi}$:

$$\int e^{-\pi t^2|\xi|^2}e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = \int\hat{f}(\xi)\varphi(\xi)d\xi = \int f(y)\hat{\varphi}(y)dy = f * g_t(x).$$

Now we take the limit as $t \to 0$:

1. Right hand side: since $\int g(y)dy = 1$, $\{g_t\}$ is an approximate identity. Thus $f * g_t \to f$ in the L^1 norm.
2. Left hand side: since $\hat{f} \in L^1$, the function is dominated by $|\hat{f}|$. By the Dominated Convergence Theorem, for every x:

$$\lim_{t\to 0} \int e^{-\pi t^2|\xi|^2} e^{2\pi i\xi\cdot x} \hat{f}(\xi) d\xi = \int e^{2\pi i\xi\cdot x} \hat{f}(\xi) d\xi = (\hat{f})^\vee(x).$$

Since the LHS converges pointwise to $(\hat{f})^\vee$ and the RHS converges to f in L^1 (which implies pointwise convergence of a subsequence a.e.), the limits must be equal almost everywhere.

It follows that $f = (\hat{f})^\vee$ a.e. Since $(\hat{f})^\vee$ is the Fourier transform of an L^1 function $(\hat{f})$, it is continuous, completing the proof.

Corollary 14.3.3

If $f \in L^1$ and $\hat{f} = 0$, then $f = 0$ a.e.

Corollary 14.3.4: Fourier isomorphism on Schwartz Space

$\mathcal{F}$ is an isomorphism of $\mathcal{S}$ onto itself.

***Proof*:**
By corollary 14.2.16, $\mathcal{F}$ maps S continuously into itself, and hence so does $f \mapsto f^\vee$, since $f^\vee(x) = \hat{f}(-x)$. By the Fourier inversion theorem, these maps are inverse to each other.

Corollary 14.3.5: Fourier Transform Has Order 4

Let $\mathfrak{F}$ represent the Fourier transform. Then:

$$\mathfrak{F}^4 = I$$

and 4 is the smallest such number

***Proof*:**
First:

$$\begin{aligned}
\mathfrak{F} \circ \mathfrak{F}(f)(x) &= \mathfrak{F}(\mathfrak{F}(f)) \\
&= \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i\xi\cdot x} d\xi \\
&= \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i\xi\cdot(-x)} d\xi \\
&= f(-x) && \text{Fourier inversion formula}
\end{aligned}$$

Hence applying it it again gives the desired result.

We shall give a name to the representation of $f(x)$ in terms of its inversion:

Definition 14.3.6: Fourier Integral

Let $f \in L^1$ and $\hat{f} \in L^1$. Then the *Fourier integral* at x is:

$$f(x) = \int \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi$$

An important result using the Fourier inversion theorem is the dual of proposition 14.2.14:

Corollary 14.3.7: Inverse Regularity and Decay

1. **Decay ⇒ Smoothness:** If $(1+|\xi|)^k \hat{f}(\xi) \in L^1(\mathbb{R}^n)$ for some integer $k \geq 1$, then $f \in C^k(\mathbb{R}^n)$. The derivative is given by:

$$\partial^\alpha f(x) = \int_{\mathbb{R}^n} (2\pi i \xi)^\alpha \hat{f}(\xi) e^{2\pi i x \cdot \xi} \, d\xi$$

for all multi-indices $|\alpha| \leq k$.

2. **Smoothness ⇒ Decay:** If $\hat{f} \in C^k(\mathbb{R}^n)$ and $\partial^\alpha \hat{f} \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$, then f decays rapidly:

$$|f(x)| \leq \frac{C}{(1+|x|)^k}$$

for some constant C.

***Proof*:**
The Inverse Fourier Transform is defined by:

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i x \cdot \xi} \, d\xi.$$

This operator differs from the forward Fourier transform only by the sign of the exponential ($+i$ instead of $-i$). Since $|e^{i\theta}| = |e^{-i\theta}| = 1$, the absolute values used in the norm estimates and convergence arguments of **??** remain unchanged.

Thus, the proof follows immediately by applying **??** to the function $\hat{f}$ (with a trivial sign change in the derivative formula).

At last we are in a position to derive the analogue of the orthonormal basis as we have done in theorem 14.2.4:

Theorem 14.3.8: The Plancherel Theorem

If $f \in L^1 \cap L^2$, then $\hat{f} \in L^2$; and $\mathcal{F}|_{L^1 \cap L^2}$ extends uniquely to a unitary isomorphism on L^2.

***Proof*:**
Let $X = \{f \in L^1 : \hat{f} \in L^1\}$. Since $\hat{f} \in L^1$ implies $f \in L^\infty$, we have $X \subset L^2$ by proposition 8.1.13, and X is dense in L^2 because $\mathcal{S} \subset X$ and $\mathcal{S}$ is dense in L^2 by proposition 8.6.8.

Given $f, g \in X$, let $h = \overline{\hat{g}}$. By the properties of the Fourier transform under complex conjugation and inversion, we have $\hat{h} = \bar{g}$. Now, using the multiplication formula (Parseval's relation for L^1):

$$\int f\bar{g} = \int f\hat{h} = \int \hat{f}h = \int \hat{f}\overline{\hat{g}}.$$

Thus $\mathcal{F}|_X$ preserves the L^2 inner product. In particular, by taking $g = f$, we obtain $\|\hat{f}\|_2 = \|f\|_2$. Since $\mathcal{F}(X) = X$ by the inversion theorem, $\mathcal{F}|_X$ extends by continuity to a unitary isomorphism on L^2.

It remains only to show that this extension agrees with $\mathcal{F}$ on all of $L^1 \cap L^2$. But if $f \in L^1 \cap L^2$ and $g(x) = e^{-\pi|x|^2}$ as in the proof of the inversion theorem, we have $f * g_t \in L^1$ by Young's inequality and $(f * g_t)^\wedge \in L^1$ because $(f * g_t)^\wedge(\xi) = e^{-\pi t^2|\xi|^2}\hat{f}(\xi)$ and $\hat{f}$ is bounded. Hence $f * g_t \in X$; moreover, by theorem 8.6.10, $f * g_t \to f$ in both the L^1 and L^2 norms. Therefore $(f * g_t)^\wedge \to \hat{f}$ both uniformly and in the L^2 norm, completing the proof.

From this, we upgrade The Hausdorff-Young Inequality over $\mathbb{R}^n$:

Theorem 14.3.9: The Hausdorff-Young Inequality

Suppose that $1 \leq p \leq 2$ and q is the conjugate exponent to p. If $f \in L^p(\mathbb{R}^n)$, then

$$\hat{f} \in L^q(\mathbb{R}^n) \qquad \text{and} \qquad \|\hat{f}\|_q \leq \|f\|_p$$

Folland goes on an interesting point about trying to make an $\mathbb{R}^n$ input function periodic which leads to the Poisson Summation Formula; I'm skipping this for now

(here is a theorem I want to add since it related to the analytic theorem)

Theorem 14.3.10: Fourier Decay and Analyticity

Let f be a smooth function such that its Fourier transform $\hat{f}$ satisfies the exponential decay estimate

$$|\hat{f}(\xi)| \leq Ce^{-R2\pi|\xi|}$$

for some constants $C, R > 0$. Then f has a global derivative growth rate α satisfying

$$\alpha \leq \frac{1}{R}.$$

Consequently, by the analyticity theorem of *Elementary Analysis* [4], f is analytic with radius of convergence at least R.

***Proof*:**
Recall that the Fourier transform diagonalizes the differentiation operator. Specifically, the Fourier transform of the k-th derivative is given by $\widehat{f^{(k)}}(\xi) = (2\pi i\xi)^k \hat{f}(\xi)$. By the Fourier inversion formula, we can express the k-th derivative as:

$$f^{(k)}(x) = \int_{-\infty}^{\infty} \widehat{f^{(k)}}(\xi) e^{2\pi i x\xi}\, d\xi = \int_{-\infty}^{\infty} (2\pi i\xi)^k \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

We wish to bound the magnitude $M_k = \sup_x |f^{(k)}(x)|$. Applying the triangle inequality and the assumed decay bound on $\hat{f}$:

$$\begin{aligned} |f^{(k)}(x)| &\leq \int_{-\infty}^{\infty} |2\pi\xi|^k |\hat{f}(\xi)|\, d\xi \\ &\leq \int_{-\infty}^{\infty} (2\pi|\xi|)^k C e^{-R|\xi|}\, d\xi \\ &= 2C(2\pi)^k \int_0^{\infty} \xi^k e^{-R\xi}\, d\xi. \end{aligned}$$

The integral on the right is a standard form related to the Gamma function (specifically, $\int_0^\infty t^k e^{-t} dt = k!$). Making the substitution $u = R\xi$, we have $\xi = u/R$ and $d\xi = du/R$:

$$\int_0^{\infty} \xi^k e^{-R\xi}\, d\xi = \frac{1}{R^{k+1}} \int_0^{\infty} u^k e^{-u}\, du = \frac{k!}{R^{k+1}}.$$

Substituting this back into our estimate for M_k:

$$M_k \leq 2C(2\pi)^k \frac{k!}{R^{k+1}} = \left(\frac{2C}{R}\right) k! \left(\frac{2\pi}{R}\right)^k.$$

Now we compute the derivative growth rate α:

$$\begin{aligned} \alpha &= \limsup_{k\to\infty} \sqrt[k]{\frac{M_k}{k!}} \\ &\leq \limsup_{k\to\infty} \sqrt[k]{\frac{2C}{R}\left(\frac{2\pi}{R}\right)^k} \\ &= \frac{2\pi}{R}. \end{aligned}$$

(Note: The factor of 2π appears due to the chosen convention of the Fourier transform. If the decay rate R is defined relative to angular frequency, the factor vanishes). In either case, α is finite and inversely proportional to the decay constant R. Thus, the derivative growth rate is bounded, and f is analytic.

I don't know if I wanna continue this section for now, about Cecaro sums and etc

14.4 Point-wise Convergence of Fourier Series

We have established that for $f \in L^2(\mathbb{T}^n)$, the Fourier series converges to f in the L^2-norm. However, norm convergence does not imply point-wise convergence. We might ask: if we pick a specific point x_0, does the partial sum sequence $S_N(f)(x_0)$ converge to $f(x_0)$?

$$S_N(f)(x) = \sum_{|k|\le N} \hat{f}(k)e^{2\pi ik\cdot x}$$

Surprisingly, for general integrable functions, the answer is often no. The mechanism governing this convergence is the *Dirichlet Kernel*.

Recall that convolution in the spatial domain corresponds to multiplication in the frequency domain. The partial sum $S_N(f)$ is the convolution of f with the inverse Fourier transform of the characteristic function of the range $[-N, N]$.

Definition 14.4.1: Dirichlet Kernel

The *Dirichlet Kernel* D_N on $\mathbb{T}^1$ is defined as:

$$D_N(x) = \sum_{k=-N}^{N} e^{2\pi ikx} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}$$

Consequently, the partial sums of the Fourier series are given by:

$$S_N(f)(x) = (f * D_N)(x) = \int_{\mathbb{T}} f(x-t)D_N(t)dt$$

The equality in the above equation is a simple application of the geometric series with ratio $r = e^{2\pi ix}$:

$$\begin{aligned}
\sum_{k=-N}^{N} r^k = r^{-N}\frac{1-r^{2N+1}}{1-r} &= \frac{r^{-N}-r^{N+1}}{1-r} \\
&= \frac{r^{-(N+1/2)}-r^{N+1/2}}{r^{-1/2}-r^{1/2}} \quad \text{(multiplying top and bottom by } r^{-1/2}\text{)} \\
&= \frac{e^{-2\pi i(N+1/2)x}-e^{2\pi i(N+1/2)x}}{e^{-\pi ix}-e^{\pi ix}} \\
&= \frac{-2i\sin((2N+1)\pi x)}{-2i\sin(\pi x)} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}
\end{aligned}$$

The convergence of $S_N(f)$ depends on whether D_N is a "good kernel" (an approximate identity). While $\int D_N = 1$, it fails the absolute integrability condition: $\int |D_N| \to \infty$ as $N \to \infty$. Specifically, $\|D_N\|_1 \sim \frac{4}{\pi^2}\log N$. This logarithmic growth allows for pathological behavior.

Theorem 14.4.2: Failure of Pointwise Convergence

1. (Du Bois-Reymond) There exists $f \in C(\mathbb{T})$ whose Fourier series diverges at a prescribed point. More precisely, the set of such f is residual in $C(\mathbb{T})$.
2. As a consequence, there exists $f \in C(\mathbb{T})$ whose Fourier series diverges at a dense set of points.
3. (Kolmogorov, 1923) There exists $f \in L^1(\mathbb{T})$ whose Fourier series diverges *everywhere*.

***Proof*:**
We prove (1) and (2) using the principle of uniform boundedness (theorem 7.2.11); item (3) requires a separate and more delicate construction. This argument was previewed in example 7.2.12.

Step 1: Setting up the functionals. Fix a point $x_0 \in \mathbb{T}$ (for concreteness, $x_0 = 0$). For each $N \in \mathbb{N}$, define the linear functional $\Lambda_N : C(\mathbb{T}) \to \mathbb{C}$ by:

$$\Lambda_N(f) = S_N f(0) = \int_{\mathbb{T}} f(t) D_N(t)\, dt$$

where we used the fact that D_N is even (definition 14.4.1). Each Λ_N is a bounded linear functional on $C(\mathbb{T})$ (equipped with the supremum norm $\|\cdot\|_\infty$), since:

$$|\Lambda_N(f)| \leq \int_{\mathbb{T}} |f(t)||D_N(t)|\, dt \leq \|f\|_\infty \cdot \|D_N\|_{L^1(\mathbb{T})}$$

Step 2: Computing the operator norms. We claim $\|\Lambda_N\| = L_N$ where $L_N = \|D_N\|_{L^1(\mathbb{T})}$ is the N-th *Lebesgue constant.* The upper bound $\|\Lambda_N\| \leq L_N$ was shown above. For the lower bound, let $g = \operatorname{sgn}(D_N)$. Since D_N is continuous with finitely many zeros, g is a piecewise constant function with finitely many jumps, and hence can be uniformly approximated by continuous functions f_ϵ with $\|f_\epsilon\|_\infty \leq 1$ and:

$$|\Lambda_N(f_\epsilon)| = \left|\int_{\mathbb{T}} f_\epsilon(t) D_N(t)\, dt\right| > L_N - \epsilon$$

Since $\epsilon > 0$ was arbitrary, $\|\Lambda_N\| = L_N$.

Step 3: The Lebesgue constants diverge. We show $L_N \to \infty$. Using $|D_N(x)| = \left|\frac{\sin((2N+1)\pi x)}{\sin(\pi x)}\right|$ and the bound $|\sin(\pi x)| \leq \pi|x|$:

$$L_N = \int_{-1/2}^{1/2} |D_N(x)|\, dx \geq 2\int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{\pi x}\, dx$$

Substituting $u = (2N+1)\pi x$:

$$L_N \geq \frac{2}{\pi}\int_0^{(N+1/2)\pi} \frac{|\sin u|}{u}\, du \geq \frac{2}{\pi}\sum_{k=1}^{N}\int_{(k-1)\pi}^{k\pi} \frac{|\sin u|}{u}\, du$$

On $[(k-1)\pi,\, k\pi]$, we have $u \le k\pi$ and $\int_0^\pi |\sin u|\, du = 2$, giving:

$$L_N \ge \frac{2}{\pi}\sum_{k=1}^{N} \frac{2}{k\pi} = \frac{4}{\pi^2}\sum_{k=1}^{N}\frac{1}{k} \sim \frac{4}{\pi^2}\log N \xrightarrow{N\to\infty} \infty$$

Step 4: Applying the uniform boundedness principle. The space $C(\mathbb{T})$ is a Banach space and $\sup_N \|\Lambda_N\| = \sup_N L_N = \infty$. By the *contrapositive* of the principle of uniform boundedness, the pointwise boundedness hypothesis must fail: there cannot hold $\sup_N |\Lambda_N(f)| < \infty$ for every $f \in C(\mathbb{T})$. Hence there exists $f \in C(\mathbb{T})$ such that:

$$\sup_N |S_N f(0)| = \sup_N |\Lambda_N(f)| = \infty$$

and so the Fourier series of f diverges at $x_0 = 0$. By translation (replacing f with $\tau_{x_0} f$ and noting that $\|\tau_{x_0} f\|_\infty = \|f\|_\infty$), the same holds at any prescribed point $x_0 \in \mathbb{T}$.

In fact, the argument gives more. Inspecting the proof of the uniform boundedness principle (theorem 7.2.11), the sets $E_n = \{f \in C(\mathbb{T}) \mid \sup_N |S_N f(0)| \le n\}$ are closed in $C(\mathbb{T})$, and since $\sup_N \|\Lambda_N\| = \infty$, each E_n has empty interior. Hence $\bigcup_n E_n = \{f \mid \sup_N |S_N f(0)| < \infty\}$ is meager, and the set of f whose Fourier series diverges at 0 is *residual.* This proves (1).

Step 5: Divergence at a dense set. For (2), let $\{x_k\}_{k=1}^\infty$ be a countable dense subset of $\mathbb{T}$. By (1), for each k the set:

$$R_{x_k} = \left\{ f \in C(\mathbb{T}) \;\middle|\; \sup_N |S_N f(x_k)| = \infty \right\}$$

is residual. A countable intersection of residual sets is residual by the Baire Category Theorem (proposition 7.2.1), so $\bigcap_k R_{x_k}$ is residual and in particular nonempty. Any f in this intersection has a Fourier series diverging at every point of the dense set $\{x_k\}$.

Notice the non-constructive nature of this argument: we proved the *existence* of a continuous function with divergent Fourier series without ever exhibiting one. The uniform boundedness principle, itself a consequence of the Baire Category Theorem, guarantees the existence of such pathological functions purely from the growth of the Lebesgue constants. The construction of an explicit example is considerably more involved (see the original work of Du Bois-Reymond, 1876).

The divergence of the Lebesgue constants has a second striking consequence, this time about the range of the Fourier transform as a map between Banach spaces. Recall that we denote by $c_0(\mathbb{Z})$ the Banach space of all sequences $(a_k)_{k\in\mathbb{Z}}$ of complex numbers satisfying $a_k \to 0$ as $|k| \to \infty$, equipped with the supremum norm $\|(a_k)\|_\infty = \sup_k |a_k|$.

For $f \in L^1(\mathbb{T})$, the bound $|\hat{f}(k)| \le \|f\|_1$ shows that $\hat{f} \in \ell^\infty(\mathbb{Z})$. In fact, $\hat{f} \in c_0(\mathbb{Z})$: trigonometric polynomials are dense in $L^1(\mathbb{T})$ (by Stone-Weierstrass and the density of $C(\mathbb{T})$ in $L^1(\mathbb{T})$), their Fourier transforms are finitely supported, and $\|\hat{f} - \hat{p}\|_\infty \le \|f - p\|_1$, so $\hat{f}$ is a uniform limit of sequences with finite support, hence decays to zero. We may thus view the Fourier transform as a bounded linear map $\mathcal{F} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$. It is natural to ask whether every sequence decaying to zero arises as the Fourier transform of some integrable function. The open mapping theorem answers this decisively:

Proposition 14.4.3: Non-Surjectivity of the Fourier Transform on L^1

The Fourier transform $\mathcal{F} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is injective, bounded, and has dense image, but is *not* surjective. In particular, $\mathcal{F}(L^1(\mathbb{T}))$ is a proper dense subspace of $c_0(\mathbb{Z})$ that is not closed.

***Proof*:**
Boundedness was established above ($\|\hat{f}\|_\infty \leq \|f\|_1$).

Injectivity. If $\hat{f}(k) = 0$ for all $k \in \mathbb{Z}$, then $\langle f, E_k \rangle = 0$ for all k, where $E_k(x) = e^{2\pi i k x}$. Since finite linear combinations of the E_k are uniformly dense in $C(\mathbb{T})$ (by Stone-Weierstrass, as used in theorem 14.2.4), it follows that $\int_{\mathbb{T}} f(x)\overline{g(x)}\, dx = 0$ for every $g \in C(\mathbb{T})$, whence $f = 0$ a.e.

Dense image. Every finitely supported sequence in $c_0(\mathbb{Z})$ is the Fourier transform of a trigonometric polynomial (which is in $L^1(\mathbb{T})$), and finitely supported sequences are dense in $c_0(\mathbb{Z})$.

Non-surjectivity. Suppose for contradiction that $\mathcal{F}$ is surjective. Since $\mathcal{F} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is then a continuous bijection between Banach spaces, the Banach isomorphism theorem (corollary 7.2.8) furnishes a bounded inverse. That is, there exists a constant $C > 0$ such that:

$$\|f\|_{L^1} \leq C\|\hat{f}\|_\infty \qquad \text{for all } f \in L^1(\mathbb{T})$$

Apply this to the Dirichlet kernel $D_N \in L^1(\mathbb{T})$. Since $\hat{D}_N(k) = 1$ for $|k| \leq N$ and $\hat{D}_N(k) = 0$ for $|k| > N$, we have $\|\hat{D}_N\|_\infty = 1$. The bound above then gives $\|D_N\|_{L^1} \leq C$ for all N. But as computed in the proof of theorem 14.4.2:

$$\|D_N\|_{L^1} \geq \frac{4}{\pi^2} \sum_{k=1}^{N} \frac{1}{k} \xrightarrow{N\to\infty} \infty$$

a contradiction. Hence $\mathcal{F}$ is not surjective, and since its image is dense but not all of $c_0(\mathbb{Z})$, the image is not closed.

This result illustrates a theme we will encounter repeatedly: the space L^1 is "too small" for the Fourier transform to be well-behaved. The failure of surjectivity means that there exist sequences decaying to zero that cannot arise as the Fourier coefficients of any integrable function. Put differently, the constraints imposed by integrability on the Fourier side are strictly stronger than decay at infinity. Characterizing the precise image $\mathcal{F}(L^1(\mathbb{T})) \subsetneq c_0(\mathbb{Z})$ remains an open problem.

Note also that the open mapping theorem is used here in its contrapositive form: the non-openness of $\mathcal{F}$ (manifested through the unbounded ratio $\|D_N\|_1/\|\hat{D}_N\|_\infty$) forces the failure of surjectivity. The same Dirichlet kernel that produces divergent Fourier series (theorem 14.4.2) is the obstruction to surjectivity, a coincidence that is not accidental: both phenomena stem from the logarithmic growth of the Lebesgue constants.

This tells us that continuity alone is *not* strong enough to guarantee that we can recover the function point-wise from its frequencies using the standard summation. To guarantee convergence, we need slightly more regularity than continuity. We look for conditions that cancel out the oscillations of the Dirichlet kernel.

Theorem 14.4.4: Pointwise Convergence Tests

Let $f \in L^1(\mathbb{T})$.

1. *Dini's Criterion (Local):* If f satisfies a local Hölder condition at x (or more generally, $\int_{-1/2}^{1/2} |\frac{f(x+t)-f(x)}{t}| dt < \infty$), then $S_N(f)(x) \to f(x)$.
2. *Dirichlet-Jordan Test (Global):* If f is of *Bounded Variation* on $\mathbb{T}$, then for every x:

$$\lim_{N\to\infty} S_N(f)(x) = \frac{f(x^+) + f(x^-)}{2}$$

***Proof*:**
We prove Dini's Criterion. We wish to show $S_N(f)(x) - f(x) \to 0$. Using the fact that $\int D_N(t)dt = 1$, we write:

$$S_N(f)(x) - f(x) = \int_{-1/2}^{1/2} (f(x-t) - f(x))D_N(t)dt$$

Substitute the explicit formula for $D_N(t)$:

$$= \int_{-1/2}^{1/2} \frac{f(x-t) - f(x)}{\sin(\pi t)} \sin((2N+1)\pi t)dt$$

Define $g(t) = \frac{f(x-t)-f(x)}{t} \cdot \frac{t}{\sin(\pi t)}$. The term $\frac{t}{\sin(\pi t)}$ is bounded and smooth near $t = 0$. By the Dini condition, $\frac{f(x-t)-f(x)}{t}$ is integrable near 0. Thus $g(t) \in L^1(\mathbb{T})$. The expression becomes:

$$\int_{-1/2}^{1/2} g(t) \sin((2N+1)\pi t)dt$$

This is simply the imaginary part of the $(2N+1)$-th Fourier coefficient of $g(t)$. By the Riemann-Lebesgue Lemma, $\hat{g}(k) \to 0$ as $k \to \infty$. Thus the integral vanishes as $N \to \infty$.

If we cannot guarantee convergence for continuous functions, we can modify our summation method. Instead of partial sums, we take the average of partial sums (Cesàro means).

Theorem 14.4.5: Fejér's Theorem

Let $\sigma_N(f) = \frac{1}{N+1}\sum_{n=0}^{N} S_n(f)$ be the Cesàro means. Then $\sigma_N(f) = f * K_N$, where K_N is the *Fejér Kernel*:

$$K_N(x) = \frac{1}{N+1}\left(\frac{\sin((N+1)\pi x)}{\sin(\pi x)}\right)^2$$

Unlike the Dirichlet kernel, $K_N(x) \geq 0$ and $\int K_N = 1$. Consequently, if $f \in C(\mathbb{T})$, then $\sigma_N(f) \to f$ *uniformly*.

***Proof*:**
The formula for K_N follows from summing the geometric series for D_n. The key properties are: 1. $K_N(x) \geq 0$. 2. $\int_{-1/2}^{1/2} K_N(x)dx = 1$. 3. For any $\delta > 0$, $\int_{|x|\geq\delta} K_N(x)dx \to 0$ as $N \to \infty$ (Mass concentration).

Let f be continuous on $\mathbb{T}$. Since $\mathbb{T}$ is compact, f is uniformly continuous. Given $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x-t) - f(x)| < \epsilon/2$ whenever $|t| < \delta$.

$$\begin{aligned} |\sigma_N(f)(x) - f(x)| &= \left| \int_{-1/2}^{1/2} (f(x-t) - f(x))K_N(t)dt \right| \\ &\leq \int_{-1/2}^{1/2} |f(x-t) - f(x)|K_N(t)dt \end{aligned}$$

Split the integral into region $A = \{|t| < \delta\}$ and $B = \{|t| \geq \delta\}$. On A: $|f(x-t) - f(x)| < \epsilon/2$. Since $\int K_N = 1$, the integral over A is $\leq \epsilon/2$. On B: $|f(x-t) - f(x)| \leq 2\|f\|_\infty$. The integral is bounded by $2\|f\|_\infty \int_{|t|\geq\delta} K_N(t)dt$. Using the explicit formula, for $|t| \geq \delta$, $K_N(t) \leq \frac{1}{N+1}\frac{1}{\sin^2(\pi\delta)}$. As $N \to \infty$, this goes to 0. Thus for large N, the second term is $< \epsilon/2$. The convergence is uniform because δ depended only on the uniform continuity of f, not on x.

14.4.6 Historical Note For a long time, it was unknown if $f \in L^2(\mathbb{T})$ converged pointwise. In 1966, Lennart Carleson proved that if $f \in L^2(\mathbb{T})$, then $S_N(f)(x) \to f(x)$ for *almost every x*. This was later extended by Hunt to all L^p for $p > 1$. This is a deep and difficult result, requiring a careful decomposition of the operator to avoid the logarithmic divergence of L^1.

14.5 *From Harmonic Analysis to Number Theory

The Fourier analysis of this book is the opening chapter of a long story that runs through modern number theory. The abelian Pontryagin duality of part IV is the GL_1 case of the Langlands program, and its non-abelian and adelic generalizations organize much of arithmetic. Rather than develop these here, this starred section records the directions and where each is taken up; every entry is a forward pointer, and the pertinent volumes carry the development.

- *Unitary representations and the abstract Plancherel theorem* generalize the Fourier transform and Pontryagin duality to non-abelian locally compact groups, in a companion volume on unitary representation theory.
- *Tate's thesis* recasts the analytic continuation and functional equation of the Riemann zeta and Dirichlet L-functions as Fourier analysis on the adeles, a single global LCA group; it belongs to the theory of automorphic forms.
- The *Selberg and Arthur trace formulas* are the non-abelian replacement for the Poisson summation formula; the *Satake isomorphism*, *Eisenstein series and the spectral decomposition*, and *Langlands functoriality* are the further structures of the automorphic theory.
- The *Peter-Weyl theorem* and the *Plancherel formula* for compact and reductive groups are the harmonic analysis that makes the automorphic spectrum computable.

14.6 *Kakeya Sets and the Fourier Transform

The Kakeya problem originated in geometric measure theory, but its influence extends into the heart of harmonic analysis. The connection is through the Fourier transform: the geometric arrangement of tubes in physical space is dual to the arrangement of frequency caps on the sphere in frequency space, and the Kakeya conjecture places fundamental constraints on how efficiently these caps can be combined. This section develops this connection, starting with Fefferman's 1971 theorem on the ball multiplier (the result that first brought Besicovitch sets into Fourier analysis) and proceeding to the restriction conjecture and its relatives. The tools used here—the Fourier transform on L^1 and L^2 (definition 14.2.11, theorem 14.3.8), Young's inequality (theorem 8.6.6), and the L^p theory of the Hilbert transform (theorem 15.3.4)—are all developed earlier in this book.

In one dimension, the operation of truncating the Fourier transform to an interval is well-controlled: if $f \in L^p(\mathbb{R})$ for $1 < p < \infty$, then the function whose Fourier transform is $\hat{f} \cdot \chi_{[-R,R]}$ is again in L^p with norm bounded by $C_p\|f\|_p$. This is essentially the content of the L^p boundedness of the Hilbert transform (theorem 15.3.4), since the multiplier $\chi_{[-R,R]}$ can be expressed in terms of $\operatorname{sgn}(\xi)$ (the Hilbert transform multiplier) via $\chi_{[-R,R]}(\xi) = \frac{1}{2}(\operatorname{sgn}(\xi + R) - \operatorname{sgn}(\xi - R))$. The natural question is whether this extends to higher dimensions: if $f \in L^p(\mathbb{R}^n)$ for $n \geq 2$, is the function obtained by truncating $\hat{f}$ to a ball again in L^p?

Definition 14.6.1: Ball Multiplier

Let $R > 0$ and $n \geq 1$. The *ball multiplier* (or *disc multiplier* when $n = 2$) is the operator $S_R : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ defined by

$$\widehat{S_R f}(\xi) = \hat{f}(\xi) \cdot \chi_{B(0,R)}(\xi)$$

where $B(0, R) = \{\xi \in \mathbb{R}^n \mid |\xi| \leq R\}$ is the closed ball of radius R in frequency space. In physical space, $S_R f = f * \check{\chi}_{B(0,R)}$, where $\check{\chi}_{B(0,R)}$ is the inverse Fourier transform of the indicator function of the ball.

On $L^2(\mathbb{R}^n)$, the ball multiplier is bounded for all n with operator norm 1: by theorem 14.3.8, $\|S_R f\|_2 = \|\hat{f} \cdot \chi_{B(0,R)}\|_2 \leq \|\hat{f}\|_2 = \|f\|_2$. The question is whether S_R extends to a bounded operator on $L^p(\mathbb{R}^n)$ for $p \neq 2$. By duality (S_R is self-adjoint), boundedness on L^p implies boundedness on $L^{p'}$ where $1/p + 1/p' = 1$, so the question reduces to $1 < p < 2$ and its dual $2 < p < \infty$.

For $n = 1$, as noted above, S_R is bounded on $L^p(\mathbb{R})$ for all $1 < p < \infty$. This follows from the Hilbert transform connection and theorem 15.3.4. For $n \geq 2$, the situation is fundamentally different:

Theorem 14.6.2: Fefferman's Theorem on the Ball Multiplier

For $n \geq 2$, the ball multiplier S_R is bounded on $L^p(\mathbb{R}^n)$ if and only if $p = 2$.

The L^2 boundedness is immediate from Plancherel. The content of the theorem is that S_R is *un*bounded on L^p for every $p \neq 2$. The proof that S_R fails to be bounded uses the existence of Besicovitch sets (theorem 5.3.4), establishing the first direct connection between the Kakeya problem and Fourier analysis.

Proof Key ideas:
The proof constructs a sequence of functions $f_N \in L^p(\mathbb{R}^n)$ with $\|f_N\|_p = 1$ but $\|S_R f_N\|_p \to \infty$, using Besicovitch-type geometry.

Step 1: Angular decomposition. Decompose the ball $B(0, R)$ in frequency space into $N \sim \delta^{-(n-1)}$ thin sectors $\{\Omega_j\}_{j=1}^N$, each subtending a solid angle $\sim \delta^{n-1}$ and centered around a direction $e_j \in S^{n-1}$. Define the sector multiplier $S_j f = (f \cdot \chi_{\Omega_j})^\vee$, so that $S_R f \approx \sum_j S_j f$ (up to errors from the boundary of the ball and the smooth decomposition).

Step 2: Wave packet interpretation. Each sector multiplier S_j localizes $\hat{f}$ to a thin region of frequency space oriented along e_j. By the uncertainty principle (proposition 14.2.14), a function whose Fourier transform is supported in a δ-neighborhood of a cap on S^{n-1} centered at e_j is spread in physical space along a tube of length $\sim 1/\delta$ and width ~ 1 in direction e_j. The operator S_j thus maps functions to "wave packets" concentrated along tubes in direction e_j. The physical-space geometry of these wave packets is controlled by the frequency-space geometry of the sectors, and the tube directions $\{e_j\}$ are δ-separated by construction.

Step 3: Constructing the test function. Choose a Besicovitch set $K \subseteq \mathbb{R}^n$ with $m(K) < \epsilon$ containing a unit segment in every direction (theorem 5.3.4). For each j, let g_j be a wave packet concentrated on a δ-tube T_j in direction e_j that lies inside K_δ (the δ-neighborhood of K). The key properties are:

1. Each g_j has $\|g_j\|_p \sim \delta^{(n-1)/p}$ (the L^p norm of a function supported on a tube of volume $\sim \delta^{n-1}$).
2. The g_j have approximately disjoint Fourier supports (since the sectors Ω_j are disjoint), so $\|\sum_j g_j\|_2^2 \approx \sum_j \|g_j\|_2^2$ by Plancherel.
3. In physical space, the supports of g_j overlap inside K_δ, which has measure $\sim \epsilon$.

Step 4: Norm comparison. Let $f_N = \sum_j g_j$. By property (2) and the fact that the g_j have comparable magnitudes, $\|f_N\|_2^2 \sim N \cdot \delta^{n-1} \sim 1$. However, since $S_R f_N \approx f_N$ (the Fourier transforms of the g_j are already supported inside $B(0, R)$), the L^p norm of $S_R f_N$ is controlled from below by the concentration of f_N inside K_δ. For $p < 2$, Hölder's inequality gives

$$\|f_N\|_p \le |K_\delta|^{1/p-1/2} \|f_N\|_2 \sim \epsilon^{1/p-1/2}$$

For this to be bounded while $\|S_R f_N\|_p = \|f_N\|_p$ stays comparable to $\|f_N\|_2 \sim 1$, the ratio $\|S_R f_N\|_p / \|f_N\|_p \gtrsim \epsilon^{1/2-1/p}$, which diverges as $\epsilon \to 0$. For $p > 2$, the dual argument gives the same conclusion.

This shows S_R cannot be bounded on $L^p(\mathbb{R}^n)$ for $p \neq 2$, since the operator norm would have to exceed $\epsilon^{1/2-1/p}$ for arbitrarily small ϵ.

The proof reveals the mechanism by which Besicovitch sets obstruct L^p boundedness: the wave packets g_j are concentrated in a set of small measure (the Besicovitch set), which inflates their L^p norm for $p \neq 2$ relative to their L^2 norm. The ball multiplier cannot "see" this concentration because it acts in frequency space, where the wave packets are spread out over disjoint sectors. The tension between frequency-space orthogonality and physical-space concentration is the core of the argument.

Example 14.6.3: Wave Packets and Tubes

In $\mathbb{R}^2$, let $\delta = 1/N$ and consider a function g whose Fourier transform is a smooth bump supported in a δ-cap on S^1 near direction $e = (1, 0)$. The Fourier support is a region of dimensions $\sim 1 \times \delta$ (radial $\times$ angular). By the uncertainty principle, g is concentrated in physical space on a tube of dimensions $\sim 1 \times 1/\delta = 1 \times N$ (perpendicular $\times$ parallel to e). The L^2 norm of g is $\sim \delta^{1/2}$ (since $\|\hat{g}\|_2 \sim (\text{area of Fourier support})^{1/2} \sim \delta^{1/2}$), while its L^p norm for $p < 2$ is $\sim \delta^{1/2} \cdot (\text{tube volume})^{1/p-1/2} \sim \delta^{1/p}$. The ratio $\|g\|_p/\|g\|_2 \sim \delta^{1/p-1/2}$, which grows as $\delta \to 0$ when $p < 2$: a single wave packet already shows the norm distortion, and summing over N wave packets in a Besicovitch configuration amplifies it.

Fefferman's theorem demonstrates that the Kakeya geometry imposes a hard limit on Fourier multiplier estimates: the ball multiplier fails for $p \neq 2$ precisely because Besicovitch sets exist. The theorem resolved a question that had been open since the work of Bochner in the 1930s, and it shifted the focus of harmonic analysis toward understanding how the geometry of frequency truncation interacts with L^p norms.

The failure of the ball multiplier does not mean that frequency truncation is impossible for $p \neq 2$; it means that the *sharp* truncation $\chi_{B(0,R)}$ is the wrong tool. Two natural modifications lead to active research programs. The first is to *smooth* the truncation, replacing the sharp indicator with a function that decays continuously near $|\xi| = R$; this leads to the Bochner–Riesz conjecture. The second is to *restrict* the Fourier transform to the boundary sphere S^{n-1} rather than the solid ball; this leads to the restriction conjecture. Both modifications exploit the curvature of the sphere (which the flat faces of a cube, for instance, would not provide), and both are intimately connected to the Kakeya conjecture.

The restriction problem asks: can the Fourier transform $\hat{f}$ of an L^p function be meaningfully restricted to a curved surface $\Sigma \subseteq \mathbb{R}^n$ (like the sphere S^{n-1})? For a flat surface (such as a hyperplane), this is impossible in general: $\hat{f}$ for $f \in L^p(\mathbb{R}^n)$ with $p > 1$ is an $L^{p'}$ function, not a continuous function, and restricting an $L^{p'}$ function to a set of Lebesgue measure zero is not meaningful. For curved surfaces, the situation is fundamentally different. The curvature of Σ causes the Fourier transform of the surface measure σ to decay at infinity (by stationary phase), and this decay translates into regularity of the extension operator $(g\,d\sigma)^\vee$. The decay rate governs the range of L^p spaces for which restriction is possible: stronger curvature gives faster decay, which in turn gives a larger range of admissible p.

Theorem 14.6.4: Tomas–Stein Restriction Theorem

Let σ denote the surface measure on $S^{n-1} \subseteq \mathbb{R}^n$. For $1 \leq p \leq \frac{2(n+1)}{n+3}$:

$$\|\hat{f}\|_{L^2(S^{n-1},d\sigma)} \leq C\|f\|_{L^p(\mathbb{R}^n)}$$

for all $f \in \mathcal{S}(\mathbb{R}^n)$ (and hence for all $f \in L^p(\mathbb{R}^n)$ by density).

The exponent $p_0 = 2(n+1)/(n+3)$ is called the *Tomas–Stein exponent*. For $n = 2$, $p_0 = 6/5$; for $n = 3$, $p_0 = 4/3$. The theorem says that the Fourier transform of an L^{p_0} function, despite being only an $L^{p_0'}$ function in general, has enough regularity to be restricted to the sphere as an L^2 function.

Proof Sketch:
The proof uses the TT^* method, a duality argument that reduces the estimate to a convolution bound.

Step 1: Duality. The restriction estimate $\|\hat{f}\|_{L^2(d\sigma)} \leq C\|f\|_{L^p}$ is equivalent, by duality, to the *extension estimate*:

$$\|(g\,d\sigma)^\vee\|_{L^{p'}(\mathbb{R}^n)} \leq C\|g\|_{L^2(S^{n-1},d\sigma)}$$

for all $g \in L^2(S^{n-1})$, where $(g\,d\sigma)^\vee(x) = \int_{S^{n-1}} g(\omega)e^{2\pi i x\cdot\omega}\,d\sigma(\omega)$ is the inverse Fourier transform of the measure $g\,d\sigma$.

Step 2: TT^* reduction. To bound $\|(g\,d\sigma)^\vee\|_{p'}$, consider the operator $T : L^2(S^{n-1}) \to L^{p'}(\mathbb{R}^n)$ defined by $Tg = (g\,d\sigma)^\vee$. The TT^* method reduces the $L^2 \to L^{p'}$ bound to an $L^p \to L^{p'}$ bound on the composition TT^*. A computation shows that $TT^*f = f*\hat{\sigma}$, where $\hat{\sigma}(x) = \int_{S^{n-1}} e^{2\pi i x\cdot\omega}\,d\sigma(\omega)$ is the Fourier transform of the surface measure.

Step 3: Decay of $\hat{\sigma}$. The key analytic input is the decay estimate $|\hat{\sigma}(x)| \lesssim (1+|x|)^{-(n-1)/2}$, which holds by stationary phase (the curvature of S^{n-1} causes cancellation in the oscillatory integral). This decay estimate, combined with Young's inequality (theorem 8.6.6) for convolutions, gives

$$\|f*\hat{\sigma}\|_{p'} \leq \|\hat{\sigma}\|_r\|f\|_p$$

where $1+1/p' = 1/r+1/p$, and $\|\hat{\sigma}\|_r < \infty$ when $r(n-1)/2 > n$, i.e., $r > 2n/(n-1)$. Optimizing gives the exponent $p_0 = 2(n+1)/(n+3)$.

The Tomas–Stein theorem is sharp in its method (TT^* cannot give a better exponent) but is not expected to be sharp in its conclusion. The restriction conjecture predicts that the estimate should hold for a larger range of exponents:

Conjecture 14.6.5: Restriction Conjecture

For $n \geq 2$ and $1 \leq p < \frac{2n}{n+1}$, there exists $C = C(n,p) > 0$ such that

$$\|\hat{f}\|_{L^p(S^{n-1},d\sigma)} \leq C\|f\|_{L^p(\mathbb{R}^n)}$$

for all $f \in L^p(\mathbb{R}^n)$. More generally, the estimate is expected to hold with $L^p(S^{n-1})$ replaced by $L^q(S^{n-1})$ for appropriate (p,q) pairs.

The restriction conjecture implies the Kakeya set conjecture. The implication goes through wave packet decomposition: if the Kakeya conjecture failed (i.e., Besicovitch sets in $\mathbb{R}^n$ could have Hausdorff dimension $< n$), then one could construct families of wave packets concentrated on a low-dimensional set whose restriction norms would violate the conjectured bounds, using the same mechanism as in Fefferman's proof. The precise implication is: the restriction conjecture for the paraboloid $\left\{(\xi,|\xi|^2) \;\middle|\; \xi \in \mathbb{R}^{n-1}\right\} \subseteq \mathbb{R}^n$ implies the Kakeya set conjecture in $\mathbb{R}^n$. The converse is not known (and is likely false), so the restriction conjecture is strictly stronger.

The restriction conjecture fits into a hierarchy of conjectures, each implying the next. The hierarchy involves progressively "smoother" ways of truncating the Fourier transform near the sphere:

Definition 14.6.6: Bochner–Riesz Means

For $\alpha \geq 0$ and $R > 0$, the *Bochner–Riesz mean of order* α is the operator $S_R^\alpha : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ defined by

$$\widehat{S_R^\alpha f}(\xi) = \hat{f}(\xi)\left(1 - \frac{|\xi|^2}{R^2}\right)_+^\alpha$$

where $(t)_+ = \max(t, 0)$. When $\alpha = 0$, $S_R^0 = S_R$ is the ball multiplier.

The parameter α controls the smoothness of the frequency cutoff: $\alpha = 0$ gives the sharp indicator function (which fails for $p \neq 2$ by Fefferman's theorem), while increasing α makes the cutoff progressively smoother near $|\xi| = R$. The Bochner–Riesz conjecture asserts that sufficient smoothness restores L^p boundedness:

Conjecture 14.6.7: Bochner–Riesz Conjecture

For $n \geq 2$, S_R^α is bounded on $L^p(\mathbb{R}^n)$ for $\alpha > \max(n|1/p - 1/2| - 1/2, 0)$ and $1 < p < \infty$. The critical index $\alpha(p) = n|1/p - 1/2| - 1/2$ is sharp.

At the top of the hierarchy sits the local smoothing conjecture, which concerns solutions to the wave equation. The half-wave operator $e^{it\sqrt{-\Delta}}$ maps initial data f to the solution $u(x,t) = e^{it\sqrt{-\Delta}} f(x)$ of $i\partial_t u + \sqrt{-\Delta}\, u = 0$ (here $\sqrt{-\Delta}$ is the operator with Fourier multiplier $|\xi|$, so $\widehat{u}(\xi, t) = e^{2\pi i t|\xi|}\hat{f}(\xi)$). At any fixed time t, Sobolev embedding (theorem 15.5.3) gives $\|u(\cdot, t)\|_{L^p} \leq C\|f\|_{H^{s(p)}}$ where $s(p) = n(1/2 - 1/p)$. The local smoothing conjecture predicts that *averaging over a time interval* gains nearly $1/2$ of a derivative:

Conjecture 14.6.8: Local Smoothing Conjecture

For $n \geq 2$, $p > \frac{2n}{n-1}$, and every $\epsilon > 0$:

$$\left(\int_1^2 \|e^{it\sqrt{-\Delta}} f\|_{L^p(\mathbb{R}^n)}^p \, dt\right)^{1/p} \leq C_\epsilon \|f\|_{H^{s(p)-1/2+\epsilon}(\mathbb{R}^n)}$$

where $s(p) = n(1/2 - 1/p)$. The gain of $1/2 - \epsilon$ derivatives over the fixed-time estimate reflects the dispersive effect of time-averaging: oscillations in different directions cancel over time, producing additional regularity.

The conjecture was formulated by Sogge in 1991 and resolved in dimension $n = 2$ by Guth, Wang, and Zhang in 2020. It remains open for all $n \geq 3$.

The four central conjectures in this area form a chain of implications, each building on the geometric and analytic content of the previous one:

$$\text{Local Smoothing} \Rightarrow \text{Bochner–Riesz} \Rightarrow \text{Restriction} \Rightarrow \text{Kakeya}$$

The locl smoothing conjecture (**??** 14.6.8) sits at the top of the hierarchy: it is the most difficult and implies all the others. The chain of implications has been established over decades of work by Stein, Fefferman, Córdoba, Bourgain, Wolff, Tao, and many others. Each implication is nontrivial and passes through specific geometric-analytic mechanisms:

1. Restriction ⇒ Kakeya: the restriction estimate for the sphere controls the L^p norm of superpositions of wave packets, and if Besicovitch sets of low dimension existed, one could construct wave packet sums violating this control (as in Fefferman's proof).

2. Bochner–Riesz ⇒ Restriction: the Bochner–Riesz means S_R^α provide a smooth approximation to the restriction operator, and L^p bounds on S_R^α at the critical index imply restriction estimates by a limiting argument as $\alpha \to 0$.

3. Local Smoothing ⇒ Bochner–Riesz: the local smoothing estimate for the wave equation controls the space-time L^p norm of wave propagation, and the Bochner–Riesz means can be recovered as time-averages of the wave solution.

The two-dimensional cases of the Kakeya, restriction, and Bochner–Riesz conjectures were settled decades ago (by Davies, Zygmund/Fefferman, and Carleson–Sjölin respectively), while the two-dimensional local smoothing conjecture was resolved only in 2020 by Guth, Wang, and Zhang. In three dimensions, the Wang–Zahl theorem (theorem 5.6.1) settles the Kakeya conjecture at the bottom of the chain, but the restriction, Bochner–Riesz, and local smoothing conjectures remain open. The gap between the Kakeya set conjecture and the restriction conjecture is precisely the gap between the set estimate and the maximal function estimate discussed in **??** 5.4.6: the maximal function version of Kakeya, not just the set version, is the input needed for the restriction implication.

14.6.9 Remark: Decoupling and Modern Developments In 2015, Bourgain and Demeter proved the "ℓ^2 decoupling conjecture" for the paraboloid. The decoupling theorem provides a way to decompose a function whose Fourier transform is supported near a curved surface into "decoupled" pieces with approximately independent L^p behavior. The theorem has had wide-reaching consequences: it implies the full restriction conjecture for the paraboloid in the $L^2 \to L^p$ range (improving on Tomas–Stein), resolves the main conjecture in Vinogradov's mean value theorem (a problem in analytic number theory open since the 1930s), and provides new bounds for exponential sums.

The proof of the decoupling theorem uses an "induction on scales" argument that is structurally analogous to the Wang–Zahl approach to the Kakeya conjecture: both reduce a problem at scale δ to a problem at scale $\sqrt{\delta}$ using a bilinear or multilinear decomposition. This structural parallel suggests that the techniques may eventually be unified into a common framework for problems involving the interaction of curvature and combinatorial geometry.

For a comprehensive treatment of decoupling, see Demeter, *Fourier Restriction, Decoupling, and Applications* (Cambridge, 2020).

The material in this section connects the geometric constructions of chapter 5 to the analytic machinery of the Fourier transform. The central message is that the Kakeya problem is not an isolated question in geometric measure theory but a fundamental constraint on how the Fourier transform interacts with L^p spaces in multiple dimensions. The tools developed throughout this book—Lebesgue measure, Hausdorff dimension, the Fourier transform, L^p estimates, and Calderón–Zygmund theory—converge at this point, and the open problems (restriction, Bochner–Riesz, local smoothing) represent some of the central challenges in modern analysis.

Exercise 14.6.1

1. Verify that S_R is bounded on $L^2(\mathbb{R}^n)$ with operator norm 1 for all n. (*Hint*: use theorem 14.3.8.)

2. Show that for $n = 1$, the ball multiplier S_R is bounded on $L^p(\mathbb{R})$ for all $1 < p < \infty$ by expressing $\chi_{[-R,R]}(\xi)$ in terms of the Hilbert transform multiplier $-i\,\mathrm{sgn}(\xi)$ and applying theorem 15.3.4.

3. In the TT^* proof of the Tomas–Stein theorem, verify that $TT^*f = f * \hat{\sigma}$ where T is the extension operator and T^* is the restriction operator. (*Hint*: compute $\langle TT^*f, g\rangle$ using the definitions.)

4. Show that the Bochner–Riesz conjecture for $\alpha = 0$ reduces to the statement that S_R is bounded on L^p, recovering Fefferman's theorem as the assertion that the conjecture fails at the endpoint $\alpha = 0$ for $p \neq 2$.

5. For $n = 1$, verify that the Bochner–Riesz conjecture holds trivially: S_R^α is bounded on $L^p(\mathbb{R})$ for all $\alpha \geq 0$ and $1 < p < \infty$. (*Hint*: the multiplier $(1 - \xi^2/R^2)_+^\alpha$ is a bounded function with bounded variation on $\mathbb{R}$, so the result follows from the Marcinkiewicz multiplier theorem or directly from the L^p boundedness of S_R via subordination.)

6. Show that the restriction estimate $\|\hat{f}\|_{L^2(S^{n-1})} \leq C\|f\|_{L^p}$ cannot hold for $p > \frac{2n}{n+1}$ by testing on the function $f = \chi_{B(0,R)}$ and examining the behavior as $R \to \infty$. (*Hint*: compute $\hat{f}(\xi)$ for $\xi \in S^{n-1}$ using the stationary phase estimate $\hat{f}(\xi) \sim R^{(n+1)/2}$ on the sphere, and compare $\|\hat{f}\|_{L^2(S^{n-1})}$ with $\|f\|_{L^p} = |B(0,R)|^{1/p} \sim R^{n/p}$.)

15

Tempered Distributions and Sobolev Spaces

The foundational theory of distributions (test functions, distributions as continuous functionals, and the distributional derivative) is developed in the Functional Analysis part, chapter 9. This chapter adds the Fourier layer: tempered distributions and the Fourier transform, the H^s Sobolev scale, and singular integrals.

15.1 Tempered Distributions

In the Fourier transform chapter, we established the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ as the natural domain for the Fourier transform: the map $\varphi \mapsto \hat{\varphi}$ is an isomorphism of $\mathcal{S}$ onto itself. We now extend this theory to distributions, obtaining a framework that allows us to take the Fourier transform of objects like δ, constant functions, and polynomials.

The key observation is that $C_c^\infty \subsetneq \mathcal{S}$, and so the dual of $\mathcal{S}$ is *smaller* than $\mathcal{D}'$: every continuous linear functional on $\mathcal{S}$ restricts to one on C_c^∞, but not every distribution extends continuously to $\mathcal{S}$ (since $\mathcal{S}$ has a stronger topology than C_c^∞). The distributions that do extend are precisely those of "at most polynomial growth", which is exactly the right condition for the Fourier transform to make sense.

Definition 15.1.1: Tempered Distribution

A *tempered distribution* is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The space of all tempered distributions is denoted $\mathcal{S}'(\mathbb{R}^n)$, equipped with the weak* topology.

Concretely, $F \in \mathcal{S}'$ means: $F : \mathcal{S} \to \mathbb{C}$ is linear, and for each sequence $\varphi_j \to \varphi$ in $\mathcal{S}$ (meaning $\sup_x |x^\alpha \partial^\beta (\varphi_j - \varphi)(x)| \to 0$ for all α, β), we have $\langle F, \varphi_j \rangle \to \langle F, \varphi \rangle$. The analogue of Proposition 9.2.2 holds: $F \in \mathcal{S}'$ if and only if there exist $C > 0$ and $N \in \mathbb{N}$ such that

$$|\langle F, \varphi \rangle| \leq C \sum_{|\alpha|,|\beta| \leq N} \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta \varphi(x)|$$

for all $\varphi \in \mathcal{S}$. The proof is analogous to the one for $\mathcal{D}'$ (using the Schwartz space seminorms $\|\varphi\|_{\alpha,\beta} = \sup |x^\alpha \partial^\beta \varphi|$ in place of the $C_c^\infty(K)$ seminorms).

The following is a nice simplification for testing for tempered distribution

Proposition 15.1.2: Tempered Distribution Criterion

Let F be a tempered distribution. Then there exists a positive integer $N \in \mathbb{N}$ such that:

$$|F(\varphi)| \leq c\|\varphi\|_N \qquad \forall \varphi \in \mathcal{S}$$

***Proof*:**
Suppose otherwise. Then it must be that for each $n \in \mathbb{N}$, there is a $\psi_n \in \mathcal{S}$ with $\|\psi_n\|_n = 1$ and $|F(\psi_n)| \geq n$. Let

$$\varphi_n = \frac{\psi_n}{\sqrt{n}}$$

Then when $n \geq N$, $\|\varphi_n\|_N \leq \|\varphi_n\|_n$, and so:

$$\|\varphi_n\|_N \leq n^{-1/2} \xrightarrow{n \to \infty} 0$$

but then:

$$|F(\varphi_n) \geq n^{1/2} \xrightarrow{n \to \infty} \infty$$

contradicting continuity.

Note this is not the same as order as the norm combines polynomial growth information and growth-order information.

Since $C_c^\infty(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$ is dense, every tempered distribution restricts to a distribution, giving an inclusion $\mathcal{S}' \hookrightarrow \mathcal{D}'$. The inclusion is strict: for example, e^{x^2} defines a distribution (via $\varphi \mapsto \int e^{x^2} \varphi$, which converges for $\varphi \in C_c^\infty$ since φ has compact support) but not a tempered distribution (the integral $\int e^{x^2} \varphi$ may diverge for $\varphi \in \mathcal{S}$ that only decays rapidly).

Example 15.1.3: Tempered Distributions

1. *L^p functions.* Every $f \in L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$ defines a tempered distribution via $\varphi \mapsto \int f\varphi$. Indeed, $|\int f\varphi| \leq \|f\|_p \|\varphi\|_q$ by Hölder, and $\|\varphi\|_q$ is controlled by finitely many Schwartz seminorms. More generally, any function f with $|f(x)| \leq C(1 + |x|)^N$ for some C, N (i.e., of polynomial growth) and locally integrable defines a tempered distribution.

2. *The delta function and its derivatives.* Since $|\partial^\alpha \varphi(0)| \leq \sup |\partial^\alpha \varphi| \leq \|\varphi\|_{0,\alpha}$, the distributions $\varphi \mapsto \partial^\alpha \varphi(x_0)$ are tempered.

3. *p.v.$(1/x)$.* We showed earlier that $|\langle \text{p.v.}(1/x), \varphi \rangle| \leq C(\|\varphi\|_\infty + \|\varphi'\|_\infty)$. Since these are

Schwartz seminorms (with $\alpha = 0$), p.v.$(1/x) \in \mathcal{S}'(\mathbb{R})$.

4. *Polynomials.* Any polynomial $p(x)$ defines a tempered distribution. The estimate is $|\int p\varphi| \leq C \sup |x^N \varphi(x)|$ for large enough N.

The Fourier transform on $\mathcal{S}$ satisfies $\int \hat{f} g = \int f \hat{g}$ for $f, g \in \mathcal{S}$ (this is immediate from Fubini's theorem and the definition $\hat{f}(\xi) = \int f(x) e^{-2\pi i x\cdot\xi}\, dx$). This identity is exactly of the transpose form we exploited in Section 3: the "transpose" of the Fourier transform is the Fourier transform itself (on $\mathcal{S}$). This motivates:

Definition 15.1.4: Fourier Transform On $\mathcal{S}'$

For $F \in \mathcal{S}'(\mathbb{R}^n)$, define $\hat{F} \in \mathcal{S}'(\mathbb{R}^n)$ by

$$\langle \hat{F}, \varphi \rangle = \langle F, \hat{\varphi} \rangle \qquad (\varphi \in \mathcal{S}).$$

This is well-defined because $\hat{\varphi} \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$ (by Corollary 14.3.4), and F is a continuous functional on $\mathcal{S}$. Moreover, $\hat{F}$ is continuous on $\mathcal{S}$ since $\varphi_j \to \varphi$ in $\mathcal{S}$ implies $\hat{\varphi}_j \to \hat{\varphi}$ in $\mathcal{S}$ (the Fourier transform is a continuous map on $\mathcal{S}$).

The operational rules of the Fourier transform extend immediately to tempered distributions by the transpose principle:

Proposition 15.1.5: Properties Of The Fourier Transform On $\mathcal{S}'$

Let $F \in \mathcal{S}'(\mathbb{R}^n)$. Then:

1. $\widehat{\partial^\alpha F}(\xi) = (2\pi i \xi)^\alpha \hat{F}(\xi)$ as tempered distributions.
2. $\widehat{x^\alpha F} = \left(\frac{-1}{2\pi i}\right)^{|\alpha|} \partial^\alpha \hat{F}$.
3. (Fourier inversion) If we define $\check{F}$ by $\langle \check{F}, \varphi \rangle = \langle F, \check{\varphi} \rangle$ where $\check{\varphi}(\xi) = \hat{\varphi}(-\xi)$, then $\check{\hat{F}} = \hat{\check{F}} = F$.
4. The Fourier transform is an isomorphism from $\mathcal{S}'$ to $\mathcal{S}'$.

***Proof*:**
Each identity is verified by testing against $\varphi \in \mathcal{S}$ and using the corresponding identity on $\mathcal{S}$.

For (1): $\langle \widehat{\partial^\alpha F}, \varphi \rangle = \langle \partial^\alpha F, \hat{\varphi} \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \hat{\varphi} \rangle$. But by the properties of the Fourier transform on $\mathcal{S}$, $\partial^\alpha \hat{\varphi}(\xi) = \widehat{(-2\pi i x)^\alpha \varphi}(\xi)$, so:

$$= (-1)^{|\alpha|} \langle F, \widehat{(-2\pi i x)^\alpha \varphi} \rangle = (-1)^{|\alpha|} \langle \hat{F}, (-2\pi i x)^\alpha \varphi \rangle = \langle (2\pi i x)^\alpha \hat{F}, \varphi \rangle.$$

Replacing the dummy variable x by ξ, this reads $\widehat{\partial^\alpha F} = (2\pi i \xi)^\alpha \hat{F}$.

Part (2) is proved similarly, and (3) follows from Fourier inversion on $\mathcal{S}$ (theorem 14.3.2): $\langle \check{\hat{F}}, \varphi \rangle = \langle \check{F}, \hat{\varphi} \rangle = \langle F, \check{\hat{\varphi}} \rangle = \langle F, \varphi \rangle$, using $\check{\hat{\varphi}} = \varphi$. Part (4) follows from (3).

The payoff of tempered distribution is obvious: we can now compute Fourier transforms of more functions/distributions. Let us do so for the most important cases.

Example 15.1.6: Fourier Transforms Of Important Distributions

1. *The delta function.* For $\varphi \in \mathcal{S}$:

$$\langle \hat{\delta}, \varphi \rangle = \langle \delta, \hat{\varphi} \rangle = \hat{\varphi}(0) = \int \varphi(x)\, dx = \langle 1, \varphi \rangle.$$

   Therefore $\hat{\delta} = 1$: the Fourier transform of a point mass is the constant function. By Fourier inversion, $\hat{1} = \delta$: the Fourier transform of the constant function 1 is the delta function. This encodes the principle that "maximally concentrated in space $\Leftrightarrow$ maximally spread in frequency".

2. *The sign function.* From $\operatorname{sgn}'(x) = 2\delta$ and the identity $\widehat{f'} = 2\pi i \xi \hat{f}$, we get $2\pi i \xi \cdot \widehat{\operatorname{sgn}} = 2\hat{\delta} = 2$, and so $\widehat{\operatorname{sgn}}(\xi) = \frac{1}{\pi i \xi}$ in $\mathcal{S}'(\mathbb{R})$. More precisely, $\widehat{\operatorname{sgn}} = \frac{1}{\pi i}\,\text{p.v.}(1/\xi)$.

3. *The Heaviside function.* Since $H = \frac{1}{2}(\operatorname{sgn} + 1)$:

$$\hat{H} = \frac{1}{2}\widehat{\operatorname{sgn}} + \frac{1}{2}\hat{1} = \frac{1}{2\pi i}\,\text{p.v.}\frac{1}{\xi} + \frac{1}{2}\delta.$$

4. *Polynomials.* From $\hat{1} = \delta$ and the identity $\widehat{x^\alpha f} = (2\pi i)^{-|\alpha|}(-\partial)^\alpha \hat{f}$, we get

$$\widehat{x^\alpha} = \left(\frac{-1}{2\pi i}\right)^{|\alpha|} \partial^\alpha \delta.$$

   So the Fourier transform of a monomial is a derivative of δ.

15.2 Distributions With Compact Support

We introduced the notation $\mathcal{E}'(U)$ earlier for the set of distributions on U with compact support. This space turns out to have a very clean characterization: it is the dual of $C^\infty(U)$ (with *no* support restriction on the test functions).

Proposition 15.2.1: Characterization Of $\mathcal{E}'$

Let $F \in \mathcal{D}'(U)$. The following are equivalent:

1. F has compact support.
2. F extends (uniquely) to a continuous linear functional on $C^\infty(U)$.

Moreover, every continuous linear functional on $C^\infty(U)$ arises in this way.

***Proof*:**
(1) ⇒ (2): Suppose $\operatorname{supp}(F) \subseteq K$ for some compact $K \subseteq U$. Choose $\chi \in C_c^\infty(U)$ with $\chi = 1$ on a neighborhood of K. For any $\psi \in C^\infty(U)$, define $\langle F, \psi\rangle := \langle F, \chi\psi\rangle$. This is well-defined since $\chi\psi \in C_c^\infty(U)$, and it is independent of the choice of χ: if χ' is another such cutoff, then $\chi - \chi' = 0$ on a neighborhood of $K = \operatorname{supp}(F)$, so $\langle F, (\chi - \chi')\psi\rangle = 0$. Continuity on $C^\infty(U)$ follows from the continuity estimate on $C_c^\infty(\operatorname{supp}(\chi))$.

(2) ⇒ (1): Suppose F extends to a continuous functional on $C^\infty(U)$ but $\operatorname{supp}(F)$ is not compact. Then for each j, there exists $\varphi_j \in C_c^\infty(U)$ with $\operatorname{supp}(\varphi_j) \cap \{|x| > j\} \neq \emptyset$, $\operatorname{supp}(\varphi_j) \cap \operatorname{supp}(\varphi_k) = \emptyset$ for $j \neq k$ (by choosing supports far apart), and $\langle F, \varphi_j\rangle = 1$. But $\psi = \sum_j \varphi_j/j$ converges in $C^\infty(U)$ (since the supports are locally finite), and $\langle F, \psi\rangle = \sum_j \langle F, \varphi_j\rangle/j = \sum 1/j = \infty$, contradicting continuity.

The "moreover" part: given a continuous linear functional L on $C^\infty(U)$, its restriction to $C_c^\infty(U)$ is a distribution (since $C_c^\infty(K) \hookrightarrow C^\infty(U)$ is continuous for each compact K). The argument above shows this distribution has compact support, and L is its unique extension.

Proposition 15.2.2: Finite Order Of Compactly Supported Distributions

Every $F \in \mathcal{E}'(U)$ has *finite order*: there exists $N \in \mathbb{N}$ such that the continuity estimate

$$|\langle F, \varphi\rangle| \leq C \sum_{|\alpha| \leq N} \|\partial^\alpha \varphi\|_{L^\infty}$$

holds for *all* $\varphi \in C^\infty(U)$ (not just those supported in a particular compact set). The smallest such N is called the *order* of F.

***Proof*:**
Since $\operatorname{supp}(F) \subseteq K$ for some compact K, we have $\langle F, \varphi\rangle = \langle F, \chi\varphi\rangle$ for any cutoff $\chi \in C_c^\infty(U)$ with $\chi = 1$ near K. By the distribution continuity check (Proposition 9.2.2) applied on $\operatorname{supp}(\chi)$:

$$|\langle F, \varphi\rangle| = |\langle F, \chi\varphi\rangle| \leq C_{\operatorname{supp}(\chi)} \sum_{|\alpha| \leq N} \|\partial^\alpha(\chi\varphi)\|_{L^\infty}.$$

By the Leibniz rule, $\|\partial^\alpha(\chi\varphi)\|_{L^\infty} \leq C' \sum_{|\beta| \leq |\alpha|} \|\partial^\beta \varphi\|_{L^\infty(\operatorname{supp}(\chi))}$. The result follows.

The inclusion chain $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ now has a clean interpretation:

1. $\mathcal{D}' = (C_c^\infty)'$: all distributions, no restrictions.
2. $\mathcal{S}' = (\mathcal{S})'$: "at most polynomial growth" distributions—these are the ones compatible with the Fourier transform.
3. $\mathcal{E}' = (C^\infty)'$: compactly supported distributions—these are the most "localized".

A fact we will use later: if $F \in \mathcal{E}'(\mathbb{R}^n)$, then $\hat{F}$ is not a tempered distribution but an actual smooth function. In fact, $\hat{F}(\xi) = \langle F_x, e^{-2\pi i x\cdot\xi}\rangle$ (where the subscript indicates the variable of the pairing) is well-defined for all ξ since $x \mapsto e^{-2\pi i x\cdot\xi}$ is in C^∞ and F extends to C^∞. Moreover, $\hat{F}$ extends to an *entire* function on $\mathbb{C}^n$, and satisfies the Paley–Wiener estimate $|\hat{F}(\xi + i\eta)| \leq C(1 + |\xi| + |\eta|)^N e^{2\pi R|\eta|}$

when $\operatorname{supp}(F) \subseteq B_R$ (compare with Corollary 14.2.15, which established the analogous result for functions).

15.3 Singular Integrals And Calderón–Zygmund Theory

In Section 9.5, we encountered the principal value distribution p.v.$(1/x)$. This object is the kernel an important operators in harmonic analysis: the *Hilbert transform*. In this section, we study the Hilbert transform from the distributional perspective and then generalize to the Calderón–Zygmund theory of singular integral operators.

Definition 15.3.1: Hilbert Transform

For $f \in \mathcal{S}(\mathbb{R})$ (or more generally $f \in L^p(\mathbb{R})$ for $1 \leq p < \infty$), the *Hilbert transform* of f is defined by

$$Hf(x) = \text{p.v.}\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{f(y)}{x-y}\,dy = \frac{1}{\pi}\lim_{\varepsilon \to 0^+}\int_{|x-y|>\varepsilon} \frac{f(y)}{x-y}\,dy.$$

Equivalently, $Hf = f * K$ where $K = \frac{1}{\pi}\,\text{p.v.}(1/x)$ is the distributional convolution.

The Hilbert transform arises naturally in complex analysis (it relates the real and imaginary parts of a function holomorphic in the upper half-plane), in signal processing (it performs a 90ř phase shift), and in PDE theory (it is the prototypical example of a singular integral operator). Its distributional nature is essential: the kernel $1/(\pi x)$ is not locally integrable, so the convolution must be interpreted in the principal value sense.

The Fourier-analytic characterization of H is remarkably clean:

Theorem 15.3.2: Fourier Multiplier Characterization Of H

For $f \in \mathcal{S}(\mathbb{R})$:

$$\widehat{Hf}(\xi) = -i\,\operatorname{sgn}(\xi)\,\hat{f}(\xi).$$

That is, H is a Fourier multiplier with symbol $m(\xi) = -i\,\operatorname{sgn}(\xi)$.

***Proof*:**
Since $Hf = \frac{1}{\pi}\text{p.v.}(1/x) * f$ and convolution becomes multiplication under the Fourier transform (for tempered distributions), we have $\widehat{Hf} = \frac{1}{\pi}\widehat{\text{p.v.}(1/x)} \cdot \hat{f}$. From Example 15.1.6, $\widehat{\operatorname{sgn}}(\xi) = \frac{1}{\pi i}\text{p.v.}(1/\xi)$. Taking the inverse Fourier transform of both sides: $\operatorname{sgn}(x) = \frac{1}{\pi i}\,(\text{p.v.}(1/\cdot)\check{)}(x)$, which gives $\widehat{\text{p.v.}(1/x)}(\xi) = -\pi i\,\operatorname{sgn}(\xi)$. Therefore

$$\widehat{Hf} = \frac{1}{\pi}(-\pi i)\operatorname{sgn}(\xi)\hat{f}(\xi) = -i\,\operatorname{sgn}(\xi)\hat{f}(\xi).$$

Since $|-i\,\operatorname{sgn}(\xi)| = 1$ for $\xi \neq 0$, an immediate consequence is:

Corollary 15.3.3: L^2 Boundedness Of H

The Hilbert transform is an isometry on $L^2(\mathbb{R})$: $\|Hf\|_{L^2} = \|f\|_{L^2}$. Moreover, $H^2 = -\mathrm{Id}$ on L^2, so $H^{-1} = -H$.

***Proof*:**

By Plancherel: $\|Hf\|_{L^2}^2 = \|\widehat{Hf}\|_{L^2}^2 = \int |\operatorname{sgn}(\xi)|^2 |\hat{f}(\xi)|^2\, d\xi = \|\hat{f}\|_{L^2}^2 = \|f\|_{L^2}^2$. For the second claim: $\widehat{H^2 f} = (-i \operatorname{sgn})^2 \hat{f} = -\hat{f}$, so $H^2 f = -f$.

The result which lies at the heart of Calderón–Zygmund theory is that H extends to a bounded operator on L^p for all $1 < p < \infty$. This can be proved using the Riesz–Thorin interpolation theorem (theorem 8.5.2) to interpolate between the L^2 bound and a suitable weak-type estimate.

Theorem 15.3.4: L^p Boundedness Of The Hilbert Transform

For $1 < p < \infty$, the Hilbert transform extends to a bounded operator $H : L^p(\mathbb{R}) \to L^p(\mathbb{R})$. That is, there exists $C_p > 0$ such that $\|Hf\|_{L^p} \le C_p \|f\|_{L^p}$ for all $f \in L^p(\mathbb{R})$. The operator H is *not* bounded on L^1 or L^∞.

***Proof*:**

(Sketch) We already know H is bounded on L^2 with constant 1. For $p = 1$, one proves the *weak-type* $(1, 1)$ *estimate*: $m(\{x : |Hf(x)| > \lambda\}) \le C\|f\|_{L^1}/\lambda$ for all $\lambda > 0$. This is the delicate part, and it uses the Calderón–Zygmund decomposition of f into a "good" part (bounded by λ) and a "bad" part (concentrated on a set of measure $\lesssim \|f\|_{L^1}/\lambda$). Once this is established, the Marcinkiewicz interpolation theorem (or direct Riesz–Thorin interpolation between L^2 and the weak L^1 endpoint) gives boundedness on L^p for $1 < p \le 2$. Duality (since $H^* = -H$) extends this to $2 \le p < \infty$. The failure at $p = 1$ and $p = \infty$ can be seen from explicit examples (e.g., $H(\chi_{[0,1]})$ has a logarithmic singularity).

15.3.1 Calderón–Zygmund Kernels

The Hilbert transform is the one-dimensional prototype of a much larger class of operators. In $\mathbb{R}^n$ for $n \ge 2$, the analogous objects are the *Riesz transforms* $R_j f = c_n\, \mathrm{p.v.}(x_j/|x|^{n+1}) * f$, which are Fourier multipliers with symbol $-i\xi_j/|\xi|$. The general framework encompassing all of these is the Calderón–Zygmund theory.

Definition 15.3.5: Calderón–Zygmund Kernel

A *Calderón–Zygmund kernel* on $\mathbb{R}^n$ is a function $K : \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ satisfying:

1. (Size condition) $|K(x)| \le C|x|^{-n}$ for all $x \neq 0$.
2. (Smoothness condition) $|\nabla K(x)| \le C|x|^{-n-1}$ for all $x \neq 0$.
3. (Cancellation condition) $\sup_{0<r<R} \left| \int_{r<|x|<R} K(x)\, dx \right| < \infty$.

The associated *Calderón–Zygmund operator* is $Tf = \text{p.v.}(K) * f$, interpreted in the distributional sense.

The size condition says K has a singularity of order $|x|^{-n}$ at the origin—exactly on the boundary of local integrability in $\mathbb{R}^n$, which is why the principal value is necessary. The smoothness condition controls the regularity away from the origin, and the cancellation condition ensures that the principal value limit exists.

Theorem 15.3.6: Calderón–Zygmund Theorem

Let T be a Calderón–Zygmund operator on $\mathbb{R}^n$. If T is bounded on $L^2(\mathbb{R}^n)$, then T extends to a bounded operator on $L^p(\mathbb{R}^n)$ for all $1 < p < \infty$, and satisfies the weak-type $(1,1)$ estimate

$$m(\{x : |Tf(x)| > \lambda\}) \le \frac{C}{\lambda} \|f\|_{L^1}.$$

The proof of this theorem is one of the landmark achievements of 20th-century harmonic analysis. The key tool is the *Calderón–Zygmund decomposition*, which splits a function into "good" and "bad" parts at a given height λ, and then estimates each part separately. The good part is handled by the assumed L^2 bound, while the bad part is estimated using the smoothness and cancellation conditions on the kernel K. The full proof can be found in Stein's *Singular Integrals and Differentiability Properties of Functions*, Chapter II. The connection to distribution theory is fundamental: the operators are defined by convolution with singular distributions, and their Fourier multiplier characterization (obtained via the Fourier transform on $\mathcal{S}'$) is the primary tool for establishing L^2 boundedness.

15.3.7 From Hilbert Transforms to the Langlands Program The Hilbert transform and Calderón–Zygmund theory may seem like purely analytic tools, but they have connections to number theory and representation theory. In the theory of automorphic forms, the intertwining operators that relate different representations of a reductive group are singular integrals of Calderón–Zygmund type, defined on the group manifold rather than $\mathbb{R}^n$. The L^p boundedness of these operators is closely related to the analytic continuation of Eisenstein series, a central construction in the Langlands program. In this way, the distributional framework developed here—principal values, Fourier multipliers, and singular kernels—extends far beyond classical analysis into the arithmetic world.

15.4 Homogeneous Distributions

These seem interesting, may cover them.

15.5 Sobolev Spaces

Sobolev spaces spaces quantify "how many derivatives a function has" in the L^2 sense, and they form the natural setting for weak solutions to differential equations. The theory of tempered distributions and the Fourier transform from the preceding sections will play a central role.

Definition 15.5.1: Sobolev Spaces (H^s)

For $s \in \mathbb{R}$, the *Sobolev space* $H^s(\mathbb{R}^n)$ is defined by

$$H^s(\mathbb{R}^n) = \left\{ f \in \mathcal{S}'(\mathbb{R}^n) : (1 + |\xi|^2)^{s/2} \hat{f}(\xi) \in L^2(\mathbb{R}^n) \right\},$$

equipped with the inner product

$$\langle f, g \rangle_{H^s} = \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(\xi)} (1 + |\xi|^2)^s \, d\xi$$

and corresponding norm $\|f\|_{H^s} = \langle f, f \rangle_{H^s}^{1/2}$.

The weight $(1 + |\xi|^2)^{s/2}$ is called the *Japanese bracket* and is sometimes written $\langle \xi \rangle^s$. The intuition is: requiring $(1 + |\xi|^2)^{s/2} \hat{f} \in L^2$ means that $\hat{f}$ must decay like $|\xi|^{-s}$ faster than a typical L^2 function. Since high-frequency components of $\hat{f}$ correspond to rapid oscillations (i.e., derivatives) of f, imposing more decay on $\hat{f}$ at high frequencies is equivalent to requiring f to have more derivatives. When s is a non-negative integer, this coincides with the classical requirement that f and its first s partial derivatives all lie in L^2.

Proposition 15.5.2: Properties of H^s

1. $H^s(\mathbb{R}^n)$ is a Hilbert space for every $s \in \mathbb{R}$.
2. $H^0(\mathbb{R}^n) = L^2(\mathbb{R}^n)$, with equality of norms (by Plancherel).
3. If $s > t$, then $H^s(\mathbb{R}^n) \hookrightarrow H^t(\mathbb{R}^n)$ continuously: $\|f\|_{H^t} \le \|f\|_{H^s}$.
4. $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$ for every s.
5. For $s \ge 0$ an integer, $f \in H^s(\mathbb{R}^n)$ if and only if $\partial^\alpha f \in L^2(\mathbb{R}^n)$ for all $|\alpha| \le s$ (where derivatives are taken in the distributional sense). The norms are equivalent:
$$\|f\|_{H^s} \asymp \left(\sum_{|\alpha| \le s} \|\partial^\alpha f\|_{L^2}^2 \right)^{1/2}.$$
6. (Duality) $(H^s)^* \cong H^{-s}$ via the L^2 pairing $\langle f, g\rangle = \int \hat{f}\overline{\hat{g}}$.
7. Differentiation is a bounded operator: $\partial^\alpha : H^s \to H^{s-|\alpha|}$ with $\|\partial^\alpha f\|_{H^{s-|\alpha|}} \le C\|f\|_{H^s}$.

***Proof*:**

(1) The map $f \mapsto (1+|\xi|^2)^{s/2}\hat{f}$ is an isometry from H^s into L^2. Since L^2 is complete and the map is an isometry onto a closed subspace (in fact, onto all of L^2, since for any $g \in L^2$ the function f with $\hat{f} = g/(1+|\xi|^2)^{s/2}$ lies in H^s and maps to g), H^s is complete.

(2) is immediate from the definition: $(1+|\xi|^2)^0 = 1$, so $\|f\|_{H^0} = \|\hat{f}\|_{L^2} = \|f\|_{L^2}$.

(3) For $s > t$: $(1+|\xi|^2)^{t/2} \le (1+|\xi|^2)^{s/2}$, so $\|f\|_{H^t} \le \|f\|_{H^s}$.

(4) Given $f \in H^s$, let $\chi_R \in C_c^\infty$ with $\chi_R = 1$ on B_R and $0 \le \chi_R \le 1$. Define f_R by $\hat{f}_R = \chi_R \hat{f}$. Then $\hat{f}_R \in C_c^\infty \subset \mathcal{S}$, so $f_R \in \mathcal{S}$ (by the Fourier isomorphism on $\mathcal{S}$). And $\|f - f_R\|_{H^s}^2 = \int_{|\xi|>R} |\hat{f}|^2(1+|\xi|^2)^s \, d\xi \to 0$ as $R \to \infty$ by dominated convergence.

(5) For integer $s \ge 0$: $\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$, so $\|\partial^\alpha f\|_{L^2}^2 = (2\pi)^{2|\alpha|} \int |\xi^\alpha|^2 |\hat{f}|^2 \, d\xi$. Since $\sum_{|\alpha| \le s} |\xi^\alpha|^2 \asymp (1+|\xi|^2)^s$ (both are polynomials of degree $2s$ with the same leading behavior), the norm equivalence follows.

(6) The pairing $\langle f, g\rangle = \int \hat{f}\overline{\hat{g}}$ gives $|\langle f, g\rangle| \le \|f\|_{H^s}\|g\|_{H^{-s}}$ by Cauchy–Schwarz. That every continuous linear functional on H^s is of this form follows from the Riesz representation theorem on the Hilbert space L^2 after conjugating by the isometry $f \mapsto \langle \xi \rangle^s \hat{f}$.

(7) $\|\partial^\alpha f\|_{H^{s-|\alpha|}}^2 = \int |(2\pi i\xi)^\alpha|^2 |\hat{f}|^2 (1+|\xi|^2)^{s-|\alpha|} \, d\xi \le C \int |\hat{f}|^2 (1+|\xi|^2)^s \, d\xi = C\|f\|_{H^s}^2$, since $|\xi^\alpha|^2(1+|\xi|^2)^{-|\alpha|} \le C$.

The most important result about Sobolev spaces is the embedding theorem, which says that if s is large enough, then functions in H^s are not just L^2 but actually continuous (and even classically differentiable).

Theorem 15.5.3: Sobolev Embedding Theorem

If $s > k + n/2$ for some non-negative integer k, then $H^s(\mathbb{R}^n) \hookrightarrow C^k(\mathbb{R}^n)$ (the space of k-times continuously differentiable functions that vanish at infinity). Moreover, the embedding is continuous: there exists $C = C(s, k, n) > 0$ such that

$$\sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^\infty} \leq C\|f\|_{H^s}$$

for all $f \in H^s(\mathbb{R}^n)$.

***Proof*:**
It suffices to prove the case $k = 0$ (i.e., $s > n/2$ implies $H^s \hookrightarrow C^0$ with $\|f\|_\infty \leq C\|f\|_{H^s}$); the general case follows by applying this to $\partial^\alpha f \in H^{s-|\alpha|}$ with $s - |\alpha| > n/2$ (which holds when $s > k + n/2$ and $|\alpha| \leq k$).

By the Fourier inversion formula (applied to $f \in \mathcal{S}$, then extended by density): $f(x) = \int \hat{f}(\xi)e^{2\pi i x \cdot \xi}\, d\xi$, and so

$$|f(x)| \leq \int |\hat{f}(\xi)|\, d\xi = \int |\hat{f}(\xi)|(1 + |\xi|^2)^{s/2} \cdot (1 + |\xi|^2)^{-s/2}\, d\xi.$$

By Cauchy–Schwarz:

$$|f(x)| \leq \left(\int |\hat{f}(\xi)|^2(1 + |\xi|^2)^s\, d\xi\right)^{1/2} \cdot \left(\int (1 + |\xi|^2)^{-s}\, d\xi\right)^{1/2} = \|f\|_{H^s} \cdot C_s.$$

The constant $C_s = \left(\int (1 + |\xi|^2)^{-s}\, d\xi\right)^{1/2}$ is finite precisely when $2s > n$, i.e., $s > n/2$: switching to polar coordinates, the integral behaves like $\int_0^\infty r^{n-1}(1 + r^2)^{-s}\, dr$, which converges if and only if $2s - n + 1 > 1$.

This gives $\|f\|_\infty \leq C_s\|f\|_{H^s}$ for $f \in \mathcal{S}$. Since $\mathcal{S}$ is dense in H^s, every $f \in H^s$ is the H^s-limit of a sequence $f_j \in \mathcal{S}$, and the estimate $\|f_j - f_k\|_\infty \leq C_s\|f_j - f_k\|_{H^s} \to 0$ shows that f_j converges uniformly. The uniform limit is a continuous function that agrees with f as an element of L^2. The vanishing at infinity follows from the density of C_c^∞ in H^s.

The threshold $s = n/2$ is sharp, as the following example shows.

Example 15.5.4: The Step Function (The Boundary Case)
On $\mathbb{R}$, the threshold for $H^s(\mathbb{R}) \hookrightarrow C^0(\mathbb{R})$ is $s > 1/2$. Consider the Heaviside function $H(x) = \mathbf{1}_{[0,\infty)}(x)$. We showed that $\hat{H}(\xi) = \frac{1}{2}\delta(\xi) + \frac{1}{2\pi i}\text{p.v.}(1/\xi)$, and more precisely, for a smooth truncation $H_R(x) = H(x)\chi_{[-R,R]}(x)$ (where χ is a smooth cutoff), one has $|\hat{H}_R(\xi)| \sim 1/|\xi|$ for large $|\xi|$.

Then $\|H_R\|_{H^s}^2 \sim \int |\xi|^{-2}(1 + |\xi|^2)^s\, d\xi \sim \int |\xi|^{2s-2}\, d\xi$, which converges near $\xi = 0$ (no issue) but diverges at infinity when $2s - 2 \geq -1$, i.e., $s \geq 1/2$.

Therefore $H \notin H^{1/2}(\mathbb{R})$, and a fortiori $H \notin H^s(\mathbb{R})$ for $s \geq 1/2$. The jump discontinuity at $x = 0$ prevents H from lying in H^s for s at or above the embedding threshold. This illustrates the sharpness of the Sobolev embedding theorem: at the boundary $s = n/2$, the embedding into C^0 fails, and discontinuous functions can sneak in.

15.5.5 Hearing the Shape of a Fractal Mark Kac's famous question "Can one hear the shape of a drum?" asks whether the eigenvalues of the Laplacian on a bounded domain Ω determine Ω up to isometry. In 1911, Hermann Weyl showed that the eigenvalue counting function $N(\lambda) = \#\{\lambda_j \le \lambda\}$ satisfies the asymptotic law $N(\lambda) \sim c_n \operatorname{vol}(\Omega)\lambda^{n/2}$ as $\lambda \to \infty$—so one can at least "hear" the volume and dimension.

Sobolev spaces enter the picture through the connection between eigenvalue asymptotics and regularity. The Laplacian $-\Delta$ on Ω with Dirichlet boundary conditions is a self-adjoint operator on $L^2(\Omega)$, and its eigenfunctions lie in $H^s(\Omega)$ for all s (by elliptic regularity). The Sobolev embedding theorem then controls the L^∞ norms of eigenfunctions in terms of their H^s norms, which in turn controls the growth of the eigenvalue counting function.

When $\partial\Omega$ is a fractal (say, the von Koch snowflake), Weyl's law acquires a correction term involving the Minkowski content and dimension of the boundary. In Lapidus's theory of "fractal drums", the modified Weyl law takes the form $N(\lambda) = c_n \operatorname{vol}(\Omega)\lambda^{n/2} - c_{n-1}\mathcal{M}^d(\partial\Omega)\lambda^{d/2} + o(\lambda^{d/2})$, where d is the Minkowski dimension of $\partial\Omega$ and $\mathcal{M}^d$ is the Minkowski content. The spectral theory of the Laplacian, Sobolev regularity, and the fractal geometry of the boundary are thus intimately intertwined. In this way, distribution theory reaches from the abstract world of generalized functions all the way to the geometry of fractals.

15.6 Applications To Partial Differential Equations

Distributions were invented by Laurent Schwartz in the 1940s precisely to provide a rigorous framework for the generalized solutions that physicists had been using informally for decades. In this section, we develop the basic theory of constant-coefficient linear PDEs from the distributional perspective, culminating in the Malgrange–Ehrenpreis theorem: every such equation has a fundamental solution.

15.6.1 Fundamental Solutions

Consider a constant-coefficient linear differential operator

$$P(\partial) = \sum_{|\alpha| \le m} a_\alpha \partial^\alpha, \qquad a_\alpha \in \mathbb{C}.$$

The associated *symbol* is the polynomial $p(\xi) = \sum_{|\alpha| \le m} a_\alpha (2\pi i \xi)^\alpha$, chosen so that $\widehat{P(\partial)f} = p \cdot \hat{f}$ for $f \in \mathcal{S}'$. Given a source term g, we wish to solve the equation $P(\partial)u = g$.

Definition 15.6.1: Fundamental Solution

A *fundamental solution* (or *Green's function*) for $P(\partial)$ is a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ satisfying

$$P(\partial)E = \delta.$$

If such an E exists and g is a distribution for which the convolution $E * g$ is well-defined (for example, if $g \in \mathcal{E}'$ or $g \in L^p$ with compact support), then $u = E * g$ satisfies $P(\partial)u = P(\partial)(E * g) = (P(\partial)E) * g =$

$\delta * g = g$, so u is a distributional solution. Thus, the problem of solving all constant-coefficient PDEs reduces to finding a single distribution E.

Example 15.6.2: Fundamental Solutions Of Classical Operators

1. *The Laplacian in* $\mathbb{R}^n$ *(*$n \geq 3$*).* For $\Delta = \sum_{j=1}^n \partial_j^2$, the fundamental solution is $E(x) = c_n|x|^{2-n}$, where $c_n = 1/(n(2-n)\omega_n)$ and ω_n is the volume of the unit ball. This is locally integrable (since $2-n > -n$ for $n \geq 3$), and the identity $\Delta E = \delta$ is verified by a direct computation: for $\varphi \in C_c^\infty$, $\langle \Delta E, \varphi\rangle = \langle E, \Delta\varphi\rangle = c_n \int |x|^{2-n}\Delta\varphi(x)\,dx$, and integrating by parts on $\{|x| > \varepsilon\}$ and letting $\varepsilon \to 0$ yields $\varphi(0)$ (the boundary terms on the sphere $|x| = \varepsilon$ contribute $\varphi(0)$ in the limit by the mean value property and the computation of the surface integral).

2. *The Laplacian in* $\mathbb{R}^2$. Here $E(x) = \frac{1}{2\pi}\log|x|$, and the verification is similar.

3. *The heat operator.* For $P(\partial) = \partial_t - \Delta_x$ on $\mathbb{R}^{n+1}$, the fundamental solution is the heat kernel $E(x,t) = (4\pi t)^{-n/2}e^{-|x|^2/4t}$ for $t > 0$, extended by 0 for $t < 0$.

15.6.2 The Malgrange–Ehrenpreis Theorem

The examples above might suggest that finding fundamental solutions requires ad hoc computations tailored to each operator. The remarkable theorem of Malgrange (1955) and Ehrenpreis (1954), proved independently, shows that this is not the case: *every* nonzero constant-coefficient operator has a fundamental solution. This is one of the great triumphs of distribution theory.

The proof uses the Fourier transform to convert the PDE $P(\partial)E = \delta$ into an algebraic problem: taking Fourier transforms gives $p(\xi)\hat{E}(\xi) = 1$, so we need to "divide by p" in $\mathcal{S}'$. The difficulty is that p may have real zeros. The key algebraic lemma shows that this division is always possible.

Lemma 15.6.3: Division By Linear Factors In $\mathcal{S}'(\mathbb{R})$

Let $a \in \mathbb{C}$. For every $f \in \mathcal{S}'(\mathbb{R})$, the equation $(\xi - a)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R})$.

***Proof*:**
If $a \notin \mathbb{R}$, then $1/(\xi - a)$ is a C^∞ function on $\mathbb{R}$ with all derivatives bounded by powers of $1/|\operatorname{Im}(a)|$, so $u = f/(\xi - a)$ (defined by $\langle u, \varphi\rangle = \langle f, \varphi/(\xi - a)\rangle$) is a well-defined tempered distribution.

If $a \in \mathbb{R}$, we may assume $a = 0$ by translation (replacing $\varphi(\xi)$ by $\varphi(\xi + a)$). We must solve $\xi \cdot u = f$, that is:

$$\langle u, \xi\varphi(\xi)\rangle = \langle f, \varphi(\xi)\rangle \qquad \text{for all } \varphi \in \mathcal{S}.$$

The map $\varphi \mapsto \xi\varphi$ sends $\mathcal{S}$ into itself, but is not surjective: its range is $\mathcal{S}_0 = \{\psi \in \mathcal{S} : \psi(0) = 0\}$, a closed subspace of codimension 1.

We define u on $\mathcal{S}_0$ as follows. For $\psi \in \mathcal{S}_0$, the function $\psi(\xi)/\xi$ is smooth on $\mathbb{R}\setminus\{0\}$ and extends smoothly to all of $\mathbb{R}$ (since $\psi(0) = 0$: write $\psi(\xi) = \xi\int_0^1 \psi'(t\xi)\,dt$, so $\psi(\xi)/\xi = \int_0^1 \psi'(t\xi)\,dt \in C^\infty$). Moreover, $\psi/\xi \in \mathcal{S}$: for large $|\xi|$, $|\xi^k\partial^j(\psi/\xi)| \lesssim |\xi|^{k-1}|\partial^j\psi| + \text{lower} \to 0$ since $\psi \in \mathcal{S}$, and near $\xi = 0$, smoothness was just established.

So for $\psi \in \mathcal{S}_0$, define $\langle u, \psi\rangle = \langle f, \psi/\xi\rangle$. This satisfies $\langle u, \xi\varphi\rangle = \langle f, \varphi\rangle$ for all $\varphi \in \mathcal{S}$ (since

$\xi\varphi \in \mathcal{S}_0$ and $(\xi\varphi)/\xi = \varphi$).

To extend u to all of $\mathcal{S}$, fix any $\varphi_0 \in \mathcal{S}$ with $\varphi_0(0) = 1$. Every $\psi \in \mathcal{S}$ decomposes uniquely as $\psi = (\psi - \psi(0)\varphi_0) + \psi(0)\varphi_0$, where $\psi - \psi(0)\varphi_0 \in \mathcal{S}_0$. Define

$$\langle u, \psi\rangle = \langle f, (\psi - \psi(0)\varphi_0)/\xi\rangle + c \cdot \psi(0)$$

for an arbitrary constant $c \in \mathbb{C}$. This is continuous on $\mathcal{S}$ (both $\psi \mapsto \langle f, (\psi - \psi(0)\varphi_0)/\xi\rangle$ and $\psi \mapsto \psi(0)$ are continuous), and satisfies $\xi u = f$ by construction. The constant c reflects the non-uniqueness: if u_1 and u_2 both solve $\xi u = f$, then $\xi(u_1 - u_2) = 0$, so $u_1 - u_2$ is supported at $\{0\}$, hence is a linear combination of δ and its derivatives—but the equation $\xi v = 0$ in $\mathcal{S}'(\mathbb{R})$ forces $v = c\delta$ for some c (since $\langle v, \xi\varphi\rangle = 0$ for all φ, meaning v vanishes on $\mathcal{S}_0$, and $\mathcal{S}_0$ has codimension 1 with the complementary functional being evaluation at 0, i.e., δ).

Lemma 15.6.4: Division By Polynomials In $\mathcal{S}'(\mathbb{R})$

Let $p(\xi)$ be a nonzero polynomial on $\mathbb{R}$. For every $f \in \mathcal{S}'(\mathbb{R})$, the equation $p(\xi)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R})$.

***Proof*:**
Factor $p(\xi) = c\prod_{j=1}^{m}(\xi - a_j)$ where $a_1, \ldots, a_m \in \mathbb{C}$ are the roots (counted with multiplicity). Apply lemma 15.6.3 inductively: first solve $(\xi - a_m)u_1 = f$, then $(\xi - a_{m-1})u_2 = u_1$, and so on. After m steps, we obtain $u = u_m/c$ with $p(\xi)u = f$.

To extend this to several variables, we use the following observation: division by a polynomial in $\mathbb{R}^n$ can be reduced to division in one variable by treating the remaining variables as parameters.

Lemma 15.6.5: Division By Polynomials In $\mathcal{S}'(\mathbb{R}^n)$

Let $p(\xi) = p(\xi_1, \ldots, \xi_n)$ be a nonzero polynomial on $\mathbb{R}^n$. For every $f \in \mathcal{S}'(\mathbb{R}^n)$, the equation $p(\xi)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R}^n)$.

***Proof*:**
We proceed by induction on n. The case $n = 1$ is lemma 15.6.4.

For $n \geq 2$, write $\xi = (\xi_1, \xi') \in \mathbb{R} \times \mathbb{R}^{n-1}$. After a linear change of coordinates (which preserves $\mathcal{S}'$), we may assume that p has degree m in ξ_1:

$$p(\xi_1, \xi') = a_m(\xi')\xi_1^m + a_{m-1}(\xi')\xi_1^{m-1} + \cdots + a_0(\xi')$$

where a_m is a nonzero polynomial in ξ' (this can always be arranged by a generic rotation).

We proceed in two stages. First, perform polynomial long division in ξ_1: using the Euclidean algorithm for polynomials in ξ_1 (with coefficients that are distributions in ξ'), we can reduce to the case where we divide by $a_m(\xi')$, which is a polynomial in $n-1$ variables, and then divide by a monic polynomial in ξ_1 (whose roots are the roots of p as a polynomial in ξ_1).

More precisely, here is the argument. Define the "partial Fourier transform" in ξ_1 and use the one-dimensional division lemma for each fixed ξ'. The subtlety is ensuring the result depends measurably (and in fact, in a tempered fashion) on ξ'.

We present the clean version. The polynomial p can be written as

$$p(\xi_1, \xi') = a_m(\xi') \prod_{j=1}^{m} (\xi_1 - r_j(\xi')),$$

where $r_j(\xi')$ are the roots in $\mathbb{C}$ (which are algebraic functions of ξ'). To solve $pu = f$, it suffices to solve $a_m(\xi')v = f$ (where v and f are tempered distributions on $\mathbb{R}^n$, and $a_m(\xi')$ acts by multiplication in the ξ' variables), and then solve $\prod_j (\xi_1 - r_j(\xi'))w = v$ by dividing by each linear factor in ξ_1 successively.

For the first step: $a_m(\xi')$ is a nonzero polynomial on $\mathbb{R}^{n-1}$. By the inductive hypothesis, the division $a_m(\xi')v = g$ can be solved in $\mathcal{S}'(\mathbb{R}^{n-1})$ for any tempered distribution g on $\mathbb{R}^{n-1}$. Extending this to $\mathbb{R}^n$ (treating ξ_1 as a parameter) solves $a_m(\xi')v = f$ in $\mathcal{S}'(\mathbb{R}^n)$.

For the second step: each factor $(\xi_1 - r_j(\xi'))$ involves division in the ξ_1 variable. When $r_j(\xi')$ is not real for a given ξ', the division is trivial (divide by a nonvanishing smooth function). When $r_j(\xi')$ is real, we apply the one-dimensional division lemma. The argument from lemma 15.6.3 extends to the parametric setting: the decomposition $\psi(\xi_1) = (\psi(\xi_1) - \psi(r_j(\xi'))\varphi_0(\xi_1)) + \psi(r_j(\xi'))\varphi_0(\xi_1)$ depends smoothly on the parameter ξ' (using the smooth dependence of roots on coefficients away from coalescence points, and handling the coalescence points by a partition of unity argument).

The full technical details require verifying that each step produces a tempered distribution (not a distribution), which follows from the polynomial growth of all quantities involved. We refer to theorem 7.3.2 in Hörmander's *The Analysis of Linear Partial Differential Operators I* for the complete argument in the parametric case.

With the division lemma in hand, the Malgrange–Ehrenpreis theorem is an immediate consequence.

Theorem 15.6.6: The Malgrange–Ehrenpreis Theorem

Let $P(\partial) = \sum_{|\alpha| \le m} a_\alpha \partial^\alpha$ be a nonzero constant-coefficient linear differential operator on $\mathbb{R}^n$. Then there exists a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ such that

$$P(\partial)E = \delta.$$

***Proof*:**

Let $p(\xi) = \sum_{|\alpha| \le m} a_\alpha (2\pi i \xi)^\alpha$ be the symbol of $P(\partial)$, so that $\widehat{P(\partial)f} = p \cdot \hat{f}$ for $f \in \mathcal{S}'$. Taking Fourier transforms, the equation $P(\partial)E = \delta$ becomes

$$p(\xi)\hat{E}(\xi) = \hat{\delta}(\xi) = 1.$$

By lemma 15.6.5, the equation $p(\xi)u = 1$ has a solution $u \in \mathcal{S}'(\mathbb{R}^n)$ (taking $f = 1$, which is a tempered distribution). Define $E = \check{u}$ (the inverse Fourier transform of u). Then $E \in \mathcal{S}' \subset \mathcal{D}'$ and

$$\widehat{P(\partial)E} = p \cdot \hat{E} = p \cdot u = 1 = \hat{\delta},$$

so $P(\partial)E = \delta$ by the injectivity of the Fourier transform on $\mathcal{S}'$.

Let us pause to appreciate what we have proved. The Malgrange–Ehrenpreis theorem says that the purely algebraic problem of dividing by a polynomial in $\mathcal{S}'$ translates, via the Fourier transform, into the analytical statement that every constant-coefficient PDE has a distributional fundamental solution. The entire proof rests on three results: the Fourier transform on $\mathcal{S}'$ (Section 15.1), the algebraic structure of polynomial division, and the one-dimensional division lemma (lemma 15.6.3), which itself is a simple consequence of the codimension-1 structure of the subspace $\{\psi \in \mathcal{S} : \psi(0) = 0\}$.

Corollary 15.6.7: Solvability Of Constant-Coefficient PDEs

Let $P(\partial)$ be a nonzero constant-coefficient operator and let $g \in \mathcal{E}'(\mathbb{R}^n)$ (a compactly supported distribution). Then the equation $P(\partial)u = g$ has a solution $u \in \mathcal{D}'(\mathbb{R}^n)$.

***Proof*:**
Let E be the fundamental solution from the Malgrange–Ehrenpreis theorem. Since $g \in \mathcal{E}'$, the convolution $u = E * g$ is well-defined as a distribution (the compact support of g ensures that $\langle E * g, \varphi\rangle = \langle E_x, \langle g_y, \varphi(x+y)\rangle\rangle$ converges for every $\varphi \in C_c^\infty$). Then

$$P(\partial)u = P(\partial)(E * g) = (P(\partial)E) * g = \delta * g = g.$$

Example 15.6.8: Revisiting The Laplacian
For $P(\partial) = \Delta = \sum_j \partial_j^2$ on $\mathbb{R}^n$, the symbol is $p(\xi) = -4\pi^2|\xi|^2$. The equation $-4\pi^2|\xi|^2\hat{E} = 1$ cannot be solved by simply writing $\hat{E} = -1/(4\pi^2|\xi|^2)$, since this function is not locally integrable near $\xi = 0$ for $n \leq 2$. However, the division lemma constructs a distributional solution. In this case, the solution can also be obtained by regularization: for instance, $\hat{E}(\xi) = \lim_{\varepsilon\to 0^+} -1/(4\pi^2(|\xi|^2 + \varepsilon))$, where the limit is taken in $\mathcal{S}'$. The inverse Fourier transform recovers the classical fundamental solution: $E(x) = c_n|x|^{2-n}$ for $n \geq 3$ and $E(x) = \frac{1}{2\pi}\log|x|$ for $n = 2$.

15.6.9 Distribution Theory in Context The Malgrange–Ehrenpreis theorem is an existence theorem: it guarantees a fundamental solution exists, but says nothing about its regularity, uniqueness (fundamental solutions are not unique in general—one can always add a solution of the homogeneous equation), or how it depends on the operator. For *elliptic* operators (those whose symbol satisfies $|p(\xi)| \geq c|\xi|^m$ for large $|\xi|$), the theory of elliptic regularity shows that distributional solutions are automatically smooth wherever the source term is smooth. For *hyperbolic* operators (like the wave operator), the fundamental solution has singularities propagating along characteristic surfaces. These deeper questions—regularity, propagation of singularities, and the fine structure of solutions—form the subject of microlocal analysis, which extends distribution theory by localizing in both position and frequency. The wavefront set of a distribution, a subset of the cotangent bundle $T^*\mathbb{R}^n$, provides the precise language for these phenomena, and opens the door to the modern theory of pseudodifferential operators and Fourier integral operators.

Part IV

Abstract Harmonic Analysis on LCA Groups

The second part is the structural counterpart of the first. The Fourier transform of part III diagonalizes the translation-invariant operators on $\mathbb{R}^n$, and the same idea applies verbatim to any locally compact abelian group, with the frequencies $\xi \in \mathbb{R}^n$ replaced by the characters of the group. The dual group $\hat{G}$ collects these characters, the Fourier transform carries functions on G to functions on $\hat{G}$, and Pontryagin duality $\hat{\hat{G}} \cong G$ is the statement that the passage loses nothing. The non-abelian generalization, in which characters give way to higher-dimensional unitary representations, is the province of the companion theory of unitary representations, for which this part is the abelian model.

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Abstract Harmonic Analysis on LCA Groups

This part carries the Fourier transform from Euclidean space to its natural home, a locally compact abelian group. The classical transform of Part I is the special case $G = \mathbb{R}^n$; the general theory replaces $\mathbb{R}^n$ by an arbitrary LCA group, its frequencies by the characters of that group, and reveals the structural fact hiding behind Fourier inversion and Plancherel, the Pontryagin duality between a group and its dual.

16.1 Locally Compact Groups and Haar Measure, Recalled

The two ingredients this theory rests on are developed in full elsewhere in the series; this section fixes notation and recalls what is used.

The point-set theory of topological and locally compact groups is [7]: a topological group is a group carrying a topology for which multiplication and inversion are continuous, and it is locally compact when the identity has a neighbourhood with compact closure. The standing examples are $\mathbb{R}^n$ and $\mathbb{R}^\times$, the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and its powers $\mathbb{T}^n$, the discrete groups $\mathbb{Z}^n$, and the p-adic groups $\mathbb{Z}_p$ and $\mathbb{Q}_p$; all are abelian, which is the standing hypothesis of this part.

The invariant measure that makes integration possible is Haar measure, constructed and proved unique up to a positive scalar in chapter 6: every locally compact Hausdorff group carries a nonzero left-invariant Radon measure μ (theorem 6.1.8), and any two differ only by scale (theorem 6.2.1). On an abelian group left and right translation coincide, so a Haar measure is automatically bi-invariant and the group is unimodular. The normalizations of the standard examples, all instances of the general theorem, are Lebesgue measure on $\mathbb{R}^n$; arc length of total mass 1 on $\mathbb{T}$; $d\mu(t) = dt/|t|$ on $\mathbb{R}^\times$; and counting measure on a discrete group. Fixing such a μ, the Haar integral $\int_G f\, d\mu$ is the invariant averaging on which the Fourier transform is built, and left- and right-uniform continuity of functions in $C_c(G)$ (lemma 6.1.1) is used without further comment.

16.2 Pontryagin Duality

In order to define the Fourier transformation on groups, we require the notion of *topological groups* which will allow us to define a Borel σ-algebra on which we will define a measure known as the *Haar measure*. With a measure, we may define the notion of *integrable function on a group G*. We next require the generalization of the domain of $\hat{f}$. When working with $\mathbb{T}^n$, the answer was "obvious" in that we had $\hat{f} : \mathbb{Z}^n \to \mathbb{R}$ since $\hat{f}$ enumerated the orthonormal basis of $L^2(\mathbb{T}^n)$. When we generlized the Fourier transform to functions $f \in L^p(\mathbb{R}^n)$ (for $1 \leq p \leq 2$), we had that the domain of $\hat{f}$ was $\mathbb{R}^n$, and in general the domain of $\hat{f}$ transformed like *functionals*. We will explore this concept when we define *Pontryagin Duality*.

After having introduced the relevant concepts, we shall introduce Fourier transforms on locally compact (Hausdorff) abelian group. Due to the Pontryagin duality, many classical results of Fourier Analysis are readily recovered. We shall not go over too much detail as this is more an essay than the beginning of a course in Harmonic Analysis, however some interesting highlights will be added.

The paper will end with some interesting uses and consequence of the Fourier transform and of the Pontryagin duality, including how it demonstrates compact abelian groups are just as hard to classify as abelian groups, how the Fourier Transform shows up in Representation Theory of finite groups, how Pontryagin Duality allows us to classify locally compact abelian groups through their characters.

A quick disclaimer that I will be assuming some 1st-year graduate analysis knowledge for this paper as I will have to touch on topics covered in those courses.

Let G be a locally compact (Hausdorff) abelian group. If G was also a topological vector space (i.e. a topological field k acts continuously on G so that G becomes a topological vector space), then we would have a natural notion of duality since G would be a vector space over some underlying field k. For general group, no such "natural dualizing" object exists. Instead, we may try to find some interesting group so that the contravariant functor between locally compact abelian groups (which we'll denote **LCA**):

$$\mathrm{Hom}(-, H) : \mathbf{LCA} \to \mathbf{LCA} \tag{16.1}$$

has properties similar to what we are used to in the dual theory we have in Functional Analysis (ex. reflexivity or properties of adjoints) and that we recover some classical results from Fourier Theory. Thanks to work by Lev Pontryagin, we have that a natural choice for H in equation (16.1) is the circle group with the usual subspace topology of $\mathbb{R}^2$. We shall henceforth label this group as $\mathbb{T}$.

Definition 16.2.1: Pontryagin Dual

Let G be a locally compact abelian group and $\mathbb{T}$ the circle group (with multiplication). Then

$$\widehat{G} := \text{Hom}(G, \mathbb{T})$$

where $\text{Hom}(G, \mathbb{T})$ is the collection of all continuous homomorphisms, is the *Pontryagin dual* of G endowed with the topology of uniform convergence of compact sets[a]. An element of $\widehat{G}$ is called a *character* of G.

[a]If you are familiar with complex analysis, this topology is commonly used for the convergence of holomorphic functions

For a better understanding of the topology on $\widehat{G}$, consider The collection of sets

$$W_K^V := \left\{ \chi \in \widehat{G} \;\middle|\; \chi(K) \subseteq V \right\}$$

where $K \subseteq G$ is a compact subset and $V \subseteq \mathbb{T}$ is a neighborhood containing the identity. Then the basis for the topology of $\widehat{G}$ is the collection of all W_K^V along with their translations. For examples of such convergence, recall power series centered at 0 ([1]) on $\mathbb{C}$ with radius R converge uniformly in $\overline{B_r(0)}$ when $r < R$, but it does not necessarily converge uniformly on $r = R$. Hence, power series technically converge in the *compact-open* topology[2].

Example 16.2.2: Pontryagin Dual

1. Let $G = \mathbb{Z}/n\mathbb{Z}$ with the discrete topology. Then:

$$\widehat{\mathbb{Z}/n\mathbb{Z}} = \text{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{T}) = \mathbb{Z}/n\mathbb{Z}$$

   To see this, notice that any $f \in \widehat{G}$ is determined by where it maps 1. Since G has finite order, $f(1)$ must have finite order. Necessarily, $f(1)$ maps to a root of unity $e^{\frac{2\pi i}{k}}$ where $1 \leq k \leq n$. Now it is easy to finish the computations to see that $\widehat{Z/n\mathbb{Z}} = \mathbb{Z}/n\mathbb{Z}$

2. Let $G = \mathbb{R}$. Then:

$$\widehat{\mathbb{R}} = \text{Hom}(\mathbb{R}, \mathbb{T}) = \mathbb{R}$$

   To see this, we will slightly modify a proof presented in Foland's *Real Analysis* Theorem 8.19. Let $\chi \in \widehat{\mathbb{R}}$. Let $a \in \mathbb{R}$ such that $\int_0^a \chi(t)dt \neq 0$. Such an a exists, or else by the Lebesgue Differentiation Theorem $\chi = 0$ a.e. Setting $A = \left(\int_0^a \chi(t)dt\right)^{-1}$, we get:

$$\chi(x) = \chi(x)AA^{-1} = A\int_0^a \chi(x)\chi(t)dx = A\int_0^a \chi(x+t)dt = A\int_x^{x+a} \chi(t)dt$$

   Since the function inside the integral continuous, χ is in fact C^1 (in fact C^∞, but we will not need this fact). Now, notice that:

$$\chi'(x) = A[\chi(x+a) - \chi(x)] = B\chi(x)$$

[1]they may be centered anywhere, but we will center them at zero for simplicity
[2]To be precise, they are locally uniformly convergent, which since $\mathbb{C}$ is locally compact is equivalent to uniform convergence on compact sets

where $B = A(\chi(a) - 1)$. This is now a simple ODE, which can be solved to show we get an exponential (and is unique by the uniqueness of a solution to an ODE). For an alternative approach, notice that

$$\frac{d}{dx}(e^{-Bx}\chi(x)) = 0$$

so $e^{-Bx}\chi(x)$ is constant. Since $\chi(0) = 1$, we have $\chi(x) = e^{Bx}$, and since $|\chi(x)| = 1$, B must be purely imaginary meaning $B = 2\pi i\xi$ for some $\xi \in \mathbb{R}$, completing the proof.

We may easily generalize this proof for when $G = \mathbb{R}^n$. In this case, the character's will be of the form $\chi(x) = e^{2\pi i \xi \cdot x}$ where $\xi \cdot x$ is the dot product.

3. Let $G = \mathbb{Z}$. Then:

$$\widehat{\mathbb{T}} = \mathrm{Hom}(\mathbb{T}, \mathbb{T}) = \mathbb{Z}$$

and similarly:

$$\widehat{\mathbb{Z}} = \mathrm{Hom}(\mathbb{Z}, \mathbb{T}) = \mathbb{T}$$

For the first, recall that $\mathbb{T} \cong \mathbb{R}/\mathbb{Z}$. Then looking at the above proof, we would consider χ to be periodic with period 1 and $e^{2\pi i\xi} = 1$ if and only if $\xi \in \mathbb{Z}$. A similar argument works for the second case.

Looking at this from a second perspective, from usual Fourier Analysis we know that the Fourier transform is a map $\mathcal{F} : L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$, and so it should be expected that $\mathbb{Z}$ and $\mathbb{T}$ are Pontryagin dual to each other.

It is natural to ask if it is always the case that that $\widehat{\widehat{G}} = G$ which demonstrates the dual spaces does not loose or introduce new information but re-paramaterizes what information G already has. We know that in general duality theory, this is not the case (the dual of the dual may be much larger than the original object, think of $C_b(X)$). If it is the case, that means that the dual $\widehat{G}$ of G contains all the required information about G. For Pontryagin duality, it is always reflexive:

Theorem 16.2.3: Pontryagin Duality Theorem

Let G be a locally compact abelian group. Then there is a natural isomorphism:

$$G \mapsto \widehat{\widehat{G}}$$

mapping $g \mapsto \mathrm{ev}_g$ where $\mathrm{ev}_g : \widehat{G} \to G$ is defined by $\mathrm{ev}_g(\chi) = \chi(g)$.

We delay the proof until the Fourier transform for groups is in hand; it is carried out in full in sections 16.2.2 and 16.2.3 and completed below. Note that in general $G \ncong \widehat{G}$ (think of $L^p(X) \ncong L^q(X) = (L^p(X))^*$), however if G is finite and abelian then $G \cong \widehat{G}$ (though not canonically). The essential take away from Pontryagin duality is that $\widehat{G}$ contains enough information on G to recover G.

16.2.4 Remark The map in theorem 16.2.3 can in fact be augmented to be a covariant functor (even an exact functor, i.e. it preserves short exact sequences). Hence, we may look at the study of extending group or the homology of groups in the Pontryagin dual.

The statement that the Fourier transform changes the basis from "positions" $\{\delta_x\}$ to "waves" $\{e^{ikx}\}$ hints at a broader duality: it changes the very nature of the "points" that constitute our space.

Let us define a function f by its values at geometric points. In the language of distributions, we can write any function as a continuous sum of Dirac deltas:

$$f(x) = \int_{\mathbb{T}^1} f(y)\delta_y(x)\,dy$$

Here, the building blocks are points $y \in \mathbb{T}^1$. A "point" is simply a linear functional that evaluates a function at a specific location: $\mathrm{ev}_y(f) = f(y)$.

Now consider the Fourier Transform. It maps a function f on the group G to a function $\hat{f}$ on the dual group $\widehat{G}$ (the set of characters). What are the "points" of this new space? They are the irreducible representations (the frequencies).

$$\hat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\,dx$$

When we transform a geometric point δ_y, it becomes a function on the dual group:

$$\widehat{\delta_y}(\chi) = \overline{\chi(y)}$$

This reveals that the Fourier Transform is an isomorphism between two different kinds of spaces:

- The *Physical Space* G, whose points are geometric locations x.
- The *Spectral Space* $\widehat{G}$, whose points are irreducible representations χ.

This is a specific instance of a grander theme in mathematics known as *Gelfand Duality*. In this framework, any commutative C^*-algebra (like the algebra of continuous functions) corresponds to a topological space (its spectrum). The Fourier Transform is the map that identifies the algebra of functions under convolution ($L^1(G)$) with the algebra of functions under pointwise multiplication ($C_0(\widehat{G})$). Effectively, we trade the geometry of the group for the geometry of its representation theory.

16.2.1 Generalizing the Fourier Transform

We may now combine all we have defined into the following

Definition 16.2.5: L^1 Fourier Transform On Group

Let $f \in L^1(G)$. Then define the *Fourier Transform* of f to be $\widehat{f} : \widehat{G} \to \mathbb{C}$ via

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}d\mu(x)$$

This definition is the proper generalization of Fourier transform. Let's say $G = \mathbb{R}$ with the usual Lebesgue measure. Then as we saw in example 16.2.2, χ is the map $\xi \mapsto e^{-2\pi i\xi}$ so that

$$\widehat{f}(\chi) = \int_{\mathbb{R}} f(x)e^{2\pi i\xi}dx$$

16.2.6 Remark the Fourier transform $f \mapsto \widehat{f}$ is the *Galfand Transformation* of f. If we take $A(\widehat{G}) = \left\{\widehat{f} \;\middle|\; f \in L^1(G)\right\}$, then $A(\widehat{G})$ is a separating sub-algebra of $C_0(\widehat{G})$. It is also not to hard to see that it is self-adjoint, and so by the Stone-Weiestrass theorem it is dense in $C_0(\widehat{G})$.

By our usual Fourier theory, we recover some important properties, namely:

Proposition 16.2.7: Properties Of Abstract Fourier Transform

Let $f, g \in L^1(G)$ for a locally compact abelian group G with Haar measure μ, and let $\chi \in \widehat{G}$. Then, writing $L_a f(x) = f(a^{-1}x)$ and $f^*(x) = \overline{f(x^{-1})}$:

1. $\widehat{f * g} = \widehat{f}\,\widehat{g}$;
2. $\widehat{L_a f}(\chi) = \overline{\chi(a)}\,\widehat{f}(\chi)$ for every $a \in G$;
3. $\overline{\widehat{f}} = \widehat{f^*}$;
4. $\widehat{f}$ is continuous with $\|\widehat{f}\|_\infty \le \|f\|_1$, and it vanishes at infinity (the Riemann-Lebesgue lemma, established in theorem 16.2.9).

***Proof*:**
For (1), Fubini's theorem and the multiplicativity $\overline{\chi(x)} = \overline{\chi(y)}\,\overline{\chi(y^{-1}x)}$ give

$$\widehat{f * g}(\chi) = \int_G \int_G f(y)\, g(y^{-1}x)\, \overline{\chi(x)}\, d\mu(y)\, d\mu(x) = \int_G f(y)\overline{\chi(y)}\Big(\int_G g(y^{-1}x)\overline{\chi(y^{-1}x)}\, d\mu(x)\Big) d\mu(y) = \widehat{f}(\chi)\,\widehat{g}(\chi).$$

For (2), the substitution $x \mapsto ax$ and invariance of μ give $\widehat{L_a f}(\chi) = \int_G f(a^{-1}x)\overline{\chi(x)}\, d\mu(x) = \int_G f(x)\overline{\chi(ax)}\, d\mu(x) = \overline{\chi(a)}\,\widehat{f}(\chi)$. For (3), using $\overline{\chi(x^{-1})} = \chi(x)$ and the invariance of μ under $x \mapsto x^{-1}$ (valid as G is unimodular),

$$\widehat{f^*}(\chi) = \int_G \overline{f(x^{-1})}\,\overline{\chi(x)}\, d\mu(x) = \int_G \overline{f(x)}\,\chi(x)\, d\mu(x) = \overline{\int_G f(x)\,\overline{\chi(x)}\, d\mu(x)} = \overline{\widehat{f}(\chi)}.$$

Finally (4): $|\widehat{f}(\chi)| \le \int_G |f|\, d\mu = \|f\|_1$, continuity is dominated convergence, and the vanishing at infinity is the content of theorem 16.2.9.

16.2.2 The Group Algebra and Bochner's Theorem

The Fourier transform of definition 16.2.5 is best read not in isolation but as the Gelfand transform (theorem 12.1.15) of a commutative Banach algebra attached to G. Making this identification precise is what brings the deeper theorems of the subject - inversion, Plancherel, and duality itself - within reach. The algebra is the group algebra.

Proposition 16.2.8: The Group Algebra

Let G be a locally compact abelian group with Haar measure μ. Equipped with convolution

$$(f * g)(x) = \int_G f(y)\, g(y^{-1}x)\, d\mu(y)$$

and the involution $f^*(x) = \overline{f(x^{-1})}$, the space $L^1(G)$ is a commutative Banach $*$-algebra. It has a unit precisely when G is discrete, and always admits an approximate unit.

***Proof*:**
Tonelli's theorem (theorem 2.5.9) and the invariance of μ give

$$\|f * g\|_1 \le \int_G \int_G |f(y)|\, |g(y^{-1}x)|\, d\mu(y)\, d\mu(x) = \|f\|_1\, \|g\|_1,$$

so convolution is well defined and submultiplicative; associativity and the identities $\|f^*\|_1 = \|f\|_1$ and $(f * g)^* = g^* * f^*$ are further applications of Fubini and invariance. Commutativity is the substitution $y \mapsto xy^{-1}$, valid because G is abelian and μ invariant. A unit would act as the point mass at e, which lies in L^1 only when $\{e\}$ is open, i.e. when G is discrete; otherwise, for a neighbourhood basis (U_α) at e the normalized indicators $u_\alpha = \mu(U_\alpha)^{-1}\mathbf{1}_{U_\alpha}$ satisfy $u_\alpha * f \to f$ in L^1 by uniform continuity of translation (lemma 6.1.1), an approximate unit.

The characters of this algebra are exactly the characters of the group; this is what fuses the abstract Gelfand machinery with the concrete Fourier transform.

Theorem 16.2.9: The Spectrum of the Group Algebra

Let G be a locally compact abelian group. The assignment

$$\chi \longmapsto \omega_\chi, \qquad \omega_\chi(f) = \int_G f(x)\, \overline{\chi(x)}\, d\mu(x) = \widehat{f}(\chi),$$

is a homeomorphism of the dual group $\widehat{G}$ onto the Gelfand spectrum $\Delta(L^1(G))$. Under it the Gelfand transform of $L^1(G)$ is the Fourier transform $f \mapsto \widehat{f}$; consequently $\widehat{f} \in C_0(\widehat{G})$ with $\|\widehat{f}\|_\infty \le \|f\|_1$ for every $f \in L^1(G)$, and the image $A(\widehat{G}) = \left\{ \widehat{f} \;\middle|\; f \in L^1(G) \right\}$ is a dense $*$-subalgebra of $C_0(\widehat{G})$.

***Proof*:**
Each $\chi \in \widehat{G}$ gives a character: $\omega_\chi(f * g) = \widehat{f * g}(\chi) = \widehat{f}(\chi)\widehat{g}(\chi) = \omega_\chi(f)\omega_\chi(g)$ by proposition 16.2.7, and $\omega_\chi \neq 0$. Conversely a character $\omega \in \Delta(L^1(G))$ is a nonzero element of $L^1(G)^* = L^\infty(G)$ (proposition 12.1.14), so $\omega(f) = \int_G f\, \overline{\psi}\, d\mu$ for a unique $\psi \in L^\infty(G)$ with $\|\psi\|_\infty \le 1$. Writing $\omega(f * g) = \omega(f)\omega(g)$ as a double integral and comparing kernels forces $\psi(xy) = \psi(x)\psi(y)$ for almost every (x, y); convolving ψ with an approximate unit yields a continuous representative satisfying the identity everywhere, so ψ agrees a.e. with a continuous homomorphism $\chi\colon G \to \mathbb{C}^\times$. From $\|\psi\|_\infty \le 1$ and $\chi(x)\chi(x^{-1}) = 1$ we get $|\chi| \equiv 1$, so $\chi \in \widehat{G}$ and

$\omega = \omega_\chi$. A comparison of subbasic neighbourhoods matches the compact-open topology on $\widehat{G}$ with the weak-$*$ topology on $\Delta(L^1(G))$. Transporting the Gelfand transform of theorem 12.1.15 through this homeomorphism turns $f \mapsto (\omega \mapsto \omega(f))$ into $f \mapsto \widehat{f}$, giving $\widehat{f} \in C_0(\widehat{G})$ and the norm bound; the range is a subalgebra by proposition 16.2.7, separates points, and satisfies $\overline{\widehat{f}} = \widehat{f^*}$, so it is dense in $C_0(\widehat{G})$ by the Stone-Weierstrass theorem ([3]).

In particular this is the rigorous form of the Riemann-Lebesgue lemma of proposition 16.2.7: $\widehat{f}$ vanishes at infinity because it is a Gelfand transform. We isolate next the class of functions that parametrizes the inversion theorem.

Definition 16.2.10: Positive-Definite Function

A function $\varphi\colon G \to \mathbb{C}$ is *positive-definite* if for every finite set $x_1, \ldots, x_n \in G$ and every $c_1, \ldots, c_n \in \mathbb{C}$,

$$\sum_{i,j=1}^{n} c_i\, \overline{c_j}\, \varphi(x_j^{-1} x_i) \geq 0.$$

The case $n = 1$ gives $\varphi(e) \geq 0$, and $n = 2$ gives $\varphi(x^{-1}) = \overline{\varphi(x)}$ and $|\varphi(x)| \leq \varphi(e)$, so a positive-definite function is bounded with $\|\varphi\|_\infty = \varphi(e)$. The basic examples are the Fourier-Stieltjes transforms $\check{\nu}(x) = \int_{\widehat{G}} \chi(x)\, d\nu(\chi)$ of finite positive measures ν on $\widehat{G}$: the defining sum is $\int_{\widehat{G}} |\sum_i c_i\, \chi(x_i)|^2\, d\nu(\chi) \geq 0$. Bochner's theorem is the converse.

Theorem 16.2.11: Bochner's Theorem

Let G be a locally compact abelian group. A continuous function $\varphi\colon G \to \mathbb{C}$ is positive-definite if and only if there is a unique finite positive Radon measure ν on $\widehat{G}$ with

$$\varphi(x) = \int_{\widehat{G}} \chi(x)\, d\nu(\chi) \qquad (x \in G).$$

***Proof*:**
The Fourier-Stieltjes transform of a finite positive measure is continuous and positive-definite by the remark above, and any representing ν then has total mass $\nu(\widehat{G}) = \check{\nu}(e) = \varphi(e)$; this is the "if" direction. For the converse, integrating definition 16.2.10 against $f \in C_c(G)$ in each variable gives the continuous Hermitian form

$$Q(f) = \int_G \int_G \varphi(y^{-1}x)\, f(x)\, \overline{f(y)}\, d\mu(x)\, d\mu(y) \geq 0. \tag{$*$}$$

Cauchy-Schwarz for Q, applied against an approximate unit and iterated through the spectral-radius identity $\|\widehat{f}\|_\infty = r(f)$ of theorem 12.1.15, yields the estimate

$$\left| \int_G f\, \overline{\varphi}\, d\mu \right| \leq \varphi(e)\, \|\widehat{f}\|_\infty \qquad (f \in L^1(G)). \tag{$**$}$$

By $(**)$ the number $\int_G f\, \overline{\varphi}\, d\mu$ depends only on $\widehat{f}$ and is dominated by its sup-norm, so

$$L(\widehat{f}) := \int_G f(x)\, \overline{\varphi(x)}\, d\mu(x)$$

is a well-defined bounded functional on the dense subalgebra $A(\widehat{G}) \subseteq C_0(\widehat{G})$ (theorem 16.2.9), hence extends to $C_0(\widehat{G})$. It is positive: for $g \in L^1(G)$,

$$L(|\widehat{g}|^2) = L(\widehat{g^* * g}) = \int_G (g^* * g)\,\overline{\varphi}\,d\mu = Q(\overline{g}) \geq 0,$$

the last equality by expanding $g^* * g$ and substituting, and $Q(\overline{g}) \geq 0$ by $(*)$; such squares are dense in the nonnegative cone of $C_0(\widehat{G})$. The Riesz-Markov theorem (theorem 4.2.1) represents L as $L(h) = \int_{\widehat{G}} h\,d\nu$ for a finite positive Radon measure ν. Unwinding, for every $f \in L^1(G)$,

$$\int_G f\,\overline{\varphi}\,d\mu = \int_{\widehat{G}} \widehat{f}\,d\nu = \int_G f(x)\,\overline{\Big(\int_{\widehat{G}} \chi(x)\,d\nu(\chi)\Big)}\,d\mu(x),$$

so φ and the continuous function $\check{\nu}$ pair identically against all of $L^1(G)$ and therefore coincide. Uniqueness of ν is the density of $A(\widehat{G})$ in $C_0(\widehat{G})$.

Bochner already forces the dual group to be rich enough to see every point of G.

Corollary 16.2.12: Characters Separate Points

Let G be a locally compact abelian group. For every $x \in G$ with $x \neq e$ there is a character $\chi \in \widehat{G}$ with $\chi(x) \neq 1$.

***Proof*:**
Since G is Hausdorff there is an open neighbourhood U of e with $Ux \cap U = \emptyset$. Choose $h \in C_c(G)$, $h \neq 0$, supported in U, and set $\varphi = h^* * h$, a continuous positive-definite function with $\varphi(e) = \|h\|_2^2 > 0$. Substituting $z = y^{-1}$,

$$\varphi(x) = \int_G \overline{h(z)}\,h(zx)\,d\mu(z) = 0,$$

since $h(z)h(zx) = 0$ pointwise ($z \in U$ and $zx \in U$ are incompatible). By Bochner (theorem 16.2.11) $\varphi = \check{\nu}$ for a positive measure ν, so $\int_{\widehat{G}} \chi(x)\,d\nu(\chi) = \varphi(x) = 0 \neq \varphi(e) = \nu(\widehat{G}) = \int_{\widehat{G}} 1\,d\nu(\chi)$. Were $\chi(x) = 1$ throughout the support of ν the two integrals would agree; hence $\chi(x) \neq 1$ for some $\chi \in \widehat{G}$.

16.2.3 Fourier Inversion and Plancherel

One of the most important results in Fourier Analysis is the ability to recover f from $\widehat{f}$ under suitable conditions, that is the *Fourier Inversion Theorem.* This result naturally generalizes. Note that since $f \in L^1(G)$, we may treat f as being define μ-a.e., and so we may only recover f μ-a.e. in the following theorem unless f has more regularity:

Theorem 16.2.13: Fourier Inversion Theorem For Groups

Let G be a locally compact abelian group with Haar measure μ. Then there is a unique Haar measure $\widehat{\mu}$ on $\widehat{G}$ such that whenever $f \in L^1(G)$ and $\widehat{f} \in L^1(\widehat{G})$, then μ-a.e. we have:

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x)\, d\widehat{\mu}(\chi)$$

If f is continuous, then it is true for all $x \in G$.

***Proof*:**
Write $P^1(G)$ for the integrable continuous positive-definite functions and $\mathcal{B}(G)$ for their linear span, a dense subspace of $L^1(G)$ containing every $g^* * g$ with $g \in C_c(G)$. Bochner's theorem (theorem 16.2.11) attaches to each $u \in P^1(G)$ a finite positive measure ν_u on $\widehat{G}$ with $u = \check{\nu}_u$, and the closing step of its proof records

$$\int_{\widehat{G}} \widehat{k}\, d\nu_u = \int_G k\, \overline{u}\, d\mu \qquad (k \in L^1(G),\ u \in P^1(G)). \tag{†}$$

A common Haar measure. For $u, v \in P^1(G)$ the measures $\widehat{u}\, d\nu_v$ and $\widehat{v}\, d\nu_u$ coincide. Testing each against $\widehat{w}$ ($w \in L^1(G)$; recall $A(\widehat{G})$ is dense in $C_0(\widehat{G})$) and applying (†) with $k = w * u$ and with $k = w * v$,

$$\int_{\widehat{G}} \widehat{w}\, \widehat{u}\, d\nu_v = \int_G (w * u)\, \overline{v}\, d\mu = \langle w * u, v\rangle, \qquad \int_{\widehat{G}} \widehat{w}\, \widehat{v}\, d\nu_u = \langle w * v, u\rangle,$$

and the right-hand sides agree because a positive-definite function is self-adjoint ($u^* = u$, $v^* = v$, from definition 16.2.10) and $L^1(G)$ is commutative:

$$\langle w * u, v\rangle = \langle w, v * u^*\rangle = \langle w, u * v^*\rangle = \langle w * v, u\rangle.$$

For each $\chi \in \widehat{G}$ some $u \in P^1(G)$ has $\widehat{u}(\chi) > 0$ (as $A(\widehat{G})$ is dense in $C_0(\widehat{G})$, some $\widehat{g}$ is nonzero at χ, and $u = g^* * g$ has $\widehat{u}(\chi) = |\widehat{g}(\chi)|^2$). Hence the prescription $d\widehat{\mu} = d\nu_u / \widehat{u}$ on $\{\widehat{u} > 0\}$ is unambiguous and defines a positive Radon measure $\widehat{\mu}$ on $\widehat{G}$ with

$$d\nu_u = \widehat{u}\, d\widehat{\mu} \qquad (u \in P^1(G)).$$

Multiplying u by a character χ_0 keeps it positive-definite and translates both $\widehat{u}$ and ν_u by χ_0, so $\widehat{\mu}$ is translation-invariant: a Haar measure on $\widehat{G}$, unique up to the scale this construction fixes (theorem 6.2.1).

Inversion. For $u \in P^1(G)$, substituting $d\nu_u = \widehat{u}\, d\widehat{\mu}$ into $u = \check{\nu}_u$ gives $u(x) = \int_{\widehat{G}} \widehat{u}(\chi)\, \chi(x)\, d\widehat{\mu}(\chi)$. More generally, for any $f \in L^1(G)$ and $u \in P^1(G)$, writing $u = \check{\nu}_u$ and applying Fubini,

$$(f * u)(x) = \int_{\widehat{G}} \widehat{f}(\chi)\, \chi(x)\, d\nu_u(\chi) = \int_{\widehat{G}} \widehat{f * u}(\chi)\, \chi(x)\, d\widehat{\mu}(\chi),$$

so inversion holds for every convolution $f * u$. If now $\widehat{f} \in L^1(\widehat{G})$, take an approximate unit $u_\alpha = e_\alpha^* * e_\alpha \in P^1(G)$ ($e_\alpha \in C_c(G)$): then $f * u_\alpha \to f$ in $L^1(G)$ while $\widehat{f * u_\alpha} = \widehat{f}\, \widehat{u_\alpha} \to \widehat{f}$ in $L^1(\widehat{\mu})$, and passing to the limit gives $f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\, \chi(x)\, d\widehat{\mu}(\chi)$ for μ-a.e. x, and everywhere when f is continuous.

The construction yields a fact used repeatedly below.

Corollary 16.2.14: Injectivity of the Fourier Transform

If $f \in L^1(G)$ has $\widehat{f} \equiv 0$, then $f = 0$.

Proof:

For every $u \in P^1(G)$ the convolution $f * u \in L^1(G)$ has $\widehat{f * u} = \widehat{f}\,\widehat{u} \equiv 0 \in L^1(\widehat{G})$, hence $f * u = 0$ by inversion (theorem 16.2.13). Taking $u = u_\alpha = e_\alpha^* * e_\alpha$ an approximate unit gives $f = \lim_\alpha f * u_\alpha = 0$.

Another very important result is Fourier Analysis on L^2;

Theorem 16.2.15: Plancherel Inversion Theorem For Groups

Let G be a locally compact abelian group, μ a Haar measure on G, and $\widehat{\mu}$ the dual measure on $\widehat{G}$ given in theorem 16.2.13. If $f\colon G \to \mathbb{C}$ is continuous with compact support, then $\widehat{f} \in L^2(\widehat{G})$ and

$$\int_G |f(x)|^2 d\mu(x) = \int_{\widehat{G}} \left|\widehat{f}(\chi)\right|^2 d\widehat{\mu}(\chi)$$

In particular, the Fourier transform is an L^2_μ isometry from $C_c(G)$ to $L^2_{\widehat{\mu}}(\widehat{G})$.

Proof:

For $f \in C_c(G)$ the function $f^* * f$ is continuous, integrable, and positive-definite, with transform $\widehat{f^* * f} = |\widehat{f}|^2$ and value $(f^* * f)(e) = \|f\|_2^2$ at the identity, so it lies in $\mathcal{B}(G)$. Evaluating the inversion theorem (theorem 16.2.13) at $x = e$ therefore gives

$$\|f\|_2^2 = (f^* * f)(e) = \int_{\widehat{G}} |\widehat{f}(\chi)|^2\, d\widehat{\mu}(\chi) = \|\widehat{f}\|_2^2.$$

Thus $f \mapsto \widehat{f}$ is a linear isometry of $(C_c(G), \|\cdot\|_2)$ into $L^2(\widehat{G})$, which (as $C_c(G)$ is L^2-dense) extends uniquely to an isometry $\mathcal{F}\colon L^2(G) \to L^2(\widehat{G})$; it is onto, hence unitary, as shown next.

That the isometry

$$\mathcal{F}\colon L^2_\mu(G) \to L^2_{\widehat{\mu}}(\widehat{G})$$

is *onto* – a *unitary operator* – follows from the module structure of its range. Let $R = \mathcal{F}(L^2(G))$, closed because $\mathcal{F}$ is an isometry of a complete space. For $g \in L^1(G)$ and $\xi \in L^2(G)$, Young's inequality gives $g * \xi \in L^2(G)$ and $\mathcal{F}(g * \xi) = \widehat{g}\,\mathcal{F}\xi$, so $\widehat{g}\,R \subseteq R$; since $A(\widehat{G}) = \{\widehat{g} \mid g \in L^1(G)\}$ is dense in $C_0(\widehat{G})$ (theorem 16.2.9), R is invariant under multiplication by all of $C_0(\widehat{G})$. A closed subspace of $L^2(\widehat{\mu})$ stable under multiplication by $C_0(\widehat{G})$ has the form $R = \mathbf{1}_E\, L^2(\widehat{\mu})$ for some measurable $E \subseteq \widehat{G}$: for $\psi \in R$ the products $\varphi\psi$ ($\varphi \in C_0(\widehat{G})$) are L^2-dense in $\mathbf{1}_{\{\psi \neq 0\}} L^2(\widehat{\mu})$, because $C_0(\widehat{G})$ is dense in $L^2(|\psi|^2\, d\widehat{\mu})$, a finite Radon measure. Every $\widehat{f} = \mathcal{F}f$ ($f \in L^1 \cap L^2$) lies in R, hence vanishes a.e. off E; were $\widehat{\mu}(\widehat{G} \setminus E) > 0$, a point χ_0 in the support of $\mathbf{1}_{\widehat{G}\setminus E}\widehat{\mu}$ would admit $f \in L^1 \cap L^2(G)$ with $\widehat{f}(\chi_0) \neq 0$ (take $f = f_0 * u$ with $\widehat{f_0}(\chi_0) \neq 0$ from theorem 16.2.9 and $u \in P^1(G)$ with $\widehat{u}(\chi_0) > 0$), so $\widehat{f} \neq 0$ on a neighbourhood of χ_0 meeting $\widehat{G} \setminus E$ in positive measure, a contradiction. Hence $E = \widehat{G}$ up to a null set, $R = L^2(\widehat{G})$, and $\mathcal{F}$ is unitary. When G is compact, $L^2(G) \subseteq L^1(G)$, so the L^1 transform already covers L^2.

The adjoint of this unitary is the inverse Fourier transform.

Theorem 16.2.16: Dual Group Fourier Transform

Let G be a locally compact abelian group with Haar measure μ and let $\mathcal{F}$ be the L^2 Fourier transform. Then the adjoint of the Fourier Transform restricted to continuous functions on compact support is the Inverse Fourier Transform

$$L^2_{\widehat{\mu}}(\widehat{G}) \to L^2_{\mu}(G)$$

***Proof*:**
For $f \in C_c(G)$ and $g \in C_c(\widehat{G})$, Fubini's theorem (theorem 2.5.9) gives

$$\langle \mathcal{F}f, g\rangle_{\widehat{G}} = \int_{\widehat{G}} \widehat{f}(\chi)\,\overline{g(\chi)}\,d\widehat{\mu} = \int_G f(x)\,\overline{\int_{\widehat{G}} g(\chi)\,\chi(x)\,d\widehat{\mu}(\chi)}\,d\mu(x) = \langle f, \check{g}\rangle_G,$$

so $\mathcal{F}^*g(x) = \int_{\widehat{G}} g(\chi)\,\chi(x)\,d\widehat{\mu}(\chi)$ is the inverse Fourier transform of theorem 16.2.13, recognized as the adjoint of $\mathcal{F}$. Since $\mathcal{F}$ is unitary (theorem 16.2.15), this adjoint $\mathcal{F}^* = \mathcal{F}^{-1}$ is its two-sided inverse.

Notable example would be when $G = \mathbb{T}$ so that $\widehat{G} = \mathbb{Z}$ and the Fourier transform computes the Fourier series of periodic functions. If G is finite, then we will get the *discrete Fourier Transform* from this process.

16.2.4 Pontryagin Duality

The machinery now assembled proves theorem 16.2.3 in full: the canonical map $\alpha\colon G \to \widehat{\widehat{G}}$, $\alpha(x)(\chi) = \chi(x)$, is an isomorphism of topological groups.

***Proof*:**
That α is a continuous homomorphism is immediate: $\alpha(x)$ is the evaluation $\chi \mapsto \chi(x)$, a character of $\widehat{G}$, and $x \mapsto \alpha(x)$ is continuous for the compact-open topologies.

Injectivity. If $\alpha(x) = 1$ then $\chi(x) = 1$ for every $\chi \in \widehat{G}$, so $x = e$ by corollary 16.2.12. Hence α is injective.

Embedding. A comparison of subbasic neighbourhoods shows α is a homeomorphism onto its image, and since G is locally compact that image $\alpha(G)$ is a closed subgroup of $\widehat{\widehat{G}}$.

Surjectivity. The Plancherel theorem (theorem 16.2.15) makes both $\mathcal{F}_G\colon L^2(G) \to L^2(\widehat{G})$ and $\mathcal{F}_{\widehat{G}}\colon L^2(\widehat{G}) \to L^2(\widehat{\widehat{G}})$ unitary. Pull the transform of $\psi \in L^2(\widehat{G})$ back along α: since

$\alpha(x)(\chi) = \chi(x)$,

$$\big(\mathcal{F}_{\widehat{G}}\psi\big)\big(\alpha(x)\big) = \int_{\widehat{G}} \psi(\chi)\,\overline{\chi(x)}\,d\widehat{\mu}(\chi) = \overline{\mathcal{F}^*_G(\overline{\psi})(x)},$$

the inverse transform of theorem 16.2.16. As $\mathcal{F}^*_G = \mathcal{F}^{-1}_G$ is unitary, $\big\|(\mathcal{F}_{\widehat{G}}\psi)\circ\alpha\big\|_{L^2(\mu)} = \|\psi\|_{L^2(\widehat{\mu})} = \|\mathcal{F}_{\widehat{G}}\psi\|_{L^2(\widehat{\widehat{\mu}})}$. The left-hand side is $\int_{\widehat{\widehat{G}}} |\mathcal{F}_{\widehat{G}}\psi|^2\,d(\alpha_*\mu)$, and as ψ runs through $L^2(\widehat{G})$ the transform $\mathcal{F}_{\widehat{G}}\psi$ runs through all of $L^2(\widehat{\widehat{\mu}})$; hence

$$\int_{\widehat{\widehat{G}}} |F|^2\,d(\alpha_*\mu) = \int_{\widehat{\widehat{G}}} |F|^2\,d\widehat{\widehat{\mu}} \qquad (F \in L^2(\widehat{\widehat{\mu}})),$$

so $\alpha_*\mu = \widehat{\widehat{\mu}}$. The pushforward $\alpha_*\mu$ is carried by the closed subgroup $\alpha(G)$, whereas the Haar measure $\widehat{\widehat{\mu}}$ charges every nonempty open set; thus $\widehat{\widehat{G}} \setminus \alpha(G)$ is open and null, hence empty. Therefore $\alpha(G) = \widehat{\widehat{G}}$, and α is a topological isomorphism.

16.2.5 Interesting Uses

Pontryagin has a couple of very interesting consequences, perhaps a good starting point is that the classification of compact abelian groups is just as difficult as the classification of general abelian groups using the following result:

Theorem 16.2.17: Compactness Criterion For LCA Group

Let G be a locally compact abelian group. Then G is compact if and only if $\widehat{G}$ is discrete. Conversely, G is discrete if and only if $\widehat{G}$ is compact.

***Proof*:**
The fact that G is compact implies $\widehat{G}$ is discrete and that G is discrete implies $\widehat{G}$ is compact comes straight from the topology of G and $\widehat{G}$ (namely that it is the compact-open topology). To get the converse of both, we simply apply Pontryagin duality.

Proposition 16.2.18: Unit In $L^1(G)$

Let G be a locally compact abelian group with Haar measure μ. Then $L^1(G)$ has a unit if and only if G is discrete.

***Proof*:**
We know that $L^1(G)$ always has an approximate identity (think of good kernels). The identity of $L^1(G)$ would be an element such that $f * \mathrm{id} = \mathrm{id} * f = f$, so $\widehat{f}\cdot\widehat{\mathrm{id}} = \widehat{f}$. We see through this formula that $\widehat{\mathrm{id}}$ would have to be the Dirac-delta distribution at 1 with norm 1. This distribution is a function if and only if G is discrete.

Representation Theory

Another place this comes in is within representation theory. Recall in [14, chapter 24.4] there was a quick word on Fourier transforms in representation theory. In classical representation theory of finite groups, we define the character to be $\chi : G \to \mathbb{C}^*$. Since G is finite, G maps to a root the roots of unity in $\mathbb{T}$, and hence the notion of character in classical representation theory aligns with our definitions. Then we may define the inner product of characters to be:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g) \overline{\chi_2}(g)$$

which will give a way to decompose representations into irreducible components via irreducible characters. If G is abelian, then may translate the decomposition theorem for representations of finite abelian groups to the decomposition of$L^2(G)$ via

$$L^2(G) = \bigoplus_{i=1}^{n} E_n$$

and in the infinite case we would have take the closure

$$L^2(G) = \widehat{\bigoplus_{i \in \mathbb{N}} E_n}$$

where E_n are the subspaces given by the usual orthonormal basis for L^2 when doing Fourier analysis. Hence, representation theory and Fourier analysis are linked!!

Class Field Theory

For those how have seen some class field theory, they have seen the Artin Reciprocity map. Recall that the idele group is a locally compact group, and so it is precisely the context in which we expect Pontryagian duality to be applied! In the local case, we have the Artin map $K^* \cong \mathrm{Gal}_K(K^{\mathrm{ab}})$ between two locally compact groups; taking the Pontryagin dual will preserve this mapping and let us work with the characters of these groups! In the global case using the Adèle ring and Idele group, we shall not get an isomorphism in the Pontryagin dual but a perfect pairing.

Part V

Probability

(I want to eventually include a part that rigorously covers probability here. I already have a "Elementary Probability Theory" textbook, so this will be the more advanced material that would require a proper book on real analysis)

Definition 16.2.19: Kernel

Let $(X, \mathcal{M})$, $(Y, \mathcal{N})$ be measurable spaces. Then a *kernel* is a function:

$$\mu : X \times \mathcal{N} \to [0, \infty]$$ check this codomain

where:

1. For a fixed $B \in \mathcal{N}$, $x \mapsto \mu(x, B)$ is $\mathcal{M}$-measurable
2. For a fixed $x \in X$, the map $B \mapsto \mu(x, B)$ is a measure on

In other words, μ can be thought of as a map from X to the space of measurable function on Y.
If clear from context, a kernel is denoted $\mu : X \to Y$.

Example 16.2.20: Kernel

1. Let $\delta_x : X \to X$ denote the Dirac measures δ_x.

Part VI

Appendix, Index, Bibliography

A

Decay vs. Smooth Table for Fourier

This table represents the relationship between the smoothness of a function and the decay of its Fourier transform:

Function Space	**Symbol**	**Property of** $f(x)$	**Property of** $\hat{f}(\xi)$ **(Dual)**
The "Nice" Spaces (High Regularity)			
Real Analytic	$C^{\omega}(\mathbb{R})$	**Holomorphic Extension:** Analytic in a strip	**Exponential Decay:** $\lvert\hat{f}(\xi)\rvert \leq Ce^{-a\lvert\xi\rvert}$ ($\iff$)
Compact Smooth	$C_c^{\infty}(\mathbb{R})$	**Extreme Locality:** Zero outside a box	**Extreme Regularity:** Entire Analytic Function
Schwartz	$\mathcal{S}(\mathbb{R})$	**Rapid Decay:** Faster than polynomial	**Rapid Decay:** Faster than polynomial ($\iff$)
Finite Smoothness	$C^k(\mathbb{R})$	**Finite Regularity:** k derivatives exist	**Polynomial Decay:** Decays like $1/\lvert\xi\rvert^k$
The "Rough" Spaces (Low Regularity)			
Wiener Algebra	$A(\mathbb{R})$	**Continuous + Integrable:** f is bounded, continuous	**Absolute Convergence:** $\hat{f} \in L^1$ ($\implies f \in C^0$)
Square Integrable	$L^2(\mathbb{R})$	**Finite Energy:** Isometry	**Finite Energy:** Isometry ($\iff$ Plancherel)
Integrable	$L^1(\mathbb{R})$	**Finite Area:** (Riemann-Lebesgue)	**Slow Decay:** Continuous, $\to 0$ ($\not\Longrightarrow L^1$)
Distributions (Compact Support)	$\mathcal{E}'(\mathbb{R})$	**Singular:** Dirac Deltas, etc.	**Polynomial Growth:** $\lvert\hat{f}(\xi)\rvert \leq C(1+\lvert\xi\rvert)^N$ ($\iff$)

Table A.1: The Duality of Regularity and Decay in Real Analysis

This next table illustrates the "Hierarchy of Continuous Regularity". It answers the question: How jagged is the function, and how much high-frequency energy is required to describe that jaggedness?

As you move down the table, the function becomes "rougher" (more singular), and consequently, its Fourier Transform $\hat{f}(\xi)$ decays slower.

Function Class	**Property**	**Example** $f(x)$	**Decay of** $\hat{f}(\xi)$
High Regularity (Fast Decay)			
Real Analytic (C^ω)	Holomorphic Strip (Super-Smooth)	$e^{-\pi x^2}$ (Gaussian)	**Exponential** $O(e^{-a\lvert\xi\rvert})$
Smooth (C_c^∞)	infinitely diff.	$e^{-1/(1-x^2)}$ (Bump Func)	**Rapid Decay** $O(\lvert\xi\rvert^{-N})$ for all N
Finite Smoothness (C^k)	k derivatives (Continuous $f^{(k)}$)	$x^k\lvert x\rvert$ (Splines)	**Polynomial** $o(\lvert\xi\rvert^{-k})$
Medium Regularity (Algebraic Decay)			
Lipschitz ($C^{0,1}$)	Bounded Derivative (Finite Slope)	$\lvert x\rvert$ (Triangle Wave)	**Linear** $O(\lvert\xi\rvert^{-1})$
Hölder ($C^{0,\alpha}$)	Infinite Derivative (Vertical Cusp)	$\lvert x\rvert^\alpha$ ($0 < \alpha < 1$)	**Fractional** $O(\lvert\xi\rvert^{-\alpha})$
Bounded Variation (BV)	Finite Total Variation (Step / Monotonic)	$\text{sign}(x)$ (Square Wave)	**Linear** $O(\lvert\xi\rvert^{-1})$
Low Regularity (Slow Decay)			
Wiener Algebra ($A(\mathbb{T})$)	Abs. Conv. Fourier (Always Continuous)	$\sum \frac{\sin(nx)}{n^2}$ (Weierstrass)	**Summable** $\hat{f} \in \ell^1$
Continuous (C^0)	**Not** BV (Wild Oscillations)	$x\sin(1/x)$ (Damped Osc.)	**Arbitrarily Slow** $o(1)$ (Riemann-Lebesgue)
Integrable (L^1)	Finite Area	–	**Arbitrarily Slow** $o(1)$ (Riemann-Lebesgue)
Distributions ($\mathcal{D}'$)	Singular (Not a function)	$\delta(x)$ (Impulse)	**Growth** $O(\lvert\xi\rvert^N)$

Table A.2: The Precise Spectrum of Continuity and Fourier Decay (Corrected)

In this hierarchy, the decay rate $1/|\xi|^s$ tells us exactly how "smooth" the function is.

1. Integer Powers ($s = k$): These correspond to existence of Weak Derivatives. A function has a weak derivative if its slope is well-behaved enough to have finite energy (L^2), even if it has kinks or jumps. For example, the absolute value function $|x|$ has no classical derivative at 0, but its "weak derivative" is the Step Function (which is perfectly valid in physics as a force field).
2. Fractional Powers ($s = k.5$): These correspond to the Geometry of the Singularity. They tell us whether a sharp point is a "needle" (Cusp), a "corner" (Lipschitz), or a "smooth turn" (Curvature). The higher the power s, the more "damped" the high frequencies are, and the harder it is for the function to make sudden changes.

Decay Rate	**Smoothness Class**	**Allowed Singularity**	**Physical Intuition**
$1/\lvert\xi\rvert$	L^2 (Finite Energy) Discontinuous	**Jump** (Step Function)	**Teleportation** Position breaks instantly.
$1/\lvert\xi\rvert^{1.5}$	C^0 (Continuous) Infinite Slope	**Cusp** (Vertical Tangent)	**Lightning Rod** Sharpest possible point.
$1/\lvert\xi\rvert^2$	$C^{0,1}$ (Lipschitz) Finite Slope	**Corner** (Kink)	**Brick Wall** Velocity breaks instantly.
$1/\lvert\xi\rvert^{2.5}$	C^1 (Smooth) Continuous Slope	**Curvature Singularity** (Infinite Curvature)	**Tight Turn** Acceleration explodes.
$1/\lvert\xi\rvert^3$	$C^{1,1}$ (Bounded Curvature) Continuous Velocity	**Acceleration Jump** (Finite Curvature Jump)	**Kick** Force turns on instantly.

Table A.3: The Fine Structure of Fourier Decay and Singularities

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